Organization Management

Use **VCF Automation** as a cloud administrator (or organization administrator) to set up projects for application teams, allocate resources, and apply governance.

In VCF Automation, an organization is a top level entity that represents a line of business or a tenant. Each organization operates in a secure and isolated environment where users can access only allocated services and resources. Organizations consume resources and cloud services from a shared pool of resources. The resources allocated to organizations can be further allocated to application teams which are modeled in VCF Automation as projects. Organizations have users with varying levels of permissions depending on their role and duties.

Organizations in VCF Automation are created by the provider administrator in the VCF Automation Provider Management UI. The provider administrator can create two types of organizations: an All Apps organization and a VM Apps organization. VM Apps organizations can run VM-based applications, while All Apps organizations are enabled to run a comprehensive set of built-in cloud services. For more information about creating VCF Automation organizations in the VCF Automation Provider Management UI, see Managing Organizations.

Regardless of the type, each organization has its own organization UI where users log in to consume resources and services. Every organization has its own identity source that they connect to and can import users from.

All Apps Organizations in VCF Automation

All Apps organizations offer application teams a comprehensive set of built-in cloud services to provision VMs, Kubernetes, networking, volumes, Secret Store, databases, Harbor container registries, external DNS, certificates, and AI workloads. These cloud services can be consumed on a self-service basis through a curated catalog or self-service interfaces.

Organizations enable line-of-business or tenant organization administrators to further organize and govern resources that are allocated to them among application teams through the following capabilities:

- Resource management for project teams.
- Centralized content management for the organization.
- Policy-based governance on usage of resources.
- Quota-based management of networking resources.
- Workflow orchestration and extensibility through event subscriptions.
- Integration with identity providers of their choice.

For information on working with All Apps organizations, see the Managing Organizations for All Apps in guide.

Provider Consumption Organizations in VCF Automation

A Provider Consumption organization (PCO) is a special type of All Apps organization in VCF Automation that the provider administrator enables in the VCF Automation Provider Management UI. The provider administrator can enable only one Provider Consumption organization. For more information about PCOs, see Create a Provider Consumption Organization.

The PCO is intended for use when the provider wants to have their own consumption service for their users and/or they want to share catalog items with other organizations under their management.

PCOs are administered in the same way as other All Apps organizations, but they offer some additional capabilities on top of what a standard All Apps organization provides.

- PCOs come with a preconfigured VCF Operations orchestrator instance that is embedded in the organization UI, which you access on the **Orchestrate** tab.
- As a PCO administrator, you can publish VCF Operations orchestrator objects as catalog items to other organizations that use the same embedded VCF Operations orchestrator instance.
- You can publish catalog items to other organizations that use VM images from a provider content library.

For more information about managing your VCF Operations orchestrator integration, see Managing VMware Cloud Foundation Operations Orchestrator.

VM Apps Organizations in VCF Automation

VM Apps organizations help existing VMware Aria Automation users transition to VCF Automation 9.0 without any impact to infrastructure, automation, or end-user experience.

In VM Apps organizations, the allocation of resources, like cloud accounts, cloud zones, profiles, image mappings and so on, is contained in the organization itself. The organization administrator defines and works with the infrastructure; the provider administrator does not allocate resources to organization of this type.

> **Important:**
>
> If you upgrade an existing VMware Aria Automation 8.x deployment to VCF Automation 9, your organizations are automatically mapped to VCF Automation for VM Apps organizations. You continue to use vIDB and VMware Aria Suite Lifecycle to manage vIDM to maintain legacy authentication.

For information on working with VM Apps organizations, see the Managing Organizations in for VM Apps guide.

Best Practices for Organizations in VCF Automation

When to use All Apps vs VM Apps?

- All Apps organizations are used when organizations are ready to leverage the comprehensive set of built-in cloud services to provision VMs, Kubernetes, and so on. The underlying infrastructure requires vSphere Supervisor clusters and is managed in the VCF Automation Provider Management UI. Talk to your provider administrator about how you can leverage these capabilities.
- VM Apps organizations are intended for current users of VMware Aria Automation who want to continue their journey with minimal disruptions. The VM Apps organization is fully self-contained with built-in infrastructure management and consumption capabilities.

When to use blueprints vs the IaaS self-service console?

- Blueprints and the catalog offer a controlled manner to build and curate infrastructure. They are prescriptive in terms of the provisioned resources and so are suitable to deploy applications in production. They are also suitable for end users who do not want to concern themselves with details of the infrastructure.
- The IaaS self-service console offers more flexibility and is more suitable for development purposes.

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

Consuming Administration Services in VCF Automation

As an organization administrator, you can use functionality in the VCF Automation organization UI that is additionally installed by your provider administrator.

- Managing VKS Clusters with VKS Cluster Management
- Using VMware Cloud Foundation Encryption Management in a VCF Automation organization

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Managing Organizations in VMware Cloud Foundation Automation

With VMware Cloud Foundation Automation (VCF Automation), you, as an organization administrator, and your application teams can view and consume infrastructure that the provider administrator made available to your organization. As an organization administrator, you create projects, manage users, and define resource limits for running workloads.

The underlying infrastructure in VCF Automation is based on VMware vSphere Supervisor, which serve as the foundational structure for resource allocation. Supervisor-based infrastructure supports modern application use cases for

provisioning containers and VMs. Provider administrators allocate infrastructure resources to organizations using regional quotas, while organization administrators use namespaces to distribute the allocated infrastructure to different projects within the organization.

Infrastructure Concepts

VCF Automation enables private cloud functionality by providing a multi-tenant management plane. Rather than abstracting the underlying infrastructure, VCF Automation routes organizations to their specific resources and services through a centralized interface.

Regions. A region acts as a unified networking and management boundary, comprising all supervisors linked to a common NSX Manager. Regions enable cross-vCenter network, resource, and IaaS service management.

Zones. A zone represents one or multiple vSphere clusters that provide the compute, memory, and storage resources for the organization.

Region Quota. A region quota defines the compute and storage resources that are allocated to an organization in a given region. The organization can receive quotas from multiple regions. The provider administrator decides from which regions to give a quota to an organization. The region quota can be split across more than one zone in a region. A region quota includes the following:

- CPU and memory utilization limits which represent the upper bound of quota in a region.
- CPU and memory reservations which represent the guaranteed capacity that is allocated in a region.
- VM classes which represent the T-shirt-sized classes of compute that are available to the organization.
- Storage classes which represent the storage policies and the quota that is assigned in a region.

Resources for Application Teams

Resources allocated to the organization in the form of a region quota can be further subdivided and allocated to application teams.

Project. Projects are logical representations of users and user groups and the resource pools that those users are entitled to. The resource pools that projects use are called namespaces. Resources, like VMs and Kubernetes clusters, are provisioned on namespaces.

Namespace. A vSphere Namespace is a resource pool that comprises CPU, memory, and storage resources that are assigned to projects inVCF Automation. Namespaces derive their quota from the overall region quota of the organization and come with controls such CPU, memory, and storage usage. Each namespace is bound to the scope of the project that it is associated with. Each project can contain multiple namespaces. Because the region quota can be split across multiple vSphere zones, the resource pool of a namespace can also be split across multiple vSphere zones. As an organization administrator, you can configure which zones a namespace can derive its resource limits from.

Namespace Class. Namespace classes are templates for defining guardrails on parameters that are used to create a namespace, like CPU limits, storage limits, permissible VM and storage classes, and content libraries. Namespace classes are a prerequisite for creating namespaces. In the namespace class, you can set default values for CPU and memory limits and reservations. If a namespace class has a zone defined, default values for namespace limits and reservations are used from the zone section. As an organization administrator, you create namespace classes that fit your project needs. You can create one or more namespaces using the default namespace classes that come preconfigured in your organization. You can also override those defaults in a given project to suit specific configurations.

An example of an organization quota:

- Region *`region-one`* has three vSphere Zones - *`vs-zone-1`*, *`vs-zone-2`*, and *`vs-zone-3`*.
- Organization *`org-1`* has a region quota of 25 GHz of CPU in *`vs-zone-1`* and 50 GHz of CPU in *`vs-zone2`* for a total quota of 75 GHz of CPU in *`region-one`*.

Networking Concepts

VPC. Virtual private clouds (VPCs) are the foundational networking element used in VCF Automation to enable applications to communicate with each other and the outside world. A VPC provides an isolated domain of networking resources that can be associated with one or more namespaces. As an organization administrator or a project

administrator, you create VPCs for your organization. In addition, organization administrators and advanced project users can use the Networking Service in the IaaS Console to create, update, and manage resources within a given VPC.

Connectivity Profile. A connectivity profile is a template for VPC configuration. You can provide one or more external and private Transit Gateway IP address blocks that are used by VPCs configured with a specific connectivity profile. When the provider administrator allocates the regional networking configuration to the organization, a default connectivity profile is created. As an organization administrator, you can create additional connectivity profiles to use in your organization.

Subnets. A subnet is a subdivision of a VPC that defines a range of IP addresses. Organization administrators can create subnets for their organization. Using the Network service, advanced project users can create subnets in VPCs that are associated with their namespaces.

Transit Gateways. A transit gateway provides both inter-VPC communication within the organization and connection to the outside network through external connections. In addition to using the default transit gateways that the organization comes with, organization administrators can create transit gateways for their organization.

Consumption Concepts

Application teams can provision resources on a self-service basis in VCF Automation using the catalog or the services. Catalog items are generated from blueprints that are curated and maintained by the organization administrator, while cloud services can be accessed directly without blueprints. Both methods of provisioning IT resources require content libraries that host the VM images.

Blueprints and Catalog. Blueprints are infrastructure templates that define the resources and cloud services you want to create to run an application. Organization administrators create the blueprints and publish them as catalog items for end users. End users interact with form entries of catalog items to request resources. These forms can be customized to hide unnecessary fields or to create dynamic entries.

Services. Users can provision cloud services directly on namespaces from the VCF Automation user interface, the VCF Automation APIs, and CLIs without the need for blueprints and the catalog.

Content Library. Content libraries host VM images that application teams use when deploying resources. Organizations can either use content libraries that are provided by the provider administrator or create their own organization-specific content libraries.

Policies. Policies provide governance on resources like VMs and VKS clusters that are provisioned from cloud services. Policies can be applied to the whole organization or to individual projects. VCF Automation supports two types of policies:

- Policies that are applied to resources provisioned from the catalog, such as Approval, Day 2, and Lease policies.
- Policies that are applied to resources provisioned through the self-service IaaS console.

Getting Started with Organizations in VCF Automation

As an administrator of a VCF Automation organization, you manage user access, network configurations, including VPCs and subnets, and policies for your organization. You also create namespaces and manage provider-assigned quotas for the resources associated with the namespaces in your organization.

Using the VCF Automation Organization User Interface

An overview of the VCF Automation organization user interface includes the following basic functions.

> **Note:**
>
> This overview covers what organization administrators with full access to the user interface see. Depending on your user rights, you may not see and access all tabs.

- The **Overview** tab shows you important infrastructure and consumption metrics in your organization. It also provides quick links to your IaaS services. VCF Automation 9.1.1 introduces the Get Started guided checklist on the Overview tab.

- The **Build & Deploy** tab is your content hub and development home. Administrators can create blueprints and content libraries, publish workflows to the catalog, and create extensibility events. Project users can deploy cloud services from catalog items or can access the IaaS Services console.
- The **Manage & Govern** tab is where administrators manage the slices of infrastructure that the provider allocated to the organization. That includes projects, namespaces, namespace classes, policies, and networking.
- The **Orchestrate** tab is where you access the VCF Operations orchestrator integration that your provider administrator configured for the organization.
- The **Administer** tab is where you manage expenses, identity providers, users and user roles, integrations, certificates and passwords, email notifications, and so on.
- The **Inbox** tab is where you can see and respond to policy-based approval requests and deployment requests that require user input.

Review Assigned Resources

On the **Manage & Govern** tab, review the following pages to familiarize yourself with the interface and to review the default objects created for your organization.

- **Projects.** All organizations are created with a default project. The default project is empty and does not have any namespaces or users.
- **Regions.** The Regions page lists all the regions where the organization has a quota in. Organizations can have a quota in one or many regions. Your provider administrator assigns the regional quota to your organization. Quota in a region can come from one or many vSphere Zones within that region.
  Click the double arrow next to a region name to review the region details. The details page contains information about available CPU and memory from each zone in the region.
- **Namespace Class.** Namespace classes are templates for creating namespaces. These templates can be used to standardize namespace attributes, like utilization limits, reservations, VM classes, storage classes, and content libraries. Your organization comes preconfigured with three default namespace classes (small, medium, and large), which serve as example templates. The only different attributes among these built-in templates are the CPU and Memory limits. Organization administrators can use these templates as-is or can modify them to suit their needs.
  Click the namespace class name to inspect the limits, VM class, and storage class presets.
- **Networking.** Review the default VPC (virtual private cloud). A VPC is an isolated domain of networking resources.
  You can also review the Connectivity Profiles tab. A connectivity profile is a template for creating VPCs.

Using the VCF Automation API Help Center

The VCF Automation API Help Center is a built-in UI which documents the API interface for the organization. Users can also execute and test APIs from the API Help Center.

You can access the API Help Center from the VCF Automation organization UI. Click your user profile in the upper right corner and then click **API Help Center**.

VCF Automation API Help Center has the following components.

- **Automation APIs**. The Automation API section provides details about various automation-specific components that are used to administer the organization, such as blueprints, catalog, instances, projects, custom forms, and so on. This documentation includes OpenAPI specifications. Each section defines the available endpoints, required input/output formats, and how to authenticate requests.
- The **Admin APIs** are used to create VCF Automation resources like projects and provision additional resources. It brings automation-specific capabilities into the Kubernetes management framework.
- The **Consumption APIs** are used to create and manage infrastructure resources, like virtual machines, networks, storage, and so on. This allows users to work with infrastructure using Kubernetes-native methods.
- **Terraform Provider**. The Terraform Provider section provides details about using Terraform provider to interact with VCF Automation. For more information on the supported Terraform providers, see Terraform Configuration in VCF Automation.
- **SDK Generation & Usage**. The VCF Automation SDK provides a standardized way to generate APIs for various components. You can explore examples that demonstrate its usage for various use cases to get started with the SDK.

Set Up Your VCF Automation Organization

As the organization administrator, you set up your organization using the following procedure.

1. Log in to the organization UI.
   When you log in to the organization for the first time, you must get the login credentials for the local account from your provider administrator. After you configure identity providers, you can use those credentials to log in.
2. Add users to your organization.
   You can onboard an identity provider and import users from there. VCF Automation supports LDAP, OIDC, and SAML-compliant identity providers. As you add users to the organization, you assign roles to them.
   a. Configure an identity provider. See Managing Identity Providers in VCF Automation Organizations.
   b. Import users and user groups. See Add Users and Assign Roles in VCF Automation.
3. Set up networking.
   VPCs are required networking constructs for namespaces. Connectivity profiles define and templatize common configuration elements for VPCs. You can use the default connectivity profile and VPC or you can create new ones.

   Note:

   In VLAN-backed regions in VCF Automation 9.1.1, before you configure networking, verify the following with your provider administrator:

   - A Distributed VXLAN Connection is configured for your region.
   - External IP address blocks is allocated to your organization.

   a. Create a connectivity profile. See Create a Connectivity Profile in VCF Automation.
   b. Create a VPC. See Create a Virtual Private Cloud in VCF Automation.

   Note:

   In VLAN-backed regions in VCF Automation 9.1.1, after VPC creation, verify that your VPC connectivity profile does not include private TGW IP blocks. In VLAN-backedregions, these fields are not available and do not appear in the UI. Default SNAT is not created automatically in VLAN-backed regions. SNAT is not supported in VLAN-backed mode.
4. Set up a content library.
   VCF Automation has two types of content libraries, provider content libraries and organization content libraries. You can use the content library provided to you by your provider administrator or you can create your own.
   a. Create a content library. See Managing Content Libraries in VCF Automation.
   b. Upload VM images to the content library and publish them to the catalog. See Add a VM Image to a Content Library in VCF Automation.
5. Create a namespace class.
   To create namespaces, you need a namespace class. You can use the default namespace classes and override the values in them at the time of creating a new namespace. You can edit any or all of the default namespace classes. Alternatively, you can create your own namespace classes. See Create a Namespace Class in VCF Automation.
6. Configure a project.
   You can use the default project and change the name and description or you can create your own projects.
   a. Create the project. See Create a Project in VCF Automation.
   b. Projects are created without namespaces. To create a namespace, see Create a Namespace in a VCF Automation Project.
   c. Add project users. See Assign Project Roles to Users and Groups in VCF Automation.
7. Create blueprints and publish them to the catalog for project users to request and consume. See Creating Blueprints in VCF Automation.
8. Publish VCF Operations orchestrator workflows to the catalog. See Publish a VCF Operations orchestrator Workflow to the Catalog in VCF Automation.
9. Set up event subscriptions. See Managing Event Subscriptions in VCF Automation.
10. Set up policies to apply governance to provisioned resources. See Managing Policies in VCF Automation.

Using the Get Started Guided Checklist

VCF Automation 9.1.1 introduces a Get Started checklist for users who have not yet completed the required organization setup. By following the steps in the Get Started checklist, users can provision their first workloads. The steps in the checklist depend on the role of the user.

- For Organization Administrators, the steps include connecting an identity provider, viewing projects, defining the resource library, provisioning workloads, adding users and assigning project roles, and enabling projects for namespace creation.
- For Project Administrators, the steps include viewing projects, viewing namespaces for workload provisioning, creating a vSphere namespace, and provisioning workloads.
- For Project Advanced Users, the steps include viewing projects, viewing namespaces for workload provisioning, and provisioning workloads.
- For Project Users, the steps include viewing projects, viewing namespaces for workload provisioning, and launching catalog items to deploy workloads.

Users can dismiss the checklist at any time during a session. After completion or dismissal, users can reopen the checklist in a new browser session.

Managing Identity Providers in VCF Automation Organizations

As an organization administrator, you can use one or more external identity providers to import and manage users and groups in your organization. You can configure an LDAP server connection, a SAML integration, and an OpenID Connect (OIDC) integration.

VCF Automation supports user login with an identity provider directly. Any third-party provider with OIDC, SAML, AD/LDAP protocols can be configured to use Single Sign-on to VCF Automation for identity and access management.

Configure Your VCF Automation Organization to Use a Custom LDAP Connection

As an organization administrator, you can configure your VCF Automation organization to use an LDAP server as a source of users and groups.

When adding or editing a new LDAP server endpoint, you can test a remote connection to an endpoint and to establish a trust relationship. VCF Automation adds any certificate that you decide to trust to a centralized certificate storage area.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Identity Providers**.
3. On the **Identity Providers** page, click the **LDAP** tab.
4. If you are configuring LDAP for the first time, click **Configure** and select a configuration option.

   If you are reconfiguring an existing LDAP connection, click **Edit** and proceed to the next step.

Option	Description
Do not use LDAP.	Use this option if you want to add VCF Automation users who are private to the organization. If you select this option, you cannot create or import user groups.
VCF Automation system LDAP service	Use when an LDAP connection is configured by your provider administrator and the organization is a member of the cloud provider company. If you select this option, enter a distinguished name for the organization and click **Save**.
Custom LDAP service	Used when you want the organization to use its own directory service. Before you can use this option, you must configure your cloud system LDAP service to link to their LDAP service. If you select this option, proceed to Step 5.

5. Click the **Custom LDAP** tab.
6. Click **Edit** and enter the required information for the LDAP connection.

Required Information	Description
Server	The host name or IP address of the LDAP server.
Port	The port number on which the LDAP server is listening. For LDAP, the default port number is 389. For LDAPS, the default port number is 636.
Base distinguished name	The base distinguished name (DN) is the location in the LDAP directory where VCF Automation to connect. To connect at root level, enter only the domain components, for example, `DC=example,DC=com`. To connect to a node in the domain tree structure, enter the distinguished name for that node, for example, `OU=ServiceDirector,DC=example,DC=com`. Connecting to a node limits the scope of the directory available to VCF Automation.
Connector type	The type of your LDAP server. Can be **Active Directory** or **OpenLDAP**.
Use SSL	If your server is LDAPS, select this check box.
Authentication method	Simple authentication consists of sending the user's DN and password to the LDAP server. If you are using LDAP, the LDAP password is sent over the network in plain text. If you want to use Kerberos, you must configure the LDAP connection by using the vCloud API.
User name	Enter the full LDAP distinguished name (DN) of a service account with domain admin rights. VCF Automation uses this account to query the LDAP directory and retrieve user information. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.
Password	The password for the service account that connects to the LDAP server. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.

7. Click the **User Attributes** tab, examine the default values for the user attributes, and, if your LDAP directory uses different schema, modify the values.
8. Click the **Group Attributes** tab, examine the default values for the group attributes, and, if your LDAP directory uses different schema, modify the values.
9. If you want to customize the **Sign in with LDAP** button label that appears on the VCF Automation login page, enter a new custom button text.
10. Click **Save**.
11. If you selected the **Use SSL** check box, and if the certificate of the LDAPS server is not yet trusted, on the **Trust Certificate** window, confirm if you trust the certificate presented by the server endpoint.
12. To test the LDAP connection settings and the LDAP attribute mappings:
    a) Click **Test**.
    b) Enter the password of the LDAP server user that you configured and click **Test**.

       If connected successfully, a green check mark is displayed.

       If a user name is entered for the test, the retrieved user and group attribute values are displayed in a table. The values that are successfully mapped to LDAP attributes are marked with green check marks. The values that are not mapped to LDAP attributes are blank and marked with red exclamation marks.
    c) To exit, click **Cancel**.
13. To synchronize VCF Automation with the configured LDAP server, click **Sync**.

    VCF Automation synchronizes the user and group information with the LDAP server regularly depending on the synchronization interval that you set in the general system settings.

    Wait a few minutes for the synchronization to finish.

On the Access Control page, you can import users and groups from the newly configured LDAP server and assign roles to them.

Configure Your VCF Automation Organization to Use a SAML Identity Provider

If you want to import users and groups from a SAML identity provider to your VCF Automation organization, you must configure the organization with this SAML identity provider. Imported users can log in to the organization with the credentials established in the SAML identity provider.

- Verify that you have access to a SAML 2.0 compliant identity provider.
- Supply the VCF Automation service provider metadata to the SAML identity provider administrator and obtain an XML file with the following metadata from your SAML identity provider in return.
  - The location of the single sign-on service
  - The location of the single logout service
  - The location of the service's X.509 certificate

  For information on configuring and acquiring metadata from a SAML provider, consult the documentation for your SAML provider.

To configure VCF Automation with a SAML identity provider, you establish a mutual trust by exchanging SAML service provider and identity provider metadata.

To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

When an imported user attempts to log in, the system extracts the following attributes from the SAML token, if available, and use them for interpreting the corresponding pieces of information about the user.

- `email address = "EmailAddress"`
- `user name = "UserName"`
- `full name = "FullName"`
- `user's groups = "Groups"`
- `user's roles = "Roles"` (this attribute is configurable)

Group information is used if the user is not directly imported but is expected to log in by virtue of membership in imported groups. A user can belong to multiple groups, so can have multiple roles during a session.

If an imported user or group is assigned the **Defer to Identity Provider** role, the roles are assigned based on the information gathered from the `Roles` attribute in the token. If a different attribute is used, this attribute name can be configured using API and only the `Roles` attribute is configurable. If the **Defer to Identity Provider** role is used, but no role information can be extracted, the user can log in but has no any rights to perform any activities.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Identity Providers**.
3. On the **Identity Providers** page, click the **SAML** tab.
4. Click **Configure**.

   The current SAML settings appear.
5. From the **Service Provider** tab, download the VCF Automation SAML service provider metadata.
   a) Enter an Entity ID for the organization.

      The Entity ID uniquely identifies your organization to your Identity Provider.
   b) Examine the certificate expiration date and, if expiring soon, regenerate the certificate by clicking **Regenerate**.

      The certificate is included in the SAML metadata, and is used for both encryption and signing. Either or both of these might be required depending on how trust is established between your organization and your SAML IDP.
   c) Click **Retrieve Metadata**.

      Your browser downloads the SAML service provider metadata, an XML file which you must provide to your identity provider.

6. On the **Identity Provider** tab, upload the SAML metadata that you previously received from your identity provider.
   a) Select **Use SAML Identity Provider**.
   b) Either click the **Browse** icon and upload the file, or copy and paste its content in the **Metadata XML** text box.
7. If you want to customize the **Sign in with SAML** button label that appears on the VCF Automation login page, enter a new custom button text.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
8. Click **Save**.

Configure Your VCF Automation Organization to Use an OpenID Identity Provider

If you want to import users and groups from an OpenID Connect (OIDC) identity provider to your VCF Automation organization, you must configure your organization with this OIDC identity provider. Imported users can log in to the organization with the credentials established in the OIDC identity provider.

OAuth is an open federation standard that delegates user access. OpenID Connect is an authentication layer on top of the OAuth 2.0 protocol. By using OpenID Connect, clients can receive information about authenticated sessions and end users. The OAuth authentication endpoint must be reachable from VCF Automation which makes it more suitable when you use public identity providers or provider-managed ones.

> **Important:**
>
> Endpoint misconfigurations can lead to user inability to complete the SSO logout from VCF Automation. See KB article 394731.

You can allow organizations to generate and issue API access tokens that applications can use on their behalf.

You can configure VCF Automation to automatically refresh your OIDC key configurations from the JSON Web Key Set (JWKS) endpoint you provide. You can configure the frequency of the key refresh process and the rotation strategy that determines whether VCF Automation adds new keys, replaces the old keys with new, or the old keys expire after a certain period.

To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

VCF Automation generates audit events for both successful and failed key refreshes under the event topic `com/vmware/vcloud/event/oidcSettings/keys/modify`. The audit events for failed key refreshes include additional information about the failure.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Identity Providers**.
3. On the **Identity Providers** page, click the **OIDC** tab.
4. If you are configuring OIDC for the first time, copy the client configuration redirect URI and use it to create a client application registration with an identity provider that complies with the OpenID Connect standard, for example, VMware Cloud Foundation Identity Broker.

   You need this registration to obtain a client ID and a client secret that you must use during the OIDC identity provider configuration.

5. Click **Configure**.
6. Verify that OpenID Connect is active, and enter the client ID and client secret information from the OIDC server registration.
7. Optional: To use the information from a well-known endpoint to automatically fill in the configuration information, turn on the **Configuration Discovery** toggle and enter a URL at the site of the provider that VCF Automation can use to sent authentication requests to.
8. Click **Next**.
9. If you did not use **Configuration Discovery** in Step 6, enter the information in the **Endpoints** section.
   a) Enter the endpoint and issuer ID information.
   b) If you are using VMware Cloud Foundation Identity Broker as an identity provider, select **SCIM** under Account Details Retrieval.

      For other identity providers, you can leave the default **User Info** selection.
   c) If you want to combine claims from the `UserInfo` endpoint and the ID Token, turn on the **Prefer ID Token** toggle.

      The identity providers do not provide all the required claims set in the `UserInfo` endpoint. By turning on the **Prefer ID Token** toggle, VCF Automation can fetch and consume claims from both sources.
   d) Enter the maximum acceptable clock skew.

      The maximum clock skew is the maximum allowable time difference between the client and server. This time compensates for any small time differences in the time stamps when verifying tokens. The default value is 60 seconds.
   e) Click **Next**.
10. If you did not use **Configuration Discovery** in Step 6, enter the scope information, and click **Next**.

    VCF Automation uses the scopes to authorize access to user details. When a client requests an access token, the scopes define the permissions that this token has to access user information.

    The default scopes are `openid`, `profile`, and `email`. You can also add `user` to scope.
11. If you are using **User Info** as an access type, map the claims and click **Next**.

    You can use this section to map the information VCF Automation gets from the user info endpoint to specific claims. The claims are strings for the field names in the VCF Automation response.
12. If you want VCF Automation to automatically refresh the OIDC key configurations, turn on the **Automatic Key Refresh** toggle.
    a) If you did not use **Configuration Discovery** in Step 6, enter the **Key Refresh Endpoint**.

       The **Key Refresh Endpoint** is a JSON Web Key Set (JWKS) endpoint and it is the endpoint from which VCF Automation fetches the keys.
    b) Select how often the key refresh occurs.

       You can set the period in hourly increments from 1 hour up to 30 days.
    c) Select a **Key Refresh Strategy**.

Option	Description
Add	Add the incoming set of keys to the existing set of keys. All keys in the merged set are valid and usable. For example, your existing set of keys includes keys A, B, and D. Your incoming set of keys includes keys B, C, and D. When the key refresh occurs, the new set includes keys A, B, C, and D.
Replace	Replace the existing set of keys with the incoming set of keys. For example, your existing set of keys includes keys A, B, and D. Your incoming set of keys includes keys B, C, and D. When the key refresh occurs, key C replaces key A. The incoming keys B, C, and D become the new set of valid keys without any overlap with the old set.

Option	Description
Expire After	You can configure an overlap period between the existing and incoming sets of keys. You can configure the overlapping time using the **Expire Key After Period**, which you can set in hourly increments from 1 hour up to 1 day. The key refresh runs start at the beginning of every hour. When the key refresh occurs, VCF Automation tags as expiring the keys in the existing set of keys that are not included in the incoming set. At the next key refresh run, VCF Automation stops using the expiring keys. Only keys included in the incoming set are valid and usable. For example, your existing set of keys includes keys A, B, and D. The incoming set includes keys B, C, and D. If you configure the existing keys to expire in 1 hour, there is 1 hour of overlap during which both sets of keys are valid. VCF Automation marks key A as expiring and until the next key refresh run, keys A, B, C, and D are usable. At the next run, key A expires and only B, C, and D continue working.

13. If you did not use **Configuration Discovery** in Step 6, upload the private key that the identity provider uses to sign its tokens.
14. If you want to customize the **Sign in with OIDC** button label that appears on the VCF Automation login page, enter a new custom button text.

    You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
15. Click **Save**.

- Subscribe to the `com/vmware/vcloud/event/oidcSettings/keys/modify` event topic.
- Verify that the **Last Run** and the **Last Successful Run** are identical. The runs start at the beginning of the hour. The **Last Run** is the time stamp of the last key refresh attempt. The **Last Successful Run** is the time stamp of the last successful key refresh. If the time stamps are different, the automatic key refresh is failing and you can diagnose the problem by reviewing the audit events.

Managing Predefined User Roles in VCF Automation

Roles assigned to users and groups are an essential part to identity and access management in VCF Automation. You can leverage identity providers with VCF Automation organizations to enable single sign-on to provision users and groups. Once the users and groups are added to the VCF Automation organization, they must be assigned roles to enable specific permissions and access to different features within VCF Automation.

What are User Roles in VCF Automation

VCF Automation has several levels of user roles. These different levels control access to the organization, the projects that produce or consume the blueprints, catalog items, access to IaaS services, and the ability for users to use or see individual parts of the user interface. These different levels give organization administrators different tools to apply any level of granularity that is required by their operational needs.

As the organization administrator, after you configure an identity provider for your organization, you can import users and groups and manage their role assignments using the **Access Control** page. There are two types of predefined roles in VCF Automation: organization roles and project roles. In addition, you can create custom roles for specific use cases.

Predefined Organization Roles

The organization roles are global. The predefined organization-level roles are Organization Administrator, Organization Auditor, and Organization User.

Organizational roles define permissions within the organization. Organization administrators can add and manage other users and create custom roles at the organization level. Custom roles allow you to tailor permissions to specific user requirements.

Role	General permissions
Organization Administrator	Can access the organization UI and add users and groups to the organization. Organization administrators can: • Manage projects and project users • Configure integrations, such as Active Directory, Bitbucket, GitHub, and GitLab • Manage blueprints and catalog items • Manage namespaces and associated resources • Fully access namespaces that they did not create • Manage new VPCs and associated resources • See all projects and all deployments • Create and manage content libraries • Create and manage policies on resources and deployments • Create and manage event subscriptions
Organization User	Can log in to the organization UI. Must be added to a project.
Organization Auditor	Read-only version of the organization administrator.
Defer to Identity Provider	User permissions are determined based on information received from the identity provider for the organization.

Predefined organization-level roles are managed by the provider administrator in the VCF Automation Provider Management UI. For a full list of what rights are given to each role, see Predefined VCF Automation Roles and Their Rights in the Provider Management documentation.

Custom Organization Roles

As an organization administrator, you can create custom roles for your organization.

The custom roles have two categories, view and manage.

- View. A user assigned to a role with this permission can see all the items for all projects across the organization in the selected sections of the user interface. This role is useful for users who need to see accounts, configurations, or assigned values.
- Manage. A user assigned to a role with this permission can see all the items and has full add, edit, and delete permissions for all projects across the organization in the selected sections of the user interface.

These permissions extend the privileges that are granted by the other roles and are not restricted by project membership. For example, you can expand a project administrator's permissions to manage parts of the infrastructure or give an organization auditor an ability to review and respond to approvals requests.

For instructions on how to create a custom role, see Create a Custom Role in for All Apps.

Project Roles

In addition to the organizational roles, VCF Automation has project roles. Project members can view and manage project resources depending on their project role. Project roles include Project Administrator, Project Advanced User, Project User, and Project Auditor.

These roles are assigned automatically on a per-project basis in VCF Automation. The project roles can vary between projects. The same user can be an administrator in one project and an advanced user in another project.

Role	General permissions
Project Administrator	Project administrators leverage the infrastructure that is created by the organization administrator to ensure that their project members have the resources they need for their development work. Project administrators can: • View all content for their project, including namespaces, deployments, and so on • Perform all Day 2 actions on resources • Manage users in the project and assign roles • Manage project-level settings and administrator content • Create namespaces if the organization administrator set up self-service for the project • Create resources using the IaaS services console in namespaces they created or have access to • Manage catalog approval requests
Project Advanced User	Advanced users work within their projects to request catalog items and create resources using the services console. They can perform day 2 actions on all resources.
Project User	Project users can only access the catalog and request catalog items. They can perform day 2 actions on their own deployments. They cannot use the IaaS services console.
Project Auditor	Project auditors are restricted to read-only access.

For a full list of what permissions each project role has, see Predefined Roles and Permissions in VCF Automation.

Add Users and Assign Roles in VCF Automation

You can add role assignments for users in VCF Automation.

- Verify that users have been imported from a valid identity provider instance that is associated with your VCF Automation deployment. See Managing Identity Providers in VCF Automation Organizations.
- Review the VCF Automation organization roles and project roles permissions in Predefined Roles and Permissions in VCF Automation. You must understand what each user role can see and do.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Access Control**.
3. Select the **Users** tab and click **Import Users**.
4. From the **Source** drop-down menu, select your identity provider type.

   If you configured only one identity provider, this option is hard-coded.
5. Search for the users you want to assign roles to.

   You can import a CSV list.
6. Assign one or more roles to the selected users.
7. If you want the roles of the users to become a combination of all group and assigned roles, turn on the **Inherit roles from groups** toggle.
8. Click **Save**.

- You can edit roles assigned to users that have been imported into VCF Automation at any time. To edit a user's role, click the vertical ellipsis next to their name on the **Users** page and click **Edit**.

Add Group and Assign Roles in VCF Automation

You can add role assignments for groups in VCF Automation.

- Verify that user groups have been imported from a valid identity provider instance that is associated with your VCF Automation deployment. See Managing Identity Providers in VCF Automation Organizations.

- Review the VCF Automation organization roles and project roles permissions in Predefined Roles and Permissions in VCF Automation. You must understand what each user role can see and do.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Access Control**.
3. Select the **Groups** tab and click **Import Groups**.
4. From the **Source** drop-down menu, select your identity provider type.

   If you configured only one identity provider, this option is hard-coded.
5. Use the search function to find the group that you are adding and select it.
6. Assign an organization role.
7. Click **Save**.

- You can edit roles assigned to groups that have been imported into VCF Automation at any time. To edit a group's role, click the vertical ellipsis next to the group's name on the **Groups** page and click **Edit**.

Disable and Delete Users in VCF Automation

As an organization administrator, you can disable and/or delete users and user groups from your VCF Automation organization.

Disabling a user or user group deactivates the user's access to the organization. They are no longer able to log in and access VCF Automation features. Disabling a user does not remove them from the organization and you can reactivate their access at a later time.

Deleting a user removes them from the organization. You must import the user again if you want to give them access to VCF Automation. Before you can delete a user, you must disable their access first.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Access Control** > **Users**.
3. Click vertical ellipsis next to the name of the user you want to delete and then click **Disable**.
4. Click **Disable** to confirm.
5. Select **Administer** > **Access Control** > **Users**.
6. Click vertical ellipsis next to the name of the user you want to delete and then click **Delete**.
7. Click **Delete** to confirm.
8. Optional: Select the **Transfer the user's objects to me** checkbox.

   Transferring user objects to the administrator ensures that objects are not orphaned when a user is deleted.
9. Click **Delete** to confirm.

Create a Custom Role in VCF Automation

As a VCF Automation organization administrator, you manage user access using the default organization roles. However, you might also want to create custom roles so that selected users can perform specific tasks or see content that is outside of their organization roles.

- Review the custom organization role descriptions so that you know more about how you can refine the permissions for your users. See Managing Predefined User Roles in VCF Automation.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Access Control** > **Roles** and click **New**.
3. Enter a unique name that you can identify when you assign users to the role and, optionally, a description.
4. Select the check boxes that correspond to the permissions you want the users to have over the resources.
5. Click **Save**.

Assign the custom role to users and groups in your organization. See Add Users and Assign Roles in VCF Automation

Predefined Roles and Permissions in VCF Automation

As an organization administrator, you assign organization roles and project roles to users in your organization to determine what they can see and do in the organization. As an organization administrator, you have full permission on all areas of the user interface.

Overview Tab

UI Context	Task	Organization Administrator	Organization User	Organization Auditor	Project Administrator	Project User	Project Advanced User	Project Auditor
Access the VCF Automation organization UI		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Overview page		Yes	Yes	Yes	Yes	Yes	Yes	Yes
My Account (User Profile)		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Notifications (Bell icon)	View notifications	Yes		Yes				
	Manage notifications	Yes						

Build & Deploy Tab

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
Instances	View catalog instances	Yes	Yes	Yes, your projects	Yes, your instances only	Yes, your projects	Yes, your projects
	View virtual machine instances	Yes	Yes	Yes, your projects	Yes, your instances only	Yes, your projects	Yes, your projects
	View Kubernetes cluster instances	Yes	Yes	Yes, your projects	Yes, your instances only	Yes, your projects	Yes, your projects
	View volume instances	Yes	Yes	Yes, your projects	Yes, your instances only	Yes, your projects	Yes, your projects
	View custom resource instances	Yes	Yes	Yes, your projects	Yes, your instances only	Yes, your projects	Yes, your projects

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
	Run day 2 actions on resources (VMs, Kubernetes clusters, volumes) and instances	Yes		Yes, your projects	Yes, your instances only	Yes, your projects	
	View blueprint topology on deployed instances	Yes		Yes, your projects	Yes, your instances only	Yes, your projects	
Catalog	View catalog	Yes	Yes	Yes, your projects	Yes, your projects	Yes, your projects	Yes, your projects
	Manage catalog	Yes		Yes, your projects			
	Share catalog items with other organizations	Yes, only if administering a provider consumption organization					
	Publish blueprints and their custom forms to the catalog	Yes					
	Request items in the catalog	Yes		Yes, your projects	Yes, your projects	Yes, your projects	
Content Hub - Blueprint Design	View blueprints and their custom forms	Yes	Yes	Yes, your projects			Yes, your projects
	Create and manage blueprints and their custom forms	Yes					
	View blueprint requests	Yes	Yes	Yes, your projects			Yes, your projects
	Manage blueprint requests	Yes		Yes, your projects			
	View custom resources	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization				

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
	Create and manage custom resources	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization					
	View resource actions	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization				
	Create and manage resource actions	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization					
	View exports	Yes	Yes	Yes, your projects			
	View imports	Yes	Yes	Yes, your projects			
Content Hub - Content Libraries	View content libraries	Yes	Yes				
	Create and manage content libraries	Yes					
Content Hub - VM Images	View VM images	Yes	Yes				
	Upload and delete VM images	Yes					
	Publish VM images to the catalog	Yes					
Content Hub - Workflows	View workflows	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization				

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
	Publish workflows to the catalog	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization					
Event Subscriptions	View extensibility events	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization				
	Create and manage extensibility subscriptions	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization					
	Deactivate subscriptions	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization					
	View subscriptions	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization				
	View event topics	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization				
Services	Access all services available for the namespace	Yes	Yes	Yes, namespaces for your projects		Yes, namespaces for your projects	Yes, namespaces for your projects

Manage & Govern Tab

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
Projects	View projects	Yes	Yes	Yes, your projects		Yes, your projects	Yes, your projects
	Create and delete projects	Yes					
	Edit project, including add users to the project and assign project roles	Yes					
Policies	View policy definitions, assignments, and decisions	Yes	Yes				
	Create and manage policies	Yes					
	View policy enforcements	Yes	Yes				
	View approval requests	Yes	Yes	Yes, your projects			Yes, your projects
	Approve and reject requests	Yes		Yes, your projects			
Regions	View regions	Yes	Yes				
Namespaces	View namespaces	Yes	Yes	Yes, your projects		Yes, your projects	Yes, your projects
	Create and manage namespaces	Yes		Yes, your projects, if the organization administrator enabled namespace self-service for the project			
Namespace Classes	View namespace classes	Yes	Yes				
	Create and manage namespace classes	Yes					
Networking - Subnets	View subnets	Yes	Yes	Yes, your namespaces		Yes, your namespaces	Yes, your namespaces
	Create and manage subnets	Yes		Yes, your namespaces using the Network Service		Yes, your namespaces using the Network Service	

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
Networking - Virtual Private Clouds	View VPCs, including connectivity profiles	Yes	Yes				
	Create and manage VPCs	Yes		Yes Cannot edit VPC settings. Can only create new resources in the VPC		Yes Cannot edit VPC settings. Can only create new resources in the VPC	
Networking - Connectivity Profiles	View connectivity profiles	Yes	Yes				
	Create and manage connectivity profiles	Yes					
Networking - DHCP Settings	View DHCP settings for available regions	Yes					
	Edit DHCP settings for available regions	Yes					
Networking - Transit Gateways	View transit gateways	Yes	Yes				
	Create and manage transit gateways	Yes					
Networking - External Connections	View centralized and distributed connections	Yes	Yes				
Networking - IP Address Blocks	View IP blocks	Yes	Yes				
	Create and manage IP blocks	Yes					
Networking - IP Quotas	View IP quotas	Yes	Yes				
	Create and manage IP quotas	Yes					
Networking - IP Allocations	View IP allocations	Yes	Yes				
	Request allocation of IP/ CIDR	Yes					

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
Avi Load Balancer - Service Engine Group	View service engine groups	Yes, if the provider administrator configured Avi Load Balancer for the organization	Yes, if the provider administrator configured Avi Load Balancer for the organization				
	Create service engine groups	Yes, if the provider administrator configured Avi Load Balancer for the organization					
Firewall	View distributed firewall policies						
	Create and manage distributed firewall policies and rules						
	View transit gateway and VPC gateway policies and rules						
	Create and manage distributed firewall policies and rules						
	View security profiles	Yes	Yes				
	Assign security profiles to VPCs	Yes					
	View groups						
	Create groups						
	View network services	Yes					
Kubernetes Management	View VKS cluster management	Yes	Yes	Yes, your projects			Yes, your projects
	Manage VKS clusters	Yes		Yes, your projects			

Administer Tab

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
Bills	View billing	Yes	Yes				
Access Control	View users, groups, and roles	Yes	Yes				
	Import users and groups, assign roles, and create and manage custom roles	Yes					
	Identity Providers	Yes	Yes				
	Certificate Management	Yes	Yes				
Activity - Audit	View audit logs	Yes	Yes				
Connections - Integrations	View Git integrations	Yes	Yes				
	Manage Git integrations	Yes					
	View VCF Operations orchestrator integration	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization and assigned the rights to the user	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization and assigned the rights to the user				
	Manage VCF Operations orchestrator integration	Yes, only when the provider administrator published the Orchestrator Rights Bundle for the organization and assigned the rights to the user					
Connections - Identity Providers							
Configuration - Settings	View settings	Yes	Yes				
	Manage settings	Yes					

				Organization User			
UI Context	**Task**	**Organization Administrator**	**Organization Auditor**	**Project Administrator**	**Project User**	**Project Advanced User**	**Project Auditor**
Certificates & Secrets - Certificate Management							
Certificates & Secrets - Secrets	View secrets	Yes	Yes				
	Manage secrets	Yes					
Administer - Feature Flags	Allow users to configure their environment using feature flags	Yes					

Create a Service Account in Your VCF Automation Organization

In VCF Automation, you can use service accounts to control access from third-party applications. As the organization administrator, you create service accounts and you can revoke access at any time.

Service accounts access VCF Automation programatically using an API token that is bound to a device ID. The API token does not expire and is renewed upon each request.

Service accounts cannot be used to access the VCF Automation organization UI.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Access Control** > **Service Accounts** and click **New**.
3. Enter a name for the service account.
4. From the **Assign Role** drop-down menu, select a role for the service account.

   The list of available roles comprises global and custom organization roles.
5. Enter a software ID for the service account or generate and enter one using the **Generate Software ID** button.

   Service accounts must have software IDs which are unique identifiers, in UUID format, representing the application that connects to VCF Automation.

   For larger solutions, to retain control over the identity of your service accounts, do not use the **Generate Software ID** option, and generate your own software ID.
6. (Optional) Enter the software version of the system using the service account.

   The software version is an optional vendor-specified informational piece of metadata that is associated with the service account. To track when a piece of software changes, VCF Automation uses the software version. The software version might be useful for identifying a service account.
7. (Optional) Enter a client URI.

   The client Uniform Resource Identifier (URI) is a URL to the webpage of the vendor and provides information about the client.
8. Optional: To keep the refresh token valid, turn off the **Require Rotation** toggle.

   By default, **Require Rotation** is enabled for service accounts. This means that after each access token request, the current refresh token becomes invalid. Instead, a new refresh token is provided which then must be used for subsequent requests.

Note:

You cannot change this setting later.

9. Click **Next**.
10. Review the service account information, and click **Finish**.

Grant Access to a Service Account in Your VCF Automation Organization

After you create a service account and the application requests authorization to receive an access token, you can grant the token by using the VCF Automation organization UI.

1. Copy the client ID from the service account details.
2. Verify that the application requesting the account makes an OAuth 2.0 Device Authorization Grant RFC-compliant request to the `https://site.cloud.example.com/oauth/provider/device_authorizationAPI` endpoint. For more information on device authorization requests, see RFC 8628 section 3.1.

   Once the application requests access, the service account status changes to Requested. The application receives the device code, user code, and some additional information.

   Run the following curl command to initiate the authorization request. Replace the *`client_id`* with your service account client ID.

   ```
   curl --location 'https://<organization-host>/oauth/tenant/exampleTenant/device_authorization' \
   --header 'Accept: application/json;version=40.0' \
   --header 'Content-Type: application/x-www-form-urlencoded' \
   --data-urlencode 'client_id=<client-id>'
   ```

   The system returns a `user_code` and a `device_code`. You use the `user_code` in the organization to grant access.

   Sample response:

   ```
   {
      "device_code": "<device-code>",
      "user_code": "<user-code>",
      "verification_uri": "<organization-host>/automation/#/administer/access",
      "expires_in": 3600,
      "interval": 60
   }
   ```

   The device must poll at the frequency that is specified in the above response (in seconds) `/oauth/provider/token` as per the RFC. The device must use the device code until it receives the tokens from VCF Automation, or the request times out.

   Sample request:

   ```
   curl --location 'https://<organization-host>/oauth/tenant/exampleTenant/token' \
   --header 'Content-Type: application/x-www-form-urlencoded' \
   --data-urlencode 'grant_type=urn:ietf:params:oauth:grant-type:device_code' \
   --data-urlencode 'client_id=<client-id>' \
   --data-urlencode 'device_code=<device-code>'
   ```

   The response includes the `access_token`. The application must use this token for all subsequent API requests.

   ```
   {
      "access_token": "<access-token>",
      "token_type": "Bearer",
      "expires_in": 3600,
      "refresh_token": "<refresh-token>"
   }
   ```

   If you do not confirm or deny an access request, the user code times out. The timeout period appears in the response of the device authorization request.

   VCF Automation grants a primary API token to the application only if the application and the administrator use the device code and user code corresponding to each other.
3. Get the user code from the application. You must enter the code in step 4.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Access Control** > **Service Accounts**.
3. Click **Review Access Requests**.
4. Enter the user code for the application that you obtained in prerequisite 3, click **Lookup**, and verify the requested access details.

   The user code is valid for one hour.
5. Grant access to the application.

   If you deny access to the application, the service account status changes back to `Created`.

You can now use the service account to request tokens from VCF Automation.

You can use the access token as bearer token for API requests. The access token expires after one hour and after that you can use the refresh token to generate a new access token.

The refresh token for service accounts does not expire. However, the refresh token becomes invalid if one access token was generated and the new refresh token must be used for the next request instead. To disable this behavior, disable “Require Rotation” as explained in the related section above.

Managing Networking in VCF Automation Organizations

As an organization administrator, you can view what networking infrastructure is assigned to your organization by the provider administrator and you can manage networking infrastructure within the organization.

VPC Network Stacks

When your provider administrator creates a region, they set a network stack for the region. The network stack of a region determines which networking features are available. You can view the network stack of a specific zone within the available regions on the Regions page in your organization.

VCF Automation organizations can use two types of network stacks:

- **Full-stack VPC.** Workload domains use Tunnel Endpoint (TEP) overlay networking, where traffic between hosts is encapsulated in VXLAN tunnels. Supports private subnets, private IP address blocks, NAT, and default SNAT.
- **VLAN-backed VPC.** Workload domains are configured as VLAN-backed, meaning traffic between hosts travels directly over the physical VLAN fabric without overlay encapsulation. Supports public subnets and VLAN extension subnets only. No private IP addressing, NAT, or SNAT.

  Note:

  VCF Automation 9.1.1 introduces support for organizations in VLAN-backed (TEP-less) networking. If your provider administrator configured your region with VLAN-backed workload domains, your organization uses VLAN-backed VPC networking.

All zones in a region must use the same network stack, either full-stack VPC or VLAN-backed VPC. Zone types cannot be mixed within the same organization region.

The following table summarizes the networking capabilities that are available to you depending on the type of network stack in your organization.

Capability	Full-stack VPC (VCF Automation 9.1+)	VLAN-backed VPC (VCF Automation 9.1.1+)
External connections	Centralized or Distributed VLAN	Distributed VLAN only
Private subnets	Supported	Not supported
Public subnets	Supported	Supported
VLAN extension subnets	Supported	Supported
Private TGW IP blocks	Supported	Not available

Capability	Full-stack VPC (VCF Automation 9.1+)	VLAN-backed VPC (VCF Automation 9.1.1+)
Private VPC IP address blocks	Supported	Not available
NAT (VPC and TGW)	Supported	Not available
Default SNAT	Supported	Not available
IPSec VPN	Supported (centralized connections only)	Not available
External IP address blocks	Supported	Supported
Load balancing	Supported	Supported (via Virtual Network Appliances)

For more information about supported workload networking modes, see Workload Connectivity Detailed Design in the VMware Cloud Foundation *Design Guide*.

Virtual Private Clouds (VPCs)

A virtual private cloud (VPC) provides an isolated domain of networking resources that can be associated with one or more namespaces. You can create VPCs for your organization and associate each VPC with one or more namespaces. You can create a new VPC for each namespace or you can use the same VPC for multiple namespaces.

> **Important:**
>
> VCF Automation is compatible only with supervisors with NSX VPCs. For more information about networking, see Virtual Private Cloud in the NSX documentation.

Each VPC can leverage three different types of IP blocks that each have different levels of reachability:

- Private CIDRs are local to the VPC and require NAT to connect to any resources outside of the VPC. These CIDRs do not need to be unique across multiple VPCs.
- Transit gateway (TGW) private IP blocks are only advertised within the VCF Automation organization and its TGW. The available IP blocks are defined by the organization administrator. Project administrators can allocate subnets from within the defined IP blocks for use in their VPCs. These IPs have direct reachability to other VPCs within the organization using the TGW private IP space.
- External IP blocks are managed by the provider administrator. IPs in this space are used for outbound Source NAT to enable external reachability for workloads using private IPs. Organization administrators can assign subnets within the external IP blocks by the allocations defined by the provider.

The Network Service can be used to create, update, manage, and delete resources within the VPC by organization administrators and advanced project users.

Connectivity Profiles

A connectivity profile is a template that allows simplified configuration of new VPCs. A default connectivity profile is configured by the provider administrator to simplify the initial organization configuration. You can provide one or more external and private transit gateway IP address blocks that are used by VPCs configured with a specific connectivity profile.

In full-stack VPC regions, a connectivity profile can include external IPv4 blocks and private TGW IPv4 blocks. In VLAN-backed VPC regions, only external IPv4 blocks are available.

Transit Gateways (TGWs)

A transit gateway (TGW) provides both inter-VPC communication within the organization and connection to the outside world through external connections. The provider administrator defines external connections and makes them available to your organization.

One or more VPCs are connected to the transit gateway, and that connection is defined by a VPC connectivity profile. Each VPC has connected workloads and, optionally, a private subnet. SNAT rules translate addresses from this private subnet to a public address in the IP spaces block. This infrastructure enables the organization and its workloads to connect to external networks.

The type of external connections available to a transit gateway depends on the region's network stack.

- In full-stack VPC regions, a transit gateway can connect through one of two types of external connections:
  - Centralized connections link to an NSX Tier-0 or Tier-0 VRF gateway, with IP addresses advertised from those gateways.
  - Distributed VLAN connections use VLAN fabric for high-performance distributed routing.

  A transit gateway can have one or more centralized connections or one or more distributed connections, but cannot mix centralized and distributed connections on the same gateway.
- In VLAN-backed VPC regions, transit gateways use a distributed VXLAN connection (DVC) for external connectivity instead of centralized or distributed VLAN connections. A DVC uses EVPN-VXLAN fabric through a Virtual Network Appliance (VNA) cluster and is provisioned by the provider administrator.

Every organization comes with a default transit gateway per region, created by the provider administrator. Organization administrators can also create transit gateways as needed. See Create a Transit Gateway in VCF Automation.

You can also manage IP addresses that the provider administrator made available to you and create IP addresses that are specific to your organization. See IP Address Management in VCF Automation Organizations.

Subnets

A subnet is a subdivision of a VPC that defines a range of IP addresses.

There are three primary subnet types available to organization administrators depending on whether your region uses full-stack VPC (standard TEP-based) or VLAN-backed VPC (TEP-less) networking.

- Private subnets are scoped to the VPC and routed with other subnets inside the same VPC, they have no direct path to the outside network. If outside communication is required, you must configure NAT rules on the transit gateway. Private CIDRs are local to the VPC and do not need to be unique across multiple VPCs. Private subnets that are scoped to the transit gateway allow inter-VPC connectivity below the transit gateway without NAT. Private subnets are available in full-stack VPC regions only.

  Note:

  Private subnets are not available in VLAN-backed VPC regions in VCF Automation 9.1.1. Private IP addressing is not supported in VLAN-backed mode because the physical VLAN fabric does not allow private IPs.
- Public subnets are accessible from outside the environment, from other VPCs, and also from customers and workloads above the transit gateway. IP addresses are allocated from external IP blocks managed by the provider administrator and assigned to the organization's connectivity profile. Public subnets are available in both full-stack VPC and VLAN-backed VPC regions.
- VLAN extension subnets extend the broadcast domains of the physical fabric directly to the organization so that organization users can connect their workloads to an existing VLAN without overlay encapsulation. The gateway resides in the physical network, not inside the VPC. VLAN extension subnets can only be public. VLAN extension subnets are available in both full-stack VPC and VLAN-backed VPC regions. In VLAN-backed mode, VLAN extension subnets map directly to a Distributed VXLAN Connection (DVC) and use the DVC's CIDR for IP allocation. The entire CIDR block is allocated to the subnet.

Subnets can be shared. Organization administrators can create private or public subnets for their organizations and share them across namespaces, allowing multiple namespaces to access the same network. Provider administrators can create VLAN extension subnets and share them with multiple organizations.

Project users can use the Network service to create subnets in VPCs that are associated with their namespaces. These subnets are used for that namespace only.

In VLAN-backed mode, shared subnets serve as the default network for pods and VMs in namespaces. When creating a namespace in a VLAN-backed VPC region, you must add at least one shared subnet and can optionally designate it as the default network.

Avi Load Balancer

In VCF Automation organizations that have regions with Avi Load Balancer enabled, each namespace is associated with a designated Avi service engine group (SEG), which defines how Avi service engines are sized, placed, and made highly available.

Depending on requirements, the provider administrator can extend Avi load-balancing capabilities to your VCF Automation organization by enabling service engine groups for one or multiple regions that are available organization. Service engine groups allow configuration and management of load-balancing resources for applications.

Each namespace is associated with a SEG. Applications that are deployed within a namespace automatically gain access to load-balancing capabilities. Project users can deploy load balancers using the Avi Load Balancer service and access Avi analytics to monitor performance and troubleshoot their applications.

There are two modes in which the organization interacts with SEGs:

- Provider-managed SEGs. The provider administrator creates, manages, and distributes service engine groups to the organization.
- Organization-managed SEGs. The provider administrator delegates the creation and management of SEGs to the organization administrator, giving them more control over how resources are allocated. See Create a Service Engine Group in the Avi Load Balancer for VMware Cloud Foundation 9.1 guide.

For more information about using the Avi Load Balancer integration with VCF Automation, see the Avi Load Balancer for VMware Cloud Foundation 9.1 guide.

vDefend Firewall

VCF Automation 9.1 integrates with vDefend Firewall. Depending on the organization structure and responsibilities, the provider administrator manages firewalls for the organization or they can delegate this task to the organization administrator.

Organization administrators or users with appropriate privileges can then create firewall policies for the following:

- Secure inbound and outbound connectivity to project resources.
- Enforce firewall rules to secure applications between VPCs and within individual VPCs.
- Secure connectivity between applications across VPCs and within a VPC.
- Connect to other organizations or shared services in other organizations.
- Secure connectivity to VPCs across projects.

Organization administrators can configure two types of vDefend Firewall policies.

- vDefend Distributed Firewall policies to control inter- and intra-VPC communications.
- vDefend Gateway Firewall policies on NSX transit gateways and VPCs to control inbound/outbound communication with the provider gateway.

Project administrators can also configure gateway firewalls for VPCs dedicated to namespaces in their projects.

For more information about using vDefend Firewall in VCF Automation, see the VCF Automation Integration with vDefend Firewall guide.

Best Practices for Organization Networking

- Effectively using VPCs. VPCs offer networking isolation. As an organization administrator, you can create VPCs and reserve them for use by certain namespaces and projects or share VPCs with all organization users. You can review the following sections for examples for effectively using VPCs to fit your organization needs.
  - For information about using a single VPC for the entire organization, see Multi-Tenancy Consumption Pattern 1: Resource Isolation with Single VPC for Multiple Businesses in One Supervisor Zone in the VMware Cloud Foundation *Design Guide*.
  - For information about how each Line of Business, represented as a VCF Automation project, can use a dedicated VPC, see Multi-Tenancy Consumption Pattern 2: Resource Isolation with Multiple VPCs for Multiple Lines of Business in One Supervisor Zone in the VMware Cloud Foundation *Design Guide*.
- Planning for IP blocks and quotas. To learn more about workload networking and how to plan for IP blocks, see Understanding Workload Networking in the VMware Cloud Foundation *Design Guide*.

Create a Subnet in a VCF Automation Organization

As an organization administrator, you can create subnets in your organization and share them across namespace so that multiple namespaces can leverage the same network.

- Review what regions are available to your organization on the **Regions** page.
- Verify that a connectivity profile is defined for the region in which you want to create the subnet. If the region does not have a connectivity profile, you can create one at the time of creating the subnet. See Create a Connectivity Profile in VCF Automation.
- Verify that you have a VPC. See Create a Virtual Private Cloud in VCF Automation.

In addition to the VPC that a namespace is mapped to, each namespace can have subnets created from other VPCs that might be used across namespaces. While connectivity within the VPC is guaranteed, subnets from multiple VPCs might not reach one another, which allows workflows involving virtual machines with multiple NICs.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Subnets** and click **New Subnet**.
3. Select a region.

   You can select from regions that the provider administrator made available to your organization.

   Important:

   If the region you select has VLAN-backed VPC networking, the UI automatically detects VLAN-backed mode. The networking settings that are available to you change depending on the type of region you selected.
4. Select a VPC from the region.

   Based on your VPC selection, a name is generated for the subnet. You can provide a new name.
5. Optional: Change the name that was generated for the subnet.
6. For VLAN-backed regions, leave the **VLAN Connection** toggle on, select a distributed external connection, and continue with Step 10b.

   If you turn off the **VLAN Connection** toggle, you configure the subnet manually. Continue with Step 8.
7. Select an access mode to apply to the subnet.

   Note:

   In VLAN-backed regions in VCF Automation 9.1.1, only public subnets are available.

Option	Description
Private - VPC	The subnet is scoped to the VPC and routed with other subnets inside the same VPC, but not to the outside network. If outside communication is required, you must configure NAT rules. See Create a NAT Rule for a Transit Gateway in VCF Automation.
Private - Transit Gateway	The subnet allows inter-VPC connectivity below the transit gateway without NAT. If workloads must be reachable to an outside network, IP translation is required.
Public	The subnet is accessible from outside the environment, from other VPCs, and also from customers and workloads above the transit gateway.

8. Specify IP blocks.

   If there is at least one IP block configured in the selected VPC, by default, the IP block is set to `Any`, which means VCF Automation picks any of the available IP blocks based on the allocation mode you select in the next step.

   You can also use a specific IP block if you want to ensure the CIDR is picked from that block only.

   For public access mode, you can select from IP blocks attached to the connectivity profile of the VPC and VLAN extension IP blocks that are not related to any VPC. An IP block configured with VLAN extension allocates the entire CIDR block to the subnet and is intended for VLAN extension use cases.
9. If you want to auto-allocate the IP block, turn on the toggle.
   - If you enable automatic IP block allocation, select an IP block size.

- If you disable automatic IP block allocation, enter the IP CIDR.

10. Configure advanced settings.
    a) To enable gateway connectivity, turn on the toggle.

       When gateway connectivity is on, the subnet is connected to the VPC gateway.

       In scenarios that require fully subnet isolation, turn the gateway connectivity toggle off.

       Note:

       In VLAN-backed regions in VCF Automation 9.1.1, you cannot change gateway connectivity.
    b) To enable static IP allocation, turn on the toggle.

       If you do not want to enable static IP allocation, turn off the toggle and specify DHCP settings.
11. Click **Create**.

- Share the subnet with a namespace. Workloads that are provisioned on the namespace can use the subnet.
  - To add the subnet to an existing namespace, see Edit Namespace Settings in VCF Automation.
  - To create a new namespace using the subnet, see Create a Namespace in a VCF Automation Project.
- Once you share the subnet with a namespace, verify that the subnet is listed for that namespace. In the Services section on the **Build & Deploy** tab, select the namespace, and open the Network service. On the **Subnets** tab, you can view a list of associated subnets and can create subnets that are scoped to this namespace only.
- Project users can now use the subnet when provisioning workloads to the namespace.

Create a Connectivity Profile in VCF Automation

As an organization administrator, you can create, update, and delete VPC connectivity profiles in VCF Automation.

- Review what regions are available to your organization on the **Regions** page. You create the connectivity profile in a region that is allocated to your organization.

Your organization comes with a default connectivity profile. As an organization administrator, you can create additional connectivity profiles to use in your organization if, for example, you need to configure different external and/or private IP address blocks.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Virtual Private Clouds** > **Connectivity Profiles** and click **New Connectivity Profile**.
3. Enter a name.
4. Select a region.

   Note:

   If the region you select has VLAN-backed VPC networking, the UI automatically detects VLAN-backed mode. The networking settings that are available to you change depending on the type of region you selected. Some options might not be available to you.
5. Select a transit gateway.

   A transit gateway is automatically picked based on the region you selected.
6. Optional: Select external IPv4 blocks.

   In full-stack VPC regions, you can select up to five IPv4 blocks.

   In VLAN-backed VPC regions, only external IPv4 blocks allocated by your provider administrator are available for selection. Continue with Step 10.
7. Optional: In full-stack VPC regions, select private TGW IPv4 blocks.

   You can select up to five IPv4 blocks.

   In VLAN-backed VPC regions, private TGW IPv4 blocks are not supported. Continue with Step 10.

8. In full-stack VPC regions, to enable a default outbound SNAT rule, turn on the toggle.

   You cannot use outbound SNAT for vSphere Kubernetes Service or any services that require workloads to reach out by default.

   In VLAN-backed VPC regions, SNAT is not supported. Continue with Step 10.

9. If you enabled default outbound SNAT, select external IP blocks.
10. Click **Create**.

- To edit existing connectivity profiles, click the vertical ellipsis next to the connectivity profile name and then click **Edit**.

  Note:

  Changes to the connectivity profile affect the northbound connectivity configuration of all VPCs that use this connectivity profile.

- Use the connectivity profile to create a VPC. See Create a Virtual Private Cloud in VCF Automation.

Create a Virtual Private Cloud in VCF Automation

As an organization administrator, you can create, update, and delete virtual private cloud (VPC) configurations in your organization.

- Review what regions are available to your organization on the **Regions** page. You create the VPC in a region that is allocated to your organization.
- Verify that a connectivity profile is defined for the region in which you want to create the VPC. If the region does not have a connectivity profile, you can create one at the time of creating the VPC. See Create a Connectivity Profile in VCF Automation.
- If you want to apply an IP quota to the VPC, verify that you have an IP quota defined. See IP Address Management in VCF Automation Organizations.

The VPC is associated with a region where namespaces are provisioned. A single VPC can be shared by multiple namespaces.

In full-stack VPC regions, VPCs use standard TEP overlay networking and support the full range of IP addressing, subnet types, and network services. When you create a VPC in a full-stack VPC region, you can configure private VPC IPv4 CIDR blocks, private TGW IPv4 blocks, external IPv4 blocks, NAT rules, gateway firewall, static routes, and VPN on the transit gateway, as well as NAT and gateway firewall on the VPC itself. You can create private subnets, public subnets, and VLAN extension subnets.

In VLAN-backed VPC region, the available configuration options are restricted because the physical VLAN fabric does not support private IP addressing or overlay encapsulation. Some fields and services are not available and do not appear in the UI during VPC creation.

As an organization administrator, you can delegate project administrators the ability to either consume specific VPCs or create their own VPCs within constraints that you set.

Namespace users can request external IPs for 1:1 NAT in order to have direct external access to a VM.

Your organization comes with a preconfigured VPC and a connectivity profile. You can use these defaults to connect networking workloads, or you can create new ones using the steps in this procedure.

1. Log in to the VCF Automation organization as an organization administrator or a project administrator with the necessary permissions.
2. Select **Manage & Govern** > **Networking** > **Virtual Private Clouds** and click **New VPC**.
3. Enter the VPC name.
4. Select a region.

   You can select from regions that the provider administrator made available to your organization.

5. Select an existing connectivity profile defined for the region you selected in the previous step, or create a new connectivity profile.
   a) If you choose to create a new connectivity profile, enter a name.
   b) Select a transit gateway.
   c) Optional: For full-stack VPC regions, select external IPv4 blocks and Private TGW IPv4 blocks.

      Note:

      Private TGW IPv4 blocks are not available in VLAN-backed VPC regions. Only external IPv4 blocks can be selected.
6. For full-stack VPC regions, specify private VPC IPv4 CIDR blocks.

   Note:

   For VLAN-backed VPC regions, skip step 6. Private VPC IPv4 CIDR blocks are not supported.
7. Expand the **Advanced Settings** section.
8. Optional: Select IP quotas to apply to the VPC.
9. Optional: To enable load balancing for the VPC, turn on the toggle.

   If you disable load balancing, you cannot use vSphere Kubernetes Service.
10. Click **Create**.

To edit existing VPCs, click the vertical ellipsis next to the VPC name and then click **Edit**.

Edit DHCP Settings for a VPC in VCF Automation

As an organization administrator, you can edit existing DHCP service configurations for every region your organization has quota in.

For DHCP server profile configurations, only the default system profiles are exposed and can be edited. You cannot delete the default system profiles. Multiple DHCP configurations are not permitted per organization.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Virtual Private Clouds** and click the **DHCP Settings** tab.
3. Select a region.
4. Click **Edit Configuration**.
5. To disable the DHCP server config, turn off the toggle.
6. Enter DNS server IPs.
7. Specify lease time in seconds.
8. Enter NTP server IPs.
9. To enable the DHCP relay config, turn on the toggle and enter external DHCP server IPs.
10. Click **Save**.

Create a NAT Rule for a VPC in VCF Automation

As an organization administrator, you can create NAT rules for VPCs in your organization.

Your organization comes with a default VPC that has a default Source NAT rule. The default NAT is read only, but you can add new rules for the default VPC.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Virtual Private Clouds**.
3. On the **VPCs** tab, select the VPC you want to create the rule for.
4. On the details page, select the **NAT** tab.
5. Click **Add NAT Rule**.
6. Enter a name for the rule.
7. Select an action.

   Based on the action you select, the infographic on the right changes to visualize the rule.

   - SNAT translates a source IP address of outbound packets so that packets appear as originating from a different network. Supported on Tier-0/Tier-1 gateways running in active-standby mode.
   - DNAT translates the destination IP address of inbound packets so that packets are delivered to a target address into another network.
   - Reflexive NAT translates addresses passing through a routing device.
8. To enable the NAT rule, turn on the toggle.
9. In the Source section, enter the source IP and port.

   This defines the source network of the packets leaving the network for **SNAT**, **NO SNAT**, and **Reflexive** rules.
10. In the Destination section, enter the destination IP, port, and protocol.

    This defines the source network of the packets leaving the network for **DNAT** and **NO DNAT** rules.

    The destination IP has to be allocated from the external IP block associated with the VPC. You can use an IP block that was allocated to your organization or you can request a new one.

    a) Click **Request IP**.

       The Request IP/CIDR window appears.
    b) Select an IP address block to use for allocation.
    c) To enable auto-allocation from the IP block, turn on the toggle.

       If you disable auto-allocation, manually specify an IP address. Click View IPs to view available IP addresses you can use.
    d) Click **Request**.
11. In the Translation section, enter an IPv4 address or an IP address range in CIDR format, and port.
12. Expand the Advanced Settings section.
13. To enable logging, turn on the toggle.

    Logs are only available to organization administrators.
14. Enter a priority value.

    By default, priority is set to `0`. A lower value indicates a higher priority.

    No SNAT and **No DNAT** rules must have a higher priority than other rules.
15. Configure a firewall.

    You can choose

    - **Match External Address**. Applies the firewall to the external address of the NAT rule.
    - **Match Internal Address**. Applies the firewall to the internal address of the NAT rule.
    - **Bypass**. Policy-based VPN traffic bypasses the NAT rule.
16. Click **Create**.

You can edit and delete the NAT rules that you created.

Create a Transit Gateway in VCF Automation

As an organization administrator, you can create, edit, and delete transit gateways in your organization. You connect transit gateways to VPCs.

- Review what regions are available to your organization on the **Regions** page. You create the transit gateway in a region that is allocated to your organization.
- Review what external connections are available to your organization on the **Transit Gateways** > **External Connections** page.
- Verify that a connectivity profile is defined for the region in which you want to create the VPC. If the region does not have a connectivity profile, you can create one at the time of creating the VPC. See Create a Connectivity Profile in VCF Automation.

Your organization comes with a default transit gateway in each region assigned by your provider administrator. You can use the default TGW or you can create new transit gateways for your organization using the steps in this procedure. You cannot delete the default transit gateway in your organization.

On the **External Connections** tab, you can view the external connections that are associated with transit gateways in your organization. External connections are always shared by the provider administrator.

Depending on the networking stack for the region that is assigned to your organization, three types of external connections are available.

Type	Description	Region Type	VCF Automation Version
Centralized	Connects to an NSX Tier-0 or Tier-0 VRF gateway. IP addresses are advertised from those gateways.	Full-stack VPC	VCF Automation 9.1 and later
Distributed VLAN	Connects to a VLAN for high-performance distributed routing.	• Full-stack VPC • VLAN-backed VPC	VCF Automation 9.1 and later
Distributed VXLAN	Connects over EVPN-VXLAN fabric via a Virtual Network Appliance (VNA) cluster.	Full-stack VPC	VCF Automation 9.1.1

Depending on the region type and what your provider administrator shared, only certain connection types may be available to your organization.

- In full-stack VPC regions, a transit gateway can have one or more centralized connections, or one or more distributed connections, but not both centralized and distributed connections at the same time.
- In VLAN-backed VPC regions, a transit gateway uses distributed VXLAN connections exclusively.

For more information about external connections, see Managing External Connections in VCF Automation.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Transit Gateways** and click **New Transit Gateway**.
3. Enter a name and, optionally, a description.
4. Select a region.
5. Next to External Connections, click **Select**.
6. Select the required connections and click **Apply**.

    - In full-stack regions, depending on your allocation, you can choose from centralized or distributed connections.
    - In VLAN-backed regions, select a distributed VLAN connection.

7. To create a new VPC connectivity profile for this transit gateway, select the check box.
   a) Enter external IP blocks.
   b) In full-stack regions, enter a private transit gateway IP block or create a new one.
      1. Enter a name.
      2. A region is selected.
      3. Enter CIDRs.
      4. Configure advanced settings.
      5. Click Create.
8. Click **Create**.

Create a NAT Rule for a Transit Gateway in VCF Automation

As an organization administrator, you can create NAT rules for transit gateways for external address binding.

Note:

NAT rules are not supported for VLAN-backed regions.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Transit Gateways**.
3. Select the TGW you want to create the rule for.
4. On the details page, select the **NAT** tab.
5. Click **Add NAT Rule**.
6. Enter a name for the rule.
7. Select an action.

   Based on the action you select, the infographic on the right changes to visualize the rule.
8. To enable the NAT rule, turn on the toggle.
9. In the Source section, enter the source IP and port.
10. In the Destination section, enter the destination IP, port, and protocol.

    The destination IP has to be allocated from the external IP block associated with the VPC. You can use an IP block that was allocated to your organization or you can request a new one.
    a) Click **Request IP**.

       The Request IP/CIDR window appears.
    b) Select an IP address block to use for allocation.
    c) To enable auto-allocation from the IP block, turn on the toggle.

       If you disable auto-allocation, manually specify an IP address. Click View IPs to view available IP addresses you can use.
    d) Click **Request**.
11. In the Translation section, enter an IPv4 address or an IP address range in CIDR format, and port.
12. Expand the Advanced Settings section.
13. To enable logging, turn on the toggle.
14. Enter a priority value.

    A lower value indicates a higher priority. The default value is 0.

15. Configure a firewall.
16. Click **Create**.

You can edit and delete the NAT rules you created.

Configure IPSec VPN for a Transit Gateway in VCF Automation

As an organization administrator, you can enable IPSec VPN in VCF Automation for transit gateways in your organization.

Transit gateways support both policy-based and route-based VPNs with preshared keys.

The IPsec VPN defines the local endpoint and supports creating IPsec sessions for secure site-to-site or remote access communication.

> **Note:**
>
> IPSec VPN is supported if the organization has at least one region configured with a centralized connection. IPSec VPN is not supported for distributed connections that link to a distributed transit gateway.

For more information about IPSec VPN, see the official NSX documentation.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Transit Gateways**.
3. Select the transit gateway for which you want to configure a VPN.
4. On the details page, select the **VPN** tab.
5. Click **Configure**.
6. To enable the VPN, turn on the toggle.
7. To enable session sync, turn on the toggle.
8. Select IKE log level.
9. Expand the Global Bypass Rules section.
10. Enter local networks.
11. Enter remote networks.
12. Click **Save**.

    A window appears. If you want to create an IPSec VPN session, click Yes to continue.

You can create an IPSec VPN session. See Create an IPSec VPN Session in VCF Automation.

Create an IPSec VPN Session in VCF Automation

As an organization administrator, you can create an IPSec VPN Session in VCF Automation for transit gateways in your organization.

Verify that you configured IPsec VPN for the transit gateway. See Configure IPSec VPN for a Transit Gateway in VCF Automation.

> **Note:**
>
> In organizations with full-stack VPC networking, IPSec VPN is supported if the organization has at least one region configured with a centralized connection. IPSec VPN is not supported for distributed connections that link to a distributed transit gateway. IPSec VPN is not supported in organizations with VLAN-backed VPC networking.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Transit Gateways**.
3. Select the transit gateway you want to configure a VPN for.
4. On the details page, select the **VPN** tab in the left-hand navigation.
5. On the VPN tab, click the **IPSec Sessions** tab.
6. Click **New IPSec Session**.
7. Enter a name and, optionally, a description for the session.

   The name must have a prefix of the parent resource name, followed by a colon, for example, *`default@us-east:ipsec-session-1`*.
8. To enable admin status, turn on the toggle.
9. Select a session type.
   - Policy-based sessions encrypt traffic based on defined policies using specific source/destination IPs and protocols.
   - Route-based sessions use a virtual tunnel interface (VTI) to encrypt all traffic routed through it. Dynamic routing is not supported.
10. In the Peer Authentication section, enter the pre-shared key.

    VCF Automation supports only pre-shared key (PSK) authentication for IPSec sessions. Pre-shared keys are used for authentication between the local and remote endpoints.
11. Click **Next** to configure endpoints.
12. In the Local Endpoint section, configure the local endpoint for the IPSec session.
    a) Enter the local IP address or create a new one.

       Define the local gateway IP to establish the VPN session with a remote site and the internal IP subnets that sends or receives traffic over the VPN.

       1. Click **Create New**.
       2. Enter a name.
       3. Select an IP address or request one.
          a. Click **Request IP**.
          b. Select an IP address block to use for allocation.
          c. To enable auto-allocation from the IP block, turn on the toggle.
             If you disable auto-allocation, manually specify an IP address. Click View IPs to view available IP addresses you can use.
          d. Click **Request**.
       4. Specify a local ID.
       5. Click **Create**.

    b) Enter local networks.
13. In the Remote Endpoint section, configure the remote endpoint for the IPSec session.
    a) Enter a remote IP address.
    b) Enter remote networks for policy-based sessions. For route-based sessions, enter a local tunnel interface.
    c) Enter a remote ID.

       The remote ID is the identifier presented by the remote peer for authentication. It can be an IP address, FQDN, or a custom ID.
14. Click **Next** to configure advanced settings.
15. Optional: Configure security profiles.

    Security profiles define the local gateway IP used for establishing the VPN session with a remote site and the internal IP subnets that send or receive traffic over the VPN.

16. Optional: Select a connection initiation mode.
17. Optional: In the TCP MSS Clamping section, turn on the toggle.

    This reduces the maximum segment size (MSS) payload of the TCP session during the IPSec connection.
18. Click **Create**.

Create VPN Local Endpoint in VCF Automation

As an organization administrator, you can create VPN local endpoints in VCF Automation. Local endpoints define the local gateway IP used for establishing VPN sessions with a remote site and the internal IP subnets that send or receive traffic over the VPN.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Transit Gateways**.
3. Select a transit gateway for which you have configured a VPN.
4. On the details page, select the **VPN** tab in the left-hand navigation.
5. On the VPN tab, click the **Local Endpoints** tab.
6. Click **New Local Endpoint**.
7. Enter a name for the endpoint.
8. Select an IP address or request one.
   a) Click Request IP.
   b) Select an IP address block to use for allocation.
   c) To enable auto-allocation from the IP block, turn on the toggle.
   d) If you disable auto-allocation, manually specify an IP address.
   e) Click View IPs to view available IP addresses you can use.
   f) Click Request.
9. Enter a local ID.

   Identity of the endpoint, which can be set to DN, FQDN, or unique string ID that identifies the endpoint. If left blank, the IP address is used as identity. When certificates are used, identity is automatically derived from the certificate associated with the endpoint.
10. Click **Create**.

Create Custom VPN Profiles in VCF Automation

As an organization administrator, you can create custom profiles for VPN connections you configured.

For more information, see the official NSX documentation.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Transit Gateways**.
3. Select a transit gateway for which you have configured a VPN.
4. On the details page, select the **VPN** tab in the left-hand navigation.
5. On the VPN tab, click the **Profiles** tab.
6. Create an IKE profile.

   Internet Key Exchange (IKE) profiles provide information about the algorithms that are used to authenticate, encrypt, and establish a shared secret between network sites when you establish an IKE tunnel.

a) Click **New IKE Profile**.
b) Enter a name.
c) Select the IKE version.
d) Select encryption algorithms.
e) Select digest algorithms.
f) Select Diffie-Hellman algorithms.
g) Enter SA lifetime in seconds.
h) Click **Create**.

7. Create an IPSec profile.

   Internet Protocol Security (IPSec) profiles provide information about the algorithms that are used to authenticate, encrypt, and establish a shared secret between network sites when you establish an IPSec tunnel.

   a) Click **New IPSec Profile**.
   b) Enter a name.
   c) Select encryption algorithms.
   d) Select digest algorithms.
   e) To activate PFS, turn on the toggle.
   f) Select Diffie-Hellman algorithms.
   g) Enter SA lifetime in seconds.
   h) Select the DF bit.
      - Copy
      - Clear
   i) Click **Create**.

8. Create a DPD profile.

   DPD (Dead Peer Detection) profiles provide information about the number of seconds to wait in between probes to detect if an IPSec peer site is alive or not.

   a) Click **New DPD Profile**.
   b) Enter a name.
   c) Select DPD probe mode.
      - Periodic
      - OnDemand
   d) In the **DPD Probe Interval** text box, enter the number of seconds you want the NSX Edge node to wait before sending the next DPD probe.
   e) In the **Retry Count** text box, enter the number of retries allowed.
   f) Click **Create**.

Create a Static Route for a Transit Gateway in VCF Automation

As an organization administrator, you can create static routes for transit gateways you configured in your organization.

> **Note:**
>
> Static routes are only supported on centralized transit gateways.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **Transit Gateways**.
3. Select a transit gateway.
4. On the details page, select the **Static Routes** tab in the left-hand navigation.
5. Click **Add Static Route**.
6. Enter a name.
7. Enter a network CIDR.
8. In the **Add Next Hops** section, select a connection.

   You can use TGW attachment or route-based IPSec sessions.
9. Enter admin distance.
10. Click **Add**.
11. Click **Create**.

IP Address Management in VCF Automation Organizations

As an organization administrator, you can view external IP address blocks assigned to your organization by the provider administrator. You can also create private TGW IP address blocks and IP quotas specific to your organization.

Prerequisites

- Review what regions are available to your organization on the **Regions** page.

Create an IP Address Block

You can view the IP address blocks are assigned to your organization by the provider administrator. You can configure a private TGW IP address block to use only within your organization and apply it to a specific VPC.

> **Note:**
>
> The default private TGW IP address block assigned by the provider administrator cannot be edited directly because it is used in the default connectivity profile for the organization. You can either create a new TGW IP address block or remove the default one from the connectivity profile, edit it, and then add it back.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **IP Management** > **IP Address Blocks** and click **New Private - TGW IP Block**.
3. A default name is generated.
   You can change the name.
4. Select a region.
5. Specify a CIDR block.
   For example, `10.0.0.0`.
6. Expand the Advanced Settings section.
7. Specify IP address ranges.
8. Specify IPs to exclude.
9. Click **Create**.

Next, you can add the private TGW IP block to a connectivity profile.

Create an IP Quota

You can configure an IP quota to control IP usage across VPCs in your organization.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **IP Management** > **IP Quotas** and click **New IP Quota**.
3. A default name is generated.
   You can change the name.
4. Select a region.
5. Specify the scope of the IP quota and IPv4 blocks.
   a. If you select the **External scope**, select external IPv4 blocks.
   b. If you select **Private - Transit Gateway**, select Private - TGW IPv4 blocks.
6. Enter a number of single IPs.
7. Enter a number of CIDRs.
8. Select the maximum size of the subnet.
9. Select the VPCs you want to apply the quota to.
10. Click **Create**.

Request an IP

You can view IP allocations and request new IPs or CIDRs to use in transit gateways and VPCs in your organization.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Networking** > **IP Management** > **IP Allocations** and click **Request IP/CIDR**.
3. Select a region.
4. Select the allocation scope.
   You can allocate the IP block to the whole organization or a specific VPC. If you choose the latter, select the VPC.
5. Select a request type.
   You can request an IP address or a CIRD.
6. If you select IP Address, select an IP block from those associated with the VPC connectivity profile.
7. To enable auto-allocation, turn on the toggle.
8. Click **Request**.

Managing Projects in VCF Automation

Projects in VCF Automation organizations provide a logical grouping of users and the infrastructure that they are assigned. Projects are used to represent real-world teams which are in charge of development or operations. Each project has one or more namespaces assigned to it. Each namespace is an envelope of resources where applications can be deployed. Access controls for projects are governed by pre-defined roles.

Project Concepts

Namespaces and Projects. Organization administrators can create and attach one or more namespaces to a project. Administrators can also delete a namespace at any time, which dissociates the namespace from the project. Every organization comes with a default project. The default project doesn’t have users or namespaces. The organization administrator can change the name, description of the default project, and add users and namespaces to the default project.

If the organization administrator enables namespace self-service for a given project, the project administrators can also create namespaces in that project. Self-service namespaces have their own view and manage permissions.

Namespace Class. A namespace class is a template that is used to create namespaces. All components of a namespace, including limits on CPU, memory, and storage utilization, VM classes and storage classes, and content libraries can be defined as a reusable template. Namespace classes are created and maintained by organization administrators.

Namespaces and Resource Limits. Infrastructure quota assigned to an organization can be subdivided and assigned to project teams in the form of namespaces. Namespaces are isolated resource pools in which resources like VMs, VKS clusters, and volumes can be provisioned. Organization administrators can override the resource limits specified in a namespace class during namespace creation.

Namespaces and Zones. The resource limits of a namespace can be satisfied from capacity in one or more vSphere zones. When creating a namespace, organization administrators can select which zones the namespace can use.

Namespaces and Organization Quotas. Namespaces count towards the organization quota. The sum of resource limits of all created namespaces cannot exceed the organization quota.

Best Practices for Projects

How many projects and namespaces you create depends on your organization needs.

- **Projects**. Multiple projects are used when isolation boundaries between project teams are required. Only project members can access deployments and resources available to a project. Fewer projects are desired when isolation requirements can be traded off for a simpler operating model. For more information, see Understanding Organizations and Projects in the VMware Cloud Foundation *Design Guide*.
- **Namespaces**. Multiple namespaces are used when strict resource limits are needed for each application. Namespaces are logical constructs that can be used to define application boundaries. Each application can be deployed on a namespace with prescribed resource limits for CPU, memory, and storage. For instance, a group of VMs and VKS clusters that make up an application can be logically grouped in a namespace. For more information, see Understanding Supervisors, Regions, Zones & vSphere Namespaces in the VMware Cloud Foundation *Design Guide*.

Create a Project in VCF Automation

As an organization administrator, you create a project to which you add users and namespaces so that project members can deploy VMs and other resources to the associated zones. You can assign a project administrator or you can operate as the project administrator.

This procedure is based on creating an initial project that includes only the basic configurations. As your application team deploys their resources, you might modify the project.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Projects** and click **New**.
3. Enter the project name and click **Next**.
4. (Optional) Add users or user groups and assign roles.

   At this stage, you can add, for example, a project administrator only and let them add more users to the project.

   User roles based on groups allow users to come and go from your identity provider rather than requiring the project administrator to constantly update the individual users in the project. For example, if deployment ownership is based on groups, you can change the ownership of a deployment to the group. Any notifications are sent to all the group members, allowing anyone to respond.

   For more information about user roles, see Organization and Project Role Permissions in for All Apps.
5. Click **Next**.
6. Review your project and click **Create Project**.

Add namespaces to your project. Namespaces can be added to a project at any time. See Create a Namespace in a VCF Automation Project.

Assign Project Roles to Users and Groups in VCF Automation

As an organization administrator, you assign project roles to users in your organization so that they can view and provision project resources.

- Users and user groups have been imported from a valid identity provider instance that is associated with your VCF Automation organization. Managing Identity Providers in for All Apps
- Users and user groups have been assigned organization roles. Add Users Role Assignments in for All Apps
- Verify that you have a project. See Create a Project in for All Apps.
- For a more detailed task and role list for various roles, see Organization and Project Role Permissions in for All Apps.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Projects**.
3. Open the project to which you want to add users.
4. On the **Users** tab, click **Add Users** or **Add Groups**.
5. Enter the names or email addresses of the users or groups you want to add.

   For example, tony@mycompany.com, sylvia@mycompany.com.

   If you are adding users, you can import a CSV list.
6. Assign a role to the users or groups.
7. Click **Add**.

Managing Namespaces in VCF Automation Projects

In VCF Automation, resources like networks, namespace classes, and namespace services are owned and provisioned by the organization administrator. Project administrators can consume these shared resources in a self-service manner within the boundaries of their own project.

In order for project administrators to be able to create namespaces for their projects, the organization administrator first sets up self-service for the project. The self-service settings govern how a namespace is created by the project administrator, including what shared resources can be used and what templates are available for them to create their own resources, like dedicated VPCs.

As an organization administrator, you enable self-service namespace creation in a project. Once self-service is set up for the project, project administrators can create namespaces within the available limitations. These namespaces can be captured as blueprints and released to the catalog for project users to deploy.

Project administrators create namespaces for their projects using the allocated resource configurations, like regions, namespace classes, VPCs, and subnets. They cannot modify the assigned namespace classes, VPCs, and regions. Project administrators can then capture the namespaces they created for their projects as blueprints and release them to the VCF Automation catalog.

Project users can request these catalog items to deploy namespaces. Their deployments are also limited to what was allocated to the project.

Create a Namespace Class in VCF Automation

To enable the creation of new namespaces, you use a namespace class as a parameterized template that defines the settings and requirements for the namespace. As an organization administrator, you can create namespace classes to define the settings and requirements for namespaces that you add for your organization.

Review what regions are available to your organization on the Regions page.

The namespace class references the VM classes and storage classes and optional content libraries to attach to the namespace. When you specify VM classes and storage classes, all the available VM classes and storage classes that are associated with the selected zones are added to the created namespace.

When your organization is first created, the following namespace classes are created by default.

Namespace Class	CPU Limit	Memory Limit
small	10 GHz	10 GB (9.77 GB useable)
medium	20 GHz	20 GB (19.53 GB useable)
large	40 GHz	40 GB (39.06 GB useable)

These default namespace classes include all VM classes that are available in the region.

You can use the default namespace classes as they are or you can modify them as needed. You can also create new namespace classes using the following procedure.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Namespace Class** and click **New**.
3. Enter a name and, optionally, a description for the namespace class.
4. Click **Next**.
5. Define CPU and memory limits.

   You configure limits and reservations per zone.

   To apply resource limits and reservations across all available zones in bulk, select the **Add all zones and apply the same limits and reservations to all zones** and enter the respective values.

   a) Click **Add**.
   b) Select a zone.
   c) For limits, enter the maximum amount of CPU and memory that can be consumed in each namespace.
   d) Optional: For reservations, enter the minimum amount of CPU and memory that you want to guarantee for each namespace.
   e) Click **Save**.
6. Click **Next**.
7. Add VM classes.

   VM classes define how much memory, CPU, or certain capabilities a VM has.

   a) In the VM Classes section, click **Add**.
   b) Select the classes that you want to use.

      To apply classes in bulk, turn on the **Select all available VM classes** toggle.
   c) Click **Add**.
8. Add storage classes.

   Storage classes define the storage policies and the quota that is assigned in a region.

   When you select a storage class, the namespace class is automatically associated with a region that is allocated to your organization. Later on, when you create a namespace using this namespace class, the region is automatically selected.

   a) In the Storage Classes section, click **Add**.
   b) Select a storage class.
   c) Define a storage limit.
   d) To apply the storage limit to all available classes in bulk, turn on the **Select all available storage classes** toggle.
   e) Click **Add**.
9. Click **Next**.
10. Optional: Add a content library.

    A content library is automatically assigned to the namespace class. The namespace class may also specify the names of configured content libraries that are not automatically attached so that they are also attached.

    Organization content libraries can be set to automatically attach to namespaces or not. If there are no content libraries set to not attach to namespaces, the content library selector is empty. Provider content libraries are always automatically shared with namespaces and namespace classes.

a) Click **Add**.
b) Select content libraries.
c) Click **Add**.

11. Click **Next**.
12. Review your configuration and click **Create**.

You can now create a namespace using your new namespace class. See Create a Namespace in a VCF Automation Project.

Enable Namespace Self-Service for Projects in Your VCF Automation Organization

As an organization administrator, you can allow project administrators to create namespaces in the projects that they manage. You use a namespace class to define the limits to be applied to a new namespace. Project administrators can then create namespaces using the available resources.

- Review what regions are available to your organization on the Regions page. Namespaces are created in a region that is allocated to the organization.
- Verify you have the necessary type of networking.
  - If you plan to allow the creation of namespace with shared VPCs, verify that you have a VPC. See Create a Virtual Private Cloud in VCF Automation.
  - If you plan to allow the creation of namespaces with dedicated VPCs, verify that you have a connectivity profile. See Create a Connectivity Profile in VCF Automation.
- Verify that you have a namespace class. See Create a Namespace Class in for All Apps.
- Verify that you have a project. See Create a Project in for All Apps.
- Verify that you assigned a project administrator to the project for which you want to configure self-service.

By giving project administrators the abiity to create namespaces on a self-service basis, you allow them to operate independently. Namespace self-service is enabled on a project basis.

There is no limit to the number of namespaces that project administrators can create in the project as long as the sum total of resources used by all namespaces does not exceed the quota for the organization.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Projects**.
3. Select the project for which you want to enable self-service.
4. Click the **Namespace Self-Service** tab.
5. Click **Configure Namespace Self-Service**.
6. Select the regions where project administrators can create namespaces and click **Next**.
7. Select namespace classes that can be used to create namespaces and click **Next**.
8. Select a networking type and click **Next**.

   You can select one or both options.

   a) To allow the usage of shared VPCs in the project, select the check box and then select from the available VPCs.
   b) To allow the creation of dedicated VPCs in the project, select the check box and then select from the list of available connectivity profiles.
9. Optional: Select shared subnets that project workloads can connect to and click **Next**.
10. Optional: Select infrastructure policies for the namespace and click **Next**.

    Infrastructure policies ensure that VMs provisioned in the namespace are placed appropriately under the underlying infrastructure to meet specific requirements. Infrastructure policies are created and assigned to your organization by the provider administrator.

Project administrators creating namespaces can choose which of the available policies to apply. Project users provisioning workloads in these namespaces can select which policies to apply to their deployments.

11. Select service engine groups (SEGs) to assign to the project and click **Next**.

    SEGs allow configuration and management of load-balancing resources for applications. Each namespace is associated with a SEG so applications that are deployed within a namespace automatically gain access to load-balancing capabilities.

    You can configure service engine groups only if the region you selected in Step 6 has Avi Load Balancer enabled. Otherwise, you don't see the option to configure SEGs.

12. Review your configuration and click **Submit**.

Project administrators can now create namespaces in this project.

Edit Self-Service Namespace Settings for a Project in Your VCF Automation Organization

As an organization administrator, you can configure the self-service namespace settings for projects in your organization. You assign the resources that project administrators use when creating namespaces which defines the infrastructure components that are available within the project scope.

When you configure self-service settings for a project, you assign regions, namespace classes, and networking resources that the project can use. With these settings in place, you create guardrails for all namespaces created within this project, while giving project administrators flexibility.

You can add multiple resources at a time. When you no longer want a resource to be available, you can remove it.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Manage & Govern** > **Projects**.
3. Select the project for which you want to modify the settings.
4. Click the **Namespace Self-Service** tab.
5. Associate a region with the project.
   a) Under Associated Regions, click **Add**.
   b) Select a region.
   c) Click **Add**.
6. Define the namespace class.
   a) Under Associated Namespace Classes, click **Add**.
   b) Select namespace classes that can be used to create namespaces.
   c) Click **Add**.
7. Define VPCs.

   You can select from VPCs that are available to the regions associated with the project.

   a) Under Associated VPCs, click **Add**.
   b) To filter VPCs by regions, click the funnel icon in the Region column and enter the region name.
   c) Select a VPC.
   d) Click **Add**.
8. Add connectivity profiles.

   You can select from connectivity profiles that are available to the regions associated with the project. Project administrators use the connectivity profiles to create dedicated VPCs for their namespaces.

a) Under Associated Connectivity Profiles, click **Add**.
b) To filter connectivity profiles by regions, click the funnel icon in the Region column and enter the region name.
c) Select a VPC.
d) Click **Add**.

9. Add subnets.

   You can select from subnets that you created and subnets that your provider administrator shared with your organization.

   In VLAN-backed organizations, only public subnets can be used in VPCs. Because private subnets are unavailable, you must assign at least one pre-created public subnet as the default network for pods and virtual machines in the namespace.

   Note:

   You cannot remove a subnet from a namespace while workloads are using it. To remove a subnet, first remove or migrate all workloads that depend on it.

   a) Under Associated Subnets, click **Add**.
   b) Select a subnet.
   c) Click **Add**.

10. Add infrastructure policies.

    If the provider administrator defined infrastructure policies for the organization, you can make infrastructure policies available to the project.

11. Add Avi Load Balancer service engine groups (SEGs).

    You can add service engine groups only if Avi Load Balancer is configured for a region associated with the project. You can select from service engine groups that are available to the region.

    For regions without Avi Load Balancer, you don't see the option to add service engine groups.

    a) Under Associated Service Engine Groups, click **Add**.
    b) Select a SEG.
    c) Click **Add**.

Project administrators can now create namespaces for their projects.

As an organization administrator, you can also set resource limits on individual namespaces. See Edit Namespace Settings in VCF Automation.

Create a Namespace in a VCF Automation Project

As a project administrator, you can create namespaces in the projects that you manage. You can then publish them as blueprints to the catalog where users in your project can request and deploy them.

- Verify that your organization administrator enabled self-service namespaces for your project. See Enable Namespace Self-Service for Projects in Your VCF Automation Organization.
- Review the available resources for your project on the **Self-Service Settings** tab.

1. Log in to the VCF Automation organization as a project administrator.
2. Select **Manage & Govern** > **Projects**.

   As an organization administrator, you can select **Manage & Govern** > **Namespaces** and click **New Namespace**. In this case, you must also select a project to associate with the namespace.

3. Select the project for which you want to create a namespace.
4. On the **Namespaces** tab, click **New Namespace**.
5. Enter a name and description for the namespace.

   A unique suffix is added to the name you enter. For example, if the name you entered is `mynamespace`, upon creation, the namespace might be called `mynamespace-abcxyz`.

   The name must be up to 63 characters long.
6. Select a namespace class.

   The namespace inherits the configuration of the namespace class.
7. Select a region.

   The region groups the resources that are allocated to the organization.

   Note:

   If the region you select has VLAN-backed VPC networking, the UI automatically detects VLAN-backed mode and hides some of the networking settings.
8. Select at least one zone.
9. Select a networking type.
   - Shared VPC. Select a VPC that was created by the organization administrator. The VPC can be shared between multiple namespaces across the organization. Project administrators can create subnets for namespaces in shared VPCs.
   - Dedicated VPC. Create a VPC that is dedicated only to this namespace. You can edit VPC settings like NAT, gateway firewall, and static routes. You can also create subnets for namespaces associated with this VPC.

     If you want to use a dedicated VPC for this namespace, you must also select a connectivity profile and enter IP blocks.

     Note:

     Dedicated VPCs are not available in VLAN-backed organizations. The namespace must use a shared VPC.
10. Optional: Attach shared VPC subnets.

    You can share a subnet with the namespace later on as a day 2 action.

    Shared subnets are created by the organization administrator in other VPCs and are shared with the project when the organization administrator configures self-service for the project.

    In VLAN-backed VPCs, you must add at least one shared subnet to the namespace. If the Shared Subnets list is empty, the UI displays the following warning: "Set at least one shared subnet for this TEP-less Namespace, otherwise workloads in this Namespace will have no networking."
11. Optional: Set default networks.

    You can set default subnets so that workloads can be configured without specifying a network. These subnets must belong to the VPC used by the namespace where workloads are provisioned. You cannot designate a default network on a subnet that belongs to a different VPC.
    - Default VM Network. The subnet the namespace uses by default when a VM is provisioned without an explicit network.
    - Default Pod Network. The subnet the namespace uses by default for pods.
    - Default VM and Pod. The subnet serves as both VM and pod default network.
12. Select infrastructure policies to apply to the namespace.

    If your organization has at least one compatible infrastructure policy available, you can assign policies to the namespace.

13. Associate the namespace with a service engine group.

    If your organization has an Avi Load Balancer configured, you can add service engine groups (SEGs) to the project. SEGs define how the Avi service engines are sized, placed, and made highly available.

14. Review the namespace summary on the right.

15. Click **Create**.

- Once your namespace is created, you can view a summary about the namespace by clicking the namespace name. In the Services section, you can access all services that are enabled for the namespace, for example, VM Service, Kubernetes Services, Volume Service, and so on. To access a service, click the respective tile.
- View and add metadata for the namespace in the **Annotation** section. See Working with Metadata for VCF Automation Resources.
- To manage a namespace, go to the **Build & Deploy** tab and select the namespace from the dropdown in the Services section. The **Overview** page shows you a utilization summary and all services that are enabled for the namespace. On the Overview page you can capture the namespace as a blueprint and manage your recent captures. Use the **Monitor** tab on the Overview page to view metrics about the namespace.
- On the **Subnets** tab on the Network Service page, you can view the shared subnets attached to the namespace.
- Capture the namespace as a blueprint. You can then release the blueprint in the VCF Automation catalog so that users in your project can request and deploy it. See Capturing Namespaces as Blueprints in VCF Automation.

Edit Namespace Settings in VCF Automation

As an organization administrator, you can edit the CPU, memory, and storage utilization limits and reservations for existing namespaces to accommodate changing application team demands.

You can increase or decrease the resources that are available to a namespace at any time. You can add content libraries, VM classes, and subnets, add storage classes and update storage class limits, and add and remove zones and change limits and reservations for each zone.

Updating a namespace might take a long time to complete depending on the changes made and the state of the underlying infrastructure. Some specific zone changes might be especially time-consuming, for example:

- If you decrease the CPU and memory limits of a namespace that has workloads running that exceed the updated capacity, the update request is paused until the workloads are deleted and the target capacity becomes available.
- If you delete a zone that has workloads running, the request is paused until all workloads are deleted.

1. Log in to the VCF Automation organization as an organization administrator.
2. Navigate to the namespace you want to edit using one of the following paths.
   - Select **Manage & Govern** > **Namespaces**. Click the namespace name to open the details page.
   - Select **Manage & Govern** > **Projects**. Click the project that contains the namespace and then click the **Namespaces** tab. Click the vertical ellipsis next to the namespace name and then click **Edit**.
3. Next to **Annotations**, click **Edit** to add metadata for the namespace.

   For more information on metadata, see Working with Metadata for VCF Automation Resources.

4. Next to **Subnets**, click **Edit** to add or remove shared subnets.

   You can add subnets that the provider administrator shared with you or subnets that you created for your organizations. See Create a Subnet in a VCF Automation Organization.

5. In the **Capacity** section, edit zone limits.
   a) Click **Resource Limits**.
   b) Select capacity type.

   - To edit resource limits across zones in bulk, select **Apply the same limits and reservations across all zones** and enter the updated values.
   - To edit resource limits for specific zones, select **Apply different limits and reservations for each zone** and enter the updated values for the desired zones.

6. In the **VM Classes** section, add or remove VM classes.

   You cannot edit VM classes.

7. In the **Storage Classes** section, add, edit, or remove storage classes.
8. In the **Infrastructure Policies** section, add or remove infrastructure policies.

   Infrastructure policies define constraints and requirements for resource placement and configuration across the regions and zones that are allocated to your organization.

   At least one compatible policy must be available to the namespace.

In the Services section, you can access all services that are enabled for the namespace, for example, VM Service, Kubernetes Services, Volume Service, and so on. To access a service, click the respective tile.

Setting up the Content Hub in VCF Automation

The Content Hub houses blueprints, VM images and content libraries, VCF Operations orchestrator workflows, and other objects that you manage and publish to the catalog where your application teams can request them for deployment. The requirements and process for setting up your VCF Automation catalog depends on the content that you are providing to your users.

Working with the Content Hub

The Content Hub contains the following sections.

Blueprint Design	The Blueprint Design page is where you create VCF Automation blueprints. You can design blueprints from scratch or you can import OVA and YAML files from VMware Cloud Director or other VCF Automation organizations. See Managing Blueprints in VCF Automation.
VM Images	VM images and related binaries are stored in content libraries and can be reused in regions that are mapped to one or multiple vCenter instances.
Content Libraries	Content libraries are collections of VM images and other files, like OVA/OVF, ISO images, and scripts, that you can reuse in different scenarios with allocation dependent on your business needs.
Workflows	A workflow is a set of actions, decisions, and results that are run sequentially to perform a specific task, such as provisioning VMs, backing up, managing infrastructure, and so on. Workflows can run according to a predefined schedule or they can be triggered by certain events. Workflows can request input that is provided by a user, by another workflow, or externally.

Publishing Content to the Catalog

The Content Hub is where you create and store your VCF Automation content. For users in your projects to be able to access and deploy this content, you must publish it to the VCF Automation catalog.

Ensure that the content is working properly before you add it to the catalog.

See Publishing and Sharing Content in the VCF Automation Catalog.

Working with blueprints

As an administrator authoring blueprints, you can update the name, description, catalog icon of the blueprint, and you can version the blueprint.

Blueprint-backed catalog items are available for all project members to request and deploy when a blueprint version is published to the catalog. Any changes to the blueprint are reflected in the corresponding catalog item. The blueprints are refreshed every six hours.

You can also edit the request form of the catalog item, in which case the changes are reflected in the blueprint.

Managing Content Libraries in VCF Automation

Content libraries are centralized repositories of VM images, like OVA/OVF, ISO images, and scripts, that are used to deploy workloads. VCF Automation content libraries can be created by provider administrators and organization administrators. Content libraries created by provider administrators are called provider content libraries. Organization administrators can create two types of content libraries, organization content libraries and project content libraries.

Provider Content Libraries

Provider content libraries are configured by the provider administrator in the VCF Automation Provider Management UI. Provider content libraries are available to all organizations on a read-only basis. Provider administrators can create one or more content libraries, each serving a specific need.

Organization and Project Content Libraries

Organization administrators can create and manage their own content libraries that they scope to either the whole organization or to individual projects. Content libraries created for the organization are not accessible to other organizations managed by the same provider administrator.

To allow project teams to operate independently, like create and manage project- or application-specific images, organization administrators can create a project content library for their project.

The following table provides a comparison between organization content libraries and project content libraries.

	Organization Content Library	Project Content Library
Scope	Organization	Single or multiple projects
Supported file types	• OVA • OVF • ISO	• OVA • OVF
Created and managed by	• Organization administrator • Provider administrator when launching the organization from the Provider Management UI	• Organization administrator • Provider administrator when launching the organization from the Provider Management UI
Project permissions	Read only. All projects in the organization can consume content from this library	• Read only. Associated projects with read permissions can consume content from this library • Read/write. Associated projects with read/write permissions can consumer, upload, and capture content from this library
Subscribe to an external library	Yes. Must be enabled by the provider administrator	Yes. Must be enabled by the provider administrator
Subscribe to a Canonical library	Yes. Must be enabled by the provider administrator	Yes. Must be enabled by the provider administrator

Content Library Concepts

Regions

- Content libraries can be made available in one or more regions in VCF Automation.
- Provider content libraries are automatically available to all organizations that have quota in a region where the content library is available.
- Organization content libraries and project content libraries are explicitly made available in every region where the organization has quota in. For example, if the organization has quota in Region1 and Region2, a content library can be available in Region1 only and so workloads that are provisioned in Region2 do not have access to the content library.

Storage

- Storage consumed by organization content libraries count towards the organization storage quota.

- Organization content libraries subscribed to external sources use storage allocated to the organizational quota as the remote images are downloaded and stored locally in VCF. This could also lead to increased storage utilization than expected.
- Sometimes a region may be backed by more than a single vCenter. In such cases, content libraries created in such regions may be replicated in each vCenter, leading to higher than expected storage utilization. Check with your provider administrator to understand how this might affect your storage utilization.

Namespaces

- Organization content libraries are generally available to all namespaces created by all projects across all regions that were selected when the content library was created. In some cases, organization administrators might want to create content libraries specific only to a namespace. This is achieved by first turning off the toggle **Assign to all current and future namespaces** when creating the organization content library. When a new namespace class is created, the organization administrator can add a content library which makes it available only to namespaces created from that namespace class.
- Project content libraries are available only to namespaces that belong to the same project.

Create an Organization Content Library in VCF Automation

As an organization administrator, you can create a content library for your organization to which you can then upload VM images.

Only ISO and OVA/OVF file types are supported as content library items.

You can also create an organization content library in VCF Automation by subscribing to an external publisher if your provider administrator enabled those options for your organization. Subscribed content libraries are read-only. You cannot upload new items to the library and you cannot delete items from it. You can, however, delete the whole library from VCF Automation. You can subscribe to externally published VCSP-compliant content libraries.

When you create a subscribed library, VCF Automation checks the total size of the items in the library and library creation fails if the items exceed the storage quota for your organization. Automatic or scheduled synchronization of subscribed libraries is not supported.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Content Libraries**, and click **New**.
3. Enter a name and description for the content library.

   After you create the content library, you can update the name and description of the content library only through the VCF Automation API.
4. Next to **Type**, select **Organization Content Library**.

   Only organization administrators can add, edit, and delete images from this content library. All non-administrator users have read permissions only.
5. If you want to assign the content library to all current and future namespaces, turn on the toggle.

   You cannot change this setting later.
6. If you want to create a subscribed library, turn on the **Subscribe to an external library** toggle.
   a) Click **Next**.
   b) Enter the URL of the external content library.
   c) If the content library is password-protected, enter the password for it.
7. Click **Next**.
8. Select one or more regions and a corresponding storage class for each region.
9. Click **Next**.
10. Review your settings and click **Confirm**.

You created an organization content library that is available in each selected region.

To troubleshoot issues with content libraries in VCF Automation, you need VCF Automation Provider Management POD logs. You can download the log bundles from the component-specific area on the **Build** > **Lifecycle** > **VCF Management** page in VCF Operations. Debug logs are in the `/opt/vmware/vcloud-director/logs/vcloud-container-debug.log` file.

Upload VM images to the content library. See Add a VM Image to a Content Library in VCF Automation.

Create a Project Content Library in VCF Automation

As an organization administrator, you can create a content library that you can scope to individual projects for better resource isolation and management.

Project content libraries allow you to make VM images available to specific users and resources within a project, as opposed to organization or provider content libraries which are always available to all users within the organization that the library is shared with. Project content libraries cannot be shared with other organizations.

When a content library is shared with a given project, all namespaces within that project can access the library. The namespace association also means that the library can be accessed by the VM Service.

By default, each project can be associated with up to 100 project content libraries. Provider administrators can configure this limit for the organization. See Create a Content Library.

Project content libraries can span multiple regions and multiple vCenter instances within a single region. In case a region is backed by more than one vCenter, project content libraries created in such regions may be replicated in each vCenter, leading to higher than expected storage utilization. Check with your provider administrator to understand how this might affect your storage utilization.

After an image is uploaded to a project content library, it can be published to the catalog, referenced in a blueprint, or directly deployed from the VM Service. The images are only available within the scope of the projects that the content library is shared with. Project content libraries support OVA/OVF images. Images from project content libraries can also be copied to a provider content library so that they become available to all organizations, if the provider administrator enabled this option for the organization.

> **Important:**
>
> Project content libraries do not support ISO files, vApp OVA files, or multi-VM OVA files.

Because project content libraries are owned by the organization, storage cost is incurred by the organization as a whole.

1. Log in to the VCF Automation as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Content Libraries**, and click **New**.
3. Enter a name and description for the content library.
4. Next to **Type**, select **Project Content Library**.
5. If you want to create a subscribed library, turn on the **Subscribe to an external library** toggle.
6. Click **Next**.
7. Select the type of subscribed library you want to create.
   a) Click **Next**.
   b) Enter the URL of the external content library.
   c) If the content library is password-protected, enter the password for it.
8. Click **Next**.
9. Configure project permissions.

   The content library is accessible to all namespaces associated with each project that is part of the scope. You can change the associated projects at any time.

   You can configure read and/or write permissions for all current and future projects that you associate with the content library or you can configure permissions for individual projects.

For subscribed content libraries, you can assign only read permissions to projects. Project users will not be able to upload VM images to the content library.

10. Click **Next**.
11. Select one or more regions and a corresponding storage class for each region.

    Note:

    Changing the storage class used by any of the regions that the library is attached to will result in an error.

12. Click **Next**.
13. Review your settings and click **Confirm**.

- Upload VM images to the content library. See Add a VM Image to a Content Library in VCF Automation.

Subscribe to Ubuntu OS Images from a Canonical Content Library in VCF Automation

As an organization administrator, you can create a content library that is subscribed to Ubuntu VM images from a Canonical content library.

- The provider administrator must enable the option to subscribe to Canonical for your organization.

VCF Automation content libraries can subscribe to Canonical, integrated with VCF, to get pre-verified Canonical content without the need for any additional configuration. Supported content includes Canonical Ubuntu Pro images which can be used to provision licensed Ubuntu VMs using the VM Service.

You create a content library for Canonical Ubuntu VM images to automate the image procurement pipeline. VCF Automation connects directly to the VCF Software Depot to synchronize validated images into local storage on a 90/60-day schedule.

Canonical publishes updates every 15 days. To fetch the newest releases, you can manually sync your VCF Automation project content library as needed.

1. Log in to the VCF Automation as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Content Libraries**, and click **New**.
3. Enter a name and description for the content library.
4. Next to **Type**, select **Organization Content Library** or **Project Content Library**.
5. For organization content libraries, if you want to assign the library to all current and future namespaces, turn on the toggle.
6. Turn on the **Subscribe to a library** toggle.
7. Click **Next**.
8. Next to type, select **Canonical**.
9. Click **Next**.
10. Configure project permissions.

    The content library is accessible to all namespaces associated with each project that is part of the scope. You can change the associated projects at any time.

    For Canonical libraries, you can assign only read permissions for all current and future projects that you associate with the content library or you can configure permissions for individual projects. Project users in these projects will not be able to upload VM images to the content library.

11. Click **Next**.
12. Select one or more regions and a corresponding storage class for each region.
13. Click **Next**.
14. Review your settings and click **Confirm**.

Add a VM Image to a Content Library in VCF Automation

As an organization administrator, you add a VM image to a content library. As a project administrator, you can add VM images to project content libraries in your project. You can then publish the VM image to the catalog.

Every VM image you upload has a unique identifier (VMI) which you can use in blueprints or to provision VMs directly.

Before you begin, ensure that you have a VM image in a supported format, saved on your local machine.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **VM Images**.
3. Click **Upload**.
4. Click the arrow next to **Select Source** and select a local file to upload.
   Organization content libraries support OVA/OVF and ISO files. Project content libraries support OVA or OVF files.

   Note:

   While you can upload an ISO image to an organization content library, you cannot publish the image directly to the catalog. You must create a blueprint containing the image first, deploy the blueprint, and then convert it to an OVF blueprint.
5. Select a content library to which to upload the VM image.
6. Enter a name and, optionally, a description for the image.
7. Click **Submit**.

It can take a few minutes for the file to get uploaded. Once the upload is complete, verify that the image status is `Ready`.

Provider administrators can copy images from project content libraries to a provider content library so that the images become available to all organizations.

Next, publish the VM image to the catalog for project users can request it. See Publish a VM Image to the VCF Automation Catalog.

Users with the Project Administrator or Project Advanced User role can use the image to deploy a VM using the VM Service.

Copy a VM Image to a Provider Content Library in VCF Automation

As an organization administrator in VCF Automation, you can copy images from a project content library in your organization to a provider content library. Then, the provider administrator can share the images with other organizations.

- Verify that the provider administrator gave you the necessary permissions to copy VM images to provider content libraries. See Enable Cross-Organization Content Sharing in the VCF Automation Provider Management UI.
  Provider consumption organizations have the necessary permissions by default.
- Verify that you have a project content library that contains the VM image you want to copy to your provider library.

1. Log in to the VCF Automation organization as an organization administrator.
2. Navigate to the desired VM image using one of the following paths.
   - Select **Build & Deploy** > **Content Hub** > **VM Images** and locate the VM image.
   - Select **Build & Deploy** > **Content Hub** > **Content Libraries** and open the project content library that contains the VM image. Click the **VM Images** tab and locate the VM image.

3. Click the vertical ellipsis next to the VM image name.
4. Click **Copy to Provider Content Library**.
5. Select a library to add the image to.

   You can share images with one provider content library at a time. If you want to add an image to multiple provider libraries, repeat this procedure as needed.
6. Click **Copy**.

In VCF Automation Provider Management, verify that the image was added to the provider content library.

Publishing and Sharing Content in the VCF Automation Catalog

As an organization administrator, you publish content to the VCF Automation catalog to make it available to project users.

Publish a Blueprint to the Catalog in VCF Automation

As an organization administrator, you publish blueprints to the VCF Automation catalog for project users to request them and provision resources.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Blueprint Design** > **Blueprints**.
3. Open the blueprint that you want to publish.
4. On the blueprint design canvas, click **Version**.
5. Rename the blueprint version as necessary.
6. Select the **Publish this version to the catalog** check box.
7. Click **Create**.
8. Close the blueprint.

Verify that the blueprint is available in the catalog. Catalog users can now request the blueprint for deployment.

By default, the blueprint becomes available to users that are members of the project that the blueprint is associated with. You can share the catalog item with other projects within your organization or with other VCF Automation organizations. See Share Blueprints with Other VCF Automation Projects and Organizations.

Share Blueprints with Other VCF Automation Projects and Organizations

As an organization administrator, you can share VCF Automation blueprints with other VCF Automation projects and organizations.

Verify that the provider administrator gave you the necessary permissions to share content with other organizations. See Enable Cross-Organization Content Sharing in the VCF Automation Provider Management UI. If you do not have these permissions, you can still share content between projects within your organization.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Blueprint Design** > **Blueprints**.
3. Select the blueprint that you want to share.
   a) To create a blueprint, click **New From** > **Black Canvas**.
   b) To edit an existing blueprint, click the blueprint name. In the blueprint design canvas, click **All Actions** > **Settings**.
4. Change the blueprint name to something that will be meaningful to your applications team.
5. Associate the blueprint with a project.
   - If you are creating a blueprint, select a project.
   - If you are editing an existing blueprint, it is already associated with a project.

6. Share the catalog item for the blueprint with other teams. You have the following options:
   - If you want to allow other projects in the organization to access the catalog item, select the check box. The catalog item becomes available to all other projects.
   - If you want to allow other organizations to access the catalog item, select the check box and then select which organizations you want to share the item with. To share the catalog item with other organizations in bulk, turn on the **Select all current and future organizations** toggle.

     Note:

     When you share a catalog item with other organizations, the catalog item is not associated with any projects in those organizations so it's not yet available to project users. The administrator must first add the catalog item to a project. See Add a Catalog Item to a Project in VCF Automation.
7. Save your settings.
   - If you are creating a blueprint, click **Create**.
   - If you are editing an existing blueprint, click **Save**.

Publish a VM Image to the VCF Automation Catalog

As an organization administrator, you can publish VM images to the catalog in VCF Automation.

You can publish OVA/OVF images directly to the catalog. For ISO images, direct publishing to the catalog is not supported. Instead, you must create a blueprint for the image with a volume claim, deploy the blueprint, and then convert it to an OVF blueprint.

1. Log in to the organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **VM Images**.

   You can also access the VM Images page by navigating to **Build & Deploy** > **Content Hub** > **Content Libraries**. Then, open the desired content library, and select the **VM Images** tab.
3. Click the vertical ellipsis next to the VM image name, and then click **Publish to Catalog**.
4. Change the default workflow name to something that will be meaningful to your application team.
5. Optional: Add a description and an icon for the catalog item.
6. Select a project for which you want to publish the catalog item.
7. Optional: (Optional) Share the catalog item for the VM image with other teams. You have the following options:
   - If you want to allow other projects in the organization to access the catalog item, select the check box.

     Note:

     To share catalog items backed by images from a project content library with other projects, the image must be available in those projects.
   - If you want to allow other organizations to access the catalog item, select the check box and then select which organizations you want to share the item with. To share the catalog item with other organizations in bulk, turn on the **Select all current and future organizations** toggle.

     Important:

     Only images from provider content libraries can be shared as catalog items with other organizations. See Copy a VM Image to a Provider Content Library in VCF Automation.

     When you share a catalog item with other organizations, the catalog item is not associated with any projects in those organizations so it's not yet available to project users. The administrator must first add the catalog item to a project. See Add a Catalog Item to a Project in VCF Automation.

8. Optional: To add more storage for the VM image, turn on the **Add storage** toggle.
   a) Select a storage class.
   b) Add volumes.
9. Optional: Toggle **Enable cloud init configuration** for supported images and provide the bootstrap configuration.
10. Click **Publish**.

Catalog users can now request the image for deployment.

Publish a VCF Operations orchestrator Workflow to the Catalog in VCF Automation

As an organization administrator in VCF Automation, you can publish VCF Operations orchestrator workflows as catalog items for project users in your organization.

- Verify that your provider administrator has created a VCF Operations orchestrator integration for your organization. See Integrate VCF Automation with VCF Operations orchestrator.
- Verify that you have the necessary permissions to access the VCF Operations orchestrator integration.
  When your provider administrator creates a VCF Operations orchestrator integration for your organization and assigns you the necessary rights to manage VCF Operations orchestrator content, you see the Orchestrate tab in the VCF Automation organization UIUI, you can access the VCF Operations orchestrator integration on the Integrations page, and you can see the workflows listed on the Content Hub page.

  Note:

  In provider consumption organizations (PCOs), a VCF Operations orchestrator integration is enabled and accessible by default. A PCO is a type of VCF Automation organization that the provider administrator uses to consume resources and perform other tasks, like sharing content across multiple organizations that they manage. See Create a Provider Consumption Organization in VCF Automation.
- Verify that the provider administrator gave you the necessary permissions to share workflows with other organizations. See Enable Cross-Organization Content Sharing in the VCF Automation Provider Management UI. If you don't have these permissions, you can still share content between projects within your organization.

For more information about VCF Operations orchestrator workflows, see Building VCF Operations Orchestrator Workflows.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Workflows**.
3. Select the workflow that you want to publish and click **Publish to Catalog**.
4. Change the default workflow name to something that will be meaningful to your applications team.
5. Add a description and a custom icon for the catalog item.
6. Associate the workflow with a project in the PCO.
7. Share the catalog item for the workflow with other teams. You have the following options:
   - If you want to allow other projects in the organization to access the catalog item, select the check box.
   - If you want to allow other organizations to access the catalog item, select the check box and then select which organizations you want to share the item with. To share the catalog item with other organizations in bulk, turn on the **Select all current and future organizations** toggle.

     Note:

     When you share a catalog item with other organizations, the catalog item is not associated with any projects in those organizations so it's not yet available to project users. The administrator must first add the catalog item to a project. See Add a Catalog Item to a Project in VCF Automation.
8. Click **Publish**.

Add a Catalog Item to a Project in VCF Automation

As an organization administrator, you can add or remove project associations for items that you published to the VCF Automation catalog.

When creating blueprints or adding other content, such as VM images or workflows, in a VCF Automation organization, each item must be associated with a project. You can then share the item with other projects in your organization.

When you work with shared content that originated in a different VCF Automation organization, the catalog item is imported in your organization without a project association. For your project users to be able to request the catalog item, you must add it to a project.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Catalog**.
3. Turn on the **Show items without a project** toggle.
4. Locate the item whose project associations you want to edit.

   Notice that the **Request** button is grayed out. The catalog item cannot be requested.
5. Slick **Actions** > **Add/Remove from Projects**.
6. Select or deselect a project.

   You can select multiple projects at a time.
7. Click **Apply**.

Managing Blueprints in VCF Automation

A blueprint is a declarative template used to define and automate the deployment of infrastructure, applications, and services. It allows users to model complex cloud resources, such as VMs, networks, storage, and software components. Blueprints orchestrate the deployment of multiple resources in the VCF platform. They act as a reusable design for deploying resources and services consistently.

Blueprint Features

- **Blueprint designer.** The Blueprint Designer offers a single interface to create blueprints for various resource types, including single machines, multi-machine deployments, and multi-tiered applications.
- **Declarative Infrastructure-as-Code.** Blueprints define the desired state of the infrastructure using YAML or a visual designer, rather than specifying step-by-step instructions.
- **Reusability and versioning.** Blueprints can be versioned, imported, exported (YAML format), and shared across projects or organizations, promoting reusability and collaboration.
- **Self-Service provisioning.** VCF Automation provides self-service portals where users can deploy and manage resources based on predefined blueprints.
- **Inputs and Customization.** Blueprint inputs enable customization during deployment, allowing users to select options like machine size or image.
- **Governance.** To ensure consistency, reusability, and governance across environments.
- **Day 2 operations.** Blueprints support day 2 operations, allowing for updates, modifications, and management of existing deployments.

Blueprint Concepts

- **Versioning.** Blueprints can be versioned. At deployment time, you can select any of your versions to deploy.
- **Resource objects.** Resource objects represent the infrastructure components and services that are provisioned and managed in a blueprint.
- **Dependencies between resources in blueprints.** When you deploy a VCF Automation blueprint, one resource might need another resource to be available first.
- **Instances.** Instances are created when blueprints are deployed from the catalog.
- **Day 2 actions.** Actions that can be performed on the provisioned resource objects.

- **Property groups.** Instead of defining the same properties repeatedly across multiple blueprints, property groups enable you to define them once and then link them to various resources. This promotes consistency and simplifies management, as updates to a property group are automatically reflected in all linked templates.
- **Property bindings.** Sometimes, a resource property needs a value found in a property of another resource. For example, a backup server might need the operating system image of the database server that is being backed up, so the database server must exist first.
- **Custom forms.** Allows users to design and customize request forms for catalog items, providing a tailored user experience for submitting requests. They can be designed with a drag-and-drop interface and configured with various properties, including conditional logic and external data sources.
- **Custom resources.** VCF Automation supports the use of custom resources within blueprints to extend automation capabilities beyond native resource types. These custom resources are built upon VCF Operations orchestrator workflows and can represent a resource.

When to Use Blueprints

Blueprints allow you to streamline and automate the deployment process, ensuring consistency and reducing manual effort.

With blueprints, you can model complex IT services by visually designing infrastructure components like VMs, networks, and storage. You can use blueprints when you want to hide the complexities of the infrastructure provisioning from the end user.

You can publish blueprints to the self-service VCF Automation catalog which allows users to request and deploy services based on predefined templates.

Best Practices for Designing Blueprints

- Leverage input properties in blueprints to allow users to customize deployments while maintaining a degree of control. Consider providing default values for required fields to improve user experience and prevent errors.
- Establish clear and consistent naming conventions for blueprints, catalog items, and other resources to improve manageability and organization.
- Using an iterative method to build and version blueprints helps teams adopt Infrastructure-as-Code (IaC) practices. Start with a minimum viable blueprint (MVB) and add features iteratively. Additionally, use tools like Git for version control.

Sharing Blueprints with Projects and Organizations

When you first create a blueprint, you associate it with a project so that when you publish the blueprint to the catalog, users in the selected project can see request and deploy the blueprint. You have the option of making the published blueprint available to other projects in the organization either when you create the blueprint or at a later time.

If you are managing a Provider Consumption organization, you can share blueprints with all other organizations under the same provider administrator or you can select specific organizations. After the blueprint is shared, whenever PCO administrator publishes a blueprint to the catalog, the catalog item is shared with the selected organizations. That means one blueprint backs multiple catalog items, one in each organization.

When sharing blueprints with other organizations, consider the following limitations.

- Blueprints that contain any of the following resource types cannot be shared:
  - property groups;
  - custom resources;
  - secrets;
  - namespaces;
  - VM images from provider content libraries.
- If a blueprint is shared with an organization, a blueprint version with restrictions cannot be published to the catalog.
- If a blueprint is not shared with any organizations yet, and it has versions that have restrictions and have been published to the catalog, the blueprint cannot be updated and shared with any organization.

Creating Blueprints in VCF Automation

You use VCF Automation to create blueprints, which are the fundamental way to define and automate the deployment and management of resources, such as virtual machines, networks, and volumes in a namespace.

> **Important:**
>
> Blueprints in VCF Automation organizations are specific to VMware Cloud Foundation elements. Public cloud capabilities are supported in cloud templates in VCF Automation for VM Apps organizations. See Managing Organizations in for VM Apps.

How blueprints work

VCF Automation blueprint creation is an infrastructure-as-code process. You start by adding resources in the design canvas. Then, you complete the details using the YAML code editor. The code editor allows you to type code directly or enter values in a form.

Designing blueprints

You use the Blueprint Design page to create and design VCF Automation blueprint specifications for the machines and applications that you want to provision, including property groups, custom resources, and resource actions.

Working with the Blueprint Design Page

To create a blueprint from scratch, go to **Build & Deploy** > **Content Hub** > **Blueprint Design**. Then, click **New from** > **Blank canvas**.

Note that other options allow you to create the blueprint from another, existing Automation blueprint to which you have access.

Once you have created the blueprint, you can populate it by adding and configuring resources as shown below.

1. Locate resources.
2. Drag resources to the canvas.
3. Connect resources.
4. Configure resources by editing the blueprint code.

Selecting and adding resources to the design canvas

Resources appear at the left of the design page for selecting and dragging.

Connecting resources

Use the design canvas graphical controls to connect resources. Resources must be compatible for a connection.

To connect, hover over the edge of a resource to reveal the connection bubble. Then, click and drag the bubble to the target resource and release.

If your resources have a context-driven relationship, you can drop them in a nested configuration with the following rules:

- `CCI.Supervisor.Namespace` Drop directly onto canvas.
- `CCI.Supervisor.Resource` Drop onto a `CCI.Supervisor.Namespace` resource or directly on canvas.

In the code editor, additional code for the source resource appears in the target resource code.

Editing blueprint code

The code editor allows you to type, cut, copy, and paste code directly. If you're uncomfortable editing code, you can click a resource that's already in the design canvas, click the code editor **Properties** tab, and enter values there. Property values that you enter appear in the code as if you had typed them directly.

Note that you can copy and paste code from one blueprint to another.

Code completion help

Adding resources to the blueprint and connecting those resources in the design canvas only creates starter code. To fully configure your blueprint resources, edit the blueprint code.

The code editor allows you to type code directly or enter property values into a form. To help with direct code creation, the editor includes syntax completion and error checking features.

Other ways to create blueprints

In addition to building a blueprint from a blank canvas, you can take advantage of existing code.

- Cloning
  To clone a blueprint, go to **Blueprint Design**, select a source, and click **Clone**. You clone a blueprint to create a copy based on the source, then assign the clone to a new project or use it as starter code for a new application.
- Uploading and downloading
  You can upload, download, and share blueprint YAML code in any way that makes sense for your site. You can also modify blueprint code by using external editors and development environments.

  Note:

  A good way to validate shared blueprint code is to inspect it in the code editor on the design page.
- Integrating VCF Automation with a repository
  An integrated git source control repository can make blueprints available to qualified users as the basis for a new deployment. See Integrate for All Apps with Git.
- You can also get code samples from the YAML that is generated when you create a new VM using the Virtual Machine Service.

Creating bindings and dependencies between resources in blueprints in VCF Automation

When you deploy a VCF Automation blueprint, one resource might need another resource to be available first.

Important:

Arrows only indicate a dependency, not a connection. To connect resources so that they communicate, see Creating Blueprints in for All Apps.

Explicit dependencies

Sometimes, a resource needs another to be deployed first. For example, a database server might need to exist first, before an application server can be created and configured to access it.

An explicit dependency sets the build order at deployment time, or for scale-in or scale-out actions. You can add an explicit dependency using the graphical design canvas or the code editor.

- Design canvas option—draw a connection starting at the dependent resource and ending at the resource to be deployed first.
- Code editor option—add a `dependsOn` property to the dependent resource, and identify the resource to be deployed first.
  An explicit dependency creates a solid arrow in the canvas.

Property bindings

Sometimes, a resource property needs a value found in a property of another resource. For example, a backup server might need the operating system image of the database server that is being backed up, so the database server must exist first.

Also called an implicit dependency, a property binding controls build order by waiting until the needed property is available before deploying the dependent resource. You add a property binding using the code editor.

- Edit the dependent resource, adding a property that identifies the resource and property that must exist first.

A property binding creates a dashed arrow in the canvas.

Versioning Blueprints in VCF Automation

As an organization administrator creating blueprints, you can safely capture a snapshot of a working design before risking further changes. At deployment time, you can select any of your versions to deploy.

Restoring an older version

From the design page, click **Version History**.

On the left, select an older version to inspect it in the canvas and code editor. When you find the version that you want, click **Restore**. Restoring overwrites the current draft without removing any named versions.

Publishing a Blueprint Version to the Catalog

You can publish a blueprint version to the VCF Automation catalog using one of two ways.

- When you create a blueprint version, select **Publlish to catalog**.
- From the design page, click **Version History**. On the left, select a version and release it.

You can't release a Current Draft to the catalog until you version it.

Comparing Blueprint Versions

When changes and versions accumulate, you might want to identify differences among them.

From the Version History view, select a version, and click **Diff**. Then, from the **Diff against** drop-down, select another version to compare to.

Note that you can toggle between reviewing code differences or visual topology differences.

Cloning a Blueprint

Although it's not the same as saving a version, from the design page, **All Actions** > **Clone** makes a copy of the current blueprint for alternative development.

Specifying formatVersion in Blueprints in VCF Automation

After versioning your blueprint, you can change the `formatVersion` value in your YAML to support additional deployment features.

The `formatVersion` appears at the top of your YAML and its value determines what you can specify in your blueprint.

- `formatVersion: 1` includes blueprint specifications for inputs and resources. It automatically applies to all basic blueprints.
- `formatVersion: 2` adds blueprint specifications for metadata, variables, and outputs. It automatically applies to the blueprints that are deployed using Private AI Automation Services, but supports any kind of deployment.

As an administrator, you can use the specifications included with `formatVersion: 2` to make deployments more accessible to your users. The following code samples show examples of how to use the `formatVersion: 2` blueprint specifications in your YAML.

Metadata blueprint specification

Use metadata to hide or unhide the day 2 operations that appear for your users.

```
metadata:
deploymentSettings:
  disableUpdateDay2Action: true
  hideDisabledDay2Actions: true
```

- When `disableUpdateDay2Action` is true, users do not see the **Update** day 2 operation in the Actions menu. If the user does not have permission to update, the option appears unavailable (dimmed).
- When `hideDisabledDay2Actions` is true, any day 2 operation that has been deactivated for the user will not appear in the Actions menu.

Variables blueprint specification

Use `variables` to specify values that are reused multiple times within a blueprint, such as in dynamic configurations. A variable definition can include plain strings, inputs, and refer to other variables. You reference the variables that you define in the resources and outputs sections of the blueprint.

```
variables:
applications:
  - name: App1
    port: 3000
    showAppInfo: true
  - name: App2
    port: 4000
    showAppInfo: false
```

Outputs blueprint specification

Use `outputs` to define the deployment information that you want to make available to your user. All outputs are displayed on the deployment details page under **User Events**, except for `__deploymentOverview` values which appear under the deployment **Overview**.

The outputs example includes:

- `variables` defined in the variables blueprint specification
- `resources` defined in a resources section and included in the following code sample
- `__deploymentOverview` written in Markdown

```
outputs:
secret1name:
  value: ${resource.secret-data1.object.metadata.name}
secret2name:
  value: ${resource.secret-data2.object.metadata.name}
  __deploymentOverview:
  value: |
    ### Deployment details
    %{if starts_with(resource.CCI_Supervisor_Namespace_1.name, 'dummy')}
      This is a dummy namespace with name ${resource.CCI_Supervisor_Namespace_1.name}
    %{else}
      This is a real namespace with name ${resource.CCI_Supervisor_Namespace_1.name}
    %{endif}

    %{for app in variable.applications}
      ##### App details
      %{if app.showAppInfo}
        App name - ${app.name}, App port - ${app.port}
      %{else}
        App info is hidden because showAppInfo is ${app.showAppInfo}
      %{endif}
    %{endfor}

    #### Handle bars bindings.

    The below values will update dynamically if the values change.
```

```
    Note that we use '\{\{' to indicate handle bars expressions

    secret 1 resource version - {{resource.secret-data1.object.metadata.resourceVersion}}
resources:
CCI_Supervisor_Namespace_1:
  type: CCI.Supervisor.Namespace
  properties:
    name: ${input.namespaceName}
    regionName: private-ai-foundation-dsdunnjz
    className: vpaif-quickstart-3
secret-data1:
  type: CCI.Supervisor.Resource
  properties:
    context: ${resource.CCI_Supervisor_Namespace_1.id}
    manifest:
      apiVersion: v1
      kind: Secret
      metadata:
        name: nvaie-apikey
      type: Opaque
      stringData:
        username: $oauthtoken
        password: ${input.ngcPortalAccessKey}
secret-data2:
  type: CCI.Supervisor.Resource
  properties:
    context: ${resource.CCI_Supervisor_Namespace_1.id}
    manifest:
      apiVersion: v1
      kind: Secret
      metadata:
        name: nvaie-apikey1
      type: Opaque
      stringData:
        username: $oauthtoken
        password: ${input.ngcPortalAccessKey}
```

The following construct examples are provided in the `__deploymentOverview`.

- The `if` construct is a conditional expression based on boolean inputs.

```
%{if starts_with(resource.CCI_Supervisor_Namespace_1.name, 'dummy')}
  This is a dummy namespace with name ${resource.CCI_Supervisor_Namespace_1.name}
%{else}
  This is a real namespace with name ${resource.CCI_Supervisor_Namespace_1.name}
%{endif}
```

- The `for` loop enables iterating over arrays. For every app listed in the array, `if` conditions are evaluated and variable values are assigned and displayed as output.

```
%{for app in variable.applications}
  ##### App details
  %{if app.showAppInfo}
    App name - ${app.name}, App port - ${app.port}
  %{else}
    App info is hidden because showAppInfo is ${app.showAppInfo}
  %{endif}
```

```
%{endfor}
```

- In the handlebars expression, variable bindings update dynamically as the actual values change. The variable is enclosed in double curly braces `{{}}` and in this example, the expression is a dot-separated path.

```
secret 1 name - {{resource.secret-data1.object.metadata.resourceVersion}}
```

Reusing a Group of Properties in VCF Automation

When you have VCF Automation properties that always appear together, you can assemble them into a property group.

You can quickly add a property group to different VCF Automation designs, which saves the time of adding the same multiple properties one by one. In addition, you have a single place to maintain or modify the set of properties, which ensures their consistent application.

Only users with the Organization Administrator role may create, update, or delete a property group. The administrator can share a property group with an entire organization or limit its use to only within a project.

> **CAUTION:**
>
> A property group might be included in many blueprints, including ones that are already released to the catalog. Changes to a property group can affect other users.

There are two types of property groups.

- Inputs
  Input property groups gather and apply a consistent set of properties at user request time. Input property groups can include entries for the user to add or select, or they might include read-only values that are needed by the design.
  Properties for the user to edit or select can be readable or encrypted. Read-only properties appear on the request form but can't be edited. If you want read-only values to remain totally hidden, use a constant property group instead.
- Constants
  Constant property groups silently apply known properties. In effect, constant property groups are invisible metadata. They provide values to your VCF Automation designs in a way that prevents a requesting user from reading those values or even knowing that they're present. Examples might include license keys or domain account credentials.

The two property group types are handled very differently by VCF Automation. When you create a property group, you must first select whether to create inputs or constants. You can't create a blended property group nor convert an existing set of properties and their property group from one type to the other.

Modifying a Property Group

Changes to a property group affect every blueprint that uses it. When the changed version of the blueprint is released, those changes also affect catalog users.

The property group editing pages show the number of blueprints that include the property group. To see which blueprint would be affected by a change, click the number.

Before modifying a property group, make sure that the change is acceptable to everyone who is creating or updating deployments based on the blueprints listed.

Deleting a Property Group

Deleting a property group causes errors in every blueprint that uses it. You cannot delete a property group before you manually remove it from all of the blueprints in which it is included.

To remove a property group from a blueprint, open and edit the blueprint in the design canvas. Use the code editor to delete the associated `propgroup` entry or entries in the `resources` section of the code.

> **Note:**
>
> You cannot delete a property group if it is included in a versioned blueprint. Versioned blueprints are read-only.

Input Property Groups in VCF Automation

VCF Automation input property groups usually include related settings for the user to enter or select. They might also include read-only values needed by the blueprint design.

Create an input property group

1. Go to **Build & Deploy** > **Blueprint Design** > **Property Groups**, and click **New**.
2. Select **Input Values**.
3. Name and describe the new property group.

Name	Property group names must be unique within a given organization. Only letters, numbers, and underscores are permitted.
Display Name	Add a heading for the entire group of properties, which appears on the request form.
Description	Explain what this set of properties is for.
Scope	Decide whether an administrator may share the property group with the whole organization. Otherwise, only one project can access the property group. Although you can always add or modify properties in the group, the scope is permanent and can't be changed later.
Project	When the scope is project-only, this project can access the property group.

4. To add a property to the group, click **New Property**.
   The panel for adding a new property is very similar to the Inputs tab of the VCF Automation design page code editor.

Name	Free-form name for the individual property. Only letters, numbers, and underscores are permitted.
Display Name	Add an individual property name to appear on the request form.
Type	String, Integer, Number, Boolean (T/F), Object, or Array.
Default Value	Preset value entry that appears in the request form. The presence of default values affects whether input is optional or required.
Encrypted	When selected, obscures the value when entering it into the request form and in the subsequent deployment. Encrypted properties can't have a default value.
Read-only	An uneditable but visible value in the request form. Requires a default.
More Options	Options that vary according to property type. Expand the drop-down, add any additional settings, and click **Create**.

5. Add more properties to the group, and click **Create** when finished.

Adding the property group to blueprint inputs

Even for a long list of property inputs, you only need to add the property group to make them all part of the request form.

Enter the inputs code directly into the blueprint design page, and take advantage of the automatic prompting as you type `$ref: /ref/p...` in the code editor. Your code might look similar to the following example.

```
inputs:
  pgmachine:
    type: object
    title: Machine Properties
    $ref: /ref/property-groups/machine
  pgrequester:
    type: object
```

```
    title: Requester Details
    $ref: /ref/property-groups/requesterDetails
```

Binding blueprint resources to the property group

To make use of property group input values, add `propgroup` bindings under the resource.

Depending on what kind of values are in a property group, you might want to reference them individually. You can enter them separately, by property group name and property name.

You can also quickly add an entire set of values to a resource by referencing an entire property group.

The following example shows how you reference an input property from a property group.

```
pve-vm:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
        apiVersion: vmoperator.vmware.com/v1alpha3
        kind: VirtualMachine
        metadata:
          name: virtual-machine-1-${env.shortDeploymentId}
        spec:
          className: small
          imageName: ${input.pgmachine.image}
```

Property groups in the custom form editor

Property groups are not editable like other fields in the custom form designer.

Constant Property Groups in VCF Automation

VCF Automation constants allow you to silently apply known key-value pairs to your designs.

How constants work

The key appears in the blueprint code, and the value becomes part of deployments that are based on that blueprint. Constants require the `propgroup` binding under the resource.

The `propgroup` binding is only used with constant property groups, not input property groups.

Secret properties

If you expect to add a secret property to a property group, create the secret property before proceeding.

Create a constant property group

1. Go to **Build & Deploy** > **Blueprint Design** > **Property Groups**, and click **New**.
2. Select **Constant Values**.
3. Name and describe the new property group.

Name	Property group names must be unique within a given organization. Only letters, numbers, and underscores are permitted.
Display Name	Leave blank. No heading appears on the request form.
Description	Explain what this set of constants is for.

Scope	Decide whether an administrator may share the property group with the whole organization. Otherwise, only one project can access the property group. Although you can always add or modify properties in the group, the scope is permanent and can't be changed later. Secrets—If you expect to add a secret property to the property group, you must use single project scope. Secret properties are saved only at the project level.
Project	When the scope is project-only, this project can access the property group.

4. To add a constant property to the group, click **New Property**.
5. Enter a name that acts as the key, and a description.
6. Select a property type.
7. Enter the constant value that you want, and click **Create**.
   - String, integer, and number types use direct entry.
   - For a secret string value, select from the list of secret properties for the project.
   - The boolean type uses a selection box to indicate true.
   - For the object or array type, replace `null` with the value that you want.
8. Add more constants to the group, and click **Save** when finished.

Binding blueprint resources to the property group

To make use of property group input values, add `propgroup` bindings under the resource.

Depending on what kind of values are in a property group, you might want to reference them individually. You can enter them separately, by property group name and property name.

You can also quickly add an entire set of values to a resource by referencing an entire property group.

The following example shows how you reference an input property from a property group.

```
pve-vm:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
        apiVersion: vmoperator.vmware.com/v1alpha3
        kind: VirtualMachine
        metadata:
          name: virtual-machine-1-${env.shortDeploymentId}
        spec:
          className: small
          imageName: ${input.pgmachine.image}
```

Creating Custom Forms for Blueprints in VCF Automation

To create useful forms based on input parameters, you can design how the information appears at request time, how the parameter values are populated, and add any specialized constraints.

Creating a Custom Request Form

Each blueprint version is coupled with a custom form. You use the Blueprint Design page to create and edit custom forms.

1. Go to **Build & Deploy** > **Content Hub** > **Blueprint Design**.
2. Open the blueprint for which you want to customize the request form.
3. On the blueprint canvas, click **Create Custom Form**.
4. Drag any custom elements onto the design canvas.

5. Configure each element using the properties pane. For more about the fields properties, see Custom form designer field properties in for All Apps.
   Custom forms support data validation by adding constraints to a field.
6. Use the Actions menu options to import or export the form, or import or export a CSS file.
7. When you are ready, click **Create**.

To preview the request form, on the blueprint canvas, click **Version History**, select the blueprint version, and click the **View Form** tab.

Importing and Exporting Custom Forms between Blueprints

You might find, after you develop a custom form, that you want to use part or all of it with another blueprint. You can export a form from one blueprint and import it into another blueprint, and then continue customizing the form for the new blueprint.

To share the custom forms, you can click **Actions** on the custom form designer and select one of the following options.

Action Menu Item	Description
Import form	Imports a JSON or YAML file.
Export form	Exports your current custom form as a JSON file.
Export form as YAML	Exports your current custom form as YAML. Export the file as YAML when you want to move a custom form from one instance to another. For example, from your test environment to your production environment. If you prefer to edit the form as YAML, you can export the form, edit it, and then import it back into the blueprint.

Adding Your Own Style Sheet to a Custom Form

You can use a custom cascading style sheet to refine how the text appears on the screen. You must create the CSS file outside of VCF Automation. You can export and import a CSS file from one blueprint to another.

Action Menu Item	Description

Import CSS	Imports a CSS file that enhances the catalog request form.
	The file might be similar to the following example.
	`#input_<field_ID> {`
	`font-size: 20px;`
	`font-weight: bold;`
	`color: red;`
	`width: 600px;`
	`}`
	`#input_<field_ID> {`
	`font-size: 20px;`
	`font-weight: bold;`
	`font-style: italic;`
	`width: 600px;`
	`}`
	In this example, replace <field_ID> with the actual field IDs from the custom form. You can locate the values by selecting the field in the form, and then you can see the value in the properties pane, beneath the field name. For example, `Field ID: deploymentName` or `Field ID: textField_fe7cf66a`.

Export CSS	Exports your customized CSS.
Remove CSS	Discards your custom CSS. The discarded CSS is not recoverable.

Adding External Validation to a Custom Form

You can also validate catalog request forms by adding an external validation. External validations are run once the user submits a form request from the UI or the API. External validations use VCF Operations orchestrator action scripts that return a string value when the input parameter value validation fails. The returned string value is displayed as an error message in the catalog request form. If the input parameter value is valid, the external validation must return nothing.

You can add an external validation from the **Validations** tab in the form editor. You drag a VCF Operations orchestrator validation element to the canvas and add a VCF Operations orchestrator action that you want to use for validation. External validations can be related to more than one field at a time.

Custom Form Designer Field Properties in VCF Automation

The field properties in VCF Automation determine how the fields looks and what default values are presented to the user. You can also use the properties to define rules that ensure that the users provide a valid entry when they request the item in the catalog.

You configure each field individually. Select the field and edit the field properties.

Value source

For many of the properties, you can select from various value source options. Not all source options are available for all field types or properties.

- **Constant.** The value does not change. Depending on the property, the value might be a string, an integer, a regular expression, a complex value, for example, based on columns in a datagrid or another complex field, or selected from a limited list, for example, Yes or No. For example, you can provide 1 as a default value integer, select No for the Read-only property, or provide the regular expression to validate a field entry.
- **Conditional value.** The value is based on one or more conditions. The conditions are processed in the order listed. If more than one condition is true, the last condition that is true determines the behavior of the field for that property. For example, you can create a condition that determines if a field is visible based on the value in another field.
- **External source.** The value is based on the results of a VCF Operations orchestrator action. For example, calculate cost based on a scripted VCF Operations orchestrator action.
- **Bind field.** The value is the same as the field to which it is bound. The available fields are limited to the same field type. For example, you bind default value for an authentication needed check box field to another check box field. When one target field check box is selected in the request form, the check box on the current field is selected.
- **Computed value.** The value is determined based on how the operator processes the selected fields and values. Text fields use the concatenate operator. Integer fields use the selected add, subtract, multiply, or divide operations. For example, you can configure an integer field to convert megabytes to gigabytes using the multiply operation.

Field appearance

You use the appearance properties to determine whether the field appears on the form and what label and custom help you want to provide to your catalog users.

Table 1818: Appearance Tab Options

Option	Description
Label and type	Provide a label and select a display type. The available display types depend on the element. Some elements support multiple text types and others only support integers. Possible values: • Array Input • Checkbox • Combobox • Data Grid • Date Time • Decimal • Drop Down • Dual List • Image • Integer • Link • Multi Select • Multi Value Picker • Object Field • Password To ensure that passwords are encrypted in the deployment request details page, the input property in the cloud template must include `encrypted:true`. • Radio Group • Text • Text Area • Text Field • Tree Picker The reference type must be an inventory object. • Value Picker Drop-down and data grid fields include a **Placeholder** setting. The entered value appears as an internal label or instructions in the drop-down menu, or as a general label or instructions in the data grid.
Visibility	Show or hide a field on the request form. • **Constant.** Select Yes to display the field on the form. Select No to hide the field. • **Conditional value.** Visibility is determined by the first expression that is true. For example, a field is visible if a check box is selected on a form. • **Bind field.** Visibility of the field is bound to another check box field. If the visibility for a field is set to No, the field constraints, like regular expressions and required fields, are disregarded.
Read-only	Prevent users from changing the field values. • **Constant.** Select Yes to display the value but prevent changes. Select No to allow changes. • **Conditional value.** Status is determined by the first expression that is true. For example, a field is read-only if the value in a storage field is greater than 2GB.
Rows per page	For data grid elements only. Enter the number of rows.

Option	Description
Custom help	Provide information about the field to your users. This information appears in signpost help for the field. You can use simple text or HTML, including href links. For example, `<a href="https://techdocs.broadcom.com/us/en/vmware-cis/vcf.html">VMware Cloud Foundation documentation</a>`.

Field values

You use the values properties to provide any default values.

Table 1819: Values Tab Options

Option	Description
Columns	For the data grid element only. Provide the label, ID, and value type for each column in your table. The default value for the data grid must include the header data that matches the defined columns. For example, if you have user_name ID for one column and user_role ID for another, then the first row is user_name,user_role.
Default value	Populates the field with a default value based on the value source. Possible value sources depend on the field. • **Constant.** The entered string. • **Orchestrator action.** Value is based on the results of the selected VCF Operations orchestrator action.
Value option	Populates a drop-down, multi-select, radio group, or value picker fields. • **Constant.** The format for the list is Value|Label,Value|Label,Value|Label. For example, `2|Small,4|Medium,8|Large`. • **External source.** Value is based on the results of the selected VCF Operations orchestrator action.
Step	For integer or decimal fields, define the incremental or decremental values. For example, if the default value is 1 and you set the step value to 3, then the allowed values are 4, 7, 10, and so on.

Field constraints

You use the constraint properties to ensure that the requesting user provides valid values in the request form.

Table 1820: Constraints Tab Options

Option	Description
Required	The requesting user must provide a value for this field. • **Constant.** Select Yes to require that the requesting user provides a value. Select no if the field is optional.
Regular expression	Provide a regular expression that validates the value and a message that appears when the validation fails. The regular expressions must follow JavaScript syntax. • **Constant.** Provide a regular expression. For example, for an email address, the regular expression might be `^[A-Za-z0-9._%+-]+@[A-Zaz0-9.-]+\\.[A-Za-z]{2,}$` and the validation error message is `The email address format is not valid. Please try again.`
Minimum value	Specify a minimum numeric value. For example, a password must have at least 8 characters. Provide an error message. For example, `The password must be at least 8 characters.` • **Constant.** Enter the integer.
Maximum value	Maximum numeric value. For example, a field is limited to 50 characters. Provide an error message. For example, `This description cannot exceed 50 characters.` • **Constant.** Enter the integer.

VCF Automation blueprint designs that prepare for day 2 changes

In addition to the day 2 actions already associated with VCF Automation resource types, you have design options that let you prepare in advance for custom updates that users might need to make.

> **CAUTION:**
>
> To change a deployment, you can edit its blueprint and reapply it, or you can use day 2 actions. However, in most cases you should avoid mixing the two approaches.
>
> Lifecycle day 2 changes such as power on/off are usually safe, but others require caution, such as when adding disks.
>
> For example, if you add disks with a day 2 action, and then take a mixed approach by reapplying the blueprint, the blueprint could overwrite the day 2 change, which might remove disks and cause data loss.

Day 2 preparation can involve either direct use of blueprint code, or the VCF Automation blueprint designer.

- You can use inputs in blueprint code so that, when you update the deployment or deployed resource, the interface prompts for fresh values.
- You can use VCF Automation to design a custom action based on a VCF Operations orchestrator workflow. Running the custom action results in the workflow or action making changes to the deployment or deployed resource.

For custom day 2 resource actions you can use the schema for deployment resource types to specify criteria for your resource action. For example, you can use the deployment name property to define which deployments can use the create resource action.

> **Important:**
>
> Bindings don't work for CCI.Supervisor.Resource objects. You can still create a day 2 operation, but the resource should be selected again as in input.

Sample Blueprints in VCF Automation

Blueprints in VCF Automation can be almost limitless in combination and application. Often, an example of successful code is your best starting point for further development.

Use the following blueprint examples as a guideline. Make substitutions in the blueprint samples to apply your own settings in terms of resource types, values, and so on.

Blueprint with a Simple VM

```
formatVersion: 2
inputs:
  vm-name:
    type: string
resources:
  CCI_Supervisor_Namespace_1:
    type: CCI.Supervisor.Namespace
    properties:
      name: ns02-k58f3
      existing: true
  CCI_Supervisor_Resource_1:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
        apiVersion: vmoperator.vmware.com/v1alpha3
        kind: VirtualMachine
        metadata:
          name: ${input.vm-name}
          labels:
            vm-selector: ${input.vm-name}
        spec:
          className: best-effort-xsmall
          imageName: vmi-d3529fcf6a96cba68
          storageClass: wcpglobal-storage-profile
          powerState: PoweredOn
```

Blueprint with Multiple VMs

```
formatVersion: 2
inputs:
  vm-name:
    type: string
resources:
  CCI_Supervisor_Namespace_1:
    type: CCI.Supervisor.Namespace
    properties:
      name: test-ns-md-03-v67ch
      existing: true
  CCI_Supervisor_Resource_1:
    type: CCI.Supervisor.Resource
    allocatePerInstance: true
    properties:
      count: 2
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
```

```
        apiVersion: vmoperator.vmware.com/v1alpha3
        kind: VirtualMachine
        metadata:
          name: ${input.vm-name}-${count.index}-${env.shortDeploymentId}
          labels:
            vm-selector: ${input.vm-name}-${count.index}-${env.shortDeploymentId}
        spec:
          className: best-effort-xsmall
          imageName: vmi-e171c7cf3fdb69c4f
          storageClass: wcpglobal-storage-profile
          powerState: PoweredOn
```

Blueprint with a VM with a Kubernetes Cluster

Note:

Ensure that the CIDR blocks defined in `spec.clusterNetwork.pods.cidrBlocks` and `services.cidrBlocks` do not overlap with your vCenter/ESXi management network, DNS/NTP servers, or external subnets.

```
formatVersion: 1
inputs: {}
resources:
  CCI_Supervisor_Namespace_1:
    type: CCI.Supervisor.Namespace
    properties:
      name: ns-mr6r6
      existing: true
  Kubernetes_Cluster_1:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
        apiVersion: cluster.x-k8s.io/v1beta1
        kind: Cluster
        metadata:
          name: k-cluster-9k4i-${env.shortDeploymentId}
          namespace: ns-mr6r6
        spec:
          clusterNetwork:
            pods:
              cidrBlocks:
                - 10.95.0.0/16
            services:
              cidrBlocks:
                - 10.96.0.0/12
            serviceDomain: cluster.local
          topology:
            class: builtin-generic-v3.3.0
            version: v1.32.0---vmware.6-fips-vkr.2
            variables:
              - name: vmClass
                value: best-effort-xsmall
              - name: storageClass
                value: wcpglobal-storage-profile
            controlPlane:
              replicas: 1
```

```
              metadata:
                annotations:
                  run.tanzu.vmware.com/resolve-os-image: os-name=photon, content-library=cl-369d95257749dd297
            workers:
              machineDeployments:
                - class: node-pool
                  name: kubernetes-cluster-9k4i-np-cep4
                  replicas: 1
                  metadata:
                    annotations:
                      run.tanzu.vmware.com/resolve-os-image: os-name=photon, content-
library=cl-369d95257749dd297
  Virtual_Machine_1:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.Kubernetes_Cluster_1.id}
      manifest:
        apiVersion: vmoperator.vmware.com/v1alpha3
        kind: VirtualMachine
        metadata:
          name: virtual-machine-1-${env.shortDeploymentId}
        spec:
          className: best-effort-small
          imageName: vmi-00544472f3ab8ba74
          powerState: PoweredOn
          storageClass: wcpglobal-storage-profile
      wait:
        conditions:
          - type: VirtualMachineCreated
            status: 'True'
```

Blueprint with a VPC and a Namespace Using the VPC

```
formatVersion: 2
resources:
  VPC:
    type: CCI.VPC
    properties:
      generateName: ${to_k8s_name(env.deploymentName, 63)}
      regionName: test-region
      privateIPs: []
  vpcAttachment_config:
    type: CCI.VPC.Configuration
    properties:
      kind: VPCAttachment
      apiVersion: vpc.nsx.vmware.com/v1alpha1
      configs:
      - generateName: vpc-attachment
        spec:
          vpcConnectivityProfileName: vpc-connectivity-profile-test
      vpc: ${resource["VPC"].id}
  Namespace:
    type: CCI.Supervisor.Namespace
    dependsOn:
    - vpcAttachment_config
    properties:
      generateName: ${to_k8s_name(env.deploymentName, 63)}
      regionName: test-region
```

```
      className: test-class
      vpcName: ${resource["VPC"].name}
      sharedSubnetNames: []
```

Blueprint with Four VMs with Defined Affinity/Anti-Affinity Compute Policies

```
aformatVersion: 1
inputs:
  imageName:
    type: string
    default: vmi-e6ce40b36a693e740
  storageClass:
    type: string
    default: vsan-default-storage-policy
resources:
  CCI_Supervisor_Namespace_1:
    type: CCI.Supervisor.Namespace
    properties:
      name: ns-mkt2-core01-w9mps
      existing: true
  Virtual_Machine_Group_11:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
        apiVersion: vmoperator.vmware.com/v1alpha5
        kind: VirtualMachineGroup
        metadata:
          name: virtual-machine-group-11
        spec:
          powerOffMode: TrySoft
          powerState: PoweredOn
          suspendMode: TrySoft
          bootOrder:
            - members:
                - kind: VirtualMachine
                  name: virtual-machine-11
                - kind: VirtualMachine
                  name: virtual-machine-21
                - kind: VirtualMachine
                  name: virtual-machine-31
                - kind: VirtualMachine
                  name: virtual-machine-41
      wait:
        conditions:
          - type: Ready
            status: 'True'
  Virtual_Machine_11:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
        apiVersion: vmoperator.vmware.com/v1alpha5
        kind: VirtualMachine
        metadata:
          name: virtual-machine-11
          labels:
```

```
          app: database
      spec:
        groupName: virtual-machine-group-11
        className: best-effort-medium
        imageName: ${input.imageName}
        powerState: PoweredOn
        storageClass: ${input.storageClass}
        affinity:
          vmAffinity:
            requiredDuringSchedulingPreferredDuringExecution:
              - labelSelector:
                  matchLabels:
                    app: database
                topologyKey: kubernetes.io/hostname
    wait:
      conditions:
        - type: VirtualMachineCreated
          status: 'True'
Virtual_Machine_21:
  type: CCI.Supervisor.Resource
  properties:
    context: ${resource.CCI_Supervisor_Namespace_1.id}
    manifest:
      apiVersion: vmoperator.vmware.com/v1alpha5
      kind: VirtualMachine
      metadata:
        name: virtual-machine-21
        labels:
          app: database
      spec:
        groupName: virtual-machine-group-11
        className: best-effort-medium
        imageName: ${input.imageName}
        powerState: PoweredOn
        storageClass: ${input.storageClass}
        affinity:
          vmAffinity:
            requiredDuringSchedulingPreferredDuringExecution:
              - labelSelector:
                  matchLabels:
                    app: database
                topologyKey: kubernetes.io/hostname
Virtual_Machine_31:
  type: CCI.Supervisor.Resource
  properties:
    context: ${resource.CCI_Supervisor_Namespace_1.id}
    manifest:
      apiVersion: vmoperator.vmware.com/v1alpha5
      kind: VirtualMachine
      metadata:
        name: virtual-machine-31
        labels:
          app: web
      spec:
        groupName: virtual-machine-group-11
        className: best-effort-medium
        imageName: ${input.imageName}
```

```
          powerState: PoweredOn
          storageClass: ${input.storageClass}
          affinity:
            vmAntiAffinity:
              requiredDuringSchedulingPreferredDuringExecution:
                - labelSelector:
                    matchLabels:
                      app: web
                  topologyKey: kubernetes.io/hostname
      wait:
        conditions:
          - type: VirtualMachineCreated
            status: 'True'
  Virtual_Machine_41:
    type: CCI.Supervisor.Resource
    properties:
      context: ${resource.CCI_Supervisor_Namespace_1.id}
      manifest:
        apiVersion: vmoperator.vmware.com/v1alpha5
        kind: VirtualMachine
        metadata:
          name: virtual-machine-41
          labels:
            app: web
        spec:
          groupName: virtual-machine-group-11
          className: best-effort-medium
          imageName: ${input.imageName}
          powerState: PoweredOn
          storageClass: ${input.storageClass}
          affinity:
            vmAntiAffinity:
              requiredDuringSchedulingPreferredDuringExecution:
                - labelSelector:
                    matchLabels:
                      app: web
                  topologyKey: kubernetes.io/hostname
      wait:
        conditions:
          - type: VirtualMachineCreated
            status: 'True'
```

Capturing Namespaces as Blueprints in VCF Automation

As an organization administrator or a project administrator, you can capture the contents of an existing namespace and create a reusable blueprint from it that you make available to your project users. The resulting blueprint captures multiple resources in the namespace, including virtual machines, their hardware, Guest OS settings, networking (subnets, VPC configurations), and storage (PVCs), secrets, and so on.

What Is a Capture

A capture is a generated blueprint that is used for defining and provisioning complete applications and environments. Captures extract the live configuration, state, and dependency graph of all VMs within the environment, and convert them into a deployable definition. The capture includes namespaces, VPCs, VMs and disks from content library images, networking resources like subnets, shared subnets, and load balancers, and so on. Captured VMs are saved to a project content library.

The resulting captured blueprint can be reused, enabling rapid environment cloning, migration, or recovery without the need to manually define or export infrastructure details. Captures are an alternative to traditional, manually authored VCF Automation blueprints and provide a way to create reusable deployments directly from existing workloads. You can reuse these blueprints to deploy identical or differentiated environments from the same blueprint.

Captures are suitable for use in hands-on labs, training environments, and similar scenarios where reusable, preconfigured environments are required.

How Do Captures Work

You publish blueprints to the VCF Automation catalog for entitled project users to request and deploy.

When a project administrator captures a namespace, VCF Automation generates a blueprint that represents the discovered namespace settings and IaaS resource configurations, including:

- Namespace based on the namespace class and blueprint specification
- New project-scoped VPC or using existing bounded organization-scoped VPC
- Includes VM PVCs and content library VM images with specified VM class and disks
- VM customizations (Cloud-init, LinuxPrep, Sysprep)
- Network attachments (subnets) for VMs to VPC segments
- VM groups for VM boot order
- Networking resources

The project administrator publishes the blueprint to the catalog. The custom form for the blueprint can also be modified. The blueprint can be exported, imported, and redeployed in different projects, enabling consistent replication of the captured environment.

When a user requests the blueprint from the catalog, they provide parameters and customizations during request time.

> **Note:**
>
> Namespaces that contain VKS clusters, Avi load balancers, or vSphere pods cannot be captured.

Capture Types

You can capture the state of an existing namespace as a blueprint. You can capture namespaces as identical copies or you can enable request-time VM customization.

- Identical Copy. You can capture a namespace keeping the exact settings of all VMs within the namespace, like Mac, IpAddress, BIOS, and other settings. See Capture an Identical Copy of a Namespace in VCF Automation.
- Copy with customizations. See Capture a Namespace with Customizations in VCF Automation.

Managing Captures

- If you have a capture in progress, you cannot create a new capture out of the same namespace before the first one is complete.
- You can delete namespace capture requests on the Recent Captures page.
- You can run day 2 operations on deployed instances of the captured blueprint. For a list of supported actions, see Day 2 Actions on Catalog-Provisioned Resources Provided with VCF Automation.

Capture an Identical Copy of a Namespace in VCF Automation

In VCF Automation, you can capture an identical copy of a namespace, where a blueprint is generated that recreates the contents of a namespace. The blueprint includes a namespace with similar settings as the source, self-service VPC and all of its associated resources, and virtual machine configurations including hardware, GuestOS, networking, and storage. This mechanism ensures that every deployment is a fully independent, exact replica of the original source.

- Verify that you have an active namespace. The namespace can only be associated with a dedicated VPC and may contain VPC private subnets.

- Verify that the organization administrator configured self-service namespaces for the project. See Enable Namespace Self-Service for Projects in Your VCF Automation Organization.
- All virtual machines within the namespace must be initialized and powered off.
- All virtual machines must belong to a single VM group. The namespace must contain only one VM group. See Create a VM Group Using Self Service in VCF Automation.
- If you have any migrated VMs, these VMs do not have any VM classes associated with them so all VM classes must be created before you start a capture. During capture, VCF Automation detects the right VM class based on the size of each VM.
- Verify that the organization administrator configured at least one project content library associated with the namespace. See Create a Project Content Library in VCF Automation.

When you capture an identical copy of a namespace, the resulting blueprint contains the following.

1. Identical namespace settings, including the region where the namespace is deployed, the namespace class used to create the namespace, and subnets which are shared with the namespace.
2. Identical VPC settings and associated resources. Captured properties include the connectivity profile of the VPC and private IP CIDRs.
3. VM hardware settings. Some of the captured hardware properties include:
   a. the unique identifier for the VM instance within the namespace;
   b. the VM class instance which specifies CPU, memory, and other core resources;
   c. the virtual BIOS of the VM which represents the lowest-level hardware identifier that the guest operating system typically sees;
   d. IP and MAC of the VM NIC cards, which are present in the SubnetPort associated with the VM NIC card.
4. VM GuestOS settings. The bootstrap field in all VMs is cleared so that guest customizations do not rerun. Secret resources that might be referenced in the GuestOS customization are not recreated.
5. Networking. Captured properties include network interfaces and subnets and subnet sets to which the VM NIC cards are connected.
6. VM storage. Captured properties include storage class, volumes, disk capacity, and disk volume provisioning mode.

1. Log in to the VCF Automation organization as a project administrator.
2. Select **Build & Deploy**.
3. Under Services, select the namespace that you want to capture.
4. On the **Overview** page, click **Capture as a Blueprint**.
5. Configure the blueprint details.
   a) Enter a name and description for the blueprint.
   b) Select a content library where you want to save the captured images.
   c) Select **Capture as identical copy**.
   d) If you want to publish the capture to the catalog, select the checkbox.

      You cannot publish the capture to the catalog if there are any issues with the capture.
   e) Click **Next**.
6. Review your configuration and click **Capture**.

- View the generated blueprint and confirm that the blueprint includes all resources from the original namespace.
- Publish the blueprint to the catalog so that users in your project can request and deploy the namespace. See Publish a Blueprint to the Catalog in VCF Automation.

Capture a Namespace with Customizations in VCF Automation

VCF Automation provides capabilities for customizing guest operating systems, including setting passwords, defining hostnames, and executing scripts on both Windows and Linux hosts by leveraging various bootstrap types to handle guest customization.

To enable individual VM customization within the blueprint, you select the option to capture the namespace with customizations when requesting the capture.

Supported customizations include the following options:

- Hostname. Assign a unique hostname to each VM.
- Password. Set or auto-generate guest OS administrator or root passwords. For an example configuration, see Capture a Namespace with an Auto-Generate Password Customization in VCF Automation.
- Scripts. Run pre- or post-customization scripts for Windows and Linux. Windows VMs can use GuiRunOnce commands or custom PowerShell scripts via Sysprep. Linux VMs can execute shell scripts via LinuxPrep.
- Domain configuration. For Windows VMs, join an Active Directory domain. For an example configuration, see Capture a Namespace with a Join Domain Customization in VCF Automation.

Capture a Namespace with an Auto-Generate Password Customization in VCF Automation

As a project administrator, you can capture a namespace as a blueprint, specifically with an auto-generate password configuration so that when users log in to the VM for the first time, they are prompted to change their password.

- Verify that you have an active namespace with a VPC and a content library. See Create a Namespace in a VCF Automation Project
- Verify that the organization administrator configured self-service namespaces for the project. See Enable Namespace Self-Service for Projects in Your VCF Automation Organization.
- Within the namespace, provision and configure one or more virtual machines.
- All virtual machines within the namespace must be initialized and powered off to ensure data consistency during capture.
- All virtual machines must belong to a single VM group. The namespace must contain only one VM group. See Create a VM Group Using Self Service in VCF Automation.
- Verify that the organization administrator configured at least one project content library associated with the namespace. See Create a Project Content Library in VCF Automation.

1. Log in to the VCF Automation organization as a project administrator.
2. Select **Build & Deploy**.
3. Under Services, select your namespace.
4. On the **Overview** page, click **Capture as a Blueprint**.
5. Configure blueprint details.
   a) Enter a name and description for the blueprint.
   b) Select a content library where you want to save the captured images.
   c) Select **Capture namespace with customizations**.
   d) Define compute and storage settings.

      You can allow catalog users to pick a different VM class and storage class for each VM.
   e) To enable guest customization options for catalog users, under **Guest OS Settings**, select the check box.
   f) Select a VM and click **Configure**.

Setting	Sample value
Host name	Enter a host name.
Password	Select **Auto generate password**. You can enable the option for the user to change the password when they log in for the first time.

   g) Click **Configure**.
   h) Optional: If you want to publish the capture to the catalog, select the check box.

      You can publish the blueprint to the catalog later on.

      You cannot publish the capture to the catalog if there are any issues with the capture.

i) Click **Next**.

6. Review your configuration and click **Capture**.
7. Once the capture is complete, inspect the resulting blueprint on the **Blueprint Design** page.

When a user requests the catalog item and the VM is powered on, it runs the guest customization and a password is automatically generated for the customized VM.

Note:

Capture time depends on storage size and performance.

- View the generated blueprint and verify that all selected customizations are correctly applied. Confirm that the blueprint includes all resources from the original namespace.
- Run a Get VM Password action on the VM. See Day 2 Actions on Catalog-Provisioned Resources Provided with VCF Automation.
- Publish the blueprint to the catalog. See Publish a Blueprint to the Catalog in VCF Automation.

Capture a Namespace with a Join Domain Customization in VCF Automation

As a project administrator, you can capture a namespace as a blueprint, specifically integrating Windows domain join configurations. Deploying this blueprint automatically enrolls new virtual machines into the specified Active Directory domain.

- Verify that you have an active namespace with a VPC and a content library. See Create a Namespace in a VCF Automation Project
- Verify that the organization administrator configured self-service namespaces for the project. See Enable Namespace Self-Service for Projects in Your VCF Automation Organization.
- Within the namespace, provision and configure one or more virtual machines.
- All virtual machines within the namespace must be initialized and powered off to ensure data consistency during capture.
- All virtual machines must belong to a single VM group. The namespace must contain only one VM group. See Create a VM Group Using Self Service in VCF Automation.
- Verify that the organization administrator configured at least one project content library associated with the namespace. See Create a Project Content Library in VCF Automation.
- (Optional) Verify that your organization administrator configured an Active Directory integration and enabled the **Override profile** setting in all applicable Active Directory profiles for the namespace. See Integrate VCF Automation with Active Directory.

Important:

Join domain is supported for Windows machines only.

1. Log in to the VCF Automation organization as a project administrator.
2. Select **Build & Deploy**.
3. Under Services, select your namespace.
4. On the **Overview** page, click **Capture as a Blueprint**.
5. Configure the blueprint details.
   a) Enter a name and description for the blueprint.
   b) Select a content library where you want to save the captured images.
   c)
   d) Select **Capture namespace with customizations**.
   e) Define compute and storage settings.

   You can allow catalog users to pick a different VM class and storage class for each VM.

f) Under **Guest OS Settings**, select the check box.
g) Select a VM and click **Configure**.
h) Configure guest customizations options for the blueprint.

Setting	Sample value
Host name	Enter a name.
Bootstrap type	SysPrep
Password	None
Join Domain	Select the check box.
Domain name	Enter the Active Directory domain to join. Ensure that you enter the fully qualified domain name, for example, `server.example.com`.
Administrator name and password	Enter administrator credentials. The user name and password of the Active Directory user who has the rights to join systems to the Active Directory domain.

i) Click **Configure**.
j) Optional: If you want to publish the capture to the catalog, select the check box.

   You can publish the blueprint to the catalog later on.

   You cannot publish the capture to the catalog if there are any issues with the capture.
k) Click **Next**.

6. Review your configuration and click **Capture**.
7. Once the capture is complete, inspect the resulting blueprint on the Blueprint Design page.

   A SysPrep is populated in the blueprint YAML of the VM that contains the domain name and administrator credentials.

Publish the blueprint to the catalog. See Publish a Blueprint to the Catalog in VCF Automation.

When a user requests the catalog item and the VM is powered on, it runs the guest customization and joins the specified Active Directory domain.

Importing or Exporting Blueprints in VCF Automation

In VCF Automation, you can export and import content to move artifacts from your development or staging environments to a production environment or between different organizations or projects.

Importing Blueprints

You can import .OVA/.OVF or .YAML files as blueprints, for example, exported from another VCF Automation organization or a VMware Cloud Director or a vCenter environment.

> **Note:**
>
> OVAs containing ISO media files are not supported during import.

By default, you can have a maximum of 5 concurrent import requests per organization. You can configure the import limit.

For more information, see the following topics:

- Import a Blueprint into VCF Automation
- Import a VMware Cloud Director OVA as a Blueprint in VCF Automation

Exporting Blueprints

You can export VCF Automation blueprints and associated VM images. you can create a single zip file containing the blueprint content and the payloads for all the dependent resources used in the blueprint which can be re-imported into another tenant to re-create the same blueprint with all of the associated resources.

Note:

When you export blueprints containing VM images that are ISO files, these images are skipped during export.

Only valid blueprints can be exported. You can select which blueprint version you want to export. By default, you can have a maximum of five concurrent export requests per organization. You can configure the export limit.

For more information, see Export a Blueprint from VCF Automation.

Import a Blueprint into VCF Automation

In VCF Automation, you can import OVA/OVF and YAML files, for example, that you exported from another VCF Automation organization.

- Verify that you have the Organization Administrator, Project Administrator, or Project Advanced User role.
- Verify that the organization administrator configured self-service namespaces for the project.
- Verify that you have the file you want to import saved on your local machine.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Blueprint Design** > **Blueprints**.
3. On the **Blueprints** page, click **New From** > **Import**.
4. Enter a name and description for the blueprint and select a project where you want to import it.
5. Select the file to import. You can preview the contents of the file.
6. Optional: Upload a custom icon for the blueprint.
7. Click **Next**.
8. On the **Image Settings** page, select the content library where you want to upload the images.

   Review the list of images identified from blueprint content. By default, images are included in the blueprint, unless you disable this option.
9. Click **Next**.
10. On the **Summary** page, review your inputs and any potential warnings.

    If any errors appear, you cannot continue the import. You must close the dialogue and restart with a valid OVA file.
11. Click **Import** to begin uploading the files and start the asynchronous import task.

    You can monitor progress on the **Imports** tab.

Upon successful completion, the import request is linked to a blueprint. If the imported bundle contains a custom form, the blueprint also has a custom form attached.

Import a VMware Cloud Director OVA as a Blueprint in VCF Automation

In VCF Automation, you can import OVA/OVF files, for example, that you exported from VMware Cloud Director and create deployable blueprints out of them.

- Verify that you have the Organization Administrator, Project Administrator, or Project Advanced User role.
- Verify that the organization administrator configured self-service namespaces for the project.
- Verify that you have the OVF/OVA file you want to import saved on your local machine.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Blueprint Design** > **Blueprints**.
3. On the **Blueprints** page, click **New From** > **Import**.
4. Enter a name and description for the blueprint and select a project where you want to import it.
5. Select the file to import. You can preview the contents of the file.
6. Optional: Upload a custom icon for the blueprint.
7. Click **Next**.
8. On the **Configuration** page, configure the namespace that VCF Automation will create for the blueprint.

   The dropdown lists only shows values as per the self-service namespace settings that the organization administrator configured for the selected project.

   a) Select a namespace class to configure the size of the application namespace which will be deployed by the imported blueprint.
   b) Select a region where the application namespace will be deployed by the imported blueprint.
   c) Select a connectivity profile of the VPC that will be created within this namespace.
9. Click **Next**.
10. On the **VM Images** page, select the content library where you want to upload the images.
11. Click **Next** to view a summary of your inputs and any potential warnings.

    Note:

    If any errors appear on the **Summary** page, you cannot continue the import. You must close the dialogue and restart with a valid OVA.
12. Click **Import** to begin uploading the files and start the asynchronous import task.

    You can monitor progress on the **Imports** tab.

Upon successful completion, the import request is linked to a blueprint. You can create a custom form for the blueprint. See Creating Custom Forms for Blueprints in VCF Automation.

Export a Blueprint from VCF Automation

You can export blueprints from VCF Automation and import them into another organization or project to recreate the same blueprint with all of its associated resources.

- Verify that you have the Organization Administrator, Project Administrator, or Project Advanced User role.

When you export a blueprint, you create a single .zip file containing the blueprint and the payloads for all dependent resources used in the blueprint. The following resources are included in the download bundle:

- Blueprint
- Custom form and custom form style
- VM Image files - .vmdk, .nvram
- .ovf

Export as OVA is only supported for blueprints in a valid state.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Build & Deploy** > **Content Hub** > **Blueprint Design** > **Blueprints**.
3. On the **Blueprints** page, select a blueprint.
4. Click **Export**.
5. Select which blueprint version you want to export.
6. The images referenced by the blueprint are loaded.
7. Choose whether to include images in the export bundle.
8. Click **Export**.

   You can monitor progress on the **Exports** tab.
9. When the export is ready, click **Download** for a particular export.

   An OVA file is generated.

   You can download each export multiple times until the bundle expires.

Managing Event Subscriptions in VCF Automation

You can extend your application lifecycles by using VCF Operations orchestrator workflows with event subscriptions.

When a triggering event occurs in your environment, the subscription is initiated and the specified workflow is run. You can view system events on the event log. Subscriptions are project-specific, meaning they are linked to blueprints and deployments through the specified project.

Provider administrators can use an MQTT client to subscribe to messages about Provider Management events and tasks. See Subscribe to VCF Automation Provider Management Events, Tasks, and Metrics by Using an MQTT Client.

Blocking event topics

Some event topics support blocking events. The behavior of a subscription depends on whether the topic supports these event types and how you configure the subscription.

Event subscriptions can use two broad types of event topics: non-blocking and blocking event topics. The event topic type defines the behavior of the event subscription.

Non-blocking event topics	Non-blocking event topics only allow you to create non-blocking subscriptions. Non-blocking subscriptions are triggered asynchronously and you cannot rely on the order that the subscriptions are triggered in.
Blocking event topics	Some event topics support blocking. If a subscription is marked as blocking, all messages that meet the set conditions are not received by any other subscriptions with matching conditions until the runnable item of the blocking subscription is run. Blocking subscriptions run in priority order. The highest priority value is 0 (zero). If you have more than one blocking subscription for the same event topic with the same priority level, the subscriptions run in a reverse alphabetical order based on the name of the subscription. After all blocking subscriptions are processed, the message is sent to all the non-blocking subscriptions at the same time. Because the blocking subscriptions run synchronously, the changed event payload includes the updated event when the subsequent subscriptions are notified. You can use blocking event topics to manage multiple subscriptions that are dependent on each other. For example, you can have two provisioning workflow subscriptions where the second subscription depends on the results of the first subscription. The first subscription changes a property during provisioning, and the second subscription records the new property, such as a machine name, in a file system. The ChangeProperty subscription is prioritized as 0 and the RecordProperty is prioritized as 1 because the second subscription uses the results of the first subscription. When a machine is provisioned, the ChangeProperty subscription begins running. Because the RecordProperty subscription conditions are based on a post-provisioning condition, an event triggers the RecordProperty subscription. However, because the ChangeProperty workflow is a blocking workflow, the event is not received until it is finished. When the machine name is changed and the first workflow subscription is finished, the second workflow subscription runs and records the machine name in the file system.

For blocking event topics, you can add a recovery runnable item to the subscription. The recovery runnable item in a subscription runs if the primary runnable item fails. For example, you can create a workflow subscription where the primary runnable item is a workflow that creates records in a CMDB system such as ServiceNow. Even if the workflow subscription fails, some records might be created in the CMDB system. In this scenario, a recovery runnable item can be used to clean up the records left in the CMDB system by the failed runnable item.

For use cases that include multiple subscriptions that are dependent on each other, you can add a `ebs.recover.continuationproperty` to the recovery runnable item. With this property, you can direct if the system must continue with the next subscription in your chain, if the current subscription fails.

Event Log

The **Event Log** tab on the **Event Subscriptions** page displays a list of all events that have occurred within your environment.

You can view the event logs on the **Events** tab. You can also filter the list of events by one or more properties. To view additional details of an individual event, select the event ID.

Event Topics Provided with VCF Automation

VCF Automation includes predefined event topics.

You can see the list of all the events that get generated in the system and you can subscribe to those events. You can trigger different workflows for different events. You can subscribe to VM Service events for operations on VMs. For example, if an event is generated after a VM is successfully created, you can subscribe to that event and trigger a VCF Operations orchestrator workflow that takes the VM and registers it in CMDB.

Event topics are the categories that group similar events together. When assigned to a subscription, event topics define which event triggers the subscription. The following event topics are provided by default with VCF Automation. All topics can be used to add or update custom properties or tags of the resource. If a VCF Operations orchestrator workflow fails, the corresponding task fails as well.

Event Topic	Blockable	Description
`Approval Events`	Yes	Issued when an action that requires an approval is triggered.
`Template configuration`	No	Issued when a blueprint configuration event, such as the creation or deletion of a blueprint, occurs. This event topic can be useful for notifying external systems of such events.
`Template version configuration`	No	Issued when a new blueprint versioning event occurs, such as the creation, release, de-release, or restoration of a version. This event topic can be useful with integrations of third-party version control systems.
`Custom resource post provision`	Yes	Issued for post provisioning events triggered by custom resource operations. **Note:** This event topic is usable only for VCF Operations orchestrator-based custom resources.
`Custom resource pre provision`	Yes	Issued for pre-provisioning events triggered by custom resource operations. **Note:** This event topic is usable only for VCF Operations orchestrator-based custom resources.
`Deployment action completed`	Yes	Issued after a deployment action is finished.
`Deployment action requested`	Yes	Issued before a deployment action is finished.

Event Topic	Blockable	Description
`Deployment completed`	Yes	Issued after the deployment of a blueprint or catalog request.
`Deployment onboarded`	No	Issued when a new deployment is onboarded.
`Deployment requested`	Yes	Issued before the deployment of a blueprint or catalog request.
`Deployment resource action completed`	Yes	Issued after the deployment of a resource action.
`Deployment resource action requested`	Yes	Issued before the deployment of a resource action.
`Deployment resource completed`	Yes	Issued after the provisioning of a deployment resource.
`Deployment resource requested`	Yes	Issued before the provisioning of a deployment resource.
`IaaS resource event`	No	Issued when a VM is created, updated, or deleted. This event topic is usable for VMs provisioned either from the catalog or the VM Service in the services console. Events for VMs are filtered using the following filter. `event.data.object.involvedObject.kind == 'VirtualMachine' && event.data.object.reason == 'CreateSuccess' // CreateFailure UpdateFailure UpdateSuccess DeleteSuccess DeleteFailure` The payload of the event contains additional information about whether a VM was created, updated, or deleted.
`Project Lifecycle Event Topic`	No	Issued when a project is created, updated, or deleted.
`Provider resource action cud`	No	Issued when a resource action is created, updated, or deleted.

Create an Event Subscription in VCF Automation

By using a VCF Operations orchestrator integration, you can create subscriptions to extend your applications.

- Verify that you have the organization administrator user role.
- Verify that you have a VCF Operations orchestrator integration. See Integrate VCF Automation with VCF Operations orchestrator.

Event subscriptions allow you to extend your applications by triggering workflows or actions at specific life-cycle events. You can also apply filters to your subscriptions to set Boolean conditions for the specified event. For example, the event and workflow only triggers if the Boolean expression is `'true'`. This is helpful for scenarios where you want to control when events or workflows are triggered.

1. Select **Build & Deploy** > **Subscriptions** > **New**.
2. Enter the details of your subscription.
3. Under **Status**, verify that the subscription is enabled.
4. Select an event topic.
5. Set conditions for the event topic.

   Conditions can be created by using a JavaScript syntax expression. This expression can include Boolean operators, such as `"&&" (AND)`, `"||" (OR)`, `"^" (XOR)`, and `"!" (NOT)`. You can also use arithmetic operators, such as `"==" (equal to)`, `"!=" (not equal to)`, `">=" (greater than or equal)`, `"<=" (less than or equal)`, `">" (greater than)`, and `"<" (lesser than)`. More complex Boolean expressions can be built out of simpler expressions. To access the event payload according to the specified topic parameters, use 'event.data' or any of the event header properties: `sourceType`, `sourceIdentity`, `timeStamp`, `eventType`, `eventTopicId`, `correlationType`, `correlationId`, `description`, `targetType`, `targetId`, `userName`, and `orgId`.

6. Select a runnable workflow for your subscription.
7. If applicable, configure the blocking behavior for the event topic.
8. To define the project scope of the subscription, deselect **Any Project** and click **Add Projects**.
9. Click **Save**.

Your subscription is created. When an event, categorized by the selected event topic occurs, the linked VCF Operations orchestrator workflow action is initiated and all subscribers are notified.

After creating your subscription, you can create or deploy a blueprint to link and use the subscription. You can verify the status of the workflow run and monitor runs and workflow status on the **Orchestrate** tab.

Create an Event Subscription to Manage Deployment Expiry in VCF Automation

You can manage expired deployments and their resources by using the `Expire` action alongside existing event topics.

After a deployment lease in your environment expires, you can use event topics to perform tasks, such as stopping the back up or monitoring of any deployment resources. To perform these day 2 operations, the VCF Automation API uses a system-level `Expire` action. This action is triggered automatically by the system whenever a deployment lease in your organization expires. The `Expire` action trigger precedes the power off event for any resources associated with that deployment.

The Expire action is included in the payload of existing event topics, such as `Deployment action requested` and `Deployment action completed`, and uses the `deploymentid` parameter to perform pre-expiry and post-expiry tasks associated with the deployment resources.

Because Expire is a deployment-level action, you must subscribe to event topics related to deployment actions, like `Deployment action requested` or `Deployment action completed`.

The Expire action is triggered approximately 10 to 15 minutes after your deployment lease expires. The system does not trigger lease expiry events prior to the actual lease expiry. The Expire action is a system-level action and users cannot trigger the events associated with it manually.

For the current use case, you are using the `Deployment action requested` event topic along with the Expire action to back up a virtual machine in your deployment as a template. For this case, the back up is performed by using a VCF Operations orchestrator workflow but the same task can also be performed by using an extensibility action as the runnable item of the subscription.

1. Select **Build & Deploy** > **Subscriptions** > **New**.
2. Enter a name for the subscription.
3. Under **Status**, verify that the subscription is enabled.
4. Under **Event Topic**, select the **Deployment action requested** event topic.
5. Toggle on the **Condition** option and add a filter for the expiry action.

   ```
   event.data.actionName == 'Expire'
   ```

   Note:

   The **Deployment action requested** event topic can be triggered by different deployment day 2 operations, such as changing the deployment lease duration. Adding the lease expiry action filter guarantees that the subscription is triggered only for expiry events.
6. Under **Workflow**, add the VCF Operations orchestrator workflow.

   The schema of this sample workflow includes a scriptable task and a workflow element which includes the `Clone virtual machine, no customization` workflow which comes preconfigured with VCF Operations orchestrator. The scriptable task element includes the following sample script:

   ```
   System.log("Lease expiry action triggered to clone a VM...")

   System.log("Deployment Id is: " + inputProperties.deploymentId);
   inputHeaders = new Properties();
   ```

```
deploymentId = inputProperties.deploymentId;
pathUriVariable = "/deployment/api/deployments/" +deploymentId +"/resources";
var restClient = vRAHost.createRestClient();
var request = restClient.createRequest("GET", pathUriVariable, null);
var keys = inputHeaders.keys;
for(var key in keys){
    request.setHeader(keys[key], inputHeaders.get(keys[key]));
}
var response = restClient.execute(request);
System.log("Content as string: " + response.contentAsString);
var content = response.contentAsString;
var obj = JSON.parse(content);

var object = new Properties(obj);
var contentJson = object.content;
for (var i = 0; i < contentJson.length; i++) {
    var resources = contentJson[i];

    var resourceProperties = resources.properties;
    System.log("Resource name is: " + resourceProperties.resourceName)
    resourceName = resourceProperties.resourceName;
}

var query = "xpath:name='" + resourceName + "'";
      var vms=Server.findAllForType("VC:VirtualMachine", query);
      vcVM=vms[0];

      System.log("VM input is: " + vcVM);
      dataStoreOutput = datastore
      template= true;
      name="test-vm-name"
```

7. Decide whether to set the subscription as blocking or non-blocking.

   Note:

   Making the subscription blocking means that the power off event for the deployment resources is triggered only after the runnable item, in this case the lease expiry workflow, finishes its run successfully. Making the subscription non-blocking means that power off event is triggered for the deployment resources regardless of the status of the workflow run.

8. To finish editing the subscription, click **Save**.

After the subscription is triggered by the lease expiry event and the workflow run is successful, navigate to vSphere and validate that your virtual machine is converted to a blueprint.

Subscribe to Events and Tasks in Your VCF Automation Organization Using an MQTT Client

As an organization administrator or a project user, you can use an MQTT client to subscribe to messages about events and tasks in your VCF Automation organization.

1. Log in to your VCF Automation organization by using the OpenAPI endpoint.
   a) Make an API versions request to VCF Automation to obtain the VCF Automation API login URL.

      ```
      GET http://vcfa.example.com/api/versions
      ```

   b) POST a request to the login URL to create a login session.

      ```
      POST https://vcfa.example.com/cloudapi/1.0.0/sessions/provider
      ```

2. To establish a WebSocket connection, set the Sec-WebSocket-Protocol property to mqtt, set the client to connect to the /messaging/mqtt path, add an authorization header, and follow the standard MQTT connect flow.

   You receive the JWT token from the standard login request to VCF Automation. You can leave the user name and password empty.

   ```
   Sec-WebSocket-Protocol: mqtt
   Authorization: Bearer {JWT_token}
   ```

3. Once the connection is established successfully, subscribe to topics through the MQTT client.

   ```
   publish/{user_org_id}/{user_id}
   ```

   Example:

   ```
   publish/debd63a0-6eae-11ea-8c7b-0050561776be/d19fd8ff-6eae-11ea-bb42-0050561776c8
   ```

   You can use wildcards to access all organization topics.

   ```
   publish/{user_org_id}/+
   ```

Managing Custom Resources in VCF Automation Blueprints

When you create a blueprint in VCF Automation, the resource type palette includes a number of default resource types. You might have use cases where you want to create blueprints based on an expanded list of resource types. You can create custom resource types, add them to the design canvas, and create blueprints that support your design and deployment needs.

> **Important:**
>
> Before you begin, ensure that a VCF Operations orchestrator instance is associated with your organization.

As an organization administrator, you have two ways to create custom resources.

- **VCF Operations orchestrator inventory type.** You can create custom resources based on existing VCF Operations orchestrator inventory objects, with or without leveraging the Dynamic Types plug-in for VCF Operations orchestrator. Dynamic type is required if the desired resource does not exist out-of-the-box in VCF Operations orchestrator as a SDK inventory object type. You can expose any VCF Operations orchestrator resource type, be it native, provided by a plug-in in VCF Operations orchestrator, or created using the Dynamic Types plug-in.
- **New schema**. You can define custom resource types that are based on workflows with any kind of input and output. Every input and output becomes part of the schema and its value is shown in the deployment. Because this custom resource type is based on your own custom outputs instead of VCF Operations orchestrator inventory objects, you must provide a READ workflow in addition to the Create and Delete workflows. You also define what inputs that workflow has.
  For the Read and Delete workflows, you can map the inputs and outputs to schema properties. This is necessary because the schema property names might not match the input/outputs of the Read and Delete workflows.

Custom resource name and resource type

The custom resource name identifies your custom resource inside the blueprint resource type palette.

The resource type of a custom resource must begin with `Custom.` and each resource type must be unique. Although the inclusion of `Custom.` is not validated in the text box, the string is automatically added if you remove it.

To add custom resources to blueprints, you must activate the custom resource. You can make the custom resource available to blueprints associated with a single project or make it available to all projects.

Lifecycle actions for custom resources

You define the following lifecycle actions for your custom resources:

- **Create**: this action is called when a deployment is started.
- **Read**: this action is used to retrieve the latest state of the deployed resource. The READ workflow is run every few minutes to get the latest status of every provisioned custom resource. You also define what inputs that workflow has,

for example, you can just pass the resource ID as an input. The Read action is required only for new schema-based custom resources.

- **Destroy**: this action is called when a deployment is deleted.

> **Note:**
>
> You can edit the VCF Operations orchestrator workflow associated with your lifecycle actions by clicking the **Change** option for the action.

VCF Operations orchestrator custom resources

Each VCF Operations orchestrator custom resource is based on a SDK inventory type and is created by a VCF Operations orchestrator workflow that has an output which is an instance of your desired SDK type. Primitive types, such as `Properties`, `Date`, `string`, and `number` are not supported for the creation of custom resource types.

> **Note:**
>
> SDK object types can be differentiated from other VCF Operations orchestrator property types by the colon (":") used to separate the plug-in name and the type name.

You can use the built-in workflows in VCF Operations orchestrator, or you can create your own. You can create custom resources by clicking on the **Resource Definition** option in the custom resource editor and selecting **Orchestrator inventory type**.

VCF Operations orchestrator custom resource inventory type

The inventory type property defines the type of your custom resource.

When you define your custom resource types, you can define the scope of the availability of the selected external type. The selected external type can be:

- Shared across projects.
- Available only for the selected project.

When you select a Create workflow in your custom resource type, the inventory type drop-down appears underneath it. The drop-down includes inventory type properties, that are selected from the output parameters of the VCF Operations orchestrator workflow. The selected workflow output properties included in the drop-down must be non-array SDK object types such as `VC:VirtualMachine`.

> **Note:**
>
> When creating custom workflows that use the dynamic type plug-in, verify that their variables are defined by using the `DynamicTypesManager.getObject()` method.

You can only have one custom resource type with a specific inventory type value per defined scope. For example, if you create a custom resource in your project that uses `VC:VirtualMachine` as an inventory type, you cannot create another custom resource for the same project that uses the same inventory type. You also cannot create two shared custom resources that use the same inventory type.

VCF Operations orchestrator lifecycle action validation

When you add Create, Destroy, and Read workflows as lifecycle actions to your custom resource, VCF Operations orchestrator validates that the selected workflows have correct input and output property definitions.

- The `Create` workflow must have an output parameter that is an SDK object type, such as `SSH:Host` or `SQL:Database`. If the selected workflow does not pass the validation, you cannot add Read or Destroy workflows, or save your changes to the custom resource. The Create workflow you select should output the desired type. For the old CR types(based on the Inventory), it is possible to create a deployment with CR by executing the create wf used in the CR as a Catalog item
- The `Destroy` workflow must have an input parameter that is an SDK object type that matches the inventory type of the custom resource.
- The `Read` workflow must have both an input and output parameter that is an SDK object type that matches the inventory type of the custom resource.

Custom resource property schema

You can edit and view the custom resource properties schema by selecting the **Schema Properties** tab. The schema includes the name, data type, property type, and, if it is available, the description of a given property. The schema also defines if a specific property is required or optional in the blueprint.

When you add VCF Operations orchestrator workflows to your custom resource, their input and output parameters are added as properties.

From this tab, you can also modify and format the properties of your VCF Operations orchestrator-based custom resources. For example, you can change the display name of a given property, add constraints, or make it read- or write-only for the deployment.

> **Note:**
>
> When adding constraints to either the item section of array fields or properties section of objects fields in the properties schema, verify that you have validated these constraints as incorrectly applied constraints can cause issues with the custom resource. For example, when adding a maximum constraint to a numbers array, you must verify that this constraint does not break the property's default value.

You can edit the property schema for custom resources by navigating to the **Schema Properties** tab and using either the **Code** or **Form** tab.

- **Code**: Edit the property schema by using YAML content. When selecting this option, the property schema is defined by using the JSON Schema.
- **Form**: Hide non-required and non-computed properties from the schema by clicking **Remove Property**.

Day 2 Operation Custom Request Forms

You can streamline the request form of the day 2 operations included in your custom resource by adding and modifying different types of resource properties.

For example, you can bind the value of an input parameter in your request form to an external source, such as a VCF Operations orchestrator action that retrieves a deployment name or project name. You can also bind the value of a specific input parameter to the computed value of two other text boxes included in the same request form.

> **Note:**
>
> This functionality is available for both custom resources and resource actions. You can customize the value of the input properties of your request form from the **Values** tab of the **Request Parameters** page of the custom resource or resource action editor.

Day 2 Operation Request Form Validation

You can validate the request form of your day 2 operations by adding an external validation. By using an external validation, you prevent the user from submitting the request form until the validation parameters are satisfied. You can add external validation from the **Validations** tab of the **Request Parameters** page of the custom resource or resource action editor. After selecting the tab, you can drag an **Orchestrator validation** element to the canvas and add a VCF Operations orchestrator action that you want to use for validation.

For example, you can create a custom resource that includes a day 2 operation for changing a user password. For such a use case, you can add a VCF Operations orchestrator action with `newPassword` and `confirmPassword` input parameters that use the `SecureString` type.

> **Note:**
>
> This is a sample script for validating a user password. For your own use case, you can decide to use a different script.

```
if (newPassword != confirmPassword) {
    return 'passwords are different';
}
if (newPassword.lenght < 7) {
```

```
    return 'password must be at least 10 symbols';
}
return null;
```

Managing Policies in VCF Automation

Governance is one of the key aspects of building a cloud-like experience for an organization. Application teams in a VCF Automation organization can use the self-service catalog and the services console to provision resources depending on their project needs and permissions. As an organization administrator, you can impose guardrails around those provisioning requests. You use VCF Automation policies to enforce the governance through approval workflows and policy templates. Each VCF Automation policy is a set of rules or parameters that are applied to VCF Automation projects to govern deployment requests from application teams, freeing the organization administrator for other tasks.

Policy Types

You can create two general categories of policies in VCF Automation:

- Policies that apply to catalog requests.
  - **Approval Policy**. An approval policy adds a layer of governance before the requested resources can be deployed. Approval policies control who must approve resource deployment before the request is provisioned. If an approval policy is configured, approvers can view the list under the **Inbox** section for pending requests for their approval. See Configure Approval Policies in VCF Automation.
  - **Day 2 Action Policy**. Day 2 action policies control what group of users can run the selected actions on deployed resources. See Configure Day 2 Action Policies in VCF Automation.
  - **Lease Policy**. Lease policies define the maximum amount of time a deployment can exist. When a deployment expires, it is destroyed and its resources are reclaimed. A lease policy reduces the need for the organization administrator to intervene manually to reclaim resources, allowing administrators to specify when they want deployed resources to expire. See Configure Lease Policies in VCF Automation.
- Policies that apply to service requests.
  - **IaaS Resource Policy**. Organization administrators can configure IaaS resource policies to govern requests by application teams to provision services, like the Virtual Machine Service, and to apply granular control over the type of provisioned resources and their configurations. IaaS resource policies offer a policy-as-code capability to enforce custom constraints on IaaS resources, beyond the built-in policy templates. See Configure IaaS Resource Policies in VCF Automation.
- Provider-managed policies.
  - **Infrastructure Policy**. The provider administrator assigns infrastructure policies to your organization to define zone placement criteria for workloads, such as namespaces and VMs. Some infrastructure policies might be mandatory for your organization, which means they are always enforced when a new namespace is created. If the requested infrastructure resources in the selected zones are not compatible with the policies, the namespace creation fails. If a policy is not mandatory, when creating a namespace, you can choose whether you want to apply that policy to the namespace.

Policy Scope and Criteria

VCF Automation policies can be scoped to the entire organization or a selection of projects in the organization. Organization administrators can further refine the policy scope by defining policy criteria. For example, to apply a policy to all the deployments except the deployments named pipeline test, add criteria and construct an expression similar to `name not equal to pipeline test`. Policy criteria can be based on project name, namespace name, namespace class, and so on. For details on constructing policy criteria, see Configure Deployment Criteria in Policies in VCF Automation.

Organization administrators can configure multiple policies for resources, projects, or namespaces. First, policies are evaluated, ranked, and, where applicable, merged to produce an effective policy.

On the **Policies** page, organization administrators can create and delete policies, and view a list of defined policies, policy assignments, and policy enforcements.

- **Policy Definition.** A list of all the policies defined for your organization. You can view policy details or delete a policy which is no longer required. To create a policy, click **New Policy** and select one of the policy types.

- **Policy Assignments.** On the Policy Assignment tab, you can view policies that are effectively applied to resources, namespaces, and projects, based on the defined policy criteria.
- **Policy Enforcement**. Use the Policy Enforcement tab to audit when policies are applied. It lists the enforcement date, policy name, and policy status.
- **Infrastructure Policies**. You can view the infrastructure policies that your provider administrator assigned to your organization.

To get you started, use the full use cases that are provided for each policy type. The use cases guide you through the process of creating more than one policy. The use case provides contextual explanations of the choices and the desired behavior.

The use cases are followed by more in-depth information about how multiple policies are processed.

Configure Approval Policies in VCF Automation

Approval policies are a level of governance that you add to exercise control over deployment and day 2 action requests before they are run. You define approval policies so that you, or others that you designate, review requests before resources are consumed or destroyed. The approval policy use cases in this procedure are an introduction that you can use as you explore your governance options.

If you have only a small team adding and deploying catalog items, then approval policies might be less useful. As you make the catalog available to a larger group of developers and general consumers, you can use the approval policies to ensure that someone reviews a request before the resources are consumed or changes are made to the provisioned items.

For example, you have a catalog item that is important, but it consumes a significant amount of resources. You want one of your IT administrators to review any deployment requests to ensure that the request is needed. Another example applies to day 2 actions. Making changes to a deployment that is used by many might be devastating. You want the project administrator who manages the deployment for that team to review all changes to the deployed catalog instance.

Who works with or is affected by approval policies

- Organization administrator. Configures the policies.
- Catalog consumers. Users who request catalog items or day 2 actions to which one or more policies apply.
- Users deploying blueprints in VCF Automation. Users who request blueprints or day 2 actions to which one or more policies apply.
- Designated approvers. Users who must review and then approve or reject a request. You can grant approver rights to selected users and user groups, or you can choose from the following approver roles.
  - Project Administrators. Administrators of projects within the policy scope are automatically assigned as approvers. If a project does not have a dedicated administrator, the approval policy is not applied to that project.
  - Project Auditor. Members of projects within the policy scope who are assigned the Auditor role. Auditor access rights are limited to approving and rejecting deployment requests for a project. If a project does not have a dedicated auditor, the approval policy is not applied to that project.

What happens when approval policies are enforced

Multiple approval policies might be enforceable. The approval policies are evaluated, and an enforced policy is applied to the request. When there are multiple valid policies, where the approvers are different people, all the approvers are added. When you have multiple policies, it is important to understand this process. For more information, see Approval policy goals and enforcement examples.

1. Approval policies are defined.
2. A user requests a catalog item or day 2 action. At request time, VCF Automation evaluates the catalog item to see if any policies apply.
3. An approval policy is enforced.
   a. The deploy card displays the status. For example, Create - Approval Pending.
   b. An email notification is sent to the requester.
   c. An email notification is sent to the approvers.

The deployment does not begin deploying and consuming infrastructure resources, or make changes to a deployed system, until the request is approved. The requesting user is notified by email that the request is waiting for approval.

d. The approvers respond to the request using the Approvals page.

4. The approval process is completed.
   a. If the request is rejected, the requesting user is notified and the deployment request is canceled.
   b. If the request is approved, the deployment proceeds.
   c. It is possible that the enforced policy is configured to automatically approve or reject a request if the approver does not take any action.

How can I use the deployment criteria

To limit what items or activities the policy applies to, you can define the deployment criteria. For more about the criteria, see Configure deployment criteria in policies in for All Apps.

Prerequisites

- An approver who must have the Organization Administrator or Project Administrator role.
- Verify that the email notification server is defined. See Add an email server in for All Apps to send notifications.

Procedure

As you review the approval policies use case and create your own policy, consult the signpost help on the key text boxes for more information.

1. Select **Manage & Govern** > **Policies** > **Definitions** > **New Policy** > **Approval Policy**.
2. Configure Approval Policy 1.
   As an administrator, you have an important catalog item that also consumes a significant amount of your resources. You want several managers to review any deployment requests to ensure that the request is really needed and that the resources exist to support it.
   1. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to all projects in your organization.
Criteria	`Catalog Item equals CompanyApplication`

   2. Define the approval behavior.

Setting	Sample Value
Approval type	Select **User based**. You select the users and user groups who are the request approvers.
Approver mode	All You want all your IT managers to agree that the deployment request does not waste resources.
Approvers	{GroupName1}@YourCompany, {ApproverName1}@YourCompany, {ApproverName2}@YourCompany The approval request is sent to all members of the user group. Only one member of the group must approve the request. If one approver rejects the request, the request is denied and the user is notified.

Setting	Sample Value
Auto expiry decision	Reject The request will be automatically rejected after the number of days specified under **Auto expiry trigger**.
Auto expiry trigger	3 The number of days that the approvers have to respond before the action under **Auto expiry decision** is triggered.
Actions	Deployment.Create You can select one or more actions that this policy applies to.

In this scenario, if any catalog consumer requests this catalog item, Approver 1, Approver 2, and any one member of User Group 1 must approve the request within 3 days or the request is rejected.

3. Configure Approval Policy 2.

   As an administrator, you have a project where you want the project administrators to approve any changes to deployments that might have catastrophic consequences. For example, deleting the deployment.

   Define when the approval policy is valid.

Setting	Sample Value
Scope	Project The policy is applied to deployments associated with the project you selected.
Criteria	None

   a. Define the approval behavior.

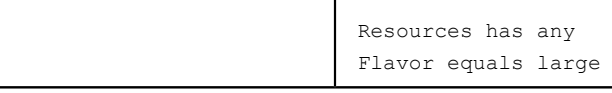
Setting	Sample Value
Approval type	Select **Role based**.
Approver role	Project Administrators If a project does not have a dedicated administrator, the approval policy is not applied to requests associated with that project.
Approver mode	Any
Auto expiry decision	Reject
Auto expiry trigger	7
Actions	Deployment.Delete, Deployment.PowerOff, Deployment.Update, and any of the component-specific power, reboot, and delete actions.

   In this scenario, when a member of one of the scoped projects submits a request to run the listed actions on a deployment, the request is rejected after seven days if the project administrator does not respond.

4. Configure Approval Policy 3.

   As an administrator, you want to maintain a little control over resource consumption. For example, when a user requests a catalog item where the size is large, you want to evaluate and approve the request. Size is defined by the flavor mappings.

   a. Define when the approval policy is valid.

Setting	Sample Value
Scope	Organization
Criteria	

b. Define the approval behavior.

Setting	Sample Value
Approval type	Select **User based**.
Approver mode	Any
Approvers	{AdminName}@YourCompany
Auto expiry decision	Reject The possible consumption of your cloud resources means that you do not want to inadvertently deploy the item without approval.
Auto expiry trigger	5
Actions	Deployment.Create and any applicable *.Machine.Resize actions. For example, CCI.Supervisor.Resource.VirtualMachine.Resize.

In this scenario, when any user submits a request for a large deployment or to resize a deployment to large, the request is rejected after 5 days if the cloud administrator does not respond.

What to do next

- For more information about how approval policies are processed, see Approval policy goals and enforcement examples.

Configure Day 2 Action Policies in VCF Automation

You define day 2 action policies so that you can control what changes your users can make to deployments and their component resources. By creating a list of permitted actions that all or some users can run on deployments, you ensure that the users cannot initiate any destructive or costly changes. The use cases related to day 2 actions policies are an introduction to the procedure.

When you entitle users to run day 2 actions, you select the individual actions that they can run. You are creating an inclusion list, not an exclusion list.

When does a day 2 actions policy go into effect

- If you do not have any Day 2 Action policies defined, then no governance is applied and all users have access to all the actions. This initial lack of governance as you are starting out ensures that you and your users can exercise the day two actions in VCF Automation without the need to understand day 2 policies.
- After you determine that you are ready to control who has access to what actions, you add governance in the form of a single Day 2 Action policy. When the first policy goes into effect, the Day 2 Action policies are enforced for all users in VCF Automation. As a result, only the users for whom the first policy is true can run the selected actions. All others are excluded. They are excluded because the actions policies include the trusted users. By excluding all others, you are able to craft the policies to match your governance goals.
- To entitle other users, you must create policies that entitle them to run the actions you select.

When are day 2 actions policies applied

To focus when the Day 2 Actions policies are applied, you can configure scope, role, and criteria. These configurations control what deployments the policy is applied to and who can run the actions when the policy is enforced.

- What deployments the policy is applied to.
  - Scope determines whether the policy is applied to deployments at the organization or project level.
  - Criteria narrow the scope of the policy to particular aspects of deployments.
- Who can run what actions on those deployments.
  - Role entitles the members of the selected role, within the selected scope and criteria, to run the selected actions. The role can be project administrator, project advanced user, project user, or a named custom role.

Day 2 policies are enforced when a user tries to manage a deployment using the Actions menu on the deployment or on the component resources.

Selecting day 2 actions

As you review the day 2 actions policies use case, you must also select the actions. You must select the actions that support your deployments.

Prerequisites

- For a list of possible actions, see What actions can I run on deployments or IaaS resources in for All Apps.
- For more information about constructing deployment criteria, see Configure deployment criteria in policies in for All Apps.

Procedure

1. Select **Manage and Govern** > **Policies** > **Definitions** > **New Policy** > **Day 2 Actions Policy**.
2. Configure Day 2 Policy 1.
   As an administrator, you want to control storage costs by restricting the ability of users to request snapshots.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy applied to all deployments in your organization.
Criteria	None
Enforcement type	Soft This enforcement type allows you to create other policies related to the snapshot actions that override this policy. Soft policies are overridden by hard policies.
Entitlement type	Role based

   b. Select the actions that the users can run, but do not select any snapshot actions.
      You explicitly entitle users to run actions. To exclude users from running snapshot actions, ensure that the actions are not selected.

   In this scenario, none of the project members in your organization are entitled to create snapshots. Nor can your project administrators. Your next step is to create a policy that entitles the project administrators to create and manage snapshots.
3. Configure Day 2 Policy 2.
   As an administrator, you want to give the project administrators the ability to create and manage snapshots.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to all deployments in your organization.
Criteria	None
Enforcement type	Soft This enforcement type allows you to create other policies related to the snapshot actions that override this policy.
Entitlement type	Role based
Role	Administrator This role applies the policy to the project administrators.

   b. Select the snapshot actions that you want the administrators to run.

Project administrators are also entitled to run any actions that the members of their projects are entitled to run. You do not need to give them permission to member actions.

In this scenario, the project administrators are entitled to run the snapshot-related actions and all the actions that their project members are entitled to run.

4. Configure Day 2 Policy 3.

   As a project administrator, you have two developers who are doing work that potentially makes a deployment unusable. You want to entitle them to snapshot and revert without your intervention. You entitle the two project members to use the snapshot actions.

   a. Define when the policy is valid.

Setting	Sample Value
Scope	Project MT5 This policy is applied to deployments associated with this project.
Criteria	`Catalog Item equals Multi-tier five machine with LB` Based on this criteria expression, only the deployments for a catalog item named Multi-tier five machine with LB are considered for policy enforcement.
Enforcement type	Hard This enforcement type ensures that the policy is enforced based on the definition.
Entitlement type	User based
Users	Add users. `jan@mycompany.com, kris@mycompany.com` Only the deployments where Jan or Kris deployed a catalog item are considered for policy enforcement.

   b. Select the snapshot actions that you want the specified users to run.

      Project administrators are also entitled to run any actions that the members of their projects are entitled to run.

   In this scenario, Jan and Kris can use the snapshot actions on the Multi-tier 5 Machines with LB catalog item that either of them deploy. Although other members of the project can see the deployment, only Jan, Kris, and the project administrator can use the snapshot actions.

What to do next

- For more examples of how the policies are processed and enforced, see How are policies processed in for All Apps.
- Configure policies that are relevant to your organizations and projects.

Day 2 Actions on Catalog-Provisioned Resources Provided with VCF Automation

After deploying catalog items, users can run day 2 actions to modify and manage the resources. The available actions depend on the resource type and whether the action is supported on a particular namespace.

As an organization administrator or a project administrator, you can set up Day 2 Actions policies to entitle users to specific actions. See How do I entitle deployment users to day 2 actions using policies.

Action	Applies to these resource types	Description
Add Disk	Machines	Add additional disks to existing virtual machines.
Create	Deployments	Create a deployment.
Create Snapshot	Machines	Create a snapshot of the virtual machine. If you are allowed only two snapshots in vSphere and you already have them, this command is not available until you delete a snapshot.

Action	Applies to these resource types	Description
	VM Groups	Create a snapshot of the VM group.
Get Password	Machines	Get the password for the virtual machine.
Change Lease	Deployments	Change the lease expiration date and time. When a lease expires, the deployment is destroyed and the resources are reclaimed. Lease expirations are controlled with lease policies.
Delete	Deployments	Destroy a deployment. All the resources are deleted and then reclaimed. If a delete fails, you can run the delete action on a deployment a second time. During the second attempt, you can select Ignore Delete Failures. If you select this option, the deployment is deleted, but the resources might not be reclaimed. You should check the systems n which the deployment was provisioned to ensure that all resources are removed. If they are not, you must manually delete the residual resources on those systems.
Delete Snapshot	Machines	Delete a snapshot of the virtual machine.
	VM Groups	Delete a snapshot of the VM group.
Edit Deployment	Deployments	Edit deployment name, description
Get Password	Machines	Retrieve VM credentials.
Power On	Deployments	Power on the deployment. If the resources were suspended, normal operation resumes from the point at which they were suspended.
	Machines	Power on the machine. If the machine was suspended, normal operation resumes from the point at which the machine was suspended.
	VM Groups	Power on all members of the VM group, potentially with controlled delays for staggered application starts. If a machine was suspended, normal operation resumes from the point at which the machine was suspended.
Power Off	Deployments	Power off the deployment after first attempting to shutdown the guest operating systems. If the soft power off fails, a hard power off still runs.
	Machines	Power off the machine after first attempting to shut down the guest operating systems. If the soft power off fails, the hard power off still runs.
	VM Groups	Power off all members of the VM group after first attempting to shut down the guest operating systems. If the soft power off fails, the hard power off still runs.
Remote Console	Machines	Launches the remote console in the VMware Remote Console (VMRC) client.

Action	Applies to these resource types	Description
Remove Disk	Machines	Remove disks from existing virtual machines. If you run the day 2 action on a deployment that is deployed as vSphere machines and disks, the disk count is reclaimed as it applies to project storage limits. Storage limits consider the actual capacity for thick and thin resource provisioning so that you cannot over-provision using thin provisioning. The project storage limits do not apply to additional disks that you added after deployment as a day 2 action.
Resize	Machines	Increase or decrease the CPU and memory of a virtual machine by selecting a different VM class. The VM must be powered off first before you apply this day 2 action.
Revert Snapshot	Virtual Machines	Revert to a previous snapshot of the machine. You must have an existing snapshot to use this action.
	Virtual Machine Groups	Revert to a previous snapshot of the VM group. You must have an existing snapshot to use this action.
Suspend	Virtual Machines	Pause the machine so that it cannot be used and does not consume any system resources other than the storage it is using.
	Virtual Machine Groups	Pause all members of the VM group so that they cannot be used and do not consume any system resources other than the storage they are using.
Web Console	Machines	Launches the web console in the browser.

Configure Lease Policies in VCF Automation

By using policy-based leases, you reduce the need to intervene manually to reclaim resources. You define lease policies so that you can control the amount of time that a deployment is available to your users. The lease policy use cases in this procedure provide a beginning point for learning about and implementing policies for your organization.

If you do not have any lease policies defined, then the deployments never expire. To reclaim the resources, you must manually destroy the deployments.

When does a lease policy go into effect

- If the policy scope is Organization, then all the deployments in your organization are managed based on the defined policies.
- If the policy scope is a project, then the deployments that are associated with that project are managed based on the defined lease. Other projects are not affected.

When are lease policies applied

Lease policies are applied when you:

- Create or update a lease policy. After lease policies are applied, they continuously evaluate the deployments in the background to ensure that they are in compliance with the defined leases.

- Request a catalog item. The maximum lease and maximum total lease values go into effect when the deployment is created.

Lease-specific options

As you review the lease policies use case, you must also configure lease-specific options. The following descriptions provide a brief summary. Consult the signpost help for more information.

- Maximum Lease (days). The number of days that the deployment resources are active without being renewed. If they are not renewed, the lease expires and the deployment is destroyed. If a grace period is specified, the user can renew the lease for up to the same number of days that the lease has been active.
- Maximum Total Lease (days). The combined total number of days that the deployment can be active, including lease renewals. Each renewal cannot exceed the maximum lease, and the cumulative renewal value cannot exceed the maximum total lease. After the total lease is reached, the deployment is destroyed and the resources within that deployment are reclaimed.
- Grace period (days). The number of days the user has to renew an expired lease before the deployment is destroyed. The grace period is not included in the total lease days. If you don't define a grace period, it defaults to 1 day.

What happens when I update an existing lease policy

You can increase the maximum lease, the maximum total lease, and the grace period of your lease policies. The updated policy is applied to new deployments only.

Note that increasing the policy parameters does not extend the expiry date of existing deployments. Existing deployments expire on the same date as originally scheduled.

If you decrease the lease values, however, the deployment expiry date is affected and existing deployments expire earlier than originally scheduled.

What happens when a deployment expires

When a deployment is about to expire, the deploying user receives an email notification. If the user doesn't extend the lease, the deployment expires and is scheduled for deletion. The deployment owner can see the scheduled deletion date in the deployment details. If the deletion date is set to Soon, then the deployment is already queued for deletion.

Expired virtual machines are powered off within several minutes after expiry. Unless the lease is extended, a powered-off machine can't be powered back on in VCF Automation. The machine can be powered back on manually in the original environment, in which case VCF Automation registers the machine as powered on. Even though the deployment is already expired, the machine is not automatically powered off for a second time until the grace period of the lease policy is elapsed and the deployment is deleted.

Procedure

In this use case, there are three policy definitions that illustrate how you can construct policies and the results when they are enforced. The last policy is not enforced, but the reasons are provided in the scenario results.

1. Select **Manage & Govern** > **Policies** > **Definitions** > **New Policy** > **Lease Policy**.
2. Configure Lease Policy 1.
   As an administrator, you want to control costs by limiting the starting lease time for all deployments to 30 days, with the option to renew the lease for a total of 90 days.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to everyone in your organization.
Criteria	None
Enforcement type	Soft This enforcement type allows you to create other policies related to this lease that override this policy.

b. Define the lease.

Setting	Sample Value
Maximum lease (days)	30
Maximum total lease (days)	90
Grace period (days)	10

In this scenario, the deployment is shut down after 30 days and an email is sent to the user. During the grace period, the user extends the lease by 30 days. After the lease expires again, the user renews it for another 30 days. At the end of the third extension, the lease reaches the maximum total lease period of 90 active days and the user cannot extend it anymore. The deployment is shut down and destroyed 10 days later.

3. Configure Lease Policy 2.

   As an administrator, you want to control costs by limiting the lease time on an expensive blueprint to two weeks. For this example, the blueprint name is `Multi-tier 5 machine with LB`.

   a. Define when the policy is valid.

Setting	Sample Value
Scope	Project MT5 This policy applied to deployments associated with this project.
Criteria	`Blueprint equals Multi-tier 5 machine with LB` Based on this criteria expression, only deployments for the referenced blueprint are considered for policy enforcement.
Enforcement type	Soft This soft enforcement still overrides the organization policy of 90 days in Policy 1 because the values are more meaningful at the project level.

   b. Define the lease policy.

Setting	Sample Value
Maximum lease (days)	14
Maximum total lease (days)	28
Grace period (days)	3

   In this scenario, both policies are applied, but Policy 2 takes precedence over Policy 1 because it is more specific. When applied, the deployment is shut down after 14 days. If the user does not extend the lease, it is destroyed three days later. If the user extends the lease for up to another 14 days, the deployment is shut down at the end of the second extension and it is destroyed three days later.

4. Review the configuration of Lease Policy 3.

   As a project manager, you realize that one of your developers is working on a complex application. The developer requires the `Multi-tier 5 Machines with LB` blueprint and another blueprint, `Distributed Database Across Clouds`, but for a longer lease than defined in Policy 2.

   Unless you understand how the policies are processed based on how they are defined, you might encounter unexpected results. Policy 3 is an example of how processing and precedence affect the result.

   This policy, as provided, will not be enforced. This example provides an opportunity for you to see how leases are applied and enforced when there is more than one that applies.

a. Define when the policy is valid.

Setting	Sample Value
Scope	Project MT5 This policy is applied to deployments in this project.
Criteria	`(Blueprint equals Multi-tier five machine with LB OR Catalog Item equals Distributed Database Across Clouds) AND Created By equals jan@mycompany.com` You use Catalog Item because the item is not a blueprint.
Enforcement type	Soft This soft enforcement still overrides the organization policy of 90 days in policy 1 because the values are more meaningful at the project level.

b. Define the lease policy.

Setting	Sample Value
Maximum lease (days)	21
Maximum total lease (days)	50
Grace period (days)	3

In this scenario, Lease Policy 2 is applied, not Lease Policy 3.

- Lease 3 has a lease time that is less than or equal to 21 days, and the policy is applied. Lease 2 has a lease time that is less than or equal to 14 days, and the policy is applied.
- Lease 2 is applicable and it does not violate the lease 3 policy. But, lease 2 is more restrictive, so it takes precedence. Lease policy 2 is more restrictive because it is for a shorter period of time.
- When both lease definitions are true and applicable, the more restrictive policy is the one that is enforced.

5. To resolve the unexpected behavior in Lease Policy 3, you can implement one of the following solutions.
   - To ensure that you can provide Jan with the needed policy, change the enforcement type to hard.
   - Alternatively, you could create a new project with access to the same resources, and then create Lease Policy 3 for that project. While this solution isolates the working policy, you must maintain a parallel project. The effort needed to set up and maintain the content sources, content sharing, and so on, are time consuming and subject to error.

What to do next

- For more examples of how the lease policies are processed and enforced, see How are policies processed in for All Apps.
- Configure policies that are relevant to your organizations and projects. If you are just getting started with lease policies, begin with one lease policy at the organization level.
- To send an email to the deploying user, configure the email server for notifications. See Add an email server in for All Apps to send notifications.
- If your organization is integrated with VCF Operations orchestrator, you can manage expired deployments and their resources by using event subscriptions. See Create an Event Subscription to Manage Deployment Expiry in for All Apps.

Configure IaaS Resource Policies in VCF Automation

You define IaaS resource policies to validate and control what namespace resources deployments can consume when users request catalog items or create resources using the services console in VCF Automation. The use cases in this procedure are an introduction to configuring IaaS resource policies.

Before your users can generate workloads on namespaces, you must establish policies to ensure the conformity and security of the workloads. An IaaS resource policy outlines a set of guidelines applicable to the objects on the namespace. These objects can include built-in items like VMs, Kubernetes clusters, PodVMs, or new objects introduced through the installation of additional services. As an organization administrator, you can configure and enforce these rules across various namespaces.

When you create an IaaS resource policy, it is applied to both newly created and existing namespaces within a project or the organization, based on namespace specifications.

What can I control with IaaS resource policies

As an organization administrator using IaaS resource policies, you control whether a user can create the following:

1. Resources for all services running in a namespace.
2. Resources for a sub-set of services, such as the VM Service only or the Kubernetes Service only. For example, you can allow only specific VMs with a certain VM class.
3. Control the number of worker or control nodes for Kubernetes clusters. For example, you might want to enforce a limit of three control nodes for a production cluster.
4. Enforce a minimum Kubernetes version for Kubernetes clusters.

What happens when IaaS resource policies are enforced

- When the policy is enforced, users can provision resources within the specified constraints.
- Multiple IaaS resource policies can be enforceable. VCF Automation does not merge overlapping IaaS policies. Both policies are applied. As the organization administrator, you are responsible for defining policies that do not conflict with each other. For example, if one policy states that all VM names must start with `vm-1` and another policy states that all VM names must start with `vm-2`, then both of these policies are applied. Because a VM name cannot start with both `vm-1` and `vm-2`, no VM is provisioned in the namespace.
- IaaS policies are not enforced on deployments that request no resources, such as a VCF Operations orchestrator workflow deployment.

When are IaaS resource policies applied

- A user requests a catalog item.
- A user provisions resources using the IaaS services console.

How many IaaS policies can I create

- You can create up to 10 IaaS resource policies for your organization unless your provider administrator set a different limit for your organization. Policies can be scoped to an entire organization or to a specific project. The limit of policies per project is half of the organization limit.
- Up to 50 IaaS resource policies can be created per supervisor across all organizations that might share the supervisor. When a new policy is created or when a new namespace is created that matches the criteria of an existing policy, a per-supervisor limit is enforced.

See VMware Configuration Maximums for information about the maximum number of IaaS resource policies that can be created per supervisor.

What IaaS policy templates are available

The following IaaS policy templates are available out of the box. In addition, you can define your own policies.

IaaS policies are backed by Kubernetes validating admission policies. Modifying the policy templates requires use of the Common Expression Language (CEL), which is used in the Kubernetes API. For more information about validating admission policies, see the official Kubernetes documentation.

Policy template	Description
Select Enforce single-control-node Kubernetes cluster (development)	Ensures a Kubernetes cluster has only 1 control node.
Enforce multi-control-node Kubernetes cluster (production)	Ensures a Kubernetes cluster has a minimum of 3 control nodes.
Enforce minimum Kubernetes version for Kubernetes cluster	Enforces a minimum version specified in a template.
Disallow Kubernetes cluster resource	Does not allow provisioning of Kubernetes cluster resources.
Disallow VM resource	Does not allow provisioning of VM resources.
Label enforcement	Requires a specific label for a resource.
Kubernetes cluster restricted pod security level	Requires cluster to have restricted pod security level
Kubernetes cluster baseline pod security level	Requires Kubernetes cluster to have baseline pod security level

Using the Assignments tab

After you create an IaaS policy, you can use the **Assignments** tab to get an overview of all IaaS policies affecting a given namespace, to track which policies were applied successfully, and to review any errors.

IaaS Policy limitations

1. `Parameterization` is not supported.
2. `AuditAnnotations` is not supported.
3. `messageExpression` is not supported.

Procedure

1. Select **Manage & Govern** > **Policies** > **New Policy** > **IaaS Resource Policy**.
2. Configure IaaS Policy 1.
   As an administrator, you want to allow your users to create or update clusters that have only 1 node.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to everyone in your organization.
Criteria	None

   b. Load the policy template.

```
failurePolicy: Fail
matchConstraints:
  resourceRules:
    - apiGroups:
        - run.tanzu.vmware.com
      apiVersions:
        - "*"
      operations:
        - CREATE
        - UPDATE
      resources:
        - tanzukubernetesclusters
    - apiGroups:
        - cluster.x-k8s.io
```

```
      apiVersions:
        - "*"
      operations:
        - CREATE
        - UPDATE
      resources:
        - clusters
  validations:
    - expression: object.spec.topology.controlPlane.replicas == 1
      messageExpression: "'Control plane nodes must be 1'"
```

c. Preview the policy enforcement.
d. Click **Create**.

In this scenario, if a user tries to provision a cluster with more than 1 control plane nodes, their request fails.

3. Configure IaaS Policy 2.
   As an administrator, you want to enforce a label on VM resources that are consumed by a specific project.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Project This policy is applied to everyone in the selected project.
Criteria	None

b. Load the policy template.

```
failurePolicy: Fail
matchConstraints:
  resourceRules:
   - apiGroups: ["vmoperator.vmware.com"]
     apiVersions: ["*"]
     operations: ["CREATE", "UPDATE"]
     resources: ["virtualmachines"]
validations:
- expression: object.metadata.labels.environment == 'prod'
  messageExpression: "'All virtual machines should have label environment=prod'"
```

c. Preview the policy enforcement.
d. Click **Create**.

In this scenario, if a user tries to provision a virtual machine without the label `environment=prod`, their request fails.

4. Configure IaaS Policy 3.
   As an administrator, you don't want VM resources to be provisioned by a specific project.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Project This policy is applied to resources provisioned in the selected project.
Criteria	None

b. Load the policy template.

```
failurePolicy: Fail
matchConstraints:
  resourceRules:
    - apiGroups:
        - "vmoperator.vmware.com"
      apiVersions:
```

```
            - "*"
          operations:
            - CREATE
          resources:
            - "virtualmachines"
      validations:
        - expression: False
          messageExpression: "'Virtual Machines are prohibited to be provisioned in the namespace.'"
```

c. Preview the policy enforcement.
d. Click **Create**.

In this scenario, VM resources cannot be provisioned on namespaces in the scoped project.

Configure Deployment Criteria in Policies in VCF Automation

The deployment criteria narrows the scope of a policy so that it is applied only to the deployments where the criteria is true. For example, you can use the deployment criteria to create a policy that is applied only to a particular catalog item or blueprint.

Constructing deployment criteria

You use the graphical interface to construct the deployment criteria expression. To construct complex expressions, you can use AND and OR. You can also group expressions as parenthetical operators. For more about how the expressions are processed, see the *Order of operations for the expression section* below.

Here is an example of an expression.

```
Deployment equals Multi-tier five machine with LB AND (Owned By equals jan@mycompany.com OR Owned By
 kris@mycompany.com)
```

Policy criteria properties

To create functional policy criteria, you must understand the syntax.

The criteria text box has various drop-down menus that provide the available properties and operators. How you construct your expression depends on the available values and on the order of operations.

The drop-down menus include the following properties. Some properties vary between policy types.

Property	Description	Available in these policy types	Supports these operators
`Template/Blueprint`	Identifier for the blueprint that was used to create the deployment.	• Approvals • Day 2 • Lease	• equals • not equal to
`Catalog Item`	Identifier for the catalog item that was used to request the deployment. Use `Catalog Item` rather than `Template` when your policy can include catalog items based on any blueprint, extensibility workflow, or other content type. For example, VM images deployed from the catalog have a `Catalog Item`.	• Approvals • Day 2 • Lease	• equals • not equal to
`Deployment`	Identifier for the deployment. Use `Deployment` when you want to apply the policy to existing deployments.	• Approvals • Day 2 • Lease	• equals • not equal to

Property	Description	Available in these policy types	Supports these operators
`Created By`	Name of the user who requested the deployment. The format is username@mycompany.com. This user is the user who requested the deployment.	• Day 2 • Lease	• equals • not equal to • matches Regex • contains
`Name`	Deployment name. Use `Name` rather than `Deployment` when you want to apply the policy to existing policies and policies that can be created in the future that match the specified deployment name expression.	• Approvals • Day 2 • Lease	• equals • not equal to • matches Regex • contains
`Namespace Name`	Name of the namespace.	• IaaS Resource	• equals • not equal to
`Namespace Class`	Namespace class.	• IaaS Resource	• equals • not equal to
`Owned By`	Name of the current deployment owner.	• Approvals • Day 2 • Lease	• equals • not equal to • matches Regex • contains
`Owner Type`	Deployment owner type. Ownership can be based on users or groups.	• Approvals • Day 2 • Lease	• equals • not equal to
`Region`	Region.	• IaaS Resource	• equals • not equal to
`Requested By`	Name of the user who requested a day 2 action. The format is username@mycompany.com. When creating approval policies, the `Requested By` criteria is the user who requested a day 2 action, not the user who requested the deployment. The user who requested the deployment is the `Created By` criteria.	• Approvals	• equals • not equal to • matches Regex • contains
`Resources`	Resources that are part of a deployment. You can define the deployment criteria based on the following resources. • Cloud Zone • Disks • Kind • Resource Property • Resource Type	• Approvals • Day 2 • Lease	

Using the *`contains`* and *`matches Regex`* operators

The `contains` and `matches Regex` operators define a search for a specified set of characters within a property. You can apply these operators to string based properties that do not support a drop-down, such as `createdBy`, `name`, and `ownedBy`.

The `contains` operator searches for all instances of the value you specify in any context. The value input text box is case sensitive and space sensitive. If you want to account for context variation, you must set a value for each additional variant. Use the `contains` operator for simple searches for a limited number of values.

The `matches Regex` operator provides great flexibility when you use it for complex searches that must account for a lot of context variation. The regular expressions must follow ECMAScript syntax. When defining regular expressions, do not enter the forward slashes (`/`) at the beginning and at the end of the value.

The following table provides examples of expressions using the two operators and compares how they might be used to achieve the same goal.

Example with the `contains` operator	Example with the `matches Regex` operator	Field value matches
`Name contains test`	`Name matches Regex test*`	All deployment names that contain *`test`* in lowercase. For example, *`test deployment`*, *`mytest`*, *`test-123`*, and so on.

`Name contains test`
`OR`
`Name contains Test`
	`Name matches Regex (t|T)est.*`	All deployment names that contain *`test`* or *`Test`*.

`(group)`
`Created By contains admin@`
`(group)`
`AND`
`Created By contains .com`
`OR`
`Created By contains .org`
`(group)`
`AND`
`Name contains test`
`OR`
`Name contains test-`
`OR`
`Name contains Test`
`OR`
`Name contains Test-`
`OR`
`Name contains deploy`
`OR`
`Name contains Deploy`
	`Created By matches Regex admin@\S+\.((com)|(org))`
	`AND`
	`Name matches ((t|T)est)|(d|D)epl.*.`
		All deployments that are created by users whose email address starts with *`admin@`* and ends with *`.com`* or *`.org`*.
		All deployment names that contain *`test`* and/or *`deploy`* in any configuration. For example, *`test deployment`*, *`testdeployment`*, *`Test-Deployment`*, and so on.

Order of operations for the expression

An expression is processed in the following order. Groups are illustrated as parentheses.

1. Expressions in groups
2. AND
3. OR

Use the following examples to understand the order.

- X OR Y AND Z. In this example, Y AND Z is evaluated before X OR Y. Next, the X OR is evaluated against the results of Y AND Z.
- (X OR Y) AND Z. In this example, X OR Y is evaluated before AND because the expression in the group is always evaluated first. Next the AND Z is evaluated against the results of X OR Y.

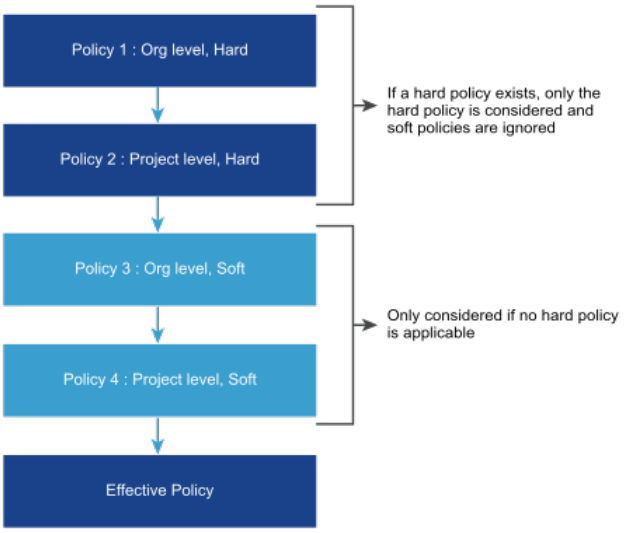
How are Policies Processed in VCF Automation

Policies are processed based on the policy definition. In particular, the scope and the enforcement level determine which policy is valid when you have multiple policies that might apply to a single deployment.

This article provides general information about policy processing, but it also includes more details for the different types of policies.

How policies are ranked based on organization level and enforcement type

When a user, who is a member of a project, creates a deployment, there might be more than one policy that applies to that deployment.

	To evaluate the policies, the system first identifies and ranks policies. 1. Are there any hard policies at either the organization and project level. If there are hard and soft policies, then only the hard policies are considered and ranked. If there are only soft policies, then the soft policies are ranked. 2. The ranking of all hard or all soft policies is ordered by scope, with organizational policies having a higher rank than project policies. 3. The final discriminating characteristic is the creation date, with older dates be ranked higher than newer dates.

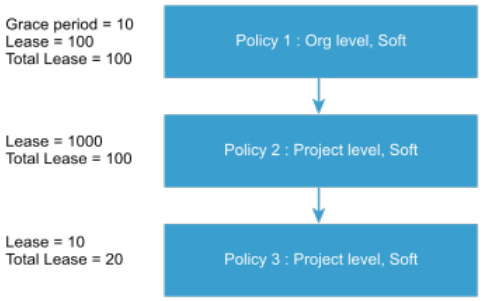
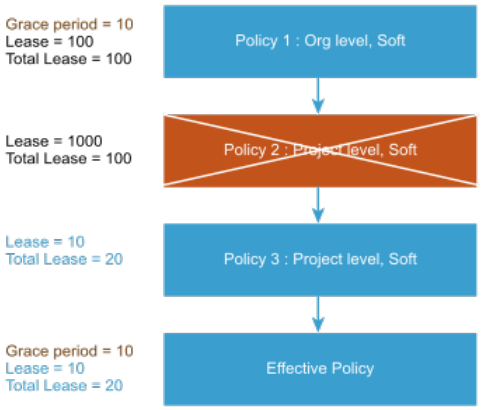
How policies are processed based on organization level and enforcement type

The policies are evaluated, ranked, and, where applicable, merged to produce an effective policy. An effective policy produces the intended results but is not always a specific named policy.

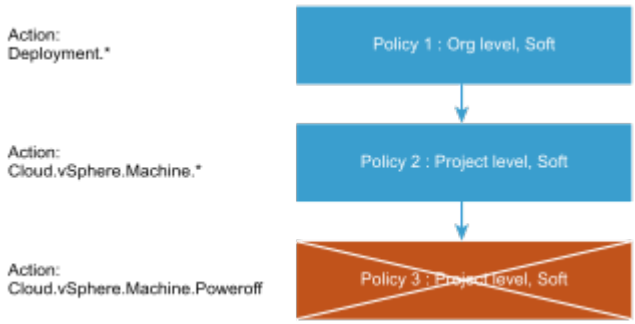
This section includes the following examples:

- Lease policies
- Day 2 actions policies

Review the following lease policy examples.

	After identifying the policies to be considered and ranking them, the policies are then evaluated to identify the merge order. • The highest ranking policy becomes the baseline. The second-level policy is applied on top of it, and so on. • If a policy is incompatible with the preceding policies, then it is discarded from consideration. For example, the values are higher they the preceding policies. • Any policy that is discarded is ignored. To see which policy is applied, select **Content and Policies** > **Policies** > **Enforcement**, locate the deployment, and review the decision notes.
	Rather than applying one policy and excluding all the others above, the policies are merged and might include values from more than one individual policy. In this example, the merging process excludes Policy 2 from consideration because the values are higher than Policy 1. Next, Policy 3 is evaluated against Policy 1. The Lease and Total Lease values in Policy 3 are lower than Policy 1, so those values, in addition to the Grace period, become part of the effective policy.

Review the following day 2 actions policy examples.

	After identifying the policies to be considered and ranking them, the policies are then evaluated to identify the merge order. • The highest ranking policy becomes the baseline. The second-level policy is applied on top of it, and so on. • If a policy is enforced by preceding policies, for example, policy 3, then it is discarded from consideration. • Any policy that is discarded is ignored. To see which policy is applied, select **Manage & Govern** > **Policies** > **Enforcement**, locate the deployment, and review the decision notes.

Lease policy management goal considerations

Now that you know how lease policies are processed, identify your policy management goals. By understanding how the policies are processed, you can meet your management goals without creating an excessive and unmanageable number of policies.

When deciding how to implement your policies, consider the following scenarios.

- Lease policy goals and enforcement examples
- Day 2 policy goals and enforcement examples

Table 1821: Lease policy goals and enforcement examples

Management goal	Configuration Example	Behavior
Meaningful default organization-level policy that still allows the project-level policy values to influence the applied values.	Organization policy = Soft • Grace period: 10 • Lease: 100 • Total Lease: 100 Project 1 policy 1= Soft • Lease: 20 • Total Lease: 50 Project 2 policy 1= Soft • Lease: 10 • Total Lease: 30	A member of project 1 requests a catalog item. Project 2 is not considered because it is not applicable to project 1 deployments. The merged effective policy is: • Grace period: 10 • Lease: 20 • Total Lease: 50
Always default to the organization-level policy.	Organization policy = Hard • Grace period: 10 • Lease: 100 • Total Lease: 100 Project 1 policy 1= Soft • Lease: 20 • Total Lease: 50	A member of project 1 requests a catalog item. Project 1 policy 1 is not considered because the hard organization level project is a higher rank and the soft policy is not considered. The effective policy is: • Grace period: 10 • Lease: 100 • Total Lease: 100
All policies are defined at the project-level, with no organization-level default policy.	Project 1 policy 1 = Soft • Grace period: 10 • Lease: 100 • Total Lease: 100 Project 1 policy 2= Soft • Lease: 20	A member of project 1 requests a catalog item. They are both soft policies, and they are both for project 1. The values are merged. The effective policy is: • Grace period: 10 • Lease: 20 • Total Lease: 100

The day 2 actions policies are used in these examples.

Table 1822: Day 2 policy goals and enforcement examples

Management goal	Configuration Example	Behavior
Meaningful default organization-level policy that still allows the project-level policy values to influence the applied values.	Organization policy = Soft • Actions : Deployment.* Project 1 policy 1= Soft • Actions: VirtualMachine.* Project 2 policy 1= Soft • Actions: VirtualMachine.*	A member of project 1 requests a catalog item. Project 2 is not considered because it is not applicable to project 1 deployments. The merged effective policy is: • Action : {Deployment.* ,VirtualMachine.*}
Always default to the organization-level policy.	Organization policy = Hard • Action : Deployment.* Project 1 policy 1= Soft • Action : VirtualMachine.*	A member of project 1 requests a catalog item. Project 1 policy 1 is not considered because the hard organization level project is a higher rank and the soft policy is not considered. The effective policy is: • Action : {Deployment.* }

Management goal	Configuration Example	Behavior
All policies are defined at the project-level, with no organization-level default policy.	Project 1 policy 1 = Soft • Actions : Deployment.ChangeLease Project 1 policy 2= Soft • Action : Deployment.Delete	A member of project 1 requests a catalog item. They are both soft policies, and they are both for project 1. The values are merged. The effective policy is: • Action : {Deployment.ChangeLease , Deployment.Delete}

Approval policy goals and enforcement examples

The approval policy evaluation follows this process.

1. A request for a deployment or day 2 action is submitted.
2. The approval service queries for policies that apply to the project that is requesting a catalog item or changing a deployed item.
3. All the applicable project- and organization-level scope policies are returned.
4. The approval policies are filtered based on the deployment criteria. Deployment criteria apply to deployments and day 2 actions.
5. If no matching policies are found, no approval is required and the deployment process proceeds.
6. If there are matching policies, for example, AP1, AP2, APn, then an approval item is created as:
   - Enforced policies = AP1, AP2, APn.
   - Approvers = A union of all the approvers in all the enforced policies.
   - Auto expiry = Reject, if any policy has a reject value; otherwise, approve.
   - Expiry = Minimum number of days of any of the enforced policies.

The following table provides a sample of multiple policies. The description of how they are processed is below the table.

Policy	Configuration example
AP1	Scope = Organization Auto expiry = Approve Expiry = 7 days
AP2	Scope = Project 1 Auto expiry - Approve Expiry = 3 days
AP3	Scope = Project 1 Auto expiry = Reject Expiry = 4 days
AP4	Scope = Project 2 Auto expiry = Approve Expiry = 5 days

Based on the policies and configuration examples above, the following information explains how a Project 1 request is processed.

1. The scope evaluation returns AP1, AP2, and AP3. AP4 is not included because it is a Project 2 policy.
2. Assuming that AP1, AP2, and AP3 satisfy the deployment and action criteria, then the approval item includes the following values:
   - Approvers = Any or all the approvers from AP1, AP2, and AP3 are added as approvers.
   - Auto expiry = Reject. AP3 provides the more restrictive behavior.
   - Expiry = 3 days. AP2 provides the lowest value.

Provisioning Resources in VCF Automation

In VCF Automation, resources like VMs and VKS clusters are provisioned in namespaces either through blueprints or through the IaaS Services console.

Provisioning Resources Using Blueprints

When using blueprints, namespaces and the desired resources are dragged and dropped on to the blueprint canvas. The resulting blueprint is then published to the catalog where end users can request it for deployment.

Blueprints are recommended when the organization administrator needs better control over the exact resource that are provisioned by end-users. Blueprints are also used when the process of provisioning resources needs to be repeatable.

Provisioning Resources Using the IaaS Services Console

When using the IaaS Services console, the desired namespace is selected from the **Build & Deploy** page. End users can create the desired resources directly on the selected namespace.

The self-service IaaS Services console enables end users to provision resources without depending on blueprints. The IaaS Services console is suitable for development needs when agility and flexibility of provisioning resources are more important than repeatability.

Requesting an Item from the Catalog in VCF Automation

Catalog items are blueprints, VM images, and other infrastructure objects that project users can request for deployment. At request time, the information that the requesting user must provide or configure depends on how the blueprint was designed. When an item is deployed, it is provisioned based on the namespaces that are associated with the selected project.

Who Works with the Catalog

As the organization administrator, you provide the catalog items. The items that are available to users depend on their project membership.

- If a user is a member of one project, they can see only the catalog items for that project.
- If a user is a member of several projects, they can see the catalog items for those projects.
- If a project does not have any configured catalog items, the Catalog section does not appear for members of that project.

Projects also determine user options at deployment time.

For more information on working with the catalog, see Request an Item from the Catalog in VCF Automation for All Apps in the *Consumer Interfaces in VMware Cloud Foundation* guide.

Managing Deployed Instances in VCF Automation

Catalog consumers use the **Instances** page in VCF Automation to view and manage all of their provisioned resources across their projects. Resources include deployed catalog items, virtual machines, Kubernetes clusters, and other custom resources.

For more information on working with instances, see Managing Deployed Instances in VCF Automation for All Apps in the *Consumer Interfaces in VMware Cloud Foundation* guide.

Using the Self-Service IaaS Console in VCF Automation

VCF Automation provides application teams with access to modern services enabled by vSphere Supervisor and cloud native applications. Users can consume necessary infrastructure by provisioning resources including VMs, Kubernetes, volumes, load balancers and networking objects, storage, and others, in a public cloud-like fashion, without depending on the organization administrator to create and distribute blueprints through the catalog.

Who Works with the IaaS Services Console

The IaaS Services console in the VCF Automation organization UI is available to users with the *Project Administrator* or *Project Advanced User* role.

What IaaS Services Are Supported

Supported services include VM Service, VM Image Service, Kubernetes Service, Network Service, Volume Service, Private AI Service, Avi Load Balancer, and extensible VCF services like Secrets, Key Management, and Data Service.

IaaS Services are scoped to a namespace. Users can use the namespace selector to switch between namespaces to which they have access. What namespaces are available to users depends on their project membership.

For more information about using the self-service IaaS Console, see the following topics in the *Building Your Applications* guide:

- Provision a VM Using the Self-Service in VCF Automation
- Provision a Kubernetes Cluster Using Self-Service in VCF Automation

Administering VCF Automation Organizations

As an administrator, you use the **Administer** tab in the organization UI to monitor and manage critical infrastructure and user management aspects of a VCF Automation organization.

Integrating VCF Automation with Other Applications

Integrations enable you to add external systems to VCF Automation. Integrations include Git, Active Directory, and VCF Operations orchestrator.

Integrate VCF Automation with Git

VCF Automation supports integration with various flavors of Git repositories so that you can manage blueprints and action scripts under source control. This functionality facilitates auditing and accountability of processes around deployment.

VCF Automation supports different flavors of Git integration as described in the following list. Each of these options is a separate integration.

- GitHub cloud
- GitLab cloud
- GitLab Enterprise
- GitHub cloud, GitHub Enterprise on-premises
- GitLab cloud GitLab Enterprise on-premises
- BitBucket on-premises

You must have an appropriate local Git repository configured with access for all designated users in order to set up Git integration with VCF Automation. Also, you must save your blueprints in a specific structure in order for them to be detected by Git. To create an integration with GitLab or GitHub, select **Administer** > **Connections** > **Integrations** and then make the appropriate selection. You will need the URL and token for the target repository.

When Git integration is configured with an existing repository, all blueprints associated with selected projects become available to qualified users. You can use these blueprints with an existing deployment or as the basis of a new deployment. When you add a project, you must select some properties regarding where and how it is stored in Git.

You can save actions to a Git repository directly from VCF Automation. You can version action scripts either directly to Git or you can create versions in VCF Automation. If you create a version of an action in VCF Automation, then it is automatically saved to Git as a version. Blueprints are a bit more complicated, because you cannot directly add them to a Git integration from VCF Automation. You must save them directly to a Git instance, and then you can retrieve them from Git when working with the blueprint management page in VCF Automation.

Before you begin

You must create and save your blueprints in a specific structure in order for them to be detected by GitLab or GitHub.

- Configure and store blueprints to be integrated with GitLab correctly. Only valid blueprints are imported into GitLab.
  - Create one or more designated folders for the blueprints.
  - All blueprints must be stored within `blueprint.yaml` files.
  - Ensure that the top of your blueprints include the name: and version:properties.
- Extract an API key for the applicable repository. In your Git account, select your login in the upper right corner, and click **Developer Settings**. Navigate to **Personal Access Tokens**, and then name your token and set an expiration date. Then, select API and create the token. Copy the resulting value and save it.

The following guidelines must be observed for all blueprints used with Git integration.

- Each blueprint must reside in a separate folder.
- All blueprints must be named `blueprint.yaml`.
- All blueprint YAML files must use name and version fields.
- Only valid blueprints are imported.
- If you update a draft blueprint imported from Git, and its content differs from that in the top version, the draft will not be updated in subsequent syncs and a new version is created. If you want to update a blueprint and also allow further sync's from Git, then you must create a new version after final changes.

Configure a GitLab Blueprint Integration in VCF Automation

This procedure demonstrates configuring GitLab integration in VCF Automation so that you can work with blueprints in the repository and automatically download saved blueprints that are associated with designated projects. To use blueprints with GitLab, you must create a connection to an appropriate GitLab instance, and then save the desired blueprints to that instance.

- Extract an API key for the applicable repository. In your GitLab account, select your login in the upper right corner, and navigate to the Settings menu. Select Access Tokens, then name your token, set an expiration date. Then, select API and create the token. Copy the resulting value and save it.

You must have an appropriate local Git repository configured with access for all designated users in order to set up Git integration with VCF Automation. Also, you must create and save your blueprints in a specific structure in order for them to be detected by GitLab.

- Configure and store blueprints to be integrated with GitLab correctly. Only valid blueprints are imported into GitLab.

When GitLab integration is configured with an existing repository, all blueprints associated with selected projects become available to qualified users. You can use these blueprints with an existing deployment or as the basis of a new deployment. When you add a project, you must select some properties regarding where and how it is stored in GitLab.

> **Note:**
>
> You cannot push new or updated blueprints to the Git repository from VCF Automation. To add blueprints to a repository, developers must use the Git interface.

If you update a draft blueprint imported from Git, and its content differs from that in the Git version, the draft will not be updated in subsequent syncs and a new version is created. If you want to update a blueprint and also allow further sync's from Git, then you must create a new version in Git after final changes.

After you set up your blueprints for use with GitLab and collect required information, you must set up integration with your GitLab instance. Then, you can import the designated blueprints into GitLab.

1. Set up integration with your GitLab environment in VCF Automation.
   a) Select **Administer** > **Integrations**.
   b) Select **GitLab**.
   c) Enter an appropriate name and description.
   d) Enter the **URL** for your GitLab instance.
   e) Enter the **Token**, also known as an API key, for the specified GitLab instance. See the prerequisites above for information about extracting the token from your GitLab instance.
   f) Click **Validate** to verify the connection.
   g) Click **Add**.
2. Configure the GitLab connection to accept blueprints in an appropriate repository.
   a) Select **Administer** > **Integrations** and choose the appropriate GitLab integration.
   b) Select **Projects**.
   c) Select **New Project** and create a name for the project.
   d) Enter the **Repository** path within GitLab. Typically, this is the user name of the main account appended to the repository name.
   e) Enter the appropriate GitLab **Branch** that you want to use.
   f) If applicable, enter a **Folder** name. If left blank, all folders are available.
   g) Click **Next** to finish adding the repository.

      When you click **Next**, an automated synchronization task is initiated that imports blueprints into the platform.

      When the synchronization tasks are complete, a message indicates that the blueprints have been imported.

You can now retrieve blueprints from GitLab.

Configure a GitHub Integration in VCF Automation

You can integrate the GitHub cloud-based repository hosting service in VCF Automation.

- You must have access to GitHub.
- Configure and store blueprints to be integrated with GitHub correctly. Only valid blueprints are imported into GitHub.

You need a valid GitHub token to configure GitHub integration in VCF Automation.

See the GitHub documentation for information about creating and locating your token.

1. Select **Administer** > **Integrations** and click **Add Integration**.
2. Select **GitHub**.
3. Enter the required information on the GitHub configuration page.
4. Click **Validate** to verify the connection.
5. Click **Add**.
6. Configure the GitHub connection to accept blueprints in an appropriate repository.
   a) Select **Administer** > **Integrations** and choose the appropriate GitHub integration.
   b) Select **Projects**.
   c) Select **New Project** and create a name for the project.
   d) Enter the **Repository** path within GitHub. Typically, this is the user name of the main account appended to the repository name.
   e) Enter the appropriate GitHub **Branch** that you want to use.If applicable, enter a **Folder** name. If left blank, all folders are available.
   f) Click `Next` to finish adding the repository.

      An automated synchronization task is initiated that imports blueprints into the platform.

      When the synchronization tasks are complete, a message indicates that the blueprints have been imported.

GitHub is available for use in VCF Automation blueprints. You can now retrieve blueprints from GitHub.

Configure a Bitbucket Integration in VCF Automation

VCF Automation supports integration with Bitbucket for use as a Git-based repository for blueprints.

- Set up an on premises Bitbucket Server deployment with one or more blueprint-based projects that you want to use with your deployments. Bitbucket Cloud is not supported.
- Create or designate a VCF Automation project to associate your Bitbucket integration.
- Blueprint files to be synched to a Bitbucket integration must be named `blueprint.yaml`.

In VCF Automation, you can work with blueprints (also called templates) using the Bitbucket integration. You must synchronize projects that you want to work with before using a Bitbucket integration. You cannot write back blueprints from the integration. If you want to create new versions of blueprint files, you must do so manually.

1. Select **Administer** > **Integrations** and click **Add Integration**.
2. Select **Bitbucket**.
3. Enter the server URL and Bitbucket credentials.
4. Click **Validate** to verify the connection.
5. Click **Add**.
6. Open the Bitbucket integration and select the **Projects** tab on the main page to associate a project with this Bitbucket integration.
7. Select the project to associate with this Bitbucket integration.
8. Click **Next** to add a Repository to Bitbucket project and indicate the type of repository you are adding and then specify the **Repository** name and **Branch**, as well as the **Folder**.
9. Click **Add**.

   If you want to add one or more repositories to a project, click **Add Repository**.

The Bitbucket integration is configured with the specified repository configuration, and you can view and work with blueprints contained in configured repositories. When you add a project to a Bitbucket integration, a synch operation runs to pull the latest versions of blueprint files from the designated repository. The History tab on the Bitbucket integration page shows records of all synch operations for the integration. By default, files are automatically synched every 15 minutes, but you can manually synch a file by selecting it and clicking **SYNC** at any time.

You can work with blueprints on the Blueprint Design page.

Integrate VCF Automation with VCF Operations orchestrator

As a provider administrator, you integrate a VCF Automation organization with VCF Operations orchestrator so that the organization administrator and project users can use VCF Operations orchestrator workflows in VCF Automation.

> **Important:**
>
> Organization administrators cannot create an integration with VCF Operations orchestrator in the organization that they manage. Organization administrators must request that the provider administrator creates the VCF Operations orchestrator integration for the organization.

VCF Operations orchestrator embedded in Provider Consumption organizations in VCF Automation

A VCF Operations orchestrator instance is embedded and configured by default as part of the Provider Consumption organization in VCF Automation. Only one VCF Operations orchestrator can be embedded in a Provider Consumption organization. That instance can also be added as an integration in VCF Automation for VM Apps organizations. The embedded instance is not supported in All Apps organizations.

An external VCF Operations orchestrator instance can be used as a replacement of the embedded VCF Operations orchestrator instance in the Provider Consumption organization in scenarios where the resource load on VCF Automation is very high.

You access the embedded VCF Operations orchestrator instance on the **Orchestrate** tab in the Provider Consumption organization UI.

The service provider can publish workflows from the embedded VCF Operations orchestrator instance as catalog items to VCF Automation for All Apps organizations.

> **Note:**
>
> Embedded VCF Operations orchestrator instances in VCF Automation are primarily used in the provider consumption organization. It is recommended that you integrate your All Apps organizations with an external VCF Operations orchestrator instances.

Configure Authentication for a VCF Operations orchestrator Integration in VCF Automation

When you configure an authentication provider for VCF Operations orchestrator, that VCF Operations orchestrator instance is restricted to the respective All Apps organization.

1. Log in to the VCF Operations orchestrator virtual appliance as `root`.
2. Configure the authentication provider.
   ```
   vracli vro authentication set -p tm -u SERVICE_PROVIDER_ACCOUNT -hn https://FQDN_VCF_AUTOMATION --
   tenant ALL_APPS_ORG_NAME
   ```
3. Redeploy the VCF Operations orchestrator pods.
   ```
   /opt/scripts/deploy.sh
   ```

The VCF Operations orchestrator instance is configured and associated with the All Apps organization.

Create a VCF Operations orchestrator Integration in VCF Automation

You integrate VCF Operations orchestrator into your All Apps organization to use it for extensibility automation tasks.

> **Important:**
>
> Only one external VCF Operations orchestrator instance can be integrated with the organization. Provider Consumption organizations have an embedded VCF Operations orchestrator instance by default.

1. To log in to the VCF Automation organization with `System Administrator` credentials, click **Launch Organization** on the organization details page in the VCF Automation Provider Management UI.
2. Once you are logged in to the organization UI, select **Administer** > **Connections** > **Integrations**.
3. Select **Orchestrator**.
4. Enter a name and description for the VCF Operations orchestrator integration.
5. Under **VCF Operations Orchestrator URL**, enter the fully qualified domain name (FQDN) of your external VCF Operations orchestrator instance.
   For example, `https://my_Orchestrator_FQDN.com:443`.

   > **Note:**
   >
   > If you have a clustered VCF Operations orchestrator environment, enter the URL of the load balancer.
6. To validate the integration, click **Validate**.
7. Click **Add**.
   The VCF Operations orchestrator integration is saved.
8. To verify that the integration is configured and that the workflows are added, navigate to the **Orchestrate** tab.

The VCF Operations orchestrator instance is ready and you can now enable extensibility with VCF Operations orchestrator for the All Apps organization.

Enable VCF Operations orchestrator for the VCF Automation Organization

To enable extensibility through VCF Operations orchestrator for the All Apps organization, you must publish the Orchestrator rights bundle to that organization.

1. Log in to VCF Automation Provider Management with System Administrator credentials.

2. Select **Administer** > **Access Control** > **Rights Bundles**.
3. Select the **Orchestrator Right Bundle**.
4. On the **Orchestrator Rights Bundle** page, click **Publish**.
5. Select the All Apps organization and click **Save**.
6. Verify that the organization is added to the **Published To** list.

Next, the organization administrator can manage extensibility with VCF Operations orchestrator-related authorizations for users in that organization.

Data Collection for VCF Operations orchestrator Integrations

VCF Operations orchestrator performs periodic data collection for your VCF Operations orchestrator integration.

After you configure VCF Operations orchestrator for your VCF Automation organization, you must wait about 10 minutes for data collection to sync. After that, data collection events for VCF Operations orchestrator integrations are triggered every 10 minutes.

> **Important:**
>
> As you work on VCF Operations orchestrator workflows, verify that you version up the workflow when you are finished editing it. Changes to non-versioned up workflows are not picked up by the data collector.

You can find information about the last data collection performed on a VCF Operations orchestrator integration by navigating to **Administer** > **Connections** > **Integrations** and selecting the VCF Operations orchestrator integration. You can also trigger a manual data collection event by clicking **Start Data Collection**.

Managing a VCF Operations orchestrator Integration in VCF Automation

Provider administrators manage the VCF Operations orchestrator integration in the VCF Automation organization.

Move Custom VCF Operations orchestrator Content between VCF Automation Organizations

As a provider administrator, you can delete an VCF Automation organization, and then delete the VCF Operations orchestrator integration associated with that organization. In that case, the VCF Operations orchestrator instance becomes unusable for other organizations.

When you delete an existing VCF Automation organization, you can move custom VCF Operations orchestrator content to another organization. To move your custom content, you can use one of the following three options.

Option	Steps
Option 1. Reconfigure the authentication provider of the VCF Operations orchestrator instance to point to the new organization.	1. To reconfigure the authentication provider, run the `vracli vro authentication set` command through SSH and replace the organization name added after the `--tenant` property with the name of the new organization. 2. Configure a new VCF Operations orchestrator integration. See Configure VCF Operations Orchestrator for VCF Automation All Apps Organizations. 3. After the integration with the new organization is complete, the organization administrator must log in to VCF Operations orchestrator or the organization UI and validate and update all tasks association with the OAuth tokens. The organization administrator must validate the following content: — Endpoint configurations in the VCF Operations orchestrator plug-in for VCF Automation for All Apps — Create scheduled workflow runs — Create policy runs — Cancel all active user interactions — Update the Dynamic Types plug-in configurations if they reference VCF Automation plug-in objects
Option 2. Save custom content using a remote Git repository.	1. Before deleting the old organization, perform the following steps: a. Verify that your custom content is exported to your remote Git repository. b. Export the Dynamic Types plug-in configurations as a separate package. 2. Configure a new VCF Operations orchestrator integration. See Configure VCF Operations Orchestrator for VCF Automation All Apps Organizations. 3. Register the remote Git repository in your new VCF Operations orchestrator instance. 4. Import the content to VCF Operations orchestrator by performing a PULL operation. 5. Configure the endpoint in the VCF Operations orchestrator plug-in for VCF Automation for All Apps. See Add a VCF Automation for All Apps host. 6. Import the Dynamic Types plug-in configurations package if the package is not saved in the Git server. 7. Recreate all needed content and values: — Content saved in the `SecureString` values — Policy runs — Recreate the Dynamic Types plug-in configurations — Schedule workflow runs

Option	Steps
Option 3. Export the custom content as a VCF Operations orchestrator package and import the package in the new VCF Operations orchestrator instance.	1. Create a package with your custom content and export it. 2. Export the Dynamic Types plug-in configurations as a separate package. 3. Configure a new VCF Operations orchestrator instance. See Configure VCF Operations Orchestrator for VCF Automation All Apps Organizations. 4. Import the Dynamic Types plug-in configurations package. 5. Import the package with custom content. 6. Recreate all needed content and values: – Content saved in the `SecureString` values – Policy runs – Recreate Dynamic Types plug-in configurations – Schedule workflow runs

Move from an Embedded to an External VCF Operations orchestrator Instance in a PCO

By default, an embedded VCF Operations orchestrator instance is integrated in the Provider Consumption Organization (PCO).

In cases where you use heavy automation, however, you might need to fine-tune resources available for VCF Operations orchestrator when resources provided to the embedded instance are not enough in any resource profile.

1. Deploy an external VCF Operations orchestrator instance.
2. Export custom VCF Operations orchestrator content by using option 1 or option 2 from the *Move Custom VCF Operations orchestrator Content between VCF Automation Organizations* section.
3. Configure the external VCF Operations orchestrator authentication for the PCO.
4. Log in to the VCF Automation Provider Management UI as a provider administrator.
5. In the side navigation panel, navigate to **Infrastructure** > **Organizations**.
6. Open the PCO and then click **Launch Organization Portal**.
7. In the PCO UI, click the **Administer** tab.
8. In the side navigation bar, click **Integrations**.
9. Open the VCF Operations orchestrator integration tile.
   a. Change the name of integration to indicate that the embedded VCF Operations orchestrator is no longer used.
   b. Select the newly deployed external VCF Operations orchestrator.
   c. Enter external VCF Operations orchestrator FQDN.
   d. Click **Validate**.
   e. Click **Save**.
10. Wait for the data collection to start or start it manually. Data collection is run automatically every 10 minutes.
11. Validate that extensibility automation is working properly.
12. Click the **Orchestrate** tab. You can now import the custom content using one of the options from the *Move Custom VCF Operations orchestrator Content between VCF Automation Organizations* section.

The embedded VCF Operations orchestrator instance continues to run but is no longer used for workloads.

If there are VM Apps organizations that are using the embedded VCF Operations orchestrator instance, consider replacing it with the same or another external VCF Operations orchestrator instance following the same procedure. VM Apps organizations can continue to share the embedded VCF Operations orchestrator instance for their workloads.

Remove an VCF Operations orchestrator Integration from an Organization

If the organization does not need to integrate with VCF Operations orchestrator, as a provider administrator or an organization administrator, you can remove the integration.

User	Steps
Organization administrator	1. Delete all deployments, resource actions, subscriptions, custom day 2 action definitions, blueprint associations with workflows, and other content associated with VCF Operations orchestrator. 2. Delete the VCF Operations orchestrator remote Git repository.
Provider administrator	1. Log in to the VCF Automation Provider Management U. 2. Navigate to **Access Control** > **Rights Bundle** > **Orchestrator Rights Bundle** 3. Click **Publish**. 4. To unpublish the Orchestrator Rights Bundle for the organization, deselect it from the list. 5. Click **Save**. 6. Navigate to **Organizations**, select the organization, and click **Launch Organization Portal**. 7. In the organization UI, click the **Administer** tab and navigate to **Connections** > **Integrations**. 8. Open the VCF Operations orchestrator integration and delete it.

Integrate VCF Automation with Active Directory

VCF Automation supports integration with Active Directory servers to provide creation of computer accounts in a specified Organizational Unit (OU) within an Active Directory server when provisioning a virtual machine. As an organization administrator, you can add one or more Active Directory endpoints in VCF Automation to centralize authentication, user management, and VM access and provisioning.

- Verify that you configured an LDAP connection to the Active Directory server.
- Verify that you created an organizational unit (OU) on your Active Directory.
- Verify that the Active Directory administrator user configured for the integration has the necessary permissions to create/delete/search for computer objects in the configured OU.

After you register an Active Directory server in VCF Automation, you configure profiles for the integration. An Active Directory profile defines the OU where all virtual machines deployed through the VCF Automation catalog or the VM Service are registered. A profile can be applied to the entire organization or it can be associated with specific projects. If an existing profile is changed, the updated profile is applied to new deployments while previously provisioned VMs remain unaffected.

You can scope only one profile to the entire organization at a time, regardless of the number of Active Directory integrations you have in your VCF Automation organization. The organization-scoped profile is used for all VMs provisioned in the organization. You cannot create a new organization-scoped profile if another one already exists.

If you need more granularity, you can create a project-scoped profile. You can create multiple project-scoped profiles. Each profile can be associated with multiple projects, but each project can be associated with only one profile at a time.

When a user deploys a VM in VCF Automation, the VM is registered in Active Directory OU based on the defined profile. When a VM is deleted in VCF Automation, it is unregistered from Active Directory.

1. Log in to the VCF Automation organization as an organization administrator.
2. Select **Administer** > **Connections** > **Integrations** and then click **Add Integration**.
3. Click **Active Directory**.
4. On the **Summary** tab, configure your integration.
   a) Enter a name and description for the integration.
   b) In the **Credentials and Configuration** section, enter the appropriate LDAP host and port.

      The specified LDAP host is used to call the Active Directory server to validate, register, and unregister computers.
   c) Enter the user name and password for the LDAP server.
   d) Enter the appropriate Base DN that specifies the root for the desired Active Directory resources, for example, `DC=org,DC=mycompany,DC=net`.

      You can specify only one DN per Active Directory integration.
   e) Click **Validate**.
   f) When prompted, trust the Active Directory certificate.

      The certificate is added to the trust store immediately. You can view it on the Certificate Management page.
   g) Click **Save**.

      The Active Directory integration is added.
5. Add an Active Directory profile.

   The profile defines the OU where the virtual machines are registered.

   a) Open the integration again and click the **Profiles** tab.
   b) Click **New**.
   c) Add a name for the Active Directory profile.

      Enter a unique name.
   d) Enter a relative DN, for example, `OU=QA`.

      The relative DN is the subdirectory (OU) that exists within the root domain that you specified in Step 4d.
   e) (Optional) Select the **Allow profile override** check box.

      - If the check box is not selected, the VM is automatically added to the group configured in the profile.
      - If the check box is selected, users can specify a different Active Directory domain in the blueprint, which will override the profile. Provisioned machines are added to the group specified in the blueprint.

      Important:

      Join domain is supported only for Windows machines.
   f) Select the scope.

      - **Organization**. All provisioned VMs use the domain of the organization.
      - **Project**. Add projects to which you want to apply the profile. All VMs provisioned in the selected projects are added to the specified OU.

      VCF Automation checks whether there is an existing organization-scoped profile across all profiles defined for all available Active Directory integrations.
   g) Click **Create**.

      The Active Directory profile is created.

Override an Active Directory profile in a blueprint

Active Directory profiles can be overridden through blueprints where the user can provide an alternative Active Directory domain name and credentials directly in the blueprint.

- If **Allow profile override** is enabled in the profile, users have the following options:

  - Provide an alternative domain name and credentials in the blueprint, which takes precedence over the ones specified in the profile. See Capture a Namespace with a Join Domain Customization in VCF Automation.
  - Prevent a VM from joining any Active Directory domain by adding the following annotation to the YAML of the VM.
    ```
    skip.config.service.vmware.com/ad: <reason>
    ```
- If **Allow profile override** is not enabled in the profile, but the user still provides an alternative domain name and credentials in the blueprint, VM creation fails.
- If there is no Active Directory profile configured, the user can manually define a SysPrep in the blueprint and specify a domain name and administrator credentials.

Terraform Configuration in VCF Automation

As an organization administrator, you can automate the deployment and configuration of VCF Automation infrastructure resources using Terraform.

Terraform is an Infrastructure as Code (IaC) tool developed by HashiCorp. You can use the open-source tool to define and provision your cloud infrastructure using the HashiCorp Configuration Language (HCL). Terraform uses the configuration files with the descriptions of the desired state of your infrastructure to create, manage, and delete the necessary resources across various cloud providers and platforms. Terraform providers are plug-ins that Terraform uses to interact with cloud providers, SaaS providers, third-party tools, and other APIs. Each provider defines its resources and data sources that map directly to the functionalities of the specific service.

For end-to-end operation of Terraform in VCF Automation, you can use three open-source Terraform providers.

- Terraform Provider for VCF Automation - You can use this provider to perform CRUD operations for a subset of the VCF Automation resources that belong to organizations, such as namespaces. This provider is available on GitHub and is distributed as part of the HashiCorp Terraform registry. The greenfield folder contains examples of configuration files for fresh VCF Automation installations that show how all resources interact with each other and provide guidance on how you can use the provider. The folder contains two parts. As an organization administrator, you can use the tenant part to create content libraries and namespaces in your organization.

  Note:

  The provider part is available to provider administrators to create the necessary organizations and configure the underlying infrastructure, such as vCenter instances, NSX instances, provider gateways, roles, and so on.
- Terraform Provider for Kubernetes - You can use the Kubernetes (K8S) provider to perform CRUD operations against the VCF Automation resources exposed through the VCF Automation Kubernetes API layer, such as projects.
  To see a usage example, navigate to `https://<FQDN>/automation/api-docs/#/terraform-provider`.
- Terraform Provider for VMware Aria Automation - You can use this provider to perform CRUD operations for VCF Automation resources that are not yet exposed through the VCF Automation Kubernetes API layers, such as blueprints.

Terraform providers by VCF Automation role

Role	Terraform Provider	Resource
Organization Administrator	Terraform Provider for VCF Automation	• Content library • Namespace
	Terraform Provider for Kubernetes	• VCF services projects • Content library • Virtual Private Cloud • Subnets
	Terraform Provider for VMware Aria Automation	• Blueprint • Catalog
Organization User	Terraform Provider for Kubernetes	Provisioning of IaaS services
	Terraform Provider for VMware Aria Automation	Catalog deployment

Managing Certificates in VCF Automation

You can import, download, edit, and delete certificates from VCF Automation.

Trusted Certificates

You can import certificates of servers that VCF Automation communicates with, such as vCenter, NSX Manager, and so on.

For more information, see Import Trusted Certificates Using Your VCF Automation Provider Management UI.

Certificate library

In the VCF Automation certificates library, you can import certificates used when creating entities that you must secure, such as servers, Edge gateways, and so on.

For more information, see Import Certificates to the Certificates Library Using Your VCF Automation Provider Management UI.

Managing Secrets in VCF Automation

A VCF Automation secret is a reusable, encrypted value that project users may add to their blueprint designs.

Secure access keys and credentials are typical examples of secret properties. Once created and saved, a secret's value can never be unencrypted or read.

Create a secret

1. Log in to VCF Automation as a project administrator.
2. Go to **Administer** > **Certificates** > **Secrets** and click **New**.
3. Enter a unique property name for the secret, without spaces or special characters.
   The name is the visible identifier for the secret.
4. Scope the secret to the entire organization or to specific projects.
   To assign the secret to one or more projects, click **Assign Projects**, select your projects, and click **Add**.
   You can't associate a secret with the same project twice. You also can't associate two secrets with the same name with the entire organization.
5. Enter the secret value.
   When typing, the value is obscured by default, which protects it if the screen is shared.
   If needed, you can click the eye symbol to reveal and verify a value. After it is saved, a secret value becomes encrypted in the database and can never be re-exposed.
6. Optionally, enter a longer description of the secret property.
7. Click `Create`.

Adding a secret to a blueprint

You can add a secret property as a binding in blueprint code.

Note that starting to type `'${secret.characters` reveals a selection list of secrets that have been created for the project.

If two secrets with the same name but different values exist at the project level and at the organization level, the project secret takes precedence. The organization secret is not available for selection in the blueprint.

```
properties:
  name: ourvm
  image: mint20
  flavor: small
  remoteAccess:
    authentication: publicPrivateKey
    sshKey: '${secret.ourPublicKey}'
```

```
username: root
```

Display Full User Names in VCF Automation

By default, users inVCF Automation are identified with user IDs, such as Active Directory SAM Account Names or User Principal Names (UPN). You can expose the personal names (first and last name) of your users on different pages across the organization, such as deployed catalog instances and policies.

To be legally compliant with regulations such as the California Consumer Privacy Act (CCPA), the General Data Protection Regulation (GDPR) and others, you must provide explicit consent to data compliance when you expose the names of users.

If you deactivate the feature, you revoke consent and the names of your users will no longer be displayed in the VCF Automation user interface.

> **Important:**
>
> VCF Automation employs a data at rest policy, which means that storing personal user data is necessary for performance and low latency requirements of the application, so storing the data is considered legally data compliant with or without explicit user consent.

For more information, see the Privacy and Data Protection policies.

1. Log in to the VCF Automation organization as an organization administrator.
2. Go to **Administer** > **Configuration** > **Settings** and click **Show names of users**.
3. Toggle the feature on or off.
4. Click **Save**.

Send Email Notifications to Users in VCF Automation

As an organization administrator, you can configure VCF Automation to send users notifications when specific events in VCF Automation occur.

- Verify that you configured an outbound email server. See Add an email server in VCF Automation to send notifications.
- Verify that the users that need to receive notifications have an email configured. Otherwise, they will not receive notifications.

You can send notifications for several types of events, called scenarios, such as the successful completion of a catalog request or a required approval. Users can view all their notifications in the Notification Center by clicking the bell icon in the VCF Automation header. Notifications are kept in the Notification Center for 30 days.

Email messages are sent to users in the following scenarios.

Scenario	Description
Alert Raised (Notification sent to Organization Administrator)	An alert is generated about an issue with your environment that needs your attention. The message is sent to the organization administrator. Alerts are kept in VCF Automation for 7 days. Alerts are configured in VCF Operations by the provider administrator.
Bill Generated (Notification sent to Organization Administrator)	A bill for your VCF Automation organization is generated. The message is sent to the organization administrator. Bills are configured in VCF Operations by the provider administrator.
Report Generated (Notification sent to Organization Administrator)	A report about your VCF Automation organization is generated. Reports can include information about costs, utilizations, reqion quota, audit-related information, and so on. The message is sent to the organization administrator. Reports are configured in VCF Operations by the provider administrator.

Scenario	Description
Deployment Lease Deletion	A deployment lease expired and the deployment is scheduled for deletion. The message is sent to the deployment owner 15–30 minutes before the deployment is destroyed.
Deployment Lease Expired	A deployment lease expired and the deployment is about to be deleted.
Deployment Lease Expiring	A deployment lease expires soon. The message is sent to the deployment owner three days before the lease expires.
Deployment Request Approved	A request is approved. The message is sent to the user who requested the deployment.
Deployment Request Rejected	A request is rejected. The message is sent to the user who requested the deployment.
Deployment Request Waiting for Approval (Notification sent to requester)	A request awaits approval. The message is sent to the user who requested the deployment.
Pending Approval Request (Notification sent to approver)	A request requires approval. The message is sent to the user who must approve the request.
Pending Work Item (Notification sent to assignees)	A request requires user input to proceed. The message is sent to the user who must provide input.
Pending Work Item (Notification sent to requester)	A request requires user input to proceed. The message is sent to the user who requested the deployment.
Work Item Answered	A request that required user input was answered and the deployment proceeded.
Work Item Rejected	A request that required user input was rejected and the deployment did not proceed.

1. Log in to VCF Automation as an organization administrator.
2. Click the bell icon in the header.
3. Click **View All Notifications**.
4. Click the **Configure Notifications** tab.
5. Next to each event for which you want to trigger user notifications, turn on the toggle.

   All notifications are enabled by default. If you want to disable notifications, turn off the toggle.

Users are subscribed to the notifications that you enabled.

- If a user email is changed in the identity provider, it might take one to two hours before the email address is updated in VCF Automation. In the meantime, notifications might be sent to the old email address.
- Notification recipients are limited to 100 users per notification. For identity provider groups, the limit is 10 groups per notification, where each group consists of a maximum of 500 users. If a group consists of more than 500 users, the notification is only sent to 500 users, and not to the rest of the users.

Customize Email Notifications to Users in VCF Automation

As an organization administrator, you can customize the email notifications that the users in your organization receive about events in VCF Automation.

You can change the text and the branding elements of the email, such as the logo, so that it matches the requirements of your organization.

Customize the Branding of Email Notifications

As an organization administrator, you can customize the way that email notifications look.

You can change the background color, edit the text, and add a new logo in the header and the footer of email notifications. Branding is applied across all notification templates, except for notifications for alerts, bills, and reports. If you want to change the branding for alert, bill, and report notifications, you must customize each one separately.

In this scenario, you want to replace the default logo and company name in the notifications your users receive with the logo of your company.

1. Log in to VCF Automation as an organization administrator.
2. Click the bell icon in the header.
3. Click **View All Notifications**.
4. Click the **Configure Notifications** tab.
5. Click **Edit shared branding**.
6. On the **Edit header** tab, enter the URL of your logo and the name of your organization.
7. To review you changes, click the **Preview** tab.
8. Click **Save**.

Customize the Subject and Body of Email Notifications

You can edit the subject line and the message body of each notification template and use the deployment attributes that are available for the scenario.

As an organization administrator, you want to craft an email notification that informs your developers when a cloud service is provisioned and provides details about the deployment.

In this scenario, you add the deployment name to the subject of an email notification.

1. Log in to VCF Automation as an organization administrator.
2. Click the bell icon in the header.
3. Click **View All Notifications**.
4. Click the **Configure Notifications** tab.
5. Click the arrow on the left of the notification scenario you want to customize.
   A preview of the notification email appears on the right.
6. Click **Customize**.
7. On the **Edit** tab, update the subject line and the message body of the notification as necessary.
   a. Click **Scenario attributes** > **properties**.
      The available attributes are specific to each notification scenario.
   b. Click the copy icon next to the name(string)property.
   c. Paste the property in the **Subject** field.
   You can also add the dynamic attributes in the message body. Using custom properties is not supported.
8. To review your changes, click the **Preview** tab.
   You see the placeholder text for the deployment name in the subject.
9. Click **Save**.
10. Test the updated notification.
    a. Click **Send test**.
    b. Enter an email to send the test email to.
    c. Click **Send** test.

Add an Email Server in VCF Automation to Send Notifications

As an organization administrator, you configure an email server if you want to send messages to users about events in VCF Automation. The messages are a courtesy that improves the experience of your consumers.

Verify that you know the credentials required to configure the email server. You must provide the server name and an email account that you want to be the message sender. If your email server requires authentication, you must also provide the user name and password.

This email server is for outbound messages only.

Email messages are sent to users in the following scenarios.

- A deployment lease expires soon. The message is sent to the deployment owner three days before the lease expires.
- A deployment lease expired and the deployment is about to be deleted. The message is sent to the deployment owner 15–30 minutes before it is destroyed.

1. Log in to VCF Automation as an organization administrator.
2. Click the bell icon in the header.
3. Click **View All Notifications**.
4. Click the **Email Server** tab.
5. Enter the information for each setting.

   If you need assistance on a particular setting, consult the signpost help.
6. To verify the configured settings, click **Test Connection**.
7. Click **Create**.

As the administrator, monitor the leases to ensure that the messages are sent to the deployment owners at the correct time.

Working with Audit Logs in VCF Automation

To discover status information about reportable VCF Automation actions, use audit logs.

VCF Automation creates an audit log when you perform a reportable action, such as when you deploy a blueprint or catalog item, perform a Day 2 action on a resource or deployment, or create, delete, or update a property group. Audit logs display a status value or a message for each reportable action type, depending on the action. They can also contain information about the action, such as a timestamp, message, user name, project name, and trace ID value.

Audit logging captures information for the following reportable actions:

- Catalog deployment operations
- Blueprint operations
- IaaS resource operations
- Day 2 deployment operations
- Day 2 resource action operations

  Each of the above four action types creates an initial audit log with a **Submitted** status. Based on how the action is completed, an additional log is created that displays a **Successful**, **Failed**, or **Canceled** status
- Create, delete, and update property group operations
- Create, delete, and update product feature toggle operations
- Create, delete, and update policy operations in VCF Automation - such as Day 2 actions, lease, and approval policy operations
- Create, delete, and update event subscription operations

  Each of the above four action types creates a message in the audit log display screen.

Searching, displaying, and exporting an audit log

You can filter your search for, and display of, audit logs. You can also output audit logs in CSV format.

To display audit logs, use the following procedure.

- Log in to VCF Automation as an organization administrator.
- Select the **Administer** tab and click **Activity** > **Audit Log**.
- Open and review the audit log or logs of interest.

Working with Metadata for VCF Automation Resources

As an organization administrator, you can add metadata for namespaces and virtual machines in your VCF Automation organization.

You use metadata to tag, categorize, and automate the management of VCF Automation resources, such as billing, power operations, and workflow automation.

> **Note:**
>
> Metadata is supported for namespaces and virtual machines only.

Organization resources that you manage in your organization do not incorporate provider-managed metadata. Any metadata applied to these resources by the provider administrator in the VCF Automation Provider Management is visible to you, but you cannot modify it.

In VCF Automation, most resources are represented as Kubernetes custom resources (CRs). The metadata is specified using the annotations of the custom resource as a string value. By using metadata, you can associate user-defined key=value pairs with an organization. You can use metadata in VCF Automation API query filter expressions.

You cannot edit metadata values directly. Instead, you must remove the old value and then create a new one.

> **Note:**
>
> You can edit annotations under the `vmware.com` or `*.vmware.com` domains only using the VCF Automation API.

1. Log in to VCF Automation as an organization administrator.
2. Select **Manage & Govern** > **Namespaces**.
3. Click the namespace name to open it.
4. In the **Annotation** section, click **Edit**.
5. Add the metadata and click **Add**.
6. Click **Save**.

Managing Organizations for VM Apps in VMware Cloud Foundation Automation

You use VMware Cloud Foundation Automation for VM Apps to connect to your public and private cloud providers so that you can deploy machines, applications, and services that you create to those resources. You and your teams develop cloud-templates-as-code in an environment that supports an iterative workflow, from development to testing to production. At provisioning time, you can deploy across a range of cloud vendors.

VCF Automation for VM Apps is a blueprint development and deployment service. You and your teams use the service to deploy machines, applications, and services to your cloud vendor resources.

How do you set up a VCF Automation for VM Apps organization

Before you begin, your service provider administrator creates the organization in the VCF Automation Provider Management UI. Only one new VM Apps organization can be created from the VCF Automation Provider Management UI. Additional VM Apps organizations can be created using the VCF Automation API.

For more information about creating organization in the VCF Automation Provider Management UI, see Overview of VCF Automation Provider Management.

As an organization administrator in VCF Automation for VM Apps, generally referred to as a cloud administrator, you set up the provisioning infrastructure and create the projects that group users and resources. The system administrator does not allocate infrastructure to your organization.

1. Add your cloud vendor accounts.
2. Determine which regions or datastores are the cloud zones that you want your developers deploying to.

3. Create projects that group the developers with the cloud zones.
4. Create policies that define the cloud zones.

As a VCF Automation for VM Apps administrator, you must understand the user roles and set up connections with your cloud account vendor and integration applications. When you configure the cloud accounts and integrations, you are configuring the communication between VCF Automation and those target systems.

VCF Automation also provides a single point where you can request and manage catalog items. The catalog is the simplified user interface that cloud administrators make available to users when the administrator's teams do not need full access to developing and building the templates. You use the catalog to deploy templates to cloud regions or datastores associated with projects.

As a cloud administrator, you create catalog items by importing released VCF Automation templates and Amazon Web Services CloudFormation templates that your users can deploy to your cloud vendor regions or datastores. Your users can then request and monitor the provisioning process. After deployment, users can manage the deployed catalog items throughout the deployment lifecycle.

How to use VCF Automation for VM Apps user interface

An overview of VCF Automation for VM Apps includes the following basic functions.

- The **Home** tab shows a summary of your resources, deployments, and other inventory currently managed by VCF Automation, as well as pending notifications and action items. This tab is available to administrators only.
- The **Consume** tab shows the current status of your provisioned, discovered, onboarded, and other resources. You can access resource details and day 2 actions that you use to manage your resources.
- The **Design** tab is your development home. You use the canvas and the YAML editor to develop and then deploy your machines and applications.
- The **Infrastructure** tab is where you add and organize your cloud vendor resources and users. This tab also provides information about deployed blueprints.
- The **Content and Policies** tab is where you import content to the catalog and create policies to manage deployments.
- The **Extensibility** tab is where you can extend and automate your application life cycles. You can subscribe to events that are used to trigger extensibility actions or VCF Operations orchestrator workflows.
- The **Orchestrator** tab is where you access your VCF Operations orchestrator integration. You use VCF Operations orchestrator to build and run workflows that extend automation capabilities.
- An **Alerts** tab provides notifications regarding capacity, performance, and availability for your infrastructure resources. You must have a configured integration with VCF Operations to see and use the alerts.
- The **Inbox** tab is where you approve catalog requests and respond to deployments that request user input.

Getting Started with Organizations in VCF Automation for VM Apps

As an organization administrator VCF Automation for VM Apps, you must understand the user roles and set up connections with your cloud account vendor and integration applications. When you configure the cloud accounts and integrations, you are configuring the communication between VCF Automation and those target systems.

You configure your cloud infrastructure to support template development and deployment. The infrastructure begins with cloud vendors, then you add VCF Automation users as project members and link them to the cloud account regions as projects. At this point, you can continue to develop templates, or you can turn over development to the project administrators and members.

As a project member, use VCF Automation as you iteratively develop and deploy templates, until you have a production-worthy product. The deployment locations are configured by your cloud administrator as part of the infrastructure. The administrator has the best understanding of your organizations resources and budget.

Get started with VCF Automation for VM Apps using the VCF Automation Launchpad

As a cloud administrator using VCF Automation for VM Apps for the first time, you can use the Launchpad in VCF Automation to quickly set up a cloud account, add items to the VCF Automation catalog, and apply a lease policy to the catalog items. You can then use the dashboard on the Home tab to review usage and consumption details about your organization.

What does the Launchpad do

Using the Launchpad, you do the following tasks in VCF Automation.

- Add a vCenter cloud account. Cloud accounts are the credentials that are used to collect data from and deploy resources to your vCenter instance.
- Share content in the VCF Automation catalog.
- Create a lease policy. Lease policies control how long a deployment is active.

When you run the Launchpad for the first time, you start by adding a cloud account. You can then perform the procedures as needed and in any order.

Once you complete the Launchpad, you can continue to use it whenever you need to add a cloud account, add catalog items, or create a lease policy quickly. If you want to revisit any of the procedures, they also appear as quick links at the top of the Home page.

> **Note:**
> If you want to permanently hide the Launchpad, click Mark as done.

How do I use the Home page dashboard

You use the Home page dashboard to view a summary of your VMs, deployments, and other inventory, such as cloud accounts, cloud zones, and projects, that are currently managed by VCF Automation.

Clicking the numbers in the dashboard widgets takes you to the respective pages in VCF Automation. For example, click the VMs count in the Cloud Accounts & VMs widget to view a full list of discovered virtual machines.

You can use the Notifications widget to track and manage approval requests for deployments or day 2 actions, deployment status updates, and expiring deployment leases. For more information about notifications, see *Send email notifications to Automation Service Broker users*.

Add a vCenter cloud account

1. Navigate to the **Home** tab in VCF Automation.
2. On the **Add Cloud Account** card, click **Add cloud account**.
3. Enter a name and description for the cloud account.
4. Provide credentials.
   a. Enter the vCenter server host IP address or fully-qualified domain name.
   b. Enter your vCenter server administrator username and password.
5. Click **Validate**.
6. If you need to add tags to support a tagging strategy, enter capability tags.
   You can use tags to distinguish between multiple vCenter cloud accounts when designing a template.
7. Click **Add**.

You added a vCenter cloud account. To view your cloud accounts, click the green banner in the Launchpad.

Publish content to the VCF Automation catalog

1. Click the **Publish to Catalog** card.
2. Select a cloud account.
3. Select the VM templates that you want to make available to your users.
   You can select up to 10 VM templates at a time. You can select from VM templates that are associated with the cloud account.
4. Select a project.
   If no projects exist, you can create one in this step.
   The project links your users with cloud account regions, so that they can deploy application templates with networks and storage resources to your vCenter instance.
5. (Optional) Set a preferred network and storage for deploying catalog items.

If you don't set a preference, catalog items are deployed to the network and storage profile that are configured for the cloud zone. If no network and storage profile exist in the cloud zone, then the catalog items are deployed at random.

The selected network and storage must be in the same cluster. Otherwise, the VM template will not provision.

You added VM templates to the VCF Automation catalog where users from your project can request them. To view the catalog, click the green banner in the Launchpad.

Create a lease expiration policy for deployments

1. Click the **Apply Lease Expiration** card.
2. Enter a name and description for the lease policy.
3. Define the lease.
   - **Maximum lease** is the number of days that the deployment resources are active without being renewed. If they are not renewed, the lease expires and the deployment is destroyed. If a grace period is specified, the user can renew the lease for up to the same number of days that the lease has been active.
   - **Grace period** is the number of days the user has to renew an expired lease before the deployment is destroyed. If you don't define a grace period, it defaults to 1 day.
4. Select a project.

   If no projects exist, you can create one in this step. The project links your users with cloud account regions, so that they can deploy application templates with networks and storage resources to your vCenter instance.

   The deployments that are associated with the specified project are managed based on the defined lease. Other projects are not affected.
5. To preview which resources in the selected project the lease policy will impact, click **Preview**.
6. Click **Create**.

The lease policy is applied to deployments. You can click the green banner to access your lease policy.

What are the VCF Automation for VM Apps user roles

VCF Automation users must be assigned roles. Roles define access to features within the application. When a VCF Automation for VM Apps organization is configured by the provider, an Organization Owner is assigned. All other VCF Automation roles are assigned by the Organization Owner.

After users and groups are imported or created in VCF Automation, you can manage the role assignments for single organizations using the **Access Control** page.

VCF Automation has several levels of user roles. These different level control access to the organization, the projects that produce or consume the cloud templates, catalog items, and the ability for uses to use or see individual parts of the user interface. These different levels give cloud administrators different tools to apply any level of granularity that is required by their operational needs.

There are three types of roles in VCF Automation: organization roles, project roles, and custom roles. For organizational roles, user-level roles can typically use resources whereas admin-level roles are required to create and configure resources. Organizational roles define permissions within the tenant; organizational owners have admin-level permissions while organizational members have user-level permissions. Organization owners can add and manage other users.

Organizational roles

Organizational roles define permissions within the organization; organizational owners have admin-level permissions while organizational members have user-level permissions. Organization owners can add and manage other users. The organization roles are global and apply to all services in the organization.

Role	General permissions
Organization Owner	Can access the organization UI and add users to organization. The organization owner cannot access a service unless they have a service role.
Organization Member	Can access the organization UI. The organization member cannot access a service unless they have a service role.
Assembler Administrator	A user who has read and write access to the entire user interface and API resources. This is the only user role that can see and do everything, including add cloud accounts, create new projects, and assign a project administrator.
Service Broker Administrator	A user who has read and write access to the entire user interface and API resources. This is the only user role that can perform all tasks, including creating a new project and assigning a project administrator.
Orchestrator Administrator	A user who has full view, update, and delete privileges in the VCF Operations orchestrator integration. An administrator can also access the content created by specific groups.
Assembler User	A user who does not have the Assembler Administrator role. In a VCF Automation project, the administrator adds users to projects as project members, administrators, or viewers. The administrator can also add a project administrator.
Service Broker User	Any user who does not have the Service Broker Administrator role. In a project, the administrator adds users to projects as project members, administrators, or viewers. The administrator can also add a project administrator.
Orchestrator Workflow Designer	A user who can create, run, edit, and delete their own VCF Operations orchestrator content. They can add their own content to their assigned group. The workflow designer does not have access to the administration and troubleshooting features of the VCF Operations orchestrator.
Assembler Viewer	A user who has read access to see information but cannot create, update, or delete values. This is a read-only role across all projects. Users with the viewer role can see all the information that is available to the administrator. They cannot take any action unless you make them a project administrator or a project member. If the user is affiliated with a project, they have the permissions related to the role. The project viewer would not extend their permissions the way that the administrator or member role does.
Service Broker Viewer	A user who has read access to see information but cannot create, update, or delete values. This is a read-only role across all projects. Users with the viewer role can see all the information that is available to the administrator. They cannot take any action unless you make them a project administrator or a project member. If the user is affiliated with a project, they have the permissions related to the role. The project viewer would not extend their permissions the way that the administrator or member role does.

Role	General permissions
Orchestrator Viewer	A user who has read access to see VCF Operations orchestrator features and content, including all groups and group content, but cannot create, update, run, delete values, or export content. This is a read-only role across all projects in all the services.

Project roles

In addition to the organization roles, VCF Automation has project roles. The project roles can vary between projects.

In the following table, which tells you what the project roles can see and do, remember that the service administrators have full permission on all areas of the user interface.

The descriptions of project roles will help you decide what permissions to give your users.

For more information about project role permissions, see *Project roles in VCF Automation for VM Apps*.

Role	General permissions
Project Administrator	Project administrators leverage the infrastructure that is created by the service administrator to ensure that their project members have the resources they need for their development work.
Project Member	Project members work within their projects to design and deploy cloud templates. Your projects can include only resources that you own or resources that are shared with other project members.
Project Viewer	Project viewers are restricted to read-only access, except in a few cases where they can do non-destructive things like download cloud templates.
Project Supervisor	Project supervisors are approvers for their projects where an approval policy is defined with a project supervisor approver. To provide the supervisor with context for approvals, consider also granting them the project member or viewer role.

Custom roles

The permissions are defined by the VCF Automation organization administrator.

The user must have at least a service viewer role so that they can access the user interface. The custom roles take precedence over the service roles.

For more information about custom roles, see Custom user roles in VCF Automation for VM Apps.

Organization and Project Role Permissions in VCF Automation for VM Apps

As a organization owner, you can assign users organization and project roles in VCF Automation. The roles determine what users can do or see. To determine the role that you want to assign, evaluate the tasks in the following tables.

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
Access VCF Automation							
Console	Access the VCF Automation organization UI	Yes	Yes	Yes	Yes	Yes	Yes

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
Infrastructure							
	See and open the Infrastructure tab	Yes	Yes	Yes	Yes	Yes	Yes
Administration - Projects	Create projects	Yes					
	Update, or delete values from project summary, provisioning, Kubernetes, integrations, and test project configurations.	Yes					
	Add users and groups, and assign roles in projects.	Yes		Yes. Your projects. Only via API.			
	View projects	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	Yes. Your projects
Administration - Users and Groups	View the users and groups assigned to custom roles.	Yes					
Administration - Custom Roles	Create custom user roles and assign them to users and groups.	Yes					
Administration - Custom Names	Create custom resource names.	Yes					
Administration - Secrets	Create and delete secret reusable properties.	Yes					
Administration - Settings	Turn on or off internal settings.	Yes					
Configure - Cloud Zones	Create, update, or delete cloud zones	Yes					
	View cloud zones	Yes	Yes				
	View cloud zone Insights dashboard	Yes	Yes				
	View cloud zones alerts	Yes	Yes				

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
Configure - Kubernetes Zones	Create, update, or delete Kubernetes zones	Yes					
	View Kubernetes zones	Yes	Yes				
Configure - Flavors	Create, update, or delete flavors	Yes					
	View flavors	Yes	Yes				
Configure - Image Mappings	Create, update, or delete image mappings	Yes					
	View image mappings	Yes	Yes				
Configure - Network Profiles	Create, update, or delete network profiles	Yes					
	View image network profiles	Yes	Yes				
Configure - Storage Profiles	Create, update, or delete storage profiles	Yes					
	View image storage profiles	Yes	Yes				
Configure - Pricing Cards	Create, update, or delete pricing cards	Yes					
	View the pricing cards	Yes	Yes				
Configure - Tags	Create, update, or delete tags	Yes					
	View tags	Yes	Yes				
Resources - Compute	Add tags to discovered compute resources	Yes					
	View discovered compute resources	Yes	Yes				
Resources - Networks	Modify network tags, IP ranges, IP addresses	Yes					

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
	View discovered network resources	Yes	Yes				
Resources - Security	Add tags to discovered security groups	Yes					
	View discovered security groups	Yes	Yes				
Resources - Storage	Add tags to discovered storage	Yes					
	View storage	Yes	Yes				
Resources - Kubernetes	Deploy or add Kubernetes clusters, and create or add namespaces	Yes					
	View Kubernetes clusters and namespaces	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
Activity - Requests	Delete deployment request records	Yes					
	View deployment request records	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
Activity - Event Logs	View event logs	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
Connections - Cloud Accounts	Create, update, or delete cloud accounts	Yes					
	View cloud accounts	Yes	Yes				
Connections - Integrations	Create, update, or delete integrations	Yes					
	View integrations	Yes	Yes				
Onboarding	Create, update, or delete onboarding plans	Yes					
	View onboarding plans	Yes				Yes. Your projects	

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
Extensibility							
	See and open the Extensibility tab	Yes	Yes			Yes	
Events	View extensibility events	Yes	Yes				
Subscriptions	Create, update, or delete extensibility subscriptions	Yes					
	Deactivate subscriptions	Yes					
	View subscriptions	Yes	Yes				
Library - Event topics	View event topics	Yes	Yes				
Library - Actions	Create, update, or delete extensibility actions	Yes					
	View extensibility actions	Yes	Yes				
Library - Workflows	View extensibility workflows	Yes	Yes				
Activity - Action Runs	Cancel or delete extensibility action runs	Yes					
	View extensibility action runs	Yes	Yes			Yes. Your projects	
Activity - Workflow Runs	View extensibility workflow runs	Yes	Yes				
Design							
Design	Open the Design tab	Yes	Yes	Yes.	Yes.	Yes.	Yes
Cloud Templates	Create, update, and delete cloud templates	Yes		Yes. Your projects	Yes. Your projects		
	View cloud templates	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
	Download cloud templates	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
	Upload cloud templates	Yes		Yes. Your projects	Yes. Your projects		
	Deploy cloud templates	Yes		Yes. Your projects	Yes. Your projects		
	Version and restore cloud templates	Yes		Yes. Your projects	Yes. Your projects		
	Release cloud templates to the catalog	Yes		Yes. Your projects	Yes. Your projects		
Custom Resources	Create, update or delete custom resources	Yes					
	View custom resources	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
Custom Actions	Create, update, or delete custom actions	Yes					
	View custom actions	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
Content and Policies							
	See and open the Content and Policies tab	Yes	Yes				
Content Sources	Create, update, or delete content sources	Yes					
	View content sources	Yes	Yes				
Content	Customize form and configure item	Yes					
	View content	Yes	Yes				
Policies - Definitions	Create, update, or delete policy definitions	Yes					
	View policy definitions	Yes	Yes				
Policies - Enforcement	View enforcement log	Yes	Yes				
Notifications - Email Server	Configure an email server	Yes					
Consume							

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
	See and open the Consume tab	Yes	Yes	Yes	Yes	Yes	Yes
Projects	See and search projects	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	Yes. Your projects	Yes. Your projects
Catalog	See and open the Catalog page	Yes	Yes	Yes	Yes	Yes	Yes
	View available catalog items	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
	Request a catalog item	Yes		Yes. Your projects	Yes. Your projects		
Deployments	View deployments including deployment details, deployment history, price, monitor, alerts, optimize, and troubleshooting information	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	
	Manage alerts	Yes		Yes. Your projects	Yes. your projects		
	Run day 2 actions on deployments based on policies	Yes		Yes. Your projects	Yes. Your projects		
Deployments - Resources	View all discovered resources	Yes	Yes				
	Run day 2 actions on discovered resources. Actions available only on machines and limited to power on and off for all machines, and remote console for vSphere machines.	Yes					
Deployments - All Resources	View deployed, onboarded, migrated resources	Yes	Yes	Yes. Your projects.	Yes. Your projects.	Yes. Your projects.	

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
	Run Day 2 actions on deployed, onboarded, and migrated resources based on policies	Yes	Yes	Yes. Your projects.	Yes. Your projects.		
Deployments - Virtual Machines	View discovered machines	Yes	Yes				
	Run day 2 actions on discovered machines. Actions are limited to power on and off, and remote console for vSphere machines.	Yes					

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
	Create New VM This option is available to administrators. However, if an administrator turns on the setting, then it is available to the other users roles. To activate the option, select **Infrastructure** > **Administration** > **Settings** and turn on **Create new resource**. By activating the option, VCF Automation users can create VMs based on any image and any flavor even though they are not administrators themselves. To avoid the potential overconsumption of resources, administrators can create approval policies to reject or approve any deployment requests based on the image used or the flavor or size requested.	Yes	Yes. Your projects.	Yes. Your projects.	Yes. Your projects.		
	View deployed, onboarded, and migrated resources.	Yes		Yes. Your projects.	Yes. Your projects.	Yes. Your projects.	

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
	Run day 2 actions on deployed, onboarded, and migrated resources based on policies	Yes		Yes. Your projects.	Yes. Your projects.		
Deployments - Volumes	View discovered volumes	Yes	Yes				
	No day 2 actions available						
	View deployed, onboarded, and migrated volumes	Yes	Yes	Yes. Your projects.	Yes. Your projects.	Yes. Your projects.	
	Run day 2 actions on deployed, onboarded, and migrated volumes based on policies	Yes		Yes. Your projects.	Yes. Your projects.		
Deployments - Networking and Security	View discovered networks, load balancers, and security groups	Yes	Yes				
	No day 2 actions available						
	View deployed, onboarded, and migrated networks, load balancers, and security groups	Yes	Yes	Yes. Your projects.	Yes. Your projects.	Yes. Your projects.	
	Run day 2 actions on deployed, onboarded, and migrated networks, load balancers, and security groups based on policies	Yes		Yes. Your projects.	Yes. Your projects.		
Inbox							
	See and open the Inbox tab	Yes	Yes				

UI Context	Task	Assembler/ Service Broker Administrator	Assembler/ Service Broker Viewer	Assembler/Service Broker User User must be a project administrator or member to see and do project-related tasks.			
				Project Administrator	Project Member	Project Viewer	Project Supervisor
Approvals	View approval requests	Yes	Yes	Yes	Yes	Yes	Yes
	Respond to approval requests	Yes		Yes. Your projects and the policy approver is Project Administrator	Only if you are a named approver	Only if you are a named approver	Yes. Your projects and the policy approver is Project Supervisor
User Input Requests	View user input requests	Yes	Yes	Yes	Yes		
	Respond to user input requests	Only if you are assigned to provide input	Only if you are assigned to provide input	Only if you are assigned to provide input	Only if you are assigned to provide input	Only if you are assigned to provide input	Only if you are assigned to provide input
Alerts							
	See and open the Alerts tab	Yes	Yes	Yes	Yes	Yes	
	Manage alerts	Yes		Yes. Your projects	Yes. Your projects		
	View alerts	Yes	Yes	Yes. Your projects	Yes. Your projects	Yes. Your projects	

Custom user roles in VCF Automation for VM Apps

As a VCF Automation administrator, you can create custom roles that define what users can see and do in VCF Automation. You can then assign users to those roles.

Custom user role permissions

Using VCF Automation, you can define more granular user roles and then assign users to those roles. The custom roles have two categories, view and manage.

- View. A user assigned to a role with this permission can see all the items for all projects in the selected sections of the user interface. This role is useful for users who need to see accounts, configurations, or assigned values.
- Manage. A user assigned to a role with this permission can see all the items and has full add, edit, and delete permissions for all projects in the selected sections of the user interface.

These permissions extend the privileges that are granted by the other roles and are not restricted by project membership. For example, you can expand a project administrator's permissions to manage parts of the infrastructure or give a service viewer an ability to review and respond to approvals requests.

How do I create custom user roles

1. Select **Infrastructure** > **Administration** > **Custom Roles**.
2. Click **New** and enter a unique **Name** that you can identify when you assign users to the role.
3. Select the check boxes that correspond to the permissions you want the users to have over the resources.
4. Click **Create**.
5. In the list, click the custom role name and click **Assign**.
6. Add the users or groups that you want to have this role and click **Add**.

How do I determine what custom roles the users have

To manage the users with the custom roles, you can review the users and groups.

1. Select **Infrastructure** > **Administration** > **Users and Groups**.
2. Review the Custom Roles column to locate users with the role.
3. To add or remove roles for a user, click the user's name and then modify the custom role assignments.

Changes in role assignments or permissions can take up to five minutes to take effect. Users might need to log out and then log back in so the change takes effect immediately.

Custom Role Descriptions

In most cases the role description is provided in the user interface. However, there are some extended descriptions provided in the following table.

User Interface	Permission	Description
Infrastructure		
	View Cloud Accounts.	View cloud accounts.
	Manage Cloud Accounts	Create, update, or delete cloud accounts.
	View Image Mappings	View image mappings.
	Manage Image Mappings	Create, update, or delete image mappings.
	View Flavor Mappings	View flavor mappings.
	Manage Flavor Mappings	Create, update, or delete flavor mappings.
	View Cloud Zones	View cloud zones, Insights, and alerts.
	Manage Cloud Zones	Create, update, or delete cloud zones. Manage alerts.
	View Requests	View activity requests.
	Manage Requests	Delete requests from the list.
	View Integrations	View integrations.
	Manage Integrations	Create, update, or delete integrations.
	View Projects	View projects.
	Manage Projects	Create projects. Add users and assign roles in projects. Update, or delete values from project summary, users, provisioning, Kubernetes, integrations, and test project configurations.
	View Onboarding Plans	View onboarding plans
	Manage Onboarding Plans	Create, update, run, or delete onboarding plans
Catalog		
	View Content	
	Manage Content	Add, update, delete content sources. Customize the content, including the catalog icons and request forms.
Policies		
	View Policies	View policy definitions.
	Manage Policies	Create, update, or delete policy definitions.
Deployments		
	View Deployments	View all deployments, including deployment details, deployment history, alerts, and troubleshooting information.

User Interface	Permission	Description
	Manage Deployments	View all deployments, respond to alerts, and run all day 2 actions that the day 2 policies allow an administrator to run on deployments and deployment components.
Cloud Templates		
	View Cloud Templates	View cloud templates.
	Manage Cloud Templates	Create, update, test, delete, version, share cloud templates, and release/unrelease a cloud template version.
	Edit Cloud Templates	Create, update, test, version, share cloud templates, and release/unrelease a cloud template version. The role does not have permission to delete cloud templates.
	Deploy Cloud Templates	Test and deploy any cloud template in any project.
	Deploy In-line Cloud Template Content	Deploy any cloud template in the projects that the assignees are associated with. The project roles can be administrator, member, or viewer.
	View property groups	View all property groups for all projects.
	Manage property groups	Create, update, and delete property groups in any project.
	View Custom Resources	View custom resources.
	Manage Custom Resources	Create, update or delete custom resources.
	View Resource Actions	View custom actions.
	Manage Resource Actions	Create, update, or delete custom actions
Extensibility		
	View Extensibility Resources	View events, subscriptions, event topics, actions, workflows, action runs, and workflow runs.
	Manage Extensibility Resources	Create, update, delete, and deactivate extensibility subscriptions. Create, update, or delete extensibility actions. Cancel or delete extensibility action runs.
Approvals		
	Manage Approvals	View the Approvals tab where you can approve or reject approval requests. Approver with this role will not receive an email notification about an approval request unless they are an approver in the policy.

Managing Identity Providers in VCF Automation for VM Apps

As an organization administrator, you can use one or more external identity providers to import and manage users and groups in your organization. You can configure an LDAP server connection, a SAML integration, and an OpenID Connect (OIDC) integration.

Important:

If you upgraded your VMware Aria Automation 8.x environment to VCF Automation 9, then vIDB is already configured as an identity provider.

If you are setting up a new VCF Automation 9 deployment, you must configure an identity provider for your organization before you can add users and user groups.

LDAP connections

Navigate to **Administer** > **Identity Profiver** > **LDAP** and click **Configure**. Select one of the following configuration options:

- Do not use LDAP. Use this option if you want to add VCF Automation users who are private to the organization. If you select this option, you cannot create or import user groups.
- VCF Automation system LDAP service. Use when this organization is a member of your Cloud provider company. Enter Distinguished name for the organization.
- Custom LDAP service. Use when you have arranged with an organization to use their directory service. Before you can use this option, you must configure your Cloud system LDAP service to link to their LDAP service.

To configure LDAP for your VM Apps organization, follow the procedure described in Configure an Organization LDAP Connection in Your VCF Automation in the VCF Automation Provider Management Guide.

SAML connections

To configure SAML for your VM Apps organization, follow the procedure described in Configure Your VCF Automation System to Use a SAML Identity Provider in the VCF Automation Provider Management Guide.

OIDC connections

To configure OIDC for your VM Apps organization, follow the procedure described in Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management UI in the VCF Automation Provider Management Guide.

Enable user groups in VCF Automation for VM Apps for projects

If a group is not available on the **Add Groups** page when you are adding users to projects, check the **Identity Providers** page and add the group if it is available.

- Verify that you configured an identity provider for the organization. See Managing Identity Providers in VCF Automation for VM Apps.
- If the groups are not synchronized, they are not available when you try to add them to a project. Verify that you synchronized the groups from your identity provider with your VMware Aria Suite Lifecycle instance.

To add members of a user group to a project, you must ensure that the group is synchronized with your instance and that the group is added to the organization.

1. In VCF Automation, click **Access Control** in the header right navigation.
2. Click **Groups**, and then click **Import Groups**.
3. Select an identity provider.
4. Use the search function to find the group that you are adding and select it.
5. Assign an organization role.

   At a minimum, the group must have an Organization Member role.
6. Click **Save**.

You can now add the user group to a project.

Learn more about user roles in VCF Automation for VM Apps

As a cloud administrator, you want to control the tasks that your users can perform in VCF Automation. Depending on your management goals and application development team responsibilities, there are different ways that you can configure the user roles to support those goals.

Edit User Roles in VCF Automation for VM Apps

You can edit roles assigned to users that have been imported into VCF Automation.

1. In VCF Automation, click **Infrastructure** > **Administration** > **Access Control**.
2. Select the vertical ellipsis menu next to the desired user on the **Users** tab and click **Edit**.
3. You can edit the organization roles for the user.
   - Select the drop down next to **Assigned Roles** to change the user's relationship with the organization.
   - To remove user roles, click the **X** next to the applicable service.
4. Click **Save**.

The user role assignment is updated as specified.

Edit Group Role Assignments in VCF Automation for VM Apps

You can edit role assignments for groups in VCF Automation.

Users and groups have been imported from a valid Workspace ONE Access instance that is associated with your VCF Automation deployment.

1. In VCF Automation, click **Infrastructure** > **Administration** > **Access Control**.
2. Select the vertical ellipsis menu next to the desired user on the **Groups** tab and click **Edit**.
3. Edit the role assignments for the selected group.
4. Click **Save**.

Role assignments are updated as specified.

Display Full Names of Users in VCF Automation for VM Apps

By default, users in VCF Automation are identified with user IDs, such as Active Directory SAM Account Names or User Principal Names (UPN). You can expose the personal names (first and last name) of your users on different pages across the organization, such as Resources, Deployments, and Policies.

To be legally compliant with regulations such as the California Consumer Privacy Act (CCPA), the General Data Protection Regulation (GDPR) and others, you must provide explicit consent to data compliance when you expose the names of users.

If you deactivate the feature, you revoke consent and the names of your users will no longer be displayed in the VCF Automation user interface.

> **Important:**
>
> VCF Automation employs a data at rest policy, which means that storing personal user data is necessary for performance and low latency requirements of the application, so storing the data is considered legally data compliant with or without explicit user consent.

For more information, see the Privacy and Data Protection policies.

1. Log in as an administrator.
2. Go to **Infrastructure** > **Settings**, and click **Show names of users**.
3. Toggle the feature on or off.
4. Click **Save**.

Enable Department of Defense Notice and Consent Banner in VCF Automation for VM Apps

For some government customers who upgraded from VMware Aria Automation 8.x to VCF Automation 9, an administrator must configure the standard Department of Defense (DoD) notice and consent banner in Workspace ONE Access to allow users to access VCF Automation.

The Standard Mandatory DoD Notice and Consent Banner text is as follows:

You are accessing a U.S. Government (USG) Information System (IS) that is provided for USG-authorized use only. By using this IS (which includes any device attached to this IS) you consent to the following conditions:

- The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations.
- At any time, the USG may inspect and seize data stored on this IS.
- Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose.

The following steps describe how to configure this banner in Workspace ONE Access. For more information, see the Workspace ONE Access administrative console documentation.

1. Log in to the Workspace ONE Access administrative console as an administrator.
2. In the Workspace ONE Access console, click the **Identity and Access Management** tab.
3. Click **Setup** and then click the **Connectors** tab.
4. Click the **Worker** link for each connector that you want to configure.
5. Click the **Auth Adapters** tab and then `CertificateAuthAdapter`.
6. Click the **Enable Consent Form before Authentication** check box.
7. Paste the Standard Mandatory DoD Notice and Consent Banner text into the **Consent Form Content** box.
8. Save your changes.

Create a Service Account and Get a Refresh Token in VCF Automation for VM Apps
As a cloud administrator, you can create a service account in VCF Automation and generate a refresh token for the service account.

1. Log in to VCF Automation as an administrator.
2. In the user profile section in the top-right corner, copy the organization ID.
3. Create a service account.
   a) Navigate to **Infrastructure** > **Access Control** > **Service Accounts** and click **New**.
   b) Enter a name for the service account.
   c) Select the Organization Owner, Assembler Administrator, and Service Broker Administrator roles.
   d) Click the magic wand to generate the software ID.
   e) Click **Next**.
   f) Review your configuration and click **Finish**.

   View your service account on the **Service Accounts** tab.
4. Get a refresh token.
   a) On the **Service Accounts** tab, click the vertical ellipsis next to the service account name.
   b) Deselect the `Required Rotation` check box and click **Save**.

      Note:

      If **Require Rotation** is selected, each refresh token becomes single-use, and the newly rotated token must be persisted every time. Any failure during rotation, such as not storing the updated token due to an error, breaks the integration. To avoid this issue, you must disable token rotation by deselecting the check box.
   c) Click the double arrow next to the service account name to expand the service account row and copy the client ID.
   d) To authorize the device, run the following command.

      ```
      curl -k 'https://<VCFA_PROVIDER_MANAGEMENT_UI_FQDN>/tm/oauth/tenant/<VCFA_ORGANIZATION_NAME>/
      device_authorization' \
      --header 'Content-Type: application/x-www-form-urlencoded' \
      --data-urlencode 'client_id=<ORGANIZATION_SERVICE_ACCOUNT_CLIENT_ID>'
      ```
   e) Copy the `user_code` value.

      The response might look like the following example.

      ```
      {
      ```

```
  "device_code": "qA9Ak5Jc130cw_UkXDUOAx93WJnh-fU7btAGfInL-A0",
  "user_code": "QSD1-9B64",
  "verification_uri": "https://vcfa.10.163.14.191.vcfd.broadcom.net/automation/#/infrastructure/
access",
  "expires_in": 3600,
  "interval": 60
}
```

f) On the **Service Accounts** tab, click `Review Access Requests`.
g) Paste the user code that copied in Step 4e.
h) Click **LOOKUP** button and then click **GRANT** button to approve the access request.
i) Copy the `device_code` value from the response you received in Step 4e.
j) To generate a refresh token, run the following command.

   Replace the `device_code` value with the value you copied in Step 4h.

```
curl -k 'https://<VCFA_PROVIDER_MANAGEMENT_UI_FQDN>/tm/oauth/tenant/<VCFA_ORGANIZATION_NAME>/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'client_id=<ORGANIZATION_SERVICE_ACCOUNT_CLIENT_ID>' \
--data-urlencode 'grant_type=urn:ietf:params:oauth:grant-type:device_code' \
--data-urlencode 'device_code=<DEVICE_CODE_FROM_DEVICE_AUTH_RESPONSE>'
```

k) Copy the token.

   The token might look like the following example.

```
{
  "access_token":
 "eyJraWQiOiJ1cm46dmNsb3VkOm9wZW5JZFByb3ZpZGVyS2V5OmQ0ZWQ4Y2UyLTc4YzEtNDgwYy1hMmZlLTQwNjZkOTU4YWY3MyIsImFsZyI6Il
TxMd3KVe-7desZ9m5SX6trJbsGE9hTcuMNShAL11CQKtwU9agVxwJAtG0A5K_OVDYPWdPqOtognVnUCVQVHqeOFrv7JTzhznyPeLEmEYOqaGhbOhl
  "token_type": "Bearer",
  "expires_in": 3600,
  "refresh_token": "vUM8xSpeb6r5mFKC5s3kx2Nm0JQ5IoKx"
}
```

Activate the VCF Automation for VM Apps adapter in VCF Operations. See Configuring VCF Automation for VM Apps Organization.

Adding cloud accounts to VCF Automation for VM Apps

Cloud accounts are the configured permissions that VCF Automation uses to collect data from the regions or data centers, and to deploy cloud templates to those regions.

The collected data includes the regions that you later associate with cloud zones.

When you later configure cloud zones, mappings, and profiles, you select the cloud account to which they are associated.

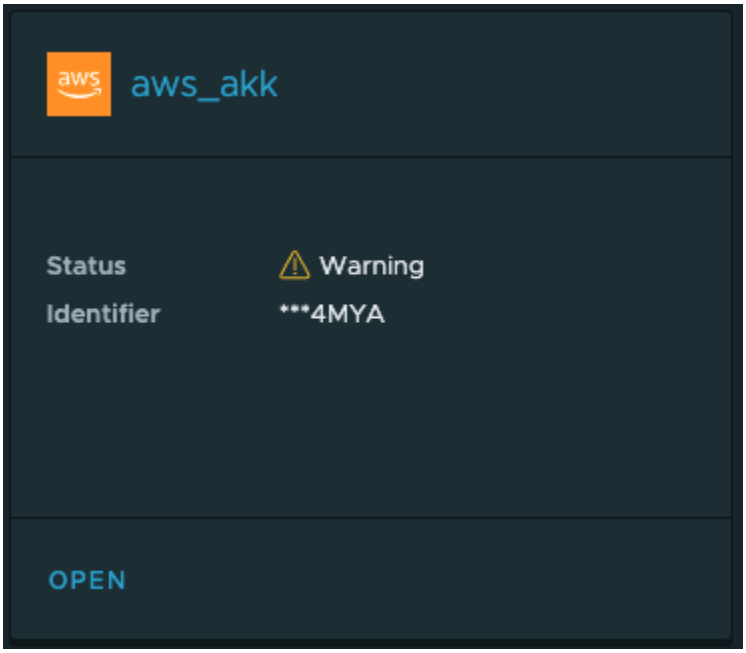
As a cloud administrator, you create cloud accounts for the projects in which team members work. Resource information such as network and security, compute, storage, and tags content is data-collected from your cloud accounts.

> **Note:**
>
> If the cloud account has associated machines that have already been deployed in the region, you can bring those machines into VCF Automation management by using an onboarding plan.

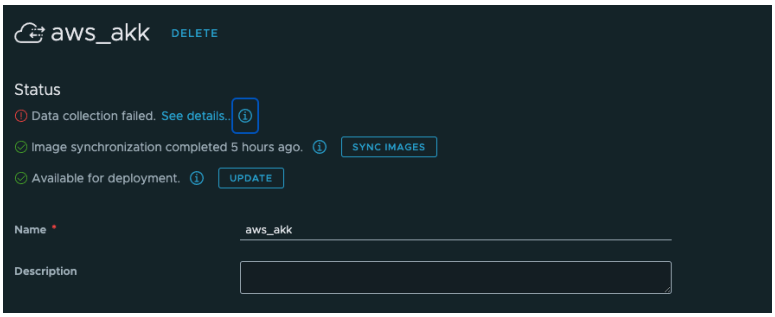
If you remove a cloud account that is used in a deployment, resources that are part of that deployment become unmanaged.

Monitoring the cloud account health

To ensure that your cloud accounts are communicating with the target system, the status of the account appears on the card based on periodic health checks.



Warnings indicate that there might be a problem that needs your attention.



In some cases, the details can help you resolve the issue. In other cases, you can run the **Update** option. It might be that the connection has been, in which case **Update** does not resolve the problem. One possible scenario is that the credentials have expired. To troubleshoot, first re-establish a connection with the updated credentials.

Credentials required for working with cloud accounts in VCF Automation for VM Apps

Before you start working in VCF Automation for VM Apps as a cloud administrator, you must gather information about your public and private cloud accounts. Use this checklist to help you set up before you begin on-boarding to the services.

vCenter cloud account credentials

This section describes the credentials that are required to add a vCenter cloud account.

Privileges are required for the vSphere agent to manage the vCenter instance. Provide an account with the following read and write privileges:

- vCenter IP address or FQDN

The permissions needed to manage VMware Cloud on AWS and vCenter cloud accounts are listed. Permissions must be enabled for all clusters in the vCenter, not just clusters that host endpoints.

To support control of VMware's Virtual Trusted Platform Module (vTPM) when deploying Windows 11 VMs, you must have the `cryptographic operations -> direct access` privilege in vCenter. Without this privilege, console access from VCF Automation to Windows 11 VMs is not possible.

For all vCenter-based cloud accounts - including NSX, vCenter, and VMware Cloud on AWS - the administrator must have vSphere endpoint credentials, or the credentials under which the agent service runs in vCenter, that provide administrative access to the host vCenter.

For more information about agent requirements, see the vSphere product documentation.

Setting	Selection
Content library To assign a privilege on a content library, an administrator must grant the privilege to the user as a global privilege. For related information, see the vSphere documentation.	• Add library item • Create local library • Create subscribed library • Delete library item • Delete local library • Delete subscribed library • Download files • Evict library item • Probe subscription information • Read storage • Sync library item • Sync subscribed library • Type introspection • Update configuration settings • Update files • Update library • Update library item • Update local library • Update subscribed library • View configuration settings
Datastore	• Allocate space • Browse datastore • Low level file operations
Datastore cluster	• Configure a datastore cluster
Folder	• Create folder • Delete folder
Global	• Manage custom attributes • Set custom attribute
Network	• Assign network
Permissions	• Modify permission
Profile-driven storage	• Profile-driven storage view To return a list of storage policies that can be mapped to a storage profile, grant the StorageProfile. View privilege to all accounts that connect VCF Automation to vCenter.
Resource	• Assign virtual machine to resource pool • Migrate powered off virtual machine • Migrate powered on virtual machine
vApp	• Import • vApp application configuration The vApp.Import application configuration is required for OVF templates and to provision VMs from the content library. The vApp.vApp application configuration is required when using cloud-init for cloud configuration scripting. This setting allows for modification of a vApp's internal structure, such as its product information and properties.

Setting	Selection
Virtual machine	Change Configuration • Add existing disk • Add new disk • Add or remove device • Advanced configuration • Change CPU count • Change memory • Change settings • Change Swapfile placement • Change resource • Extend virtual disk • Modify device settings • Remove disk • Rename • Set annotation • Toggle disk change tracking Edit Inventory • Create from existing • Create new • Move • Remove Interaction • Configure CD media • Connect devices • Console interaction • Install VMware tools • Power off • Power on • Reset • Suspend Provisioning • Clone template • Clone virtual machine • Customize guest • Deploy template • Read customization specifications Snapshot management • Create snapshot • Remove snapshot • Revert to snapshot
vSphere Tagging	• Assign or unassign vSphere tag • Assign or unassign vSphere tag on object • Create vSphere tag • Create vSphere tag category • Delete vSphere tag • Delete vSphere tag category • Edit vSphere tag • Edit vSphere tag category • Modify UsedBy field for category • Modify UsedBy field for tag

Microsoft Azure cloud account credentials

This section describes the credentials that are required to add a Microsoft Azure cloud account.

Configure a Microsoft Azure instance and obtain a valid Microsoft Azure subscription from which you can use the subscription ID.

If you are using an external HTTP Internet proxy, it must be configured for IPv4.

- General settings
  The following overall settings are required.

Setting	Description
Subscription ID	Allows you to access to your Microsoft Azure subscriptions.
Tenant ID	The authorization endpoint for the Active Directory applications you create in your Microsoft Azure account.
Client application ID	Provides access to Microsoft Active Directory in your Microsoft Azure individual account.
Client application secret key	The unique secret key generated to pair with your client application ID.

- Settings for creating and validating cloud accounts

The following permissions are needed for creating and validating Microsoft Azure cloud accounts.

Setting	Selection
Microsoft Compute	• Microsoft.Compute/virtualMachines/extensions/write • Microsoft.Compute/virtualMachines/extensions/read • Microsoft.Compute/virtualMachines/extensions/delete • Microsoft.Compute/virtualMachines/deallocate/action • Microsoft.Compute/virtualMachines/delete • Microsoft.Compute/virtualMachines/powerOff/action • Microsoft.Compute/virtualMachines/read • Microsoft.Compute/virtualMachines/restart/action • Microsoft.Compute/virtualMachines/start/action • Microsoft.Compute/virtualMachines/write • Microsoft.Compute/availabilitySets/write • Microsoft.Compute/availabilitySets/read • Microsoft.Compute/availabilitySets/delete • Microsoft.Compute/disks/delete • Microsoft.Compute/disks/read • Microsoft.Compute/disks/write
Microsoft Network	• Microsoft.Network/loadBalancers/backendAddressPools/join/action • Microsoft.Network/loadBalancers/delete • Microsoft.Network/loadBalancers/read • Microsoft.Network/loadBalancers/write • Microsoft.Network/networkInterfaces/join/action • Microsoft.Network/networkInterfaces/read • Microsoft.Network/networkInterfaces/write • Microsoft.Network/networkInterfaces/delete • Microsoft.Network/networkSecurityGroups/join/action • Microsoft.Network/networkSecurityGroups/read • Microsoft.Network/networkSecurityGroups/write • Microsoft.Network/networkSecurityGroups/delete • Microsoft.Network/publicIPAddresses/delete • Microsoft.Network/publicIPAddresses/join/action • Microsoft.Network/publicIPAddresses/read • Microsoft.Network/publicIPAddresses/write • Microsoft.Network/virtualNetworks/read • Microsoft.Network/virtualNetworks/subnets/delete • Microsoft.Network/virtualNetworks/subnets/join/action • Microsoft.Network/virtualNetworks/subnets/read • Microsoft.Network/virtualNetworks/subnets/write • Microsoft.Network/virtualNetworks/write
Microsoft Resources	• Microsoft.Resources/subscriptions/resourcegroups/delete • Microsoft.Resources/subscriptions/resourcegroups/read • Microsoft.Resources/subscriptions/resourcegroups/write
Microsoft Storage	• Microsoft.Storage/storageAccounts/delete • Microsoft.Storage/storageAccounts/read • Microsoft.Storage/storageAccounts/write • Microsoft.Storage/storageAccounts/listKeys/action is not generally required, but may be needed by users to view storage accounts.

Setting	Selection
Microsoft Web	• Microsoft.Web/sites/read • Microsoft.Web/sites/write • Microsoft.Web/sites/delete • Microsoft.Web/sites/config/read • Microsoft.Web/sites/config/write • Microsoft.Web/sites/config/list/action • Microsoft.Web/sites/publishxml/action • Microsoft.Web/serverfarms/write • Microsoft.Web/serverfarms/delete • Microsoft.Web/sites/hostruntime/functions/keys/read • Microsoft.Web/sites/hostruntime/host/read • Microsoft.web/sites/functions/masterkey/read

- Settings for action-based extensibility

  If you are using Microsoft Azure with action-based extensibility, the following permissions are required, in addition to the minimal permissions.

Setting	Selection
Microsoft Web	• Microsoft.Web/sites/read • Microsoft.Web/sites/write • Microsoft.Web/sites/delete • Microsoft.Web/sites/*/action • Microsoft.Web/sites/config/read • Microsoft.Web/sites/config/write • Microsoft.Web/sites/config/list/action • Microsoft.Web/sites/publishxml/action • Microsoft.Web/serverfarms/write • Microsoft.Web/serverfarms/delete • Microsoft.Web/sites/hostruntime/functions/keys/read • Microsoft.Web/sites/hostruntime/host/read • Microsoft.Web/sites/functions/masterkey/read • Microsoft.Web/apimanagementaccounts/apis/read
Microsoft Authorization	• Microsoft.Authorization/roleAssignments/read • Microsoft.Authorization/roleAssignments/write • Microsoft.Authorization/roleAssignments/delete
Microsoft Insights	• Microsoft.Insights/Components/Read • Microsoft.Insights/Components/Write • Microsoft.Insights/Components/Query/Read

- Settings for action-based extensibility with extensions

  If you are using Microsoft Azure with action-based extensibility with extensions, the following permissions are also required.

Setting	Selection
Microsoft.Compute	• Microsoft.Compute/virtualMachines/extensions/write • Microsoft.Compute/virtualMachines/extensions/read • Microsoft.Compute/virtualMachines/extensions/delete

Google Cloud Platform (GCP) cloud account credentials

This section describes the credentials that are required to add a Google Cloud Platform cloud account.

The Google Cloud Platform cloud account interacts with the Google Cloud Platform compute engine.

The Project Admin and Owner credentials are required for creating and validating Google Cloud Platform cloud accounts.

If you are using an external HTTP Internet proxy, it must be configured for IPv4.

The compute engine service must be enabled. When creating the cloud account in VCF Automation, use the service account that was created when the compute engine was initialized.

The following compute engine permissions are also needed, depending on the actions that the user can take.

Setting	Selection
roles/compute.admin	Provides full control of all compute engine resources.
roles/iam.serviceAccountUser	Provides access to users who manage virtual machine instances that are configured to run as a service account. Grant access to the following resources and services: • compute.* • resourcemanager.projects.get • resourcemanager.projects.list • serviceusage.quotas.get • serviceusage.services.get • serviceusage.services.list
roles/compute.imageUser	Provides permission to list and read images without having other permissions on the image. Granting the compute.imageUser role at the project level gives users the ability to list all images in the project. It also allows users to create resources, such as instances and persistent disks, based on images in the project. • compute.images.get • compute.images.getFromFamily • compute.images.list • compute.images.useReadOnly • resourcemanager.projects.get • resourcemanager.projects.list • serviceusage.quotas.get • serviceusage.services.get • serviceusage.services.list

Setting	Selection
roles/compute.instanceAdmin	Provides permissions to create, modify, and delete virtual machine instances. This includes permissions to create, modify, and delete disks, and also to configure shielded VMBETA settings. For users that manage virtual machine instances (but not network or security settings or instances that run as service accounts), grant this role to the organization, folder, or project that contains the instances, or to the individual instances. Users that manage virtual machine instances that are configured to run as a service account also need the roles/iam.serviceAccountUser role. • compute.acceleratorTypes • compute.addresses.get • compute.addresses.list • compute.addresses.use • compute.autoscalers • compute.diskTypes • compute.disks.create • compute.disks.createSnapshot • compute.disks.delete • compute.disks.get • compute.disks.list • compute.disks.resize • compute.disks.setLabels • compute.disks.update • compute.disks.use • compute.disks.useReadOnly • compute.globalAddresses.get • compute.globalAddresses.list • compute.globalAddresses.use • compute.globalOperations.get • compute.globalOperations.list • compute.images.get • compute.images.getFromFamily • compute.images.list • compute.images.useReadOnly • compute.instanceGroupManagers • compute.instanceGroups • compute.instanceTemplates • compute.instances • compute.licenses.get • compute.licenses.list • compute.machineTypes • compute.networkEndpointGroups • compute.networks.get • compute.networks.list • compute.networks.use • compute.networks.useExternalIp • compute.projects.get • compute.regionOperations.get • compute.regionOperations.list • compute.regions • compute.reservations.get • compute.reservations.list • compute.subnetworks.get • compute.subnetworks.list • compute.subnetworks.use • compute.subnetworks.useExternalIp
	• compute.targetPools.get • compute.targetPools.list • compute.zoneOperations.get • compute.zoneOperations.list

Setting	Selection
roles/compute.instanceAdmin.v1	Provides full control of compute engine instances, instance groups, disks, snapshots, and images. Also provides read access to all compute engine networking resources. **Note:** If you grant a user this role at the instance level, that user cannot create new instances. • compute.acceleratorTypes • compute.addresses.get • compute.addresses.list • compute.addresses.use • compute.autoscalers • compute.backendBuckets.get • compute.backendBuckets.list • compute.backendServices.get • compute.backendServices.list • compute.diskTypes • compute.disks • compute.firewalls.get • compute.firewalls.list • compute.forwardingRules.get • compute.forwardingRules.list • compute.globalAddresses.get • compute.globalAddresses.list • compute.globalAddresses.use • compute.globalForwardingRules.get • compute.globalForwardingRules.list • compute.globalOperations.get • compute.globalOperations.list • compute.healthChecks.get • compute.healthChecks.list • compute.httpHealthChecks.get • compute.httpHealthChecks.list • compute.httpsHealthChecks.get • compute.httpsHealthChecks.list • compute.images • compute.instanceGroupManagers • compute.instanceGroups • compute.instanceTemplates • compute.instances • compute.interconnectAttachments.get • compute.interconnectAttachments.list • compute.interconnectLocations • compute.interconnects.get • compute.interconnects.list • compute.licenseCodes • compute.licenses • compute.machineTypes • compute.networkEndpointGroups • compute.networks.get • compute.networks.list • compute.networks.use • compute.networks.useExternalIp • compute.projects.get • compute.projects.setCommonInstanceMetadata • compute.regionBackendServices.get • compute.regionBackendServices.list • compute.regionOperations.get • compute.regionOperations.list • compute.regions • compute.reservations.get

NSX cloud account credentials

This section describes the credentials that are required to add an NSX cloud account.

Provide an account with the following read and write privileges.

- NSX IP address or FQDN
- NSX user name and password

Associate the user with both the **Audit** role and the custom role, which has the specified privileges outlined below. Add this user to VCF Automation as a cloud account for seamless authentication with NSX.

The following lists the minimum privileges required for the custom role.

Category/Subcategory	Permission
Networking - Tier-0 Gateways	Read-only
Networking - Tier-0 Gateways -> OSPF	None
Networking - Tier-1 Gateways	Full Access
Networking - Segments	Full Access
Networking - VPN	None
Networking - NAT	Full Access
Networking - Load Balancing	Full Access
Networking - Forwarding Policy	None
Networking - Statistics	None
Networking - DNS	None
Networking - DHCP	Full Access
Networking - IP Address Pools	None
Networking - Profiles	Read-only
Security - Threat Detection & Response	None
Security - Distributed Firewall	Full Access
Security - IDS/IPS & Malware Prevention	None
Security - TLS Inspection	None
Security - Identity Firewall	None
Security - Gateway Firewall	None
Security - Service Chain Management	None
Security - Firewall Time Window	None
Security - Profiles	None
Security - Service Profiles	None
Security - Firewall Settings	Full Access
Security - Gateway Security Settings	None
Inventory	Full Access
Troubleshooting	None
System	None

Administrators also require access to the vCenter as described in the *vCenter cloud account credentials* section of this topic.

VMware Cloud on AWS (VMC on AWS) cloud account credentials

This section describes the credentials that are required to add an VMware Cloud on AWS (VMC on AWS) cloud account.

Provide an account with the following read and write privileges:

- The cloudadmin@vmc.local account or any user account in the CloudAdmin group
- NSX Enterprise Administrator role and access credentials
- NSX Cloud Admin access to your organization's VMware Cloud on AWS SDDC environment
- Administrator access to your organization's VMware Cloud on AWS SDDC environment
- The VMware Cloud on AWS API token for your VMware Cloud on AWS environment in your organization's VMware Cloud on AWS service
- vCenter IP address or FQDN

Administrators *also* require access to the vCenter as described in the *Add a vCenter cloud account* section of this table.

NSX integration with Microsoft Azure VMware Solution (AVS) for VCF Automation

For information about connecting NSX running on Microsoft Azure VMware Solution (AVS) to VCF Automation, including configuring custom roles, see the Microsoft product documentation.

Configure Microsoft Azure for use with VCF Automation for VM Apps

You must gather some information and complete an appropriate configuration to use a Microsoft Azure instance with a VCF Automation cloud account.

This task describes the process of configuring Microsoft Azure at a high level. Because Microsoft Azure is a third-party product, the interface can be changed or updated at any time independently of VCF Automation, the accuracy of specific steps might vary. You should regard this procedure as representative rather than definitive.

You need the following to complete this procedure:

- Subscription ID - Click the Subscriptions icon on the left toolbar in your Azure portal to view the subscription ID.
- Tenant ID - Click the Help icon and select Show Diagnostics in your Azure portal. Search for tenant and record the ID when you have located it.
- Client Application ID - You can locate this under App registrations on the Manage menu.
- Client Application Secret Key - The Client Application Secret Key functions like a password that Automation will use to authenticate with Azure. You must generate a secret key when you set up Azure. See Step 7 below for specific instructions.

1. Locate and record your Microsoft Azure subscription and tenant IDs as well as the Client Application ID and appropriate Client Application Secret Key.

   Note that the location and configuration procedures related to these components may change with different versions of Azure. See the Microsoft Azure documentation for the latest information.

2. Now you can add permissions to the account. Select API permissions in the Azure interface and add a permission. Then you can use the Select an API page, select Azure Service Management under Commonly used Microsoft APIs. Select Delegated permissions to select Delegated permissions and add the `user_impersonation` permission.

3. After you create an account to connect to Azure, you must give it the required permissions to the subscription. Navigate back to Subscriptions, select the subscription you are adding to VCF Automation and on the menu, select Access control (IAM).

4. You can create a new storage account and a resource group at this point. Alternatively, you can create these in blueprints later.

   - Storage Account - Use the following procedure to configure an account.
     a. In your Microsoft Azure portal, locate the Storage Accounts icon, currently located on the sidebar. Make sure the correct subscription is selected and click **Add**. You can also, search for storage account in the Azure search field.
     b. Enter the required information for the storage account. You will need your subscription ID.

c. Select whether to use an existing resource group or create a new one. Make note of your resource group name, as you will need it later.

Note:
Save the location of your storage account as you will need it later.

5. Create a virtual network. Alternatively, if you have a suitable existing network, you can select that one.

   If you are creating a network, you must select Use an Existing Resource Group and specify the group that you created in the preceding step. Also, select the same location that you specified previously. Microsoft Azure will not deploy virtual machines or other objects if the location doesn't match between all applicable components that the object will consume.

   a) Locate the Virtual Network icon on the left panel and click it, or search for virtual network. Make sure to select the correct subscription and click **Add**.
   b) Enter a unique name for your new virtual network and record it for later.
   c) Enter the appropriate IP address for your virtual network in the **Address space** field.
   d) Ensure that the correct subscription is selected and click **Add**.
   e) Enter the remaining basic configuration information.
   f) You can modify the other options as necessary. For most configurations, you can leave the defaults.
   g) Click **Create**.

6. Set up an Azure Active Directory application so that VCF Automation can authenticate.
   a) Locate the Active Directory icon on the Azure left menu and click it.
   b) Click **App Registrations** and select **Add**.
   c) Type a name for your application that complies with Azure name validation.
   d) Leave Web app/API as the Application Type.
   e) The Sign-on URL can be anything that is appropriate for your usage.
   f) Click **Create**.

7. Create a secret key to authenticate the application in VCF Automation.
   a) Click the name of your application in Azure.

      Make note of your Application ID for later use.
   b) Click **All Settings** in the next pane and select Keys from the settings list.
   c) Enter a description for the new key and choose a duration.
   d) Click **Save** and make sure to copy the key value to a safe location as you will be unable to retrieve it later.
   e) On the left menu, select **API Permissions** for the application and click **Add a Permission** to create a new permission.
   f) Select Azure Service Management on the Select an API page.
   g) Click **Delegated Permissions**.
   h) Under Select permissions select user_impersonation and then click **Add Permissions**.

8. Authorize your Active Directory application to connect to your Azure subscription so that you can deploy and manage virtual machines.
   a) In the left menu, click the Subscriptions icon, and select your new subscription.

      You may need to click on the text of the name to get the panel to slide over.
   b) Select the Access control (IAM) option to see the permissions to your subscription.
   c) Click **Add** under the Add a Role Assignment heading.
   d) From the Role drop down, select one of the following options:

      - **Contributor** if you need to manage all resources.
      - **Owner** if you need to manage all resources and control role-based access within VCF Automation.

e) Leave the default selection in the Assign Access to drop down.
f) Type the name of your application in the Select box.
g) Click **Save.**

It is highly recommended that you install the Microsoft Azure command line interface tools. These tools are freely available for both Windows and Mac operating systems. See the Microsoft documentation for more information about downloading and installing these tools.

When you have the command line interface installed, you can use it to authenticate your new subscription.

1. Open a terminal window and type your Microsoft Azure login. You will receive a URL and a shortcode that will allow you to authenticate.
2. In a browser, enter the code that you received from the application on your device.
3. Enter your Auth Code and click **Continue**.
4. Select your Azure account and login.
   If you have multiple subscriptions, ensure that the correct one is selected using the `azure account set --subscription <subscription-name>` command.
5. Before you proceed, you must register the `Microsoft.Compute` provider to your new Azure subscription using the `azure provider register --namespace "Microsoft.Compute"` command.
   If the command times out and generates an error the first time your run it, run it again.

When you have completed configuration, you can use the `azure vm image list` command to retrieve available Azure virtual machine marketplace image names. You can choose the desired image and record the URN provided for it and later use it in blueprints.

You must manually accept the agreement terms of the image using the Azure command line interface as shown:

```
“az vm image terms accept --urn jetware-srl:postgresql:postgresql96-ubuntu-1604:1.0.170503”
```

The following example shows how you might sign in a specific subscription using the Azure command line.

```
az account list
az login --identity --username <client_id|object_id|resource_id>
```

VCF Automation allows any subscription user to map a marketplace image. This does not indicate that the user has access to the image. The user account used to accept all Azure image term agreements must be the same one that was used to create the VCF Automation cloud account.

Create a Microsoft Azure cloud account in VCF Automation for VM Apps

As a cloud administrator, you can create a Microsoft Azure cloud account for account regions to which your team will deploy VCF Automation cloud templates.

- Verify that you have the required administrator credentials and have enabled HTTPS access on port 443. See Credentials required for working with cloud accounts in VCF Automation for VM Apps.
- Verify that you have the required user role. See What are the VCF Automation for VM Apps user roles.
- Configure a Microsoft Azure account for use with VCF Automation. See Configure Microsoft Azure for use with .
- If you do not have external Internet access, configure an Internet server proxy.

VCF Automation Microsoft Azure integrations support Azure Marketplace images. Users can specify an Azure Marketplace when they set up image mapping in VCF Automation. VCF Automation automatically picks up the plan information in any specified Marketplace image. When using Marketplace images, users must accept legal terms for plans in their Azure portal or an exception will occur when they try to deployment the cloud template and the deployment will fail.

> **Important:**
>
> Starting with VCF Automation 9.1.1, native management capabilities for public cloud resources, including Microsoft Azure, are deprecated and disabled by default in VM Apps organizations. For more information, see the VCF Automation 9.1.1 release notes.

1. Select **Infrastructure** > **Connections** > **Cloud Accounts** and click **Add Cloud Account**.
2. Select the Microsoft Azure account type and enter credentials and other values.
3. Select the appropriate **Azure environment**. The selection can be Azure Public Cloud, which is appropriate for most users, or it can be Azure US Government Cloud, which is valid for authorized government users.
4. Click **Validate**.

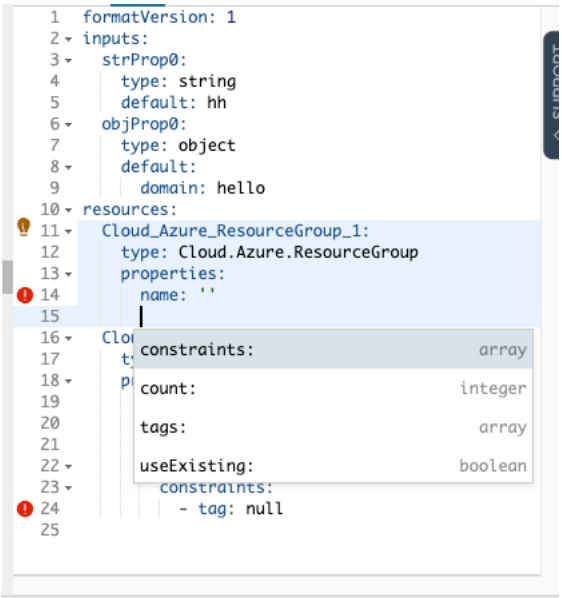
   The account regions associated with the account are collected.
5. Select the regions to which you want to provision this resource.
6. For efficiency, click **Create a Cloud zone for the selected regions**.
7. If you need to add tags to support a tagging strategy, enter capability tags. See How do I use tags to manage VCF Automation resources and deployments and Creating a tagging strategy.
8. Click **Save**.

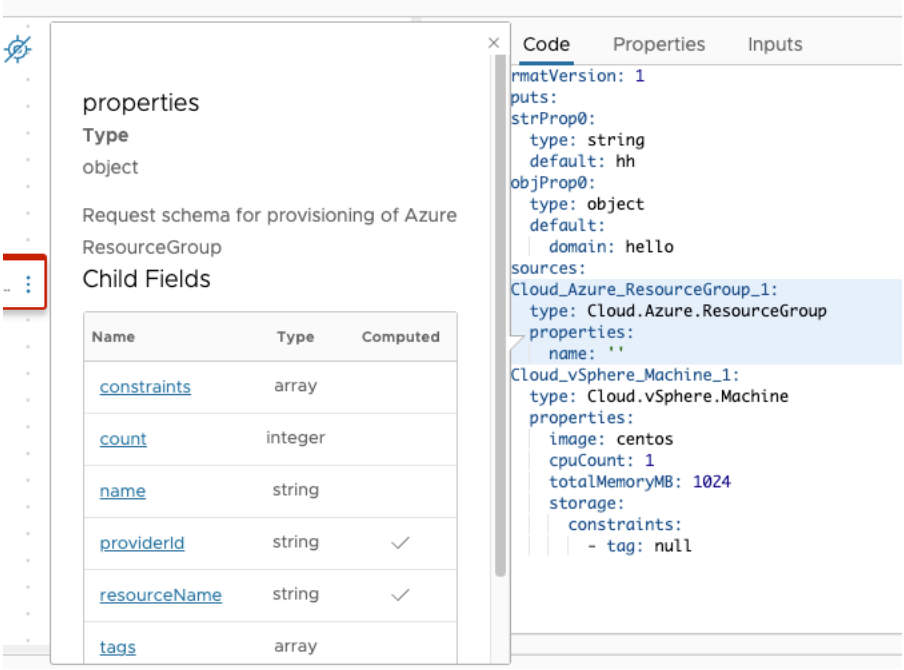
The account is added to VCF Automation , and the selected regions are available for the specified cloud zone.

Create infrastructure resources for this cloud account.

When you create a cloud template using the Azure cloud account, you can use constraints to deploy a resource group to a selected region.

The cloud template `type:cloud.Azure.ResourceGroup` option is visible in the auto-suggestion list and the properties pop-ups in the cloud template as seen in the following samples.





When you add an Azure cloud account to a cloud template, you can choose to reuse availability sets if you want. Subscriptions have a limit of 2000 availability sets and 25,000 virtual machines, so it makes sense to reuse availability sets when possible. There are two YAML properties that you can use to control how deployments use availability sets. The `availabilitySetName` property enables you to specify an availability set to use. The second property is `doNotAttachAvailabilitySet` which is set to false by default. If this property is set to true, VCF Automation will create the deployment with no availability set.

You can use availability sets or availability zones with an Azure cloud account, but not both.

You cannot create a deployment without an availability set if you use a load balancer attached to the virtual machine.

The following table describes how VCF Automation behaves depending on whether a resource group and an availability set are specified in the cloud template.

An availability set cannot exist without being part of a resource group. The availability sets in a given resource group must have unique names. Availability sets can have the same name only if they are part of different resource groups.

If you do not specify a resource group name, then VCF Automation will create a new resource group, which means that a new availability set must also be created even if a name is passed. The new set will use the name that is passed.

Table 1823:

Resource Group Specified	Availability Set Specified	Result
No	No	VCF Automation creates a new resource group and a new availability set for the virtual machine.
Yes	No	VCF Automation reuses the existing resource group and creates a new availability set for the virtual machine.
No	Yes	VCF Automation creates a new resource group and a new availability set with the specified name.
Yes	Yes	VCF Automation reuses the existing resource group. If an availability set with the specified name already exists in that group, it will also be reused. If there is no availability set with the specified name in the group, a new one is created with that name.

As an alternative, you can use availability zones with deployed Azure virtual machines rather than availability sets. If you use availability zones, you can specify an Azure availability zone for a provisioned machine, either by tagging compute resources or by using restrictions expressed through flavor mapping.

This feature enables users to provision a machine in an Azure availability zone by tagging the computes, or selecting the flavor mapping that contains the desired availability zone.

If you want to use availability zones for an Azure virtual machine, you must explicitly pass the `attachAvailabilityZone` boolean property set to `true` in the cloud template for the cloud account as shown in the following cloud template snippet.

```
formatVersion: 1
inputs: {}
resources:
   Cloud_Aure_Machine_1:
   type: Cloud Azure Machine
   properties:
     image: ubuntu
     flavor: small
     attachAvailabilityZone: true
```

If this property is not present, no availability zone are used for the provisioned machine; instead, an availability set is created. If this property is present and set to true, you can select the desired availability zone on the Compute tab of the cloud zone.

If availability zones are used in a cloud account, the IP SKU type and IP allocation method will be Static and the PublicIPAddress sku type is limited to Standard. Also, not all availability zones are supported for every flavor in every region. The following table outlines the regions that support selection of availability zones. Note that each of the supported regions has three zones. Each region in the table should support Azure availability zones, but depending on the selected size in the flavor mapping, it can be restricted to only one zone.

Region	Zones Supported
Brazil South	3
Canada Central	3
Central US	3
East US	3
East US 2	3
South Central US	3
West US 2	3
West US 3	3
US Gov Virginia	3
East Asia	3
Southeast Asia	3
Australia East	3
China North 3	3
Central India	3
Japan East	3
Korea Central	3
North Europe	3
West Europe	3
France Central	3
Germany West Central	3

Region	Zones Supported
Norway East	3
Sweden Central	3
Switzerland North	3
UK South	3
South Africa North	3

If you want to use machine provisioning in an availability zone and later have a working ssh to the machine, you must first have a correctly configured network security group, with enabled ssh, in the Azure portal. In addition, you must create a network profile with the enumerated data for the network and security groups and then use this network profile on provisioning.

VCF Automation supports Azure disk snapshots for deployed virtual machines. See Working with snapshots for Microsoft Azure virtual machine disks in VCF Operations for more information.

VCF Automation supports several boot diagnostics options for Azure deployments. Boot diagnostics supports debugging of Azure virtual machines and includes collection of log information and relevant screen shots. See Using boot diagnostics and log analytics with a Microsoft Azure virtual machine in VCF Automation for VM Apps for more information.

In addition to standard disk options, Azure integrations supports selection of Azure performance disk types for users of Azure SSD Premium managed disks. This feature enables users to maximize performance as needs demand. Not all performance disk configurations are supported. See https://learn.microsoft.com/en-us/azure/virtual-machines/disks-change-performance for more information about supported configuations.

You can configure the performance disk type during deployment configuration using the `performanceTierType` property in your cloud template. The following example demonstrates a potential implementation.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_Azure_Disk_1:
    type: Cloud.Azure.Disk
    properties:
      capacityGb: 2
      managedDiskType: Premium SSD
      performanceTier: P1
```

Using boot diagnostics and log analytics with a Microsoft Azure virtual machine in VCF Automation for VM Apps
You can invoke and configure Microsoft Azure boot diagnostics from an Azure instance in a cloud template. In addition you can also configure log analytics for an Azure virtual machine instance. Boot diagnostics is a debugging feature for Azure virtual machines that facilitates diagnostics for virtual machine boot failures. Using boot diagnostics, a user can monitor the state of a virtual machine as it is booting up by collecting serial log information and screenshots.

Boot Diagnostics

Boot diagnostics captures serial log information and screenshots and these needs to be saved to the disk. The disk can be of two types, Azure Managed Disk or Unmanaged Disk.

The `bootDiagnostics` YAML property is supported in Azure cloud templates. When this property is set to `true`, boot diagnostics are enabled on the applicable Azure virtual machine deployment.

The following YAML snippet shows an example of how the `bootDiagnostics` property is used.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_Azure_Machine_1:
    type: Cloud.Azure.Machine
    metadata:
      layoutPosition:
```

```
        - 0
        - 0
    properties:
      image: ubuntu
      flavor: small
      bootDiagnostics: true
```

Boot diagnostics can also be invoked on a deployed Azure virtual machine as a day 2 operation. Navigate to the Deployments page in VCF Automation and select the Azure deployment. The Actions menu on this page enables you to toggle between Enable Boot Diagnostics and Disable Boot Diagnostics.

After you have deployed a cloud template with boot diagnostics enabled, the VCF Automation Deployments page for the deployment will indicate that boot diagnostics are enabled. If you want to disable boot diagnostics, click the Actions menu on the Deployments page and select Disable Boot Diagnostics.

Log Analytics

Log Analytics enables you to edit and run log queries on data collected by Azure Monitor Logs, and then interactively analyze the results. You can use Log Analytics queries to retrieve records that match specific criteria to help identify trends and patterns and provide a variety of data insights. By enabling Log Analytics on a Azure virtual machine, that machine will act as a data source.

Before you can configure log analytics in an VCF Automation cloud template, you must create and configure an Azure Log Analytics workspace. You can do this using the Virtual Machines option in the Azure Monitor menu. See the Microsoft Azure documentation for more information.

To configure log analytics, you must have the Azure Workspace ID and Workspace Key. You can find these on the Agent Management tab in Azure under the Log Analytics Workspace.

The following cloud template example shows how log analytics can be configured using extensions.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_Azure_Machine_1:
    type: Cloud.Azure.Machine
    properties:
      image: ubuntu
      flavor: small
      extensions:
        - autoUpgradeMinorVersion: true
          name: test-loga
          protectedSettings:
            workspaceKey: xxxxxxxxx
          publisher: Microsoft.EnterpriseCloud.Monitoring
          settings:
            workspaceId: aaaaaaaaa
          type: OmsAgentForLinux
          typeHandlerVersion: '1.0'
```

After you have deployed a cloud template with Log Analytics enabled, you can enable or disable it using the Actions menu options on the VCF Automation Deployments page for the deployment.

Working with snapshots for Microsoft Azure virtual machine disks in VCF Operations

You can create full or incremental snapshots of Microsoft Azure managed disks.

The VCF Automation Deployments page for an Azure deployment contains an Actions menu that provides several options for creating and deleting snapshots from Azure deployments on virtual machine managed disks and on independent managed disks. The following list outlines the specific snapshot functionality that is supported.

- Create a disk snapshot - Supported for both external and compute disks. You can also create snapshots for a disk in a different resource group.
- Delete a disk snapshot - Supported for external disks only
- Encrypt snapshots using an Azure disk encryption set.
- You can provide key-value pairs as tags during snapshot creation.

Snapshots on unmanaged disks are currently not supported.

If you use encryption, the current snapshot implementation supports platform-managed key encryption. By default, the network policy allows access from everywhere, so restricting access to snapshots by using the network policy is not possible.

For more information about Microsoft Azure snapshot support, see the Microsoft product documentation.

Create an Amazon Web Services cloud account in VCF Automation for VM Apps

As a cloud administrator, you can create a VCF Automation cloud account for Amazon Web Services (AWS) for account regions to which your team will deploy VCF Automation cloud templates.

- Verify that you have the required administrator credentials and have enabled HTTPS access on port 443.
- Verify that you have the required user role. See What are the VCF Automation for VM Apps user roles.
- Verify that you have required AWS administrator credentials.
- If you do not have external Internet access, configure an Internet server proxy.

For authorized users, AWS cloud accounts support access to the AWS GovCloud configuration. This configuration supports most of the standard VCF Automation cloud account functionality with regard to project configuration, tags, and infrastructure. In VCF Automation templates, it does support use of AWS Platform as a Service (PaaS) properties.

> **Important:**
>
> Starting with VCF Automation 9.1.1, native management capabilities for public cloud resources, including Amazon Web Services (AWS), are deprecated and disabled by default in VM Apps organizations. For more information, see the VCF Automation 9.1.1 release notes.

The following procedure describes how to configure an AWS cloud account.

1. Select **Infrastructure** > **Connections** > **Cloud Accounts** and click **Add Cloud Account**.
2. Select the AWS account type, and enter credentials and other values.
3. Click **Validate**.
   The account regions associated with the account are collected.
4. Select the regions to which you want to provision this resource.

   > **Note:**
   > AWS China regions are not supported.
5. For efficiency, click **Create a Cloud zone for the selected regions**.
6. If you need to add tags to support a tagging strategy, enter capability tags. See How do I use tags to manage VCF Automation resources and deployments and Creating a tagging strategy.
7. Click **Add**.

The account is added to VCF Automation, and the selected regions are available for the specified cloud zone.

Configure infrastructure resources for this cloud account.

AWS integration supports use of GP3 and IO2 VolumeType disk options. There are some constraints on volume size and IOs values for specific volume types. The following page explains the supported configurations: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-volume-types.html.

GP3 and IO2 VolumeType selections must be configured in a storage profile before they can be used in deployments. Then, users can add them when setting up a cloud template using the `volumeType` property to specify the volume configuration. The following cloud template snippet shows an example of how the `volumeType` property is used.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_AWS_Volume_1:
    type: Cloud.AWS.Volume
    properties:
      capacityGb: 1
      volumeType: gp3
      iops: 3000
```

You can also add a valid GP3 or IO2 volume to an existing deployment as a day 2 action uisng the Actions menu on the VCF Automation Deployments page. In addition, resize, updaate and delete operations are supported as day 2 actions.

AWS configuration options in VCF Automation for VM Apps
There are several AWS configuration options that facilitate particular customer configurations, such as role based authentication.

Configure a role based cloud account for use with AWS

You can configure role-baed access to an AWS cloud account using the following procedure. VMware provides a helper JSON to facilitate this procedure that is displayed when you click Create IAM Role on the AWS create cloud account page. The JSON helper includes instructions to create and configure an IAM role for VCF Automation.

> **Note:**
> The helper JSON helps you to configure access to basic functionality such as machine creation. If you wamt to run more complex tasks using ABX extensibility actions, you must assign more permissions to the AWS role in the AWS portal.

To set up role based permissions for AWS when using the on-premise version of VCF Automation, you must create a master cloud account for trusted identity authentication. When you create a new role-based cloud account, the external ID -- which is the same as the orgId for the organization – will be populated and you can copy it and use it when you set up the role in the AWS portal. After the role is configured, you can create the AWS cloud account using the ARN ID from the AWS role that you created.

1. Create a user account in the AWS portal with `sts:AssumeRole` permissions.
2. Create the master cloud account in VCF Automation using accessKey/secretKey , For the Authentication Method, you must select the radio button for Trusted Identity for role-based authentication.
3. In the AWS portal, create a policy and paste in the snippet from the helper JSON.
4. In VCF Automation, open the cloud account wizard to create a role-based AWS cloud account. The External ID field is populated, and you can copy the ID.
5. In the AWS portal, create the AWS role with trust on the AWS master cloud account that you created in VCF Automation. You can get the externalId from the role-based cloud account that you created in VCF Automation, and then paste it into the External ID field in the AWS portal.
6. Search for the vRA Access Policy in the AWS portal, then create the role that you want to use for AWS cloud account access on this policy.
7. Then copy the ARN ID that you will need when you create the cloud account.
8. Create the AWS cloud account in VCF Automation using the ARN ID.

Create a Google Cloud Platform cloud account in VCF Automation for VM Apps

As a cloud administrator, you can create a Google Cloud Platform (GCP) cloud account for account regions to which your team deploys VCF Automation cloud templates.

- Verify that you have the required administrator credentials and have enabled HTTPS access on port 443.

- Verify that you have the required user role. See What are the VCF Automation for VM Apps user roles.
- Verify that you can access to the Google Cloud Platform JSON security key.
- Verify that you have the needed security information for your Google Cloud Platform instance. You can obtain most of this information from your instance or from the Google documentation.
- If you do not have the external Internet access, configure an Internet server proxy.

Important:

Starting with VCF Automation 9.1.1, native management capabilities for public cloud resources, including Google Cloud Platform (GCP), are deprecated and disabled by default in VM Apps organizations. For more information, see the VCF Automation 9.1.1 release notes.

1. In VCF Automation, select **Infrastructure** > **Connections** > **Cloud Accounts** and then click **Add Cloud Account**.
2. Select the Google Cloud Platform account type and enter the appropriate credentials and related information. Use the service account that was created when the source GCP account compute engine was initialized.

   To successfully create the cloud account in VCF Automation, the source GCP account must have the compute engine service enabled.

   In VCF Automation, the project ID is part of the Google Cloud Platform endpoint. You specify it when you create the cloud account. During data collection of project-specific private images, the VCF Automation GCP adapter queries the Google Cloud Platform API.
3. Click **Validate**.

   The account regions associated with the account are collected.
4. Select the regions to which you want to provision this resource.
5. For efficiency, click **Create a Cloud zone for the selected regions**.
6. If you need tags to support a tagging strategy, enter capability tags. See How do I use tags to manage VCF Automation resources and deployments and Creating a tagging strategy.
7. Click **Add**.

The account is added to VCF Automation, and the selected regions are available for the specified cloud zone.

When you configure a Google Cloud Platform cloud account and specify a GCP project, users with appropriate privileges can use GCP Virtual Private Cloud (VPC) network functionality. These users have access to local networks for that project as well as to all of the shared networks and sub networks for that host project. You can select one of these local or shared networks on the VCF Automation**Networks** page and add it to a network profile so that you can use it in a GCP deployment. Users with the following roles can use VPC networks.

- Compute Instance Administrators
- Compute Network Administrators
- Owners
- Editors

Create infrastructure resources for this cloud account.

The following paragraphs provide some information on deploying a Google Cloud Platform virtual machine from VCF Automation.

When you add a Google Cloud Platform cloud account to an VCF Automation cloud template, you can use the `useSoleTenant` YAML property to indicate that you want to deploy a virtual machine to a sole tenant node. This configuration enables you to isolate virtual machines for security, privacy or others issues.

To facilitate this functionality, Google Cloud Platform node affinity labels are converted to tags in VCF Automation, and these tags are applied on relevant VCF Automation availability zones where node groups reside. When the `useSoleTenant` property is set to true, constraint tags must be one of the node affinity labels. Also, to deploy a machine in sole tenant mode, you must include the `useSoleTenant` property in the cloud template as well as the constraint tags.

Before using this feature, you must create the appropriate node template and node affinity labels in Google Cloud Platform and then create a node group.

The following YAML example shows how the `useSoleTenant` property can be used in VCF Automation cloud templates. The constraint tags are the node affinity labels that were auto-collected from your Google Cloud Platform server.

```
resources:
  Cloud_GCP_Machine_1:
    type: Cloud.GCP.Machine
    properties:
      image: ubuntu
      flavor: c2-family
      name: demo-vm
      useSoleTenant: true
      constraints:
        -tag: 'env:prod'
        -tag: 'region:asia-east1'
```

The Google Cloud Platform cloud account supports several administrator day 2 actions on deployed virtual machines. These day 2 actions for Google Cloud Platform virtual machines include create, delete, and revert actions for snapshots and attached disks. These actions are available from the **Actions** menu on the VCF Automation**Deployments** page.

Setting up storage bucket resources for Google Cloud Platform (Classic) in VCF Automation for VM Apps
VCF Automation supports Google Cloud Platform (GCP) storage buckets allowing users to quickly and easily create and manage their storage resources. VCF Automation storage bucket support includes the creation of multi-regional/dual-regional storage buckets, restricted public access, and encryption.

> **Note:**
>
> This topic describes how to create classic storage bucket resources.

You create storage buckets using VCF Automation cloud templates and you deploy these cloud templates in the typical manner using the VCF Automation**Design** tab functionality. After you deploy a cloud template you can check the GCP portal to confirm that the storage bucket was provisioned successfully.

The following are required properties for storage bucket cloud templates.

- Storage bucket name - There are some limitations on the storage bucket name construction. See the following page for detailed information: https://cloud.google.com/storage/docs/buckets
- locationType - one of {SINGLE_REGION, DUAL_REGION, MULTI_REGION"} If the locationType is either DUAL_REGION or MULTI_REGION - regionID property is also MANDATORY There are different values of the regionId, depending on the locationType. See the following page for detailed information: https://cloud.google.com/storage/docs/locations
- storageClass - one of {STANDARD, NEARLINE, COLDLINE, ARCHIVE, REGIONAL, MULTI_REGIONAL, DURABLE_REDUCED_AVAILABILITY}

If you want to create storage buckets with restricted public access and that are encrypted by CMEK, there are some prerequisites described in the Google Cloud Platform documentation.

> **Note:**
> The customer-managed encryption key must be located in the same region as the storage bucket.

Storage bucket cloud template examples

The following cloud template examples demonstrate some typical approaches to creating storage buckets using VCF Automation cloud templates.

The following cloud template example shows how you might create a simple single region storage bucket.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_GCP_StorageBucket_1:
    type: Cloud.GCP.StorageBucket
    properties:
```

```
      name: simple-storage-bucket
      locationType: SINGLE_REGION
      storageClass: STANDARD
```

The following cloud template example demonstrates restricted public access using an encryption key.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_GCP_StorageBucket_1:
    type: Cloud.GCP.StorageBucket
    properties:
      name: encrypted-storage-bucket
      storageClass: STANDARD
      locationType: SINGLE_REGION
      kmsKey:
        kmsProjectId: gcp
        keyRingRegion: asia-east1
        keyRing: asia-key
        keyName: asia-east-key
```

The following cloud template example demonstrates how you might create a multi-region storage bucket.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_GCP_StorageBucket_1:
    type: Cloud.GCP.StorageBucket
    properties:
      name: multi-regional-sb
      locationType: MULTI_REGION
      storageClass: STANDARD
      regionId: EU
```

The following cloud template shows an example of dual-regional storage bucket implementation.

```
formatVersion: 1
inputs: {}
resources:
 Cloud_GCP_StorageBucket_1:
    type: Cloud.GCP.StorageBucket
 properties:
  name: dual-regional-sb
  storageClass: NEARLINE
  locationType: DUAL_REGION
  regionId: ASIA
  locations:
   - ASIA-EAST1
   - ASIA-SOUTHEAST1
```

Configuring load balancers with Google Cloud Platform cloud accounts (Classic) in VCF Automation for VM Apps
VCF Automation Google Cloud Platform (GCP) cloud account users can configure load balancers to support custom solutions, including high availability.

VCF Automation GCP cloud account users can configure several aspects of load balancer configuration to support specific requirements. These features can support provisioning infrastructure for SAP NetWeaver and SAP Hana, for example.

Setting up a high availability load balancer configuration with a GCP cloud account

VCF Automation users with a GCP cloud account can configure several aspects of load balancer configuration to support specific requirements, such as high availability. These features can support provisioning infrastructure for SAP NetWeaver and SAP Hana.

This configuration enables you to deploy different load balancer instances in different availability zones. To set this up, you need to configure the following components in GCP:

- Reserve virtual IP addresses
- Provision health checks
- Create a firewall rule to support the health checks

The following page describes how to set up and configure these components in GCP: https://cloud.google.com/solutions/sap/docs/netweaver-ha-config-rhel#configure_the_failover_support In addition, see the following pages for information about NetWeaver and Hana configurations:

- NetWeaver: https://cloud.google.com/solutions/sap/docs/netweaver-ha-config-rhel#configure_the_failover_support
- SAP Hana: https://cloud.google.com/solutions/sap/docs/sap-hana-ha-config-rhel#configure_the_failover_support

After you set up the GCP side, you must set up the appropriate components in VCF Automation as described below:

- Create a GCP cloud account if you don't already have one.
- Set up flavor and image mapping as appropriate for your environment.
- Create a default network profile and tag it so that you can invoke it from a cloud template.
- Create compute availability zones and tag them appropriately. These enable you to tell VCF Automation to provision instances and instance groups to specific availability zones in a cloud template.

When you have configured all of these components, you can navigate to the VCF Automation **Design** tab and create a cloud template. This template should model the appropriate load balancer configuration with specifications for health check and instances and instance groups as appropriate. For high availability purposes, you must configure multiple load balancers. The cloud template should also specify the appropriate virtual machines and the availability zones in which they should be deployed.

The following cloud template example demonstrates how one might set up a high availability load balancer configuration for NetWeaver.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_LoadBalancer_1:
    type: Cloud.LoadBalancer
    properties:
      routes:
        - healthCheckConfiguration:
            healthyThreshold: 2
            unhealthyThreshold: 2
            timeoutSeconds: 10
            intervalSeconds: 10
            port: 65000
            protocol: TCP
          protocol: TCP
          port: 1000
          instancePort: 1000
      network: ${resource.Cloud_Network_1.id}
      instances:
        - ${resource.Cloud_GCP_Machine_1.id}
        - ${resource.Cloud_GCP_Machine_2.id}
      internetFacing: false
      fallbackInstanceGroupsByInstanceNames:
        - ${resource.Cloud_GCP_Machine_2.resourceName}
```

```
Cloud_LoadBalancer_2:
  type: Cloud.LoadBalancer
  properties:
    routes:
      - healthCheckConfiguration:
          healthyThreshold: 2
          unhealthyThreshold: 2
          timeoutSeconds: 10
          intervalSeconds: 10
          port: 65000
          protocol: TCP
        protocol: TCP
        port: 1000
        instancePort: 1000
    network: ${resource.Cloud_Network_1.id}
    instances:
      - ${resource.Cloud_GCP_Machine_1.id}
      - ${resource.Cloud_GCP_Machine_2.id}
    internetFacing: false
    useInstanceGroupsFrom: ${resource.Cloud_LoadBalancer_1.resourceName}
    fallbackInstanceGroupsByInstanceNames:
      - ${resource.Cloud_GCP_Machine_1.resourceName}
Cloud_GCP_Machine_2:
  type: Cloud.GCP.Machine
  properties:
    image: image
    flavor: flavor
    networks:
      - network: ${resource.Cloud_Network_1.id}
    constraints:
      - tag: zone-b
Cloud_Network_1:
  type: Cloud.Network
  properties:
    networkType: existing
    constraints:
      - tag: default
Cloud_GCP_Machine_1:
  type: Cloud.GCP.Machine
  properties:
    image: image
    flavor: flavor
    networks:
      - network: ${resource.Cloud_Network_1.id}
    constraints:
      - tag: zone-a
```

Additional load balancer configuration options

Using cloud template configurations, VCF Automation users can configure the following additional aspects of load balancer configuration.

- Share instance groups between load balancers
- Configure failover instance groups
- Configure your load balancers to accept traffic on all ports

Share instance groups

To share instance groups between the load balancers the users must explicitly declare it in the cloud template. Because there isn't direct access to the instance groups in the cloud template, the customers must use the `useInstanceGroupsFrom` property. In this way VCF Automation will provision the first load balancer and will reuse the instance groups in the other load balancers.

The following example illustrates how you can use a cloud template to configure two load balancers where the second one uses the instance groups from the first one:

```
Cloud_LoadBalancer_1:
    type: Cloud.LoadBalancer
    properties:
      routes:
        - healthCheckConfiguration:
            healthyThreshold: 2
            unhealthyThreshold: 2
            timeoutSeconds: 10
            intervalSeconds: 10
            port: 65000
            protocol: TCP
          protocol: TCP
      network: ${resource.Cloud_Network_1.id}
      instances:
        - ${resource.Cloud_GCP_Machine_1.id}
        - ${resource.Cloud_GCP_Machine_2.id}
      internetFacing: false
  Cloud_LoadBalancer_2:
    type: Cloud.LoadBalancer
    properties:
      routes:
        - healthCheckConfiguration:
            healthyThreshold: 2
            unhealthyThreshold: 2
            timeoutSeconds: 10
            intervalSeconds: 10
            port: 65000
            protocol: TCP
          protocol: TCP
      network: ${resource.Cloud_Network_1.id}
      instances:
        - ${resource.Cloud_GCP_Machine_1.id}
        - ${resource.Cloud_GCP_Machine_2.id}
      internetFacing: false
      useInstanceGroupsFrom: ${resource.Cloud_LoadBalancer_1.resourceName}
```

Configure failover instance groups

To configure failover instance groups for each GCP load balancer, you must explicitly declare it in the cloud template. Because there isn't direct access to the instance groups in the template, you must use the `failoverInstanceGroupsByInstanceNames` property which tells VCF Automation to find the instance group by a machine that is contained in that instance group.

The following example illustrates configuration of a load balancer which has one primary and one failover instance group:

```
Cloud_LoadBalancer_1:
  type: Cloud.LoadBalancer
  properties:
    routes:
      - healthCheckConfiguration:
```

```
          healthyThreshold: 2
          unhealthyThreshold: 2
          timeoutSeconds: 10
          intervalSeconds: 10
          port: 65000
          protocol: TCP
        protocol: TCP
    network: ${resource.Cloud_Network_1.id}
    instances:
      - ${resource.Cloud_GCP_Machine_1.id}
      - ${resource.Cloud_GCP_Machine_2.id}
    internetFacing: false
    failoverInstanceGroupsByInstanceNames:
      - ${resource.Cloud_GCP_Machine_2.resourceName}
```

Configure for traffic on all ports

To configure a load balancer to accept traffic on all ports, you can omit the port property of the route configuration. You can view a sample cloud template that uses the three newly created features and provisions the infrastructure for this setup. For related information, see Configure the Cloud Load Balancing failover support.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_LoadBalancer_1:
    type: Cloud.LoadBalancer
    properties:
      routes:
        - healthCheckConfiguration:
            healthyThreshold: 2
            unhealthyThreshold: 2
            timeoutSeconds: 10
            intervalSeconds: 10
            port: 65000
            protocol: TCP
          protocol: TCP
      network: ${resource.Cloud_Network_1.id}
      instances:
        - ${resource.Cloud_GCP_Machine_1.id}
        - ${resource.Cloud_GCP_Machine_2.id}
      internetFacing: false
      failoverInstanceGroupsByInstanceNames:
        - ${resource.Cloud_GCP_Machine_2.resourceName}
      address: 10.132.0.72
  Cloud_LoadBalancer_2:
    type: Cloud.LoadBalancer
    properties:
      routes:
        - healthCheckConfiguration:
            healthyThreshold: 2
            unhealthyThreshold: 2
            timeoutSeconds: 10
            intervalSeconds: 10
            port: 65000
            protocol: TCP
          protocol: TCP
      network: ${resource.Cloud_Network_1.id}
      instances:
```

```
        - ${resource.Cloud_GCP_Machine_1.id}
        - ${resource.Cloud_GCP_Machine_2.id}
      internetFacing: false
      useInstanceGroupsFrom: ${resource.Cloud_LoadBalancer_1.resourceName}
      failoverInstanceGroupsByInstanceNames:
        - ${resource.Cloud_GCP_Machine_1.resourceName}
      address: 10.132.0.64
  Cloud_GCP_Machine_2:
    type: Cloud.GCP.Machine
    properties:
      image: image
      flavor: flavor
      networks:
        - network: ${resource.Cloud_Network_1.id}
      constraints:
        - tag: zone-c
  Cloud_Network_1:
    type: Cloud.Network
    properties:
      networkType: existing
      constraints:
        - tag: default
  Cloud_GCP_Machine_1:
    type: Cloud.GCP.Machine
    properties:
      image: image
      flavor: flavor
      networks:
        - network: ${resource.Cloud_Network_1.id}
      constraints:
        - tag: zone-b
```

Create a basic vCenter cloud account in VCF Automation

You can add a basic vCenter cloud account for the account regions to which you want to deploy VCF Automation cloud templates.

- Verify that you have the required service account credentials and have enabled HTTPS access on port 443.
- Verify that you have the cloud administrator user role. See What are the VCF Automation for VM Apps user roles.
- If you are creating the cloud account for use with a remote vSphere agent (which is possible by using the **Forward all traffic through proxy** option), verify that you have deployed and configured a VCF Automation extensibility (vREx) proxy on the target remote vCenter server.
- Verify that you have properly configured your ports and protocols to support the cloud account.

For network and security purposes, you can associate a vCenter cloud account with an NSX cloud account. An NSX cloud account can be associated to one or more vCenter cloud accounts.

1. Select **Infrastructure** > **Connections** > **Cloud Accounts** and click **Add Cloud Account**.
2. Select the vCenter cloud account type.
3. Enter the vCenter server host IP address or FQDN.
4. Enter your vCenter server administrator credentials and click **Validate**.

   All data centers that are associated with the vCenter account are data-collected.

   The following elements are data-collected, as are all vSphere tags for the following elements:

   - Machines

- Clusters and hosts
- Port groups
- Data stores

Remote vSphere agents (proxies) are also collected if they have been configured on the vCenter

You might consider using the VCF Automation extensibility proxy option to gain a more reliable connection to a distant server, to get better processing throughput, or for network isolation reasons. To use a remote vSphere agent, where you have already met the required prerequisites for deploying and configuring a VCF Automation extensibility (vREx) proxy, select the Forward all traffic through proxy check box and select the configured VCF Automation extensibility (vREx) proxy from the drop-down menu of available VCF Automation extensibility (vREx) proxies. After you select a vREx proxy, you must revalidate and save your user name and password credentials.

5. Select at least one of the available data centers on the specified vCenter server to allow provisioning for this cloud account.
6. For efficiency, create a cloud zone for provisioning to the selected data centers.

   You can also create cloud zones as a separate step according to your organization's cloud strategy.

   For information about cloud zones, see Learn more about cloud zones in VCF Automation for VM Apps.
7. Select an existing NSX cloud account.

   You can select the NSX account now, or later when you edit the cloud account.

   For information about NSX cloud accounts, see Create an NSX cloud account in VCF Automation for VM Apps.
8. Add any Site Associations.
9. If you want to add tags to support a tagging strategy, enter capability tags.

   You can add tags now, or later when you edit the cloud account. For information about tagging, see How do I use tags to manage VCF Automation resources and deployments.
10. Click **Add**.

The cloud account is added and the selected data centers are available for the specified cloud zone. Collected data such as machines, networks, storage, and volumes is listed in the **Resources** section of the **Infrastructure** tab.

Configure remaining infrastructure resources for this cloud account. See Building your VCF Automation for VM Apps resource infrastructure.

Create an NSX cloud account in VCF Automation for VM Apps

For network and security purposes, you can create an NSX cloud account and associate it with one or more vCenter cloud accounts.

- Verify that you have the required administrator credentials and have enabled HTTPS access on port 443. Also verify that you have properly configured your ports and protocols to support the cloud account.
- Verify that you have the cloud administrator user role. See What are the VCF Automation for VM Apps user roles.
- Verify that you have a vCenter cloud account to use with this NSX cloud account. See Create a basic vCenter cloud account in VCF Automation.

An NSX cloud account can be associated to one or more vCenter cloud accounts.

The association between NSX and one or more vCenter cloud accounts must be configured outside of VCF Automation, specifically in your NSX application. VCF Automation doesn't create the association between NSX and vCenter. In VCF Automation, you specify one or more configuration associations that already exists in NSX.

When you create an NSX cloud account in VCF Automation, you specify a manager type and an NSX mode. These selections cannot be changed after you create the cloud account.

You can connect to an NSX Global Manager and configure an association between an NSX Global Manager and local managers in the context of the NSX federation.

For related information about NSX options and capabilities in general, see the NSX Documentation.

To facilitate fault tolerance and high availability in deployments, each NSX endpoint represents a cluster of three NSX Managers.

- VCF Automation can point to one of the NSX Managers. Using this option, one NSX Manager receives the API calls from VCF Automation.
- VCF Automation can point to the Virtual IP of the cluster. Using this option, one NSX Manager assumes control of the VIP. That NSX Manager receives the API calls from VCF Automation. In case of failure, another node in the cluster assumes control of the VIP and receives the API calls from VCF Automation.

  For more information about VIP configuration for NSX, see the NSX documentation.
- VCF Automation can point to a load balancer VIP to load-balance the calls to the three NSX Managers. Using this option, all three NSX Managers receive API calls from VCF Automation.

  You can configure the VIP on a third-party load balancer or on an NSX load balancer.

  For large scale environments, consider using this option to split the VCF Automation API calls among the three NSX Managers.

1. Select **Infrastructure** > **Connections** > **Cloud Accounts** and click **Add Cloud Account**.
2. Select the NSX account type and specify a cloud account name and description.
3. Enter the host IP address or FQDN for the NSX Manager instance or VIP (see above for information about the expected behavior that pertains to the NSX Manager and VIP options).
4. Enter your NSX user name and password administrator credentials.
5. For **Manager type**, select either **Global** or **Local** (default).
   - Global Manager

     The Global Manager setting is only available for use with the Policy **NSX mode** setting. It is not available when using the Manager **NSX mode** setting.

     The Global setting refers to the NSX federation capabilities, including global network segments. Only NSX cloud accounts with the Global setting support NSX federation.

     When using the Global Manager setting, you are prompted to identify a Local Manager NSX cloud account and an associated vCenter cloud account.

     You cannot associate a Global Manger NSX cloud account with vCenter cloud account, as you can with an Local Manager NSX cloud account. Similar to how a Local Manager NSX cloud account can be associated to multiple vCenter cloud accounts, a Global Manager NSX cloud account can be associated to multiple Local Manager NSX cloud accounts.
   - Local Manager

     Use the Local setting to define a traditional NSX cloud account, which can be associated to one or more vSphere cloud accounts. You can associate a Global manager NSX cloud account with a Local NSX cloud accounts. Note that this is also the setting to use if you are creating a new and empty target NSX cloud account for the purposes of NSX-V to NSX migration.

   You cannot change the **Manager type** setting after you create the cloud account.
6. For **NSX mode**, **Policy** mode is automatically selected.

   The Policy mode is available for NSX 3.0 and NSX 3.1 forward. This option enables VCF Automation to use the additional capabilities available in the NSX Policy API.

   If you are using NSX with a VMware Cloud on AWS cloud account in a cloud template, the NSX cloud account must use the Policy **NSX mode**.

   The Policy setting refers to the NSX Policy API form of NSX.

   If you have existing NSX cloud accounts that were created prior to the introduction of the Policy mode in VCF Automation 8.2, they use the Manager API method. It is recommended that you replace your existing NSX cloud accounts with new NSX cloud accounts that specify the Policy API method.
7. Click **Validate** to confirm the credentials in relation to the selected NSX Manager type and NSX mode.

   The assets associated with the account are collected.

If the NSX host IP address is not available, validation fails.

8. Add any Site Associations.
9. If you want to add tags to support a tagging strategy, enter capability tags.

   You can add or remove capability tags later. See How do I use tags to manage VCF Automation resources and deployments.
10. Click **Save**.

You can create or edit a vCenter cloud account to associate with this NSX cloud account. See Create a basic vCenter cloud account in VCF Automation.

Create and configure one or more cloud zones for use with the data centers that are used by this cloud account. See Learn more about cloud zones in VCF Automation for VM Apps.

Configure infrastructure resources for this cloud account. See Building your VCF Automation for VM Apps resource infrastructure.

For samples of using NSX options in VCF Automation cloud templates, see Network, security group, and load balancer resource examples in VCF Automation.

Create a VMware Cloud Foundation (VCF) cloud account in VCF Automation for VM Apps

You can configure a VMware Cloud Foundation (VCF) as a cloud account within VCF Automation to use workload domains.

You must have an instance of VMware SDDC Manager 4.1 or higher configured as an VCF Automation integration for use with this cloud account.

A VCF cloud account enables you to incorporate a VCF workload into VCF Automation to facilitate a comprehensive hybrid cloud management solution. VCF Automation offers several entry points from which you can activate the VCF cloud account configuration page. If you access this page using the **Add Cloud Account** button on the SDDC integration Workload Domain tab, the workload is pre-selected, including the basic information for the vCenter and NSX Manager.

1. Select **Infrastructure** > **Connections** > **Cloud Accounts** and click **Add Cloud Account**.
2. Select the VCF Cloud Account type, and enter a **Name** and **Description**.
3. Select the SDDC manager instance that you are using with this cloud account.
4. Select a workload domain that you want to use with this VCF cloud account.
5. If you want to have VCF Automation use Cloud Foundation managed service credentials for vCenter and NSX, select **Use cloud foundation managed service credentials**. Later, if you want to change these credentials, you must use the VCF mechanism for password management.

   If you select this option, you can skip steps 6, 7, and 8.
6. Enter the credentials required to access the vCenter associated with this cloud account.
7. Under the NSX Manager heading, enter NSX credentials if you want to manually enter credentials for the VCF cloud account, or click Create and Validate Service Credentials if you want VCF Automation to create and validate NSX credentials.
8. Enter the credentials required to access the NSX network associated with this cloud account.
9. Click **Validate** to confirm a connection to the SDDC manager.
10. If applicable, select the data centers that you want to provision to under the Configuration heading. Click the check box if you want to create a cloud zone for the selected data centers.
11. If you use tags to support a tagging strategy, enter capability tags.

    Note:
    When capability tags are applied to a VMware Cloud Foundation (VCF) cloud account, these tags are not inherited by deployments for the underlying vCenter Server and NSX components. As a workaround, you can apply tags at the next level, such as on the cloud zones.

12. Click **Add**.

This cloud account brings the selected workload domain associated with the specified SDDC manager into VCF Automation for use.

If you want to manage additional workload domains using VCF Automation, you must repeat this process for each domain.

Building your VCF Automation for VM Apps resource infrastructure

VCF Automation resource infrastructure is where you define cloud account regions as zones into which cloud templates and their workloads can be deployed.

In addition, resource infrastructure involves creation of common mappings of images and machine sizes, and profiles that define network and storage capabilities across cloud account regions or data centers.

> **Important:**
>
> Starting with VCF Automation 9.1.1, native management capabilities for public cloud resources, including Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP), are deprecated and disabled by default in VM Apps organizations. For more information, see the VCF Automation 9.1.1 release notes.

How to add cloud zones that define VCF Automation for VM Apps target placement regions or data centers

A VCF Automation cloud zone is a set of resources within a cloud account type such as AWS or vSphere.

Cloud zones in a specific account region are where your cloud templates deploy workloads. Each cloud zone is associated with a VCF Automation project.

Select **Infrastructure** > **Configure** > **Cloud Zones** and click **Add New Zone**.

Learn more about cloud zones in VCF Automation for VM Apps

VCF Automation cloud zones are sections of compute resources that are specific to your cloud account type such as AWS or vSphere.

Cloud zones are specific to a region, you must assign them to a project. There is a many-to-many relationship between cloud zones and projects. VCF Automation supports deployment to the most popular public clouds including Azure, AWS, and GCP as well as to vSphere. See Adding cloud accounts to VCF Automation for VM Apps.

Additional placement controls include placement policy options, capability tags, and compute tags.

You can add up to 3000 cloud zones to a project. However, the maximum amount of cloud zones that can be part of a cloud template deployment can be no more than 100. At the deployment stage, the cloud zones are filtered based on assigned constraint tags or cloud account. If there are more than 100 applicable cloud zones left, the deployment fails.

Placement policy

Placement policy drives host selection for deployments within the specified cloud zone.

- default - Distributes compute resources across clusters and hosts machines based on availability. For example, all machines in a particular deployment are provisioned on the first applicable host.
- binpack - Places compute resources on the most loaded host that has enough available resources to run the given compute.
- spread - Provisions compute resources, at a deployment level, to the cluster or host with the least number of virtual machines. For vSphere, Distributed Resource Scheduler (DRS) distributes the virtual machines across the hosts. For example, all requested machines in a deployment are placed on the same cluster, but the next deployment may choose another vSphere cluster depending on the current load.

  For example, let's assume you have the following configuration:

  – DRS cluster 1 with 5 virtual machines

  - DRS cluster 2 with 9 virtual machines
  - DRS cluster 3 with 6 virtual machines

  If you request a cluster of 3 virtual machines and you select a Spread policy, they should all be placed on cluster 1. The updated loads become 8 virtual machines for cluster 1, while the loads for clusters 2 and 3 remain the same at 9 and 6.

  Then, if you request an additional 2 virtual machines, they are placed on DRS cluster 3, which will now have 8 virtual machines. The load for clusters 1 and 2 remain the same at 8 and 9.
- spread by memory - Provisions compute resources, at a deployment level, to the cluster or host with the greatest amount of free memory. However, the free memory calculation for private cloud zones is different from public cloud zones.
  - Private cloud zones - Total memory of the hosts or clusters is retrieved for the zone. The amount of allocated memory for all the managed machines is retrieved. The allocated memory is divided by the total memory to determine the ratio, which is inversely proportional to the size of the zone or cluster. The zone or cluster with the smallest ratio is considered to have the most available memory because it has the largest amount of free memory proportional to its size.
  - Public cloud zones - Total memory cannot be retrieved from the hosts or clusters. Therefore, the amount of memory allocated to all the managed virtual machines is retrieved. The cloud zones are ordered by the amount of allocated memory so that the zone with the least amount of allocated memory is prioritized.

If two cloud zones both match all the criteria needed for provisioning, then the placement logic selects the one with the highest defined project provisioning priority.

For more information about how the selected cloud zone placement policy interacts with a project placement policy, see *How do project-level placement policies affect resource allocation*.

Capability tags

Cloud template contain constraint tags to help determine deployment placement. During deployment, cloud template constraint tags are mapped to matching capability tags in cloud zones and compute resources to determine which cloud zones are available for virtual machine resource placement.

Compute resources

You can view and manage the compute resources that are available to provision workloads, such as AWS availability zones and vCenter clusters, to this cloud zone.

> **Note:**
>
> Cloud zones cannot share compute resources. Legacy cloud zones that use shared compute resources are still supported, but users are prompted to update them to conform with current standards.
>
> Cloud zones that are auto-generated during cloud account creation are associated with the underlying compute resources after data collection.

If a vCenter compute cluster is DRS-enabled, the cloud zone only displays the cluster in the list of computes and it does not display the child hosts. If a vCenter compute cluster is not DRS-enabled, the cloud zone only displays standalone ESXi hosts, if present.

Add compute resources as appropriate for the cloud zone. The Compute tab contains a filter mechanism that enables you to control how compute resources are included with cloud zones. Initially, the filter selection is Include all Compute and the list below shows available compute resources, and they are all available for use in deployments. You have two additional options for adding compute resources to a cloud zone.

- Manually select compute - Select this option if you want to manually select compute resources from the list below. After you select them, click Add Compute to add the resources to the zone. The selected resources are available for use in deployments.
- Dynamically include compute by tags - Select this option if you want to include or exclude compute resources for the zone based on tags. All compute resources are shown until you add appropriate tags that match existing tags on

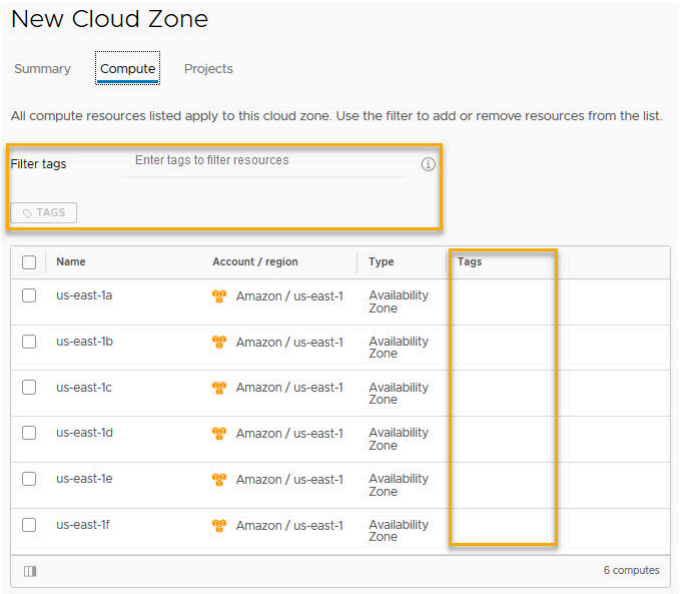
compute resources. After you add one or more tags, compute resources with tags that match the filter are included in the zone and are available for use in deployments, while those that don't match are excluded.

- Include all unassigned compute - If you select this option, the cloud zone is not updated dynamically when new computes become available. Only the computes that were available when it was created are present in the zone.

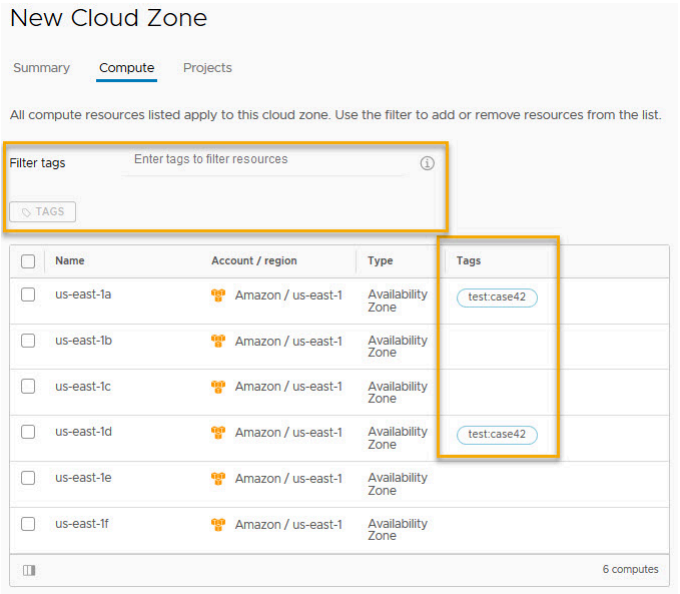
For either compute option, you can remove one or more of the compute resources shown on the page by selecting the box to the right and clicking Remove.

Compute tags help to further control placement. You can use tags to filter available compute resources to only those that match one or more tags, as shown in the following examples.

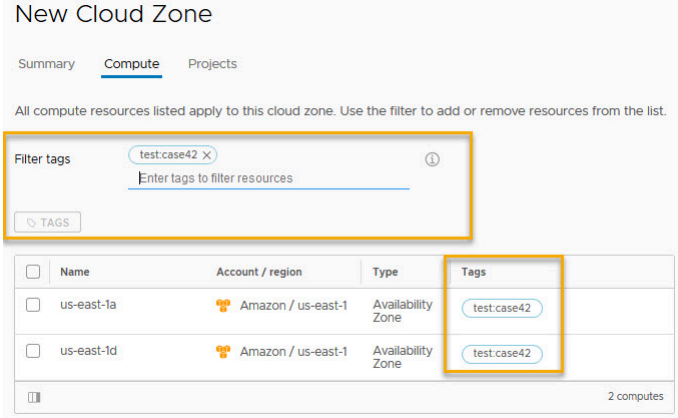
- Computes contain no tags and no filtering is used.



- Two computes contain the same tag but no filtering is used.



- Two computes contain the same tag and the tag filter matches the tag used on the two computes.



Projects

You can view which projects have been configured to support workload provisioning to this cloud zone.

Insights dashboard

If you have an associated VCF Operations application that you have configured to work with VCF Automation, you can access an **Insights** dashboard in the cloud zone. The dashboard displays capacity-related information about resources and deployments for the vSphere or VMware Cloud on AWS cloud zone, provided that the cloud accounts are configured in both VCF Automation and VCF Operations and are being monitored in VCF Operations. To learn more about the **Insights** dashboard, see Resource management and deployment optimization using VCF Operations metrics in VCF Automation.

How to add flavor mappings in VCF Automation for VM Apps to specify common machine sizing

A VCF Automation flavor map is where you use natural language to define target deployment sizes for a specific cloud account/region.

Flavor maps express the deployment sizes that make sense for your environment. One example might be *small* for 1 CPU and 2 GB memory and *large* for 2 CPUs and 8 GB memory for a vCenter account in a named datacenter and *t2.nano* for an Amazon Web Services account in a named region.

1. In VCF Automation, select **Infrastructure** > **Configure** > **Flavor Mappings** and click **New**.
2. Respond to onscreen prompts to create the flavor mapping.

Learn more about flavor mappings in VCF Automation for VM Apps

A flavor mapping groups a set of target deployment sizings for a specific cloud account/region in VCF Automation using natural language naming.

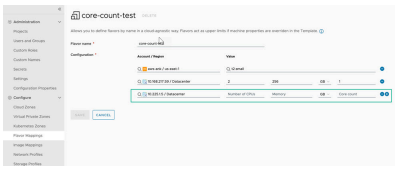
Flavor mapping lets you create a named mapping that contains similar flavor sizings across your account regions. For example, a flavor map named `standard_small` might contain a similar flavor sizing (such as 1 CPU, 2 GB RAM) for some or all available account/regions in your project. When you build a cloud template, you pick an available flavor that fits your needs.

Organize flavor mappings for your project by deployment intent.

To simplify cloud template creation, you can select a pre-configuration option when you add a new cloud account. When you select the pre-configuration option, your organization's most popular flavor mapping and image mapping for the specified region are selected.

With regard to image mapping in cloud templates that contain vSphere resources, if there are no flavor mappings defined for a vSphere cloud zone, you can configure unlimited memory and CPU by using vSphere-specific settings in the cloud template. If there are flavor mappings defined for a vSphere cloud zone, the flavor mapping serves as a limit for vSphere-specific configurations in the cloud template.

You can specify a maximum number of vCPUs per socket in your flavor mappings. During compute allocation, this specification addresses the `coreCount` property in a VCF Automation cloud template and when provisioning the deployment.



How to add image mapping in VCF Automation for VM Apps to access common operating systems

A VCF Automation image map is where you use natural language to define target deployment operating systems for a specific cloud account/region.

1. In VCF Automation, select **Infrastructure** > **Configure** > **Image Mappings** and click **New**.
2. Respond to onscreen prompts to create the image mapping.

> **Note:**
> You can create an image mapping in multiple regions by activating the **Create the same mapping in multiple regions** toggle and selecting **Add account/region**. You can also perform this operation by cloning an image mapping. To do this, select **View ungrouped** in the grid view and click **Clone**.

Learn more about image mappings in VCF Automation for VM Apps

An image mapping groups a set of predefined target operating system specifications for a specific cloud account/region in VCF Automation by using natural language naming.

Cloud vendor accounts such as Microsoft Azure and Amazon Web Services use images to group a set of target deployment conditions together, including OS and related configuration settings. vCenter and NSX-based environments, including VMware Cloud on AWS, use a similar grouping mechanism to define a set of OS deployment conditions. When you build and eventually deploy and iterate a template, you pick an available image that best fits your needs.

Organize image mappings for a project by similar operating system settings, tagging strategy, and functional deployment intent.

To simplify template creation, you can select a pre-configuration option when you add a new cloud account. When you select the pre-configuration option, your organization's most popular flavor mapping and image mapping for the specified region are selected.

When you add image information to a template, you use either the `image` or `imageRef` entry in the `properties` section of a machine component. For example, if you want to clone from a snapshot, use the `imageRef` property.

For examples of `image` and `imageRef` entries in template code, see Designing cloud templates in VCF Automation for VM Apps.

To assign a permission on a content library, an administrator must grant the permission to the user as a global permission. For related information, see the vSphere documentation.

Applying an image mapping to multiple regions

You can apply an image mapping to cloud accounts in multiple regions. Use the image mapping UI to filter and search your cloud accounts by account type (for example, AWS or Azure), and then select an image to apply to that cloud account type in multiple regions at the same time. For example, you can specify that all ubuntu20 images be available for all AWS cloud accounts in one or more specific AWS AMI regions. This capability also allows you to display, edit, and delete images mappings for one or more regions at a time.

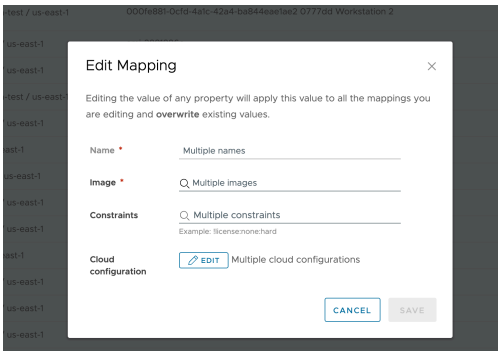
- Search and filter configurable cloud accounts and regions across properties (for example account type, region name, and so on).
- Select and deselect which regions to apply an image mapping to based on the search filter results.
- Apply an updated image mapping to multiple regions at the same time.

To display ungrouped image mappings, select **View ungrouped** in the **Filter** drop-down menu.

The **View ungrouped** option displays all of the image mappings that are not grouped by a name or region.

When selecting all items on an ungrouped page, an information label appears that allows you to select all items on all pages.

Selected image mappings can be edited or deleted. A sample **Edit Mapping** menu is shown.

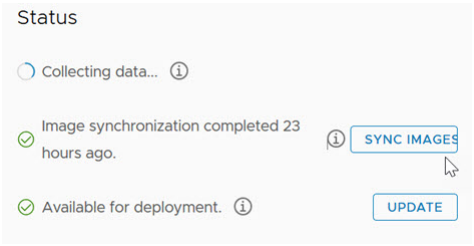


Note that some edit options are not available for certain selection scenarios. For example, a scenario in which you have selected multiple cloud account types would not allow you to edit an image (on the **Edit Mapping** menu) because the cloud account types would not have a common image. Use the following table to determine if you can edit an image (on the **Edit Mapping** menu) for your selected items.

Synchronizing images for the cloud account/region

To ensure that the images you are adding or removing for a given cloud account/region on the VCF Automation**Infrastructure** > **Configure** > **Image Mappings** page are current, run image synchronization.

1. Open the associated **Cloud Account/Region** by selecting **Infrastructure** > **Connections** > **Cloud Accounts** in VCF Automation.
2. Select the existing cloud account/region.
3. Click the **Sync Images** button and let the action complete.



4. When the action is complete, click **Infrastructure** > **Configure** > **Image Mapping**.
5. Define a new or edit an existing image mapping and select the cloud account/region from step 1.
6. Click the image synchronization icon on the **Image Mapping** page.

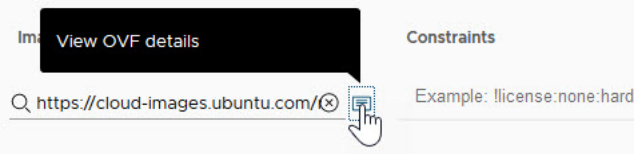


7. Configure image mappings settings for the specified cloud account/region on the **Image Mapping** page.

Viewing OVF details

You can include OVA/OVF specifications in VCF Automation template objects, such as vCenter machine components and images.

If your image contains an OVF file, you can also discover its content by hovering over the *View OVF details* option to display OVF details such as name and location. For more information about the OVF file format, see `vcenter ovf: property`. To view the OVF details, the image mapping must reside on the web server.



Note:
You cannot use OVF properties of images in vCenter content libraries when provisioning from a VCF Automation cloud template. If the OVF is hosted directly in vCenter, the *View OVF details* option is not visible or otherwise available. The *View OVF details* option is not available for images in content libraries.

Using shared and latest images from a Microsoft Azure image gallery

When creating image mappings for Microsoft Azure, you can select images from a shared Azure image gallery in the subscription. The images in the drop-down menu are data-collected and made available based on your selected region.

While shared image galleries can be used across multiple subscriptions, they cannot be listed in the image mapping drop-down menu across subscriptions. Only the images of a particular subscription are data-collected and listed in the image mappings list. To use an image from an image gallery in a different subscription, provide the image ID in the image mapping and use that image mapping in the template.

Using constraints and tags to select a suitable compute for the image

You can use image constraints to select a suitable compute based on the compute's tag. For example, if you have a `tag1` constraint on a particular image and if the image is selected during allocation, then a compute with tag `tag1` is selected for placing the machine being provisioned. If such compute does not exist, the allocation of the machine fails.

Using a cloud configuration script to control deployment

You can use a cloud configuration script in an image map, template, or both to define custom OS characteristics to be used in an VCF Automation deployment. For example, based on whether you are deploying a template to a public or private cloud, you can apply specific user permissions, OS permissions, or other conditions to the image. A cloud configuration script adheres to a `cloud-init` format for Linux-based images or a `cloudbase-init` format for Windows-based images. VCF Automation supports the cloud-init tool for Linux systems and the cloudbase-init tool for Windows.

For Windows machines, you can use any cloud configuration script format that is supported by `cloudbase-init`.

The machine resource in the following sample template code uses an image that contains a cloud configuration script, the content of which is seen in the `image` entry.

```
resources:
demo-machine:
type: Cloud.vSphere.Machine
properties:
flavor: small
image: MyUbuntu16
https://cloud-images.ubuntu.com/releases/16.04/release-20170307/ami-ubuntu-16.04-1.10.3-00-15269239.ova
cloudConfig: |
ssh_pwauth: yes
chpasswd:
list: |
${input.username}:${input.password}
expire: false
users:
- default
- name: ${input.username}
lock_passwd: false
sudo: ['ALL=(ALL) NOPASSWD:ALL']
groups: [wheel, sudo, admin]
```

```
shell: '/bin/bash'
runcmd:
- echo "Defaults:${input.username}  !requiretty" >> /etc/sudoers.d/${input.username}
```

Dynamic property evaluation works when using cloudConfig directly in a template, but isn't supported for cloudConfig in an image map.

In the template code, you use the `image` setting to reference an image that is defined as an image mapping. You use the `imageRef` setting to identify a template that contains a snapshot (for linked clones), an image template, or a content library template OVF.

What happens when an image mapping and a template contain a cloud configuration script

When a template that contains a cloud configuration script uses an image mapping that contains a cloud configuration script, both scripts are combined. The merge action processes the contents of the image mapping script first and the contents of the template script second, with consideration being given to whether the scripts are in `#cloud-config` format or not.

- For scripts that are in the `#cloud-config` format, the merge combines the contents of each module (for example `runcmd`, `users`, and `write_files`) as follows:
  - For modules where the contents are a list, the lists of commands from the image mapping and from the template are merged, excluding commands that are identical in both lists.
  - For modules where the contents are a dictionary, the commands are merged and the result is a combination of both dictionaries. If the same key exists in both dictionaries, the key from the image mapping script dictionary is preserved and the key from the template script dictionary is ignored.
  - For modules where the contents are a string, the content values from the image mapping script are kept and the content values from the template script are ignored.
- For scripts that are in a format other than `#cloud-config` or when one script is in `#cloud-config` format and the other is not, both scripts are combined in a way that the image mapping script is run first and the template script is run when the image mapping script is finished.

Add an image from a vCenter content library

When a local or publisher content library resides in a vCenter that is managed by your VCF Automation organization, content library template images appear in the image drop-down menu. The images listed include OVF and VM template images in local or publisher vCenter content libraries. Images in subscriber content libraries do not appear in the drop-down menu. The template from which a VM has been cloned is shown in the machine details section of the machine deployments user interface.

> **Note:**
>
> If the publisher content library vCenter is managed by VCF Automation, then publisher information is displayed in the image mapping selection grid in the following format: *`publisher_content_library_name`* / *`content_item_name`*
>
> To assign a permission on a content library, an administrator must grant the permission to the user as a global permission.
>
> If the publisher content library vCenter is not managed by VCF Automation, then subscriber information is displayed in the image mapping selection grid in the following format: *`subscriber_content_library_name`* / *`content_item_name`*
>
> For example, in the following scenario only the subscriber content library items are visible in the VCF Automation image mapping list:
>
> - For a vCenter named VC-1, there is a subscriber content library in the VC and a cloud account is created in VCF Automation that is associated to VC-1.
> - For a vCenter named VC-2, there is a publisher content library in the VC that the subscriber content library of VC-1 is subscribed from. However, there is no cloud account in VCF Automation that is associated to VC-2.
>
> Because VC-1 is associated to a VCF Automation cloud account, the subscriber content library is available in VCF Automation. Its contents are collected and displayed in the VCF Automation image mapping list.

However, because VC-2 is not associated to a cloud account, VCF Automation has no knowledge of its publisher content library. To display the publisher content library items in the image mapping list, you must associated a cloud account to the VC-2 vCenter.

When you deploy a template that contains a VM template image mapping, VCF Automation attempts to access the mapped image in the content library that is closest to the datastore, and then closest to the host, of the machine to be provisioned. This can include a local content library as well as a publisher or subscriber content library.

When you deploy a template that contains an OVF template image mapping, OVF images are accessed as specified in the image mapping row if the image is in a local content library or a local subscriber of a specified remote publisher content library.

The VM template in vCenter includes multiple disks known as image disks. When deploying a VM using a VM template as an image from VCF Automation, one of the image disks is selected as a boot-disk. The allocation and placement of the boot-disk can be controlled adding constraints. To add a constraint, select your cloud template and navigate to **vSphere Machine** > **Properties** > **Storage** > **Constraints**. Other image disks follow the allocation and placement decision made for the boot-disk. You cannot control the placement of each individual image disk.

More information about configuring and using cloud configuration scripts

For more information about working with cloud configuration scripts in templates, see Machine initialization in VCF Automation for VM Apps .

Batch operations for image mappings

VCF Automation supports batch operations through which you can easily manage large groups of image mappings. These operations allow you to create, clone, edit, and delete multiple image mappings. To access these batch operations, select **View ungrouped** in the grid view. For example, you can perform a batch edit on multiple image mappings by selecting **Edit**.

Note:
You cannot give the same name to multiple image mappings from the same region.

While in the image mapping editor, you can also use the **Match by name** function to create or edit multiple mappings across cloud accounts and providers based on the specified name. When performing this operation VCF Automation searches for a private image first. If no private image is available, the service searches for a public image or an image shared across data centers in vSphere.

How to add network profiles in VCF Automation

A VCF Automation network profile describes the behavior of the network to be deployed.

For example, a network might need to be Internet-facing rather than internal-only.

Networks and network profiles are cloud-specific.

Select **Infrastructure** > **Configure** > **Network Profiles** and click **New**.

Learn more about network profiles in VCF Automation

A network profile defines a group of networks and network settings that are available for a cloud account in a particular region or datacenter in VCF Automation.

You typically define network profiles to support a target deployment environment, for example a small test environment where an existing network has outbound access only or a large load-balanced production environment that needs a set of security policies. Think of a network profile as a collection of workload-specific network characteristics.

What's in a network profile

A network profile contains specific information for a named cloud account type and region in VCF Automation, including the following settings:

- Named cloud account/region and optional capability tags for the network profile.
- Named existing networks and their settings.
- Network policies that define on-demand and other aspects of the network profile.
- Optional inclusion of existing load balancers.
- Optional inclusion of existing security groups.

You determine the network IP management functionality based on the network profile.

> **Note:**
>
> VCF Automation supports one network profile for any given VM or VM cluster compute resource. If a VM or VM cluster compute resource has multiple networks and separate network profiles are selected for each, allocation will fail during template deployment. A VM or VM cluster compute resource can only support one network profile, not multiple network profiles.

Network profile capability tags are matched with constraint tags in cloud templates to help control network selection. Further, all tags that are assigned to the networks that are collected by the network profile are also matched with tags in the cloud template to help control network selection when the cloud template is deployed.

Capability tags are optional. Capability tags are applied to all networks in the network profile, but only when the networks are used as part of that network profile. For network profiles that do not contain capability tags, tag matching occurs on the network tags only. The network and security settings that are defined in the matched network profile are applied when the cloud template is deployed.

When using static IP, the address range is managed by VCF Automation. For DHCP, the IP start and end addresses are managed by the independent DHCP server, not by VCF Automation. When using DHCP or mixed network address allocation, the network utilization value is set to zero. An on-demand network allocated range is based on the CIDR and subnet size that is specified in the network profile. To support both static and dynamic assignment in the deployment, the allocated range is divided into two ranges - one for static allocation and another for dynamic allocation.

Networks

Networks, also referred to as subnets, are logical subdivisions of an IP network. A network groups a cloud account, IP address or range, and network tags to control how and where to provision a cloud template deployment. Network parameters in the profile define how machines in the deployment can communicate with one another over IP layer 3. Networks can have tags.

You can add networks to the network profile, edit aspects of networks that are used by the network profile, and remove networks from the network profile.

When you add a network to the network profile, you can select available networks from a filtered list of vSphere and NSX networks. If the network type is supported for the cloud account type, you can add it to the network profile.

In a VCF-based deployment, NSX network segments are created locally on the NSX network and are not created as global networks.

- **Network domain** or **Transport zone**
  A network domain or transport zone is the distributed virtual switch (dvSwitch) for the vSphere vNetwork Distributed PortGroups (dvPortGroup). A *`transport zone`* is an existing NSX concept that is similar to terms like *`dvSwitch`* or *`dvPortGroup`*.
  When using an NSX cloud account, the element name on the page is **Transport zone**, otherwise it is **Network domain**.
  For standard switches, the network domain or transport zone is the same as the switch itself. The network domain or transport zone defines the boundaries of the subnets within vCenter.
  A transport zone controls which hosts an NSX logical switch can reach to. It can span one or more vSphere clusters. Transport zones control which clusters and which virtual machines can participate in the use of a particular network. Subnets that belong to the same NSX transport zone can be used for the same machine hosts.
- **Domain**
  Represents the domain name for the machine. The domain name is passed to the vSphere machine customization spec.
- **IPv4 CIDR** and **IPv4** default gateway

vSphere machine components in the cloud template support IPv4, IPv6, and dual stack IP assignment for network interfaces. For example, 192.168.100.14/24 represents the IPv4 address 192.168.100.14 and its associated routing prefix 192.168.100.0, or equivalently, its subnet mask 255.255.255.0, which has 24 leading 1-bits. The IPv4 block 192.168.100.0/22 represents the 1024 IP addresses from 192.168.100.0 to 192.168.103.255.

- **IPv6 CIDR** and **IPv6** default gateway
  vSphere machine components in the cloud template support IPv4, IPv6, and dual stack IP assignment for network interfaces. For example, 2001:db8::/48 represents the block of IPv6 addresses from 2001:db8:0:0:0:0:0:0 to 2001:db8:0:ffff:ffff:ffff:ffff:ffff.
  The IPv6 format is not supported for on-demand networks. For related information, see Using network settings in network profiles and cloud templates in VCF Automation.

  Note:

  You can assign an IPv6 address space to single and dual stacked workloads when an IPv6 address is requested per integration with an external IPAM system. This capability requires that the integrated external IPAM plug-in also supports IPv6.
- **DNS servers** and **DNS search domains**
- **Support public IP**
  Select this option to flag the network as public. Network components in a cloud template that have a `network type: public` property are matched to networks that are flagged as public. Further matching occurs during cloud template deployment to determine network selection.
- **Default for zone**
  Select this option to flag the network as a default for the cloud zone. During cloud template deployment, default networks are preferred over other networks.
- **Origin**
  Identifies the network source.
- **Tags**
  Specifies one or more tags assigned to the network. Tags are optional. Tag matching affect which networks are available for your cloud template deployments.
  Network tags exist on the network item itself, irrespective of the network profile. Network tags apply to every occurrence of the network they have been added to and to all network profiles that contain that network. Networks can be instanced into any number of network profiles. Regardless of network profile residency, a network tag is associated with that network wherever the network is used.
  When you deploy a cloud template, constraint tags in a cloud template's network components are matched to network tags, including network profile capability tags. For network profiles that contain capability tags, the capability tags are applied to all the networks that are available for that network profile. The network and security settings that are defined in the matched network profile are applied when the cloud template is deployed.

Network Policies

By using network profiles, you can define subnets for existing network domains that contain static, DHCP, or a mixture of static and DHCP IP address settings. You can define subnets and specify IP address settings by using the **Network Policies** tab.

When using NSX or VMware Cloud on AWS, network policy settings are used when a cloud template requires the `networkType: outbound` or `networkType: private` or when an NSX network requires `networkType: routed`.

Depending on the associated cloud account, you can use network policies to define settings for the `outbound`, `private`, and `routed` network types and for on-demand security groups. You can also use network policies to control `existing` networks when there is a load balancer associated with that network.

Note:

For information about using a VLAN transport zone to support VLAN segment specification for private NSX networks, see Network resources in VCF Automation.

Outbound networks allow one way access to upstream networks. Private networks do not allow any outside access. Routed networks allow East/West traffic between the routed networks. The existing and public networks in the profile are used as the underlying or upstream networks.

Options for the following on-demand selections are described in the **Network Profiles** on-screen help and summarized below.

- **Do not create an on-demand network or on-demand security group**
  You can use this option when specifying an `existing` or `public` network type. cloud templates that require an `outbound`, `private`, or `routed` network are not matched to this profile.
- **Create an on-demand network**
  You can use this option when specifying an `outbound`, `private`, or `routed` network type.
  Amazon Web Services, Microsoft Azure, NSX, vSphere, and VMware Cloud on AWS support this option.
- **Create an on-demand security group**
  You can use this option when specifying an `outbound` or `private` network type.
  A new security group is created for matched cloud templates if the network type is `outbound` or `private`.
  Amazon Web Services, Microsoft Azure, NSX, and VMware Cloud on AWS support this option.

Network policy settings can be cloud account type-specific. These settings are described in the on-screen signpost help and summarized below:

- **Network domain** or **Transport zone**
  A network domain or transport zone is the distributed virtual switch (dvSwitch) for the vSphere vNetwork Distributed PortGroups (dvPortGroup). A *`transport zone`* is an existing NSX concept that is similar to terms like *`dvSwitch`* or *`dvPortGroup`*.
  When using an NSX cloud account, the element name on the page is **Transport zone**, otherwise it is **Network domain**.
  For standard switches, the network domain or transport zone is the same as the switch itself. The network domain or transport zone defines the boundaries of the subnets within vCenter.
  A transport zone controls which hosts an NSX logical switch can reach to. It can span one or more vSphere clusters. Transport zones control which clusters and which virtual machines can participate in the use of a particular network. Subnets that belong to the same NSX transport zone can be used for the same machine hosts. Transport zone types are overlay or VLAN. For information about using a VLAN transport zone for defining VLAN segments, see Network resources in VCF Automation.
- **External subnet**
  An on-demand network with outbound access requires an external subnet that has outbound access. The external subnet is used to provide outbound access if requested in the cloud template - it does not control network placement. For example, the external subnet does not affect the placing of a private network.
- **CIDR**
  CIDR notation is a compact representation of an IP address and its associated routing prefix. The CIDR value specifies the network address range to be used during provisioning to create subnets. This CIDR setting on the **Network Policies** tab accepts IPv4 notation ending in /nn and containing values between 0 - 32.
- **Subnet size**
  This option specifies the on-demand network size, using IPv4 notation, for each isolated network in a deployment that uses this network profile. The subnet size setting is available for internal or external IP address management.
  The IPv6 format is not supported for on-demand networks.
- **Distributed logical router**
  For an on-demand routed network, you must specify a distributed logical network when using an NSX-V cloud account.
  A distributed logical router (DLR) is used to route east/west traffic between on-demand routed networks on NSX-V. This option is only visible if the account/region value for the network profile is associated to an NSX-V cloud account.
- **IP range assignment**
  The option is available for cloud accounts that support NSX or VMware Cloud on AWS, including vSphere.
  The IP range setting is available when using an existing network with an external IPAM integration point.
  You can select one of the following three options to specify an IP range assignment type for the deployment network:
  - **Static and DHCP**
    Default and recommended. This mixed option uses the allocated **CIDR** and **Subnet range** settings to configure the DHCP server pool to support half of the address space allocation using the DHCP (dynamic) method and half of the IP address space allocation using the Static method. Use this option when some of the machines that are

connected to an on-demand network require assigned static IP addresses and some require dynamic IP addresses. Two IP ranges are created.

This option is most effective in deployments with machines that are connected to an on-demand network, where some of the machines are assigned static IPs and other machines have IPs dynamically assigned by an NSX DHCP server and deployments where the load balancer VIP is static.

- **DHCP (dynamic)**
  This option uses the allocated CIDR to configure an IP pool on a DHCP server. All the IP addresses for this network are dynamically assigned. A single IP range is created for each allocated CIDR.
- **Static**
  This option uses the allocated CIDR to statically allocate IP addresses. Use this option when a DHCP server is not required to be configured for the network. A single IP range is created for each allocated CIDR.

- **IP blocks**
  The IP blocks setting is available when using an on-demand network with an external IPAM integration point.
  Using the IP block setting, you can add a named IP block, or range, to the network profile from your integrated external IPAM provider. You can also remove an added IP block from the network profile.
  External IPAM is available for the following cloud account/region types:
  - vSphere
  - vSphere with NSX
  - vSphere with NSX-V
- **Network Resources - External network**
  External networks are also referred to as existing networks. These networks are data-collected and made available for selection.
- **Network Resources - Tier-0 logical router**
  NSX uses the tier-0 logical router as a gateway to networks that are external to the NSX deployment. The tier-0 logical router configures outbound access for on-demand networks.
- **Network Resources - Edge cluster**
  The specified edge cluster provides routing services. The edge cluster is used to configure outbound access for on-demand networks and load balancers. It identifies the edge cluster, or resource pool, where the edge appliance is to be deployed.
- **Network Resources - Edge datastore**
  The specified edge datastore is used to provision the edge appliance. This setting applies to NSX-V only.

Tags can be used to specify which networks are available to the cloud template.

Load Balancers

You can add load balancers to the network profile. Listed load balancers are available based on information that is data-collected from the source cloud account.

If a tag on any of the load balancers in the network profile matches a tag in a load balancer component in the cloud template, the load balancer is considered during deployment. Load balancers in a matched network profile are used when a cloud template is deployed.

For more information, see Using load balancer settings in network profiles in VCF Automation and Network, security group, and load balancer resource examples in VCF Automation.

Security Groups

When a cloud template is deployed, the security groups in its network profile are applied to the machine NICs that are provisioned. For an Amazon Web Services-specific network profile, the security groups in the network profile are available in the same network domain (VPC) as the networks that are listed on the Networks tab. If the network profile has no networks listed on its Networks tab, all available security groups are displayed.

You can use a security group to further define the isolation settings for an on-demand `private` or `outbound` network. Security groups are also applied to `existing` networks. You can also assign global security groups.

Listed security groups are available based on information that is data-collected from the source cloud account or added as an on-demand security group in a project cloud template. For more information, see Security resources in VCF Automation.

Security groups are applied to all the machines in the deployment that are connected to the network that matches the network profile. As there might be multiple networks in a cloud template, each matching a different network profile, you can use different security groups for different networks.

> **Note:**
>
> In addition to specifying a security group, you can also select NSX networks (default) or vSphere networks or both. When you deploy a cloud template, VCF Automation adds the allocated or specified security group to machine NICs that are connected to the allocated NSX network. Only machine NICs that are connected to an NSX network can be added to a vDefend group. If the machine NIC is connected to a vSphere network, the template deployment fails.

Adding a tag to an existing security group allows you to use the security group in a cloud template `Cloud.SecurityGroup` component. A security group must have at least one tag or it cannot be used in a cloud template. For more information, see Security resources in VCF Automation and Network, security group, and load balancer resource examples in VCF Automation.

More information about network profiles, networks, cloud templates, and tags

For more information about networks, see Network resources in VCF Automation.

For examples of sample network component code in a cloud template, see Network, security group, and load balancer resource examples in VCF Automation.

For more information about tags and tag strategy, see How do I use tags to manage VCF Automation resources and deployments.

For information about how to name machine NICs, see How can I configure a network interface controller name by using extensibility actions.

Using network settings in network profiles and cloud templates in VCF Automation

You use networks and network profiles in VCF Automation to help define the behavior of network provisioning for your deployments.

In VCF Automation, you can define cloud-specific network profiles. See Learn more about network profiles in VCF Automation.

Using network and network profile settings, you can control how network IP addresses are used in VCF Automation cloud templates and deployments.

IPv4 and IPv6 support in VCF Automation networks

VCF Automation networks support single stack IPv4, single stack IPV6, or dual stack IPv4 and IPv6.

IPv6 is supported for existing vSphere networks and existing NSX networks.

IPv6 is not supported for load balancers, NSX on-demand networks, or external third-party IPAM providers such as Infoblox.

You can assign an IPv6 address space to single and dual stacked workloads when an IPv6 address is requested per integration with an external IPAM system. This capability requires that the integrated external IPAM plug-in also supports IPv6.

VLAN segments and private NSX networks

You can specify VLAN segments for private NSX on-demand network when the network segments are used with a Policy API-type of NSX cloud account. For information on supported configurations and network profile requirements, see Network resources in VCF Automation.

External IPAM provider support

In addition to the supplied internal IPAM support, you can use an external IPAM provider to dynamically or statically allocate IP address for networks - as IP ranges for existing networks in your cloud template designs and deployments and IP blocks for on-demand networks in your cloud template designs and deployments. You can use external IPAM derived from an external provider integration that is based on the VCF Automation IPAM SDK - for example one of the Infobox plug-ins that are available for download from the VMware Marketplace.

Support for external IPAM providers, such as Infoblox, is available for vendor-specific IPAM integration points that you create by using the **Infrastructure** > **Connections** > **Add Integration** > **IPAM** menu sequence.

Options for defining external IPAM provider address information is available by using the **Add IPAM IP Range** option on the **Network Policies** > **Add IPAM IP Range** page.

In addition to the above VCF Automation external IPAM options, you can also specify a VCF Operations orchestrator extensibility action in a cloud template to use a Network Configure event topic. For more information about this related IPAM method, see Event topics provided with VCF Automation for VM Apps in.

Network types

A network component in a cloud template is defined as one of the following `networkType` types.

Network type	Definition
`existing`	Selects an existing network that is configured on the underlying cloud provider, such as vCenter, Amazon Web Services, and Microsoft Azure. An existing network is required by the `outbound` on-demand network. You can define a range of static IP addresses on an existing network.
`public`	Machines on a public network are accessible from the Internet. An IT administrator defines these networks. The definition of a `public` network is identical to that of an `existing` network for networks that allow network traffic to occur along public networks.
`private`	An on-demand network type. Limits network traffic to occur only between resources on the deployed network. It prevents inbound and outbound traffic. In NSX, it can be equated to on-demand NAT one-to-many.
`outbound`	An on-demand network type. Limits network traffic to occur between the compute resources in the deployment but also allows one-way outbound network traffic. In NSX, it can be equated to on-demand NAT one-to-many with external IP.
`routed`	An on-demand network type. Routed networks contain a routable IP space divided across available subnets that are linked together. The virtual machines that are provisioned with routed networks, and that have the same routed network profile, can communicate with each other and with an existing network. Routed networks are an on-demand network type that is available for NSX-V and NSX networks. Microsoft Azure and Amazon Web Services provides this connectivity by default. A `routed` network is only available for cloud template specification in a `Cloud.NSX.Network` network component.

For more information, see More about network resources in VCF Automation cloud templates.

For examples of populated cloud templates that contain network component data, see Network, security group, and load balancer resource examples in VCF Automation.

Sample network scenarios

You can expect the following behavior when you deploy a cloud template that uses the following network profile configuration.

Network type or scenario	No network profiles available for cloud zone	Network profiles available for cloud zone
No network	If no network is specified in the cloud template, a random network is selected from the same provisioning region as the compute. Preference is given to networks that are labeled as default. If no networks exist in an available provisioning region, provisioning fails.	A network is selected from a matched network profile. Preference is given to networks that are labeled as default. If none of the network profiles meet the criteria, provisioning fails.
Existing network	If the network component in the cloud template contains constraint tags, those constraints are used to filter the list of available networks. Constraint tags in the cloud template's network component are matched to network tags and, if available, network profile constraint tags. From the filtered list of networks, a single network is selected from the same provisioning region as the compute. Preference is given to networks that are labeled as default. If after filtering based on constraints there are no networks in the provisioning region, provisioning fails.	A network is selected from a matching network profile. Preference is given to networks that are labeled as default. If none of the network profiles meet the criteria, provisioning fails. Network constraints can be used to filter existing networks in the profile based on their pre-assigned tags.
Public network	If the network has constraints, those constraints are used to filter the list of available networks that have the `supports public IP` attribute set. From the filtered list of networks, a random network is selected from the same provisioning region as the compute. Preference is given to networks that are labeled as default. If after filtering based on constraints there are no public networks in the provisioning region, provisioning fails.	A network with the `supports public IP` attribute is selected from a matching network profile. Preference is given to networks that are labeled as default. Network constraints can be used to filter existing public networks in the profile based on their pre-assigned tags.
Private network	Provisioning fails because private networks require information from a network profile.	A new network or new security group is created based on settings in the matched network profile. Network constraint tags can be used to filter network profiles and networks.
Outbound network	Provisioning fails because outbound networks require information from a network profile.	A new network or new security group is created based on settings in the matched network profile. Network constraint tags can be used to filter network profiles and networks.
On-demand routed network	Provisioning fails because routed networks require information from a network profile.	For NSX-V we need DLR (Distributed Logical Router) selection. For NSX and VMware Cloud on AWS, we require similar on-demand settings as private and outbound.
Example Wordpress use case with existing or public networks	Provisioning occurs as described for an existing network or public network.	See above descriptions for existing network and public network behavior.

Network type or scenario	No network profiles available for cloud zone	Network profiles available for cloud zone
Example Wordpress use case with existing or public networks and private or outbound networks	Provisioning fails because the network requires information from a network profile.	See above descriptions for a private network and an outbound network.
Example Wordpress use case with load balancer	Provisioning fails because a load balancer requires information from a network profile. Provisioning can occur when existing load balancers are present.	A new load balancer is created based on the network profile configuration. You can specify an existing load balancer that has been enabled in the network profile. Provisioning fails if you request an existing load balancer, but none meet the constraints in the network profile.

Using security group settings in network profiles and cloud template designs in VCF Automation
You can define and change security group settings in network profiles and cloud templates.

You can use security group capabilities in the following ways:

- Existing security group specified in a network profile.
  You can add an existing security group to a network profile. When a cloud template uses that network profile, its machines are members of the security group. This method does not require that you add a security group resource to a cloud template. You can also use a load balancer in this configuration. For related information, see More about load balancer resources in VCF Automation cloud templates.
- Security group component associated to machine resource in a cloud template
  You can drag and drop a security group resource onto a cloud template and bind the security group resource to a machine NIC by using constraint tags on the existing security group in the cloud template and on the existing security group in the data-collected resource. You can also make this association by connecting the objects on the cloud template design canvas, similar to how you associate networks to machines.
  When you drag and drop a security group resource onto the cloud template design canvas, it can be of type `existing` or `new`. If it's an `existing` security group type, add a tag constraint value as prompted. If it's a `new` security group type, you can configure firewall rules.
- An existing security group allocated with tag constraints and associated with a machine NIC in the cloud template
  You can associate a security group resource with a machine NIC (in a machine resource) in the cloud template by matching tags between the two resources.
  As an example for NSX when tags are specified in the source endpoint, you can use NSX tags specified in your NSX application. You can then use an NSX tag, specified as a constraint on a network resource in a cloud template, where the network resource is connected to a machine NIC in the cloud template. NSX tags allow you to dynamically group machines by using a pre-defined NSX tag that is data-collected from the NSX source endpoint. Use a logical port when you create the NSX tag in NSX.
- Firewall rules in an on-demand security group resource in a cloud template
  You can add firewall rules to an on-demand security group in a cloud template.
  For information about available firewall rules, see More about security group and tag resources in VCF Automation cloud templates.

Learn more

For information about defining security groups in network profiles, see Learn more about network profiles in VCF Automation.

For information about viewing and changing security groups settings in infrastructure resource pages, see Security resources in VCF Automation.

For information about defining security groups in cloud templates, see More about security group and tag resources in VCF Automation cloud templates.

For examples of security group resources in cloud templates, see Network, security group, and load balancer resource examples in VCF Automation.

Using load balancer settings in network profiles in VCF Automation

You can configure load balancer settings in your network profile configuration.

You can add an existing load balancer to a network profile by using the **Load Balancer** tab.

You can add a load balancer to a cloud template design by associating it to a network profile that contains one or more load balancers or directly by using a load balancer resource in the cloud template design canvas or code.

Examples for including a load balancer VIP based on security group use in a network profile

There are two types of security groups that you can use in a network profile – an existing security group that you select from the **Security Groups** tab and an on-demand security group that you create by using an isolation policy on the **Network Policies** tab.

When a load balancer VIP is associated to a security group based on network profile settings, the security group configuration is supplied by the network profile.

The following table illustrates some sample scenarios.

Cloud template design topology - associated resources	Network profile configuration	Security group membership
One-armed load balancer with VIP on private network, and a machine on the same private network.	The selected network profile uses isolation policy defined as an on-demand security group.	The machine NIC and the load balancer VIP are added to the isolation security group.
One-armed load balancer with VIP on private network, and a machine on the same private network.	The selected network profile uses an existing security group and uses isolation policy defined as an on-demand security group.	The machine NIC and the load balancer VIP are added to the isolation security group and the existing security group.
Two-armed load balancer with VIP on a public network and machine on a private network.	The selected network profile uses an existing security group and uses isolation policy defined as an on-demand security group.	The machine NIC and the load balancer VIP are added to the isolation security group and the existing security group.
Two-armed load balancer with VIP on a public network and a machine on a private network.	The selected network profile uses an existing security group.	The machine NIC and the load balancer VIP are added to the existing security group.
Two-armed load balancer, VIP is on network 1 and the machine is on network 2.	Two network profiles: • Network profile 1: Uses an existing security group 1. • Network profile 2: Uses an existing security group 2.	The load balancer lands on network profile 1 and the machine lands on network profile 2. The load balancer VIP is added to security group 1 and the machine NIC is added to security group 2.

Learn more

For information about adding load balancer resources to a cloud template design, see More about load balancer resources in VCF Automation cloud templates.

For examples of cloud template designs that include load balancers, see Network, security group, and load balancer resource examples in VCF Automation.

How do I configure a network profile to support an on-demand network for an external IPAM integration in VCF Automation

You can configure a network profile to support blocks of IP addresses for an on-demand network when that network profile is used in a VCF Automation cloud template that uses external IPAM integration.

While the following prerequisites apply to the person who creates or edits the network profile, the network profile itself would be applicable when used by a cloud template deployment that contains an IPAM integration.

This sequence of steps is shown in the context of an IPAM provider integration workflow.

- Verify that you have cloud administrator credentials.
- Verify that you have the cloud administrator user role.
- Verify that you have an account with the external IPAM provider, for example Infoblox or Bluecat, and that you have the correct access credentials to your organization's account with the IPAM provider. In this example workflow, the IPAM provider is Infoblox.
- Verify that you have an IPAM integration point for the IPAM provider and that the IPAM package used to create the IPAM integration supports on-demand networks.
- While the Infoblox IPAM package supports on-demand networks, if you are using an external IPAM integration for a different provider, verify that their IPAM integration package supports on-demand networks.

Using an existing integration for a particular external IPAM provider, you can provision on-demand network to create of a new network in the external IPAM system.

Using this process, you configure a block of IP addresses instead of supplying a parent CIDR (as is done when using VCF Automation's internal IPAM). The IP address block is used during on-demand network provisioning to segment the new network. The IP blocks are data-collected from the external IPAM provider, provided the integration supports on-demand networking. For example, when using an Infoblox IPAM integration, IP blocks represent Infoblox network containers.

When you use an on-demand network profile and an external IPAM integration in a cloud template, the following events occur when the cloud template is deployed:

- A network is created in the external IPAM provider.
- A network is also created in VCF Automation, reflecting the new network configuration from the IPAM provider, including settings such as CIDR and gateway properties.
- The IP address for the deployed virtual machine is fetched from the newly created network.

> **Note:**
>
> You can assign an IPv6 address space to single and dual stacked workloads when an IPv6 address is requested per integration with an external IPAM system. This capability requires that the integrated external IPAM plug-in also supports IPv6.

In this on-demand networking example, you configure a network profile to allow a cloud template deployment to provision a machine to an on-demand network in vSphere by using Infoblox as the external IPAM provider.

For related information, see How do I configure a network profile to support an existing network for an external IPAM integration in VCF Automation.

1. To configure a network profile, click **Infrastructure** > **Configure** > **Network Profiles** in the VCF Automation service.
2. Click **New Network Profile**.
3. Click the **Summary** tab and specify the following sample settings:
   - Specify a vSphere cloud account/region, for example `vSphere-IPAM-OnDemandA/Datacenter`.
     This example assumes use of a vSphere cloud account that is not associated with an NSX cloud account.
   - Name the network profile, for example `Infoblox-OnDemandNP`.
   - Add a capability tag for the network profile, for example `infoblox_ondemandA`.
     Make note of the capability tag value, as you must also use it as a cloud template constraint tag to make the network profile association to be used when provisioning the cloud template.
4. Click the **Network Policies** tab and specify the following sample settings:
   - From the **Isolation policy** drop-down menu, select **On-demand network**.
     This option allows you to use external IPAM IP blocks. Depending on the cloud account, new options appear. For example, the following options appear when using a vSphere cloud account that is associated to an NSX cloud account:
     - Transport zone
     - Tier-0 logical router
     - Edge cluster

For this example, the vSphere cloud account is not associated to NSX, so the **Network domain** menu option appears.

- Leave the **Network domain** option blank.

5. Click **External** as the address management **Source**.
6. Click **Add IP Block**, which opens the **Add IPAM IP Block** page.
7. From the **Provider** menu on the **Add IPAM IP Block** page, select an existing external IPAM integration.
8. From the **Address spaces** menu, select one of the available and listed IP blocks, for example **10.23.118.0/24** and add it.

   If the IPAM provider supports address spaces, the **Address spaces** menu appears. For an Infoblox integration, address spaces are represented by Infoblox network views.
9. Select a **Subnet size**, such as **/29 (-6 IP addresses)**.
10. Click **Create**.

A network profile is created that can be used to provision an on-demand network using the specified external IPAM integration. The following sample cloud template shows a single machine to be deployed to a network that is defined by this new network profile.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      image: ubuntu
      flavor: small
      networks:
        - network: '${resource.Cloud_Network_1.id}'
          assignment: static
  Cloud_Network_1:
    type: Cloud.Network
    properties:
      networkType: private
      constraints:
        - tag: infoblox_ondemandA
```

Note:
When the cloud template is deployed, the first available network in the specified IP block is fetched and considered to be the network CIDR. If you are using an NSX network in the cloud template, you can instead set the CIDR of the network manually by using the network property `networkCidr`, as shown below, to manually set a CIDR and override the settings for IP blocks and subnet size that are specified in the associated network profile.

```
Cloud_Network_1:
       type: Cloud.Network
       properties:
         networkCidr: 10.10.0.0/16
```

How do I configure a network profile to support an existing network for an external IPAM integration in VCF Automation

to support IP address ranges for an existing network when that network profile is used in a VCF Automation cloud template that uses external IPAM integration, you can configure a network profile.

Note:

You can assign an IPv6 address space to single and dual stacked workloads when an IPv6 address is requested per integration with an external IPAM system. This capability requires that the integrated external IPAM plug-in also supports IPv6.

For related information, see How do I configure a network profile to support an on-demand network for an external IPAM integration in VCF Automation.

NSX Projects and VPCs in network profiles
VCF Automation supports discovering networks and security groups from NSX Projects and VPCs which can be added to network profiles.

What are NSX Projects and VPCs and how are they used in VCF Automation

VCF Automation supports the discovery of networks and security groups included in all NSX Projects and Virtual Private Clouds (VPCs) configured in NSX Manager. These networks and security groups can be added to network profiles which can be selected for allocation to your cloud templates.

An NSX Project enables multi-tenancy for NSX network and security objects where each project is analogous to a tenant. A NSX VPC is an additional layer of tenancy that you can configure within an existing project. Both NSX Projects and VPCs are primarily configured and managed in NSX Manager by an administrator. For more information on NSX Projects and VPCs and how to add them in your NSX Manager environment, go to NSX Multi-tenancy.

VCF Automation collects information about three types of NSX Project and VPC infrastructure objects:

- NSX Project segments
- NSX VPC subnets
- NSX Project and VPC groups

Within VCF Automation segments and subnets are treated as networks, while groups are treated as security groups.

Modifying VPC subnets can have certain limitations in VCF Automation depending on their IP configuration. Subnets with a manual IP configuration include CIDR routing which cannot be modified in VCF Automation. However, you can still update the IP range of these subnets by clicking the **Manage IP Ranges** button. Subnets with an automatic IP configuration in NSX cannot be modified in VCF Automation. You cannot create IP ranges of these subnets since the IPAM is managed by NSX and the **Manage IP Ranges** button is greyed out.

Information regarding segments and subnets is displayed under **Infrastructure** > **Resources** > **Networks**. Information regarding security groups is displayed under **Infrastructure** > **Resources** > **Security**. If a given network or security group is part of an NSX Project or VPC, this information is displayed in the **NSX Project/VPC** column.

Networks and security groups that are part of an NSX Project and/or VPC include three new custom properties.

Custom Property	Description
`NsxProjectAndVpc`	The names of the NSX Project and, if applicable for the specific network or security group, NSX VPC.
`NsxProjectId`	The ID of the NSX Project.
`NsxVpcId`	The ID of the NSX VPC.

Allocation logic for Network Interface Controllers (NICs)

VCF Automation allocation logic prioritizes security groups based on the hierarchy of the networks to which the NIC or NICs are connected. The highest priority is given to `/infra` level segments, followed by Project segments, and finally VPC subnets. `/infra` level security groups are always prioritized if available. However, if the deployment selects security groups from either NSX Projects or VPCs, certain limitations apply. If a NSX Project security group is selected, for example, all NICs assigned to this security group must land on segments in this specific NSX Project or subnets in the VPC or VPCs included under the NSX Project. If a VPC security group is selected, all NICs assigned to this security group can only land on subnets in this specific VPC. If you select multiple security groups in a network profile, only those that follow this allocation logic are selected for the NIC based on the network selection.

Note:
If you perform a day 2 operation on a deployed cloud template that changes the constraint tags of an existing network, the operation can fail even if the specified network can be found. This is caused by the new network not complying with the security groups associated with the existing NICs specified during the initial day 0 deployment. In such a scenario, you receive an error message similar to the following:

```
Operation: 'Update.Network': Unable to find a valid subnet for network 'Network_Name_' of type
 'EXISTING' with constraints '[{"tag":"public:subnetType"},{"tag":"DEV-VPC:vpc"}]' in network profile
 'NetP'. Filtered subnets [DEV-VPC-PUBLIC-SUB] NSX projects [DEV-PRO / DEV-VPC] are not compatible
 with NSX projects [DEV-PRO / DEV-VPC1, DEV-PRO] of security groups [DEV-VPC1-GRP1, DEV-PROJ-GRP].
```

Add VCF Automation storage profiles that account for different requirements

An VCF Automation storage profile describes the kind of storage to be deployed.

Storage is usually profiled according to characteristics such as service level or cost, performance, or purpose, such as backup.

Select **Infrastructure** > **Configure** > **Storage Profiles** and click **New**.

Learn more about storage profiles in VCF Automation

A cloud account region contains storage profiles that let the cloud administrator define storage for the region in VCF Automation.

What does a storage profile do

Storage profiles include disk customizations, and a means to identify the type of storage by capability tags. Tags are then matched against provisioning service request constraints to create the desired storage at deployment time.

Storage profiles are organized under cloud-specific regions. One cloud account might have multiple regions, with multiple storage profiles under each.

Vendor-independent placement is possible. For example, you might have three different vendor accounts and a region in each. Each region includes a storage profile that is capability tagged as *fast*. At provisioning time, a request containing a *fast* hard constraint tag looks for a matching *fast* capability, regardless of which vendor cloud is supplying the resources. A match then applies the settings from the associated storage profile during creation of the deployed storage item.

Note:
Different cloud storage might have different performance characteristics but still be considered the *fast* offering by the administrator who tagged it.

You can use a capability tag in your storage profile and set that tag as a soft constraint in the storage section of a virtual machine's properties in the cloud template. This action helps the VM prefer that storage profile for placement during deployment. If a matching tag is not present in the storage profile, then the default storage profile for that region is selected during deployment.

Capability tags that you add to storage profiles should not identify actual resource targets. Instead, they describe types of storage. For related information, see Storage resources in VCF Automation.

Default provisioning type

The storage profile provisioning type only establishes a default behavior. The setting doesn't necessarily affect placement and might be overridden by a property in the cloud template.

For example, you might set the storage profile for thin provisioning. In most cases, requests would create thin provisioning storage by default. However, if the cloud template has the `provisioningType` property set to eager-zero, the cloud template overrides the default of thin.

Note:
When you want exact control, it's better to add capability and constraint tags labeled for the desired provisioning type.

For the provisioning type default, a cloud template property overrides a storage profile default, and a storage profile default overrides a default from a vCenter storage policy.

Disk allocation with machines

In a project with multiple cloud zones that belong to different cloud accounts, a disk follows the machine even if the disk isn't attached to the machine. This behavior keeps the resources together to prevent failure when you decide to attach the disk later.

For example, the following design won't work. The cloud template attempts to use location constraints to separate the disk, but the deployment returns a `No matching placement` error instead.

If you need to place a disk in a different cloud account, use a separate deployment to deploy the disk.

```
resources:
  Machine1:
    type: Cloud.vSphere.Machine
      properties:
        image: ubuntu
        flavor: small
        constraints:
          - tag: 'location:siteA'
  Disk1:
    type: Cloud.vSphere.Disk
      properties:
        capacityGb: 1
        constraints:
          - tag: 'location:siteB'
```

Using affinity filters for improved storage/compute recommendations

While earlier VCF Automation releases used a storage filter that selected the first eligible storage option and passed it to downstream filters, the storage filter now passes all the eligible storage options to downstream filters. This allows the compute spread policy to consider and evaluate multiple storage profiles and provide improved storage selections.

First class and standard disks

By using the **Disk type** option on the storage profile page, or by using the VCF Automation API, you can create a storage profile to support first class disk (FCD) or standard disk storage. In effect, the first class disk option creates a vSphere storage profile.

- First class disk
  First class disks can exist independent from a vSphere virtual machine. A first class disk also has life-cycle management capabilities that can operate independently of a virtual machine. First class disks are available for vSphere 6.7 Update 2 and later, and are currently implemented in VCF Automation as an API-only feature.
  For information about FCD storage, including the capabilities that are available from the VCF Automation API, and links to the API documentation itself, see What can I do with first class disk storage in VCF Automation for VM Apps.
- Standard disk
  Standard disk storage is created and managed as an integrated component of a virtual machine.
  For information about standard disk storage, see What can I do with standard disk storage in VCF Automation for VM Apps and What can I do with persistent disk storage in VCF Automation for VM Apps.

Storage profile and datastore priority

The priority value of storage profiles and datastore or datastore clusters allows users to manually define how they are selected during deployment.

Note:
The assigned storage profile priority does not override the default storage profile. The default storage profile is selected when no constraint tags are included in the relevant cloud template. In such a case, the default storage profile is selected even if it has the lowest priority value of all eligible storage profiles.

You can set the storage profile priority by entering the desired value in the **Priority** text box. The highest priority value is zero.

Storage profile priority is considered only after the eligible storage profiles are filtered by other factors such as capability tags and available capacity.

The datastore information is found under the **Datastore** tab. Datastores included in the storage profile can be managed in the following ways:

- Show all datastores and datastore clusters included in the datacenter.
- Manually add datastores or datastore clusters that you want to be associated with the storage profile.
- Dynamically add datastores or datastore clusters based on tags.

To set the datastore priority, select one or more datastores or datastore clusters and click **Set priority**. Similarly to storage profiles, the highest priority value is zero.

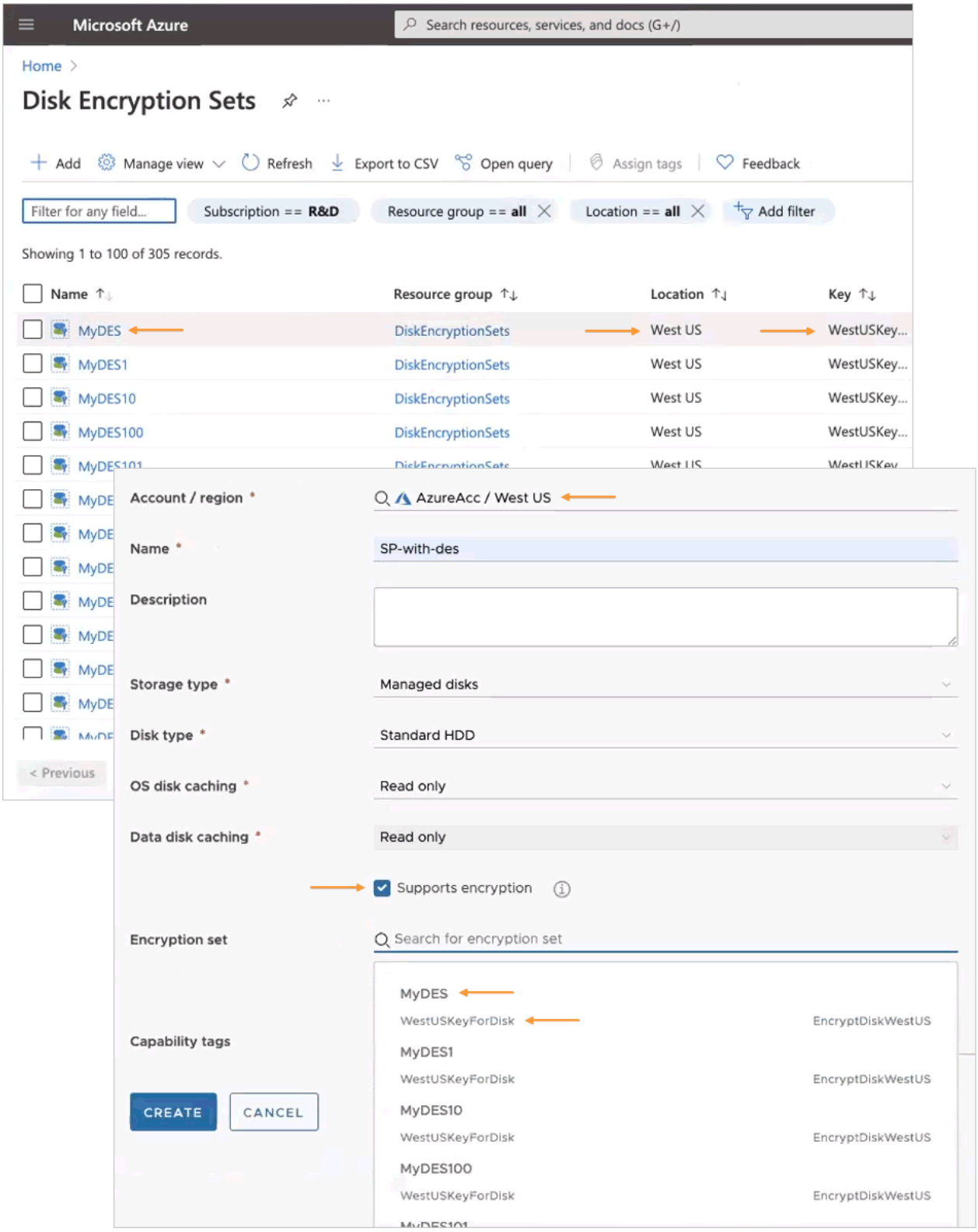
During deployment, VCF Automation first checks the priority of eligible storage profiles and then the priority of the associated datastores or clusters. If multiple storage profiles have the same priority, the latest updated storage profile is selected. If multiple datastores or datastore clusters have the same priority value, they are selected based on available capacity.

For example, your environment might include the storage profiles **profile_01** and **profile_02** with a priority value of one and **profile_03** with a priority value of two. While the first two storage profiles have the same priority, **profile_02** was updated more recently, so it is selected for deployment. After selecting the storage profile, VCF Automation checks the priority and available capacity of the datastores and datastore clusters. For example, let's say **profile_02** includes two datastores with the same priority value. In this scenario the datastore with highest available capacity is chosen.

Note:
If two storage profiles have the same priority and one of them has a compute resource specified while the other does not, higher priority is given to the storage profile with the specified compute resource.

Azure server-side disk encryption

For Azure resources, if you elect to support encryption in a managed disk storage profile, you also select disk encryption that has an associated key. Available encryption and keys correspond to the disk encryption sets configured in Azure for the location.



Using Pricing Cards in VCF Automation

VCF Automation pricing cards help cloud administrators define and assign the pricing policy for the monetary impact of your individual deployments to help you manage resources.

Note:

For pricing to work on multi-tenant environments, each VCF Automation organization must use a different VCF Operations instance.

Pricing cards define the rates for a pricing policy. The pricing policy can then be assigned to specific projects to define a total price. After creating a VCF Operations or CloudHealth endpoint, a predefined default rate card is available with a cost equal to price configuration on the **Infrastructure** > **Configure** > **Pricing Cards** tab. You can create pricing cards that apply to projects only or cloud zones. By default, all new pricing cards are applied to projects.

Note:
If you change the **All pricing cards are applied to** setting, all existing pricing card assignments are deleted. Also, if the VCF Operations endpoint is deleted from VCF Automation, all pricing cards and assignments are also deleted.

The price of a deployment over time appears on both the deployment card and project as the month-to-date price, which resets to zero at the beginning of each month. The component cost breakdowns are available in the deployment details. Providing this information at the deployment level informs the cloud administrator, but it also helps the members understand the impact their work might have on budgets and long-term development.

You can choose to display pricing information from users in VCF Automation by selecting the **Display pricing information** button. If left disabled, the pricing information is hidden from VCF Automation users.

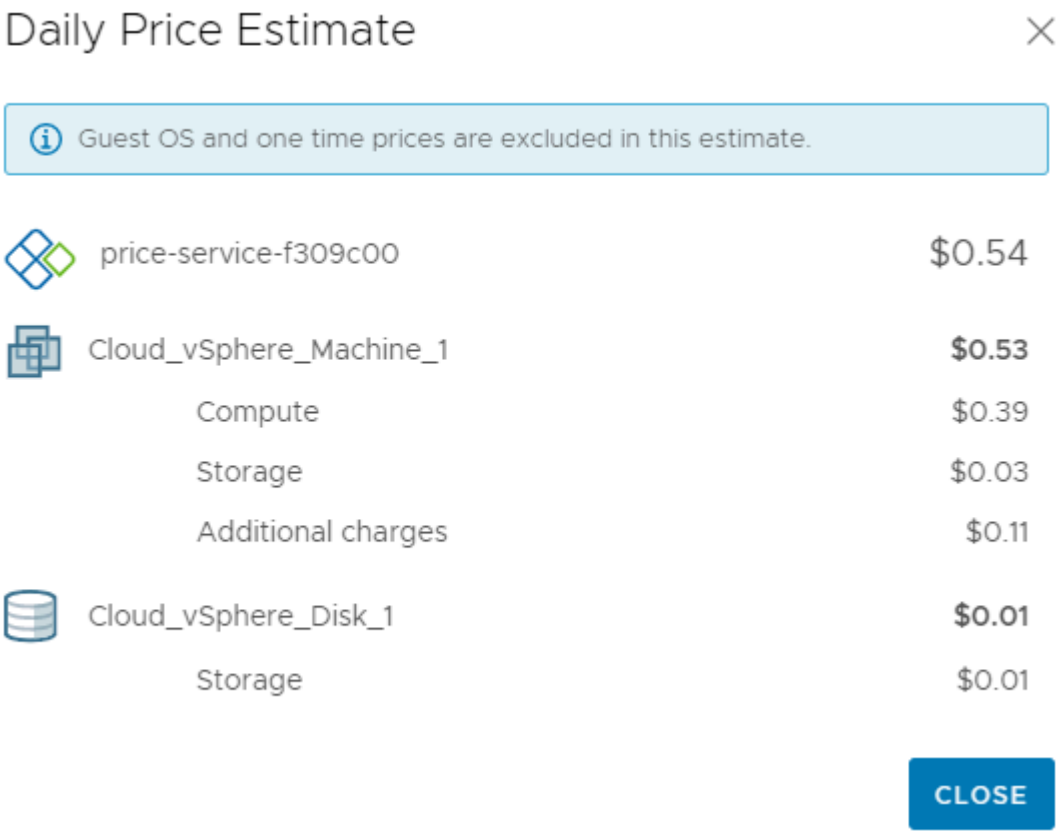
How is price calculated

The initial price that you see at the deployment level for your compute and storage resources are based on industry standard benchmark rates, and then calculated over time. The rate is applied to hosts and the service calculates the CPU and memory rates. The server recalculates the price every 6 hours.

New policies, assignments, and upfront pricing are priced during the next occurring data collection cycle. By default, the data collection cycle is run every 5 minutes. It can take up to 6 hours for new policies or changes to be updated in projects and deployments.

How do I estimate the price of my deployments and projects

Before deploying a catalog item, you can use the upfront price as a price estimate for your deployment. To view the price in VCF Automation, you must have a VCF Operations integration endpoint configured with pricing enabled and currency preset.



For an upfront price estimation, the size of boot disk per VM is always 8 GB.

The upfront price of a deployment is a daily price estimate, based on the allocation of a resource, for a given catalog item before it is deployed. After a catalog item is deployed, you can view the month-to-date price as an aggregate of the upfront price on the **Deployment** and **Infrastructure** > **Projects** tabs. Upfront pricing is supported for private cloud resources such as vSphere machine and vSphere disk, VCF Automation catalog items, and cloud agnostic items with vCenter configured for private cloud.

> **Note:**
>
> Upfront pricing is not supported for public cloud resources, or non-vSphere machine or disk private cloud resources.

To estimate the cost of your deployment, from the Catalog select a catalog item and click **Request** > **Calculate**. If the price is acceptable, click **Submit**.

You can use project pricing cards to estimate the total price of all your projects.

To estimate the cost of a project, on the Infrastructure Pricing Card page next to **All pricing cards are applied to** setting, click **Edit** and select **Projects**.

If you change the **All pricing cards are applied to** setting, all existing pricing card assignments are deleted. Create pricing cards and assignments using a cost-based approach.

Create pricing cards for vSphere and VMware Cloud on AWS in VCF Automation

Depending on the pricing strategy determined by the cloud administrator for private cloud deployments, you can create and assign a pricing card to projects or cloud zones.

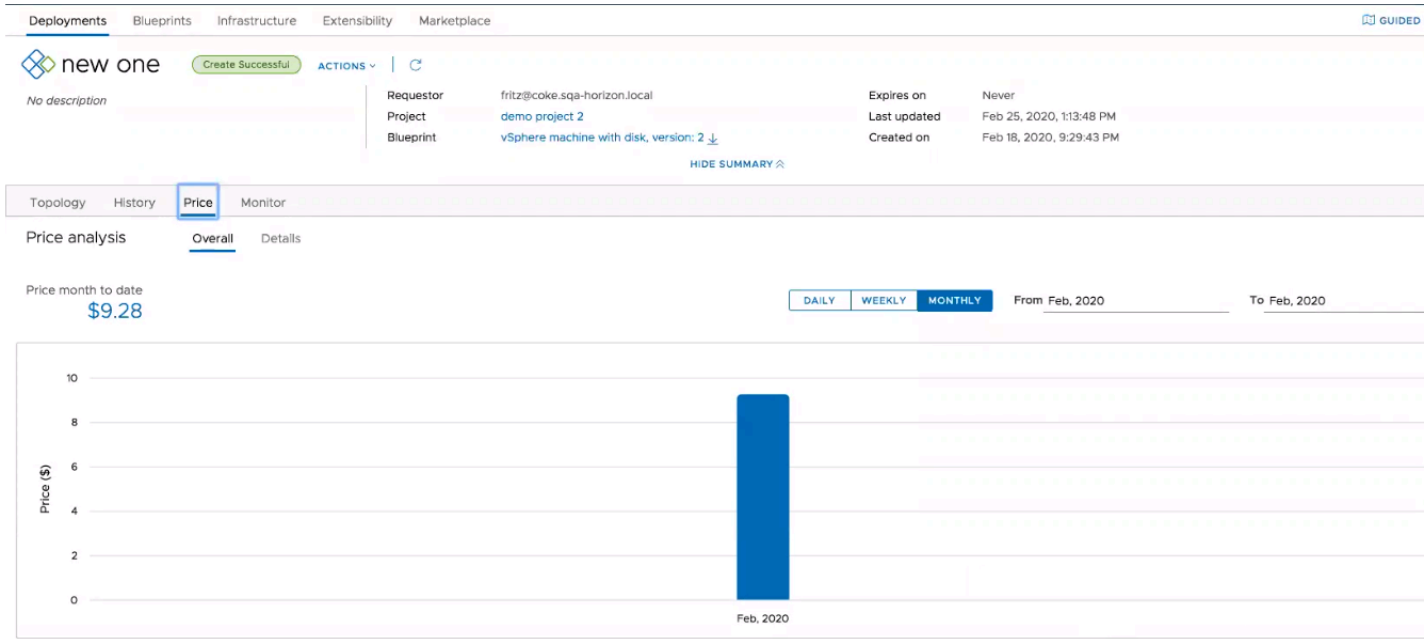
Before you can create or assign pricing cards, you must configure and enable pricing and configure currency for your VCF Automation organization in VCF Operations. When configuring VCF Operations with VCF Automation, ensure that both applications are set to the same timezone.

You must configure a VCF Operations endpoint before you can configure pricing cards. To configure the VCF Operations endpoint, navigate to **Infrastructure** > **Connections** > **Integrations** > **Add Integration**.

Note:
When multiple VCF Operations endpoints are added they must not monitor the same vCenter.

Pricing cards are customizable based on user-selected parameters. After configuring a pricing card, you can assign it to one or more projects and cloud zones determined by the pricing strategy.

You can manually refresh the price server at any time on the VCF Operations **Endpoint** page, **Infrastructure** > **Integrations** > **VCF Operations Endpoint**. In the vCenter servers section, click **Sync**. When manually refreshing the price server using the **Sync** option, the price is recalculated for all projects in the organization. Depending on how many projects your organization has this process might be intensive and take time.



After creating and assigning a pricing card, you can view the price history of your deployments and projects. To view the price history, navigate to your deployment and click **Price**. The price analysis provides an overview and detailed view of the deployment price along with the price month-to-date value. You can change the graphical representation to display the deployment price as daily, weekly, or monthly values. Also, you can specify an exact date range or month for the price history.

To view the price breakdown by cost component, click **Details**.

Prices are determined by costed component types.

Table 1824: Costed Component Types

Cloud template Component Type	Service Name/Object Type	Cloud template Resource Type	Comments
Cloud Agnostic	Machine	Cloud.Machine	If an agnostic machine is configured with vSphere, you can view deployment cost.
	Disk	Cloud.Volume	If an agnostic disk is attached to a virtual machine that is configured with vSphere, you can view deployment cost.
vSphere	vSphere machine	Cloud.vSphere.Machine	Deployed using a cloud-specific cloud template.

Cloud template Component Type	Service Name/Object Type	Cloud template Resource Type	Comments
	vSphere disk	Cloud.vSphere.Disk	Deployed using a cloud-specific cloud template attached to a virtual machine.
VMware Managed Cloud (VMC)	vSphere machine	Cloud.vSphere.Machine	VMC only supports rate-based pricing cards (cost based pricing cards are not supported).
	vSphere disk	Cloud.vSphere.Disk	

1. Navigate to **Infrastructure** > **Pricing Cards** > **New**.
2. On the **Summary** tab, enter a name and description for the pricing card. Once the policy is defined on the pricing tab, the Overview table is populated with pricing card rates.

   Note:
   The currency unit is determined by the valued selected in VCF Operations.
3. Optional. Select the **Default for unassigned projects?** check box to assign this pricing card to all unassigned projects by default.
4. Click **Pricing**, and configure the details of your pricing policy.

Table 1825: Pricing Policy Configuration

Parameter	Description
Basic Charges	Enter a name and description for your policy. Select cost or rate based. • Cost - The cost is defined in VCF Operations. If selected, a multiplication factor is required. For example, if you select 1.1 as a factor, the cost is multiplied by 1.1 resulting in a 10% increase to the calculated cost. The price equation using cost is: <cost> x <multiplication factor> = Price • Rate - If selected, you must use absolute values to determine the cost. The price equation using rate is: <Rate> = Price. Select a rate interval from the drop down list to specify how this rate is charged. In the basic charges section, you define the cost or rate for CPU, memory, storage, and additional miscellaneous costs.
Guest OSes	You can define a Guest OS charge by clicking **Add Charge**. Enter the guest OS name and define the charging method and base rate. • Recurring - enter a base rate and define recurring interval as the charge period. The absolute rate value is required and it is added to the overall price. • One time - define the one-time base rate charge. The absolute value is required and it is added as a one time price. • Rate Factor - A multiplication factor is required that is applied to the select charge category. For example, if you select CPU Charge and a rate factor of 2. The Guest OS CPU is charged as 2 times the standard cost value. You can add multiple Guest OS items with different rates by clicking **Add Charge** and configuring an additional charge policy. **Note:** Upfront charges for each Guest OS are not shown on the summary page, even though they are part of the policy.

Parameter	Description
Tags	You can define a tag charge by clicking **Add Charge**. Select the tag name and define the charging method and base rate. • Recurring - enter a base rate and define recurring interval as the charge period. The absolute rate value is required and it is added to the overall price. • One time - define the one-time base rate charge. The absolute value is required and it is added as a one time price. • Rate Factor - A multiplication factor is required that is applied to the select charge category. Select how to charge the tag based on powered on state. You can add multiple tags with different rates by clicking **Add Charge** and configuring an additional charge policy. **Note:** Additional charges in the calculated final price include on tags on VMs and does not include tags on disks and networks.
Custom Properties	You can define a Custom Property charge by clicking **Add Charge**. Enter the property name and value, and define the charging method and base rate. • Recurring - enter a base rate and define recurring interval as the charge period. The absolute rate value is required and it is added to the overall price. • One time - define the one-time base rate charge. The absolute value is required and it is added as a one time price. • Rate Factor - A multiplication factor is required that is applied to the select charge category. Select how to charge the custom property based on powered on state. You can add multiple custom properties with different rates by clicking **Add Charge** and configuring an additional charge policy.
Overall Charges	Define any additional charge you would like to add to the pricing policy. You can add both one time and recurring charges.

One time charges are not shown in the price estimate of a catalog item or on the **Summary** tab. Only the daily price estimate for a given catalog item is shown.

5. Click the **Assignments** tab and click **Assign Projects**. Select one or more projects to assign the pricing card to.

   Note:
   By default pricing cards are applied to projects. On the **Infrastructure** > **Pricing Cards** tab, you can select to apply pricing cards to cloud zones. If cloud zones were selected, you would click **Assign Cloud Zones** on the **Assignments** tab.

6. Click **Create** to save and create your pricing policy.

Your new pricing policy appears on the **Pricing Cards** page. To view or edit the policy details and configuration click **Open**.

How do I use tags to manage VCF Automation resources and deployments

Tags are a critical component of VCF Automation that drive the placement of deployments through matching of capabilities and constraints. You must understand and implement tags effectively to make optimal use of VCF Automation.

Fundamentally, tags are labels that you add to VCF Automation items. You can create any tags that are appropriate for your organization and implementation. Tags function as much more than labels though, because they control how and

where VCF Automation uses resources and infrastructure to build deployable services. Tags also support governance within VCF Automation.

Tag structure

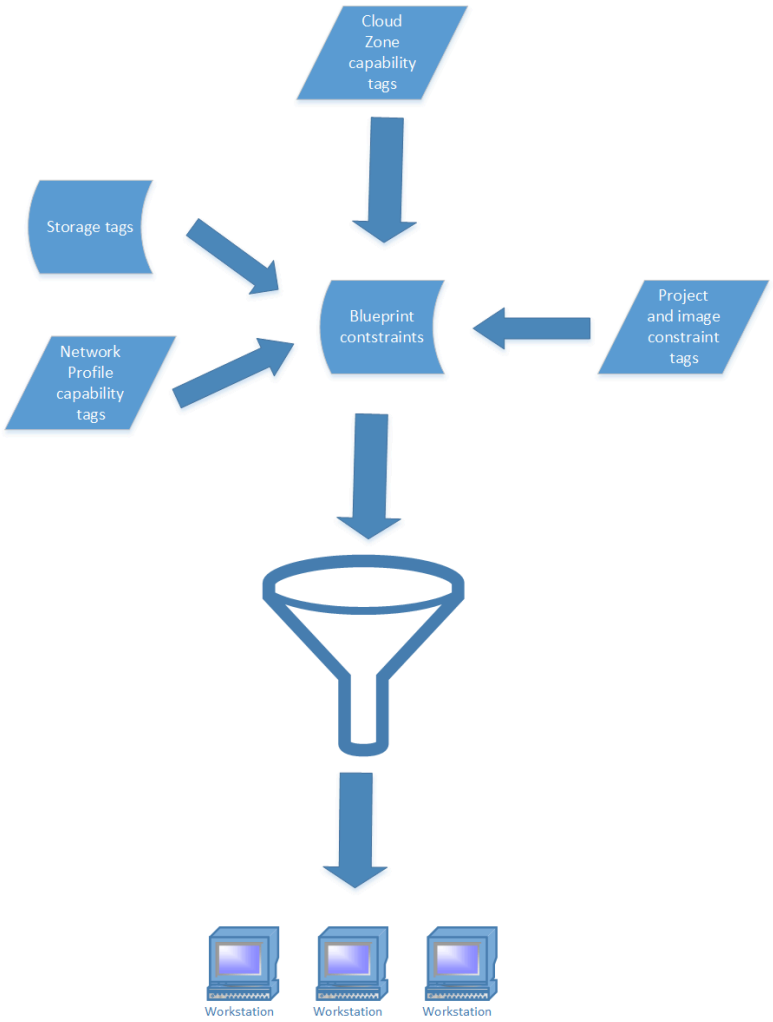
Structurally, tags must follow the `name:value` pair convention, but otherwise their construction is largely free form. Throughout VCF Automation, all tags appear the same, and tag functionality is determined by context.

For example, tags on infrastructure resources function primarily as capability tags because VCF Automation uses them to match resources with deployments. Secondarily, they also identify the resources.

Tag function

The primary function of tags is to express capabilities and constraints that VCF Automation uses to define deployments. Context determines the function of tags. Tags placed on cloud zones, network and storage profiles, and individual infrastructure resources function as capability tags and define desired capabilities for infrastructure used in deployments. Tags placed on cloud templates function as constraints that define resources for deployments. Also, cloud administrators can place constraint tags on projects to exercise a form of governance over those projects. These constraint tags are added to other constraints expressed in cloud templates.

During provisioning, VCF Automation matches these capabilities with constraints, also expressed as tags, in cloud templates to define deployment configuration. This tag-based capability and constraint functionality serves as the foundation for deployment configuration in VCF Automation. For instance, you can use tags to make infrastructure available only on PCI resources in a particular region.



On a secondary level, tags also facilitate search and identification of storage and network items and other infrastructure resources.

For example, assume that you are setting up cloud zones and you have many compute resources available. If you have tagged your compute resources appropriately, you can use the search function on the Compute tab of the Cloud Zone page to filter the resources that are associated with that particular cloud zone.

Also, the VCF Automation Tag Management page and resource configuration pages contain search functions that enable you to locate items by tag names. Using logical and human readable tags for these items is key to facilitating this search and identification function.

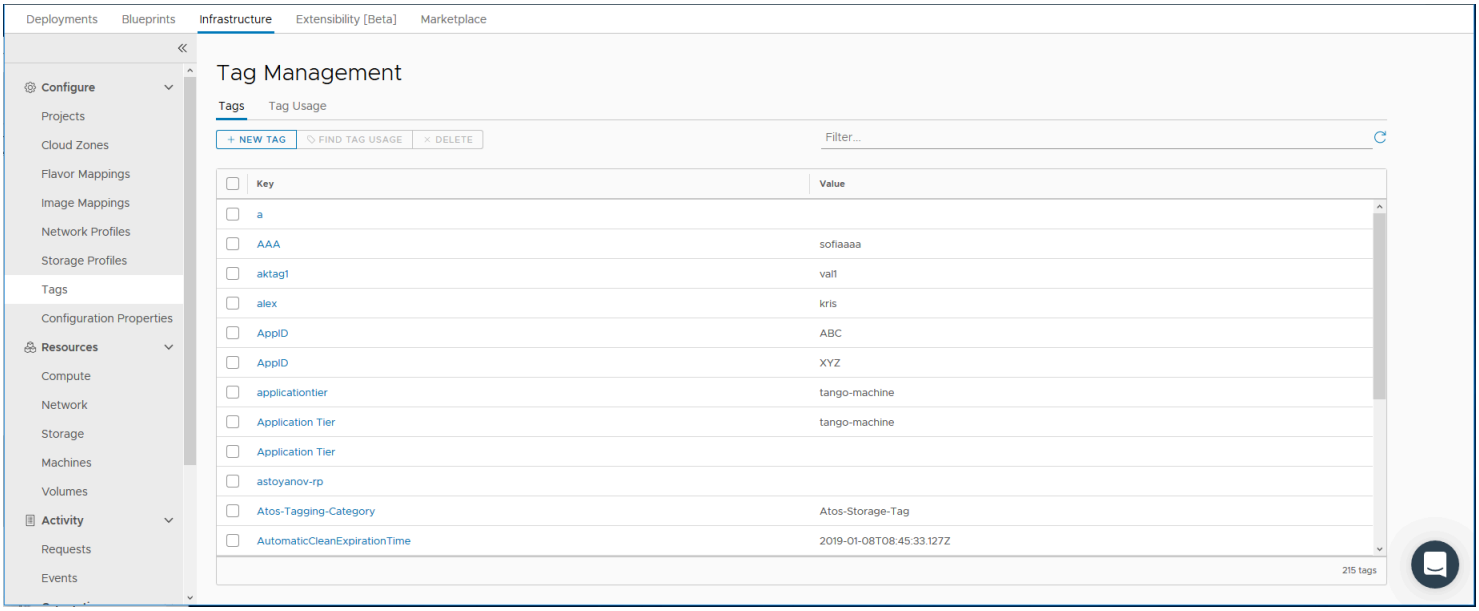
Take a look at the following video for more information and examples of tag usage: https://youtu.be/4zNQ33RyQio

External tags

VCF Automation might also contain external tags. These tags are imported automatically from cloud accounts that you associate with an VCF Automation instance. These tags might be imported from vSphere, AWS, Azure or other external software products. When imported, these tags are available for use in the same manner as user created tags.

Managing tags

You can use the **Tag Management** page in VCF Automation to monitor and manage your tags library. You can also create tags on this page. In addition, the **Tag Management** page is the only page on which you can view and identify external tags.



Tag strategy

To minimize confusion, before creating tags in VCF Automation, devise an appropriate tag strategy and tagging conventions, so that all users who create and use tags understand what they mean and how they should be used. See Creating a tagging strategy.

Creating a tagging strategy

You must carefully plan and implement an appropriate tagging strategy based on your organization's IT structure and goals to maximize VCF Automation functionality and minimize potential confusion.

While tagging serves several common purposes, your tagging strategy must be tailored to your deployment needs, structure, and goals.

Best practices for tagging

Some general characteristics of an effective tag strategy:

- Design and implement a coherent strategy for tagging that relates to the structure of your business and communicate this plan to all applicable users. A strategy must support your deployment needs, use clear human readable language, and be understandable to all applicable users.
- Use simple, clear, and meaningful names and values for tags. For instance, tag names for storage and network items should be clear and coherent so that users can readily understand what they are selecting or reviewing tag assignments for a deployed resource.
- Though you can create tags using a name with no value, as a best practice, it is more appropriate to create an applicable value for each tag name, as this makes the tag usage clear to other users.
- Avoid creating duplicate or extraneous tags. For example, only create tags on storage items that relate to storage issues.

Tagging implementation

Map out your primary considerations for a basic tagging strategy. The following list shows typical considerations to consider when mapping your strategy. Be aware that these considerations are representative rather than definitive. You might have other considerations that are highly relevant to your use cases. Your specific strategy must be appropriate for your specific use cases.

- How many different environments do you deploy to. Typically, you will create tags that represent each environment.
- How are your compute resources structured and used to support deployments.
- How many different regions or locations do you deploy to. Typically, you will create tags, at the profile level, that represents each of these different regions or locations.
- How many different storage options are available for deployments, and how do you want to characterize them. These options should be represented by tags.
- Categorize your networking options and create tags to accommodate all applicable options.
- Typical deployment variables. For example, how many different environments do you deploy to. Typically, many organizations have Test, Dev, and Production environments at a minimum. You will want to create and coordinate constraint tags and cloud zone capability tags that match so that you can easily set up deployments to one or more of these environments.
- Coordinate tags on network and storage resources so that they make logical sense in context of the network and storage profiles in which they are used. The resource tags can serve as a finer level of control over the resource deployment.
- Coordinate cloud zone and network profile capability tags, and other capability tags, with constraint tags. Typically, your administrator will create capability tags for cloud zones and network profiles first, and then other users can design cloud templates with constraints that match these capability tags.

After you understand the important considerations for your organization, you can plan appropriate tag names that address these considerations in a logical manner. Then, create an outline of your strategy and make it available to all users with privileges to create or edit tags.

As a useful implementation approach, you can begin by tagging all of your compute infrastructure resources individually. As noted, use logical categories for tag names that relate to the specific resource. For instance, you might tag storage resources as tier1, tier2, etc. Also, you might tag compute resources based on their operating system, such as Windows, Linux, etc.

After you tag resources, you can then consider the approach to creating tags for cloud zone and storage and network profiles that best suits your needs.

Using capability tags in VCF Automation

In VCF Automation, capability tags enable you to define deployment capabilities for infrastructure components. Along with constraints, they function as the basis of placement logic in VCF Automation.

You can create capability tags on compute resources, cloud zones, images and image maps, and networks and network profiles. The pages for creating these resources contain options for creating capability tags. Alternatively, you can use the Tag Management page in VCF Automation to create capability tags. Capability tags on cloud zones and network

profiles affect all resources within those zones or profiles. Capability tags on storage or network components affect only the components on which they are applied.

Typically, capability tags might define characteristics such as location for a compute resource, adapter type for a network, or tier level for a storage resource. They can also define environment location or type and any other business considerations. As with your overall tagging strategy, you should organize your capability tags in a logical manner for your business needs.

VCF Automation matches capability tags from cloud zones with constraints on cloud templates at deployment time. So, when creating and using capability tags, you must understand and plan to create appropriate cloud template constraints so that matching will occur as expected.

Tag inheritance

VCF Automation uses tag inheritance to selectively propagate tags on cloud accounts and compute resources that correspond to that cloud account.

> **Note:**
> Tag propagation behavior does not apply to storage profiles. This means that VCF Automation will not automatically select the constraint for storage profiles, so users must manually add the required constraint tag for it to be selected and applied to storage profiles.

The following example illustrates how tag inheritance works.

Compute resources

- Cluster1 with tag cluster-1
- Cluster2 with tag cluster-2
- Cluster3 with tag cluster-3

```
Vm resource:
  properties:
    constraints:
      - tag: 'cluster-01'
```

Cloud Account

vSphere cloud account with all three of the tags: cluster-1, cluster-2, and cluster-3

While consolidating tags on compute resources, VCF Automation also considers the cloud account level tags. Hence, the effective tags on the computes are cluster-1, cluster-2 and cluster-3 and this is why when we provide any of these tags as shown in the preceding example, all the compute resources become eligible for placement and the machine can land on any of the compute hosts.

As a best practice, to minimize unexpected results and tag clutter, any given tag should be applied only at the cloud account level if that tag is an appropriate capability for all subordinate compute resources.

Using constraint tags in VCF Automation

Tags added to projects and cloud templates function as constraint tags when they are used to match capability tags on infrastructure resources, profiles, and cloud zones. In the case of cloud templates, VCF Automation uses this matching functionality to allocate resources for deployments.

VCF Automation enables you to use constraint tags in two primary ways. The first way is when configuring projects and images. You can use tags as constraints to associate resources with the project or image. The second is in cloud templates where tags specified as constraints are used to select resources for deployments. Constraints applied in both of these ways are merged in cloud templates to form a set of deployment requirements that define resources available for a deployment.

How constraint tags work on projects

When configuring VCF Automation resources, cloud administrators can apply constraint tags on projects. In this way, administrators can apply governance constraints directly at the project level. All constraints added at this level are applied to every cloud template requested for the applicable project, and these constraint tags take precedence over other tags.

If constraint tags on the project conflict with constraint tags on the cloud template, the project tags take precedence, thus allowing the cloud administrator to enforce governance rules. For example, if the cloud administrators creates a `location:london` tag on the project, but a developer places a `location:boston` tag on the cloud template, the former will take precedence and the resource is deployed to infrastructure containing the `location:london` tag.

There are three types of constraints tags that users can apply on projects: network, storage, and extensibility. You can apply as many instances of each tag type as needed. Project constraints can be hard or soft. By default they are hard. Hard constraints allow you to rigidly enforce deployment restrictions. If one or more hard constraints are not met, the deployment will fail. Soft constraints offer a way to express preferences that will be selected if available, but the deployment won't fail if soft constraints are not met.

How constraint tags work in cloud templates

In cloud templates, you add constraint tags to resources as YAML code to match the appropriate capability tags that your cloud administrator created on resources, cloud zones and storage and network profiles. In addition, there are other more complex options for implementing constraint tags. For example, you can use a variable to populate one or more tags on a request. This enables you to specify one or more of the tags at request time.

Create constraint tags by using the `tag` label under a constraint heading in the cloud template YAML code. Constraint tags from projects are added to the constraint tags created in cloud templates.

VCF Automation supports a simple string formatting to make using constraints easier in YAML files:

```
[!]tag_key[:tag_value][:hard|:soft]
```

By default VCF Automation creates a positive constraint with hard enforcement. The tag value is optional, though recommended, as in the rest of the application.

> **Note:**
> Only the hard or soft delimiters are officially supported as extensions to the basic tag `key:value` format. Attempting to use other delimiters will likely cause errors.

The following WordPress with MySQL example shows YAML constraint tags that specific location information for compute resources.

```
name: "wordPressWithMySql"
components:
  mysql:
    type: "Compute"
    data:
      name: "mysql"
      # ... skipped lines ...
  wordpress:
    type: "Compute"
    data:
      name: "wordpress"
      instanceType: small
      imageType: "ubuntu-server-1604"
      constraints:
        - tag: "!location:eu:hard"
        - tag: "location:us:soft"
        - tag: "!pci"
      # ... skipped lines ...
```

How hard and soft constraints work in projects and cloud templates

Constraints in both projects and cloud templates can be hard or soft. The preceding code snippet shows examples of hard and soft constraints. By default all constraints are hard. Hard constraints allow you to rigidly enforce deployment restrictions. If one or more hard constraints are not met, the deployment will fail. Soft constraints express preferences that apply if available, but they won't cause a deployment to fail if not met.

If you have a series of hard and soft constraints on a specific resource type, the soft constraints can also serve as tie breakers. That is, if multiple resources meet a hard constraint, the soft constraints are used to select the actual resource used in the deployment.

For example, let's say that you create a hard storage constraint with a tag of `location:boston`. If no storage in the project matches this constraint, any related deployment will fail.

> **Note:**
> Soft constraints on networks can be used to select networks within a network profile, but do not affect the selection of the network profile itself.

Standard tags

VCF Automation applies standard tags to some deployments to support analysis, monitoring, and grouping of deployed resources.

Standard tags are unique within VCF Automation. Unlike other tags, users do not work with them during deployment configuration, and no constraints are applied. These tags are applied automatically during provisioning on AWS, Azure, and vSphere deployments. These tags are stored as system custom properties, and they are added to deployments after provisioning.

The list of standard tags appears below.

Table 1826: Standard tags

Description	Tag
Organization	org:`orgID`
Project	project:`projectID`
Requester	requester:`username`
Deployment	deployment:`deploymentID`
Cloud template reference (if applicable)	blueprint:`blueprintID`
Component name in cloud template	blueprintResourceName:`CloudMachine_1`
Placement Constraints: applied in cloud template, request parameters, or via IT policy	constraints:`key:value:soft`
Cloud Account	cloudAccount:`accountID`
Zone or profile, if applicable	zone:`zoneID`, networkProfile:`profileID`, storageProfile:`profileID`

How VCF Automation processes tags

In VCF Automation, tags express capabilities and constraints that determine how and where resources are allocated to provisioned deployments during the provisioning process.

VCF Automation uses a specific order and hierarchy of operations in resolving tags to create provisioned deployments. Understanding the basics of this process will help you to implement tags efficiently to create predictable deployments.

The following list summarizes the high level operations and sequence that VCF Automation uses to resolve tags and define a deployment:

- Cloud zones are filtered by several criteria, including availability and profiles; tags in profiles for the region the zone belongs to are matched at this point.
- Zone and compute capability tags are used to filter the remaining cloud zones by hard constraints.
- Out of the filtered zones, priority is used to select a cloud zone. If there are several cloud zones with the same priority, they are sorted by matching soft constraints, using a combination of the cloud zone and compute capabilities.
- After a cloud zone is selected, a host is selected by matching a series of filters, including hard & soft constraints as expressed in cloud templates.

How do I set up a simple tagging structure

This topic describes a basic approach and options for a logical VCF Automation tagging strategy. You can use these examples as a starting point for an actual deployment, or you can devise a different strategy that better suits your needs.

Typically, the cloud administrator is the primary individual responsible for creating and maintaining tags.

This topic refers to the WordPress use case described elsewhere in the VCF Automation documentation to illustrate how tags can be added to some key items. It also describes possible alternatives and extensions to the tagging examples that appear in the WordPress use case.

The WordPress use case describes how to place tags on cloud zones and storage and network profiles. These profiles are like organized packages of resources. Tags placed on profiles apply to all items within the profile. You can also create and place tags on storage resources and individual network items as well as on compute resources, but these tags apply only to the specific resources on which they are placed. When setting up tags, it is usually best to begin by tagging compute resources, and then you can add tags to profiles and cloud zones later. Also, you use these tags to filter the list of compute resources for a cloud zone.

For example, while you can place tags on storage profiles as shown in this use case, you can also place tags on individual storage policies, data stores, and storage accounts. Tags on these resources enable you to exercise finer control over how storage resources are deployed. During processing in preparation for deployment, these tags are resolved as a next level of processing after the profile tags.

As an example of how you might configure a typical customer scenario, you could place a tag of `region: eastern` on a network profile. This tag would apply to all resources within that profile. Then you could place a tag of `networktype:pci` on a pci network resource within the profile. A cloud template with constraints of eastern and pci would create deployments that use this pci network for the eastern region.

1. Tag your compute infrastructure resources in a logical and appropriate manner.

   It is particularly important that you tag compute resources in a logical manner so that you can find them using the search function on the Compute tab of the Create Cloud Zone page. Using this search function, you can quickly filter the compute resources associated with a cloud zone. If you tag Storage and Networks at the profile level, you may not need to tag individual storage and network resources.

   a) Select **Resources** > **Compute** to view the compute resources that have been imported for your VCF Automation instance.
   b) Select each compute resource as appropriate and click **Tags** to add a tag to the resource. You can add more than one tag to each resource if appropriate.
   c) Repeat the previous step for storage and network resources as appropriate.

2. Create cloud zone and network profile capability tags.

   You can use the same tags for both cloud zones and network profiles, or you can create unique tags for each item if that makes more sense for your implementation.

   In network profiles, you can place tags on the entire profile as well as on subnets within the profile. Tags applied at the profile level apply to all components, such as subnets, within that profile. Tags on subnets apply only to the specific subnet on which they are placed. During tag processing, the profile level tags take precedence over the subnet level tags.

   In this example we create three simple tags that appear throughout the use case documentation for VCF Automation cloud zone and network profile tags. These tags identify the environment for the profile components.

   - `zone:test`
   - `zone:dev`
   - `zone:prod`

3. Create storage profile tags for your storage components.

   Typically, storage tags identify the performance level of storage items, such as tier1 or tier2, or they identify the nature of storage items, such as pci.

   - `usage:general`
   - `usage:fast`

After you create a basic tagging structure, you can begin working with it and add or edit tags as appropriate to refine and extend your tagging capabilities.

How to work with resources in VCF Automation

A cloud administrator can review VCF Automation resources that are exposed through data collection.

The cloud administrator can label resources with capability tags to affect where VCF Automation cloud templates are deployed.

In addition to the views provided here, you can also manage various resources using the Resources tab.

Compute resources in VCF Automation

A cloud administrator can review compute resources that are exposed through data collection.

The cloud administrator might choose to apply tags directly to the resources to label capabilities for matching purposes in VCF Automation provisioning.

Network resources in VCF Automation

In VCF Automation, cloud administrators can view and edit the network resources that have been data-collected from the cloud accounts and integrations that are mapped to your project.

After you add a cloud account to your VCF Automation infrastructure, for example by using the **Infrastructure** > **Connections** > **Cloud Accounts** menu sequence, data collection discovers the cloud account's network and security information. That information is then available for to use in networks, network profiles, and other definitions.

Networks are the IP-specific components of an available network domain or transport zone. If you're an Amazon Web Services or Microsoft Azure user, think of networks as subnets.

You can display information about the networks in your project by using the **Infrastructure** > **Resources** > **Networks** page.

The VCF Automation**Networks** page contains information such as:

- Networks and load balancers that are defined externally in the network domain of your cloud account, for example in vCenter, NSX, or Amazon Web Services.
- Networks and load balancers that have been deployed by the cloud administrator.
- IP ranges and other network characteristics that have been defined or modified by your cloud administrator.
- External IPAM provider IP ranges for a particular address space in an provider-specific external IPAM integration.

For more information about networks, see the following information, signpost help for various settings on the **Networks** page, and Learn more about network profiles in VCF Automation.

Networks

vSphere networks, regular NSX networks, and global (federated) NSX networks are supported.

You can view and edit networks and their characteristics, for example to add tags or remove support for public IP access. You can also manage network settings such as DNS, CIDR, gateway, and tag values. You can also define new, and manage existing, IP ranges within a network.

For existing networks you can change the IP range and tag settings by selecting the network's checkbox and selecting either **Manage IP Ranges** or **Tags**. Otherwise you can select the network itself to edit its information.

Tags provide a means for matching appropriate networks, and optionally network profiles, to network components in cloud templates. Network tags are applied to every instance of that network, regardless of any network profiles in which the network may reside. Networks can be instanced into any number of network profiles. Regardless of network profile residency, a network tag is associated with that network wherever the network is used. Network tag matching occurs with other components in the cloud template after the cloud template has been matched with one or more network profiles.

Machine tags are defined in the cloud template and apply to the machine if deployed to a vCenter. Machines that are connected to an NSX local manager or global manager are also tagged in the cloud template. Note that machine tagging is different than machine NIC (network interface) tagging.

- Using global (federated) NSX networks

  NSX global networks are networks that are defined by the NSX global manager and apply to one or more NSX local managers. For global networks, existing and public networks are supported for NSX global manager and local manager cloud accounts and the vCenter cloud accounts that are associated to the local managers. Local manager representation of stretched networks is defined within a transport zone. The transport zone is an NSX local manager construct that defines the span of NSX networks for vCenter hosts and clusters.

  VCF Automation enumerates, or data collects, existing and public networks. You can create a global network by adding an existing or public network on an NSX global manager. The global network can then be consumed by all the associated local managers. Global networks can span one, all, or a subset of the associated local managers.

  You can provision a machine on a global network by using a static IP assignment. DHCP is not supported.

  You can create the following types of global networks on a global manager:

  1. Overlay - an overlay network is associated with a Tier-0/Tier-1 local manager and automatically stretches to all the sites connected to the Tier-0/Tier-1 local manager. For each local manager, the default overlay transport zone is used.
  2. VLAN - a VLAN network applies to a single local manager and the transport zone can be manually selected.

  Global networks are listed on the **Infrastructure** > **Resources** page with all the cloud accounts that they apply to.

  The following Day 2 operations are supported for global networks:

  - Reconfigure a network in a cloud template definition from a global network to a local network and vice versa.
  - Scale-out/scale-in of machines on global networks.

  For more information about using global networks in cloud templates, see More about network resources in VCF Automation cloud templates.
- Using VLAN segments in non-federated NSX networks

  You can provision NSX VLAN segments by specifying one or more VLAN IDs on a private NSX network type. Use this technique when, for example, your overall design prohibits you from provisioning overlay networks on NSX. This option requires that you select a VLAN transport zone in a supporting network profile.

  When using non-federated networks, you can provision private NSX on-demand VLAN segments when the network segments are used with a Policy API-type of NSX cloud account. VLAN segments are not connected to a Tier-1 router, therefore only private networks support VLAN segment specification. Once created, VLAN segments that are provisioned by VCF Automation can also be used as existing networks in other VMware cloud templates.

  To use VLAN segments, you must first configure the intended network profile to allow subnet isolation for the on-demand network. You must specify a VLAN transport zone in the network profile. If you specify an overlay transport zone, the network profile cannot be used for VLAN specifications. An example of VLAN transport zone selection in a network profile is shown below. For related information about configuring network profiles, see Learn more about network profiles in VCF Automation.

You specify one or more VLAN segments, or arrays of VLAN IDs, by using the `vlanIds` property in the `Cloud.NSX.Network` component in the VMware cloud template YAML. To specify multiple `vlanIds` values in the private network `Cloud.NSX.Network` component, use a separate row entry for each value. The VCF Automation API requires that you specify multiple VLAN values in a comma-separated list, but using that format in the cloud template YAML is unsupported. The supported VLAN values range between 0 to 4094. For sample cloud template YAML code, see Network, security group, and load balancer resource examples in VCF Automation.

For information about reusing networks, see How to reuse networking and security resources in VCF Automation for VM Apps.

IP Ranges

Use an IP range to define or make changes to the start and end IP address for a particular network in your organization. You can display and manage IP ranges for listed networks. If the network is managed by an external IPAM provider, you can manage IP ranges in connection with the associated IPAM integration point.

Click **New IP Range** to add an additional IP range to the network. You can specify an **internal IP range**, or if there is a valid IPAM integration available you can specify an **External IP range**.

You cannot include the default gateway in an IP range. The subnet IP range cannot include the subnet gateway value.

If you are using an external IPAM integration for a particular IPAM provider, you can use the **External IP range** to select an IP range from an available external IPAM integration point.

> **Note:**
> When an IP range from an external IPAM provider is deleted in the external IPAM application, the IP range is automatically deleted during enumeration in VCF Automation. The deleted IP range is no longer visible or available for network association in VCF Automation, thus avoiding orphaned IP address ranges.

VCF Automation allows you to apply and manage an IP address range across multiple vSphere and NSX networks. Shared IP range support is provided for both internal and external IPAM. You can set a single IP range on an NSX stretch network such that machines on that network can use IP addresses that are assigned from the single IP address even if they are deployed to different vCenters.

IP Addresses

You can see the IP addresses that are currently used by your organization and display their status, for example `available` or `allocated`. The IP addresses that are displayed are either IP addresses that are managed internally by VCF Automation or IP addresses that are designated for deployments that contain an external IPAM provider integration. External IPAM providers manage their own IP address allocation.

If the network is managed internally by VCF Automation, and not by an external IPAM provider, you can also release IP addresses.

When using internal IPAM and releasing IP addresses, for example after deleting a machine that had been using the IP addresses or clicking **Release IP address** for a selected network, there is a wait period between when the unused addresses are released and when they become available for reuse. The wait period, or release timeout period, allows the DNS cache to clear. The IP addresses can then be allocated to a new machine. By default, the IP address release wait period is 30 minutes. You can change the wait period by clicking **Infrastructure** > **Administration** > **Settings** and changing the **IP release timeout** value.

- During the release timeout period, relevant IP addresses are listed as released. When the release timeout period has expired, they are listed as available.
- The system checks every 5 minutes for newly released IP addresses, so even if the release timeout value is 1 minute it can take between 1 and 6 minutes for released IP addresses to become available, depending on when the last check was run. The 5 minute checking interval applies to all values other than 0.
- If you set the release timeout value to 0, IP addresses are released immediately and become available immediately.
- The release timeout value applies to all cloud accounts in the organization.

Load Balancers

You can manage information about available load balancers for the account/region cloud accounts in your organization. You can open and display the configured settings for each available load balancer. You can also add and remove tags for a load balancer.

For more information about using load balancers in cloud templates, see More about load balancer resources in VCF Automation cloud templates.

Network Domains

The network domains list contains related and non-overlapping networks.

Security resources in VCF Automation

After you add a cloud account in VCF Automation, data collection discovers the cloud account's network and security information and makes that information available for use in network profiles and other options.

Security groups and firewall rules support network isolation. Security groups are data-collected. Firewall rules are not data-collected.

Using the **Infrastructure** > **Resources** > **Security** menu sequence, you can view on-demand security groups that have been created in VCF Automation cloud template designs and existing security groups that were created in source

applications, such as NSX and Amazon Web Services. Available security groups are exposed by the data collection process.

You can use a tag to match the machine interface (NIC) with a security group in a cloud template definition or in a network profile. You can view the available security groups and add or remove tags for selected security groups. A cloud template author can assign one or more security groups to a machine NIC to control security for the deployment.

In the cloud template design the `securityGroupType` parameter in the security group resource is specified as `existing` for an existing security group or `new` for an on-demand security group.

Existing security groups

Existing security groups are displayed and classified in the **Origin** column as `Discovered`.

Existing security groups from the underlying cloud account endpoint, such as NSX-V, NSX, or Amazon Web Services applications, are available for use.

A cloud administrator can assign one or more tags to an existing security group to allow it to be used in a cloud template. A cloud template author can use a `Cloud.SecurityGroup` resource in a cloud template design to allocate an existing security group by using tag constraints. An existing security group requires at least one constraint tag be specified in the security resource in the cloud template design.

If you edit an existing security group directly in the source application, such an in the source NSX application rather than in VCF Automation, the updates are not visible in VCF Automation until you data collection runs and data collects the associated cloud account or integration point from within VCF Automation. Data collection runs automatically ever 10 minutes.

Existing security groups are supported for NSX global manager and local manager cloud accounts and the vCenter cloud accounts that are associated to the local managers. VCF Automation enumerates, or data collects, existing security groups and attaches them to the machine's network interfaces (NICs). You can create a global security group by adding an existing security group on an NSX global manager. The global security group can then be consumed by the associated local managers. Global security groups can span one, all, or a subset of the associated local managers.

- Global existing security groups are supported and enumerated for all defined regions.
- Global security groups are listed on the **Infrastructure** > **Resources** page with all the cloud accounts that they apply to.
- You can associate a machine interface (NIC) with an existing global security group directly in a cloud template or in the selected network profile.
- The following Day 2 operations are supported for global security groups:
  - Security group reconfiguration in a cloud template from a global to a local security group and vice versa.
  - Scale-out/scale-in of machines that are associated with global security groups.

On-demand security groups

On-demand security groups that you create in VCF Automation, either in a cloud template or in a network profile, are displayed and classified in the **Origin** column as `Managed by` VCF Automation. On-demand security groups that you create as part of a network profile are internally classified as an isolation security group with pre-configured firewall rules and are not added to a cloud template design as a security group resource. On-demand security groups that you create in a cloud template design, and that can contain express firewall rules, are added as part of a security group resource that is classified as `new`.

> **Note:**
>
> You can create firewall rules for on-demand security groups for NSX and NSX directly in a security group resource in cloud template design code. The **Applied To** column does not contain security groups that are classified or managed by a vDefend Distributed Firewall (DFW). Firewall rules that apply to applications are for east/west DFW traffic. Some firewall rules can only be managed in the source application and cannot be edited in VCF Automation. For example, ethernet, emergency, infrastructure, and environment rules are managed in NSX.

On-demand security groups are not currently supported for NSX global manager cloud accounts.

Learn more

For more information about using security groups in network profiles, see Learn more about network profiles in VCF Automation.

For information about defining firewall rules, see Using security group settings in network profiles and cloud template designs in VCF Automation.

For more information about using security groups in a cloud template, see More about security group and tag resources in VCF Automation cloud templates.

For cloud template design code samples that contain security groups, see Network, security group, and load balancer resource examples in VCF Automation.

Storage resources in VCF Automation

A cloud administrator can work with storage resources and their capabilities, which are discovered through VCF Automation data collection from associated cloud accounts.

Storage resource capabilities are exposed through tags that typically originate at the source cloud account. A cloud administrator can choose to apply additional tags directly to storage resources though, using VCF Automation. The additional tags might label a specific capability for matching purposes at provisioning time.

VCF Automation supports standard disk and first class disk capabilities. First class disk is available for vSphere only.

- What can I do with standard disk storage in VCF Automation for VM Apps
- What can I do with first class disk storage in VCF Automation for VM Apps

Capabilities on storage resources become visible as part of the definition of an VCF Automation storage profile. See Learn more about storage profiles in VCF Automation.

First class disks that have been data-collected appear on the **Consume** > **Deployments** > **Volumes** view.

Learn more about resources in VCF Automation

VCF Automation can expose additional information around data-collected resources, such as pricing cards.

How does data collection work in VCF Automation

After initial data collection, resource data collection occurs automatically every 10 minutes. The data collection interval is not configurable and you cannot manually initiate data collection.

You can discover information about resource data collection and image synchronization for an existing cloud account in the Status section of its page. Do so by selecting **Infrastructure** > **Connections** > **Cloud Accounts** and then clicking **Open** on the existing cloud account of your choice.

You can open an existing cloud account and see its associated endpoint version in the **Status** section of its page. If the associated endpoint has been upgraded, the new endpoint version is discovered during data collection and reflected in the **Status** section on the cloud account's page.

Resource data collection

Data collection occurs every 10 minutes. Each cloud account displays when its data collection last completed.

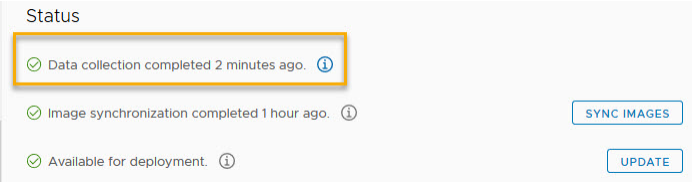


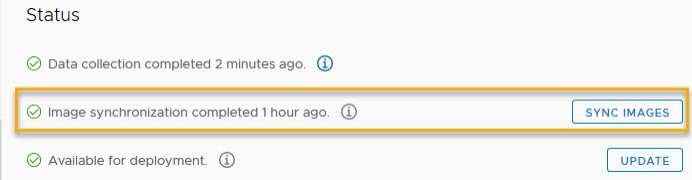
Image data collection

Image synchronization occurs every 24 hours. You can initiate image synchronization for some cloud account types. To initiate image synchronization, open the cloud account (**Infrastructure** > **Cloud Accounts** then select and open

the existing cloud account) and click the **Sync Images** button. There is no image synchronization option for NSX cloud accounts.

> **Note:**
>
> Images are internally classified as either public or private. Public images are shared and are not specific to a particular cloud subscription or organization. Private images are not shared and are specific to a specific subscription. Public and private images are automatically synchronized every 24 hours. An option on the cloud account page allows you to trigger synchronization for private images.

The cloud account page displays when image synchronization was last completed.



To facilitate fault tolerance and high availability in deployments, each NSX data center endpoint represents a cluster of three NSX managers. For related information, see Create an NSX cloud account in VCF Automation for VM Apps.

Cloud accounts and onboarding plans

When you create a cloud account, all machines that are associated to it are data-collected and then displayed on the **Resources** > **Virtual Machines** page. If the cloud account has machines that were deployed outside of VCF Automation, you can use an onboarding plan to allow VCF Automation to manage the machine deployments.

For information about adding cloud accounts, see Adding cloud accounts to VCF Automation for VM Apps.

Updating networking resources in VCF Automation after N-VDS to C-VDS migration in NSX

After NSX migration from NSX Virtual Distributed Switch (N-VDS) to converged VDS (C-VDS), you must update impacted vSphere network resources in VCF Automation to continue using those resources in new and existing cloud templates and deployments.

After N-VDS to C-VDS migration, your vSphere networks may appear to be missing from VCF Automation network profiles in which they are members. To avoid losing these vSphere type networks, and continue to allocate them in existing and new deployments, you must manually update all listed C-VDS networks in VCF Automation.

> **Note:**
>
> While users do not need the VMware Cloud on AWS Cloudadmin role to create VMware Cloud on AWS cloud accounts in VCF Automation for N-VDS, they do need that permission level to access C-VDS assets after N-VDS to C-VDS migration. Active Directory members with containerized permissions need host switch-level access (ReadOnly) to migrated C-VDS resources in VCF Automation. Cloud Administrator group (Cloudamin role) users have host switch-level permissions. VCF Automation users who are not members of the VMware Cloud on AWS Cloud Administrator group cannot access migrated C-VDS resources.
>
> - Active Directory members who are assigned the Cloud admin role in VMware Cloud on AWS prior to the N-VDS to C-VDS migration in NSX have the Cloudadmin role in VMware Cloud on AWS after N-VDS to C-VDS migration, and thus have the required access level to migrated C-VDS resources.
> - Active Directory members who are not assigned the Cloud admin role in VMware Cloud on AWS before N-VDS to C-VDS migration in NSX must be assigned the Cloud admin role after the migration.

> **Note:**
>
> This procedure is specific to actions needed in VCF Automation to update *vSphere* networks after N-VDS to C-VDS migration has been performed in NSX. There is no action needed in VCF Automation on *NSX* networks after N-VDS to C-VDS migration; *NSX* networks require no manual intervention after N-VDS to C-VDS migration.

NSX networks that are attached to vCenter cloud accounts, as well as to VMware Cloud on AWS cloud accounts, are supported and do not require the manual intervention described in this procedure. However, *NSX* networks that are attached to VMware Cloud on Dell cloud accounts may require the manual intervention described here. For related information, see VMware Cloud on AWS (VMConAWS) and VMware Cloud on Dell EMC Migration from N-VDS to VDS (82487).

While an NSX administrator can migrate NSX on VDS (N-VDS) network types to converged VDS (C-VDS) network types in NSX, this action impacts existing vSphere network resources in VCF Automation. The VCF Automation administrator can perform post-migration actions to reconcile those resources in VCF Automation with the associated changes in NSX and vCenter. Note that C-VDS, or simply VDS, is also referred to elsewhere as vSphere Virtual Distributed Switch (VDS).

For related information about NSX converged VDS (C-VDS), see VMware Knowledge Base article NSX-T on VDS (79872).

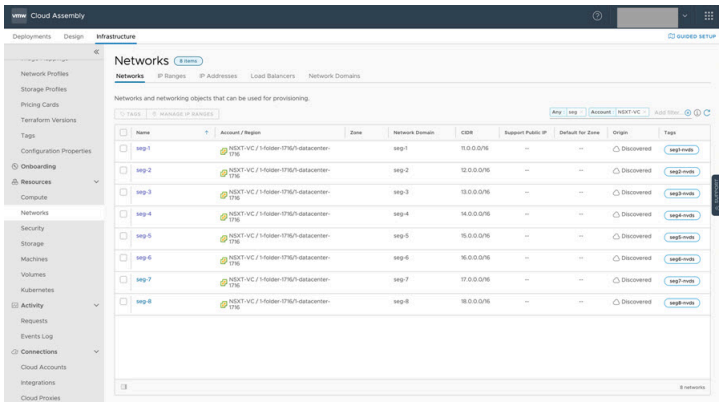
Note:

This sample scenario illustrates the steps needed to reconcile resources in a VCF Automation environment after N-VDS to C-VDS migration. You can use this example and procedure in VCF Automation 8.5 and later to reconcile changes made in vCenter after migrating from N-VDS to C-VDS in NSX.

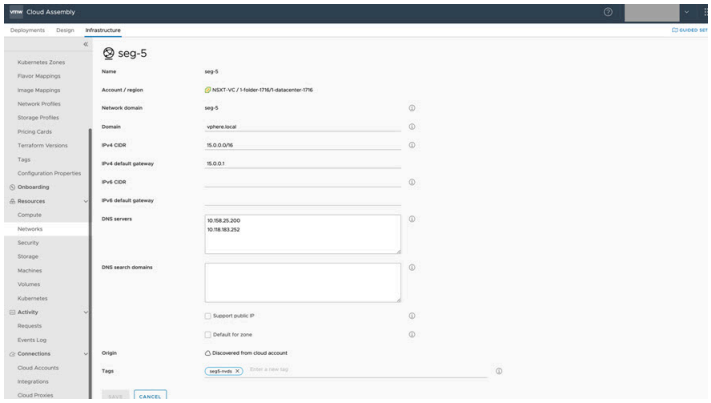
Example: VCF Automation resources pre-migration

This example illustrates sample NSX resources in a sample VCF Automation environment prior to N-VDS to C-VDS migration.

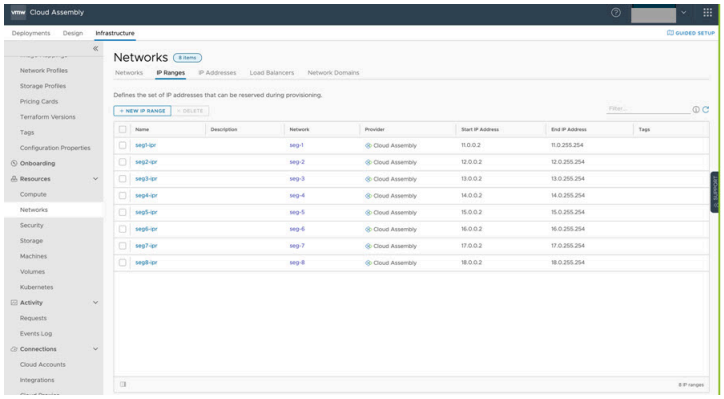
- This example contains NSX and vCenter cloud accounts.
- The example contains several vSphere networks.

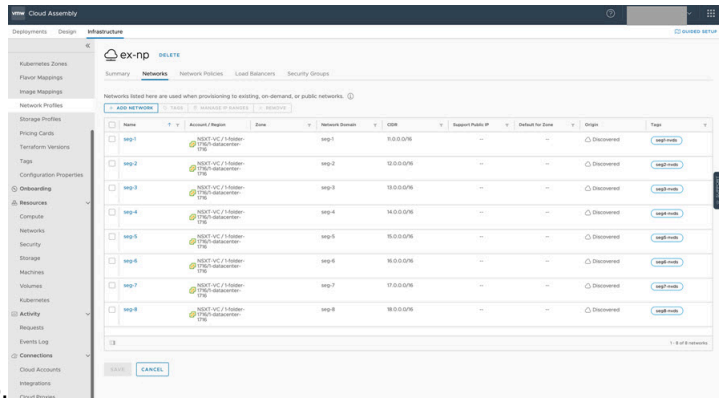


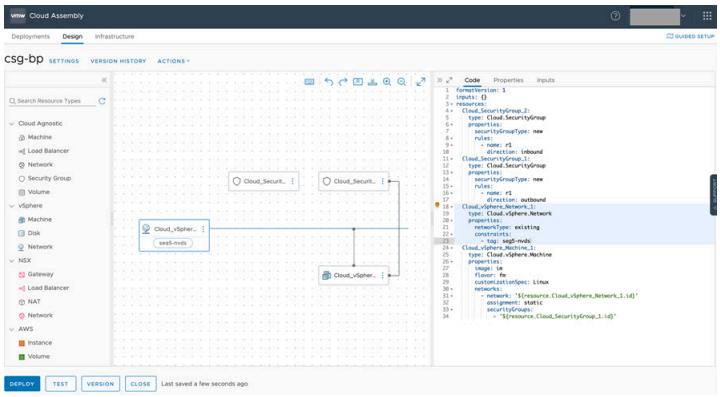
- The example network configuration contains CIDR and DNS settings.



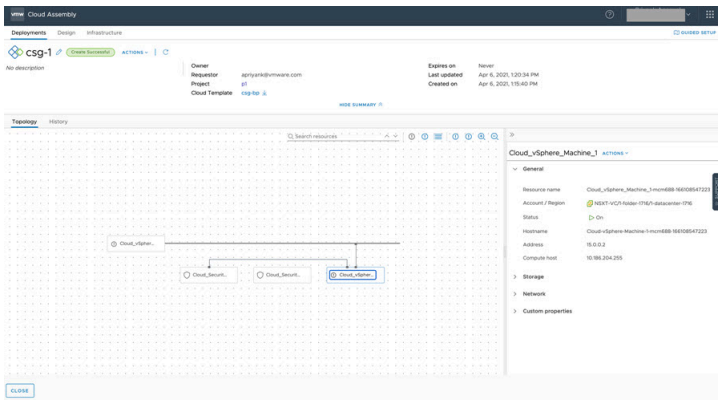
- The example also includes existing IP ranges.



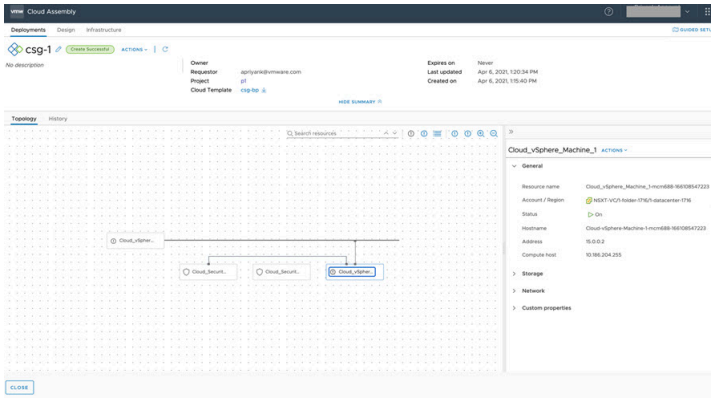
- The example contains a network profile (**ex-np**) which contains several N-VDS (N-VDS) networks, including  **seg-5**.
- In this example, the existing **seg5** network component is shown with sample cloud template syntax. The network is tagged as an N-VDS network. The example illustrates needed post-migration updates to the **seg5** network later in the workflow.



- The example cloud template generates the deployment.



- The example machine IP addresses are displayed in the sample deployment.

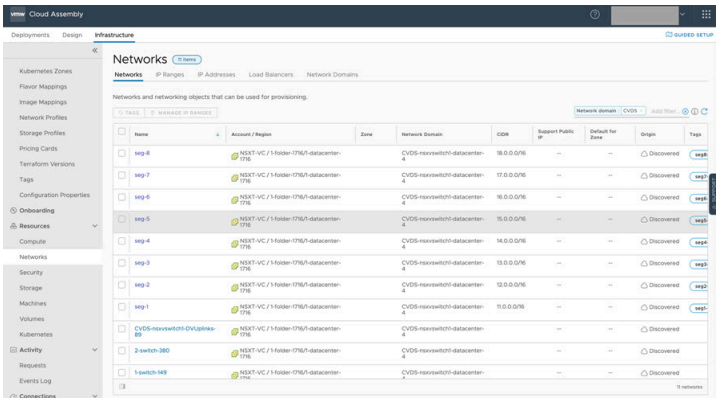


Example: Post-migration Step 1 – Run data collection after N-VDS to C-VDS migration and enumeration

In the above section, screen shots were used to illustrate the infrastructure used in an example VCF Automation environment, concluding with the output cloud template and deployment.

After you or another administrator perform N-VDS to C-VDS migration in NSX, wait at least 10 minutes to allow VCF Automation to perform its periodic data collection and enumeration process to fetch and display impacted resources in VCF Automation.

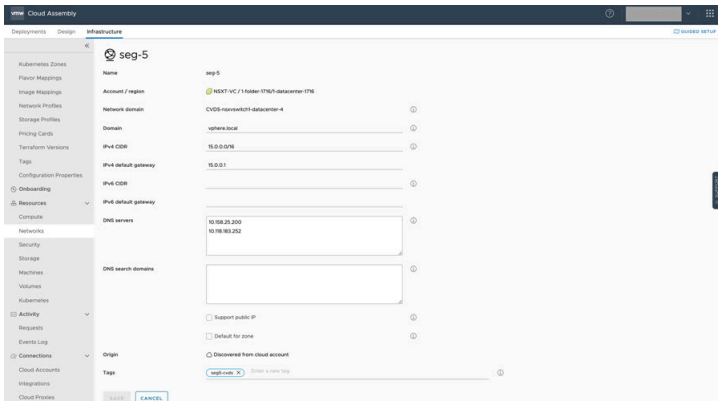
After allowing VCF Automation data collection to complete, click **Infrastructure** > **Resources** > **Networks** to view and access available C-VDS networks. Notice the **seg5** network, as shown below.



Example: Post-migration Step 2 – Add previously defined CIDR and DNS to migrated C-VDS networks

Edit a migrated C-VDS network to add CIDR and DNS details that had been specified in the pre-migration N-VDS definition and change the network tagging.

1. Add CIDR and DNS details that had been defined in its pre-migration N-VDS definition
2. Add a new tag for the sample C-VDS **seg-5** network segment, such as `seg5-cvds`.



Note that the original N-VDS **seg-5** network was tagged as *seg5-nvds*, as seen in earlier screens. The change in resource tagging details is required by network reconfiguration. VCF Automation requires that you include a different tag name in the cloud template for the C-VDS network than the tag used in the original N-VDS network. The changed tagging identifies a change in the cloud template when generating a valid redeployment.

Example: Post-migration Step 3 – Add updated IP range information

You can edit network IP ranges to IP range details that had been specified in the pre-migration N-VDS definition, by using a command line API or by using a menu sequence in VCF Automation.

- Option 1: Use the API to update IP range data, as shown in the following sample screen.

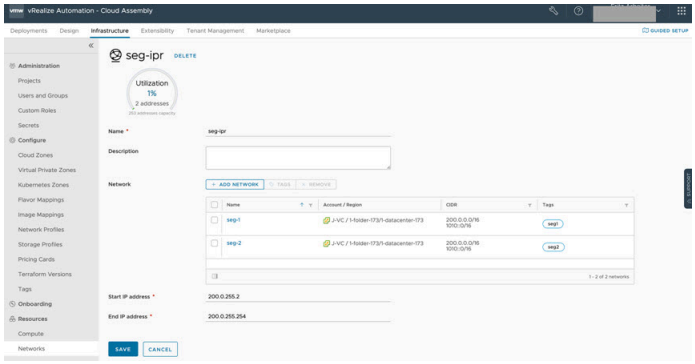
```
PATCH : {{host}}/iaas/api/network-ip-ranges/{{subnet-range-id}}

Headers :
    - Authorization : Bearer {{token}}

Payload :

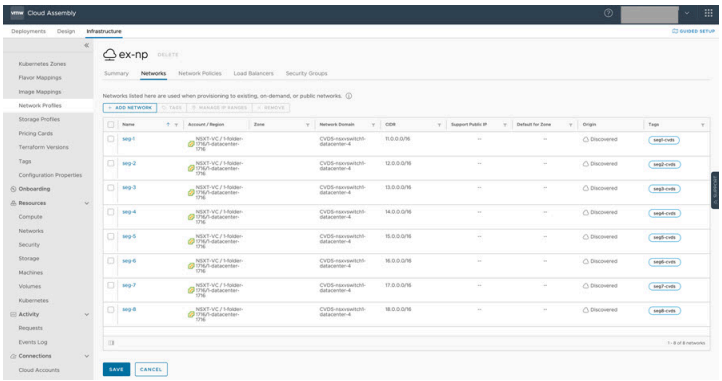
{
    "fabricNetworkIds": ["{{subnet-id}}"]
}
```

- Option 2: Use the user interface to update IP range data, as shown in the following sample screen.



Example: Post-migration Step 4 – Update network profiles to correct missing networks

Post-migration, N-VDS networks are reconciled and deleted from VCF AutomationVCF Automation after data collection and enumeration. Impacted network profiles (such as the example **ex-np**) have missing networks. To correct the missing networks issue, update each N-VDS network as a C-VDS network, as shown below.

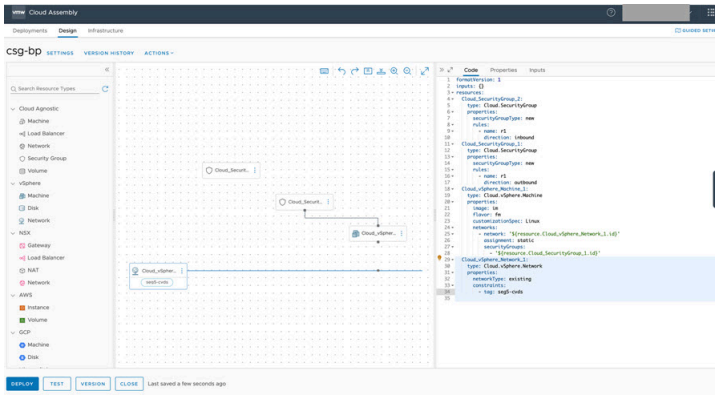


Example: Post-migration Step 5 – Update network constraints in cloud templates

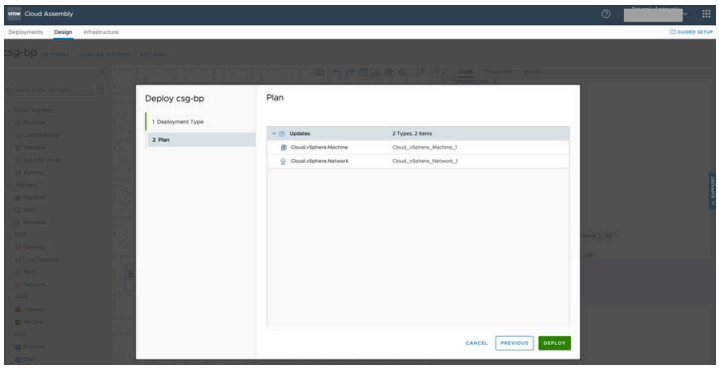
For existing deployments, you must update network constraints in cloud template to match the new C-VDS networks in the updated network profiles. Updated network constraints are also needed to perform iterative deployments and to reconfigure networks from their original vSphere N-VDS representation to vSphere C-VDS representation.

For new deployments, the specified C-VDS resources are used, thus this step is not required. Iterative deployments and network reconfiguration simply work as designed.

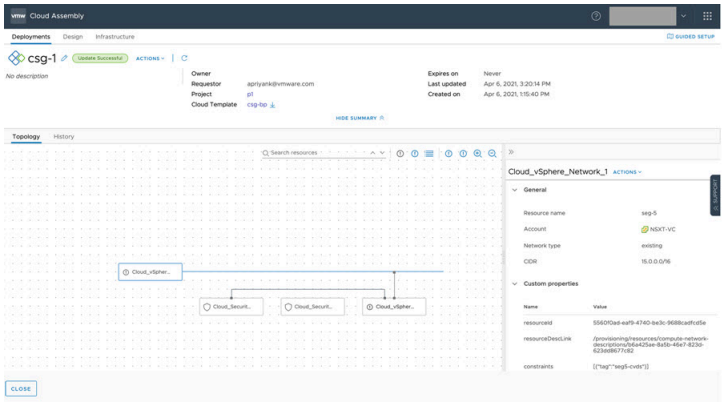
1. For this example, change network constraints in the cloud template from `seg5-nvds` to `seg5-cvds`, as shown below.



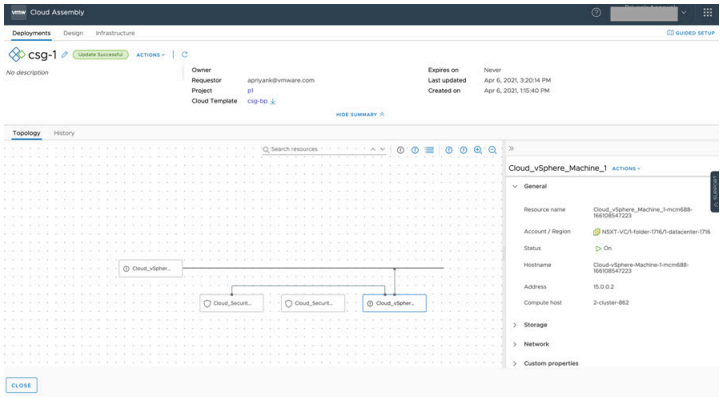
2. Perform an iterative deployment to reconfigure the network, as shown below.



3. After successful redeployment, notice that the network custom properties display the updated constraints, as shown below.



Because the IP range was updated earlier with the new C-VDS data, the machine IP address does not change in the redeployment, as shown below.



Use the Insights dashboard to monitor resource capacity and notify project owners in VCF Automation

A cloud administrator can monitor and manage infrastructure resources and deployment optimizations within each cloud zone. By visualizing real-time insights, and reviewing suggested actions for the resources you support, you can proactively help project owners manage their resource capacity and optimize their deployments.

You can use the **Insights** dashboard to explore metric data for the resources and deployments in cloud zones within the projects that you manage. Use that information, provided from a combination of VCF Automation and your integrated VCF Operations application, to make any needed adjustments to memory, CPUs, and so on, or share that information with your team so that they can be better informed and make any needed adjustments.

The Insights dashboard enables you to contact some or all of the project owners who have deployments in the cloud zone that contain reclaimable resource capacity. The cloud zone insights display reclaimable capacity for projects and deployments.

Contacted project owners see notification on their deployment's **Alerts** page. The notification contains their name and the name of (and link to) each deployment that can be optimized.

The **Insights** dashboard is available for vSphere and VMware Cloud on AWS cloud zones, provided that the cloud accounts are configured in both VCF Automation and VCF Operations and are being monitored in VCF Operations.

Prerequisites

- Review Resource management and deployment optimization using VCF Operations metrics in VCF Automation.
- Verify that you have VCF Automation cloud administrator credentials and have enabled HTTPS access on port 443.
- Verify that you have the VCF Automation cloud administrator user role.
- Configure VCF Automation integration with VCF Operations.
- Configure the VCF Automation adapter in VCF Operations.

For related information, see the VCF Operations product documentation.

About VCF Operations and the collected resource capacity metrics

VCF Operations collects capacity metrics for the same infrastructure resources that you and the teams that you support use in VCF Automation. By integrating VCF Automation with VCF Operations, the VCF Operations metric data is made available and displayed for each managed project in an **Insights** dashboard within each cloud zone.

Project data is parsed to the VCF Automation dashboard from the integrated VCF Operations application. The Insights dashboard displays the following information:

- CPU utilization percentage relative to capacity
- Memory utilization percentage relative to capacity
- Storage utilization percentage relative to capacity
- Calculated CPU and memory demand history and projected demand
- Option to contact owners of some or all of the deployments in a cloud zone that can be optimized by reclaiming resources, for example by resizing or deleting machines. Optimization data is calculated in the order of days.

The Insights dashboard is available for vSphere resources.

A trend widget displays the compute components of a cloud zone (such as clusters and hosts), their CPU GHz usage relative to CPU capacity, and their memory GB usage relative to memory capacity.

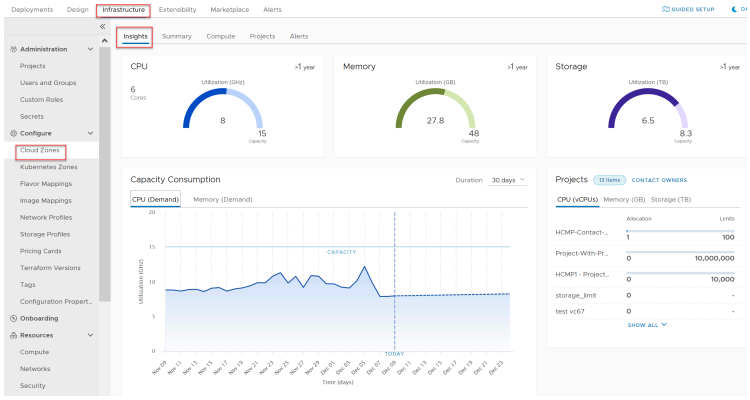
Information about the roles that are required to use alerts is available at Custom user roles in VCF Automation for VM Apps.

For related information, see Resource management and deployment optimization using VCF Operations metrics in VCF Automation.

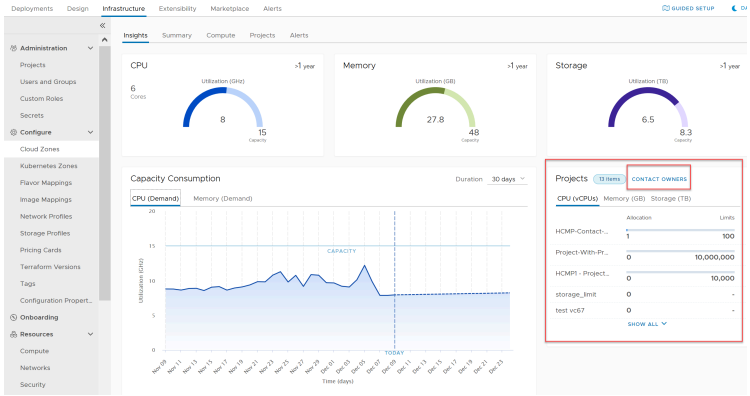
Procedure

Open a cloud zone to discover its capacity metrics and optionally fetch information about project deployments that can be optimized. Data is collected and supplied by the associated VCF Operations application.

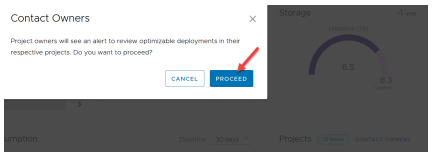
1. From VCF Automation, click **Infrastructure** > **Configure** > **Cloud Zones** and select a cloud zone.
2. Click the **Insights** tab and examine the insights dashboard.
   The following example displays CPU, memory, and storage capacity information for the resources that are used by projects in the cloud zone.



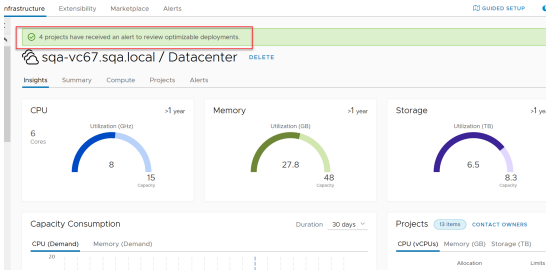
3. To notify the project owner of any deployments that can be optimized, click **Contact Owner** in the **Projects** section. Notifications appear on the **Alerts** tab page.



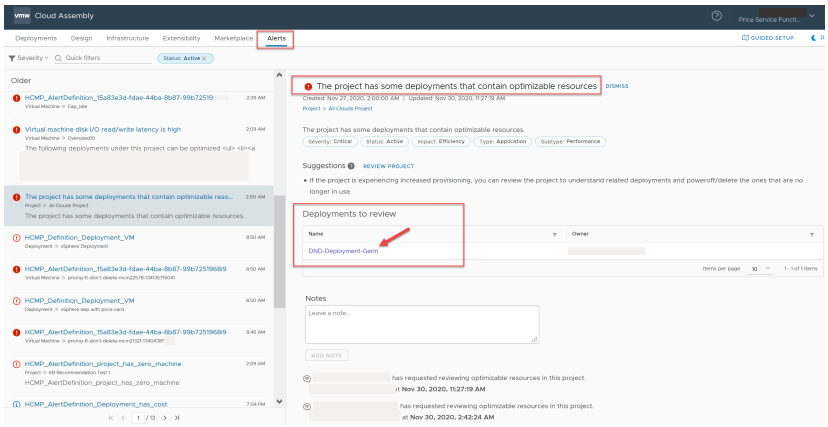
4. To fetch optimization information about each deployment for the project, click **Proceed**.



   If the project contains deployments that can be optimized, that information is conveyed to the project owner on the VCF Automation**Alerts** tab.
5. A message appears indicating the number of deployments that can be optimized.



Notification information about these resources and deployments is available to the project owner on the VCF Automation**Alerts** tab. For this example, that notification information includes the name of, and a link to, each deployment that can be optimized, as shown in the following example:



Next steps

Use the information that you have obtained from the **Insights** dashboard to make any needed adjustments to the resources that you manage. Open the **Alerts** page to obtain additional information, suggested actions, and links to deployments that can be optimized. See How to use Alerts to manage resource capacity, performance, and availability in VCF Automation.

How to use Alerts to manage resource capacity, performance, and availability in VCF Automation

As a cloud administrator, you need to know when VCF Automation capacity, performance, and availability are becoming problematic so that you can proactively react before users begin to run out of resources.

You can display a range of alerts provided by the associated VCF Operations application. Alerts are available for vSphere and VMware Cloud on AWS resource objects. Use information in alerts to modify the resources and deployments that you manage, or share that information with your team so they can modify objects that they manage.

> **Note:**
>
> To examine and act on project deployments that you should consider optimizing, see How to use Alerts to optimize deployments in VCF Automation for VM Apps.

Alerts are currently available for vSphere and VMware Cloud on AWS resource objects only. The **Alerts** tab is only available if access to VCF Operations is configured.

The VCF Automation alerts threshold values are set in VCF Operations. Some VCF Automation alerts are currently predefined. Alerts notifications are also set in VCF Operations. For information about setting alert definitions and configuring notifications, see the VCF Operationsproduct documentation.

Prerequisites

- Review Resource management and deployment optimization using VCF Operations metrics in VCF Automation.

- Verify that you have VCF Automation cloud administrator credentials and have enabled HTTPS access on port 443. See Credentials required for working with cloud accounts in VCF Automation for VM Apps.
- Verify that you have the VCF Automation cloud administrator user role. See What are the VCF Automation for VM Apps user roles.
- Configure VCF Automation integration with VCF Operations.
- Configure the VCF Automation adapter in VCF Operations.
- Configure the roles that are need to manage alerts. See Custom user roles in VCF Automation for VM Apps.
  For related information, see See Integrate VCF Automation with VCF Operations and *Application Integration* in the VCF Operations product documentation.
  Role capabilities include:
  - Cloud administrators can manage cloud zone alerts.
  - Project administrators can manage project alerts.
  - Service broker administrators can manage deployment alerts.

About VCF Operations and resource alerts

VCF Operations collects health, usage, and other metrics for the same infrastructure resources and deployments that you manage in VCF Automation. By integrating VCF Automation with VCF Operations, that monitored data is made available to you in VCF Automation by using the **Alerts** tab in the VCF Automation main menu .

The alerts data provided by VCF Operations includes heath and risk threshold concerns for cloud templates, deployments, organizations, and projects. It also contains information about deployments that can be optimized, based on the owner being contacted by an action taken on the cloud zone **Insights** tab. For related information, see How to use Alerts to optimize deployments in VCF Automation for VM Apps.

Alert details for each deployment include:

- Project name
- Deployment name (and link to the deployment) that contain resources that can be optimized
- Suggested actions
- Potential cost savings from reclamation and optimization
- Total number of virtual CPUs used by the deployment
- Total amount of RAM memory used by the deployment
- Total amount of storage used by the deployment
- Virtual machines in the deployment that are recommended for reclamation and optimization, including resource name, idle machines, powered off machines, oversized and undersized machines, underutilized machines, and machine snapshots

By using the **Contact project owners** option on the cloud zone Insights dashboard, you can see a summary of all projects that have reclaimable capacity (CPU, memory, and storage) in the cloud zone and provide an alert to some or all of the project owners.

Procedure

You can display alerts threshold information about the resources that you manage by using filtering options on the **Alerts** page. Alerts data is supplied by your associated VCF Operations application. Suggested actions are provided for each alert.

You can also select a deployment from the **Deployments to review** section to open and optimize that deployment.

1. From within the VCF Automation service, click the **Alerts** tab in the main menu.
2. To control the how alerts are displayed, experiment with the available filters. For example, select the **Resources** option from the filters drop-down menu.
3. To display alerts and suggested actions for those alerts, use quick filter options in the selector panel.
   - Display alerts about cloud zone resources.
     VCF Operations can monitor time remaining, capacity remaining, reclaimable capacity, and so on.
   - Display alerts about virtual machine resources.

The majority of virtual machine alerts pertain to on/off status, latency, and so on.
  - Display alerts about deployment resources.
    The deployment alerts pertain to reclaimable resources and right-sizing.
  - Display alerts about project resources.
    The project alerts pertain to reclaimable resources and allocation limits.
4. Explore other filter types and their quick filtering options to further control the list of alerts.
  - Use the **Impacts** quick filters of health, risk, and efficiency.
  - Use the **Severity** quick filters of critical, immediate, warning, and information.
  - Use the **Status** quick filters of active, cancelled, and dismissed.
  - Use the **Subtype** filters of availability, performance, and capacity.
  - Use the **Type** quick filters of application, hardware, infrastructure, storage, and network.
5. Take any needed actions based on alerts data and suggestions.

Next steps

To learn about other actions that are available, see How to use Alerts to optimize deployments in VCF Automation for VM Apps.

You can also display capacity **Insights** for cloud zone-based resources in projects that you manage. For information about using VCF Operations- supplied **Insights** data in VCF Automation, see Use the Insights dashboard to monitor resource capacity and notify project owners in VCF Automation.

How to use Alerts to optimize deployments in VCF Automation for VM Apps

As a cloud administrator or project owner, you can monitor and manage machine resources for best possible optimization by using data that is obtained from VCF Operations and displayed in VCF Automation.

When you connect VCF Automation with VCF Operations, you can access data-collected information about resources in the projects that you manage. Alerts and insights data is provided to inform you of various concerns about the projects that you manage, and provide an easy means to communicate optimization suggestions and supporting data collected from VCF Operations to project owners easily and efficiently without ever leaving the VCF Automation application. For example, you can see reclaimable resource capacity, with specific cost savings, for each deployment in a cloud zone. Where a cloud zone contains multiple deployments that can be optimized, you can notify some or all of the project and deployment owners.

Deployment optimization alerts can be generated from the Insights dashboard. See Use the Insights dashboard to monitor resource capacity and notify project owners in VCF Automation. You can contact project owners so that they can open a named deployment to be optimized from a link provided on the **Alerts** page. As well, project owners can open their deployments directly and use the **Optimize** tab to perform available optimization tasks. Actions that a project owner can take include reclaiming resources by deleting non-critical deployments, and stopping further provisioning within a cloud zone.

> **Note:**
>
> To learn about other resource remediation actions that you can take, see How to use Alerts to manage resource capacity, performance, and availability in VCF Automation.

Prerequisites

See How to use Alerts to manage resource capacity, performance, and availability in VCF Automation for needed credentials and configuration information for accessing VCF Operations data in VCF Automation.

To request that project owners be alerted of deployments that are optimizable, see Use the Insights dashboard to monitor resource capacity and notify project owners in VCF Automation.

About

Each deployment contains an **Optimize** tab. The following optimization parameters are available:

- Machines that can be rightsized - Displays information and actions for oversized and undersized machines in the deployment, along with optimization cost savings.
- Machines that are under-utilized - Displays information and actions for idle or powered off machines in the deployment, along with optimization cost savings.
- Machine snapshots - Displays information and actions for machine snapshots if machines in the deployment contain snapshots, along with optimization cost savings.

As an administrator, you can notify project owners that they have deployments to optimize. Notifications appear on the **Alerts** tab in VCF Automation.

The **Alerts** tab is only available if access to VCF Operations is configured. Project owners can open and optimize their deployments to respond to alerts.

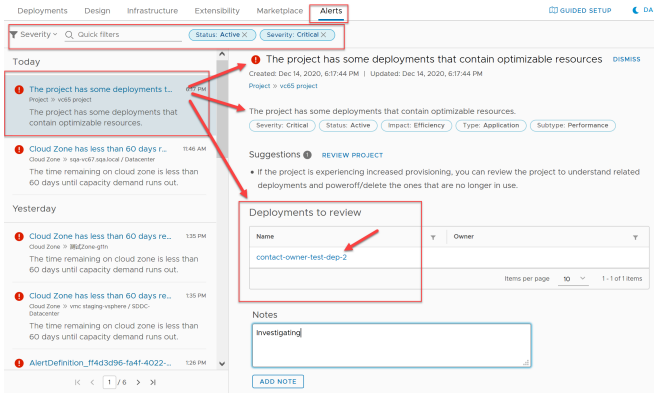
Procedure

You can display alerts threshold information about the resources that you manage by using filtering options on the **Alerts** page. Alerts data is supplied by your associated VCF Operations application. Suggested actions are provided for each alert. In this example the project owner opens their deployment from a link supplied on an alert notification. The deployment's **Optimize** tab displays available machine parameters to optimize.

1. As a project owner or administrator, click the **Alerts** tab in the main menu.



2. Find an alert that contains information about a deployment that can be optimized and click the deployment name from **Deployments to review** to open that deployment and display its **Optimize** tab.



3. When the deployment opens, click the **Optimize** tab.



4. If there are underutilized machines, examine and act on idle and powered off machines. You can power off or delete an undersized deployment.
5. If there are machines that can be rightsized, examine and act on any oversized and undersized machines in the deployment.
6. If one or more of the machines in the deployment contains a snapshot, you can delete or export each snapshot.
7. When you are finished, confirm that the deployment has been optimized to your satisfaction and close the deployment

Next steps

To learn about other actions that are available, see How to use Alerts to manage resource capacity, performance, and availability in VCF Automation.

You can also display capacity **Insights** for cloud zone-based resources in projects that you manage. For information about using VCF Operations- supplied **Insights** data in VCF Automation, see Use the Insights dashboard to monitor resource capacity and notify project owners in VCF Automation.

What can I do with standard disk storage in VCF Automation for VM Apps

Standard disks can be persistent or non-persistent.

VCF Automation supports two categories of storage – standard disk and first class disk. First class disk is only available for vSphere.

- vSphere
  vSphere supports dependent (default), independent persistent, and independent non-persistent standard disks. For related information, see What can I do with persistent disk storage in VCF Automation for VM Apps.
  When you deploy a vSphere virtual machine with one or more disks, the disk resource name depends on your custom naming conventions. However, the disk `.vmdk` file name follows the default vSphere naming convention, for example, machineName_index, and is provided in the disk **diskFile** custom property. For example, `[your datastore]Cloud_Machine_{index number}-{identifier}/Cloud_Machine-{index number}-{identifier}.vmdk`. The disk file name will be the same as the disk name. To use this convention, you must turn the feature on by making the following changes to your VCF Automation instance using the API. To make the changes, you must have a bearer token.

  ```
  https://<host>/provisioning/config/toggles
  {
      "key": "enable.vsphere.default.disk.file.naming",
      "value": "true"
  }
  ```

  When you delete a virtual machine, its dependent and independent non-persistent disks are also deleted.
  When you delete a virtual machine, its independent persistent disks are not deleted.
  You can create a snapshot of dependent and independent non-persistent disks. You cannot create a snapshot of an independent persistent disk.
  vSphere cloud zone storage limits that you defined in each project are evaluated when you deploy a cloud template that includes independent non-persistent or independent persistent disks, or when you run a day relevant day 2 action. A request might fail if the disk requested exceeds the limits.
- Amazon Web Services (AWS) EBS
  You can attach an EBS volume to an AWS compute instance or detach an EBS volume from an AWS compute instance.
  When you delete a virtual machine, its attached EBS volume is detached but not deleted.
- Microsoft Azure VHD
  Attached disks are always persistent.
  When you delete a virtual machine, you specify whether to remove its attached storage disks.
- Google Cloud Platform (GCP)
  Attached disks are always persistent.
  Persistent disks are located independently from your virtual machine instances, so you can detach or move persistent disks to keep your data even after you delete your instances.
  When you delete a virtual machine, its attached disk is detached but not deleted.

For related information, see Learn more about storage profiles in VCF Automation.

What can I do with persistent disk storage in VCF Automation for VM Apps

Persistent disks preserve valuable data from accidental deletion.

In a cloud template, under a volume, you can add the `persistent: true` property to have the disk survive VCF Automation deletions. Persistent disks aren't removed during deployment deletion nor Day 2 delete or remove disk operations.

Because of that, persistent disks can remain in your infrastructure even after a deployment deletion or disk deletion. To remove them, you can use the following techniques.

- Explicitly pass the purge flag as a query parameter using the DELETE API.

- Delete them directly from your cloud endpoint.

Note that there is no VCF Automation user interface for removing them.

What can I do with first class disk storage in VCF Automation for VM Apps

A first class disk (FCD) provides storage life-cycle management on virtual disks as a disk-as-a service or as EBS-like disk storage that allows you to create and manage disks independently of vSphere virtual machines.

VCF Automation supports two categories of storage disks – standard disk and first class disk. First class disk functionality is supported for vSphere only. VCF Automation currently provides first class disk functionality as an API-only capability.

A first class disk has its own life-cycle management capabilities that operate independently from a VM. One way that a first class disk differs from an independent persistent disk, is that you can use a first class disk to create and manage snapshots independent of a VM.

You can create a new VCF Automation storage profile to support either first class disk capabilities or standard disk capabilities. See Learn more about storage profiles in VCF Automation and Storage resources in VCF Automation.

You can also add a `Cloud.vSphere.Disk` first class disk element in your VCF Automation cloud templates and deployments to support vSphere first class disks. First class disks that have been data-collected appear on the **Resources** > **Resources** > **Volumes** page.

In vCenter, first class disks are also referred to as *Improved Virtual Disks (IVD)* or *managed virtual disks*.

Capabilities

Using VCF Automation API capabilities, you can:

- Create, list, and delete a first class disk.
- Resize a first class disk.
- Attach and detach a first class disk.
- Create and manage first class disk snapshots.
- Convert an existing standard disk to first class disk

The following scenarios are not supported:

- Provisioning VMs from snapshots on a datastore cluster.
- Owning and sharing device-based storage blocks by users and tenants.
- Creating and restoring VM snapshots.
- Attaching storage across multiple VMs and across clusters.

API use case documentation for creating and managing first class disk (FCD) storage in VCF Automation is available in the *Creating and Using a First Class Disk* section of the VCF Automation Programming Guide for your VCF Automation release.

Considerations and limitations

First class disk considerations and limitations currently include:

- First class disk is available for vSphere VMs only.
- vSphere 6.7 Update 2 or later is required to use first class disks.
- Provisioning first class disks on datastore clusters is not supported.
- Volume multi-attach is not supported for first class disks.
- First class disks with snapshots cannot be resized.
- First class disks with snapshots cannot be deleted.
- First class disk snapshot hierarchy can only be constructed by using the `createdAt` API option.
- The minimum VM hardware version required to attach a first class disk is vmx-13 (ESX 6.5 compatible).
- Cloud zone storage limits that you defined in each project are evaluated when you deploy a cloud template that includes a first class disk, or when you run a day relevant day 2 action. A request might fail if the disk requested exceeds the limits.

How to reuse networking and security resources in VCF Automation for VM Apps

Configured networking components such as networks, load balancers, and security groups, can be treated as reusable objects. These objects can be inserted into VCF Automation templates to quickly add a predefined set of network, security group, and/or load balancer settings. This capability, which is sometimes referred to as infrastructure-as-code or provider networks, allows you to reuse predefined networking and security configurations in multiple templates and/or deployments.

A configured networking component is simply an VCF Automation template that consists of only a networking component, or a security group, or a load balancer that has been configured with settings and saved as a stand-alone template. These templates can be inserted into other templates to quickly add a previously specified configuration. Only Cloud Admins can create, define, and then save a configured networking component template. However, after a Cloud Admin creates the template, the template can be inserted into (and reused as a resource chunk) within other templates by a designer who does not have Cloud Admin privileges.

The following considerations impact whether a configured networking component can be used:

- Deployment environment, such as simple, distributed, load-balanced, or HA.
- User role assignments, specifically Cloud Admin or non-Cloud Admin. Only Cloud Admins can create the reusable configured networking component template. But non-Cloud Admins can insert a configured networking component template into their own templates. However, non-Cloud Admins cannot change the configured networking component content /settings after they are inserted into their own templates.

How to manage infrastructure capacity for VCF Automation for VM Apps

As a cloud administrator, you can set values that control how resources are allocated for deployments for your entire organization.

While you can use placement policies on cloud zones and in projects to control the distribution of workloads at deployment time, the methods presented here are used to prevent overallocation and set limits at the organization level.

Each of the resource checks described in this article operate independently. They are not dependent on one another.

What to do first

- Ensure that you have the organization owner role and VCF Automation Administrator service role. You will need them to get an authentication token and run the commands. See What are the VCF Automation for VM Apps user roles.
- Get an authentication topic. See *Getting Your Authentication Token* in the API Programming guide.

Prevent memory overallocation

Some systems allow you to deploy resources even if the host or cluster does not have sufficient resources. For example, you successfully deploy a virtual machine, but you cannot turn it on due a lack of storage or memory.

To avoid deploying resources that you cannot turn on, you can set the PREVENT_COMPUTE_MEMORY_OVERALLOCATION configuration property to TRUE. This change ensures that VCF Automation tracks how much memory is allocated on each host or cluster. It then uses that value to prevent provisioning to hosts or clusters that are fully utilized, preventing overallocation. The allocated memory is calculated by adding up all the managed virtual machines in the host or cluster. Virtual machines that were provisioned outside VCF Automation, discovered, but have not yet been onboarded are not counted.

This is a global property.

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `true`.
4. For the `key`, enter `PREVENT_COMPUTE_MEMORY_OVERALLOCATION`.
5. Click **Execute**.

To reset the preventing property and allow overallocation, set the value to `false`.

Set memory allocation limits

You can set a percentage value that is used as the maximum amount of memory for a host or cluster. This value is the allocation limit. You can set a conservative value of less than 100%. You can also set a value over 100% if you have a clear understanding of how your resources are allocated and want to fine tune the limit.

For example, you have a host or cluster with 100 GB of total memory and you set the value at 50%. VCF Automation considers the host to have 50 GB of total memory. Or, you can set the value to 120%. VCF Automation then considers the host to have 120 GB of memory.

You can set a global property and a host- or cluster-specific property. Any host- or cluster-specific value setting takes precedence over the global setting. This allows you to set a global default value and then set a more refined value for particular hosts or clusters.

To set the global limit:

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `50`.
4. For the `key`, enter `DEFAULT_MAX_ALLOWED_COMPUTE_MEMORY_ALLOCATION_PERCENT`.
5. Click **Execute**.

To set a host and cluster limit:

1. To retrieve the host or cluster ID, referred to in the API as fabric computes, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Fabric%20Compute/getFabricComputes` and click` **Try it out**.
2. Locate and copy the ID for the host or cluster that you want apply the limit to.
3. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Fabric%20Compute/updateFabricCompute`.
4. Click **Try it out**.
5. Enter the host or cluster ID that you retrieved using the **Get** command.
6. Enter the property and value.
   For example, `"maximumAllowedMemoryAllocationPercent": 120`
7. Click **Execute**.

Ignore powered off VMs when calculating allocated memory

If you want to ignore powered off machines when calculating memory allocated on each host or cluster, you can set a property to exclude powered off VMs. This property affects the deployment process and day 2 power on and off actions.

This property is valid only if `PREVENT_COMPUTE_MEMORY_OVERALLOCATION` it set to `TRUE`

To set the neglect memory property:

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `true`.
4. For the `key`, enter `NEGLECT_POWERED_OFF_VMS_MEMORY_OVERALLOCATION`.
5. Click **Execute**.

To reset the property, set the value to `false`. If you turn off the property after using the capabilities, the you might see some of your deployments exceeding the limits.

Prevent storage overallocation of datastores

When you turn on the property, the amount of storage allocated on each datastore is tracked and that value is used to prevent provisioning to datastores where the provisioning request exceeds the available allocated storage.

The amount is calculated by summing the memory of all the managed disks on the datastore. Machines that are provisioned outside VCF Automation, discovered, but not yet onboarded are not included in the calculation.

Set this global property.

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `true`.
4. For the key, enter `PREVENT_COMPUTE_STORAGE_OVERALLOCATION`.
5. Click **Execute**.

Set storage allocation limits

You can set a percentage value that is used as the maximum amount of storage for a datastore. This value is the allocation limit. You can set a conservative limit of less than 100%. You can also set a value over 100% if you have a clear understanding of how your resources allocation and you want to fine tune the limit.

For example, you have a datastore with 100 GB of total storage and you set the value to 50%. VCF Automation considers the datastore to have 50 GB of total storage. Or, you can set the value to 120%. VCF Automation then considers the host to have 120 GB of storage.

You can set a global property and a datastore specific property. Any datastore-specific value setting takes precedence over the global setting. This method allows you to set a global default value and then set a more refined value for particular datastores.

To set the global limit:

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `50`.
4. For the `key`, enter `DEFAULT_MAX_ALLOWED_STORAGE_ALLOCATION_PERCENT`.
5. Click **Execute**.

To set a limit on a specific datastore:

1. To retrieve the host or cluster ID, referred to in the API as fabric computes, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Fabric%20vSphere%20Datastore/getFabricVsphereDatastore` and click **Try it out**.
2. Locate and copy the ID for the datastore that you want to apply the limit to.
3. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Fabric%20vSphere%20Datastore/updateFabricVsphereDatastore`.
4. Click **Try it out**.
5. Enter the datastore ID that you retrieved using the **Get** command.
6. Enter the property and value.
   For example, `"maximumAllowedStorageAllocationPercent":120`
7. Click **Execute**.

Prevent CPU overallocation

Some systems allow you to deploy resources even if the host or cluster does not have sufficient resources. For example, you successfully deploy a virtual machine, but you cannot turn it on due to insufficient virtual CPUs.

To avoid deploying resources that you cannot turn on, you can set the PREVENT_COMPUTE_CPU_OVERALLOCATION configuration property to TRUE. This change ensures that VCF Automation tracks how many virtual CPU threads are allocated on each host or cluster. It then uses that value to prevent provisioning to hosts or clusters that are fully utilized, preventing overallocation. The allocated CPUs is calculated by adding up all the managed virtual machines in the host or cluster. Virtual machines that were provisioned outside VCF Automation, discovered, but have not yet been onboarded are not counted.

This is a global property.

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `true`.
4. For the `key`, enter `PREVENT_COMPUTE_CPU_OVERALLOCATION`.
5. Click **Execute**.

To reset the preventing property and allow overallocation, set the value to `false`.

Set CPU allocation limits

You can set a percentage value that is used as the maximum number of virtual CPU threads for a host or cluster. This value is the allocation limit. You can set a conservative value of less than 100%. You can also set a value over 100% if you have a clear understanding of how your resources are allocated and want to fine tune the limit.

For example, you have a host or cluster with 10 CPUs and you set the value at 50%. VCF Automation considers the host to have 5 CPUs. Or, you can set the value to 120%. VCF Automation then considers the host to have 12 CPUs.

You can set a global property and a host- or cluster-specific property. Any host- or cluster-specific value setting takes precedence over the global setting. This allows you to set a global default value and then set a more refined value for particular hosts or clusters.

To set the global limit:

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `50`.
4. For the `key`, enter `DEFAULT_MAX_ALLOWED_COMPUTE_CPU_ALLOCATION_PERCENT`.
5. Click **Execute**.

To set a host and cluster limit:

1. To retrieve the host or cluster ID, referred to in the API as fabric computes, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Fabric%20Compute/getFabricComputes` and click **Try it out**.
2. Locate and copy the ID for the host or cluster that you want apply the limit to.
3. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Fabric%20Compute/updateFabricCompute`.
4. Click **Try it out**.
5. Enter the host or cluster ID that you retrieved using the **Get** command.
6. Enter the property and value.
   For example, `"maximumAllowedCpuAllocationPercent": 120`
7. Click **Execute**.

Ignore powered off VMs when calculating allocated CPUs

If you want to ignore powered off machines when calculating CPUs allocated to each host or cluster, you can set a property to exclude powered off VMs. This property affects the deployment process and day 2 power on and off actions.

This property is valid only if `PREVENT_COMPUTE_CPU_OVERALLOCATION` it set to `TRUE`

To set the neglect CPU property:

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `true`.
4. For the `key`, enter `NEGLECT_POWERED_OFF_VMS_CPU_OVERALLOCATION`.

5. Click **Execute**.

To reset the property, set the value to `false`. If you turn off the property after using the capabilities, the you might see some of your deployments exceeding the limits.

How to work with audit logs in VCF Automation for VM Apps

To discover status information about reportable VCF Automation actions, use audit logs.

VCF Automation creates an audit log when you perform a reportable action, such as when you deploy a cloud template or catalog item, perform a Day 2 action on a resource or deployment, or create, delete, or update a property group. Audit logs display a status value or a message for each reportable action type, depending on the action. They can also contain information about the action, such as a timestamp, message, user name, project name, and trace ID value.

Audit logging captures information for the following reportable actions:

- Catalog deployment operations
- Cloud template operations
- Day 2 deployment operations
- Day 2 resource action operations
  Each of the above four action types creates an initial audit log with a *`Submitted`* status. Based on how the action is completed, an additional log is created that displays a *`Successful`*, *`Failed`*, or *`Canceled`* status.
- Create, delete, and update property group operations
- Create, delete, and update product feature toggle operations
- Create, delete, and update policy operations in VCF Automation- such as Day 2 actions, lease, content sharing, approval, deployment limit, and resource quota policy operations
- Create, delete, and update extensibility subscription operations
  Each of the above four action types creates a message in the audit log display screen.

Searching, displaying, and exporting an audit log

You can filter your search for, and display of, audit logs. You can also output audit logs in CSV format.

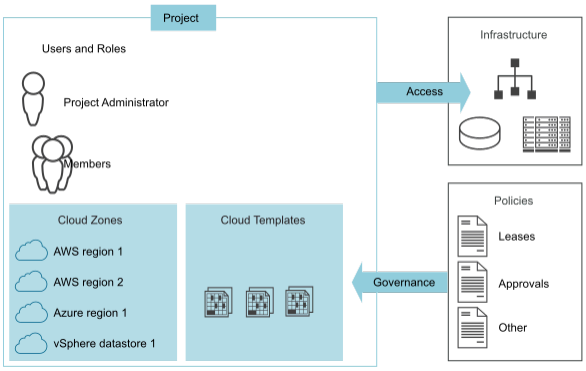
To display audit logs, use the following procedure.

1. Log in to VCF Automation as an administrator.
2. Select the **Infrastructure** tab and click **Activity** > **Audit Log**.
3. Open and review the audit log or logs of interest.

Adding and managing projects in VCF Automation for VM Apps

Projects control who has access to VCF Automation cloud templates and where the templates are deployed. You use projects to organize and govern what your users can do and to what cloud zones they can deploy cloud templates in your cloud infrastructure.

Cloud administrators set up the projects, to which they can add users and cloud zones. Anyone who creates and deploys cloud templates must be a member of at least one project.



Add a project for your VCF Automation for VM Apps development team

You create a project to which you add members and cloud zones so that the project members can deploy their cloud templates to the associated zones. As the VCF Automation for VM Apps organization administrator, you create a project for a development team. You can then assign a project administrator or you can operate as the project administrator.

- Verify that you configured the cloud zones. See Building your VCF Automation for VM Apps resource infrastructure.
- Verify that you configured the mappings and profiles for the regions that include as cloud zones for this project. See Building your VCF Automation for VM Apps resource infrastructure.
- Verify that you have the necessary permissions to perform this task. See What are the VCF Automation for VM Apps user roles.
- Determine who you are designating as the project administrator. To understand what the project administrator can do in VCF Automation, see What are the VCF Automation for VM Apps user roles.

When you create a cloud template, you first select the project to associate it with. The project must exist before you can create the cloud template.

Ensure that your projects support the business needs of the development team.

- Does the project provide the resources that support the team's goals.
- Do your project members require or expect their deployments to be shared or private. Shared deployments are available to all the project members on the Deployments page, not only the deploying member. You can change the deployment sharing state at anytime.

  When you share the deployment with project members, the members can run the same day 2 action. To manage the ability of members to run day 2 actions, you can create day 2 policies in VCF Automation. The policies apply to VCF Automation deployments.

This procedure is based on creating an initial project that includes only the basic configurations. As your development team creates and deploys their cloud templates, you might modify to the project. You can add constraints, custom properties, and other options to improve deployment efficiencies.

1. Select **Infrastructure** > **Administration** > **Projects**, and click **New**.
2. Enter the project name.
3. Click the **Users** tab.
   a) To make deployments by project members accessible only to the requesting user, turn off **Deployment sharing**. To ensure that you can assign the ownership of a deployment to another member of the project, verify that the **Deployment sharing** is turned on.
   b) Add users or user groups with assigned roles.

      User roles based on groups allows users to come and go from your Active Directory groups rather than requiring the project administrator to constantly update the individual users in the project. For example, if deployment ownership is based on groups, you can change the ownership of a deployment to the group. Any notifications are sent to all the group members, allowing anyone to respond.

4. Click the **Provisioning** tab and add one or more cloud zones.

   Add any cloud zones and virtual private zones that contain the resources that support the cloud templates deployed by the project users.

   For each zone, you can set a zone priority and you can limit the amount of resources that the project can use. For more information about limits, see Using cloud zone resource limits.

5. If the workloads requested for this project take more than two hours to deploy, enter a longer value for the **Timeout**.

   The default value is two hours.

6. Click **Create**.

7. To test your project with the project cloud zones, click **Test Configuration** on the Projects page.

   The simulation runs a standardized hypothetical deployment test against the project cloud zone resources. If it fails, you can review the details and correct your resource configuration.

Get started with cloud templates. See Designing cloud templates in VCF Automation for VM Apps.

Learn more about projects in VCF Automation for VM Apps

Projects are the connector between cloud templates and resources. The more you understand about how they work and how you can make them work for you, the more effective your VCF Automation development and deployment process will be.

Using project tags and custom properties in VCF Automation for VM Apps
As an administrator, you can add project-level governance constraints or custom properties when the requirements of the project are different from the VCF Automation cloud templates. In addition to constraint tags, you can add resource tags that are added to deployed resources during the provisioning process so that you can manage the resources.

What are project resource tags

A project resource tag operates as an standardized identifying tag that you can use to manage the deployed compute resources and ensure compliance.

The resource tags defined in a project are added to all the machine resources deployed that are part of that project along with any machine-specific tags. You can then use the standard tagging to manage the resources using other applications, for example, monitor spending cost using CloudHealth, and, importantly, to ensure compliance. The project resource tags are not added to other resources such as network or storage.

For example, as a cloud administrator, you want to use an application like CloudHealth to manage costs. You add the `costCenter:eu-cc-1234` tag to a project dedicated to developing a European Union human resources tool. When the project team deploys from this project, the tag is added to the deployed machine resources. You then configure the costing tool to identify and manage the resources that include this tag. Other projects with other cost centers would have alternative values to go with the key.

What are project constraint tags

A project constraint operates as a governance definition. It is a `key:value` tag that defines what resources the deployment request consumes or avoids in the project cloud zones.

The deployment process looks for tags for the networks and storage that match the project constraints, and deploys based on matching tags.

The extensibility constraint is used to specify which VCF Operations orchestrator integrated instance to use for extensibility workflows.

Consider the following formats when you configure project constraints.

- **key:value** and **key:value:hard**. Use this tag, in either format, when the cloud template must be provisioned on resources with the matching capability tag. The deployment process fails when no matching tag is found. For example, a cloud template deployed by the members of a project must be provisioned on a PCI-compliant network. You use `security:pci`. If no networks are found in the project cloud zones, the deployment fails, ensuring no insecure deployments.

- **key:value:soft**. Use this tag when you prefer a matching resource, but you want the deployment process to proceed without failing and can accept resources where the tag does not match. For example, you prefer that the project members deploy their cloud templates to a less expensive storage, but you do not want storage availability to interfere with their ability to deploy. You use `tier:silver:soft`. If there is no storage tagged tier:silver in the project cloud zones, the cloud template still deploys on other storage resources.
- **!key:value**. Use this tag, with hard or soft, when you want to avoid deploying to resources with a matching tag.

Importantly, the project constraint tags have a higher priority than the cloud template constraint tags and override them at deployment time. If you have a cloud template where this must never happen, you can use the `failOnConstraintMergeConflict:true` in the template. For example, if your project has a network `loc:london` constraint, but the cloud template is `loc:mumbai`, but rather than the project location taking precedence, you want the deployment to fail with a constraint conflict message, you add a property similar to the following sample.

```
constraints:
 - tag: 'loc:mumbai'
failOnConstraintMergeConflict:true
```

How might I use project custom properties

You can use a project custom property for reporting, to trigger and populate extensibility actions and workflow, and to override cloud template level properties.

Adding a custom property to a deployment allows you to use the value in the user interface or to retrieve it using the API so that you can generate reports.

Extensibility can also use a custom property for an extensibility subscription. For more information about extensibility, see Extending and automating application life cycles with extensibility.

A cloud template might have a particular property value that you want to change for a project. You can provide an alternative name and value as a custom property.

You can also encrypt the property value so that neither you nor your users can see the value that is included in the deployment. For example, you can encrypt a password that all users in the project use, but that you do not want visible. After you encrypt the value and save the project, you cannot unmask or replace the value. If you clear the **Encrypted** check box, the value is removed. You must re-enter a value.

Using cloud zone resource limits in projects in VCF Automation for VM Apps

When you configure a project, you add cloud zones. You can define the resource limits for each cloud zone, allowing you to conserve resources where needed.

The possible limits include the number of instances, memory, and CPUs.

vSphere storage limits

For vSphere cloud zones only, you can configure storage limits for deployed resources that are based on vSphere VM templates and content library items. Storage limits consider the actual capacity for thick and thin resource provisioning so that you cannot over-provision using thin provisioning. First class disks and vSphere independent disks are also considered in the storage limits. Storage limits do not apply to OVA/OVF templates that are outside of the content library.

The storage limits are evaluated when you request a deployment and when you make changes to the deployment using the resize disk, resize boot disk, remove disk, and the update count actions. Like initial provisioning, thick and thin provisioning are considered to prevent over-provisioning. These storage limits do not apply to other resource types such as AWS, Microsoft Azure, or Google Cloud Platform.

As you add each zone and apply limits, don't limit the project resources so narrowly that the members cannot deploy their cloud templates.

When your users submit a deployment request, the zones are evaluated to determine which zones have the resources to support the deployment. If more than one zone supports the deployment, then the priority is evaluated and the workload is placed on the one with the higher priority, which is the lowest integer.

Ignore powered off VMs when evaluating usage for limits

In addition to setting limits for cloud zones in the project, you can also configure the limits to ignore powered off VMs when calculating memory and CPU usage. This property affects the deployment process and day 2 power on and off actions.

For example, when you add a cloud zone to a project, you apply a limit of 5120 MB memory and 5 CPUs. You then deploy a templates that consume the amount and reach the limit. The next deployment fails. However, you know that two of the VMs are powered off and you want to ignore the usage of the powered off VMs and deploy an additional template. You can set a global property for you organization that ignores powered off machines with determining resource usage.

To ignore powered off VMs, you can set a global property.

1. To add the property, go to `https://your_automation_URL/iaas/api/swagger/ui/#/Property/patchConfigurationProperty`.
2. Click **Try it out**.
3. For the `value`, enter `true`.
4. For the `key`, enter `NEGLECT_POWERED_OFF_VMS_RESERVATION`.
5. Click **Execute**.

To reset the property, set the value to `false`. If you turn off the property after using the capabilities, the you might see some of your deployments exceeding the limits.

When you turn the property on and off, VCF Automation recalculates the CPU, memory, storage, and instance count usage.

For more information about capacity management, see How to manage infrastructure capacity for VCF Automation for VM Apps.

How do project-level placement policies affect resource allocation in VCF Automation for VM Apps

As an administrator, you can define the placement policy for projects where more than one cloud zone is eligible as the deployment target zone. For example, you might have a project where you want to deploy cloud templates based on the set priority, you might want to balance the deployed resources across multiple zones based on which one has the best VM to host ratio, or you might want to balance the VMs based on the amount of free resources in the zones. .

Allocation considerations

For a default or spread placement policy.

- If the deploying user has permission to manage cloud accounts that are in maintenance mode, the allocation process can select a cloud account that is in maintenance mode because the user might need to run a test deployment before closing the maintenance window.
- If the user does not have permission to manage cloud accounts, then the cloud accounts that are in maintenance mode are filtered out of the allocation process.
- Hosts that are in maintenance mode are counted as part of the spread ratio. To exclude a host in maintenance from the ratio calculation, you must set the power state to off.

For a spread policy.

- Ratios are calculated based on hosts. The hosts can be standalone or part of a cluster.
- If a standalone host is powered off, it is not counted as part of the ratio.
- If a host that is part of a cluster is powered off, the powered off state is not reflected in the cluster and the host is still considered when calculating the ratio.

For a spread by memory policy.

- The amount of available memory is calculated by identifying the amount of memory of all the hosts in the cloud zone and identifying the allocated memory of all the currently deployed virtual machines. However, there are variations based on whether the project is evaluating private cloud zones, public cloud zones, or a mix of private and public.
  - Projects with only private cloud zones.

    To determine available memory for private cloud zones, the total memory of the hosts in the zone is retrieved. The amount of memory allocated to all the managed virtual machines is retrieved. The allocated memory is divided by the total memory to determine the ratio, which is inversely proportional to the size of the zone or cluster. The zone

or cluster with the smallest ratio is considered to have the most available memory because it has the largest amount of free memory proportional to its size.

- Projects with only public cloud zones.

  It is not possible to determine the free memory for public cloud zones. Therefore, the amount of memory allocated to all the managed virtual machines is retrieved. The cloud zones are ordered by the amount of allocated memory so that the zone with the least amount of allocated memory is prioritized first.
- Projects with a mix of private and public cloud zones.

  To determine the amount of available memory, the private cloud zone with the largest amount of total memory, which is retrievable information, is made the baseline that is used for all the public cloud zones.

  To determine available memory for private cloud zones, the total memory of the hosts in the zone is retrieved. The amount of memory allocated to all the managed virtual machines is retrieved. The allocated memory is divided by the total memory to determine the ratio, which is inversely proportional to the size of the zone or cluster. The zone or cluster with the smallest ratio is considered to have the most available memory because it has the largest amount of free memory proportional to its size.

How to set the placement policy

If you have multiple cloud zones in a project that are equally eligible as the target for a deployment, the deployment request evaluates where to place them based on how you have the **Placement policy** configured.

1. Select **Infrastructure** > **Administration** > **Projects** and create or select a project.
2. In the project, click the **Provisioning** tab.
3. Select a policy.

Placement policy	Description
Default	Deploys the requested resources to the first cloud zone that matches the requirements. Select Default when you want the workloads deployed in the priority order and don't mind utilizing all the resources on a host. If this option is selected, the VM and Hosts values are not retrieved.
Spread	Deploys the requested resources to the cloud zone with the smallest number of virtual machines per hosts. Select Spread when you want to distribute the workloads across hosts, utilizing resources broadly across hosts. If this option is selected, the number of VMs and hosts are retrieved from the cloud zone resources and evaluated.
Spread by Memory	Deploys the requested resources to the cloud zone with the largest amount of free memory. Select Spread by Memory when you want the workloads placed on hosts with the largest amount of unallocated memory. If this option is selected, the identification of the zone with the most free memory depends on whether the project contains only private cloud zones, public cloud zones, or a mix. See the allocation considerations above.

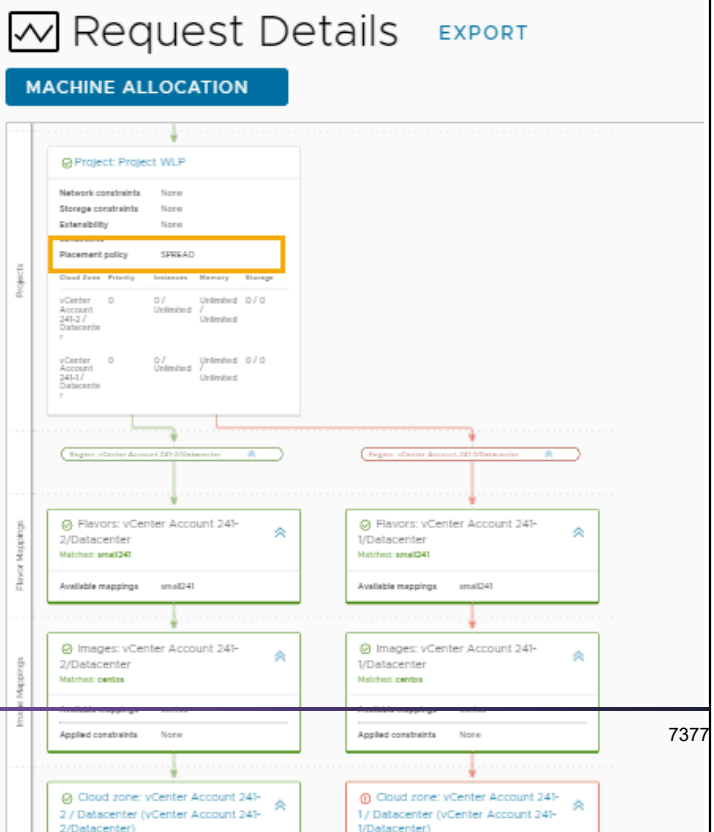
4. Click **Save**.

Review how the policy is applied

After you configure the project-level placement policy, you can view where the system plans to deploy the cloud template in a provisioning diagram.

1. Select **Design** > **Templates** and select or configure a template that uses the project to which you selected a policy.
2. Click **Test**.
3. When the test completes successfully, click **Provisioning Diagram** in the test results.

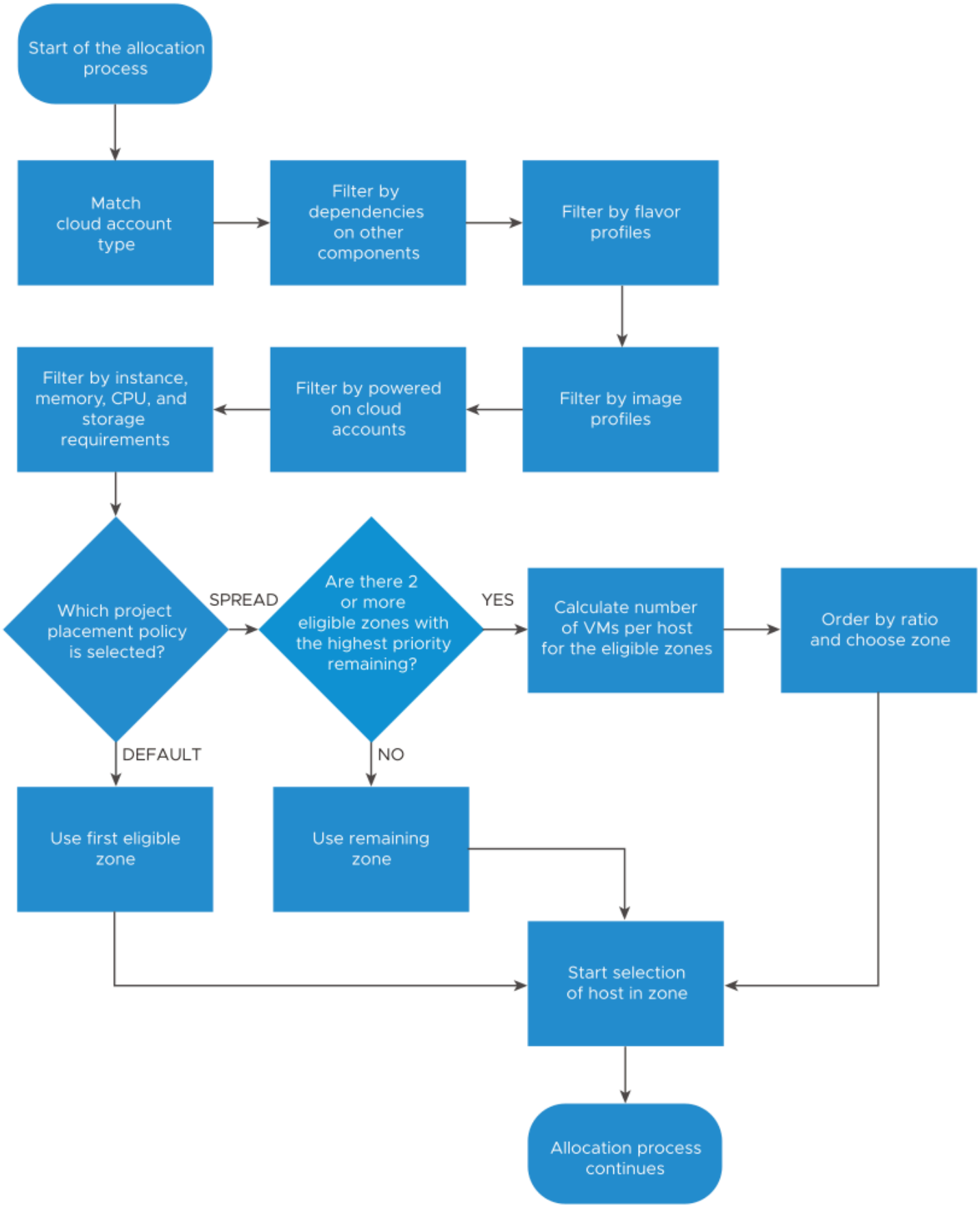
4. The diagram will resemble one of the two examples.

Policy Type	Provisioning diagram
Default	
Spread	

5. If you are ready to deploy, return to the cloud template and click **Deploy**.

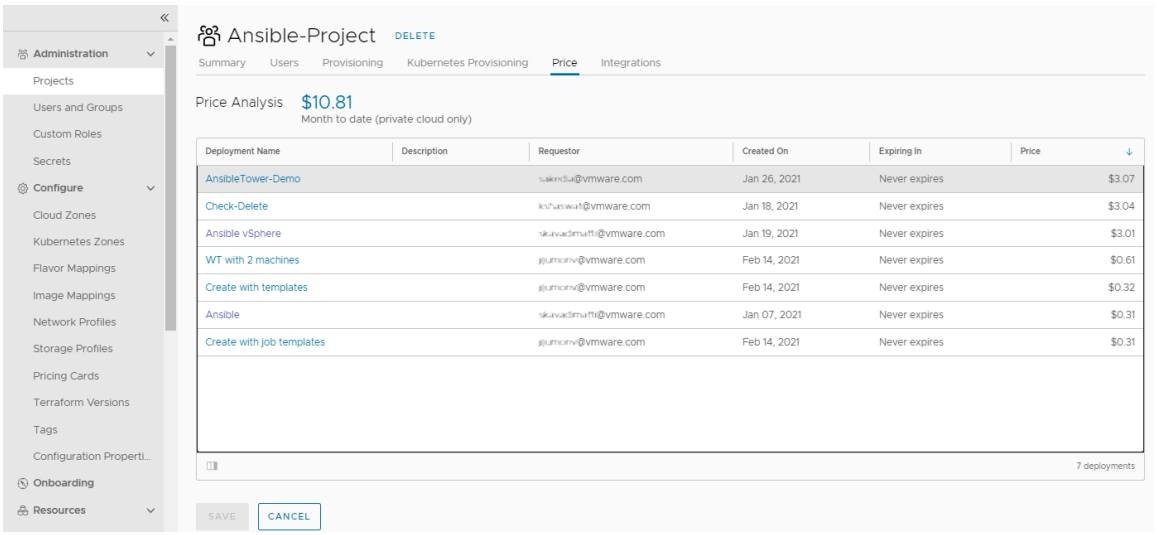
Placement policy evaluation during the allocation process

The following diagram helps you understand when the policy is evaluated during the allocation process and when the target zone and host are identified.


Start of the allocation process
Match cloud account type
Filter by dependencies on other components
Filter by flavor profiles
Filter by image profiles
Filter by powered on cloud accounts
Filter by instance, memory, CPU, and storage requirements
Which project placement policy is selected?
SPREAD
Are there 2 or more eligible zones with the highest priority remaining?
YES
Calculate number of VMs per host for the eligible zones
Order by ratio and choose zone
DEFAULT
NO
Use first eligible zone
Use remaining zone
Start selection of host in zone
Allocation process continues

What are the project prices in VCF Automation for VM Apps

The costs available in your VCF Automation projects help you manage the resource expenses associated with entire projects. The project also includes the individual deployment costs.



The cost information that you see for a project and for the individual deployments appears after at least one deployment associated with the project is provisioned. The costs are calculated and updated daily so that you can track the cost of a deployment over time. The initial values are based on industry benchmarks.

Cloud administrators can adjust the values to reflect your actual costs.

For more information, see Using Pricing Cards in VCF Automation.

How do VCF Automation projects work at deployment time

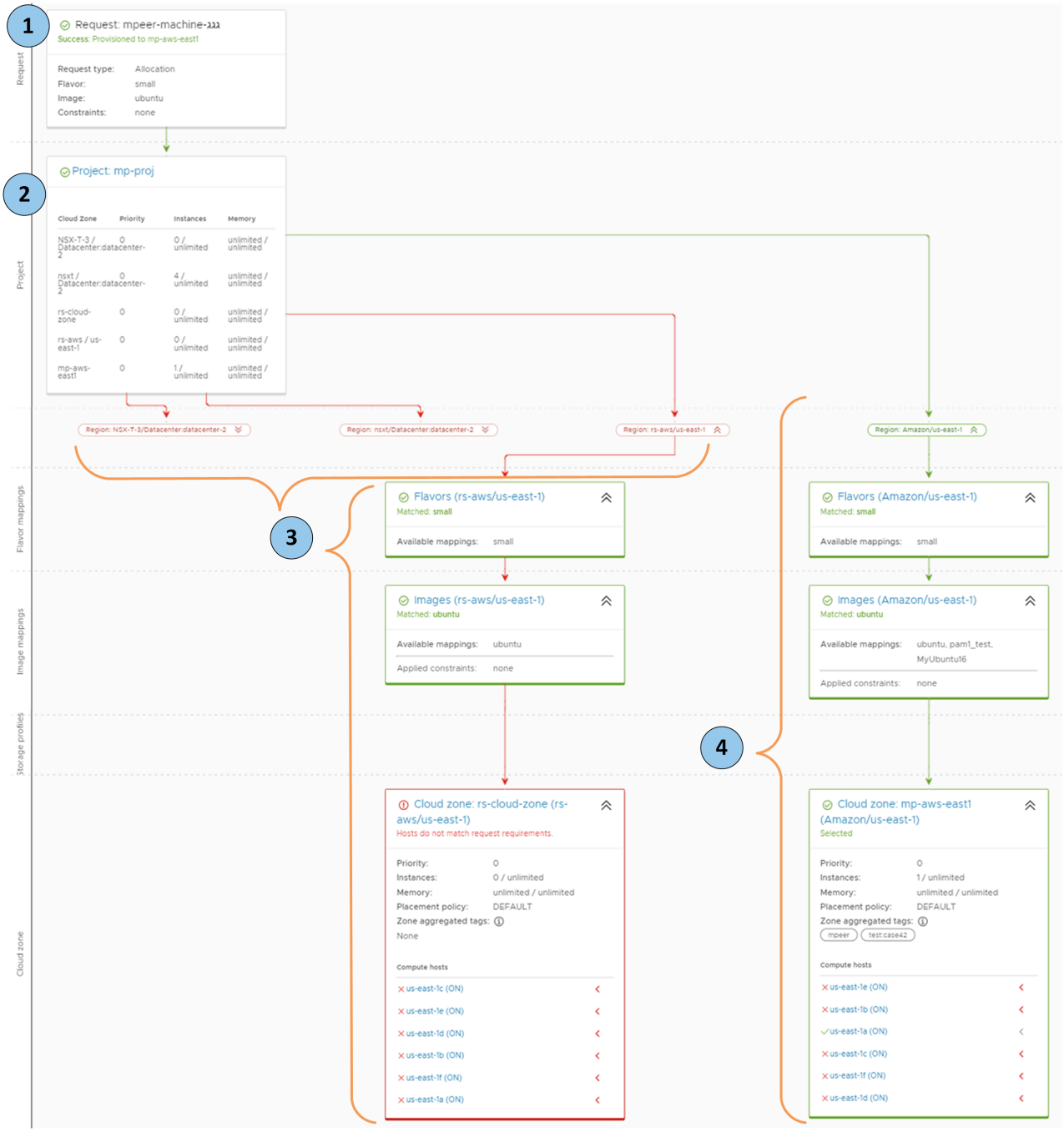
Projects control user access to the cloud zones and user ownership of the provisioned resources. Whether your are a cloud administrator or a cloud template developer, you must understand how the projects work at deployment time so that you can manage your deployments and troubleshoot any problems.

As a cloud administrator who is setting up projects for various teams, you must understand how projects determine where cloud template components are deployed. This understanding helps you create projects that support cloud template developers and to troubleshoot failed deployments.

When you create a cloud template, you first associate it with a project. At deployment time, the cloud template requirements are evaluated against the project cloud zones to find the best deployment location.

The following workflow illustrates the process.

1. You submit a cloud template deployment request.
2. The project evaluates the template and project requirements, for example, flavor, image, and constraint tags. The requirements are compared to the project cloud zones to locate a zone that supports the requirements.
3. These zones did not have the resources to support the request.
4. This cloud zone supports the request requirements and the template is deployed to this cloud zone account region.



Setting up policies in VCF Automation for VM Apps

To provide the background management of your deployments, you set up policies. Each VCF Automation policy is a set of rules or parameters that are applied to deployments, freeing the cloud administrator for other tasks.

Getting started with policies

To begin creating policies, select **Content and Policies** > **Policies** > **Definitions**. Any policy that you add is applied to current deployments and any new deployments.

To get you started, use the full use cases that are provided for each policy type. The use cases guide you through the process of creating more than one policy. The use case provides contextual explanations of the choices and the desired behavior.

The use cases are followed by more in-depth information about how multiple policies are processed.

How do I configure approval policies in VCF Automation for VM Apps

Approval policies are a level of governance that you add to exercise control over deployment and day 2 action requests before they are run. You define approval policies in VCF Automation so that you, or others that you designate, review requests before resources are consumed or destroyed. The approval policy use cases in this procedure are an introduction that you can use as you explore your governance options.

If you have only a small team adding and deploying catalog items, then approval policies might be less useful. But as you make the catalog available to a larger group of developers and general consumers, you can use the approval policies to ensure that someone reviews a request before the resources are consumed or changes are made to the provisioned items.

For example, you have a catalog item that is important, but it consumes a significant amount of resources. You want one of your IT administrators to review any deployment requests to ensure that the request is needed. Another example applies to day 2 actions. Making changes to a deployment that is used by many might be devastating. You want the project administrator who manages the deployment for that team to review all changes to the deployed catalog item.

Who works with or is affected by approval policies

- VCF Automation administrator. Configures the policies.
- Catalog consumers. Users who request catalog items or day 2 actions to which one or more policies apply.
- Users deploying cloud templates in VCF Automation. Users who request templates or day 2 actions in VCF Automation to which one or more policies apply.
- Designated approvers. Users who must review and then approve or reject a request. You can grant approver rights to selected users and user groups, or you can choose from the following approver roles.
  - Project Administrators. Administrators of projects within the policy scope are automatically assigned as approvers. If a project does not have a dedicated administrator, the approval policy is not applied to that project.
  - Project Supervisors. Members of projects within the policy scope who are assigned the Supervisor role. Supervisor access rights are limited to approving and rejecting deployment requests for a project. If a project does not have a dedicated supervisor, the approval policy is not applied to that project.

What happens when approval policies are enforced

Multiple approval policies might be enforceable. The approval policies are evaluated, and an enforced policy is applied to the request. When there are multiple valid policies, where the approvers are different people, all the approvers are added. When you have multiple policies, it is important to understand this process. For more information, see Approval policy goals and enforcement examples.

1. Approval policies are defined.
2. A user requests a catalog item or day 2 action. At request time, VCF Automation evaluates the catalog item to see if any policies apply.
3. An approval policy is enforced.
   a. The deploy card displays the status. For example, Create - Approval Pending.
   b. An email notification is sent to the requester. See How do I track my requests that require approval in VCF Automation for VM Apps.
   c. An email notification is sent to the approvers. See How do I respond to an approval request in VCF Automation for VM Apps.

      The deployment does not begin deploying and consuming infrastructure resources, or make changes to a deployed system, until the request is approved. The requesting user is notified by email that the request is waiting for approval.
   d. The approvers respond to the request using the Approvals page in VCF Automation.
4. The approval process is completed.
   a. If the request is rejected, the requesting user is notified and the deployment request is canceled.
   b. If the request is approved, the deployment proceeds.
   c. It is possible that the enforced policy is configured to automatically approve or reject a request if the approver does not take any action.

How can I use the deployment criteria

To limit what items or activities the policy applies to, you can define the deployment criteria. For more about the criteria, see How do I configure deployment criteria in policies.

Approval policy constraints

- The change lease action is not available to include in an approval policy.
- Using custom resources as resource type in the policy criteria is not supported.

Prerequisites

- An approver, who might not be a regular VCF Automation or VCF Automation user, must have one of the following combination of roles:
  - Organization member and VCF Automation service user
  - Organization member and the Manage Approvals custom role

  These roles provide the minimum level of permissions and still allow them to approve or reject a request.
- Verify that the email notification server is defined. See Add an email server in VCF Automation to send notifications.

Procedure

As you review the approval policies use case and create your own policy, consult the signpost help on the key text boxes for more information.

1. Select **Content and Policies** > **Policies** > **Definitions** > **New Policy** > **Approval Policy**.
2. Configure Approval Policy 1.
   As an administrator, you have an important catalog item that also consumes a significant amount of your cloud resources. You want several managers to review any deployment requests to ensure that the request is really needed and that the resources exist to support it.
   1. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to all projects in your organization.
Criteria	`Catalog Item equals CompanyApplication`

   2. Define the approval behavior.

Setting	Sample Value
Approval type	Select **User based**. You select the users and user groups who are the request approvers.
Approver mode	All You want all your IT managers to agree that the deployment request does not waste resources.
Approvers	{GroupName1}@YourCompany, {ApproverName1}@YourCompany, {ApproverName2}@YourCompany The approval request is sent to all members of the user group. Only one member of the group must approve the request.
Auto expiry decision	Reject The possible load on your cloud resources means that you do not want to inadvertently deploy the item without approval.

Setting	Sample Value
Auto expiry trigger	3 This value should carry you over a long weekend when the managers might not be available.
Actions	Deployment.Create

In this scenario, if any catalog consumer requests this catalog item, Approver 1, Approver 2, and any one member of User Group 1 must approve the request within 3 days or the request is rejected.

3. Configure Approval Policy 2.

   As an administrator, you have several projects where you want the project administrators to approve any changes to deployments that might have catastrophic consequences. For example, deleting the deployment.

   a. Define when the approval policy is valid.

Setting	Sample Value
Scope	Multiple Projects `Project name contains Prod` The policy is applied to deployments associated with all projects that match the scope criteria.
Criteria	None

   b. Define the approval behavior.

Setting	Sample Value
Approval type	Select **Role based**.
Approver role	Project Administrators If a project does not have a dedicated administrator, the approval policy is not applied to requests associated with that project.
Approver mode	Any
Auto expiry decision	Reject
Auto expiry trigger	7
Actions	Deployment.Delete, Deployment.PowerOff, Deployment.Update, and any of the component-specific power, reboot, and delete actions.

In this scenario, when a member of one of the scoped projects submits a request to run the listed actions on a deployment, the request is rejected after seven days if the project administrator does not respond.

4. Configure Approval Policy 3.

   As an administrator, you want to maintain a little control over resource consumption. For example, when a user requests a catalog item where the size is large, you want to evaluate and approve the request. Size is defined by the flavor mappings.

   a. Define when the approval policy is valid.

Setting	Sample Value
Scope	Organization
Criteria	`Resources has any` `Flavor equals large`

b. Define the approval behavior.

Setting	Sample Value
Approval type	Select **User based**.
Approver mode	Any
Approvers	{AdminName}@YourCompany
Auto expiry decision	Reject The possible consumption of your cloud resources means that you do not want to inadvertently deploy the item without approval.
Auto expiry trigger	5
Actions	Deployment.Create and any applicable *.Machine.Resize actions. For example, Cloud.vSphere.Machine.Resize.

In this scenario, when any user submits a request for a large deployment or to resize a deployment to large, the request is rejected after 5 days if the cloud administrator does not respond.

What to do next

- For more information about how approval policies are processed, see Approval policy goals and enforcement examples.
- For more about the consumer and approver experience, see How do I track my requests that require approval in VCF Automation for VM Apps and How do I respond to an approval request in VCF Automation for VM Apps.

Configure multi-level approval policies in VCF Automation for VM Apps

You use multi-level approvals to maintain technical constraints and capabilities for resources within the organization. In this use case, there are two approval policy definitions that illustrate how you can construct multi-level approvals and the results when they are enforced.

To manage virtual infrastructure resources and to control prices, you add two policies at different approval levels, where one approval is for machine resources and the other is for price of machine per day.

1. Select **Content and Policies** > **Policies** > **Definitions** > **New Policy** > **Approval Policy**.
2. Configure Approval Policy 1.

   You want an administrator to approve the deployment request first.

   a) Define when the policy is valid.

Setting	Sample Value
Scope	Organization

Criteria	`Resources has any`
	`CPU Count greater than 6`
	`OR`
	`Total Memory (MB) greater than 8`

b) Define the approval behavior.

Setting	Sample Value
Approval level	1 Enter a value between 1 and 99. You prioritize levels based on the order that you want them processed. When the approval policy is triggered, if the first level of approval is rejected, the request is rejected.
Approval type	Select **Role based**.
Approver role	Select **Project Administrators**.
Approver mode	Any You want a project administrator to ensure that the deployment request does not waste resources.
Auto expiry decision	Approve
Auto expiry trigger	3
Actions	Deployment.Create

If a deployment request exceeds at least one of the level 1 deployment criteria, then an administrator must approve the request. Only level 1 approvers are notified of the request. When an approver approves the request at this level, the request is routed to the next level. If an approver rejects the request, it is not routed to any higher levels and the request is denied.

3. Configure Approval Policy 2.

   To ensure that a provisioning request is within budget, create a second policy.

   a) Define when the approval policy is valid.

Setting	Sample Value
Scope	Organization
Criteria	`Deployment Creation Cost greater than 15`

   b) Define the approval behavior.

Setting	Sample Value
Approval level	2 Because you want this policy to be executed after Policy 1, enter a number that is larger than the number you specified in Policy 1.
Approval type	Select **User based**.
Approver mode	Any
Approvers	Add user groups Select the finance user group.
Auto expiry decision	Reject
Auto expiry trigger	7
Actions	Deployment.Create

In this scenario, when a user requests a virtual machine, the request is evaluated to determine whether the requested CPU and memory amounts are over the amounts specified in level 1. If they are not and the request is approved at level 1, then the level 2 condition is evaluated.

- For more information about how approval policies are processed, see Approval policy goals and enforcement examples.

- For more about the consumer and approver experience, see How do I track my requests that require approval in VCF Automation for VM Apps and How do I respond to an approval request in VCF Automation for VM Apps.

How do I entitle deployment users to day 2 actions using policies in VCF Automation for VM Apps

You define day 2 action policies so that you can control what changes your users can make to deployments and their component resources. By creating a list of permitted actions that all or some users can run on deployments, you ensure that the users cannot initiate any destructive or costly changes. The use cases related to day 2 actions policies are an introduction to the procedure.

When you entitle users to run day 2 actions, you select the individual actions that they can run. You are creating an inclusion list, not an exclusion list.

When does a day 2 actions policy go into effect

- If you do not have any Day 2 Action policies defined, then no governance is applied and all users have access to all the actions. This initial lack of governance as you are starting out ensures that you and your users can exercise the day two actions in VCF Automation and VCF Automation without the need to understand day 2 policies.
- After you determine that you are ready to control who has access to what actions, you add governance in the form of a single Day 2 Action policy. When the first policy goes into effect, the Day 2 Action policies are enforced for all users in VCF Automation and VCF Automation. As a result, only the users for whom the first policy is true can run the selected actions. All others are excluded. They are excluded because the actions policies include the trusted users. By excluding all others, you are able to craft the policies to match your governance goals.
- To entitle other users, you must create policies that entitle them to run the actions you select.

How does deployment sharing affect day 2 actions policies

Deployment sharing in projects affects how you configure the day 2 actions entitlements. If the project is not set to share, then only the requesting user can see a deployment. If the project shares deployments, then all the members of the project can see the deployment, and run any actions that they are entitled to run by a Day 2 Action policy. Deployment sharing is configured in a project. Select **Infrastructure** > **Administration** > **Projects**, then select the project and click the **Users** tab.

As you create your policies, the way that you define Day 2 Actions policies must take sharing status into consideration.

When are day 2 actions policies applied

To focus when the Day 2 Actions policies are applied, you can configure scope, role, and criteria. These configurations control what deployments the policy is applied to and who can run the actions when the policy is enforced.

- What deployments the policy is applied to.
  - Scope determines whether the policy is applied to deployments at the organization or project level.
  - Criteria narrow the scope of the policy to particular aspects of deployments.
- Who can run what actions on those deployments.
  - Role entitles the members of the selected role, within the selected scope and criteria, to run the selected actions. The role can be project administrator, project member, or a named custom role.

Day 2 policies are enforced when a user tries to manage a deployment using the Actions menu on the deployment or on the component resources.

In this use case, which is used to illustrate a collection of day 2 action policies, the assumption is that you enabled deployment sharing in the project.

Selecting day 2 actions

As you review the day 2 actions policies use case, you must also select the actions. You must select the actions that support your cloud accounts.

- Actions are cloud specific. When you are entitling the users to make changes, consider what cloud accounts the entitled users are deploying to and ensure that you select all the cloud-specific versions of the actions. For example,

add Cloud.AWS.EC2.Instance.Resize, Cloud.GCP.Machine.Resize, and Cloud.Azure.Machine.Resize to entitle users to resize those machines.

- Cloud agnostic actions, for example, Cloud.Machine.Resize, exist to accommodate resources where the on-boarding or migration process cannot identify the machine type. If you entitle users to the cloud agnostic actions, you have not entitled them to run the cloud-specific action that will make the changes to the deployed resources. The agnostic actions might appear in the action menu, but running the actions has no effect. Avoid entitling the agnostic actions and only entitle cloud-specific actions to ensure that actions are available to the users for your various cloud platforms.

Prerequisites

- For a list of possible actions, see What actions can I run on deployments or supported resources in VCF Automation for VM Apps.
- For more information about constructing deployment criteria, see How do I configure deployment criteria in policies.
- Custom roles are used in Day 2 Policy 4. Create a Deployment Troubleshooter role, but with the Manage Deployment role in the custom Deployment Troubleshooting role does not limit the members by project. The Manage Deployment role allows the assignees to see all deployments and run all actions. If the Troubleshooting Deployments role does not include Manage Deployments, then the assignees see deployments based on their project membership.

Procedure

1. Select **Content and Policies** > **Policies** > **Definitions** > **New Policy** > **Day 2 Actions Policy**.
2. Configure Day 2 Policy 1.
   As an administrator, you want to control storage costs by restricting the ability of users to request snapshots.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy applied to all deployments in your organization.
Criteria	None
Enforcement type	Soft This enforcement type allows you to create other policies related to the snapshot actions that override this policy.
Entitlement type	Role based
Role	Member This role applies the policy to all project members.

   b. Select the actions that the users can run, but do not select any snapshot actions.
      You explicitly entitle users to run actions. To exclude users from running snapshot actions, ensure that the actions are not selected.

   In this scenario, none of the project members in your organization are entitled to create snapshots. Nor can your project administrators. Your next step is to create a policy that entitles the project administrators to create and manage snapshots.
3. Configure Day 2 Policy 2.
   As an administrator, you want to give the project administrators the ability to create and manage snapshots.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to all deployments in your organization.
Criteria	None
Enforcement type	Soft This enforcement type allows you to create other policies related to the snapshot actions that override this policy.

Setting	Sample Value
Entitlement type	Role based
Role	Administrator This role applies the policy to the project administrators.

b. Select the snapshot actions that you want the administrators to run.

Project administrators are also entitled to run any actions that the members of their projects are entitled to run. You do not need to give them permission to member actions.

In this scenario, the project administrators are entitled to run the snapshot-related actions and all the actions that their project members are entitled to run.

4. Configure Day 2 Policy 3.

As a project administrator, you have two developers who are doing work that potentially makes a deployment unusable. You want to entitle them to snapshot and revert without your intervention. You entitle the two project members to use the snapshot actions.

a. Define when the policy is valid.

Setting	Sample Value
Scope	Project MT5 This policy is applied to deployments associated with this project.
Criteria	`Catalog Item equals Multi-tier five machine with LB` Based on this criteria expression, only the deployments for a catalog item named Multi-tier five machine with LB are considered for policy enforcement.
Enforcement type	Hard This enforcement type ensures that the policy is enforced based on the definition.
Entitlement type	User based
Users	Add users. `jan@mycompany.com, kris@mycompany.com` Only the deployments where Jan or Kris deployed a catalog item are considered for policy enforcement.

b. Select the snapshot actions that you want the specified users to run.

Project administrators are also entitled to run any actions that the members of their projects are entitled to run.

In this scenario, Jan and Kris can use the snapshot actions on the Multi-tier 5 Machines with LB catalog item that either of them deploy. Although other members of the project can see the deployment, only Jan, Kris, and the project administrator can use the snapshot actions.

5. Configure Day 2 Policy 4.

As an administrator, you want to assign the permissions to run most of the day 2 actions to the users who are assigned to a Deployment Troubleshooter custom role. While most custom role permissions go across projects, what users can see in the Deployments page is based on their project membership. To see the deployments, the users who are assigned the custom roles must be members of the projects that deployed them.

a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization
Criteria	None
Enforcement type	Soft This enforcement type allows you to create other policies related to the extended day 2 that override this policy.
Entitlement type	Role based

Setting	Sample Value
Role	Select the **Deployment Troubleshooter** role.

b. Select all the actions that you want the members of this custom role to be able to run.

In this scenario, all the users with the Deployment Troubleshooting role can manage all deployments and run all selected day 2 actions across projects. The Manage Deployments role grants service administrator privileges on deployments so that they can run any action that a service administrator can run. If the Deployment Troubleshooting custom role does not include the Manage Deployments role, the users can run all the selected day 2 actions for the deployments belonging to their projects.

What to do next

- For more examples of how the policies are processed and enforced, see How are policies processed in VCF Automation for VM Apps.
- Configure policies that are relevant to your organizations and projects.

How do I configure deployment leases using policies in VCF Automation for VM Apps

By using policy-based leases, you reduce the need to intervene manually to reclaim resources. You define lease policies so that you can control the amount of time that a deployment is available to your users. The lease policy use cases in this procedure provide a beginning point for learning about and implementing policies for your organization.

If you do not have any lease policies defined, then the deployments never expire. To reclaim the resources, you must manually destroy the deployments.

When does a lease policy go into effect

- If the policy scope is Organization, then all the deployments in your organization are managed based on the defined policies.
- If the policy scope is a project, then the deployments that are associated with that project are managed based on the defined lease. Other projects are not affected.

When are lease policies applied

Lease policies are applied when you:

- Create or update a lease policy. After lease policies are applied, they continuously evaluate the deployments in the background to ensure that they are in compliance with the defined leases.
- Request a catalog item in VCF Automation or a cloud template in VCF Automation. The maximum lease and maximum total lease values go into effect when the deployment is created.
- Onboard workloads or resources in VCF Automation so that you can manage them using VCF Automation, VCF Automation, or VCF Automation.

Lease-specific options

As you review the lease policies use case, you must also configure lease-specific options. The following descriptions provide a brief summary. Consult the signpost help for more information.

- Maximum Lease (days). The number of days that the deployment resources are active without being renewed. If they are not renewed, the lease expires and the deployment is destroyed. If a grace period is specified, the user can renew the lease for up to the same number of days that the lease has been active.
- Maximum Total Lease (days). The combined total number of days that the deployment can be active, including lease renewals. Each renewal cannot exceed the maximum lease, and the cumulative renewal value cannot exceed the maximum total lease. After the total lease is reached, the deployment is destroyed and the resources within that deployment are reclaimed.
- Grace period (days). The number of days the user has to renew an expired lease before the deployment is destroyed. The grace period is not included in the total lease days. If you don't define a grace period, it defaults to 1 day.

What happens when I update an existing lease policy

You can increase the maximum lease, the maximum total lease, and the grace period of your lease policies. The updated policy is applied to new deployments only.

Note that increasing the policy parameters does not extend the expiry date of existing deployments. Existing deployments expire on the same date as originally scheduled.

If you decrease the lease values, however, the deployment expiry date is affected and existing deployments expire earlier than originally scheduled.

What happens when a deployment expires

When a deployment is about to expire, the deploying user receives an email notification. If the user doesn't extend the lease, the deployment expires and is scheduled for deletion. The deployment owner can see the scheduled deletion date in the deployment details. If the deletion date is set to Soon, then the deployment is already queued for deletion.

Expired virtual machines are powered off within several minutes after expiry. Unless the lease is extended, a powered-off machine can't be powered back on in VCF Automation. The machine can be powered back on manually in the original environment, in which case VCF Automation registers the machine as powered on. Even though the deployment is already expired, the machine is not automatically powered off for a second time until the grace period of the lease policy is elapsed and the deployment is deleted.

Procedure

In this use case, there are three policy definitions that illustrate how you can construct policies and the results when they are enforced. The last policy is not enforced, but the reasons are provided in the scenario results.

1. Select **Content and Policies** > **Policies** > **Definitions** > **New Policy** > **Lease Policy**.
2. Configure Lease Policy 1.
   As an administrator, you want to control costs by limiting the starting lease time for all deployments to 30 days, with the option to renew the lease for a total of 90 days.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to everyone in your organization.
Criteria	None
Enforcement type	Soft This enforcement type allows you to create other policies related to this lease that override this policy.

   b. Define the lease.

Setting	Sample Value
Maximum lease (days)	30
Maximum total lease (days)	90
Grace period (days)	10

   In this scenario, the deployment is shut down after 30 days and an email is sent to the user. During the grace period, the user extends the lease by 30 days. After the lease expires again, the user renews it for another 30 days. At the end of the third extension, the lease reaches the maximum total lease period of 90 active days and the user cannot extend it anymore. The deployment is shut down and destroyed 10 days later.
3. Configure Lease Policy 2.
   As an administrator, you want to control costs by limiting the lease time on an expensive template to two weeks. For this example, the template name is `Multi-tier 5 machine with LB`.

a. Define when the policy is valid.

Setting	Sample Value
Scope	Project MT5 This policy applied to deployments associated with this project.
Criteria	`Cloud Template equals Multi-tier 5 machine with LB` Based on this criteria expression, only deployments for the referenced template are considered for policy enforcement.
Enforcement type	Soft This soft enforcement still overrides the organization policy of 90 days in Policy 1 because the values are more meaningful at the project level.

b. Define the lease policy.

Setting	Sample Value
Maximum lease (days)	14
Maximum total lease (days)	28
Grace period (days)	3

In this scenario, both policies are applied, but Policy 2 takes precedence over Policy 1 because it is more specific. When applied, the deployment is shut down after 14 days. If the user does not extend the lease, it is destroyed three days later. If the user extends the lease for up to another 14 days, the deployment is shut down at the end of the second extension and it is destroyed three days later.

4. Review the configuration of Lease Policy 3.

   As a project manager, you realize that one of your developers is working on a complex application. The developer requires the `Multi-tier 5 Machines with LB` template and another template, `Distributed Database Across Clouds`, but for a longer lease than defined in Policy 2.

   Unless you understand how the policies are processed based on how they are defined, you might encounter unexpected results. Policy 3 is an example of how processing and precedence affect the result.

   This policy, as provided, will not be enforced. This example provides an opportunity for you to see how leases are applied and enforced when there is more than one that applies.

   a. Define when the policy is valid.

Setting	Sample Value
Scope	Project MT5 This policy is applied to deployments in this project.

Criteria	`(Cloud Template equals Multi-tier five machine with LB`
	`OR`
	`Catalog Item equals Distributed Database Across Clouds)`
	`AND`
	`Created By equals jan@mycompany.com`
	You use Catalog Item because it is a non- VCF Automation template.

Enforcement type	Soft This soft enforcement still overrides the organization policy of 90 days in policy 1 because the values are more meaningful at the project level.

b. Define the lease policy.

Setting	Sample Value
Maximum lease (days)	21
Maximum total lease (days)	50
Grace period (days)	3

In this scenario, Lease Policy 2 is applied, not Lease Policy 3.

- Lease 3 has a lease time that is less than or equal to 21 days, and the policy is applied. Lease 2 has a lease time that is less than or equal to 14 days, and the policy is applied.
- Lease 2 is applicable and it does not violate the lease 3 policy. But, lease 2 is more restrictive, so it takes precedence. Lease policy 2 is more restrictive because it is for a shorter period of time.
- When both lease definitions are true and applicable, the more restrictive policy is the one that is enforced.

5. To resolve the unexpected behavior in Lease Policy 3, you can implement one of the following solutions.
   – To ensure that you can provide Jan with the needed policy, change the enforcement type to hard.
   – Alternatively, you could create a new project with access to the same resources, and then create Lease Policy 3 for that project. While this solution isolates the working policy, you must maintain a parallel project. The effort needed to set up and maintain the content sources, content sharing, and so on, are time consuming and subject to error.

What to do next

- For more examples of how the lease policies are processed and enforced, see How are policies processed in VCF Automation for VM Apps.
- Configure policies that are relevant to your organizations and projects. If you are just getting started with lease policies, begin with one lease policy at the organization level.
- To send an email to the deploying user, configure the email server for notifications. See Add an email server in VCF Automation to send notifications.
- If you use VCF Operations orchestrator, you can manage expired deployments and their resources by using extensibility subscriptions. See *Using extensibility subscriptions to manage deployment expiry*.

How do I configure resource quotas using policies in VCF Automation for VM Apps

Resource quota policies control the amount of resources that are available to your users. You define resource quota policies so that you limit the resources that can be consumed by each user, project, or the organization. The use cases in this procedure are an introduction to resource quota policies.

If you do not have any resource quota policies defined, then no governance is applied and users can consume resources until all available resources are used up.

As a cloud administrator, you can create one or more resource quota policies and apply them, for example, at the organization level. As users across the organization request deploy resources, resource quota policies track the consumption of resources to ensure that new deployment requests do not exceed the resource limits defined in the policies.

Defining the policy scope

As you create your policies, you must configure policy scope. The scope determines whether the policy is applied to resources at the organization or project level. For more information about policy scope, see How do I configure scope in Automation Service Broker policies.

- If the policy scope is organization, then all resources in your organization are managed based on the defined policies.
- If the policy scope is multiple projects, then the resources that are associated with the specified projects are managed based on the defined policy.
- If the policy scope is a single project, then the resources that are associated with that project are managed based on the defined policy. Other projects are not affected.

Defining scope level limits

When defining resource quotas, you must specify scope level limits for each resource. Level limits provide additional resource governance. For example, if you want to apply a resource quota policy to the whole organization, you can set the scope level to organization limits, or you can define limits for a smaller segment, such as projects or users within that organization.

You can set only one limit for a resource type per scope level in the same policy. For example, you can set a resource quota for storage consumption at the organization level and per user in the same policy. You cannot define two storage quotas at the organization level in the same policy.

Resource quota limits are dependent on the broad policy scope. If you change the scope after you define the resource quota limits, the resource quota settings are deleted and you must start over.

The scope level drop-down menu includes the following options.

Option	Description	Available at these policy scope levels
Organization Limits	Limits the amount of resources that are available for consumption at the organization level. Resource quotas with organization limits are distributed among all users or all projects in the organization.	• Organization
Organization User Limits	Limits the total amount of resources that each user can consume within the organization.	• Organization
Projects Limits	Limits the amount of resources that are available for consumption at the project level. Resource quotas with project limits are distributed among all users in the specified projects. Project limits are not cumulative. If the policy scope is set to multiple projects, the resource limits are applied per project.	• Organization • Multiple projects • Project
Projects User Limits	Limits the total amount of resources that each user who belongs to the specified projects can consume at the project level.	• Organization • Multiple projects • Project

How are resource quota policies enforced

- Multiple resource quota policies might be enforceable. The resource quota policies are evaluated, and an enforced policy is applied to the deployment request. When there are multiple policies defined for a resource at the same scope level, the resource quota with the lowest limit value is enforced. The use case in this procedure provides more information about how resource quotas are processed.
- When a resource quota policy is enforced, all existing deployment resources are evaluated against the resource quota, except for deployment requests that are in-progress. Resource use is updated after the deployment request is completed, so in-progress requests are not included in the evaluation.
- When deploying cloud templates, resource quota policies allow over-provisioning of storage because the system does not know the actual storage size of the deployment before the machine is provisioned in the endpoint. After the resource use is updated and the system recognizes that the provisioning resources exceed the resource quota limit, the policy does not allow any subsequent requests.
- Resource quota policies are enforced on the following day 2 actions: Add Disk, Change Owner, Change Project, Resize Machine, Resize Boot Disk, Resize Disk, Update Deployment.
- Resource quota policies support only VMware vSphere, Amazon Web Services, Microsoft Azure and Google Cloud Platform resources created from cloud templates.

When are resource quota policies applied

Resource quota policies are applied when:

- A user requests a catalog item in VCF Automation or a cloud template in VCF Automation.
- A user changes a deployment or its component resources.
- When you create a new policy or update an existing policy, the system can take up to two minutes to apply the changes. For example, if you create a new deployment within two minutes of updating a policy, the policy updates might not apply to the deployment request.

Procedure

In this use case, there are three policy definitions that illustrate how you can construct resource quota policies and the results when they are enforced.

1. Select **Content and Policies** > **Policies** > **Definitions** > **New Policy** > **Resource Quota Policy**.
2. Configure Resource Quota Policy 1.
   As a cloud administrator, you want to control how resources are distributed among users and projects in the organization that you administer.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to the whole organization.

   b. Define the resource quotas.

Scope Level	Resource and Limit
Organization Limits	CPU = 2000
Organization User Limits	CPU = 10
Project Limits	CPU = 200
Project User Limits	CPU = 5

   In this scenario, the total amount that is available for consumption among all users in the organization is 2000 CPU and the total amount that is available per project is 200 CPU. Each user can use up to 5 CPU in each project that they belong to, but no more than 10 CPU, combined across all their deployments. Once a scope level limits is reached, any new deployment request that exceeds this limit fails.
3. Configure Resource Quota Policy 2.
   As a project administrator, you want to control how resources are distributed among developers in several projects that you administer.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Multiple Projects Define the project criteria. For example, `Project name contains dev` This policy is applied only to projects whose name contains the phrase *dev*.

   b. Define the resource quotas.

Scope Level	Resource and Limit
Project Limits	CPU = 100
Project User Limits	CPU = 10

In this scenario, the resources that are available at each scope level are evaluated and both Policy 1 and Policy 2 are enforced. Between the two policies, the lowest limits are applied.

- Projects user limits in Policy 1 are applied because the defined value is lower than in Policy 2.
- Project limits in Policy 2 are applied because the defined value is lower than in Policy 1.
- Organization level limits defined in Policy 1 also apply to the projects specified in the scope of Policy 2.

4. Configure Resource Quota Policy 3.
   As a cloud administrator, you want to distribute resources at the project and organization level evenly among users.
   1. Define when the policy is valid.

Setting	Sample Value
Scope	Organization This policy is applied to the whole organization.

   2. Define the resource quotas.

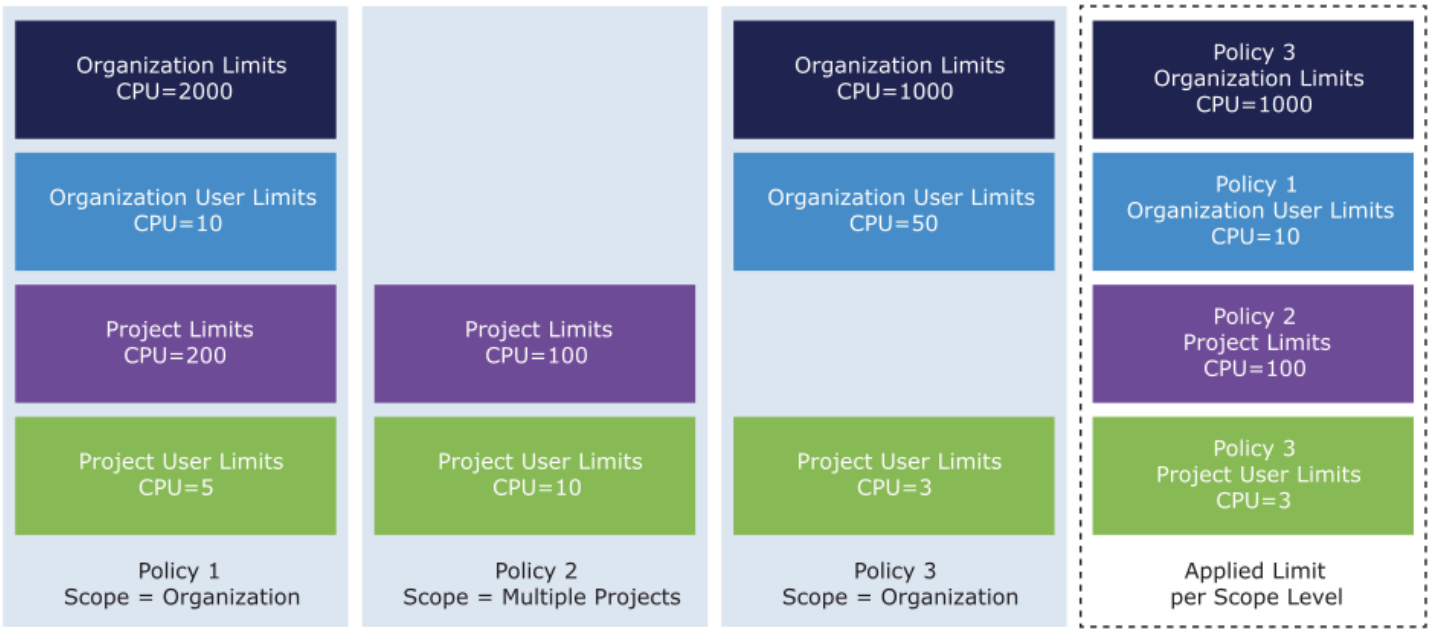
Scope Level	Resource and Limit
Organization Limits	CPU = 1000
Organization User Limits	CPU = 50
Project User Limits	CPU = 3

In this scenario, the resources that are available at each scope level are evaluated and all three policies are enforced. Again, the lowest scope level limits between the three policies are applied.

- Projects User Limits in Policy 3 are applied because the defined value is lower than in Policy 1 and Policy 2.
- Organization User Limits in Policy 3 are not applied. Instead, the limit defined in Policy 1 is applied because the value is lower.
- Organization level limits defined in Policy 3 are applied because the value is lower than in Policy 1.

Summary

Based on the configuration examples above, the following diagram summarizes how resource quotas across multiple policies are applied.



What to do next

- For more examples of how other policies are processed and enforced, see How are policies processed in VCF Automation for VM Apps.
- Configure policies that are relevant to your organizations and projects.
- Monitor provisioned resources on the My Resource Usage dashboard.

How do I limit deployment resources using policies in VCF Automation for VM Apps

You define deployment limit policies to control the amount of resources that deployments can consume when users deploy templates in VCF Automation and request catalog items in VCF Automation. The use cases in this procedure are an introduction to configuring deployment limit policies.

Deployment limits are applied to individual deployments for cloud templates or catalog items. If you want to limit resources at the user, project, or organization level, see How do I configure resource quotas using policies in VCF Automation for VM Apps.

As a cloud administrator, you can limit the total memory, CPU count, storage, and number of virtual machines that can be used per deployment. You can also limit memory, CPU count, and storage for specific resources in the deployment, for example, machines within a cloud template.

The limits apply to all deployments within the policy scope. You can use the policy criteria to narrow the scope to a specific deployment, in which case the policy applies to that deployment only.

How are deployment limit policies enforced

- When the policy is enforced, users can provision deployment resources within the specified limits.
- Multiple deployment limit policies can be enforceable. If there are multiple policies defined for a deployment, the lowest limit value is enforced for each resource.
- If there are resource quota policies and approval policies defined that affect the deployments within the policy scope, deployment limits are enforced before the other policy types.
- If a deployment requests no resources, such as a workflow deployment, the policy is not enforced on that deployment.

When are deployment limit policies applied

- A user requests a catalog item in VCF Automation or a cloud template in VCF Automation.
- A user changes a deployment or its component resources.

Deployment limit policy constraints

The storage value for some images is not calculated during allocation because the images do not contain any storage-related information. A default boot disk size of 8 GB is allocated for storage for such images. The following table provides more information about what images contain boot disk capacity information for each cloud type.

Cloud Type	Boot disk capacity unavailable	Boot disk capacity available
Azure	• Default images • Standard images The default boot disk size is 8 GB.	• Private images • Custom images
AWS	Instance store image disk sizes, including boot disk, are not counted. The default boot disk size is 8 GB.	• Public images • Private images
GCP		Public images
vSphere	• ova • ovf The default boot disk size is 8 GB. Image disks are not counted.	• VM templates • Library item ova • Library item ovf

Procedure

In this use case, there are three policy definitions that illustrate how you can construct deployment limit policies and the results when they are enforced.

1. Select **Content and Policies** > **Policies** > **Definitions** > **New Policy** > **Deployment Limit Policy**.
2. Configure Deployment Limit Policy 1.
   As a cloud administrator, you want to limit the amount of resources that deployments across the organization can use.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization The policy is applied to all deployments in the organization.
Criteria	None

   b. Define the deployment limits.

Resource	Sample Limit Value
CPU	200
VM Count	3
Memory	100 GB
Storage	240 GB

   In this scenario, any deployment across the organization can use up to 200 CPUs, 3 virtual machines, 100GB of memory, and 240GB of storage in total.
3. Configure Deployment Limit Policy 2.
   As a project administrator, you want to apply granular control over resources that are provisioned at the deployment level in a project that you manage.
   a. Define when the policy is valid.

Setting	Sample Value
Scope	Project = TestProj1 This policy is applied to all deployments in the specified project.
Criteria	None

   b. Define the deployment limits.

Resource	Sample Limit Value
CPU	15
Memory	10 GB

   In this scenario, the resources that are available for deployments within the scoped project are evaluated and both Policy 1 and Policy 2 are applied. In this case, the CPU and memory values are lower in Policy 2, so Policy 2 is enforced.
4. Configure Deployment Limit Policy 3.
   As a cloud administrator, you want to control the amount of resources that are consumed when a specific cloud template is deployed by anyone in your organization. Additionally, you want to define limits for specific deployment resources within the cloud template.

a. Define when the policy is valid.

Setting	Sample Value
Scope	Organization
Criteria	`Cloud template equals Proj1Templ1` This policy is applied to all deployments in the organization that are provisioned from the specified cloud template.

b. Define the deployment limits.

Resource	Sample Limit Value
CPU	15
VM Count	10
Memory	20 GB

c. Define the deployment resource limits.

a. Define a limit for all deployments provisioned by developers in the organization.

Setting	Sample Value
Name	Depl Resource Limit 1

Criteria	`Tags has any`
	`Key equals env`
	`AND`
	`Value equals dev`

Limits	CPU = 5 Memory = 10 GB Storage = 40 GB

b. Define limits for a machine in the cloud template.

Setting	Sample Value
Name	Depl Resource Limit 2
Criteria	`Resource Type equals Cloud.vSphere.Machine`
Limits	CPU = 4 Memory = 8 GB

In this scenario, you apply two levels of governance.

1. When a user in your organization deploys the Proj1Templ1 cloud template, the resources that are available for the whole deployment are evaluated and all three existing policies are applied. The lowest deployment limits between the three policies are enforced.
   - The CPU limit value defined in Policy 3 is applied.
   - The VM Count limit value defined in Policy 1 is applied.
   - The Memory limit value defined in Policy 2 is applied.
2. The requested deployment resources are evaluated against Depl Resource Limit 1 and Depl Resource Limit 2 and the deployment resource limits are applied. In this case, there are no other policies that limit these specific deployment resources.

What to do next

- For more examples of how other policies are processed and enforced, see How are policies processed in VCF Automation for VM Apps.

- Configure policies that are relevant to your organizations and projects.
- Monitor provisioned resources on the My Resource Usage dashboard.

How do I configure content sharing policies in VCF Automation for VM Apps

Content sharing policies control what items and actions are available in the VCF Automation catalog for users and user groups. If you have catalog items that require additional governance, you can apply content sharing policies to those items.

You can share content at the project level or at the organization level. When you add a content source or a catalog item to a content sharing policy, you allow the users and user groups specified in the policy to request the items in the VCF Automation catalog.

How are content sharing policies enforced

- You can create content sharing policies that apply to the whole organization, to selected projects, or to a single project. You can also provide additional governance at the content source or at the catalog item level for all content that is associated with a specific project.
- For the organization scope, all users can request shared catalog items. For items that are not shared, users with the Viewer and User roles can view or request catalog items only if they are members of the project that the item is associated with, based on their project role.
- Multiple content sharing policies can be created per project and per organization.

Content sharing policy constraints

- Users who are not members of the organization or the project, specified in the policy scope, can still be added to the content sharing policy when the policy is created through an API request. Such users, however, still don't have access to the catalog items associated with the project. You can limit the policy scope to adding users who belong to the selected project.

Procedure

In this use case, there are three policy definitions that illustrate how you can construct content sharing policies and the results when they are enforced.

1. Select **Content and Policies** > **Policies** > **Definitions** > **New Policy** > **Content Sharing Policy**.
2. Configure Content Sharing Policy 1.
   As an administrator, you want to grant two new users in your project access to all cloud templates that are associated with the project.
   a. Select a project to which to apply the policy.

Setting	Sample Value
Scope	Select **Project** and search for your project. This policy is applied to content associated with this project.

   b. Select what content you want to share with members of the project.

Setting	Sample Value
Content sharing	Click **Add Items** > **Content Sources** , then select the *cs-project1* content source to share with users. In this scenario, *cs-project1* contains four cloud templates.

c. Select the users you want to share the content with.

Setting	Sample Value
Entitlement type	User based
Users	Click **Add Users** and enter the emails of the two new users. *User1@company.com*, *User2@company.com* You can only select users who are associated with the scoped project.

In this scenario, all four cloud templates associated with the content source you specified become available for User 1 and User 2.

3. Configure Content Sharing Policy 2.
   You want to share a new cloud template with developers in several projects.
   a. Select the projects to which to apply the policy.

Setting	Sample Value
Scope	Select **Multiple Projects** and define project criteria. For example, `Project name contains dev` This policy is applied only to projects whose name contains the phrase *dev*.

   b. Select the content that you want to share.

Setting	Sample Value
Content sharing	Click **Add Items** > **All Content**, then select an individual cloud template to share with users.

   c. Select the users you want to share the content with.

Setting	Sample Value
Entitlement type	User based
Users	Select the **Share content with all users/groups in the project** check box.

   In this scenario, the cloud template becomes available to all users and user groups in the developer projects that are included in the policy scope.

4. Configure Content Sharing Policy 3.
   You want to grant administrators access to multiple content sources across your organization.
   a. Select a project to which to apply the policy.

Setting	Sample Value
Scope	Organization

   b. Select the content that you want to share with members of the project.

Setting	Sample Value
Content sharing	Click **Add Items** > **Content Sources** , then select the content sources you want to share. *cs-project3*, *cs-project4*

c. Select the users you want to share the content with.

Setting	Sample Value
Entitlement type	Role based
Users	Select **Administrator**. You can grant content sharing rights only to users with the Project Administrator or Project Member role as well as custom user roles.

In this scenario, all content from the selected content sources becomes available to users with the Project Administrator role across your organization.

What to do next

- For more examples of how other policies are processed and enforced, see How are policies processed in VCF Automation for VM Apps.
- Configure policies that are relevant to your organizations and projects.
- Provide content to your users. See Adding Content to the Catalog in VCF Automation for VM Apps.

How do I configure policy scope in VCF Automation for VM Apps

When you create a policy, you specify its scope to determine how the policy is applied. You can assign the policy to the whole organization, to multiple projects within the organization, or to a single project.

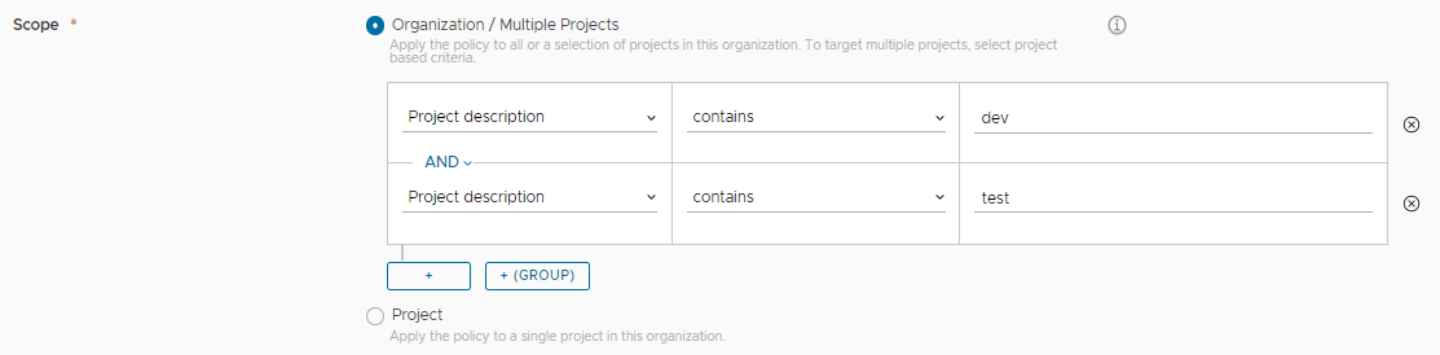
The scope options are the same for all policy types. After you create a policy, you cannot change the scope.

The following table provides more information about the application of each scope option.

Option	Project Criteria	Application
Organization/Multiple Projects	No	Organization. If no project criteria is defined, the policy is applied to all deployments in the organization.
	Yes	Multiple projects. If you define project specific criteria, the policy is applied to deployments associated with the projects that meet the specified criteria.
Project	Select a project.	Single project. The policy is applied only to deployments associated with the project that you select.

Setting policy scope to multiple projects

If you want to apply a policy to multiple projects in your organization, you set the policy scope to Organization/Multiple Projects and specify project specific criteria.



When you define project criteria, you can filter projects based on project name, description, and ID.

Project criteria work in the same way as deployment criteria. For more information about constructing criteria, see How do I configure deployment criteria in Service Broker policies.

The following table gives more information about how you can use each property to refine the scope of your policy.

Property	Supports these operators	Example

Project description	• equals
	• not equal to
	• matches Regex
	• contains
		You create a policy and you want to limit the application to developer projects in the organization that you administer.
		You set the scope to Organization/Multiple Projects and you add a project description expression that looks like the following example.
		
		`Project description contains dev`
		`AND`
		`Project description contains test`

Project ID	• equals
	• not equal to
		You want to apply a policy only to two or three projects, you set the scope to Organization/Multiple Projects and you add a project ID expression that looks like the following example.
		
		`Project ID equals proj123`
		`OR`
		`Project ID equals proj456`
		`OR`
		`Project ID equals proj789`

Project name	• equals • not equal to • matches Regex • contains	You want to limit a policy to test projects in your organization, you set the scope to Organization/Multiple Projects and you add a project name expression that looks like the following example. `Project name matches Regex (t|T)est.*`

How do I configure deployment criteria in policies

The deployment criteria narrows the scope of a policy so that it is applied only to the deployments where the criteria is true. For example, you can use the deployment criteria to create a policy that is applied only to a particular catalog item or template.

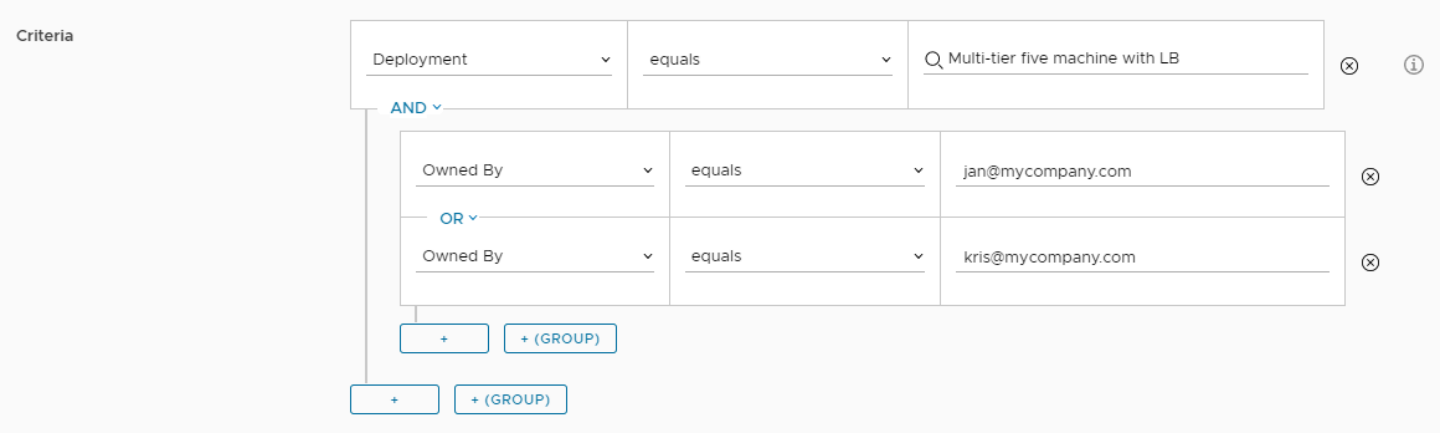
Constructing deployment criteria

You use the graphical interface to construct the deployment criteria expression. To construct complex expressions, you can use AND and OR. You can also group expressions as parenthetical operators. For more about how the expressions are processed, see the *Order of operations for the expression section* below.

Here is an example of an expression.

```
Deployment equals Multi-tier five machine with LB AND (Owned By equals jan@mycompany.com OR Owned By
 kris@mycompany.com)
```

Using the deployment criteria components, it looks like the following example.



Policy criteria properties

To create functional policy criteria, you must understand the syntax.

The criteria text box has various drop-down menus that provide the available properties and operators. How you construct your expression depends on the available values and on the order of operations.

The drop-down menus include the following properties. Some properties vary between policy types.

Property	Description	Available in these policy types	Supports these operators
`Template/Cloud template`	Identifier for the VCF Automation template that was used to create the deployment. Use `Template` rather than `Catalog Item` when your policy is specific to VCF Automation templates. For example, an Amazon Web Services template does not have a `Template`.	• Approvals • Day 2 • Lease	• equals • not equal to
`Catalog Item`	Identifier for the VCF Automation catalog item that was used to request the deployment. Use `Catalog Item` rather than `Template` when your policy can include VCF Automation catalog items based on any template, extensibility workflow, or other content type. For example, VCF Automation templates and Amazon Web Services CloudFormation templates deployed from the catalog have a `Catalog Item`.	• Approvals • Day 2 • Lease	• equals • not equal to
`Deployment`	Identifier for the deployment. Use `Deployment` when you want to apply the policy to existing deployments.	• Approvals • Day 2 • Lease	• equals • not equal to

Property	Description	Available in these policy types	Supports these operators
`Created By`	Name of the user who requested the deployment. The format is username@mycompany.com. This user is the user who requested the deployment.	• Day 2 • Lease	• equals • not equal to • matches Regex • contains
`Name`	Deployment name. Use `Name` rather than `Deployment` when you want to apply the policy to existing policies and policies that can be created in the future that match the specified deployment name expression.	• Approvals • Day 2 • Lease	• equals • not equal to • matches Regex • contains
`Namespace Name`	Name of the namespace.	• IaaS Resource • VKS Resource	• equals • not equal to
`Namespace Class`	Namespace class.	• IaaS Resource • VKS Resource	• equals • not equal to
`Owned By`	Name of the current deployment owner.	• Approvals • Day 2 • Lease	• equals • not equal to • matches Regex • contains
`Owner Type`	Deployment owner type. Ownership can be based on users or Active Directory groups.	• Approvals • Day 2 • Lease	• equals • not equal to
`Region`	Region.	• IaaS Resource • VKS Resource	• equals • not equal to
`Requested By`	Name of the user who requested a day 2 action. The format is username@mycompany.com. When creating approval policies, the `Requested By` criteria is the user who requested a day 2 action, not the user who requested the deployment. The user who requested the deployment is the `Created By` criteria.	• Approvals	• equals • not equal to • matches Regex • contains
`Resources`	Resources that are part of a deployment. You can define the deployment criteria based on the following resources. • Cloud Zone • Disks • Kind • Resource Property • Resource Type Using Avi Load Balancer resources is supported for Approvals and Lease policies only.	• Approvals • Day 2 • Lease	

Property	Description	Available in these policy types	Supports these operators
`TKR Version`	Name of the vSphere Kubernetes cluster.	• VKS Resource	• equals • not equal to • matches Regex • contains
`VKS Name`	Name of the vSphere Kubernetes cluster.	• VKS Resource	• equals • not equal to • matches Regex • contains

Using the *`contains`* and *`matches Regex`* operators

The `contains` and `matches Regex` operators define a search for a specified set of characters within a property. You can apply these operators to string based properties that do not support a drop-down, such as `createdBy`, `name`, and `ownedBy`.

The `contains` operator searches for all instances of the value you specify in any context. The value input text box is case sensitive and space sensitive. If you want to account for context variation, you must set a value for each additional variant. Use the `contains` operator for simple searches for a limited number of values.

The `matches Regex` operator provides great flexibility when you use it for complex searches that must account for a lot of context variation. The regular expressions must follow ECMAScript syntax. When defining regular expressions, do not enter the forward slashes (`/`) at the beginning and at the end of the value.

The following table provides examples of expressions using the two operators and compares how they might be used to achieve the same goal.

Example with the **`contains`** operator	Example with the **`matches Regex`** operator	Field value matches
`Name contains test`	`Name matches Regex test*`	All deployment names that contain *`test`* in lowercase. For example, *`test deployment`*, *`mytest`*, *`test-123`*, and so on.

`Name contains test`
`OR`
`Name contains Test`
	`Name matches Regex (t|T)est.*`	All deployment names that contain *`test`* or *`Test`*.

Example with the `contains` operator	Example with the `matches Regex` operator	Field value matches
(group) Created By contains admin@ (group) AND Created By contains .com OR Created By contains .org (group) AND Name contains test OR Name contains test- OR Name contains Test OR Name contains Test- OR Name contains deploy OR Name contains Deploy	Created By matches Regex admin@\S+\.((com)|(org)) AND Name matches ((t|T)est)|(d|D)epl.*.	All deployments that are created by users whose email address starts with *admin@* and ends with *.com* or *.org*. All deployment names that contain *test* and/or *deploy* in any configuration. For example, *test deployment*, *testdeployment*, *Test-Deployment*, and so on.

Order of operations for the expression

An expression is processed in the following order. Groups are illustrated as parentheses.

1. Expressions in groups
2. AND
3. OR

Use the following examples to understand the order.

- X OR Y AND Z. In this example, Y AND Z is evaluated before X OR Y. Next, the X OR is evaluated against the results of Y AND Z.
- (X OR Y) AND Z. In this example, X OR Y is evaluated before AND because the expression in the group is always evaluated first. Next the AND Z is evaluated against the results of X OR Y.

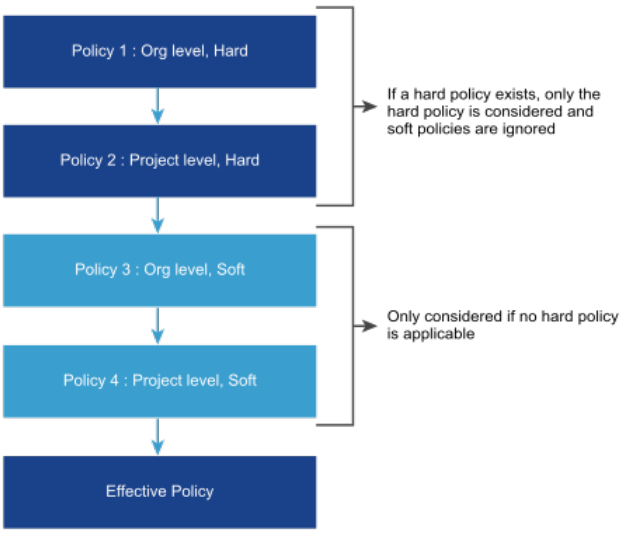
How are policies processed in VCF Automation for VM Apps

Policies are processed based on the policy definition. In particular, the scope and the enforcement level determine which policy is valid when you have multiple policies that might apply to a single deployment.

This article provides general information about policy processing, but it also includes more details for the different types of policies.

How policies are ranked based on organization level and enforcement type

When a user, who is a member of a project, creates a deployment, there might be more than one policy that applies to that deployment.

	To evaluate the policies, the system first identifies and ranks policies. 1. Are there any hard policies at either the organization and project level. If there are hard and soft policies, then only the hard policies are considered and ranked. If there are only soft policies, then the soft policies are ranked. 2. The ranking of all hard or all soft policies is ordered by scope, with organizational policies having a higher rank than project policies. 3. The final discriminating characteristic is the creation date, with older dates be ranked higher than newer dates.

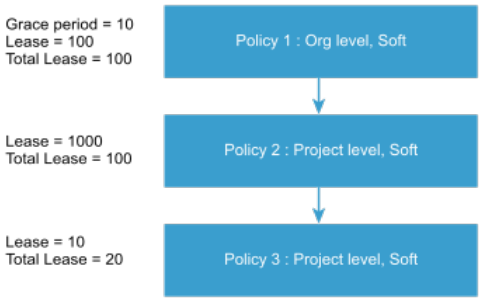
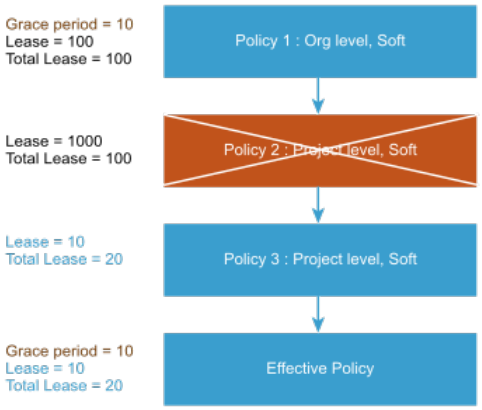
How policies are processed based on organization level and enforcement type

The policies are evaluated, ranked, and, where applicable, merged to produce an effective policy. An effective policy produces the intended results but is not always a specific named policy.

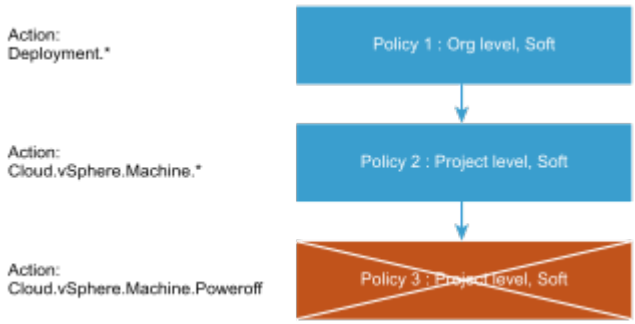
This section includes the following examples:

- Lease policies
- Day 2 actions policies

Review the following lease policy examples.

	After identifying the policies to be considered and ranking them, the policies are then evaluated to identify the merge order. • The highest ranking policy becomes the baseline. The second-level policy is applied on top of it, and so on. • If a policy is incompatible with the preceding policies, then it is discarded from consideration. For example, the values are higher they the preceding policies. • Any policy that is discarded is ignored. To see which policy is applied, select **Content and Policies** > **Policies** > **Enforcement**, locate the deployment, and review the decision notes.
	Rather than applying one policy and excluding all the others above, the policies are merged and might include values from more than one individual policy. In this example, the merging process excludes Policy 2 from consideration because the values are higher than Policy 1. Next, Policy 3 is evaluated against Policy 1. The Lease and Total Lease values in Policy 3 are lower than Policy 1, so those values, in addition to the Grace period, become part of the effective policy.

Review the following day 2 actions policy examples.

	After identifying the policies to be considered and ranking them, the policies are then evaluated to identify the merge order. • The highest ranking policy becomes the baseline. The second-level policy is applied on top of it, and so on. • If a policy is enforced by preceding policies, for example, policy 3, then it is discarded from consideration. • Any policy that is discarded is ignored. To see which policy is applied, select **Content and Policies** > **Policies** > **Enforcement**, locate the deployment, and review the decision notes.

Lease policy management goal considerations

Now that you know how lease policies are processed, identify your policy management goals. By understanding how the policies are processed, you can meet your management goals without creating an excessive and unmanageable number of policies.

When deciding how to implement your policies, consider the following scenarios.

- Lease policy goals and enforcement examples
- Day 2 policy goals and enforcement examples

Table 1827: Lease policy goals and enforcement examples

Management goal	Configuration Example	Behavior
Meaningful default organization-level policy that still allows the project-level policy values to influence the applied values.	Organization policy = Soft • Grace period: 10 • Lease: 100 • Total Lease: 100 Project 1 policy 1= Soft • Lease: 20 • Total Lease: 50 Project 2 policy 1= Soft • Lease: 10 • Total Lease: 30	A member of project 1 requests a catalog item. Project 2 is not considered because it is not applicable to project 1 deployments. The merged effective policy is: • Grace period: 10 • Lease: 20 • Total Lease: 50
Always default to the organization-level policy.	Organization policy = Hard • Grace period: 10 • Lease: 100 • Total Lease: 100 Project 1 policy 1= Soft • Lease: 20 • Total Lease: 50	A member of project 1 requests a catalog item. Project 1 policy 1 is not considered because the hard organization level project is a higher rank and the soft policy is not considered. The effective policy is: • Grace period: 10 • Lease: 100 • Total Lease: 100
All policies are defined at the project-level, with no organization-level default policy.	Project 1 policy 1 = Soft • Grace period: 10 • Lease: 100 • Total Lease: 100 Project 1 policy 2= Soft • Lease: 20	A member of project 1 requests a catalog item. They are both soft policies, and they are both for project 1. The values are merged. The effective policy is: • Grace period: 10 • Lease: 20 • Total Lease: 100

The day 2 actions policies are used in these examples.

Table 1828: Day 2 policy goals and enforcement examples

Management goal	Configuration Example	Behavior
Meaningful default organization-level policy that still allows the project-level policy values to influence the applied values.	Organization policy = Soft • Actions : Deployment.* Project 1 policy 1= Soft • Actions: Cloud.vSphere.Machine.* Project 2 policy 1= Soft • Actions: Cloud.Azure.Machine.*	A member of project 1 requests a catalog item. Project 2 is not considered because it is not applicable to project 1 deployments. The merged effective policy is: • Action : {Deployment.* ,Cloud.vSphere.Machine.*}
Always default to the organization-level policy.	Organization policy = Hard • Action : Deployment.* Project 1 policy 1= Soft • Action : Cloud.vSphere.Machine.*	A member of project 1 requests a catalog item. Project 1 policy 1 is not considered because the hard organization level project is a higher rank and the soft policy is not considered. The effective policy is: • Action : {Deployment.* }

Management goal	Configuration Example	Behavior
All policies are defined at the project-level, with no organization-level default policy.	Project 1 policy 1 = Soft • Actions : Deployment.ChangeLease Project 1 policy 2= Soft • Action : Deployment.Delete	A member of project 1 requests a catalog item. They are both soft policies, and they are both for project 1. The values are merged. The effective policy is: • Action : {Deployment.ChangeLease , Deployment.Delete}

Approval policy goals and enforcement examples

The approval policy evaluation follows this process.

1. A request for a deployment or day 2 action is submitted.
2. The approval service queries for policies that apply to the project that is requesting a catalog item or changing a deployed item.
3. All the applicable project- and organization-level scope policies are returned.
4. The approval policies are filtered based on the deployment criteria. Deployment criteria apply to deployments and day 2 actions.
5. If no matching policies are found, no approval is required and the deployment process proceeds.
6. If there are matching policies, for example, AP1, AP2, APn, then an approval item is created as:
   - Enforced policies = AP1, AP2, APn.
   - Approvers = A union of all the approvers in all the enforced policies.
   - Auto expiry = Reject, if any policy has a reject value; otherwise, approve.
   - Expiry = Minimum number of days of any of the enforced policies.

The following table provides a sample of multiple policies. The description of how they are processed is below the table.

Policy	Configuration example
AP1	Scope = Organization Auto expiry = Approve Expiry = 7 days
AP2	Scope = Project 1 Auto expiry - Approve Expiry = 3 days
AP3	Scope = Project 1 Auto expiry = Reject Expiry = 4 days
AP4	Scope = Project 2 Auto expiry = Approve Expiry = 5 days

Based on the policies and configuration examples above, the following information explains how a Project 1 request is processed.

1. The scope evaluation returns AP1, AP2, and AP3. AP4 is not included because it is a Project 2 policy.
2. Assuming that AP1, AP2, and AP3 satisfy the deployment and action criteria, then the approval item includes the following values:
   - Approvers = Any or all the approvers from AP1, AP2, and AP3 are added as approvers.
   - Auto expiry = Reject. AP3 provides the more restrictive behavior.
   - Expiry = 3 days. AP2 provides the lowest value.

Working with the catalog in VCF Automation for VM Apps

As an organization administrator, after you configure your cloud infrastructure, you can continue to develop templates, or you can turn over development to the project administrators and members.

Project members use VCF Automation to iteratively develop and deploy templates. The deployment locations are configured by the cloud administrator as part of the infrastructure. The administrator has the best understanding of your organizations resources and budget.

The VCF Automation catalog is the streamlined user interface that you provide to your development operations and other teams. You import the machine and application templates that you need, and add governance in the form of projects to control who can deploy what resources, and to control where the resources are deployed. You can also apply policies and customize the catalog request form.

Designing cloud templates in VCF Automation for VM Apps

Deployments begin with cloud templates, which are encoded specifications that define machines, applications, and services to create on cloud resources by way of VCF Automation.

How cloud templates work

Templates can target specific cloud vendors or be cloud agnostic. The cloud zones assigned to your project determine which approach you might take. Check with your cloud administrator so that you know what kind of resources make up your cloud zones.

VCF Automation template creation is an infrastructure-as-code process. You start by adding resources in the design canvas. Then, you complete the details using the code editor. The code editor allows you to type code directly or enter values in a form.

Before you create a cloud template

You can create a VCF Automation template at any time. To deploy it, however, you first need to define your cloud resource infrastructure and create a project that includes that infrastructure.

Getting started with creating and designing cloud templates in VCF Automation for VM Apps

You use the Design page in VCF Automation to create and design cloud template specifications for the machines and applications that you want to provision, including property groups, custom resources, and resource actions.

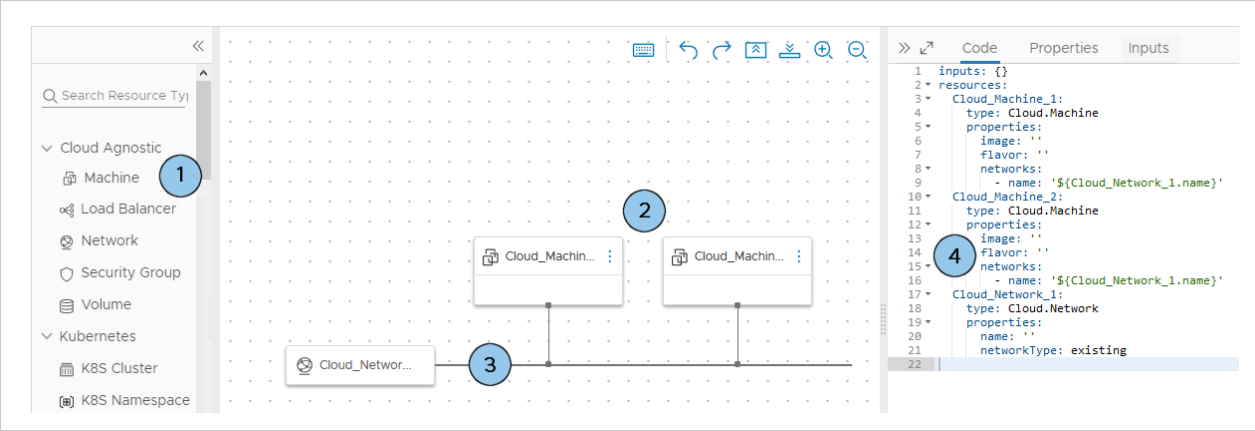
How to use the Design page

To create a cloud template from scratch, go to **Design** > **Templates**. Then, click **New from** > **Blank canvas**.

Note that other options allow you to create the template from either existing Terraform code or from another, existing VCF Automation template to which you have access.

Once you have created the template, you can populate it by adding and configuring resources as shown below.

1. Locate resources.
2. Drag resources to the canvas.
3. Connect resources.
4. Configure resources by editing the cloud template code.



Selecting and adding resources to the design canvas

Resources appear at the left of the design page for selecting and dragging.

Cloud agnostic resources	You can deploy cloud agnostic resources to any cloud vendor. At provisioning time, the deployment uses cloud specific resources that match. For example, if you expect a cloud template to deploy to both AWS and vSphere cloud zones, use cloud agnostic resources.
Cloud vendor resources	Vendor resources, such as those specific to Amazon Web Services, Microsoft Azure, Google Cloud Platform, or VMware vSphere, can only be deployed to matching AWS, Azure, GCP, or vSphere cloud zones. You can add cloud agnostic resources to a cloud template that contains cloud specific resources for a particular vendor. Just be aware of what the project cloud zones support in terms of vendor. **Important:** Starting with VCF Automation 9.1.1, native management capabilities for public cloud resources, including Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP), are deprecated and disabled by default in VM Apps organizations. For more information, see the VCF Automation 9.1.1 release notes.
Configuration management resources	Configuration management resources depend on your integrated applications. For example, a Puppet resource can monitor and enforce the configuration of the other resources.

Connecting resources

Use the VCF Automation design canvas graphical controls to connect resources.

Resources must be compatible for a connection. For example:

- Connecting a load balancer to a cluster of machines.
- Connecting a machine to a network.
- Connecting external storage to a machine.

Important:

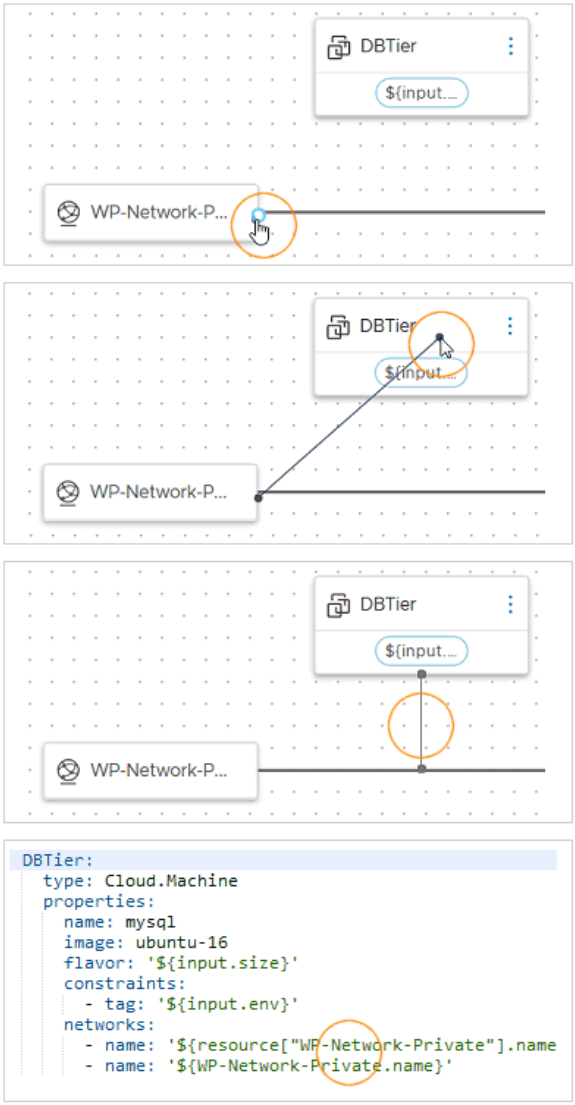
A solid line connector requires that the two resources be deployed in the same cloud zone. If you add conflicting constraints to the resources, deployment might fail.

For example, you can't deploy connected resources where constraint tags force the placement of one to a zone in us-west-1, and the other to a zone in us-east-1.

Solid or dashed arrows only indicate a dependency, not a connection. For more about dependencies, see Creating bindings and dependencies between resources in VCF Automation for VM Apps.

To connect, hover over the edge of a resource to reveal the connection bubble. Then, click and drag the bubble to the target resource and release.

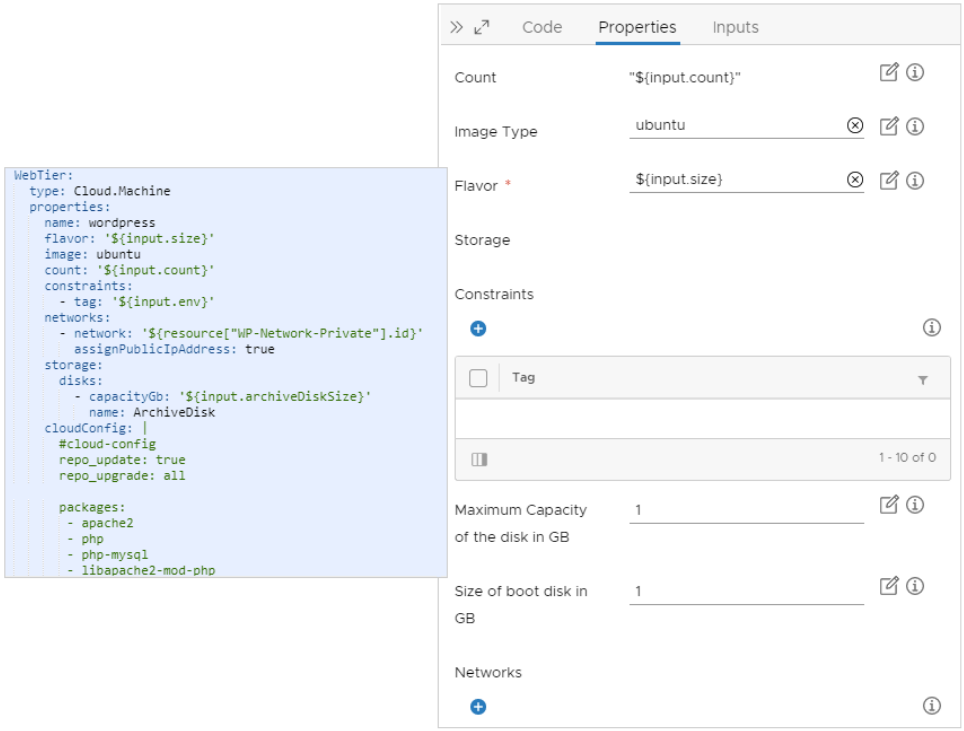
In the code editor, additional code for the source resource appears in the target resource code.



In the figure, the SQL machine and private network are connected, so they must be deployed in the same cloud zone.

Editing cloud template code

The code editor allows you to type, cut, copy, and paste code directly. If you're uncomfortable editing code, you can click a resource that's already in the design canvas, click the code editor **Properties** tab, and enter values there. Property values that you enter appear in the code as if you had typed them directly.

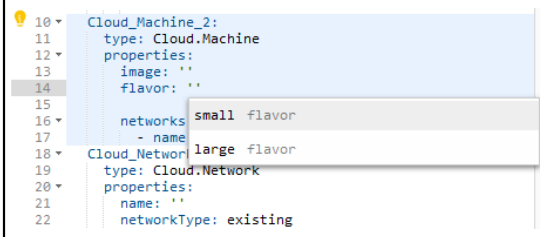
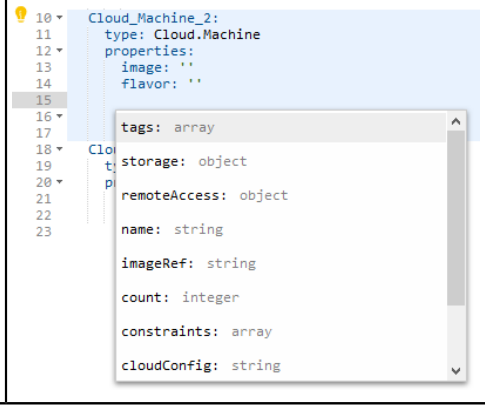


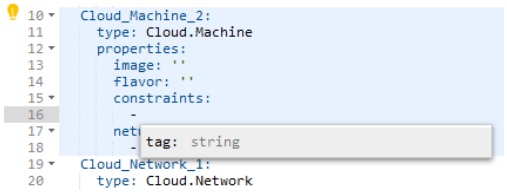
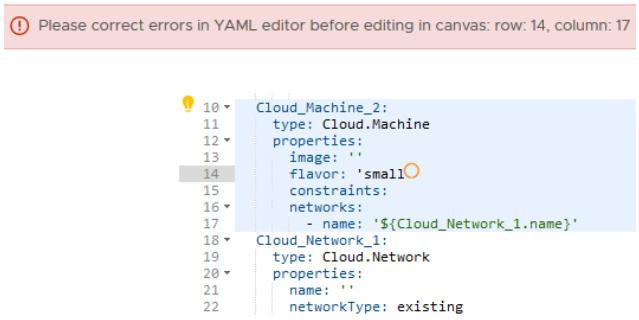
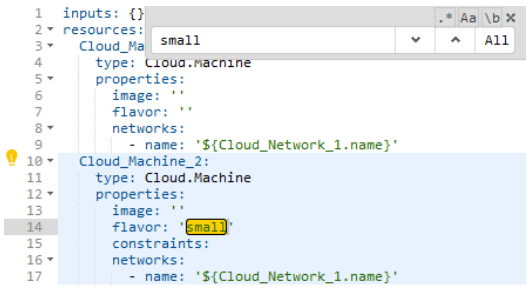
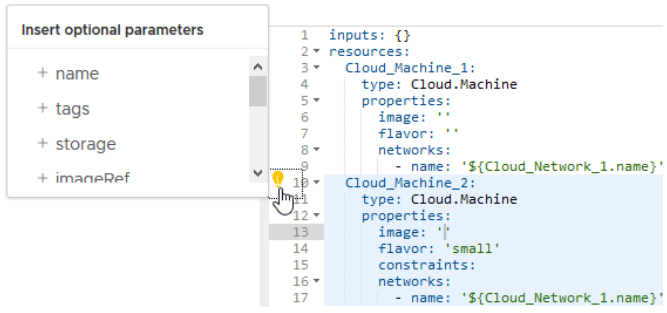
Note that you can copy and paste code from one cloud template to another.

Getting code completion help in your VCF Automation template

Adding VCF Automation resources to the template and connecting those resources in the design canvas only creates starter code. To fully configure your template resources, edit the template code.

The code editor allows you to type code directly or enter property values into a form. To help with direct code creation, the VCF Automation editor includes syntax completion and error checking features.

Editor Hints	Example
Available values	
Allowed properties	

Editor Hints	Example
Child properties	
Syntax errors	
Ctrl+F to search	
Optional parameters	

Creating bindings and dependencies between resources in VCF Automation for VM Apps

When you deploy a VCF Automation template, one resource might need another resource to be available first.

> **Important:**
>
> Arrows only indicate a dependency, not a connection. To connect resources so that they communicate, see Getting started with creating and designing cloud templates in VCF Automation for VM Apps.

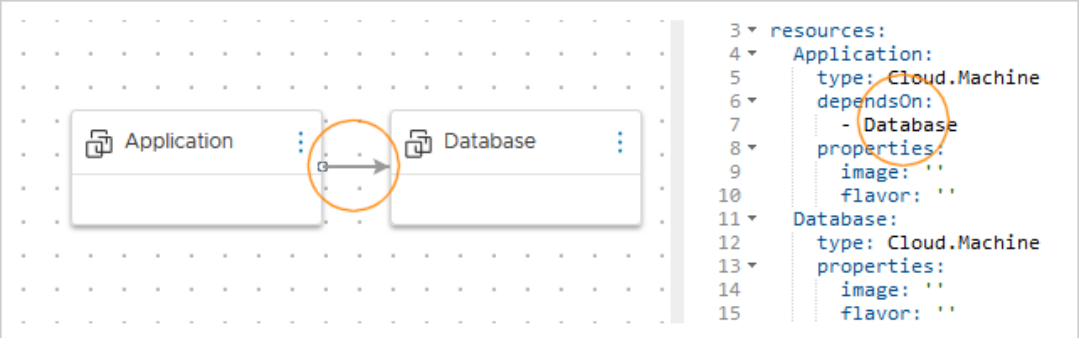
Explicit dependencies

Sometimes, a resource needs another to be deployed first. For example, a database server might need to exist first, before an application server can be created and configured to access it.

An explicit dependency sets the build order at deployment time, or for scale in or scale out actions. You can add an explicit dependency using the graphical design canvas or the code editor.

- Design canvas option—draw a connection starting at the dependent resource and ending at the resource to be deployed first.

- Code editor option—add a `dependsOn` property to the dependent resource, and identify the resource to be deployed first.
  An explicit dependency creates a solid arrow in the canvas.

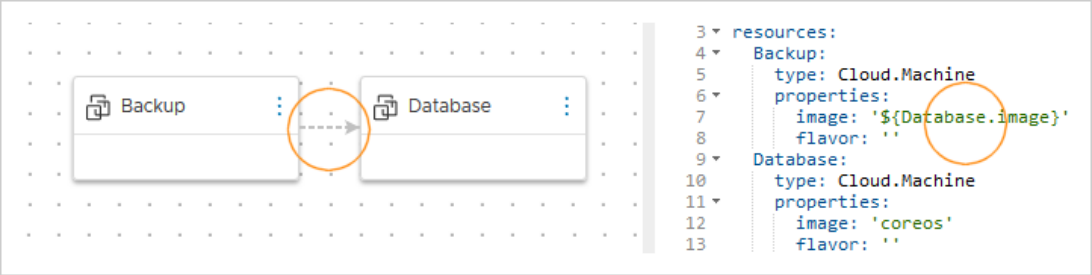


Property bindings

Sometimes, a resource property needs a value found in a property of another resource. For example, a backup server might need the operating system image of the database server that is being backed up, so the database server must exist first.

Also called an implicit dependency, a property binding controls build order by waiting until the needed property is available before deploying the dependent resource. You add a property binding using the code editor.

- Edit the dependent resource, adding a property that identifies the resource and property that must exist first.
  A property binding creates a dashed arrow in the canvas.



Versioning your templates in VCF Automation for VM Apps

As a cloud template developer, you can safely capture a snapshot of a working design before risking further changes.

At deployment time, you can select any of your versions to deploy.

Capturing a cloud template version

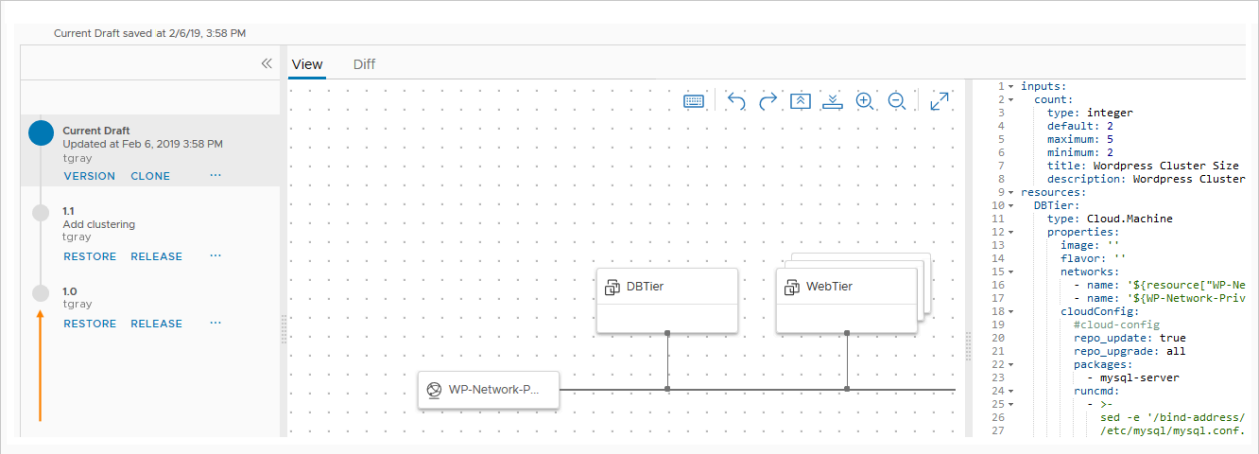
From the design page, click **Version**, and provide a name.

The name must be alphanumeric, with no spaces, and only periods, hyphens, and underscores allowed as special characters.

Restoring an older version

From the design page, click **Version History**.

On the left, select an older version to inspect it in the canvas and code editor. When you find the version that you want, click **Restore**. Restoring overwrites the current draft without removing any named versions.



Releasing a version to VCF Automation

From the design page, click **Version History**.

On the left, select a version and release it.

You can't release a Current Draft until you version it.

Reimporting the version in VCF Automation

To enable the new version for catalog users, reimport it.

In VCF Automation, go to **Content & Policies** > **Content Sources**.

In the list of sources, click the source for the project that contains the cloud template with the newly released version.

Click **Save & Import**.

Comparing cloud template versions

When changes and versions accumulate, you might want to identify differences among them.

In VCF Automation, from the Version History view, select a version, and click **Diff**. Then, from the **Diff against** drop-down, select another version to compare to.

Note that you can toggle between reviewing code differences or visual topology differences.

Figure 538: Code Differences

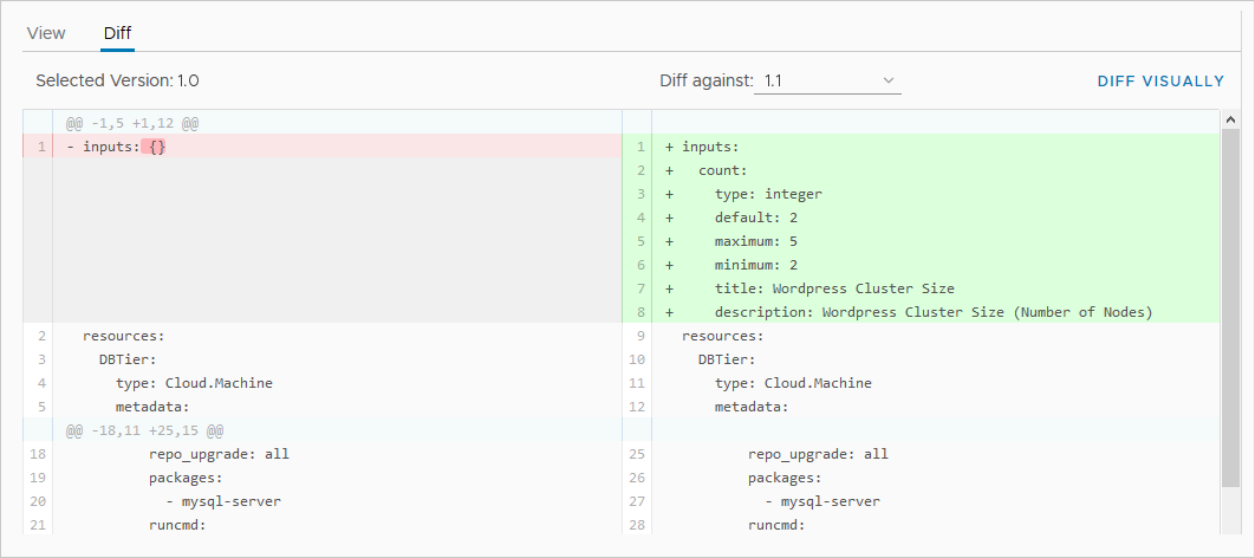
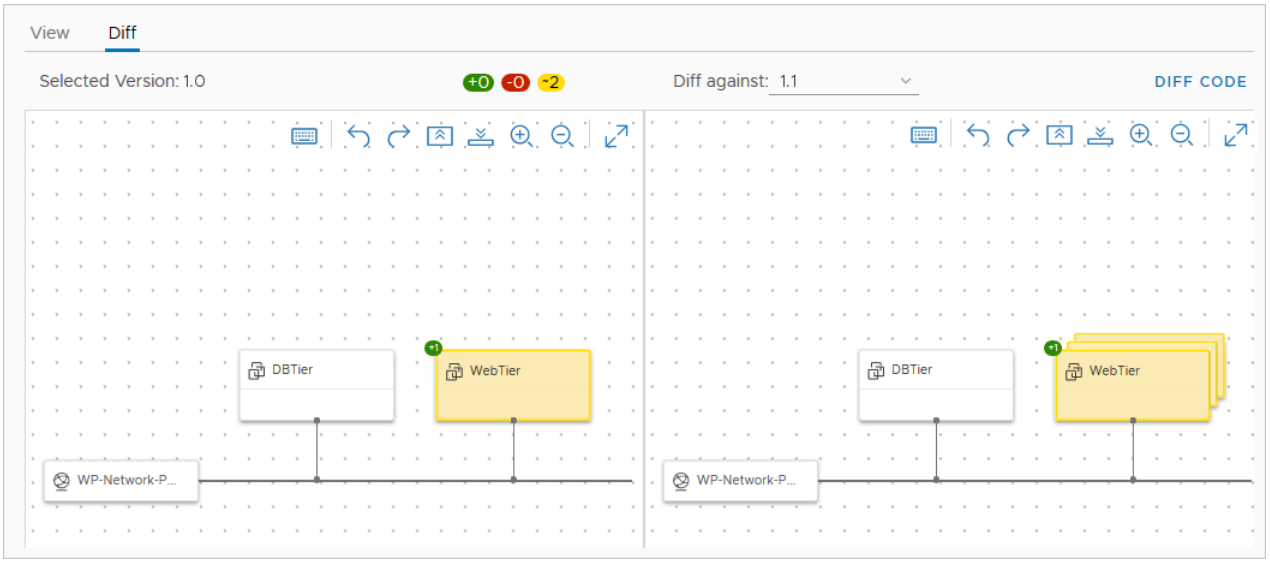


Figure 539: Visual Topology Differences



Cloning a cloud template

Although it's not the same as saving a version, from the design page, **Actions** > **Clone** makes a copy of the current template for alternative development.

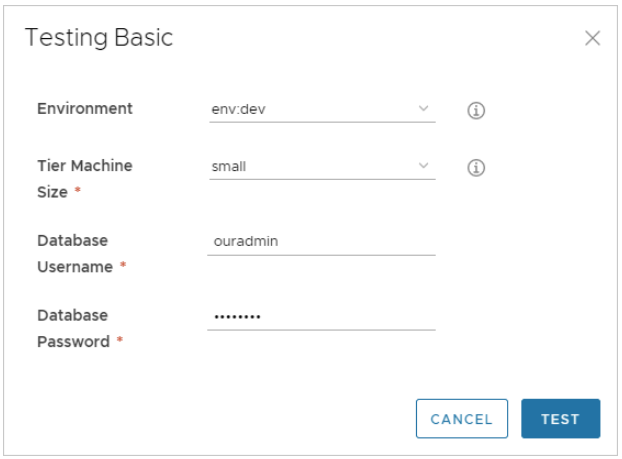
User input in requests in VCF Automation for VM Apps

As a cloud template designer, you use input parameters so that users can make custom selections at request time.

How inputs work

When users supply inputs, you no longer need to save multiple copies of templates that are only slightly different. In addition, inputs can prepare a template for day 2 operations. See How to use cloud template inputs for VCF Automation day 2 updates .

The following inputs show how you might create one cloud template for a MySQL database server, where users can deploy that one template to different cloud resource environments and apply different capacity and credentials each time.



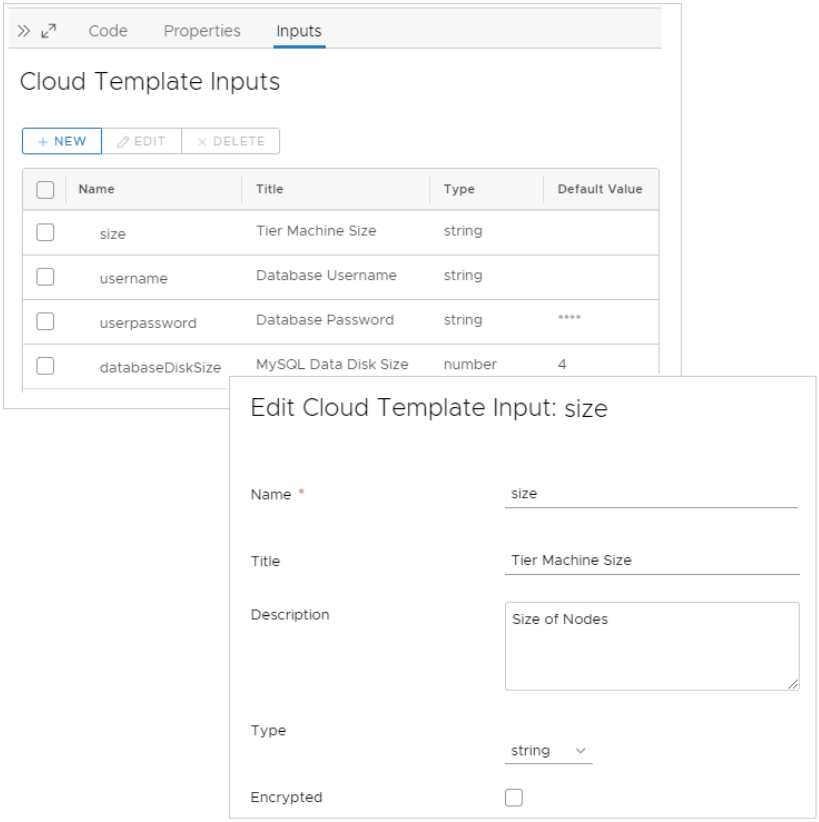
Adding input parameters

Add an `inputs` section to your template code, where you set the selectable values.

In the following example, machine size, operating system, and number of clustered servers are selectable.

```
inputs:
  wp-size:
    type: string
    enum:
      - small
      - medium
    description: Size of Nodes
    title: Node Size
  wp-image:
    type: string
    enum:
      - coreos
      - ubuntu
    title: Select Image/OS
  wp-count:
    type: integer
    default: 2
    maximum: 5
    minimum: 2
    title: Wordpress Cluster Size
    description: Wordpress Cluster Size (Number of nodes)
```

If you're uncomfortable editing code, you can click the code editor **Inputs** tab, and enter settings there. The following example shows some inputs for the MySQL database mentioned earlier.



Referencing input parameters

Next, in the `resources` section, you reference an input parameter using `${input.property-name}` syntax.

If a property name includes a space, delimit with square brackets and double quotes instead of using dot notation: `${input["property name"]}`

Important:
In cloud template code, you cannot use the word `input` except to indicate an input parameter.

```
resources:
  WebTier:
    type: Cloud.Machine
    properties:
      name: wordpress
      flavor: '${input.wp-size}'
      image: '${input.wp-image}'
      count: '${input.wp-count}'
```

Nested input

If you want to organize or categorize inputs, nested input is supported. In the following example, the CPU and memory are together under a parent heading of *level*.

Note:
A group of nested inputs is different than creating and referencing a formal property group.

```
inputs:
  cluster:
    type: integer
    title: Cluster
    default: 1
    minimum: 1
    maximum: 4
  level:
    type: object
    properties:
      cpu:
        type: integer
        title: CPU
        default: 1
        minimum: 1
        maximum: 4
      memory:
        type: integer
        title: Memory
        default: 2048
        minimum: 2048
        maximum: 4096
```

In the *resources* section, to reference the nested input, include the parent in the path.

```
resources:
  Disk_1:
    type: Cloud.vSphere.Disk
    allocatePerInstance: true
    properties:
      capacityGb: 1
      count: ${input.cluster}
  Machine_1:
    type: Cloud.vSphere.Machine
    allocatePerInstance: true
    properties:
      totalMemoryMB: ${input.level.memory}
      attachedDisks:
        - source: ${slice(resource.Disk_1[*].id, count.index, count.index + 1)[0]}
```

```
    count: ${input.cluster}
    imageRef: ubuntu
    cpuCount: ${input.level.cpu}
```

Optional versus required input

For all types except Boolean, user entry is optional by default. To require input, do one of the following:

- Set a default value.
- When you have no nested inputs, add the `populateRequiredOnNonDefaultProperties` property:

```
inputs:
  cluster:
    type: integer
    populateRequiredOnNonDefaultProperties: true
    title: Cluster
    minimum: 1
    maximum: 4
```

  Note that you can also apply this setting when referencing a formal property group:

```
inputs:
  pgmachine:
    type: object
    populateRequiredOnNonDefaultProperties: true
    $ref: /ref/property-groups/machine
```

- With nested inputs, add the `populateRequiredForNestedProperties` property:

```
inputs:
  cluster:
    type: integer
    title: Cluster
    default: 1
    minimum: 1
    maximum: 4
  level:
    type: object
    properties:
      cpu:
        type: integer
        populateRequiredForNestedProperties: true
        title: CPU
        minimum: 1
        maximum: 4
      memory:
        type: integer
        populateRequiredForNestedProperties: true
        title: Memory
        minimum: 2048
        maximum: 4096
```

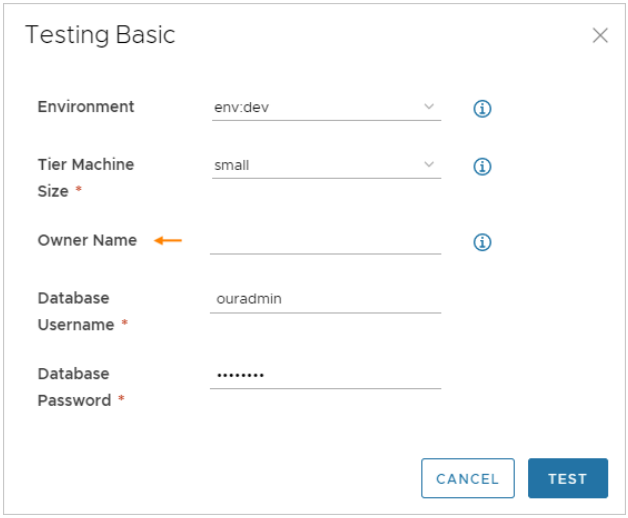
  Note that you can also apply this setting when a formal property group reference is nested:

```
  level:
    type: object
    properties:
      cpu:
        type: integer
        populateRequiredForNestedProperties: true
        title: CPU
        minimum: 1
        maximum: 4
```

```
      memory:
        type: integer
        populateRequiredForNestedProperties: true
        title: Memory
        minimum: 2048
        maximum: 4096
      pgrequester:
        type: object
        populateRequiredForNestedProperties: true
        $ref: /ref/property-groups/requesterDetails
```

Optional input—To force input to remain optional, set an empty default value using tick marks:

```
owner:
  type: string
  minLength: 0
  maxLength: 30
  title: Owner Name
  description: Account Owner
  default: ''
```



Sending inputs to VCF Operations orchestrator

To supply a cloud template input to a VCF Operations orchestrator action, take the following steps. With VCF Operations orchestrator actions, the Inputs form is easier to use than typing directly in the cloud template code.

1. Verify that the action you want exists in VCF Operations orchestrator.
2. In the template design page in VCF Automation, go to the **Inputs** tab, and click to add a new input.
3. Select a type, and select **External source**.
4. Next to Action, click the **Select** button.
5. Use the Action search field to locate and select the VCF Operations orchestrator action.
6. Enter or select any action parameters.

After you save and create the input, it appears in the cloud template code as a `$dynamicDefault` input. For example:

```
inputs:
  backupnetwork:
    type: string
    $dynamicDefault: /data/vro-actions/com.insanpaolo/getBackupNetworkVLAN?
network={{abcdef123456}}&tenant={{abcdef123456}}
```

Note:
If the VCF Operations orchestrator workflow you send inputs to runs for too long, causing a delay of more than 30 seconds, it might cause timeout issues in VCF Automation.

List of input properties

Property	Description
const	Used with oneOf. The real value associated with the friendly title.
default	Prepopulated value for the input. The default must be of the correct type. Do not enter a word as the default for an integer.
description	User help text for the input.
encrypted	Whether to encrypt the input that the user enters, true or false. Passwords are usually encrypted. You can also create encrypted properties that are reusable across multiple cloud templates. See Secret properties in VCF Automation for VM Apps.

enum	A drop-down menu of allowed values.
	Use the following example as a format guide.
	
	`enum:`
	`  - value 1`
	`  - value 2`

format	Sets the expected format for the input. For example, (25/04/19) supports date-time. Allows the use of the date picker in VCF Automation custom forms.
items	Declares items within an array. Supports number, integer, string, Boolean, or object.
maxItems	Maximum number of selectable items within an array.
maxLength	Maximum number of characters allowed for a string. For example, to limit a field to 25 characters, enter `maxLength: 25`.
maximum	Largest allowed value for a number or integer.
minItems	Minimum number of selectable items within an array.
minLength	Minimum number of characters allowed for a string.
minimum	Smallest allowed value for a number or integer.
oneOf	Allows the user input form to display a friendly name (title) for a less friendly value (const). If adding the *default* property shown above in this table, set that default to a const value, not a title. Valid for use with types string, integer, and number.
pattern	Allowable characters for string inputs, in regular expression syntax. For example, `'[a-z]+'` or `'[a-z0-9A-Z@#$]+'`
properties	Declares the key:value properties block for objects.
readOnly	Used to provide a form label only.
title	Used with oneOf. The friendly name for a const value. The title appears on the user input form at deployment time.

Property	Description
type	Data type of number, integer, string, Boolean, or object. **Important:** A Boolean type adds a blank checkbox to the request form. Leaving the box untouched does not make the input False. To set the input to False, users must check and then clear the box.
writeOnly	Hides keystrokes behind asterisks in the form. Cannot be used with enum. Appears as a password field in VCF Automation custom forms.

Additional examples

String with enumeration

```
image:
  type: string
  title: Operating System
  description: The operating system version to use.
  enum:
    - ubuntu 16.04
    - ubuntu 18.04
  default: ubuntu 16.04

shell:
  type: string
  title: Default shell
  Description: The default shell that will be configured for the created user.
  enum:
    - /bin/bash
    - /bin/sh
```

Integer with minimum and maximum

```
count:
  type: integer
  title: Machine Count
  description: The number of machines that you want to deploy.
  maximum: 5
  minimum: 1
  default: 1
```

Array of objects

```
tags:
  type: array
  title: Tags
  description: Tags that you want applied to the machines.
  items:
    type: object
    properties:
      key:
        type: string
        title: Key
      value:
        type: string
```

```
          title: Value
```

String with friendly names

```
platform:
  type: string
  oneOf:
    - title: AWS
      const: platform:aws
    - title: Azure
      const: platform:azure
    - title: vSphere
      const: platform:vsphere
  default: platform:aws
```

String with pattern validation

```
username:
  type: string
  title: Username
  description: The name for the user that will be created when the machine is provisioned.
  pattern: ^[a-zA-Z]+$
```

String as password

```
password:
  type: string
  title: Password
  description: The initial password that will be required to logon to the machine. Configured to reset on
first login.
  encrypted: true
  writeOnly: true
```

String as text area

```
ssh_public_key:
  type: string
  title: SSH public key
  maxLength: 256
```

Boolean

```
public_ip:
  type: boolean
  title: Assign public IP address
  description: Choose whether your machine should be internet facing.
  default: false
```

Date and time calendar selector

```
leaseDate:
  type: string
  title: Lease Date
  format: date-time
```

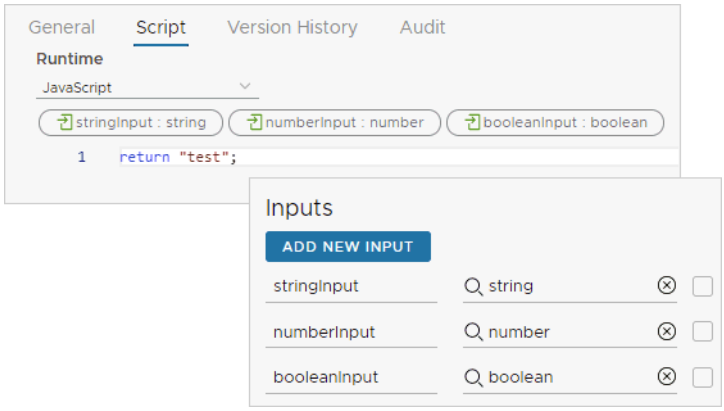
VCF Operations orchestrator actions as inputs

In an VCF Automation template, VCF Operations Orchestrator Client actions can be included as cloud template inputs.

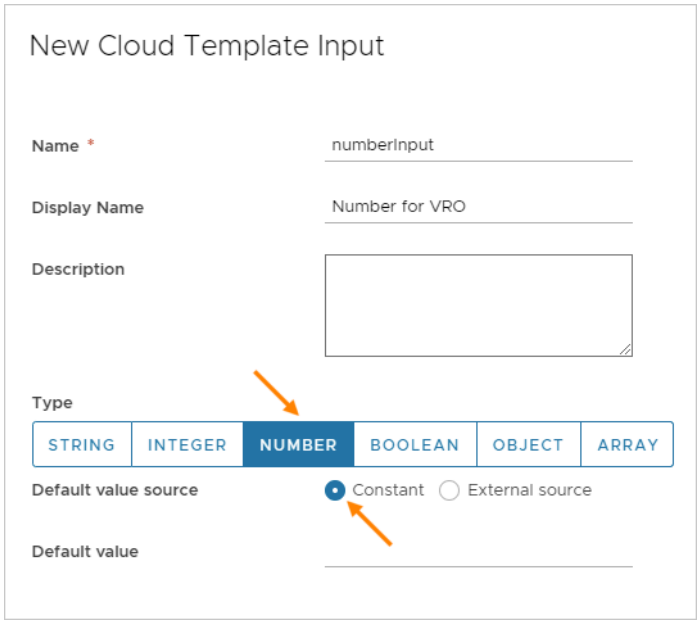
Adding an VCF Operations Orchestrator Client action to cloud template inputs

To use VCF Operations Orchestrator Client actions as cloud template inputs, follow these guidelines.

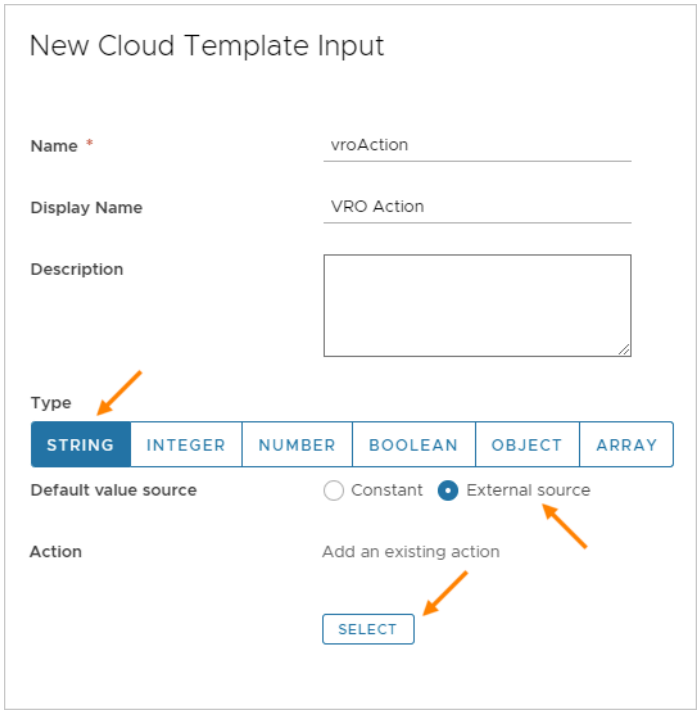
1. In the instance of VCF Operations Orchestrator Client that is embedded with VCF Automation, create an action that does what you want.
   The VCF Operations Orchestrator Client action must only include primitive string, integer, number, and boolean types. VCF Operations Orchestrator Client types are not supported.
   In this simple example, the VCF Operations Orchestrator Client action collects three inputs and returns a hard-coded string.



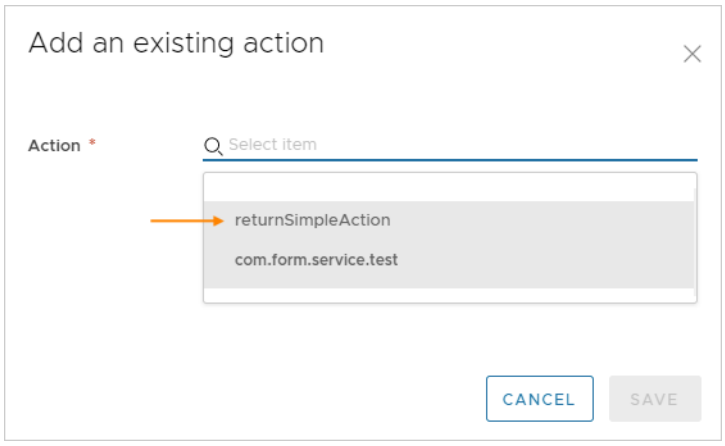
2. In VCF Automation, create or edit a cloud template.
3. In the code editor, click the **Inputs** tab, and **New Cloud Template Input**.
4. To add the VCF Operations Orchestrator Client action inputs, click the type, and click **Constant**.
   Separately add each VCF Operations Orchestrator Client action input as a new cloud template input.



5. After adding the action inputs, create another new cloud template input, click the type, click **External source**, and click **Select**.



6. In **Action**, search for and select the VCF Operations Orchestrator Client action that you created, and click **Save**.



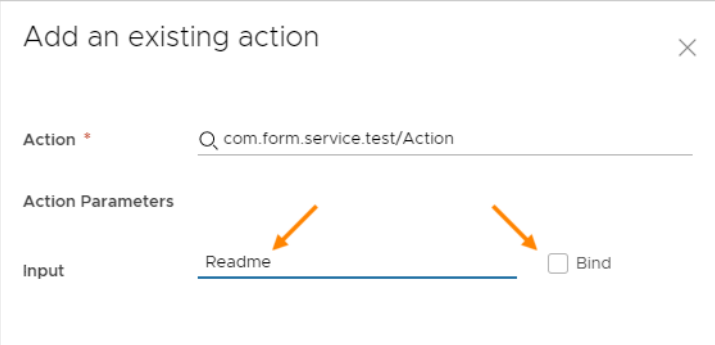
When deploying the cloud template, the VCF Operations Orchestrator Client action settings appear in the input form for the requesting user.



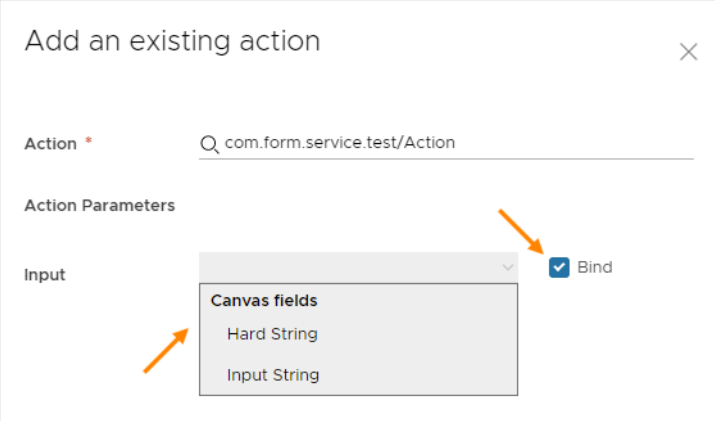
Configurable defaults

To populate the input form with default values, do one of the following when adding the VCF Operations Orchestrator Client action as the external source.

- Manually supply the default property value.
  Clear the **Bind** option, and enter the value.



- Use another property value from the inputs already in the cloud template.
  Select the **Bind** option, and select a property from the drop-down list.

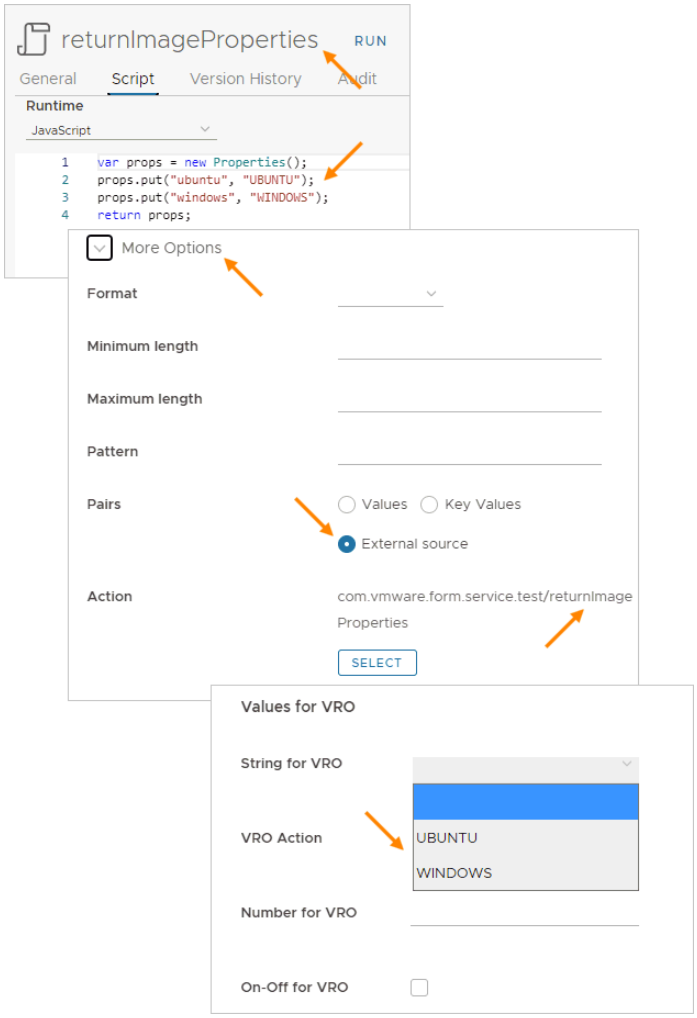


Adding VCF Operations Orchestrator Client enumerated input selections

To create an VCF Operations Orchestrator Client based selection list in an input form, do the following when adding to the cloud template inputs.

1. In VCF Operations Orchestrator Client, create an action that maps the values that you want for the list.
2. In VCF Automation, when adding the cloud template input, expand **More Options**.
3. For **Pairs**, click **External source**, click **Select**, and add the VCF Operations Orchestrator Client action that you created.

   Note:
   If you also create a default value when adding the property, that default must exactly match one of the enumerated values from the VCF Operations Orchestrator Client action.



Reusing a group of properties in VCF Automation for VM Apps

When you have VCF Automation properties that always appear together, you can assemble them into a property group.

You can quickly add a property group to different VCF Automation designs, which saves the time of adding the same multiple properties one by one. In addition, you have a single place to maintain or modify the set of properties, which ensures their consistent application.

Only users with the VCF Automation Administrator role may create, update, or delete a property group. The administrator can share a property group with an entire organization or limit its use to only within a project.

> **CAUTION:**
>
> A property group might be included in many cloud templates, including ones that are already released to the catalog. Changes to a property group can affect other users.

There are two types of property groups.

- Inputs
  Input property groups gather and apply a consistent set of properties at user request time. Input property groups can include entries for the user to add or select, or they might include read-only values that are needed by the design.
  Properties for the user to edit or select can be readable or encrypted. Read-only properties appear on the request form but can't be edited. If you want read-only values to remain totally hidden, use a constant property group instead.
- Constants

Constant property groups silently apply known properties. In effect, constant property groups are invisible metadata. They provide values to your VCF Automation designs in a way that prevents a requesting user from reading those values or even knowing that they're present. Examples might include license keys or domain account credentials.

The two property group types are handled very differently by VCF Automation. When you create a property group, you must first select whether to create inputs or constants. You can't create a blended property group nor convert an existing set of properties and their property group from one type to the other.

Input property groups in VCF Automation for VM Apps

VCF Automation input property groups usually include related settings for the user to enter or select. They might also include read-only values needed by the cloud template design.

Creating the input property group

1. Go to **Design** > **Property Groups**, and click **New**.
2. Select **Input Values**.
3. Name and describe the new property group.

Name	Property group names must be unique within a given organization. Only letters, numbers, and underscores are permitted.
Display Name	Add a heading for the entire group of properties, which appears on the request form.
Description	Explain what this set of properties is for.
Scope	Decide whether an administrator may share the property group with the whole organization. Otherwise, only one project can access the property group. Although you can always add or modify properties in the group, the scope is permanent and can't be changed later.
Project	When the scope is project-only, this project can access the property group.

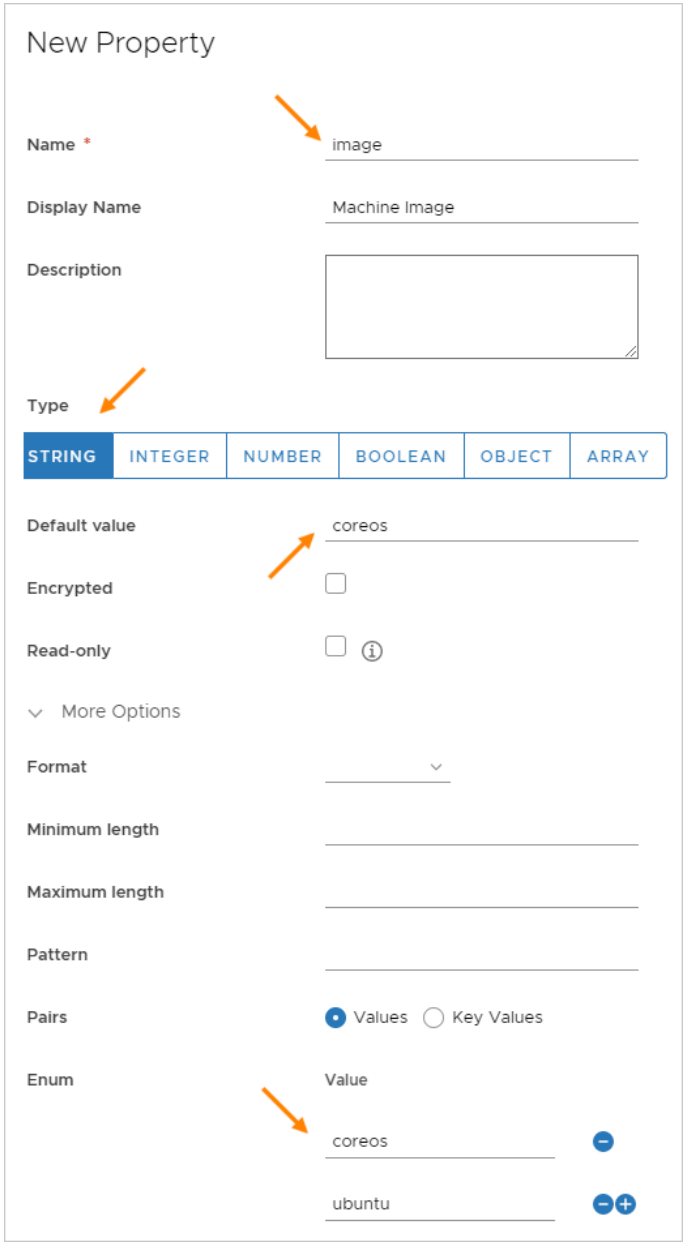
4. To add a property to the group, click **New Property**.

   The panel for adding a new property is very similar to the Inputs tab of the VCF Automation design page code editor.

Name	Free-form name for the individual property. Only letters, numbers, and underscores are permitted.
Display Name	Add an individual property name to appear on the request form.
Type	String, Integer, Number, Boolean (T/F), Object, or Array.
Default Value	Preset value entry that appears in the request form. The presence of default values affects whether input is optional or required. For more information, see User input in requests in VCF Automation for VM Apps.
Encrypted	When selected, obscures the value when entering it into the request form and in the subsequent deployment. Encrypted properties can't have a default value.
Read-only	An uneditable but visible value in the request form. Requires a default.
More Options	Options that vary according to property type. Expand the drop-down, add any additional settings, and click **Create**.

In the following example, the property being added represents the operating system image, and the requesting user can select from two.

Note:
The operating systems shown in the example figure must already be part of the configured VCF Automation infrastructure.



5. Add more properties to the group, and click **Save** when finished.

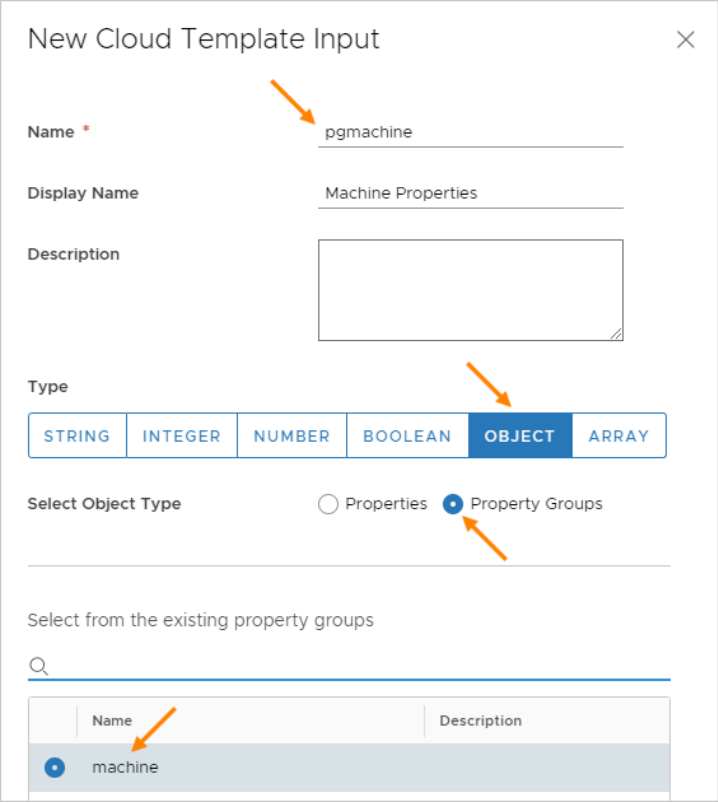


Adding the property group to cloud template inputs

Even for a long list of property inputs, you only need to add the property group to make them all part of the request form.

1. In the cloud template design page, above the editing area on the right, click the **Inputs** tab.
2. Click **New Cloud Template Input**.
3. Name and describe the property group.

Name	Enter something similar to the property group name that you created earlier.
Display Name	Enter the same heading that you created earlier for the entire group of properties, which appears on the request form.
Type	Select **Object**.
Object Type	Select **Property Group**.
Property groups list	Select the property group that you want. Only property groups that are created and available for your project appear. Note that constant property groups don't appear.



4. Click **Create**.

The process creates cloud template inputs code similar to the following example.

```
inputs:
  pgmachine:
    type: object
    title: Machine Properties
    $ref: /ref/property-groups/machine
  pgrequester:
    type: object
    title: Requester Details
    $ref: /ref/property-groups/requesterDetails
```

You may also enter code directly into the VCF Automation design page, and take advantage of the automatic prompting as you type `$ref: /ref/p...` in the code editor.

Binding cloud template resources to the property group

To make use of property group input values, add bindings under the resource.

Depending on what kind of values are in a property group, you might want to reference them individually. You can enter them separately, by property group name and property name.

```
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      image: '${input.pgmachine.image}'
      flavor: '${input.pgmachine.flavor}'
```

You can also quickly add an entire set of values to a resource by referencing an entire property group.

```
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      requester: '${input.pgrequester}'
```

Completed code

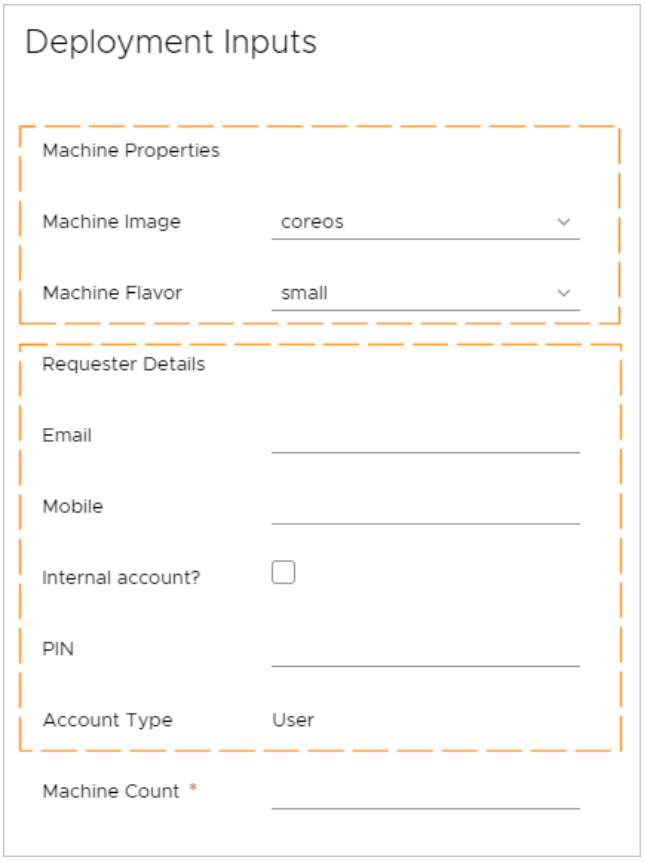
When you're finished with the inputs and resources, the finished code looks similar to the following example.

Code Properties Inputs

```
formatVersion: 1
inputs:
  pgmachine:
    type: object
    title: Machine Properties
    $ref: /ref/property-groups/machine
  pgrequester:
    type: object
    title: Requester Details
    $ref: /ref/property-groups/requesterDetails
  count:
    type: integer
    title: 'Machine Count'
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      image: '${input.pgmachine.image}'
      flavor: '${input.pgmachine.flavor}'
      count: '${input.count}'
      requester: '${input.pgrequester}'

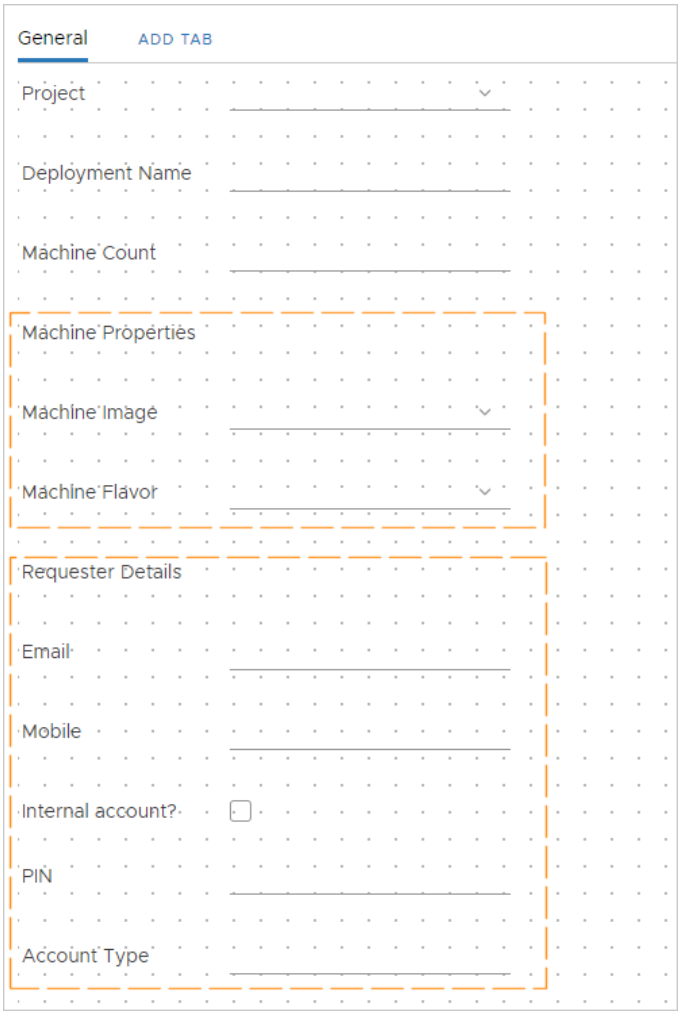
```

Upon deployment request, your property groups appear for the requesting user to complete.



Property groups in the VCF Automation custom form editor

Property groups are not editable like other fields in the custom form designer.



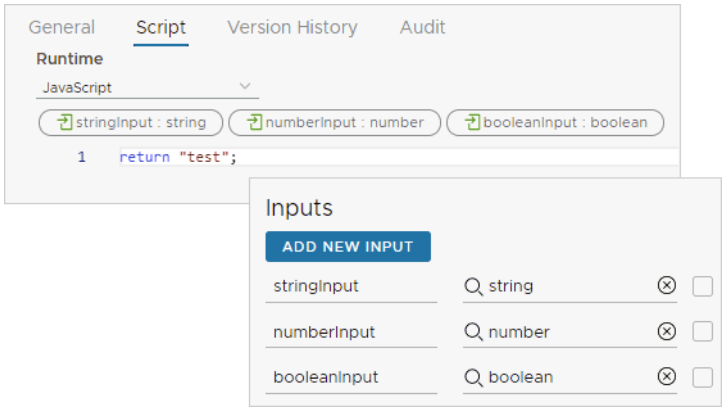
VCF Operations orchestrator actions in an input property group in VCF Automation for VM Apps

In a VCF Automation input property group, you can add dynamic interaction with VCF Operations orchestrator.

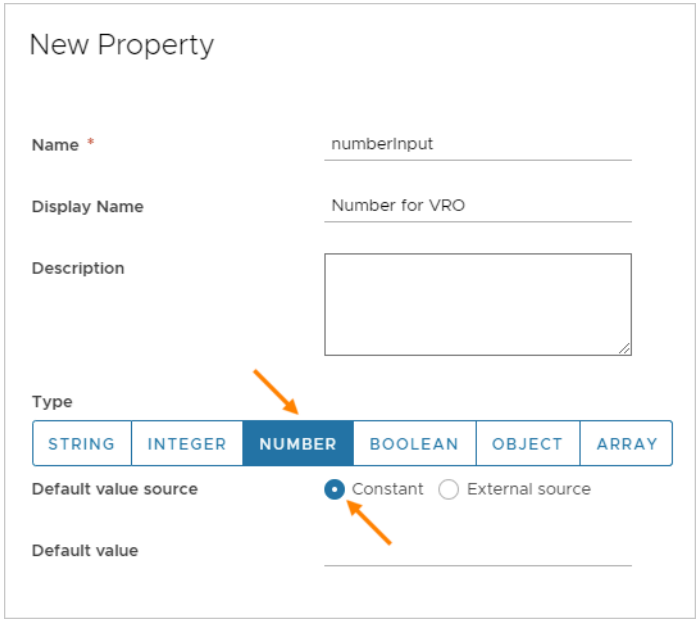
Adding a VCF Operations orchestrator action to an input property group

To add dynamic interaction with VCF Operations orchestrator to an input property group, follow these guidelines.

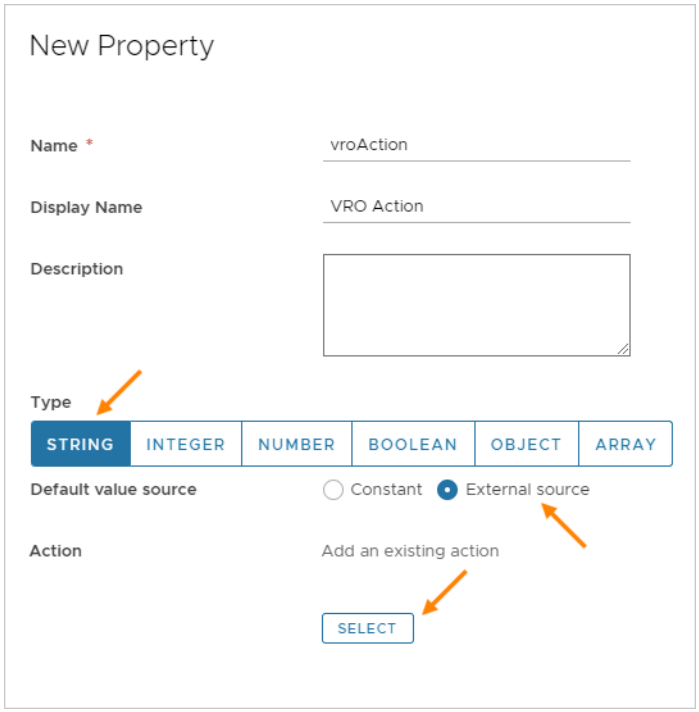
1. In the instance of VCF Operations orchestrator that is embedded with VCF Automation, create an action that does what you want.
   The VCF Operations orchestrator action must only include primitive string, integer, number, and boolean types. VCF Operations orchestrator types are not supported.
   In this simple example, the VCF Operations orchestrator action collects three inputs and returns a hard-coded string.



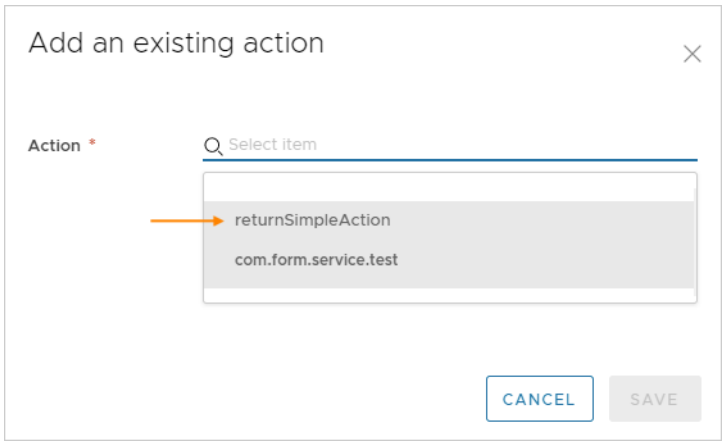
2. In VCF Automation, start the process of creating or editing an input property group. See Input property groups in VCF Automation for VM Apps if necessary.
3. To add the VCF Operations orchestrator action inputs to a property group, add new properties, click the type, and click **Constant**.
   Separately add each VCF Operations orchestrator action input.



4. After adding the inputs, add a new property, click the type, click **External source**, and click **Select**.



5. In **Action**, search for and select the VCF Operations orchestrator action that you created, and click **Save**.



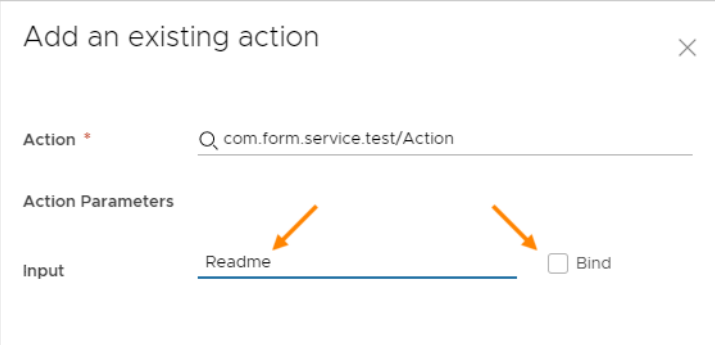
6. Save the property group, and add it to your cloud template. See Input property groups in VCF Automation for VM Apps if necessary.
   When deploying the cloud template, the VCF Operations orchestrator action property group appears in the input form for the requesting user.



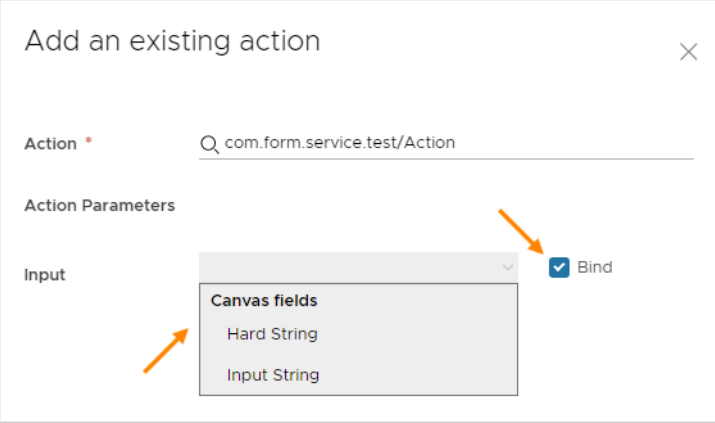
Configurable defaults

To populate the input form with default values, do one of the following when adding the VCF Operations orchestrator action as the external source.

- Manually supply the default property value.
  Clear the **Bind** option, and enter the value.



- Use another property value from the same property group.
  Select the **Bind** option, and select a property from the drop-down list.

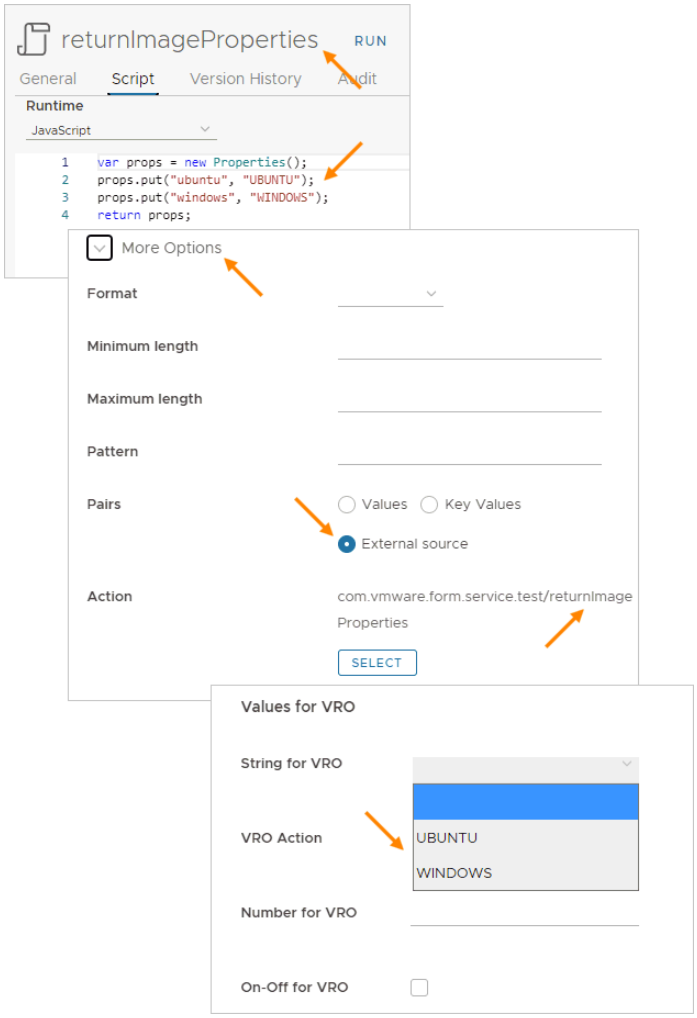


Adding VCF Operations orchestrator enumerated input selections

To create a VCF Operations orchestrator based selection list in an input form, do the following when adding to a property group.

1. In VCF Operations orchestrator, create an action that maps the values that you want for the list.
2. In VCF Automation, when adding a property to the group, expand **More Options**.
3. For **Pairs**, click **External source**, click **Select**, and add the VCF Operations orchestrator action that you created.

   Note:
   If you also create a default value when adding the property, that default must exactly match one of the enumerated values from the VCF Operations orchestrator action.



Constant property groups in VCF Automation for VM Apps

VCF Automation constants allow you to silently apply known key-value pairs to your designs.

How constants work

The key appears in the cloud template code, and the value becomes part of deployments that are based on that cloud template. Constants require the `propgroup` binding under the resource.

The `propgroup` binding is only used with constant property groups, not input property groups.

Secret properties

If you expect to add an secret property to a property group, create the secret property before proceeding. See Secret properties in VCF Automation for VM Apps.

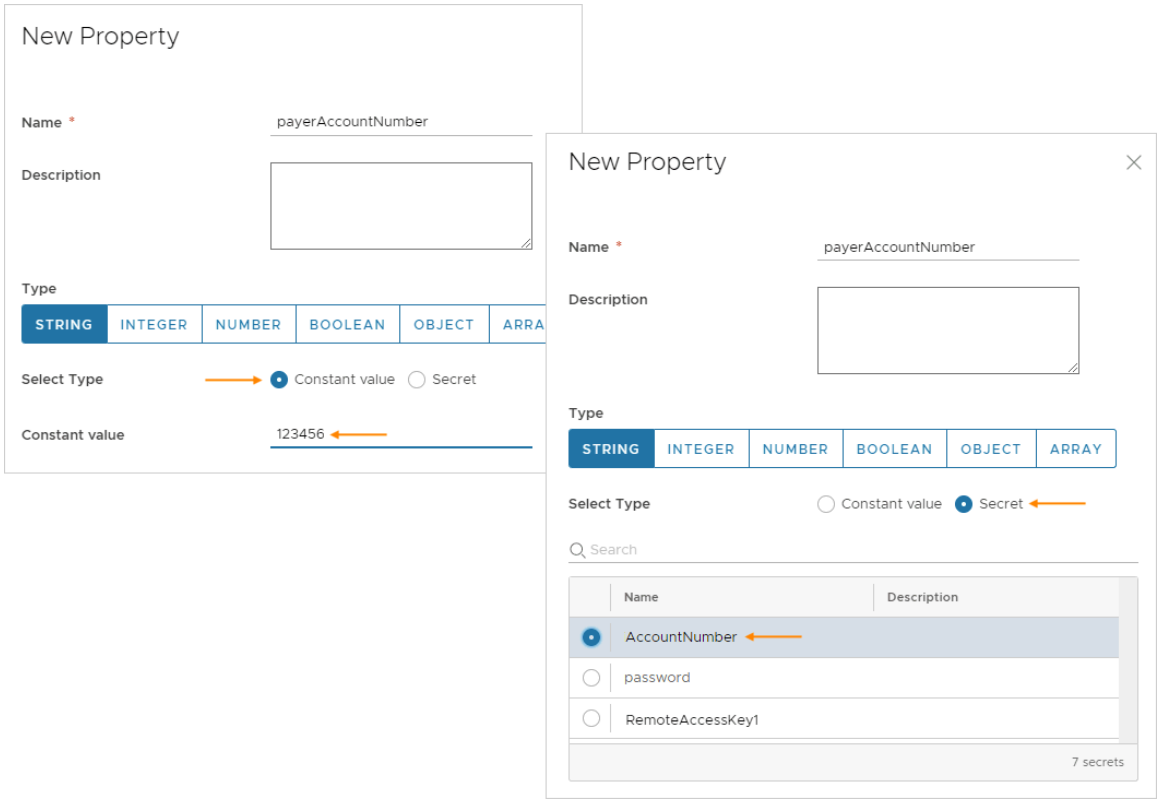
Creating the constant property group

1. Go to **Design** > **Property Groups**, and click **New**.
2. Select **Constant Values**.
3. Name and describe the new property group.

Name	Property group names must be unique within a given organization. Only letters, numbers, and underscores are permitted.
Display Name	Leave blank. No heading appears on the request form.

Description	Explain what this set of constants is for.
Scope	Decide whether an administrator may share the property group with the whole organization. Otherwise, only one project can access the property group. Although you can always add or modify properties in the group, the scope is permanent and can't be changed later. Secrets—If you expect to add a secret property to the property group, you must use single project scope. Secret properties are saved only at the project level.
Project	When the scope is project-only, this project can access the property group.

4. To add a constant property to the group, click **New Property**.
5. Enter a name that acts as the key, and a description.
6. Select a property type.
7. Enter the constant value that you want, and click **Create**.
   - String, integer, and number types use direct entry.
   - For a secret string value, select from the list of secret properties for the project.
   - The boolean type uses a selection box to indicate true.
   - For the object or array type, replace `null` with the value that you want.



8. Add more constants to the group, and click **Save** when finished.



Binding cloud template resources to the property group

To silently use constant values within a resource, add `propgroup` bindings under the resource.

You can quickly add an entire set of constants to a resource by referencing the property group itself.

```
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      payerInfo: '${propgroup.payerDetails}'
```

Alternatively, you can add individual constants from the property group to selected parts of your design.

```
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      payerAccount: '${propgroup.payerDetails.payerAccountNumber}'
      payerCost: '${propgroup.payerDetails.payerCostCenter}'
      payerFed: '${propgroup.payerDetails.payerFederal}'
```

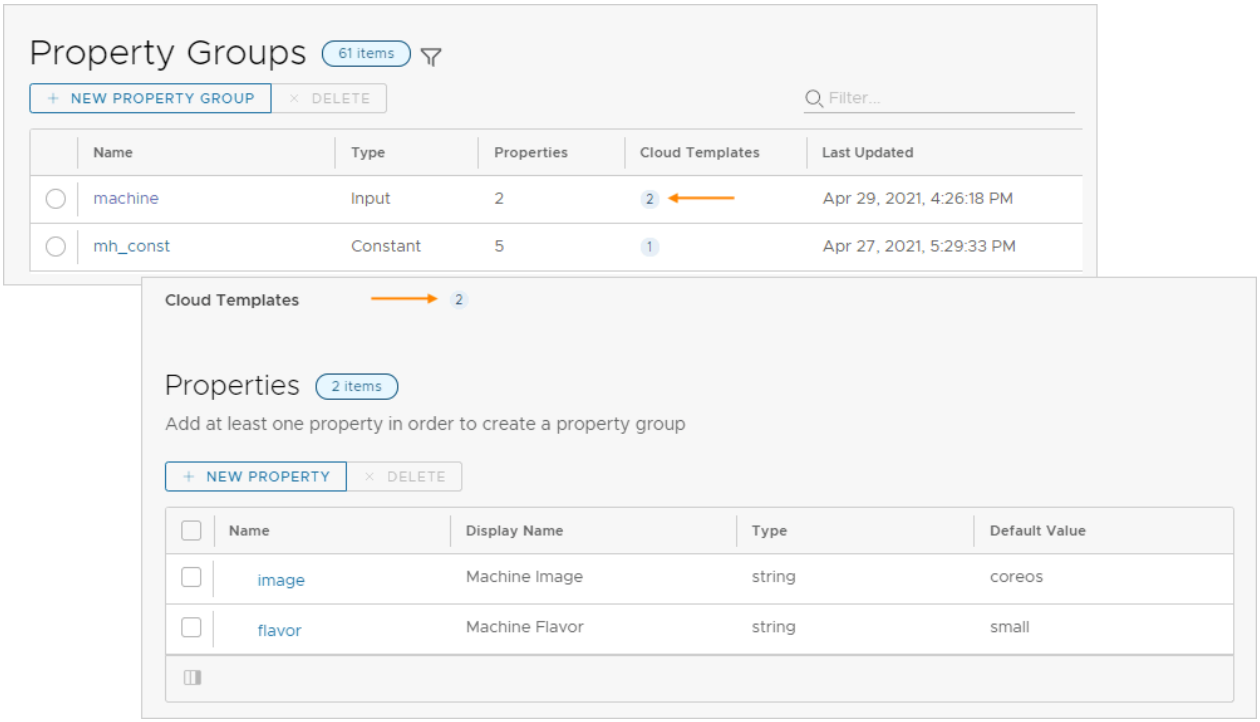
Learn more about property groups in VCF Automation for VM Apps

One VCF Automation property group might be included in many cloud templates, which affects how you need to manage property groups.

Modifying a property group

Changes to an VCF Automation property group affect every cloud template that uses it. When the changed version of the cloud template is released, those changes also affect VCF Automation catalog users.

The property group list and property group editing pages show the number of cloud templates that include the property group. To see which cloud template would be affected by a change, click the number.



Before modifying a property group, make sure that the change is acceptable to everyone who is creating or updating deployments based on the cloud templates listed.

Deleting a property group

Deleting a property group would cause errors in every cloud template that uses it.

You cannot delete a property group until you manually remove it from all of the cloud templates in which it is included. To remove a property group from a cloud template, open the cloud template in the design canvas.

- Input property groups
  Under the Inputs tab, select and remove the property group. Alternatively, use the code editor to delete the associated property group in the `inputs` section of the code.
- Constant property groups
  Use the code editor to delete the associated `propgroup` entry or entries in the `resources` section of the code.

> **Note:**
>
> You cannot delete a property group if it is included in a versioned cloud template. Versioned cloud templates are read-only.

Resource flags for requests in VCF Automation for VM Apps

VCF Automation includes several cloud template settings that adjust how a resource is handled at request time.

Resource flag settings aren't part of the resource object properties schema. For a given resource, you add the flag settings outside of the properties section as shown.

```
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    preventDelete: true
    properties:
      image: coreos
      flavor: small
      attachedDisks:
```

```
      - source: '${resource.Cloud_Volume_1.id}'
  Cloud_Volume_1:
    type: Cloud.Volume
    properties:
      capacityGb: 1
```

Resource Flag	Description
allocatePerInstance	When set to true, resource allocation can be customized for each machine in a cluster. If you're using extensibility, true causes the `compute.allocation.pre` extensibility event topic to run multiple times when deploying more than one cloud machine. The default is false, which allocates resources equally across the cluster, resulting in the same configuration for each machine. In addition, day 2 actions might not be separately possible for individual resources. Per instance allocation allows count.index to correctly apply the configuration for individual machines. For code examples, see Machine and disk clusters in VCF Automation for VM Apps.
createBeforeDelete	Some update actions require that the existing resource be removed and a new one be created. By default, removal is first, which can lead to conditions where the old resource is gone but the new one wasn't created successfully for some reason. Set this flag to true if you need to make sure that the new resource is successfully created before deleting the previous one.
createTimeout	The VCF Automation default timeout for resource allocate, create, and plan requests is 2 hours (2h). In addition, a project administrator can set a custom default timeout for these requests, applicable throughout the project. This flag lets you override any defaults and set the individual timeout for a specific resource operation. See also updateTimeout and deleteTimeout.
deleteTimeout	The VCF Automation default timeout for delete requests is 2 hours (2h). In addition, a project administrator can set a different default timeout for delete requests, applicable throughout the project. This flag lets you override any defaults and set the individual timeout for a specific resource delete operation. See also updateTimeout and createTimeout.
dependsOn	This flag identifies an explicit dependency between resources, where one resource must exist before creating the next one. For more information, see Creating bindings and dependencies between resources in VCF Automation for VM Apps.
dependsOnPreviousInstances	When set to true, create cluster resources sequentially. The default is false, which simultaneously creates all resources in a cluster. For example, sequential creation is useful for database clusters where primary and secondary nodes must be created, but secondary node creation needs configuration settings that connect the node to an existing, primary node.
forceRecreate	Not all update actions require that the existing resource be removed and a new one be created. If you want an update to remove the old resource and create a new one, independent of whether the update would have done so by default, set this flag to true.

Resource Flag	Description
ignoreChanges	Users of a resource might reconfigure it, changing the resource from its deployed state. If you want to perform a deployment update but not overwrite the changed resource with the configuration from the cloud template, set this flag to true.
ignorePropertiesOnUpdate	Users of a resource might customize certain properties, and those properties might be reset to their original cloud template state during an update action. To prevent any properties from being reset by an update action, set this flag to true.
preventDelete	If you need to protect a created resource from accidental deletion during updates, set this flag to true. If a user deletes the deployment, however, the resource is deleted.
recreatePropertiesOnUpdate	Users of a resource might reconfigure properties, changing the resource from its deployed state. During an update, a resource might or might not be recreated. Resources that aren't recreated might remain with properties in changed states. If you want a resource and its properties to be recreated, independent of whether the update would have done so by default, set this flag to true.
updateTimeout	The VCF Automation default timeout for update requests is 2 hours (2h). In addition, a project administrator can set a different default timeout for update requests, applicable throughout the project. This flag lets you override any defaults and set the individual timeout for a specific resource update operation. See also deleteTimeout and createTimeout.

Expressions in VCF Automation for VM Apps

For increased flexibility, you can add expressions to cloud template code in VCF Automation.

How expressions work

VCF Automation expressions use the `${expression}` construct, as shown in the following examples.

> **Note:**
> VCF Automation expressions aren't the same as regular expressions. See the expression syntax for VCF Automation.

The following code samples are pruned to show only the important lines. The entire, unedited cloud template appears at the end.

Examples

At deployment time, allow the user to paste in the encrypted key needed for remote access:

```
inputs:
  sshKey:
    type: string
    maxLength: 500
resources:
  frontend:
    type: Cloud.Machine
    properties:
      remoteAccess:
        authentication: publicPrivateKey
```

```
      sshKey: '${input.sshKey}'
```

For deploying to VMware Cloud on AWS, set the folder name to the required name of *Workload*:

```
inputs:
  environment:
    type: string
    enum:
      - AWS
      - vSphere
      - Azure
      - VMC
      - GCP
    default: vSphere
resources:
  frontend:
    type: Cloud.Machine
    properties:
      folderName: '${input.environment == "VMC" ? "Workload" : ""}'
```

At deployment time, tag the machine with an all-lowercase *env* tag that matches the selected environment:

```
inputs:
  environment:
    type: string
    enum:
      - AWS
      - vSphere
      - Azure
      - VMC
      - GCP
    default: vSphere
resources:
  frontend:
    type: Cloud.Machine
    properties:
      constraints:
        - tag: '${"env:" + to_lower(input.environment)}'
```

Set the number of machines in the front-end cluster to one (small) or two (large). Note that the large cluster is set by process of elimination:

```
inputs:
  envsize:
    type: string
    enum:
      - Small
      - Large
resources:
  frontend:
    type: Cloud.Machine
    properties:
      count: '${input.envsize == "Small" ? 1 : 2}'
```

Attach machines to the same *Default* network by binding to the property found in the network resource:

```
resources:
  frontend:
    type: Cloud.Machine
    properties:
```

```
      networks:
        - network: '${resource.Cloud_Network_1.name}'
  apitier:
    type: Cloud.Machine
    properties:
      networks:
        - network: '${resource.Cloud_Network_1.name}'
  Cloud_Network_1:
    type: Cloud.Network
    properties:
      name: Default
      networkType: existing
```

Encrypt access credentials submitted to the API:

```
resources:
  apitier:
    type: Cloud.Machine
    properties:
      cloudConfig: |
        #cloud-config
        runcmd:
          - export apikey=${base64_encode(input.username:input.password)}
          - curl -i -H 'Accept:application/json' -H 'Authorization:Basic :$apikey' http://example.com
```

Discover the address of the API machine:

```
resources:
  frontend:
    type: Cloud.Machine
    properties:
      cloudConfig: |
        runcmd:
          - echo ${resource.apitier.networks[0].address}
  apitier:
    type: Cloud.Machine
    properties:
      networks:
        - network: '${resource.Cloud_Network_1.name}'
```

Complete cloud template

```
inputs:
  environment:
    type: string
    enum:
      - AWS
      - vSphere
      - Azure
      - VMC
      - GCP
    default: vSphere
  sshKey:
    type: string
    maxLength: 500
  envsize:
    type: string
    enum:
```

```
      - Small
      - Large
resources:
  frontend:
    type: Cloud.Machine
    properties:
      folderName: '${input.environment == "VMC" ? "Workload" : ""}'
      image: ubuntu
      flavor: medium
      count: '${input.envsize == "Small" ? 1 : 2}'
      remoteAccess:
        authentication: publicPrivateKey
        sshKey: '${input.sshKey}'
      cloudConfig: |
        packages:
          - nginx
        runcmd:
          - echo ${resource.apitier.networks[0].address}
      constraints:
        - tag: '${"env:" + to_lower(input.environment)}'
      networks:
        - network: '${resource.Cloud_Network_1.name}'
  apitier:
    type: Cloud.Machine
    properties:
      folderName: '${input.environment == "VMC" ? "Workload" : ""}'
      image: ubuntu
      flavor: small
      cloudConfig: |
        #cloud-config
        runcmd:
          - export apikey=${base64_encode(input.username:input.password)}
          - curl -i -H 'Accept:application/json' -H 'Authorization:Basic :$apikey' http://example.com
      remoteAccess:
        authentication: publicPrivateKey
        sshKey: '${input.sshKey}'
      constraints:
        - tag: '${"env:" + to_lower(input.environment)}'
      networks:
        - network: '${resource.Cloud_Network_1.name}'
  Cloud_Network_1:
    type: Cloud.Network
    properties:
      name: Default
      networkType: existing
      constraints:
        - tag: '${"env:" + to_lower(input.environment)}'
```

Expression syntax in VCF Automation for VM Apps

The expression syntax exposes all of the available capabilities of expressions in VCF Automation templates.

> **Note:**
> VCF Automation expressions aren't the same as regular expressions (regex).

The following syntax is only partly represented in the examples shown in Expressions in VCF Automation for VM Apps.

Literals

The following literals are supported:

- Boolean (true or false)
- Integer
- Floating point
- String
  Backslash escapes double quote, single quote, and backslash itself:
  `"` is escaped as `\"`
  `'` is escaped as `\'`
  `\` is escaped as `\\`
  Quotes only need to be escaped inside a string enclosed with the same type of quote, as shown in the following example.
  ```
  "I am a \"double quoted\" string inside \"double quotes\"."
  ```
- Null

Environment variables

Environment names:

- orgId
- projectId
- projectName
- deploymentId
- deploymentName
- blueprintId
- blueprintVersion
- blueprintName
- requestedBy (user)
- requestedAt (time)

Syntax:

```
env.ENV_NAME
```

Example:

```
${env.blueprintId}
```

Resource variables

Resource variables let you bind to resource properties from other resources.

Syntax:

```
resource.RESOURCE_NAME.PROPERTY_NAME
```

Resource names cannot contain dashes or dots. Underscores are allowed.

Examples:

- `${resource.db.id}`
- `${resource.db.networks[0].address}`
- `${resource.app.id}` (Return the string for non-clustered resources, where count isn't specified. Return the array for clustered resources.)
- `${resource.app[0].id}` (Return the first entry for clustered resources.)

Resource self variables

Resource self variables are allowed only for resources supporting the allocation phase. Resource self variables are only available (or only have a value set) after the allocation phase is complete.

Syntax:

```
self.property_name
```

Example:

`${self.address}` (Return the address assigned during the allocation phase.)

Note that for a resource named `resource_x`, `self.property_name` and `resource.resource_x.property_name` are the same and are both considered self-references.

Conditions

Syntax:

- Equality operators are == and `!=`.
- Relational operators are <><= and >=.
- Logical operators are `&&||` and `!`.
- Conditionals use the pattern:
  condition-expression?*true-expression*:*false-expression*

Examples:

```
${input.count < 5 && input.size == 'small'}
```

```
${input.count < 2 ? "small":"large"}
```

Cluster count index

Syntax:

```
count.index
```

Examples:

- Return the node type for clustered resources:
  `${count.index == 0 ? "primary":"secondary"}`
- Set the size of each disk during allocation:

```
inputs:
  disks:
    type: array
    minItems: 0
    maxItems: 12
    items:
      type: object
      properties:
        size:
          type: integer
          title: Size (GB)
          minSize: 1
          maxSize: 2048
resources:
  Cloud_vSphere_Disk_1:
    type: Cloud.vSphere.Disk
    allocatePerInstance: true
    properties:
      capacityGb: '${input.disks[count.index].size}'
```

```
count: '${length(input.disks)}'
```

- For more examples, see Machine and disk clusters in VCF Automation for VM Apps.

Arithmetic operators

Syntax:

Operators are `+-/*` and `%`.

Example:

```
${(input.count + 5) * 2}
```

String concatenation

Syntax:

`${'ABC' + 'DEF'}` evaluates to `ABCDEF`.

Operators [ ] and .

The expression follows ECMAScript in unifying the treatment of the [ ] and . operators.

So, `expr.identifier` is equivalent to `expr["identifier"]`. The identifier is used to construct a literal whose value is the identifier, and then the [ ] operator is used with that value.

Example:

```
${resource.app.networks[0].address}
```

In addition, when a property includes a space, delimit with square brackets and double quotes instead of using dot notation.

Incorrect:

```
input.operating system
```

Correct:

```
input["operating system"]
```

Construction of map

Syntax:

```
${{'key1':'value1', 'key2':input.key2}}
```

Construction of array

Syntax:

```
${['key1','key2']}
```

Example:

```
${[1,2,3]}
```

Functions

Syntax:

```
${function(arguments...)}
```

Example:

```
${to_lower(resource.app.name)}
```

Table 1829: Functions

Function	Description
abs(number)	Absolute number value
avg(array)	Return average of all values from array of numbers
base64_decode(string)	Return decoded base64 value
base64_encode(string)	Return base64 encoded value
ceil(number)	Returns the smallest (closest to negative infinity) value that is greater than or equal to the argument and is equal to a mathematical integer
contains(array, value)	Check if array contains a value
contains(string, value)	Check if string contains a value
digest(value, type)	Return digest of value using supported type (md5, sha1, sha256, sha384, sha512)
ends_with(subject, suffix)	Check if subject string ends with suffix string

filter_by(array, filter)	Return only the array entries that pass the filter operation
	`filter_by([1,2,3,4], x => x >= 2 && x <= 3)`
	returns `[2, 3]`
	`filter_by({'key1':1, 'key2':2}, (k,v) => v != 1)`
	returns `[{"key2": 2}]`

floor(number)	Returns the largest (closest to positive infinity) value that is less than or equal to the argument and is equal to a mathematical integer
format(format, values...)	Return a formatted string using Java Class Formatter format and values.
from_json(string)	Parse json string
from_yaml(yamlText, decode_binary=true/false)	Convert YAML into JSON format.
join(array, delim)	Join array of strings with a delimiter and return a string
json_path(value, path)	Evaluate path against value using XPath for JSON.
keys(map)	Return keys of map
length(array)	Return array length
length(string)	Return string length

map_by(array, operation)	Return each array entry with an operation applied to it
	`map_by([1,2], x => x * 10)`
	returns `[10, 20]`
	`map_by([1,2], x => to_string(x))`
	returns `["1", "2"]`
	`map_by({'key1':1, 'key2':2}, (k,v) => {k:v*10})`
	returns `[{"key1":10},{"key2":20}]`

Function	Description

map_to_object(array, keyname)	Return an array of key:value pairs of the specified key name paired with values from another array
	`map_to_object(resource.Disk[*].id, "source")`
	returns an array of key:value pairs that has a key field called source paired with disk ID strings.
	Note that `map_by(resource.Disk[*].id, id => {'source':id})` returns the same result.
	If multiple arguments are passed to the map_to_object function, then it returns an array of nested objects, where the keys are part of the nested object, and the values are set on the innermost key.
	`map_to_object(resource.Cloud_vSphere_Machine_1[*].address, "ip", "addr")`
	returns an array of objects in the following format:
	`{ "ip" : { "addr" : <value> } }`
	The value of the innermost key is paired with the address of a machine in the Cloud_vSphere_Machine_1 cluster.
	There is one object per machine in the cluster. If there there are three machines in the cluster, then there are three entries in the object:
	`{ "ip": { "addr": <value> } , "ip": { "addr": <value> } , "ip": { "addr": <value> } }`

matches(string, regex)	Check if string matches a regex expression
max(array)	Return maximum value from array of numbers
merge(map, map)	Return a merged map
min(array)	Return minimum value from array of numbers
not_null(array)	Return the first entry which is not null
now()	Return current time in ISO-8601 format
range(start, stop)	Return a series of numbers in increments of 1 that begins with the start number and ends just before the stop number
replace(string, target, replacement)	Replace string containing target string with target string
reverse(array)	Reverse entries of array
slice(array, begin, end)	Return slice of array from begin index to end index
split(string, delim)	Split string with a delimiter and return array of strings
starts_with(subject, prefix)	Check if subject string starts with prefix string
substring(string, begin, end)	Return substring of string from begin index until end index

Function	Description
sum(array)	Return sum of all values from array of numbers
to_json(value)	Serialize value as json string
to_lower(str)	Convert string to lower case
to_number(string)	Parse string as number
to_string(value)	Return string representation of the value
to_upper(str)	Convert string to upper case
to_yaml(jsonText):	Convert JSON back to YAML format.
trim(string)	Remove leading and trailing spaces
url_encode(string)	Encode string using url encoding specification
uuid()	Return randomly generated UUID
values(map)	Return values of map

Troubleshooting

The YAML language uses a colon and space (": ") as the separator between key and value in key-value pairs. Expression syntax depends on YAML, so a space after a colon can sometimes cause an expression to fail.

For example, the space between `"win" :` and `"lin"` in the following expression causes a failure.

```
${contains(input.image,"Windows") == true ? "win" : "lin"}
```

The working expression omits the space.

```
${contains(input.image,"Windows") == true ? "win" :"lin"}
```

If an expression continues to fail, try enclosing the entire expression in tick marks as shown.

```
ezOS: '${contains(input.image,"Windows") == true ? "win" :"lin"}'
```

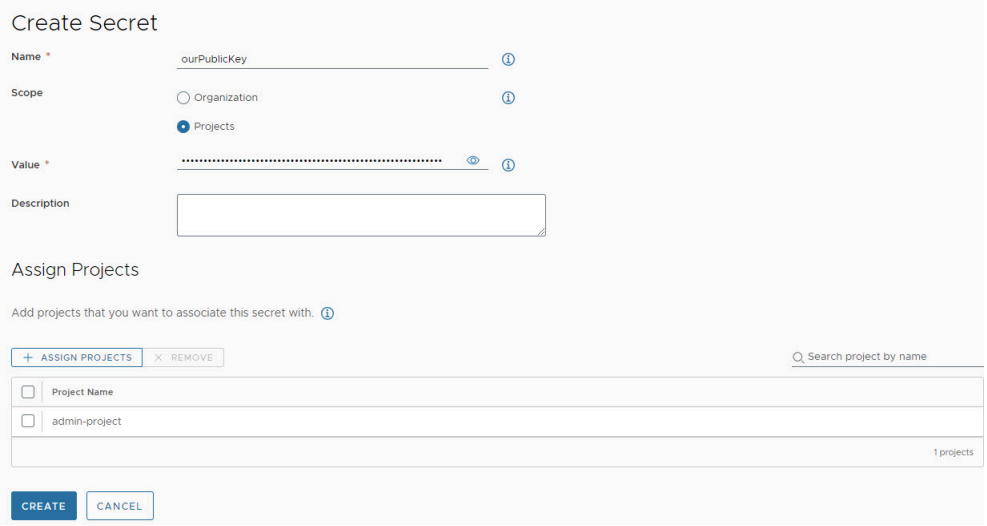
Secret properties in VCF Automation for VM Apps

A secret VCF Automation property is a reusable, encrypted value that project users may add to their cloud template designs.

Secure access keys and credentials are typical examples of secret properties. Once created and saved, a secret property value can never be unencrypted or read.

Creating a secret property

1. Log in to VCF Automation with project administrator role privileges.
2. Go to **Infrastructure** > **Administration** > **Secrets**, and click **New Secret**.
3. Enter a unique property name for the secret, without spaces or special characters.
   The name is the visible identifier for the secret.
4. Scope the secret to the entire organization or to specific projects.
   To assign the secret to one or more projects, click **Assign Projects**, select your projects, and click **Add**.
   You can't associate a secret with the same project twice. You also can't associate two secrets with the same name with the entire organization.
5. Enter the secret value.
   When typing, the value is obscured by default, which protects it if the screen is shared.
   If needed, you can click the eye symbol to reveal and verify a value. After it is saved, a secret value becomes encrypted in the database and can never be re-exposed.
6. Optionally, enter a longer description of the secret property.
7. Click **Create**.



Adding a secret property to a cloud template

Project users may add a secret property as a binding in cloud template code.

Note that starting to type the `'${secret.` characters reveals a selection list of secrets that have been created for the project.

If two secrets with the same name but different values exist at the project level and at the organization level, the project secret takes precedence. The organization secret is not available for selection in the cloud template.

```
type: Cloud.Machine
properties:
  name: ourvm
  image: mint20
  flavor: small
  remoteAccess:
    authentication: publicPrivateKey
    sshKey: '${secret.ourPublicKey}'
    username: root
```

Remote access to a deployment in VCF Automation for VM Apps

To remotely access a machine that VCF Automation has deployed, you add properties, before deployment, to the cloud template for that machine.

For remote access, you can configure one of the following authentication options.

> **Note:**
> In cases where keys need to be copied, you might also create a cloudConfig section in the cloud template, to automatically copy the keys upon provisioning. The specifics aren't documented here, but Machine initialization in VCF Automation for VM Apps provides general information about cloudConfig.

Generate a key pair at provisioning time

If you don't have your own public-private key pair for remote access authentication, you can have VCF Automation generate a key pair.

Use the following code as a guideline.

1. In VCF Automation, before provisioning, add `remoteAccess` properties to the cloud template as shown in the example.
   The username is optional. If you omit it, the system generates a random ID as the username.

Example:

```
type: Cloud.Machine
properties:
  name: our-vm2
  image: Linux18
  flavor: small
  remoteAccess:
     authentication: generatedPublicPrivatekey
     username: testuser
```

2. In VCF Automation, provision the machine from its cloud template, and bring it to a started-up state.
   The provisioning process generates the keys.
3. Locate the key name in the **Resources** > **Deployments** > **Topology** properties.
4. Use the cloud provider interface, such as the vSphere client, to access the provisioned machine command line.
5. Grant read permission to the private key.
   `chmod 600 key-name`
6. Go to the VCF Automation deployment, select the machine, and click **Actions** > **Get Private Key**.
7. Copy the private key file to your local machine.
   A typical local file path is `/home/username/.ssh/key-name`.
8. Open a remote SSH session, and connect to the provisioned machine.
   `ssh -i key-nameuser-name@machine-ip`

Supply your own public-private key pair

Many enterprises create and distribute their own public-private key pairs for authentication.

Use the following code as a guideline.

1. In your local environment, obtain or generate your public-private key pair.
   For now, just generate and save the keys locally.
2. In VCF Automation, before provisioning, add `remoteAccess` properties to the cloud template as shown in the example.
   The sshKey includes the long alphanumeric found within the public key file *key-name*.pub.
   The username is optional and gets created for you to log in with. If you omit it, the system generates a random ID as the username.
   Example:

```
type: Cloud.Machine
properties:
  name: our-vm1
  image: Linux18
  flavor: small
  remoteAccess:
     authentication: publicPrivateKey
     sshKey: ssh-rsa Iq+5aQgBP3ZNT4o1baP5Ii+dstIcowRRkyobbfpA1mj9tslf
qGxvU66PX9IeZax5hZvNWFgjw6ag+ZlzndOLhVdVoW49f274/mIRild7UUW...
     username: testuser
```

3. In VCF Automation, provision the machine from its cloud template, and bring it to a started-up state.
4. Using the cloud vendor client, access the provisioned machine.
5. Add the public key file to the home folder on the machine. Use the key that you specified in remoteAccess.sshKey.
6. Verify that the private key file counterpart is present on your local machine.
   The key is typically `/home/username/.ssh/key-name` with no .pub extension.
7. Open a remote SSH session, and connect to the provisioned machine.
   `ssh -i key-nameuser-name@machine-ip`

Supply an AWS key pair

By adding an AWS key pair name to the cloud template, you can remotely access a machine that VCF Automation deploys to AWS.

Be aware that AWS key pairs are region specific. If you provision workloads into us-east-1, the key pair must exist in us-east-1.

Use the following code as a guideline. This option works for AWS cloud zones only.

```
type: Cloud.Machine
properties:
  image: Ubuntu
  flavor: small
  remoteAccess:
     authentication: keyPairName
     keyPair: cas-test
constraints:
  - tag: 'cloud:aws'
```

Supply a username and password

By adding a username and password to the cloud template, you can have simple remote access to a machine that VCF Automation deploys.

Although it is less secure, logging in remotely with a username and password might be all that your situation requires. Be aware that some cloud vendors or configurations might not support this less secure option.

1. In VCF Automation, before provisioning, add `remoteAccess` properties to the cloud template as shown in the example.
   Set the username and password to the account that you expect to log in with.
   Example:
   ```
   type: Cloud.Machine
   properties:
     name: our-vm3
     image: Linux18
     flavor: small
     remoteAccess:
        authentication: usernamePassword
        username: testuser
        password: admin123
   ```
2. In VCF Automation, provision the machine from its cloud template, and bring it to a started-up state.
3. Go to your cloud vendor's interface, and access the provisioned machine.
4. On the provisioned machine, create or enable the account.
5. From your local machine, open a remote session to the provisioned machine IP address or FQDN, and log in with the username and password as usual.

SCSI disk placement with VCF Automation for VM Apps

To manage a SCSI disk, you must specify and know its SCSI controller and logical unit number (LUN). For a vSphere disk object, you can use VCF Automation to assign both values in the cloud template.

The ability to use different SCSI controllers is important for performance and is required for some deployment types, such as Oracle Real Application Clusters (RAC).

> **Note:**
> VCF Automation only processes virtual devices configured with the SCSI controller.

SCSI controller and LUN disk properties

To assign a SCSI controller and LUN, add the following cloud template properties:

`SCSIController`

`unitNumber`

You also have the option to omit the properties, in which case assignment follows a predictable default. VCF Automation no longer deploys SCSI disks in random order, which made them difficult to manage.

SCSI controllers and disks are numbered in order, with zero being first. Each SCSI controller can support SCSI disks with unit numbers 0–15.

> **Note:**
>
> If the SCSI controller type is VMware Paravirtual, VCF Automation 8.18.1 or later connected to vCenter 6.7 or later supports up to 64 disks per controller with unit numbers 0-64.

Option 1: Set both SCSI controller and unit number

You may fully specify both properties as shown in the following example. If so, assignment of the SCSI controller and unit number match the values that you enter.

```
resources:
 Cloud_vSphere_Machine_1:
   type: Cloud.vSphere.Machine
   properties:
     image: centos
     cpuCount: 1
     totalMemoryMB: 1024
     attachedDisks:
       - source: '${resource.Cloud_vSphere_Disk_1.id}'
       - source: '${resource.Cloud_vSphere_Disk_2.id}'
       - source: '${resource.Cloud_vSphere_Disk_3.id}'
 Cloud_vSphere_Disk_1:
   type: Cloud.vSphere.Disk
   properties:
     capacityGb: 1
     SCSIController: SCSI_Controller_2unitNumber: 0
 Cloud_vSphere_Disk_2:
   type: Cloud.vSphere.Disk
   properties:
     capacityGb: 1
     SCSIController: SCSI_Controller_2unitNumber: 1
 Cloud_vSphere_Disk_3:
   type: Cloud.vSphere.Disk
   properties:
     capacityGb: 1
     SCSIController: SCSI_Controller_3unitNumber: 4
```

Option 2: Set only the SCSI controller

You may specify the SCSI controller and omit the unit number. In this case, assignment of the SCSI controller matches the value you enter. The unit number is set to the first available unit number under that controller.

```
resources:
 Cloud_vSphere_Machine_1:
   type: Cloud.vSphere.Machine
   properties:
     image: centos
```

```
      cpuCount: 1
      totalMemoryMB: 1024
      attachedDisks:
        - source: '${resource.Cloud_vSphere_Disk_1.id}'
        - source: '${resource.Cloud_vSphere_Disk_2.id}'
        - source: '${resource.Cloud_vSphere_Disk_3.id}'
  Cloud_vSphere_Disk_1:
    type: Cloud.vSphere.Disk
    properties:
      capacityGb: 1
      SCSIController: SCSI_Controller_0
  Cloud_vSphere_Disk_2:
    type: Cloud.vSphere.Disk
    properties:
      capacityGb: 1
      SCSIController: SCSI_Controller_0
  Cloud_vSphere_Disk_3:
    type: Cloud.vSphere.Disk
    properties:
      capacityGb: 1
      SCSIController: SCSI_Controller_1
```

Option 3: Omit both properties

You may omit the SCSI controller and unit number. In this case, assignment is set to the first available SCSI controller, and the first available unit number under that controller.

```
 resources:
  Cloud_vSphere_Machine_1:
    type: Cloud.vSphere.Machine
    properties:
      image: centos
      cpuCount: 1
      totalMemoryMB: 1024
      attachedDisks:
        - source: '${resource.Cloud_vSphere_Disk_1.id}'
        - source: '${resource.Cloud_vSphere_Disk_2.id}'
        - source: '${resource.Cloud_vSphere_Disk_3.id}'
  Cloud_vSphere_Disk_1:
    type: Cloud.vSphere.Disk
    properties:
      capacityGb: 1
  Cloud_vSphere_Disk_2:
    type: Cloud.vSphere.Disk
    properties:
      capacityGb: 1
  Cloud_vSphere_Disk_3:
    type: Cloud.vSphere.Disk
    properties:
      capacityGb: 1
```

Not an option: LUN only

You cannot omit the SCSI controller and specify only a unit number. Doing so might result in a deployment where multiple SCSI controllers have a disk of that number but, for management purposes, you won't know which disk is which.

Using inputs to set the SCSI controller and LUN

To make the design more dynamic, use inputs so that the user may specify which SCSI controller and unit number at request or update time.

```
inputs:
 diskProperties:
   type: array
   minItems: 1
   maxItems: 10
   items:
     type: object
     properties:
       size:
         type: integer
       SCSIController:
         type: string
         title: SCSI Controller
         enum:
           - SCSI_Controller_0
           - SCSI_Controller_1
           - SCSI_Controller_2
           - SCSI_Controller_3
       unitNumber:
         type: integer
         title: Unit Number

resources:
 app:
   type: Cloud.vSphere.Machine
   allocatePerInstance: true
   properties:
     flavor: small
     image: centos
     attachedDisks: '${map_to_object(slice(resource.disk[*].id, 0, 4), ''source'')}'
 disk:
   type: Cloud.vSphere.Disk
   allocatePerInstance: true
   properties:
     capacityGb: '${input.diskProperties[count.index].size}'
     SCSIController: '${input.diskProperties[count.index].SCSIController}'
     unitNumber: '${input.diskProperties[count.index].unitNumber}'
     count: ${length(input.diskProperties)}
```



Machine initialization in VCF Automation for VM Apps

You can apply machine initialization in VCF Automation by running commands directly or, if deploying to vSphere-based cloud zones, through customization specifications.

How commands and customization specifications work

- Commands
  A cloudConfig section in your cloud template code holds the commands that you want to run.
- Customization specifications
  A property in your cloud template code references a vSphere customization specification by name.

Commands and customization specifications might not mix

When deploying to vSphere, proceed carefully if you attempt to combine cloudConfig and customization specification initialization. They aren't formally compatible and might produce inconsistent or unwanted results when used together.

For an example of how commands and customization specifications interact, see vSphere static IP addresses in VCF Automation for VM Apps.

vSphere customization specifications in VCF Automation templates

When deploying to vSphere based cloud zones in VCF Automation, customization specifications can apply guest operating system settings at deployment time.

Enabling the customization specification

The customization specification must exist in vSphere, at the target that you deploy to.

Edit the cloud template code directly. The following example points to an `automation-assembler-linux` customization specification for a WordPress host on vSphere.

```
resources:
  WebTier:
    type: Cloud.vSphere.Machine
    properties:
        name: wordpress
        cpuCount: 2
        totalMemoryMB: 1024
        imageRef: 'Template: ubuntu-18.04'
        customizationSpec: 'automation-assembler-linux'
        folderName: '/Datacenters/Datacenter/vm/deployments'
```

Whether to use customization specifications or cloudConfig commands

If you want the provisioning experience to match what you are currently doing in vSphere, continuing to use customization specifications might be the best approach. However, to expand to hybrid or multiple cloud provisioning, a more neutral approach is cloudConfig initialization commands.

For more about cloudConfig sections in cloud templates, see Configuration commands in VCF Automation templates.

Commands and customization specifications might not mix

When deploying to vSphere, proceed carefully if you attempt to combine embedded cloudConfig command and customization specification initialization. They aren't formally compatible and might produce inconsistent or unwanted results when used together.

For an example of how commands and customization specifications interact, see vSphere static IP addresses in VCF Automation for VM Apps.

Configuration commands in VCF Automation templates

You can add a cloudConfig section to VCF Automation template code, in which you add machine initialization commands that run at deployment time.

cloudConfig command formats

- Linux—initialization commands follow the open cloud-init standard.
- Windows—initialization commands use Cloudbase-init.

Linux cloud-init and Windows Cloudbase-init don't share the same syntax. A cloudConfig section for one operating system won't work in a machine image of the other operating system.

What cloudConfig commands can do

You use initialization commands to automate the application of data or settings at instance creation time, which can customize users, permissions, installations, or any other command-based operations. Examples include:

- Setting a hostname
- Generating and setting up SSH private keys
- Installing packages

Where cloudConfig commands can be added

You can add a cloudConfig section to cloud template code, but you can also add one to a machine image in advance, when configuring infrastructure. Then, all cloud templates that reference the source image get the same initialization.

You might have an image map and a cloud template where both contain initialization commands. At deployment time, the commands merge, and VCF Automation runs the consolidated commands.

When the same command appears in both places but includes different parameters, only the image map command is run.

See Learn more about image mappings in VCF Automation for VM Apps for additional details.

Syntax in cloudConfig commands

Faulty cloudConfig commands can result in a resource that isn't correctly configured or behaves unpredictably.

To cancel a deployment when there is a syntax error in `#cloud-config` statements, add the following property.

```
      cloudConfigSettings:
        deploymentFailOnCloudConfigRuntimeError: true
```

In the following cloud template, the `- mkdir` command isn't on a new line under `runcmd:` as required, so the directory will never be created. The `deploymentFailOnCloudConfigRuntimeError: true` property fails the deployment because of the error.

```
formatVersion: 1
inputs: {}
resources:
  Cloud_vSphere_Machine_1:
    type: Cloud.vSphere.Machine
    properties:
      image: img1
      cpuCount: 1
      totalMemoryMB: 1024
      cloudConfigSettings:
        deploymentFailOnCloudConfigRuntimeError: true
      cloudConfig: |
        #cloud-config
        runcmd:- mkdir -p /tmp/test-dir
```

If you omit the property or set it to `false`, deployment continues even if cloudConfig commands fail.

In addition, the property requires the `#cloud-config` line. If you omit the line, deployment continues regardless of the property setting.

Correct:

```
cloudConfigSettings:
  deploymentFailOnCloudConfigRuntimeError: true
cloudConfig: |
  #cloud-config
  runcmd:
   - mkdir -p /tmp/test-dir
```

Incorrect:

```
cloudConfigSettings:
  deploymentFailOnCloudConfigRuntimeError: true
cloudConfig: |
  runcmd:
   - mkdir -p /tmp/test-dir
```

Example cloudConfig commands

The following example cloudConfig section is taken from the WordPress use case cloud template code for the Linux-based MySQL server.

Note:

To ensure correct interpretation of commands, always include the pipe character `cloudConfig: |` as shown.

```
cloudConfig: |
  #cloud-config
  repo_update: true
  repo_upgrade: all
  packages:
   - apache2
   - php
   - php-mysql
   - libapache2-mod-php
   - php-mcrypt
   - mysql-client
  runcmd:
   - mkdir -p /var/www/html/mywordpresssite && cd /var/www/html && wget https://wordpress.org/
latest.tar.gz && tar -xzf /var/www/html/latest.tar.gz -C /var/www/html/mywordpresssite --strip-components 1
   - i=0; while [ $i -le 5 ]; do mysql --connect-timeout=3 -h ${DBTier.networks[0].address} -u root -
pmysqlpassword -e "SHOW STATUS;" && break || sleep 15; i=$((i+1)); done
   - mysql -u root -pmysqlpassword -h ${DBTier.networks[0].address} -e "create database wordpress_blog;"
   - mv /var/www/html/mywordpresssite/wp-config-sample.php /var/www/html/mywordpresssite/wp-config.php
   - sed -i -e s/"define( 'DB_NAME', 'database_name_here' );"/"define( 'DB_NAME',
 'wordpress_blog' );"/ /var/www/html/mywordpresssite/wp-config.php && sed -i -e s/"define( 'DB_USER',
 'username_here' );"/"define( 'DB_USER', 'root' );"/ /var/www/html/mywordpresssite/wp-config.php && sed -
i -e s/"define( 'DB_PASSWORD', 'password_here' );"/"define( 'DB_PASSWORD', 'mysqlpassword' );"/ /var/www/
html/mywordpresssite/wp-config.php && sed -i -e s/"define( 'DB_HOST', 'localhost' );"/"define( 'DB_HOST',
 '${DBTier.networks[0].address}' );"/ /var/www/html/mywordpresssite/wp-config.php
   - service apache2 reload
```

If a cloud-init script behaves unexpectedly, check the captured console output in `/var/log/cloud-init-output.log` when troubleshooting. For more about cloud-init, see the cloud-init documentation.

Commands and customization specifications might not mix

When deploying to vSphere, proceed carefully if you attempt to combine embedded cloudConfig command and customization specification initialization. They aren't formally compatible and might produce inconsistent or unwanted results when used together.

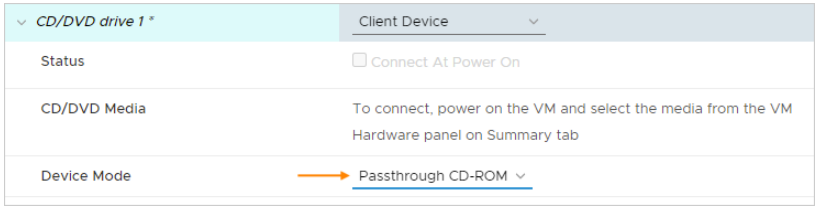
For an example of how commands and customization specifications interact, see vSphere static IP addresses in VCF Automation for VM Apps.

vSphere templates for initialization in VCF Automation for VM Apps

When your VCF Automation template deploys an image based on a vSphere template, the vSphere template must be configured in advance to support cloud-init.

To configure a vSphere template to support cloud-init, take the following steps.

1. On the virtual machine that will become the template, install cloud-init.
   For example, use `yum` to install cloud-init on CentOs, or `apt-get` to install on Ubuntu.
2. Set the CD-ROM of the virtual machine to passthrough mode.



3. From the guest operating system command line, run `cloud-init clean`.

   Note:
   When `cloud-init clean` finishes, do not modify the virtual machine any further.
4. Shut down the virtual machine and convert it to a template.

vSphere static IP addresses in VCF Automation for VM Apps

When deploying to vSphere in VCF Automation, you can assign a static IP address but must take care not to introduce conflicts between cloudConfig initialization commands and customization specifications.

Sample designs

The following designs safely apply a static IP address without any conflict between cloud template initialization commands and customization specifications. All contain the `assignment: static` network setting.

Design	Sample Cloud Template Code

Assign a static IP address to a Linux machine that has no cloud-init code	`resources:`
	`  wpnet:`
	`  type: Cloud.Network`
	`  properties:`
	`  name: wpnet`
	`  networkType: public`
	`  constraints:`
	`  - tag: sqa`
	`  DBTier:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  flavor: small`
	`  image: linux-template`
	`  networks:`
	`  - name: '${wpnet.name}'`
	`  assignment: static`
	`  network: '${resource.wpnet.id}'`

Assign a static IP address to a Linux machine with cloud-init code that doesn't contain network assignment commands.
NOTE: The vSphere customization spec is applied whether you set the customizeGuestOs property to true or omit the customizeGuestOs property.
	Ubuntu sample
	`resources:`
	`  wpnet:`
	`  type: Cloud.Network`
	`  properties:`
	`  name: wpnet`
	`  networkType: public`
	`  constraints:`
	`  - tag: sqa`
	`  DBTier:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  flavor: small`
	`  image: ubuntu-template`
	`  customizeGuestOs: true`
	`  cloudConfig: |`
	`  #cloud-config`
	`  ssh_pwauth: yes`
	`  chpasswd:`
	`  list: |`
	`  root:Pa$$w0rd`
	`  expire: false`
	`  write_files:`
	`  - path: /tmpFile.txt`
	`  content: |`
	`  ${resource.wpnet.dns}`
	`  runcmd:`
	`  - hostnamectl set-hostname --pretty ${self.resourceName}`
	`  - touch /etc/cloud/cloud-init.disabled`
	`  networks:`
	`  - name: '${wpnet.name}'`
	`  assignment: static`
	`  network: '${resource.wpnet.id}'`
	CentOS sample
	`resources:`
	`  wpnet:`
	`  type: Cloud.Network`
	`  properties:`

Design	Sample Cloud Template Code

Assign a static IP address to a Linux machine with cloud-init code that contains network assignment commands. The customizeGuestOs property must be false.	Ubuntu sample
	`resources:`
	`  wpnet:`
	`  type: Cloud.Network`
	`  properties:`
	`  name: wpnet`
	`  networkType: public`
	`  constraints:`
	`  - tag: sqa`
	`  DBTier:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  flavor: small`
	`  image: ubuntu-template`
	`  customizeGuestOs: false`
	`  cloudConfig: |`
	`  #cloud-config`
	`  write_files:`
	`  - path: /etc/netplan/99-installer-config.yaml`
	`  content: |`
	`  network:`
	`  version: 2`
	`  renderer: networkd`
	`  ethernets:`
	`  ens160:`
	`  addresses:`
	`  - ${resource.DBTier.networks[0].address}/${resource.wpnet.prefixLength}`
	`  gateway4: ${resource.wpnet.gateway}`
	`  nameservers:`
	`  search: ${resource.wpnet.dnsSearchDomains}`
	`  addresses: ${resource.wpnet.dns}`
	`  runcmd:`
	`  - netplan apply`
	`  - hostnamectl set-hostname --pretty ${self.resourceName}`
	`  - touch /etc/cloud/cloud-init.disabled`
	`  networks:`
	`  - name: '${wpnet.name}'`
	`  assignment: static`
	`  network: '${resource.wpnet.id}'`
	CentOS sample
	`resources:`
	`  wpnet:`
	`  type: Cloud.Network`
	`  properties:`
	`  name: wpnet`
	`  networkType: public`
	`  constraints:`
	`  - tag: sqa`
	`  DBTier:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  flavor: small`
	`  image: centos-template`
	`  customizeGuestOs: false`

Design	Sample Cloud Template Code
When basing the deployment on a referenced image, assign a static IP address to a Linux machine with cloud-init code that contains network assignment commands. The customizeGuestOs property must be false. In addition, the cloud template must not include the ovfProperties property, which blocks customization.	resources: wpnet: type: Cloud.Network properties: name: wpnet networkType: public constraints: - tag: sqa DBTier: type: Cloud.vSphere.Machine properties: flavor: small imageRef: 'https://cloud-images.ubuntu.com/releases/focal/release/ubuntu-20.04-server-cloudimg-amd64.ova' customizeGuestOs: false cloudConfig: | #cloud-config ssh_pwauth: yes chpasswd: list: | root:Pa$$w0rd ubuntu:Pa$$w0rd expire: false write_files: - path: /etc/netplan/99-netcfg-vrac.yaml content: | network: version: 2 renderer: networkd ethernets: ens192: dhcp4: no dhcp6: no addresses: - ${resource.DBTier.networks[0].address}/${resource.wpnet.prefixLength} gateway4: ${resource.wpnet.gateway} nameservers: search: ${resource.wpnet.dnsSearchDomains} addresses: ${resource.wpnet.dns} runcmd: - netplan apply - hostnamectl set-hostname --pretty ${self.resourceName} - touch /etc/cloud/cloud-init.disabled networks: - name: '${wpnet.name}' assignment: static network: '${resource.wpnet.id}'

Day 2 customizations

Like an initial deployment, a Day 2 action also might include network configuration. To skip customization during Day 2 actions, add the following property:

```
customizeGuestOsDay2: false
```

Designs that won't work or might produce unwanted results

- The cloud-init code doesn't contain network assignment commands, and the customizeGuestOs property is false.
  Neither initialization commands nor customization spec are present to configure network settings.
- The cloud-init code doesn't contain network assignment commands, and the ovfProperties property is set.
  Initialization commands aren't present, but ovfProperties blocked the customization spec.
- The cloud-init code contains network assignment commands, and the customizeGuestOs property is missing or set to true.
  Application of the customization spec conflicts with initialization commands.

Other workarounds for cloud-init and customization specs

When deploying to vSphere, you can also customize an image to work around cloud-init and customization spec conflicts. See the following external repository for more information.

- vSphere Image Preparation Scripts

Delayed deployment in VCF Automation for VM Apps

A virtual machine might need to be fully initialized before proceeding with VCF Automation deployment.

For example, deploying a machine that is still installing packages and starting a web server might lead to conditions where a fast user tries to reach the application before it's available.

Be aware of the following considerations when using this feature.

- The feature makes use of the cloud-init phone_home module and is available when deploying Linux machines.
- Phone home isn't available for Windows because of Cloudbase-init limitations.
- Phone home can affect deployment order like an explicit dependency, but has more flexibility around timing and processing options.
  See Creating bindings and dependencies between resources in VCF Automation for VM Apps.
- Phone home requires a cloudConfig section in the cloud template.
- Your creativity is a factor. Initialization commands might include embedded wait time between operations, which can be used in concert with phone home.
- Cloud template-based phone home won't work if the machine template already contains phone_home module settings.
- The machine must have outbound communication access back to VCF Automation.

To introduce a deployment delay in VCF Automation, add a `cloudConfigSettings` section to the cloud template:

```
cloudConfigSettings:
  phoneHomeShouldWait: true
  phoneHomeTimeoutSeconds: 600
  phoneHomeFailOnTimeout: true
```

Property	Description
phoneHomeShouldWait	Whether to wait for initialization, true or false.
phoneHomeTimeoutSeconds	When to decide whether to proceed with deployment even though initialization is still running. Default is 10 minutes.
phoneHomeFailOnTimeout	Whether to proceed with deployment after timing out, true or false. Note that even when proceeding, deployment might still fail for separate reasons.

Windows guest customization in VCF Automation for VM Apps

To have VCF Automation automatically initialize a Windows machine at deployment, prepare an image that supports Cloudbase-Init, then a cloud template that contains the appropriate commands.

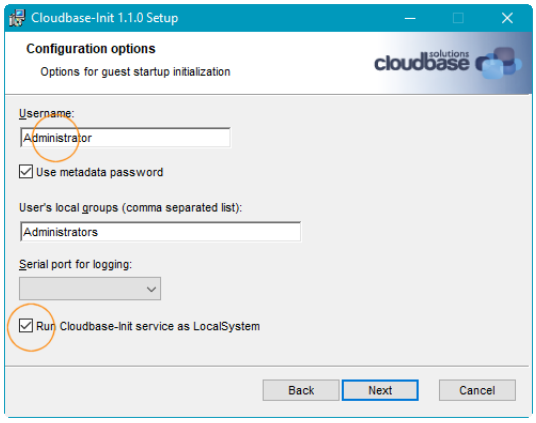
The image creation process varies depending on cloud vendor. The example shown here is for vSphere.

Windows VCF Automation image for vSphere

For VCF Automation to initialize a Windows machine deployed to vSphere, the image needs to be based on a vSphere template with Cloudbase-Init installed and configured.

Creating the image

1. Use vSphere to make and power on a Windows virtual machine.
2. On the virtual machine, log in to Windows.
3. Download Cloudbase-Init.
   https://cloudbase.it/cloudbase-init/#download
4. Start the Cloudbase-Init setup .msi file.
   During installation, enter **Administrator** as the username, and select the option to run as LocalSystem.



   Other setup selections can remain as default values.
5. Allow the installation to run, but do not close the final Completed page of the setup wizard.

   Important:
   Do not close the final page of the setup wizard.
6. With the Completed page of the setup wizard still open, use Windows to navigate to the Cloudbase-Init installation path, and open the following file in a text editor.
   `conf\cloudbase-init-unattend.conf`
7. Set `metadata_services` to `OvfService` as shown. Add the setting if it doesn't already exist.
   `metadata_services=cloudbaseinit.metadata.services.ovfservice.OvfService`
8. Save and close `cloudbase-init-unattend.conf`.
9. In the same folder, open the following file in a text editor.
   `conf\cloudbase-init.conf`
10. Set `first_logon_behaviour`, `metadata_services`, and `plugins` as shown. Add the settings if they don't already exist.

```
first_logon_behaviour=always
. . .
metadata_services=cloudbaseinit.metadata.services.ovfservice.OvfService
. . .
plugins=cloudbaseinit.plugins.windows.createuser.CreateUserPlugin,cloudbaseinit.plugins.windows.setuserpassword.SetU
. . .
```

    If you are using an IPAM provider to assign static IPs on Windows, include `cloudbaseinit.plugins.common.networkconfig.NetworkConfigPlugin` in your list of plugins to ensure proper IPAM integration.

11. Save and close `cloudbase-init.conf`.
12. On the Completed page of the setup wizard, select the options to run Sysprep and to shut down after Sysprep, then click **Finish**.

    Note:
    VMware has seen cases where running Sysprep prevents deployments of the image from working.

    When deploying, VCF Automation applies a dynamically generated customization specification, which disconnects the network interface. The pending Sysprep state in the image might cause the customization specification to fail and leave the deployment disconnected.

    If you suspect that this is happening in your environment, try leaving the Sysprep options deactivated when creating the image.



13. After the virtual machine shuts down, use vSphere to turn it into a template.

Additional details

The following table expands upon the configuration entries made during setup.

Configuration Setting	Purpose
Username, CreateUserPlugin, and SetUserPasswordPlugin	After Sysprep, first boot uses CreateUserPlugin to create the username Administrator account with a blank password. SetUserPasswordPlugin allows Cloudbase-Init to change the blank password to the remote access password that will be included in the cloud template.
First Logon Behavior	This setting prompts the user to change the password upon first login.
Metadata services	By listing only OvfService, Cloudbase-Init won't try to find other metadata services that aren't supported in vCenter. This results in cleaner log files, because the logs would otherwise fill with entries about failing to find those other services.
Plugins	By listing only plugins with capabilities supported by OvfService, logs are again cleaner. Cloudbase-Init runs plugins in the order specified.
Run as LocalSystem	This setting supports any advanced initialization commands that might require Cloudbase-Init to run under a dedicated administrator account.

Cloudbase-Init commands for Windows in VCF Automation for VM Apps

To run Windows machine initialization at deployment time, add Cloudbase-Init commands to the VCF Automation template code.

The example shown here is based on vSphere, but other cloud vendors should be similar.

Prerequisites

- Create infrastructure. In VCF Automation, add your vSphere cloud account and an associated cloud zone.
- Add flavor and image mappings, and add network and storage profiles.
  In your infrastructure, an image mapping must point to a Windows template that you created to support Cloudbase-Init. See Windows VCF Automation image for vSphere.
  If the template isn't listed, go to Cloud Accounts, and synchronize images. Otherwise, automatic synchronization runs every 24 hours.
- Add a project, add users, and make sure the users can provision to your cloud zone.

Procedure

1. In VCF Automation, go to the **Design** tab, and create a new cloud template.
2. Add a `cloudConfig` section with the Cloudbase-init commands that you want.
   The following command examples create a new file at the Windows `C:` drive and set the host name.

```
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      image: cloudbase-init-win-2016
      flavor: small
      remoteAccess:
        authentication: usernamePassword
        username: Administrator
        password: Password1234@$
      cloudConfig: |
        #cloud-config
        write_files:
          content: Cloudbase-Init test
          path: C:\test.txt
        set_hostname: testname
```

3. Add `remoteAccess` properties so that you configure the machine for initial login to Windows.
   As mentioned when you created the template, the metadata service picks up the login credentials and exposes them to CreateUserPlugin and SetUserPasswordPlugin. Note that the password must meet Windows password requirements.
4. From VCF Automation, test and deploy the cloud template.
5. After deploying, use Windows RDP and the credentials in the template to log in to the new Windows machine and verify the customization.
   In the example above, you would look for the `C:\test.txt` file, and check the system properties for the host name.

Machine and disk clusters in VCF Automation for VM Apps

VCF Automation template designs can deploy a cluster of machines and attach a cluster of disks.

To deploy clusters of machines and disks, take advantage of the `allocatePerInstance`resource flag, and `count.index` and `map_to_object`expression syntax in your cloud templates.

The following cloud template code examples can serve as guidelines for designs that deploy clusters.

Two machines that share a disk cluster

```
resources:
  app0:
    type: Cloud.Machine
    allocatePerInstance: true
    properties:
      image: ubuntu
```

```
      flavor: small
      attachedDisks: '${map_to_object(slice(resource.disk[*].id, 0,2), "source")}'
  app1:
    type: Cloud.Machine
    allocatePerInstance: true
    properties:
      image: ubuntu
      flavor: small
      attachedDisks: '${map_to_object(slice(resource.disk[*].id, 2,4), "source")}'
  disk:
    type: Cloud.Volume
    allocatePerInstance: true
    properties:
      count: 4
      capacityGb: 5
```

Variable number of machines with one disk each

```
inputs:
  count:
    type: integer
    default: 2
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    allocatePerInstance: true
    properties:
      image: ubuntu
      flavor: small
      count: '${input.count}'
      attachedDisks: '${map_to_object(slice(resource.disk[*].id, count.index, count.index + 1), "source")}'
  disk:
    type: Cloud.Volume
    allocatePerInstance: true
    properties:
      count: '${input.count}'
      capacityGb: 5
```

Variable number of machines with two disks each

```
inputs:
  count:
    type: integer
    default: 2
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    allocatePerInstance: true
    properties:
      image: ubuntu
      flavor: small
      count: ${input.count}
      attachedDisks: '${map_to_object(slice(resource.disk[*].id, 2*count.index, 2*(count.index + 1)),
 "source")}'
  disk:
    type: Cloud.Volume
    allocatePerInstance: true
```

```
    properties:
      count: ${2*input.count}
      capacityGb: 5
```

Set disk sizes at request time

```
inputs:
  disksize:
    type: array
    minItems: 2
    maxItems: 2
    items:
      type: object
      properties:
        size:
          type: integer
resources:
  app:
    type: Cloud.Machine
    allocatePerInstance: true
    properties:
      flavor: small
      image: ubuntu
      attachedDisks: ${map_to_object(slice(resource.disk[*].id, 0, 2), 'source')}
  disk:
    type: Cloud.Volume
    allocatePerInstance: true
    properties:
      count: 2
      capacityGb: ${input.disksize[count.index].size}
```

Custom naming deployed resources in VCF Automation for VM Apps

You can use custom naming templates to override the system naming of deployed resources to naming conventions that you define. The naming is applied at deployment time.

To determine if you are using the new method, select **Infrastructure** > **Administration** > **Custom Names**.

- If you see an option labeled **New Custom Name**, then you can are using the new global custom naming method. For more information about this method, see Create global custom naming for deployed resources in VCF Automation for VM Apps.
- If you see the enrollment option, then you must consider your choices. See the following section.

Why are there two custom naming methods

VCF Automation has moved to a new custom naming method. The new method provides you with greater flexibility and additional naming options. It is also much easier to manage the naming across projects.

Prior to the November 2021/8.6.1 release, VCF Automation had one method for creating custom names for deployed resources. The custom naming was defined in each project.

In the November 2021/8.6.1, VCF Automation released a beta version of a new global custom naming method. You could now create global templates for your organization and for your projects, and assign the templates to projects as needed.

Beginning in the March 2022/8.7.1 release, you will encounter one of the following scenarios:

- If you are new to VCF Automation, you can use the global custom naming method. The older, project-level method is not available to you.

- If you currently use VCF Automation but you never created any custom naming templates in any of your projects, you can use the global custom naming method. The project-level method is not available to you.
- If you currently use VCF Automation and you created at least one custom naming template in a project, you have three options:
  - You can enroll to take advantage of the new custom naming but not migrate the current custom naming templates to the new method. This is the recommended option, particularly if you have more than a few projects with already defined custom naming templates.
  - You can enroll and migrate your current project templates to the new global method. You can do this, but it is not guaranteed that all your project templates will migrate. Some might have formats that are incompatible with the global method.
  - You do not enroll and continue to use the project-by-project template method. The project-by-project method is more difficult to manage and has more limited features. To review the project-by-project method, see Create project-by-project custom names for deployed resources in VCF Automation for VM Apps.

How do I enroll

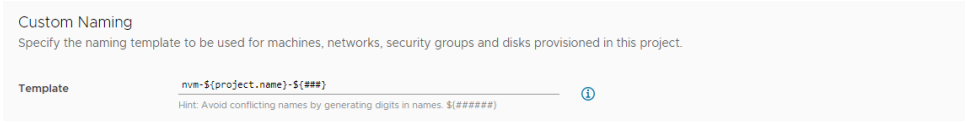
You can enroll in the global custom naming by either migrating your current custom naming project templates or enrolling but not migrating. The second option is recommended.

1. Select **Infrastructure** > **Administration** > **Custom Names**.
2. Click **Enroll Now**.
3. Select an enrollment option and click **Enroll**.

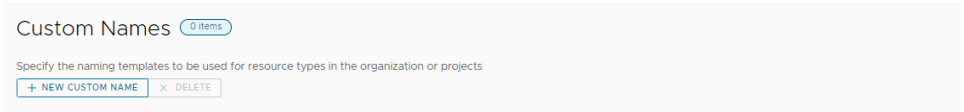
Enrollment option	Description	Considerations
Do not migrate custom names from any projects	Enrolls you in the global method. You can create templates and apply them to the organization and projects.	• After enrollment, the current templates are not longer available. If you plan to use the same format, capture the format before enrolling so that you can create similar templates using the global method.
Migrate custom names for all projects	Enrolls you in the custom naming and creates project-level templates for each project. You can manage the templates and apply them to other projects if needed.	• If you have 100 projects and you migrate the templates, you will end up with 100 variations. • Some project templates might fail migration due to unsupported formats.

4. Based on how you enrolled, review where you started from and then verifying the enrollment results.
   - If you selected **Do not migrate**, create a new template and apply it to your projects. See Create global custom naming for deployed resources in VCF Automation for VM Apps.

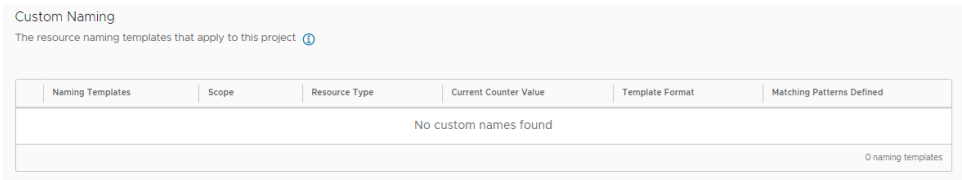
     You started with the template defined in the project.

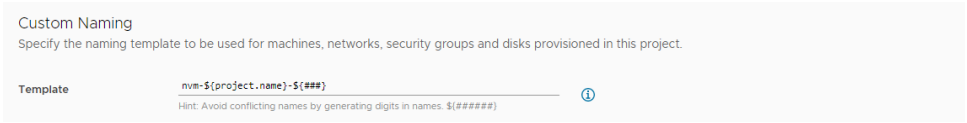
     After enrollment, you have no defined templates and you must create the templates to use going forward. Click **New Custom Name** to get started.

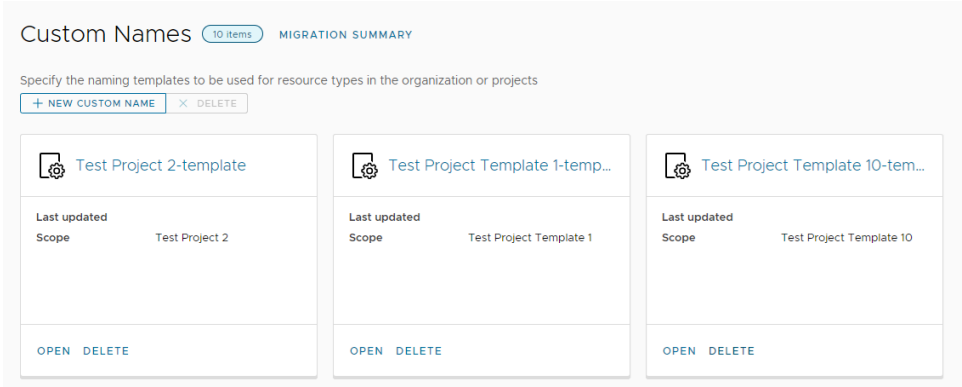
     After enrollment and until you create and assign a template to a project, the custom naming in the project is blank, as illustrated below. After you create a template and assign it to the project, the table will list applicable templates.



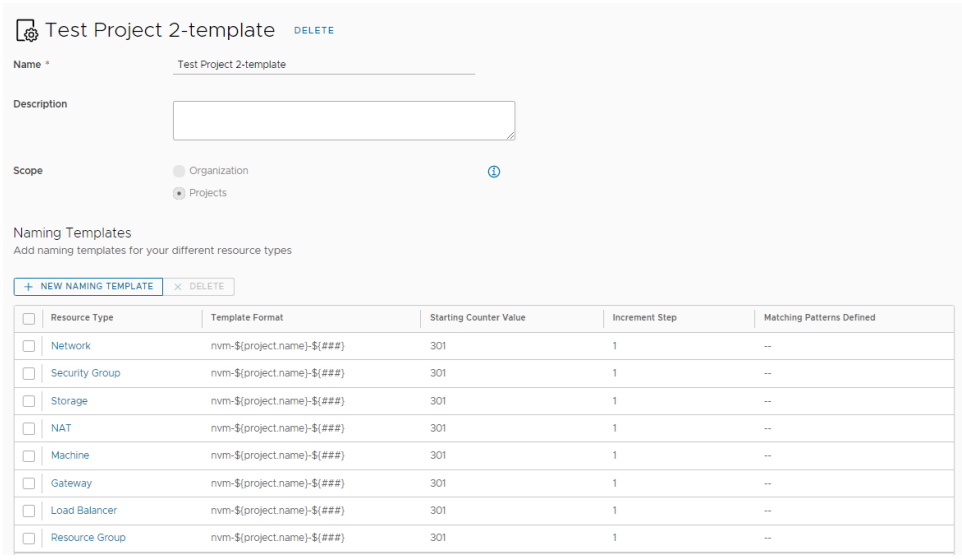
– If you selected **Migrate**, review and create any new templates that you need. For more information about managing your templates, see Create global custom naming for deployed resources in VCF Automation for VM Apps.
You started with the templates defined in the project.



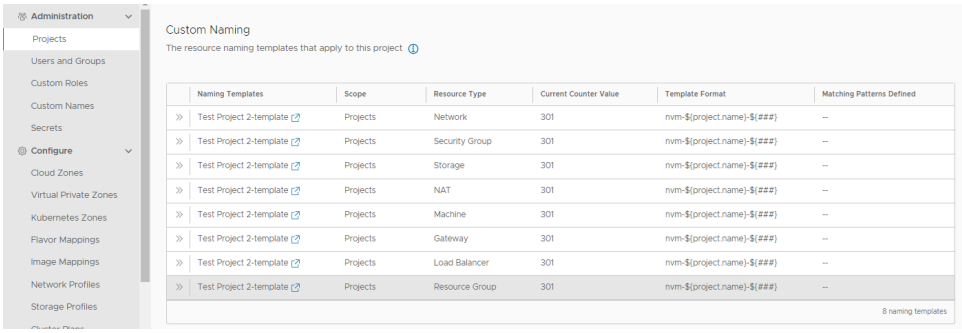
After enrollment, the defined templates for each project are added to the Custom Names page.



If you examine a template, you see the template for the different resource types. You cannot modify the template formats, but you can delete and create a template specific to the resource type.



After enrollment, the custom naming in the project is updated with the migrated custom naming template for each resource type. The list is for informational purposes. To manage the templates, use the Custom Names .



Create global custom naming for deployed resources in VCF Automation for VM Apps
As a cloud or project administrator, you have a prescribed naming convention for resources in your environment, and you want the deployed resource to follow those conventions without user interaction. You can create global naming templates for some or all deployments in VCF Automation.

You can start by creating project-level custom names or organization-level names. Project-level custom names take precedence over organization-level names. This example starts with a simple organization custom name and then adds project-level naming.

> **CAUTION:**
> If you have cloud templates or are using the API where you deploy 2000 or more resources at the same time per project, and where a custom name if applied, you might encounter a `Retries exhausted` error during deployment on some of those resources. If this scenario is common, you should not apply the new custom name to the projects or organizations where this occurs. Instead, you should use the default system naming, not the custom naming option.

What to do first

- To demonstrate the custom naming capability, these use cases need two projects. One is name Sales and one is named Marketing. For more information about creating projects, see Add a project for your VCF Automation for VM Apps development team.
- As you make choices about organization and project templates, ensure that you understand which naming conventions take precedence.
- As you create a template using the template format, the possible properties are provided.

Create an organization-level custom template

When you want a default custom name template for deployments that do not have project-level templates, create templates that are organizational in scope.

1. Select **Infrastructure** > **Administration** > **Custom Names** and click **New Custom Name**.
2. Enter a **Name**.
   This example uses `Prefix and Timestamp for Org`.
3. Select **Organization** as the Scope.
4. Click **New Naming Template** and configure the following options.

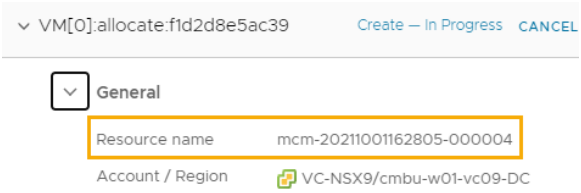
Option	Sample Value
Resource type	Machine
Validate compute name for uniqueness	Select the check box. Unique resources names ensures that you are never confused about which resource you are working with, even when working in API-based environments. The number configuration in the following template format increments to ensure uniqueness.
Template format	mcm-${timestamp}-${######}

Starting counter value	1
Increment step	1 With this configuration, all the deployments across all the projects will increment from this starting point. In this example, where the starting counter is 1 and the increment is 1, the first deployment is numbered as 2. I you need the deployment to start at 1, then set the starting counter to zero and the increment step to 1.

5. Click **Add**.
6. Continue adding templates for other resource types using the following examples.

Resource Type Name	Naming Template Format
Machine	`mcm-${timestamp}-${##}`
Network	`ntw-${timestamp}-${##}`
Storage	`stg-${timestamp}-${##}`
Load Balancer	`ldb-${timestamp}-${##}`
Resource Group	`rsg-${timestamp}-${##}`
Gateway	`gtw-${timestamp}-${##}`
NAT	`nat-${timestamp}-${##}`
Security Group	`scg-${timestamp}-${##}`

7. Click **Create**.
8. Test the name template by deploying templates that include the defined resource types.



Create a project-level custom template with advanced pattern matching

You can create a single custom name template that you can assign to different projects. You use the advanced option to set different numbering starting points for different projects.

This example uses the machine resource.

1. Select **Infrastructure** > **Administration** > **Custom Names** and click **New Custom Name**.
2. Enter a **Name**.
   This example uses `Project Name with Advanced Numbering`.
3. Select **Project** as the Scope.
4. Click **New Naming Template** and configure the following options.

Option	Sample Value
Resource type	Machine
Template format	${project.name}-${######}
Starting counter value	1

Increment step	1 With this configuration, the deployments in the assigned projects will increment from this starting point. In this example, where the starting counter is 1 and the increment is 1, the first deployment is numbered as 2. I you need the deployment to start at 1, then set the starting counter to zero and the increment step to 1. With 1 as the starting value, ProjectA starts with 2 and ProjectB starts with 2, unless the value is overridden by an advanced matching pattern value.

5. To add more refinement to the default organization naming template, click **Advanced** and click **Add Matching Pattern**.
   You can set different naming patterns to start the counter at different numbers or reset the number to 1 for each pattern. For example, the Sales project numbering starts at 100 and the Marketing project numbering starts at 200.
   a. Enter the pattern for the sales project.

Option	Sample Value
Pattern text	Sales-
Starting counter value	100

   b. Enter the pattern for the marketing project.

Option	Sample Value
Pattern text	Marketing-
Starting counter value	200

6. Click **Add**.
7. Continue adding templates for other resource types, as needed.
8. Click **Assign Projects** and add the Sales and Marketing projects.
9. Click **Create**.
10. Test the name template by first deploying a cloud template that is in the Sales project, and then one in the Marketing project.



Example of a custom name with custom properties and project-specific values

In addition to the properties provided in the template format autofill options, you can use custom properties. To use custom properties in a custom name, you must define them in a property group.

A custom property value in the cloud template take precedence over the same property value in the custom name.

This example uses costCenter as the property and SalesCost as the value for the Sales project and MktCost as the value for the Marketing project.

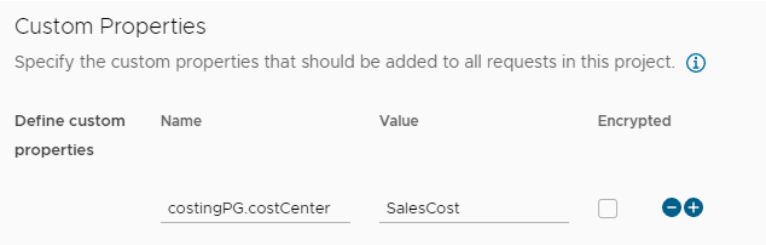
1. Create a property group named costingPG.
   a. Select **Infrastructure** > **Design** > **Property Groups** and click **New Property Group**.
   b. Click **Input Values**.
   c. Enter the **Name**`costingPG`.
   d. For the **Scope**, select **Available for any project**.
   e. Click **New Property** configure the following values and enter the name `costCenter`.

Options	Sample Values
Name	costCenter
Type	String

Default value	research

f. Click **Create**.
g. Click **Create** to save the new property group.

2. Configure custom properties for the Sales project and the Marketing project.
   These steps illustrate the process for the Sales project. You can repeat the steps and use the marketing project values.
   – Sales property name: costingPG.costCenter. Value: SalesCost.
   – Marketing property name: costingPG.costCenter. Value: MktCost.
   a. Select **Infrastructure** > **Administration** > **Projects** and open the Sales project.
   b. Click the **Provisioning** tab and locate the Custom Properties section.
   c. Enter `costingPG.costCenter` as the property name and `SalesCost` and the value.



   d. Click **Save**.
   e. Repeat the process in the Marketing project using `MktCost` and the property value.

3. Create a project-level custom name.
   This example uses the same projects that the advanced example users. You can assign custom names to only one project at a time. To test the results of this example, you can either apply the custom name to new projects or remove the Sales and Marketing projects from the Project Name with Advanced Numbering example.
   This example assumes that you have not yet assigned the Sales and Marketing projects.
   a. Select **Infrastructure** > **Administration** > **Custom Names** and click **New Custom Name**.
   b. Enter `Project Name and costingPG` as the **Name**.
   c. Select **Project** as the Scope.
   d. Click **New Naming Template** and configure the following options.

Table 1830:

Option	Value
Resource Type	Machine
Template format	${project.name}-${costingPG.costCenter}-${######}
Starting counter value	1
Increment step	1

   e. Click **Add**.
   f. Click **Assign Projects** and add the Sales and Marketing projects.
   g. Click **Create**.

4. Test the name template by first deploying a cloud template that is in the Sales project, and then one in the Marketing project.

Figure 540: Sales project resource name



Figure 541: Marketing project resource name



Example of a custom name with property group and cloud template user inputs

This example shows you how to create a custom name based on user inputs in the template. You define two of the user inputs, operating system and size, in the cloud template. One of the user inputs, mktSite, uses a site.siteCode custom property that accepts user inputs and you add it to the cloud template.

1. Define the custom property for the marketing project.
   a. Select **Infrastructure** > **Design** > **Property Groups** and click **New Property Group**.
   b. Click **Input Values**.
   c. Name and describe the property group.

Option	Value
Name	site
Scope	Available for any project. The property group must be generally available so that the property is available to use in the custom name template format.

   d. Click **New Property** configure the following values.

Option	Value
Name	siteCode
Type	String
Default value	West
More options > **Enum Value**	Enter separate values for East, West, North, and South. These values represent the possible site codes that your users might select.

2. Create the custom naming template.
   a. Select **Infrastructure** > **Custom Names** > **New Custom Name**.
   b. Enter the **Name**`Site-OS-Size for Marketing Project`.
   c. Select Projects.
      Projects is selected and the only option if you already have an organization scoped custom name.
   d. Click **New Naming Template** and configure the following options.

Option	Value
Resource type	Machine
Template format	${site.siteCode}-${resource.name}-${######} The site.siteCode property is the one that you defined in the previous step. The resource.name is defined in the cloud template in the next step.
Starting counter value	1
Increment step	1

3. Define user inputs in the template.
   The user inputs included in this example include operating system and size, and a user input based on the site.siteCode custom property.
   The cloud template in this step is simple for demonstration purposes. You can apply the example to one your existing templates.

a. Select **Design** > **New From** > **Blank canvas**.
b. Enter the **Name**`Marketing Template`.
c. In **Project** list, select `Marketing`.
d. Click **Create**.
e. Enter the code for the cloud template or use the following sample code.
   Notice that the user inputs are for size and OS. You configure the variables for each input in the flavor (`${input.size}`) and image (`${input.OS}`) machine properties. Finally, the name property is used to construct the resource name (`${input.OS}-${input.size}`) that can be used in the custom naming template.

```
formatVersion: 1
inputs:
  size:
     type: string
     enum:
       - small
       - medium
   OS:
     type: string
     enum:
       - centos
       - ubuntu
resources:
  Cloud_vSphere_Network_1:
    type: Cloud.vSphere.Network
    properties:
      networkType: existing
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      image: ${input.OS}
      flavor: ${input.size}
      name: ${input.OS}-${input.size}
      networks:
        - network: ${resource.Cloud_vSphere_Network_1.id}
```

f. To add the site.siteCode custom property as a user input, click the **Inputs** tab in the right pane and configure the following options.

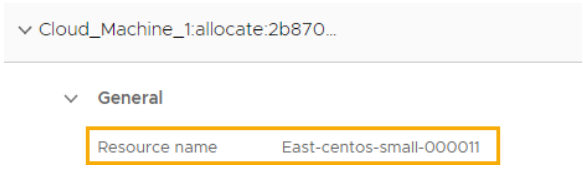
Option	Value
Name	mktSite
Display Name	Select a site
Type	Object
Select Object Type	Property Groups
Select from the existing property groups	site

g. On the Code tab, notice that the added mktSite user input and then update the YAML to include the custom property.
   For example, add `site.siteCode: ${input.mktSite.siteCode}` to the YAML.

```
formatVersion: 1
inputs:
  size:
    type: string
    title: Select the machine size
    enum:
```

```
        - small
        - medium
    OS:
      type: string
      title: Select an operating system
      enum:
        - centos
        - ubuntu
    mktSite:
      type: object
      $ref: /ref/property-groups/site
      title: Select a site
resources:
  Cloud_vSphere_Network_1:
    type: Cloud.vSphere.Network
    properties:
      networkType: existing
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      image: ${input.OS}
      flavor: ${input.size}
      name: ${input.OS}-${input.size}
      networks:
        - network: ${resource.Cloud_vSphere_Network_1.id}
      site.siteCode: ${input.mktSite.siteCode}
```

4. Deploy the template and verify the custom machine name.
   Since West is the default value, select a site other than West.



Resource naming precedence at deployment time

Based on how you configure custom naming, the names are applied to the deployed resources based on the following rules.

A common practice is to define a general organization-level naming template and then to create one or more project-level templates, depending one your project management needs.

Custom Naming Scenario	Precedence
No custom names	• The system creates the deployment resource names for all the deployment components.
Organization-level name	• Applies the custom naming property values and strings to all the deployments. • If the deploying project has a project-level custom name, the organization-level name is overridden. • If an organization-level custom name is not defined, the project-level name is used for assigned projects. • If neither an organization-level nor a project-level custom name is defined, the system creates the resource names.

Project-level names	• Applies the project-level custom naming property values and strings to all assigned projects. • If an organization-level name exists, the project-level name takes precedence over the organization name. • If an organization-level custom name is not defined, the project-level name is used for assigned projects. • If neither an organization-level nor a project-level custom name is defined, the system creates the resource names.

Working with template formats

When defining your template format, you can use strings and properties. The primary advantage of using properties is that you can use the same properties as the format for multiple projects, but the name is derived from the value properties, which can vary by project, cloud template, platform, and other properties.

The template format properties include the provided properties, discussed here, and any custom properties that you defined in the **Design** > **Property Groups**.

- Valid characters
  - Spaces are not allowed.
  - For readability, you can use special characters to separate properties. The allowed characters are _ (underscore) and - (dash).
- Provided properties

Table 1831: Provided Properties

Properties	Description
endpoint	Cloud account information, such as description, enter (AWS), or the name. Endpoint only applies to machines. Examples • `${endpoint.desc}` • `${endpoint.endpointType}` • `${endpoint.name}`
project	Project information. Example • `${project.desc}` • `${project.name}`
resource	Includes the resource name from the cloud template, custom properties that are in the cloud template or the project. Example • `${resource.name}`
timestamp	Date and time of the deployment. The numeric value is yyyy mm dd hh mm ss, but without the spaces. For example, 20210825120345. Example • `${timestamp}`
user	Example • `${user}`
username	Example • `${userName}`

Properties	Description
######	Number property. The maximum length is 19 digits. If you use two digits, ##, the maximum number before the numbering starts over is 99. If you use ###, the numbering restarts after 999. Example • `${####}`.
Custom property	Custom properties must be part of property group. The property in the custom name includes the property group name and the property. Example • `${propertygroup.property}`

Working with the counter

In a single node instance of VCF Automation, the counter increments based on the values that you define in your custom naming templates and as they are applied to the target organization and projects.

In a multi node environment, the counter attempts of increment based on the values in the template. If it encounters a contention, application of the number is attempted three time. It is possible that a request could fail after the third attempt to resolve the contention. In this case, you might see a gap in the numbering.

You might also encounter gaps in the numbering for other reasons. For example, you deploy three virtual machines. The second machine fails during provisioning. The counters for the deployed virtual machines are vm-01 and vm-03, with vm-02 missing.

If you do not use the custom naming, preferring to rely on the default naming strategy, the pools of counter values are reserved to the nodes as follows:

- Node 1: 1-300
- Node 2: 301-600
- Node 3: 601-900

This means that in a multi node environment, the number can appear to be unexpected. The number depends on which node processes the request.

List of resource types to which you can apply custom names

You can create a custom name for only one resource type or for all the resource types. Only the resource types with custom names use the template. All other resources use the default template.

For example, if you create a project-level custom name for only your machines and you have organization-level names for all the other resource types, the machine resource takes the project name and all the other resources use the organization names.

Table 1832: Sample list of resources

Custom naming resource types	Deployment resource types
Machines	• Cloud.Machine • Cloud.vSphere.Machine • Cloud.AWS.EC2.Instance • Cloud.GCP.Machine • Cloud.Azure.Machine
Networks	• Cloud.Network • Cloud.vSphere.Network • Cloud.NSX.Network

Custom naming resource types	Deployment resource types
Storage	• Cloud.Volume • Cloud.vSphere.Disk • Cloud.AWS.Volume • Cloud.GCP.Disk • Cloud.Azure.Disk
Load balancers	• Cloud.LoadBalancer • Cloud.NSX.LoadBalancer
Resource Groups	• Cloud.Azure.ResourceGroup
Gateways	• Cloud.NSX.Gateway
NAT	• Cloud.NSX.NAT
Security groups	• Cloud.SecurityGroup
Generic	• Resources that use the custom naming allocation helper The custom naming allocation helper generates custom names for your resources during the allocation process. The Generic resource type covers all possible resources and is linked exclusively to the custom naming allocation helper.

Deleting custom name templates

Deleting a template does not affect currently deployed resource naming. However, you must not delete a custom name template until you verify the impact on all the assigned projects.

You can remove a project from a naming template. The template continues to work for the other projects.

Deleting projects with custom names

If a project has a custom naming template applied, you must first remove the project from the template before you can delete the project.

Create project-by-project custom names for deployed resources in VCF Automation for VM Apps

As a cloud or project administrator, you can create an individual naming template in each VCF Automation project.

- Verify that you know the naming convention that you want to use for deployments from a project.
- This procedure assumes you have or can create a simple cloud template that you use to test your custom host prefix naming.

As a cloud or project administrator where you defined resource custom naming templates at the project level prior to March 2022/8.7.1, you have the option to continue to use the method described below or you can convert your current templates to the new global templates for your organization and projects.

- For more information about enrolling in to the global templates, see Custom naming deployed resources in VCF Automation for VM Apps.
- For more information about creating global custom naming templates, see Create global custom naming for deployed resources in VCF Automation for VM Apps.

If you never created custom naming templates or are new to VCF Automation, you only have the global option.

To proceed with this method, consider the following example. Your host naming convention is to prefix a resource as *`projectname-sitecode-costcenter-whereDeployed-identifier`*. You configure the custom naming template for the machines for each project. Some of the template variables are pulled from the system as it is deployed, other are based on project custom properties. The custom naming template for the above prefix looks similar to the following example.

```
${project.name}-${resource.siteCode}-${resource.costCenter}-${endpoint.name}-${######}
```

The identifier, provided in the template as `${######}`, shows a six digit identifier. The identifier is a counter that ensures uniqueness. The counter is global for the organization and increments across all projects, not only the current project.

When you have multiple projects, do not expect a sequence from 000123 to 000124 for deployments in you current project. You can expect an increment from 000123 to 000127.

All resource names must be unique. To ensure uniqueness, use the incremental number property. The numbers increment for all deployments, including deployments that VCF Automation names. As your system becomes more robust, and because the system applies custom names to many resource types, the numbering can appear random, but the values still ensure uniqueness. The numbers also increment when you run a test deployment.

The following list is a sample of where the custom names are applied. The list is not meant to be definitive.

Table 1833: Sample list of resources to which custom names are applied

Resource Group	Resource Types
Virtual machines	• Cloud.Machine • Cloud.vSphere.Machine • Cloud.AWS.EC2.Instance • Cloud.GCP.Machine • Cloud.Azure.Machine
Load balancers	• Cloud.LoadBalancer • Cloud.NSX.LoadBalancer
Networks	• Cloud.Network • Cloud.vSphere.Network • Cloud.NSX.Network
Security groups	• Cloud.SecurityGroup
Disks	• Cloud.Volume • Cloud.vSphere.Disk • Cloud.AWS.Volume • Cloud.GCP.Disk • Cloud.Azure.Disk
NSX	• Cloud.NSX.Gateway • Cloud.NSX.NAT
Microsoft Azure	• Cloud.Azure.ResourceGroup

In addition to the examples provided here, you can also add the user name, the image that is used, other built-in options, and simple strings. As you build the template, hints regarding possible options are provided.

Remember that some of the values you see are only use case examples. You won't be able to use them letter-by-letter in your environment. Think about where you would make your own substitutions, or extrapolate from the example values, in order to fit your own cloud infrastructure and deployment management needs.

1. Select **Infrastructure** > **Projects**.
2. Select an existing project or create a new one.
3. On the **Provisioning** tab, locate the Custom Properties section and create the properties for the site code and cost center values.

   This is where you replace the values you see here with ones pertinent to your environment.



a) Create a custom property with the name `siteCode` and the value `BGL`.
b) Add another custom property with the name `costCenter` and the value `IT-research`.

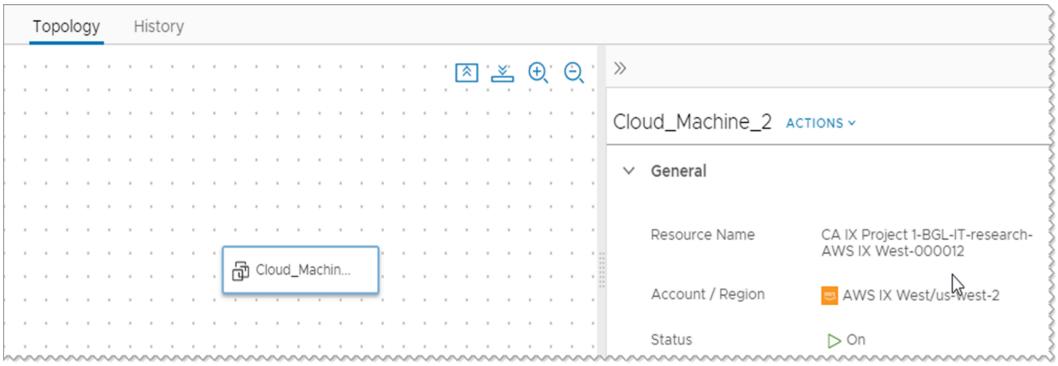
4. Locate the Custom Naming section and add the following template.

   ```
   ${project.name}-${resource.siteCode}-${resource.costCenter}-${endpoint.name}-${######}
   ```

   You can copy in the string, but if this is your first naming template, consider using the hint text and quick select as you build the template.

5. Deploy a cloud template associated with the project to verify that the custom name is applied to the resource.
   a) Click the **Design** tab, and then click a cloud template associated with the project.
   b) Deploy the cloud template.

      The **Deployments** page opens, showing your deployment in process.
   c) When deployment is completed, click the deployment name.
   d) On the **Topology** tab, notice that your custom name is the resource name in the right pane.



6. If you deployed a test cloud template to verify the naming convention, you can delete the deployment.

Create custom naming templates for your other projects.

Terraform Configuration in VCF Automation for VM Apps

As a cloud administrator, you can automate the deployment and configuration of VCF Automation infrastructure resources using Terraform. You can embed Terraform configurations as a resource in VCF Automation cloud templates.

Terraform is an Infrastructure as Code (IaC) tool developed by HashiCorp. You can use the open-source tool to define and provision your cloud infrastructure using the HashiCorp Configuration Language (HCL). Terraform uses the configuration files with the descriptions of the desired state of your infrastructure to create, manage, and delete the necessary resources across various cloud providers and platforms. Terraform providers are plug-ins that Terraform uses to interact with cloud providers, SaaS providers, third-party tools, and other APIs. Each provider defines its resources and data sources that map directly to the functionalities of the specific service.

For end-to-end operation of Terraform in VCF Automation for VM Apps, you can use the Terraform Provider for VMware Aria Automation to perform CRUD operations for resources in your VM Apps organization.

Terraform providers by VCF Automation role

Role	Terraform Provider	Resource
Organization Owner	Terraform Provider for VMware Aria Automation	• Blueprint • Catalog
Organization Member	Terraform Provider for VMware Aria Automation	Catalog deployment

Preparing a VCF Automation Terraform runtime environment

Designs that include Terraform configurations require access to a Terraform runtime environment that you integrate with VCF Automation for VM Apps.

Add a Terraform runtime

The runtime environment consists of a Kubernetes cluster that runs Terraform CLI commands to perform requested operations. In addition, the runtime collects logs and returns the results from Terraform CLI commands.

VCF Automation requires users to configure their own Terraform runtime Kubernetes cluster. Only one Terraform runtime per organization is supported. All Terraform deployments for that organization use the same runtime.

You must also set up a time sync service, such as NTP, so that the Kubernetes cluster time is synced.

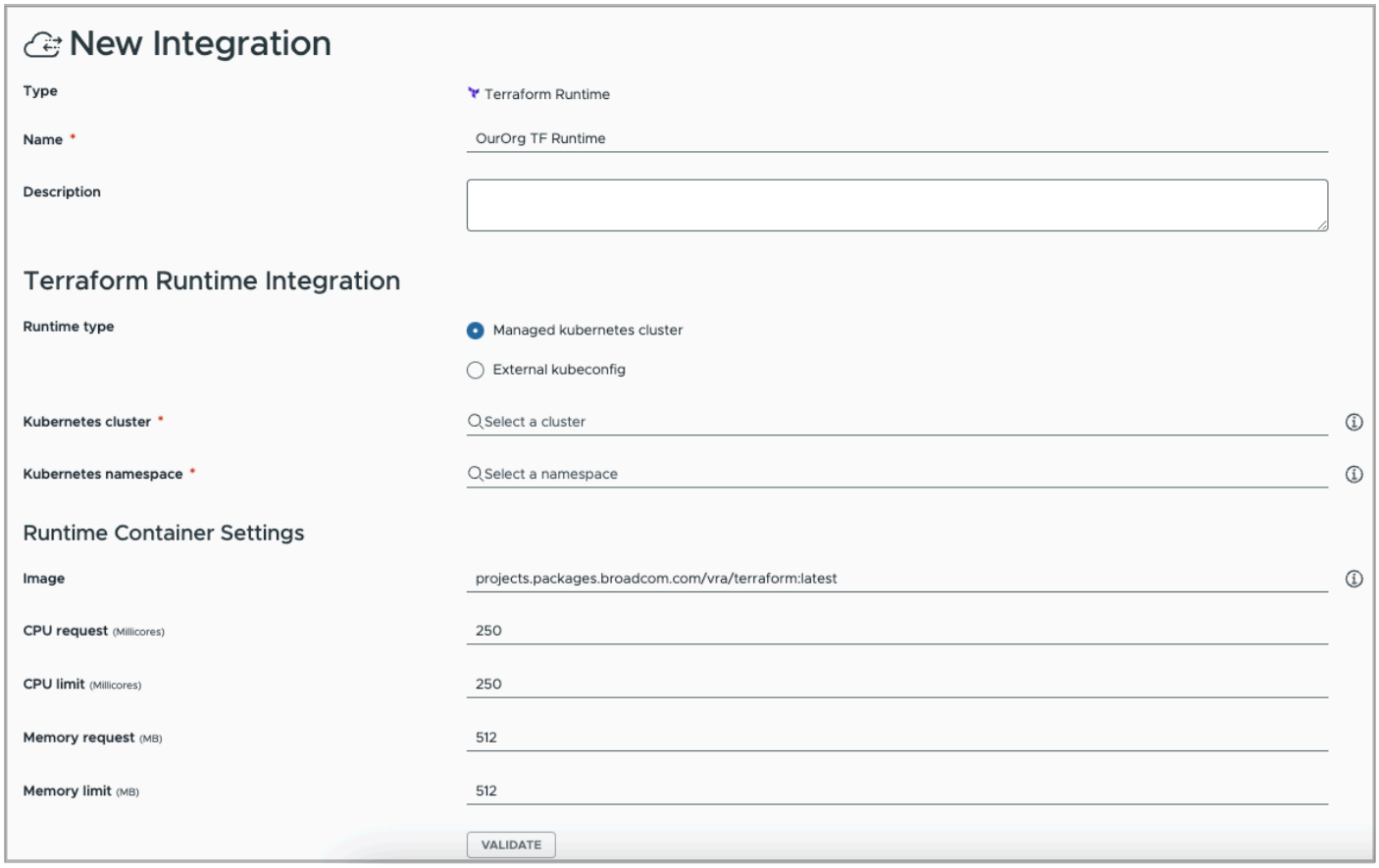
> **Note:**
>
> To run Terraform commands and states you need a Terraform runtime environment that runs on K8s. When you create your Terraform integration in VCF Automation, you can choose between a managed k8 (a k8s cluster that is already managed by VCF Automation) or an external k8s cluster, (a k8s cluster that is not managed by VCF Automation) and the namespace where the Terraform pods will be created. If you are using an external cluster, you must provide Kubeconfig access to the k8s cluster for the Terraform runtime target. Kubeconfig is a k8s standard and the Kubeconfig doesn't need to be an admin Kubeconfig. You can use a service account with minimum permissions, for example permission to run pods in a namespace and permission generate a Kubeconfig for use with the VCF Automation Terraform runtime integration.

1. Verify that you have a Kubernetes cluster on which to run the Terraform CLI.
   - All users can supply a kubeconfig file to run the Terraform CLI on an unmanaged Kubernetes cluster.
   - Enterprise license users have the option to run the Terraform CLI on a Kubernetes cluster managed by VCF Automation.

     In VCF Automation, go to **Infrastructure** > **Resources** > **Kubernetes**, and verify that you have a Kubernetes cluster.
2. If the Kubernetes cluster is newly added or modified, wait for its data collection to complete.

   Data collection retrieves the list of namespaces and other information, and might take up to 5 minutes depending on provider.
3. After data collection completes, go to **Infrastructure** > **Connections** > **Integrations** > **Add Integration**, and select the **Terraform Runtime** card.
4. Enter settings.

Figure 542: Example Terraform runtime integration



Setting	Description
Name	Give the runtime integration a unique name.
Description	Explain what the integration is for.
Terraform Runtime Integration:	
Runtime type (Enterprise only)	Enterprise license users may select whether to run the Terraform CLI on a Kubernetes cluster managed by VCF Automation or an unmanaged one.
Kubernetes kubeconfig (all users)	For an unmanaged Kubernetes cluster, paste in the entire contents of the kubeconfig file for the external cluster. This option is available for all users.
Kubernetes cluster (Enterprise only)	For Kubernetes managed by VCF Automation, select the cluster in which to run the Terraform CLI. The cluster and its kubeconfig file must be reachable. You can validate access to kubeconfig with a GET on `/cmx/api/resources/k8s/clusters/{clusterId}/kube-config`. This option is only available for Enterprise licenses.
Kubernetes namespace	Select the namespace to use within the cluster, for creating pods that run the Terraform CLI.
Runtime Container Settings:	

Setting	Description
Image	Enter the path to the container image of the Terraform version that you want to run, for example `projects.packages.broadcom.com/vra/terraform:latest` **Note:** If you previously entered a container image with `vmware.com` in the path, see KB 370092. The **VALIDATE** button does not check for the container image.
CPU request	Enter the amount of CPU for running containers. Default is to 250 millicores.
CPU limit	Enter the maximum allowable CPU for running containers. Default is to 250 millicores.
Memory request	Enter the amount of memory for running containers. Default is 512 MB.
Memory limit	Enter the maximum allowable memory for running containers. Default is 512 MB.

5. Click **VALIDATE** and adjust settings as needed.
6. Click **ADD**.

Settings are cached. After adding the integration, you can modify settings such as the cluster or namespace, but it might take up to 5 minutes for a change to be detected and for the Terraform CLI to run under the new settings.

Troubleshooting the Terraform runtime

Some Terraform configuration deployment problems might be related to the runtime integration.

Problem	Cause	Resolution
Validation fails with an error stating that the namespace is invalid.	You modified the cluster but left the previous namespace in the UI.	Always reselect a namespace after modifying the cluster selection.
The namespace drop down is empty or doesn't list newly added namespaces.	Data collection for the cluster has not completed. Data collection takes up to 5 minutes after entering or modifying the cluster and up to 10 minutes when entering or modifying the namespace.	For a new cluster with existing namespaces, wait up to 5 minutes for data collection to complete. For a new namespace in an existing cluster, wait up to 10 minutes for data collection to complete. If the problem continues, remove the cluster and re-add it under **Infrastructure** > **Resources** > **Kubernetes**.
Terraform CLI containers are created in a previous cluster, previous namespace, or with previous runtime settings, even after the integration account was updated.	The Kubernetes API client used by VCF Automation is cached for 5 minutes.	Changes might need up to 5 minutes to take effect.
Validation or a Terraform deployment operation fails with an error stating that kubeconfig is not available.	Sometimes these errors occur because the cluster isn't reachable from VCF Automation. In other cases, user credentials, tokens, or certificates are invalid.	The kubeconfig error can occur for a number of reasons and might require engagement with technical support for troubleshooting.

How to add proxy support

To have your external Kubernetes runtime cluster connect through a proxy server, follow these steps.

1. Log in to your external Kubernetes cluster server.
2. Create an empty folder.
3. In the new folder, add the following lines to a new file named Dockerfile.

```
FROM projects.registry.vmware.com/vra/terraform:latest as final
ENV https_proxy=protocol://username:password@proxy_host:proxy_port
ENV http_proxy=protocol://username:password@proxy_host:proxy_port
ENV no_proxy=.local,.localdomain,localhost
```

4. Modify the placeholder values so that the `https_proxy` and `http_proxy` environment variables include the proxy server settings that you use to access the internet.
   The *protocol* will be http or https according to what your proxy server uses, which might not match the environment variable name of `https_proxy` or `http_proxy`.
5. Save and close Dockerfile.
6. From the empty folder, run the following command. Depending on your account privileges, you might need to run the command in sudo mode.
   ```
   docker build --file Dockerfile --tag custom-terraform-runtime:1.0 .
   ```
   The command creates a local custom-terraform-runtime:1.0 Docker image.
7. In VCF Automation, under **Infrastructure** > **Connections** > **Integrations**, go to your Terraform runtime integration.
8. Create or edit the runtime container settings to use the custom-terraform-runtime:1.0 image:



VCF Automation Terraform runtime with no internet access

VCF Automation users who need to design and run Terraform integrations while disconnected from the internet can set up their runtime environment by following this example.

> **Note:**
> To obtain a source for image creation, setup involves briefly connecting to the internet. You might need to do those steps outside of your disconnected site if a temporary connection isn't possible.

This process assumes that you have your own Docker registry and can access its repositories without an internet connection.

Create the custom container image

1. Build a custom container image that includes the Terraform provider plug-in binaries.
   The following Dockerfile shows an example of creating a custom image with the Terraform GCP provider.
   The base image `projects.registry.vmware.com/vra/terraform:latest` download in the Dockerfile requires internet access to the VMware Harbor registry at projects.registry.vmware.com.
   Firewall settings or proxy settings can cause the image build to fail. You might need to enable access to releases.hashicorp.com to download the Terraform provider plug-in binaries. However, you may use your private registry to supply the plug-in binaries as an option.

```
FROM projects.registry.vmware.com/vra/terraform:latest as final

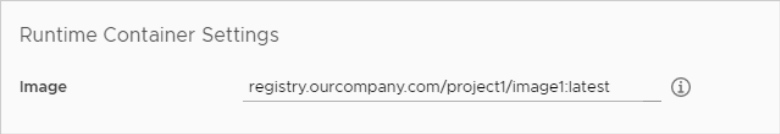
# Create provider plug-in directory
ARG plugins=/tmp/terraform.d/plugin-cache/linux_amd64
RUN mkdir -m 777 -p $plugins

# Download and unzip all required provider plug-ins from hashicorp to provider directory
RUN cd $plugins \
    && wget -q https://releases.hashicorp.com/terraform-provider-google/3.58.0/terraform-provider-
google_3.58.0_linux_amd64.zip \
    && unzip *.zip \
    && rm *.zip

# For "terraform init" configure terraform CLI to use provider plug-in directory and not download from
 internet
```

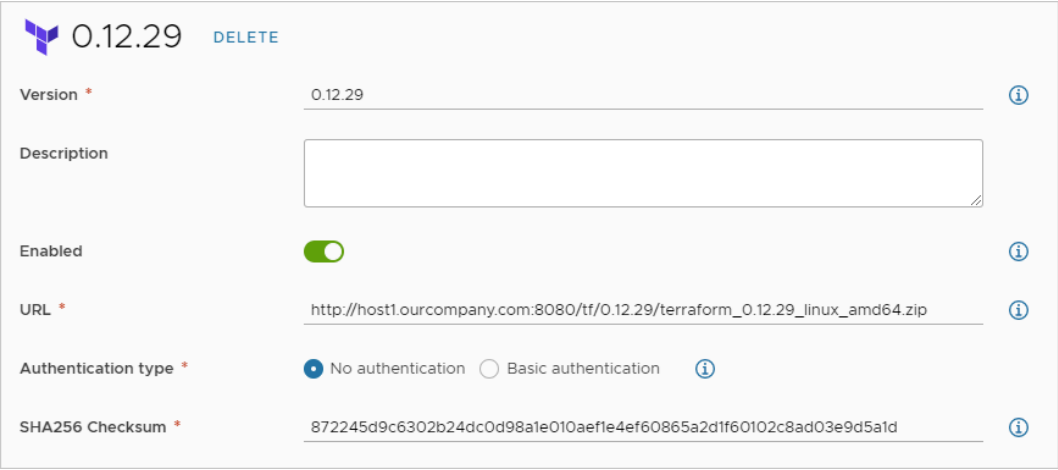
```
ENV TF_CLI_ARGS_init="-plugin-dir=$plugins -get-plugins=false"
```

2. Build, tag, and push the custom container image to your own Docker repository at your disconnected site.
3. In VCF Automation at your disconnected site, under **Infrastructure** > **Connections** > **Integrations**, go to your Terraform runtime integration.
4. Create or edit the runtime container settings to add your repository for the custom container image. The example built custom container image name is `registry.ourcompany.com/project1/image1:latest`.



Host the Terraform CLI locally

1. Download the Terraform CLI binaries.
2. Upload the Terraform CLI binaries to your local web or FTP server.
3. In VCF Automation, go to **Infrastructure** > **Configure** > **Terraform Versions**.
4. Create or edit the Terraform version so that it includes the URL to the Terraform CLI binaries hosted on your local server.
5. If your local web or FTP server requires login authentication, select **Basic authentication**, and enter username and password credentials that can access the server.
   To change the authentication type, you must have the cloud administrator role in VCF Automation.



Design and deploy Terraform configurations

With the runtime in place, you can add Terraform configuration files to git, design cloud templates for them, and deploy.

To get started, see Preparing for Terraform configurations in VCF Automation for VM Apps.

Troubleshooting

When deploying, open the deployment in VCF Automation. Under the History tab, look for Terraform events, and click **Show Logs** to the right. When your local Terraform provider is working, the following messages appear in the log.

```
Initializing provider plugins
```

```
Terraform has been successfully initialized
```

For a more robust log, you can manually edit the cloud template code to add `TF_LOG: DEBUG` as shown in the following example.

```
resources:
  terraform:
    type: Cloud.Terraform.Configuration
    properties:
```

```
providers:
  - name: google
    # List of available cloud zones: gcp/us-west1
    cloudZone: gcp/us-west1
environment:
  # Configure terraform CLI debug log settings
  TF_LOG: DEBUG
terraformVersion: 0.12.29
configurationSource:
  repositoryId: fc569ef7-f013-4489-9673-6909a2791071
  commitId: 3e00279a843a6711f7857929144164ef399c7421
  sourceDirectory: gcp-simple
```

Creating your own base image

Although VMware occasionally updates the base image at `projects.registry.vmware.com/vra/terraform:latest`, that image might be out of date and contain vulnerabilities.

To build your own base image, use the following Dockerfile instead.

```
FROM alpine:latest as final
RUN apk add --no-cache git wget curl openssh
```

Preparing for Terraform configurations in VCF Automation for VM Apps

Before you add a Terraform configuration to an VCF Automation template, set up and integrate your version control repository.

Prerequisites

For the VCF Automation on-premises product to run Terraform operations, you need the Terraform runtime integration. See Preparing a VCF Automation Terraform runtime environment.

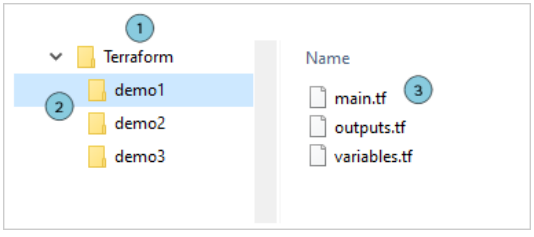
Store Terraform configuration files in a version control repository

VCF Automation supports the following version control repositories for Terraform configurations.

- GitHub cloud, GitHub Enterprise on-premises
- GitLab cloud, GitLab Enterprise on-premises
- Bitbucket on-premises

In your version control repository, create a default directory with one layer of subdirectories, each with Terraform configuration files. Create one subdirectory per Terraform configuration.

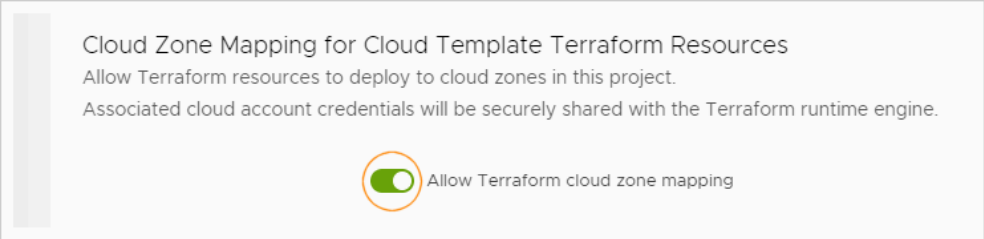
1. Default directory
2. Single subdirectory layer
3. Deployment-ready Terraform configuration files
   Don't include a Terraform state file with configuration files. If `terraform.tfstate` is present, errors occur during deployment.



Enable cloud zone mapping

If you expect to deploy to a cloud account, the Terraform runtime engine needs those cloud zone credentials.

In the project **Provisioning** tab, enable **Allow Terraform cloud zone mapping**.



Even though credentials are securely transmitted, for additional security, you should leave the option deactivated if project users don't need to deploy to a cloud account.

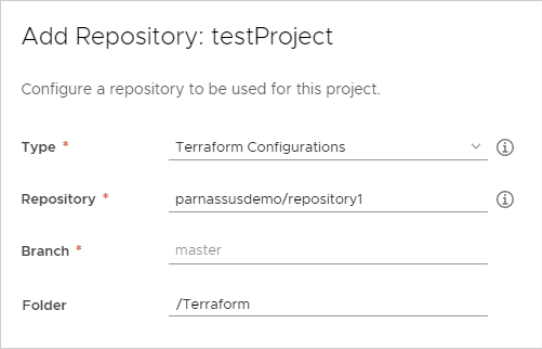
Integrate your repository with VCF Automation

In VCF Automation, go to **Infrastructure** > **Connections** > **Integrations**.

Add an integration to the repository offering type where you stored the Terraform configurations: GitHub, GitLab, or Bitbucket.

When you add your project to the integration, select the **Terraform Configurations** type, and identify the repository and branch.

Folder is the default directory of your earlier structure.



Designing for Terraform configurations in VCF Automation

With your repository and Terraform configuration files in place, you can design a VCF Automation template for them.

Prerequisites

Set up and integrate your version control repository. See Preparing for Terraform configurations in VCF Automation for VM Apps.

Enable Terraform runtime versions

You can define the Terraform runtime versions available to users when deploying Terraform configurations. Note that Terraform configurations might also include internally coded version constraints.

To create the list of allowable versions, go to **Infrastructure** > **Configure** > **Terraform Versions**.

Add Terraform resources to the design

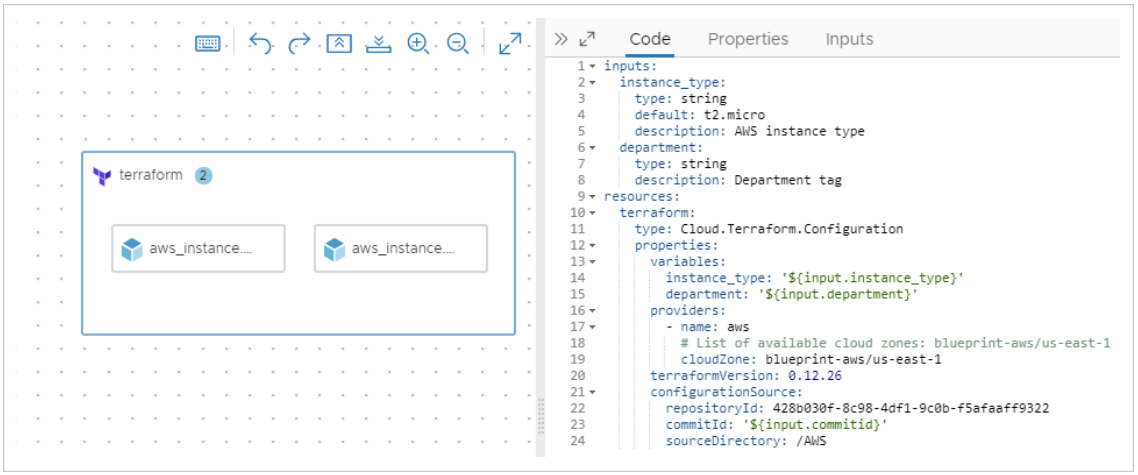
Create your cloud template that includes Terraform configurations.

1. In VCF Automation, go to **Design** > **Cloud Templates** and click **New from** > **Terraform**.
   The Terraform configuration wizard appears.

2. Follow the prompts.

Wizard Page	Setting	Value
New Cloud Template	Name	Give the design an identifying name.
	Description	Explain what the design is for.
	Project	Select the project that includes the repository integration where the Terraform configuration is stored.
Configuration Source	Repository	Select the integrated repository where you stored the Terraform configuration.
	Commit	Select a repository commit, or leave the entry blank to use the Terraform configuration from the repository head. Bitbucket Limitation—The number of selectable commits might be truncated because of the Bitbucket repository server configuration.
	Source directory	Select a subdirectory from the repository structure that you created. The example subdirectories shown in the earlier setup were demo1, demo2, and demo3. GitHub Enterprise Limitation—The source directory field is a plain text field in which you manually enter the name of the Terraform configuration subdirectory that you want.
Finalize Configuration	Repository	Verify the correct repository selection.
	Source directory	Verify the correct directory selection.
	Terraform version	Select the Terraform runtime version to run when deploying the Terraform configuration.
	Providers	If the Terraform configuration included a provider block, verify the provider and cloud zone that this cloud template will deploy to. Having no provider isn't a problem. After finishing the wizard, just edit the provider and cloud zone in the template properties to add or change the deployment target.
	Variables	Select sensitive values for encryption, such as passwords.
	Outputs	Verify the outputs from the Terraform configuration, which convert to expressions that your design code can further reference.

3. Click **Create**.
   The Terraform resource appears on the cloud template canvas, with VCF Automation code that reflects the Terraform configuration to deploy.



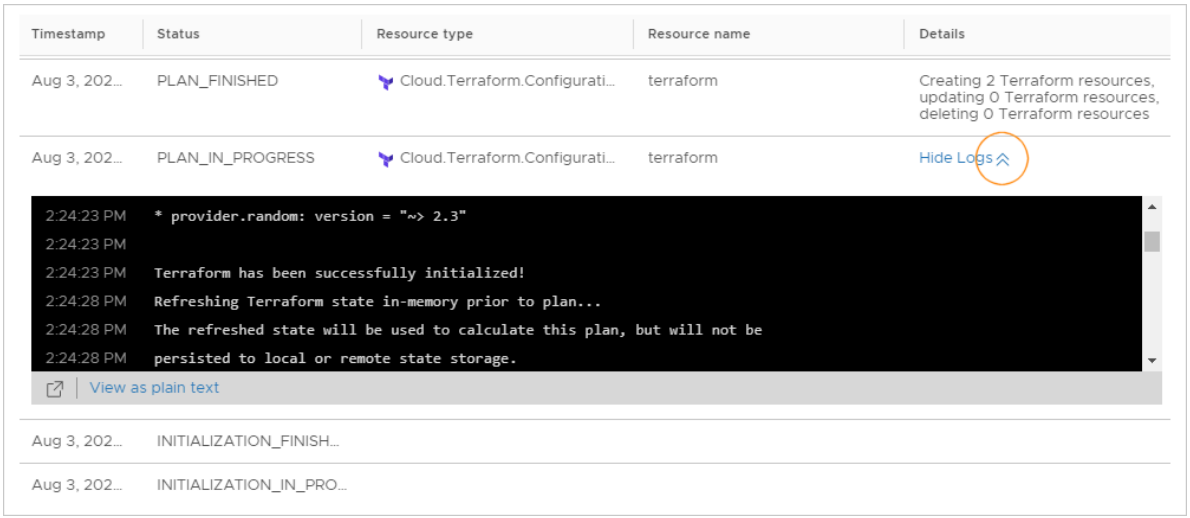
If desired, you can add other VCF Automation resources to the cloud template, to combine Terraform and non-Terraform code into a hybrid design.

> **Note:**
> Updating Terraform configurations in the repository doesn't synchronize the changes into your cloud template. Automatic synchronization can introduce security risks, such as newly added sensitive variables.
>
> To capture Terraform configuration changes, rerun the wizard, choose the new commit, and identify any new sensitive variables.

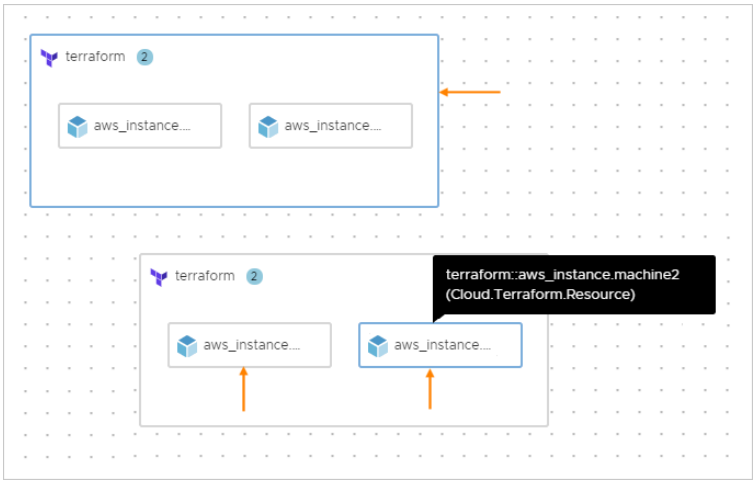
Deploy the cloud template

When you deploy the cloud template, the deployment **History** tab lets you expand an event such as an allocate or create phase, to inspect a log of messages from the Terraform CLI.

Approvals—In addition to the expected Terraform phases such as PLAN, ALLOCATE, or CREATE, VCF Automation introduces governance by means of an approval phase.



After deploying, you see an outer resource that represents the overall Terraform component, with child resources inside for the separate components that Terraform created. The parent Terraform resource controls the lifecycle of the child resources.



Using a secret VCF Automation property in a Terraform configuration

You can apply secret, encrypted values to Terraform configurations that you add to VCF Automation cloud template designs.

1. In your git repository, add a Terraform configuration source file that references the secret properties as variables.
   In this Terraform configuration source example, API and application keys are the secret variables.

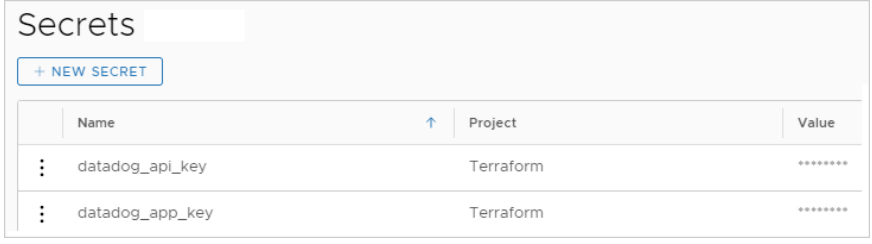
```
variable "datadog_api_key" {
  description = "Datadog API Key"
}

variable "datadog_app_key" {
  description = "Datadog App Key"
}
provider "datadog" {
  api_key = "${var.datadog_api_key}"
  app_key = "${var.datadog_app_key}"
}

# Create a new monitor
resource "datadog_monitor" "default" {
  # ...
}

# Create a new timeboard
resource "datadog_timeboard" "default" {
  # ...
}
```

2. In VCF Automation, go to **Infrastructure** > **Administration** > **Secrets**, and enter your secret property values.
   Add secret names and corresponding values. For the names, it's easiest to simply enter the same name as the variable name from your Terraform source.
   If needed, see Secret properties in VCF Automation for VM Apps for more details.



3. In VCF Automation, import the Terraform configuration for use in a cloud template.

Go to **Design** > **Cloud Templates** and click **New From** > **Terraform**.

Note:
Even though the variables appear for selection on the last page of the wizard, you do not need to set the secret variables as sensitive. Secret VCF Automation variables will already be encrypted and do not need the encryption that the wizard applies.

If needed, see Designing for Terraform configurations in VCF Automation for more details.

The example cloud template should look similar to the following code:

```
inputs:
  datadog_api_key:
    type: string
    description: Datadog API Key
  datadog_app_key:
    type: string
    description: Datadog App Key
resources:
  terraform:
    type: Cloud.Terraform.Configuration
    properties:
      variables:
        datadog_api_key: '${input.datadog_api_key}'
        datadog_app_key: '${input.datadog_app_key}'
      providers: []
      terraformVersion: 0.12.29
      configurationSource:
        repositoryId: 0fbf8f5e-54e1-4da3-9508-2b701gf25f51
        commitId: ed12424b249aa50439kr1c268942a4616bd751b6
        sourceDirectory: datadog
```

4. In the code editor, for the secret values, manually change `input` to `secret` as shown.

```
terraform:
  type: Cloud.Terraform.Configuration
  properties:
    variables:
      datadog_api_key: '${secret.datadog_api_key}'
      datadog_app_key: '${secret.datadog_app_key}'
```

5. In the `inputs:` section of the code, remove the input entries that were replaced by the bindings to secret properties.

Learn more about Terraform configurations in VCF Automation

Be aware of certain limitations and troubleshooting when you embed Terraform configurations as a resource in VCF Automation.

Limitations for Terraform configurations

- When validating a design with Terraform configurations, the TEST button checks VCF Automation syntax but not the native Terraform code syntax.
  In addition, the TEST button doesn't validate commit IDs associated with Terraform configurations.
- For a cloud template that includes Terraform configurations, cloning the template to a different project requires the following workaround.
  1. In the new project, under the **Integrations** tab, copy the `repositoryId` for your integration.
  2. Open the clone template. In the code editor, replace the `repositoryId` with the one you copied.
- In the version control repository, don't include a Terraform state file with configuration files. If `terraform.tfstate` is present, errors occur during deployment.

Supported day 2 actions for the parent Terraform resource

For the parent Terraform resource, you can view or refresh the Terraform state file. For more about the state file actions, see the comprehensive list of actions at What actions can I run on deployments or supported resources in VM Apps.

Supported day 2 actions for child resources

After deploying Terraform configurations, it might take up to 20 minutes for a day 2 action to become available on child resources.

For child resources in a Terraform configuration, only the following subset of day 2 actions are supported. For details about the actions, look them up in the comprehensive list of actions at What actions can I run on deployments or supported resources in VM Apps.

Provider	Terraform Resource Type	Supported Day 2 Actions
AWS	aws_instance	Power On
		Power Off
		Reboot
		Reset
Azure	azurerm_virtual_machine	Power On
		Power Off
		Restart
		Suspend
vSphere	vsphere_virtual_machine	Power On
		Power Off
		Reboot
		Reset
		Shutdown
		Suspend
		Create Snapshot
		Delete Snapshot
		Revert Snapshot
GCP	google_compute_instance	Power On
		Power Off
		Create Snapshot
		Delete Snapshot

Troubleshooting day 2 action availability

Out-of-the-box (OOTB) day 2 actions that are missing or deactivated might need troubleshooting.

Problem	Cause	Resolution
A Terraform resource does not have an expected OOTB day 2 action on the Actions menu.	The action might not be supported for the provider and resource type as mentioned in the previous list. Alternatively, the action might need up to 20 minutes to appear due to the timing of resource discovery and resource caching.	Check the provider and resource type in the design. Wait up to 20 minutes for data collection to complete.
A Terraform resource does not have an expected day 2 action even after the 20 minutes to account for data collection.	A resource discovery problem is preventing the action from appearing. One way that happens is when the resource is accidentally created on an out-of-project cloud zone. For example, your project only includes a cloud account and region us-east-1 cloud zone, but the Terraform configuration includes a provider block for us-west-1, and you didn't change it at design time. Another possibility is that data collection isn't working.	Check the project cloud zones against the cloud zones in the design. Go to **Infrastructure** > **Connections** > **Cloud Accounts** and check the data collection status and last successful collection time for the cloud account.
Even though there are no obvious problems with the resource state and data collection, a day 2 action is deactivated (gray).	Occasional, intermittent timing issues and data collection failures are known to occur.	The problem should resolve itself within 20 minutes.
The wrong day 2 action is deactivated, one that should be active based on the resource state. For example, Power Off is enabled, and Power On is deactivated, even though the resource was powered off using the provider interface.	Data collection timing can cause a temporary mismatch. If you change the power state from outside VCF Automation, it takes time to correctly reflect the change.	Wait up to 20 minutes.

Using custom Terraform providers in VCF Automation

If you want to use a custom Terraform provider, take the following steps.

In your git version control repository, under the Terraform directory that contains main.tf, add the following subdirectory structure and your custom Terraform provider ZIP file.

```
terraform.d/plugins/<HOSTNAME>/<NAMESPACE>/<TYPE>/terraform-provider-
<TYPE_VERSION_TARGET>.zip
```

For example, if you downloaded azurerm version 3.12.0, you create the following structure.

```
terraform.d/plugins/registry.terraform.io/hashicorp/azurerm/terraform-provider-
azurerm_3.12.0_linux_amd64.zip
```

Custom resource types for cloud templates in VCF Automation for VM Apps

When you create a cloud template in VCF Automation, the resource type palette includes resource types for the supported cloud account and integration endpoints. You might have use cases where you want to create cloud templates based on an expanded list of resource types. You can create custom resource types, add them to the design canvas, and create cloud templates that support your design and deployment needs.

Custom resource name and resource type

The custom resource name identifies your custom resource inside the cloud template resource type palette.

The resource type of a custom resource must begin with `Custom.` and each resource type must be unique. For example, you might set `Custom.ADUser` as a resource type for a custom resource that adds Active Directory users. Although the inclusion of `Custom.` is not validated in the text box, the string is automatically added if you remove it.

Extensibility action custom resources

With custom resource types, you can use extensibility actions in cloud templates to build complex applications. For example, you can use the native integration of extensibility actions with Amazon Web Services and Microsoft Azure to easily integrate with their respective services. You can create extensibility action custom resources by clicking on the **Based on** option in the custom resource editor and selecting **ABX user-defined schema**.

Lifecycle actions for extensibility action custom resources

When using a extensibility action for your custom resource, you can define the following lifecycle actions:

- **Create**: this extensibility action is called when a deployment is started.
- **Read**: this extensibility action is used to retrieve the latest state of the deployed resource.
- **Update**: this extensibility action is called when a cloud template property is updated. This action is triggered only when a property is not marked with `recreateOnUpdate`.
- **Destroy**: this extensibility action is called when a deployment is deleted.

These lifecycle actions can either be selected manually from your existing extensibility actions or generated automatically by selecting **Generate Actions**. When you select **Generate Actions** you must specify the project in which the new extensibility action will be generated in.

> **Note:**
> You can edit the extensibility actions associated with your lifecycle actions by clicking on the **Open** option next to the specific action.

VCF Operations Orchestrator Client custom resources

Each VCF Operations Orchestrator Client custom resource is based on a SDK inventory type and is created by a VCF Operations Orchestrator Client workflow that has an output which is an instance of your desired SDK type. Primitive types, such as `Properties`, `Date`, `string`, and `number` are not supported for the creation of custom resource types.

> **Note:**
> SDK object types can be differentiated from other VCF Operations Orchestrator Client property types by the colon (":") used to separate the plug-in name and the type name. For example, `AD:UserGroup` is an SDK object type used to manage Active Directory user groups.

You can use the built-in workflows in VCF Operations Orchestrator Client, or you can create your own. Using VCF Operations Orchestrator Client to create anything-as-a-service/XaaS workflows means that you can create a cloud template that adds an Active Directory user to machines at deployment time, or add a custom F5 load balancer to a deployment. You can create VCF Operations Orchestrator Client custom resources by clicking on the **Based on** option in the custom resource editor and selecting **vRO inventory**.

VCF Operations Orchestrator Client custom resource external type

The external type property defines the type of your VCF Operations Orchestrator Client custom resource. When you select a Create workflow in your custom resource type in VCF Automation, the external type drop-down appears underneath it. The drop-down includes external type properties, that are selected from the output parameters of the VCF Operations Orchestrator Client workflow. The selected workflow output properties included in the drop-down must be non-array SDK object types such as `VC:VirtualMachine` or `AD:UserGroup`.

> **Note:**
> When creating custom workflows that use the dynamic type plug-in, verify that their variables are defined by using the `DynamicTypesManager.getObject()` method.

When you define your custom resource types, you also define the scope of the availability of the select external type. The selected external type can be:

- Shared across projects.

- Available only for the selected project.

You can only have one custom resource type with a specific external type value per defined scope. For example, if you create a custom resource in your project that uses `VC:VirtualMachine` as an external type, you cannot create another custom resource for the same project that uses the same external type. You also cannot create two shared custom resources that use the same external type.

VCF Operations Orchestrator Client lifecycle action validation

When you add Create, Delete, and Update workflows as lifecycle actions to your custom resource, VCF Automation validates that the selected workflows have correct input and output property definitions.

- The Create workflow must have an output parameter that is an SDK object type, such as `SSH:Host` or `SQL:Database`. If the selected workflow does not pass the validation, you cannot add Update or Delete workflows, or save your changes to the custom resource.
- The Delete workflow must have an input parameter that is an SDK object type that matches the external type of the custom resource.
- The Update workflow must have both an input and output parameter that is an SDK object type that matches the external type of the custom resource.

Custom resource property schema

You can edit and view the custom resource properties schema by selecting the **Properties** tab. The schema includes the name, data type, property type, and, if it is available, the description of a given property. The schema also defines if a specific property is required or optional in the cloud template.

Note:
For the property schema of extensibility action custom resources, all properties are required in the cloud template.

When you add VCF Operations Orchestrator Client workflows to your custom resource, their input and output parameters are added as properties. For extensibility action custom resources, you must create the property schema of extensibility action custom resources manually in the **Properties** tab.

Note:
The properties defined in the schema of extensibility action custom resources must also be returned values in the **Read** action of your custom resource and have the same names. For example, if your schema includes the properties `domain` and `location`, the **Read** action must also return the same properties with the same names.

From this tab, you can also modify and format the properties of your VCF Operations Orchestrator Client or extensibility action based custom resources. For example, you can change the display name of a given property or add constraints.

Note:
When adding constraints to either the item section of array fields or properties section of objects fields in the properties schema, verify that you have validated these constraints as incorrectly applied constraints can cause issues with the custom resource. For example, when adding a maximum constraint to a numbers array, you must verify that this constraint does not break the property's default value.

You can edit the property schema for custom resources by navigating to the **Properties** tab and using either the **Code** or **Form** tab.

- **Code**: Edit the property schema by using YAML content. When selecting this option, the property schema is defined by using the JSON Schema.
- **Form**: For extensibility action custom resources, by clicking **New Property**, you create a new property by configuring its name, display name, description, property type, and default value. For VCF Operations Orchestrator Client custom resources, you can also hide non-required and non-computed properties from the schema by clicking **Remove Property**.

Day 2 Operation Custom Request Forms

You can streamline the request form of the day 2 operations included in your custom resource by adding and modifying different types of resource properties.

For example, you can bind the value of an input parameter in your request form to an external source, such as a VCF Operations Orchestrator Client action that retrieves a deployment name or project name. You can also bind the value of a specific input parameter to the computed value of two other text boxes included in the same request form.

> **Note:**
> This functionality is available for both custom resources and resource actions. You can customize the value of the input properties of your request form from the **Values** tab of the **Request Parameters** page of the custom resource or resource action editor.

Day 2 Operation Request Form Validation

You can validate the request form of your day 2 operations by adding an external validation. By using an external validation, you prevent the user from submitting the request form until the validation parameters are satisfied. You can add external validation from the **Validations** tab of the **Request Parameters** page of the custom resource or resource action editor. After selecting the tab, you can drag an **Orchestrator validation** element to the canvas and add a VCF Operations Orchestrator Client action that you want to use for validation.

For example, you can create a custom resource that includes a day 2 operation for changing a user password. For such a use case, you can add a VCF Operations Orchestrator Client action with `newPassword` and `confirmPassword` input parameters that use the `SecureString` type.

> **Note:**
> This is a sample script for validating a user password. For your own use case, you can decide to use a different script.

```
if (newPassword != confirmPassword) {
    return 'passwords are different';
}
if (newPassword.lenght < 7) {
    return 'password must be at least 10 symbols';
}
return null;
```

Adding VCF Operations Orchestrator Client actions as input properties

You must manually add any VCF Operations Orchestrator Client actions to your cloud template.

Create a custom resource and add it to a cloud template.

When creating a custom resource based on VCF Operations Orchestrator Client workflows, the resource schema is based on the data coming from VCF Operations Orchestrator Client. All workflow input and output properties are automatically added as custom resource properties. Any input properties associated with VCF Operations Orchestrator Client actions must be added manually.

1. Navigate to **Design** > **Cloud Templates** and select your cloud template.
2. Select the custom resource from the cloud template canvas.
3. Select the **Inputs** tab, and click **New Cloud Template Input**.
4. Enter a name for the new input parameter.
5. Under **Default value source**, select **External source**.
6. Click **Select**.
7. Enter the name of the VCF Operations Orchestrator Client action you want to add as an input parameter.
8. Verify the action parameters and click **Save**.
9. To finish adding the new input parameter, click **Create**.
10. Select the **Code** tab, and bind the input parameter to the resource property by using the `${input.prop}` method.

You have added a VCF Operations Orchestrator Client action as an input parameter to your custom resource.

How to create an VCF Automation template that adds users to Active Directory

In addition to the VCF Automation cloud template resources that you use when you create cloud templates, you can also create your own custom resources.

- Verify that you configured an VCF Operations Orchestrator Client integration. See Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps.
- Verify that the workflows that you are using for the create, update, destroy, and day 2 actions exist in VCF Operations Orchestrator Client and run successfully from there.
- In VCF Operations Orchestrator Client, locate the resource type used by the workflows. The workflows included in this custom resource must all use the same resource type. In this use case, the resource type is `AD:User`. For more information on resource type validation, see Custom resource types for cloud templates in VCF Automation for VM Apps.
- By using the built-in Active Directory workflows in your VCF Operations Orchestrator Client integration, configure an Active Directory server.
- Verify that you know how to configure and deploy a machine cloud template.

Custom resources are VCF Operations Orchestrator Client or extensibility action objects that you manage through VCF Automation with the lifecycle actions defined in the custom resource. The cloud template service automatically calls up the appropriate VCF Operations Orchestrator Client workflows or extensibility actions when the operation associated with a specific lifecycle action is triggered. You can extend the functionality of the resource type by also selecting VCF Operations Orchestrator Client workflows or extensibility actions that can be used as day 2 operations.

This use case uses built-in workflows that are provided in the VCF Operations Orchestrator Client library. It includes prescriptive values or strings to demonstrate how to perform the process. You can modify them to suit your environment.

For reference purposes, this use case uses a project named **DevOpsTesting**. You can replace this sample project with any project in your environment.

1. Create an Active Directory custom resource for adding a user in a group.

   This step adds the custom resource to the cloud template design canvas as a resources type.

   a) In VCF Automation, select **Design** > **Custom Resources**, and click **New Custom Resource**.

   b) Provide the following values.

      Remember, except for the workflow names, these are sample values.

Setting	Sample Value
Name	`AD user` This is the name that appears in the cloud template resource type palette.
Resource Type	`Custom.ADUser` The resource type must begin with `Custom.` and each resource type must be unique. Although the inclusion of `Custom.` is not validated in the text box, the string is automatically added if you remove it. This resource type is added to the resource type palette so that you can use it in the cloud template.

c) To enable this resource type in the cloud template resource type list, verify that **Activate** option is toggled on.
d) Select the **Scope** setting that makes the resource type available to any project.
e) Under **Based on**, verify that **vRO Inventory** is selected as the lifecycle action provider.
f) Select the workflows that define the resource and the day 2 actions.

Note:
The selected day 2 workflows must have an input parameter that is of the same type as the external type. The external type input is not displayed in the day 2 custom form requested by the user, as it is automatically bound to the custom resource.

Setting	Sample Value
Lifecycle Actions - Create	Select the **Create a user with a password in an organizational unit** workflow. If you have multiple VCF Operations Orchestrator Client integrations, select the workflow on the integration instance you use to run these custom resources. After selecting the workflow, the external type drop-down menu becomes available and is automatically set to `AD:User`. **Note:** An external source type can be used only once if shared and once per project. In this use case, you are providing the same custom resource for all the projects. It does mean that you cannot use `AD:User` for any other resource types for all projects. If you have other workflows that require the `AD:User` type, you must create individual custom resources for each project.
Lifecycle Actions - Destroy	Select the **Destroy a user** workflow.
Additional Actions	Select the **Change a user password** workflow. On the **Add Action** window, add a name for the action, such as `password_change` and click **Add**. To modify the action request form that the user responds to when they request the action, click the icon in the **Request Parameters** column. **Note:** For additional action workflows, verify that the workflow has a input parameter that is of the same type as the external type.

In this example, there is no appropriate application of an update workflow. A common example of an update workflow, which makes changes to the provisioned custom resource, is scaling in or scaling out a deployment.

g) Review the schema key and type values in the **Properties** tab so that you understand the workflow inputs so that you can configure the inputs in the cloud template.

The schema lists the required and optional input values defined in the workflow. The required input values are included in the cloud template YAML.

In the Create a user workflow, `accountName`, `displayName`, and `ouContainer` are required input values. The other schema properties are not required. You can also use the schema to determine where you want to create bindings to other field values, workflows, or actions. Bindings are not included in this use case.

h) To finish creating your custom resource, click **Create**.

2. Create a cloud template that adds the user to a machine when you deploy it.
   a) Select **Design** > **Cloud Templates**, and click **New from** > **Blank canvas**.
   b) Name the cloud template `Machine with an AD user`.
   c) Select the **DevOpsTesting** project and click **Create**.
   d) Add and configure a vSphere machine.
   e) From the custom resource list on the left of the cloud template design page, drag the **AD user** resource type onto the canvas.

      Note:
      You can select the custom resource by either scrolling down and selecting it from the left pane, or searching for it in the **Search Resource Types** text box. If the custom resource does not appear, click the refresh button next to the **Search Resource Types** text box.
   f) On the right, edit the YAML code to add the mandatory input values and the password.

      Add an `inputs` section in the code so that users can provide the name of the users that they are adding. In the following example, some of these values are sample data. Your values might be different.

```
inputs:
  accountName:
    type: string
    title: Account name
    encrypted: true
  displayName:
    type: string
    title: Display name
  password:
    type: string
    title: Password
    encrypted: true
  confirmPassword:
    type: string
    title: Password
    encrypted: true
  ouContainer:
    type: object
    title: AD OU container
    $data: 'vro/data/inventory/AD:OrganizationalUnit'
    properties:
        id:
            type: string
        type:
            type: string
```

   g) In the `resources` section, add `${input.`*`input-name`*`}` code to prompt for the user selection.

```
resources:
  Custom_ADUser_1:
    type: Custom.ADUser
    properties:
        accountName: '${input.accountName}'
        displayName: '${input.displayName}'
        ouContainer: '${input.ouContainer}'
        password: '${input.password}'
        confirmPassword: '${input.confirmPassword}'
```

3. Deploy the cloud template.
   a) On the cloud template designer page, click **Deploy**.
   b) Enter the **Deployment Name**`AD User Scott`.
   c) Select the **Cloud Template Version** and click **Next**.
   d) Complete the deployment inputs.
   e) Click **Deploy**.
4. Monitor the provisioning request on the **Deployments** page to ensure that the user is added to Active Directory and that the deployment is successful.

When your tested cloud template is working, you can then begin using the **AD user** custom resource with other cloud templates.

How to create an VCF Automation template that includes SSH

You can create custom resources that you can use to build cloud templates using VCF Operations Orchestrator Client workflows. In this use case, you add a custom resource that adds an SSH host. You can then include the resource in cloud templates. This procedure also adds an update workflow so that users change the SSH configuration after deployment rather than perform individual day 2 actions.

- Verify that you configured an VCF Operations Orchestrator Client integration. See Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps.
- Verify that the workflows that you are using for the create, update, destroy, and day 2 actions exist in VCF Operations Orchestrator Client and run successfully from there.
- In VCF Operations Orchestrator Client, locate the resource type used by the workflows. The workflows included in this custom resource must all use the same resource type. In this use case, the resource type is `SSH:Host`. For more information on resource type validation, see Custom resource types for cloud templates in VCF Automation for VM Apps.
- Verify that you know how to configure and deploy a machine cloud template.

Custom resources are VCF Operations Orchestrator Client or extensibility action objects that you manage through VCF Automation with the lifecycle actions defined in the custom resource. The cloud template service automatically calls up the appropriate VCF Operations Orchestrator Client workflows or extensibility actions when the operation associated with a specific lifecycle action is triggered. You can extend the functionality of the resource type by also selecting VCF Operations Orchestrator Client workflows or extensibility actions that can be used as day 2 operations.

This use case uses built-in workflows provided in the VCF Operations Orchestrator Client library. It includes prescriptive values or strings to demonstrate how to perform the process. You can modify them to suit your environment.

For reference purposes, this use case uses a project named **DevOpsTesting**. You can replace the project with one that you already have.

1. Create an SSH host custom resource for adding SSH to a cloud template.

   This step adds the custom resource to the cloud template design canvas as a resource type.
   a) In VCF Automation, select **Design** > **Custom Resources**, and click **New Custom Resource**.
   b) Provide the following values.

      Remember, except for the workflow names, these are sample values.

Table 1834:

Setting	Sample Value
Name	`SSH Host - DevOpsTesting Project` This is the name that appears in the cloud template resource type palette.
Resource Type	`Custom.SSHHost` The resource type must begin `Custom.` and each resource type must be unique. Although the inclusion of `Custom.` is not validated in the text box, the string is automatically added if you remove it. This resource type is added to the design canvas so that you can use it in the cloud template.

c) To enable this resource type in the cloud template resource type list, verify that **Activate** option is toggled on.
d) Select the **Scope** setting that makes the resource type available to the **DevOpsTesting** project.
e) Under **Based on**, verify that **vRO Inventory** is selected as the lifecycle action provider.
f) Select the workflows that define the resource.

Setting	Setting
Lifecycle Actions - Create	Select the **Add SSH Host** workflow. If you have multiple VCF Operations Orchestrator Client integrations, select the workflow on the integration instance you use to run these custom resources. After select the workflow, the external type drop-down menu becomes available and is automatically set to `SSH:Host`. An external source type can be used only once if share and once per project. In this use case, you are providing the custom resource for only the **DevOpsTesting** project. If you had other workflows that require the `SSH:Host` type, you must create individual custom resources for each project.
Lifecycle Actions - Update	Select the **Update SSH Host** workflow.
Lifecycle Actions - Destroy	Select the **Remove SSH Host** workflow.

g) Review the schema key and type values in the **Properties** tab so that you understand the workflow inputs so that you can configure the inputs in the cloud template.

   The schema lists the required and optional input values defined in the workflow. The required input values are included in the cloud template YAML.

   In the **Add SSH Host** workflow, `hostname`, `port`, and `username` are required input values. The other schema properties are not required. You can also use the schema to determine where you want to create bindings to other field values, workflows, or actions. Bindings are not included in this use case.
h) To finish creating your custom resource, click **Create**.

2. Create a cloud template that adds the SSH host when you deploy it.
   a) Select **Design** > **Cloud Templates**, and click **New from** > **Blank canvas**.
   b) Name the cloud template `Machine with SSH Host`.
   c) Select the **DevOpsTesting** project, and click **Create**.
   d) Add and configure a vSphere machine.
   e) From the custom resource list on the left of the cloud template design page, drag the **SSH Host - DevOpsTesting Project** resource type onto the canvas.

      Note:
      You can select the custom resource by either scrolling down and selecting it from the left pane, or searching for it in the **Search Resource Types** text box. If the custom resource does not appear, click the refresh button next to the **Search Resource Types** text box.

A reminder that the resource type is available because it was configured for the project. If you were creating a cloud template for another project, you cannot see the resource type.

f) On the right, edit the YAML code to add the mandatory input values.

Add an `inputs` section in the code so that users can provide the user name and the host name at deployment time. In this example, the port default is 22. In the following example, some of these values are sample data. Your values might be different.

```
inputs:
  hostname:
    type: string
    title: The hostname of the SSH Host
  username:
    type: string
    title: Username
```

g) In the `resources` section, add `${input.input-name}` code to prompt for the user selection.

```
resources:
  Custom_SSHHost_1:
    type: Custom.SSHHost
    properties:
      port: 22
      hostname: '${input.hostname}'
      username: '${input.username}'
```

3. Deploy the cloud template.
   a) On the cloud template designer page, click **Deploy**.
   b) Enter the **Deployment Name**`SSH Host Test`.
   c) Select the **Cloud Template Version** and click **Next**.
   d) Complete the deployment inputs.
   e) Click **Deploy**.
4. Monitor the provisioning request on the **Deployments** page to ensure that the SSH host is included in the deployment and that the deployment is successful.

When your tested cloud template is working, you can then begin using the `SSH Host` custom resource with other cloud templates.

VCF Automation designs that prepare for day 2 changes

In addition to the day 2 actions already associated with VCF Automation resource types, you have design options that let you prepare in advance for custom updates that users might need to make.

CAUTION:

To change a deployment, you can edit its cloud template and reapply it, or you can use day 2 actions. However, in most cases you should avoid mixing the two approaches.

Lifecycle day 2 changes such as power on/off are usually safe, but others require caution, such as when adding disks.

For example, if you add disks with a day 2 action, and then take a mixed approach by reapplying the cloud template, the cloud template could overwrite the day 2 change, which might remove disks and cause data loss.

Day 2 preparation can involve either direct use of cloud template code, or the VCF Automation design interface.

- You can use inputs in cloud template code so that, when you update the deployment or deployed resource, the interface prompts for fresh values.
- You can use VCF Automation to design a custom action based on a VCF Operations orchestrator workflow or an extensibility action. Running the custom action results in the workflow or extensibility action making changes to the deployment or deployed resource.

For custom day 2 resource actions you can use the schema for deployment resource types to specify criteria for your resource action. For example, you can use the deployment name property to define which deployments can use the create resource action.

How to use cloud template inputs for VCF Automation day 2 updates
When designing cloud templates, VCF Automation input parameters allow day 2 users to re-enter selections from the initial deployment request.

> **CAUTION:**
>
> Some property changes cause a resource to be re-created. For example, changing the `connection_string.name` under a `Cloud.Service.Azure.App.Service` deletes the existing resource and creates a new one.

For information on how to create inputs, see User input in requests in VCF Automation for VM Apps.

For a specific day 2 example, see the next section.

How to move a deployed machine to another network in VCF Automation for VM Apps
While maintaining deployments and networks, you might need the ability to relocate machines that you deployed with VCF Automation for VM Apps.

- The VCF Automation network profile must include all subnets that the machine will connect to. In VCF Automation, you can check networks by going to **Infrastructure** > **Configure** > **Network Profiles**.
  The network profile must be in an account and region that are part of the appropriate VCF Automation project for your users.
- Tag the two subnets with different tags. The example that follows assumes that `test` and `prod` are the tag names.
- The deployed machine must keep the same IP assignment type. It can't change from static to DHCP, or vice versa, while moving to another network.

For example, you might deploy to a test network first, then move to a production network. The technique described here lets you design a cloud template in advance to prepare for such day 2 actions. Note that the machine is moved. It isn't deleted and redeployed.

This procedure only applies to `Cloud.vSphere.Machine` resources. It won't work for cloud agnostic machines deployed to VCF Automation.

1. In VCF Automation for VM Apps, go to **Design**, and create a cloud template for the deployment.
2. In the inputs section of the code, add an entry that lets the user select a network.

```
inputs:
  net-tagging:
    type: string
    enum:
      - test
      - prod
    title: Select a network
```

3. In the resources section of the code, add the `Cloud.Network` and connect the machine to it.
4. Under the `Cloud.Network`, create a constraint that references the selection from the inputs.

```
resources:
  ABCServer:
    type: Cloud.vSphere.Machine
    properties:
      name: abc-server
      . . .
      networks:
       - network: '${resource["ABCNet"].id}'
  ABCNet:
```

```
type: Cloud.Network
properties:
  name: abc-network
  . . .
  constraints:
   - tag: '${input.net-tagging}'
```

5. Continue with your design, and deploy it as you normally would. At deployment, the interface prompts you to select the `test` or `prod` network.
6. When you need to make a day 2 change, go to **Resources** > **Deployments** > **Deployments**, and locate the deployment associated with the cloud template.
7. To the right of the deployment, click **Actions** > **Update**.
8. In the Update panel, the interface prompts you the same way, to select the `test` or `prod` network.
9. To change networks, make your selection, click **Next**, and click **Submit**.

How to create an VCF Automation resource action to vMotion a virtual machine

After you deploy a cloud template, you can run day 2 actions that change the deployment. VCF Automation includes many day 2 actions, but you might want to provide others. You can create custom resource actions and make them available to users as day 2 actions.

- Verify that you configured an VCF Operations Orchestrator Client integration. See Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps.
- Verify that the workflow that you are using for the day 2 action exists in VCF Operations Orchestrator Client and runs successfully there.

The custom resource actions can be based on VCF Operations Orchestrator Client workflows or extensibility actions. This example of a custom day 2 resource action is meant to introduce you to the creation process for VCF Operations Orchestrator Client based resource actions.

1. Create a custom resource action that uses vMotion to move a vSphere virtual machine from one host to another.
   a) In VCF Automation, select **Design** > **Resource Actions**, and click **New Resource Action**.
   b) Provide the following values.

      Remember, except for the workflow names, these are sample values.

Setting	Sample Value
Name	`vSphere_VM_vMotion` This is the name that appears in the resource actions list.
Display name	`Move VM` This is the name that users see in the deployment actions menu.

c) Click the **Activate** option to enable this action in the day 2 actions menu for resources that matches the resource type.
d) Select the resource type and workflow that define the day 2 action.

Setting	Sample Value
Resource Type	Select the **Cloud.vSphere.Machine** resource type. This is the resource type that is deployed as a cloud template component, not necessarily what is in the cloud template. For example, you might have a cloud agnostic machine in your cloud template, but when it is deployed on a vCenter Server, the machine is **Cloud.vSphere.Machine**. Because the action applies to the deployed type, do not use cloud agnostic types when you define your resource actions. In this example, vMotion only works for vSphere machines, but you might have other actions that you want to run on multiple resource types. You must create an action for each resource type.
Workflow	Select the **Migrate virtual machine with vMotion** workflow. If you have multiple VCF Operations Orchestrator Client integrations, select the workflow on the integration instance you use to run these custom resource actions.

2. Create a binding for the VCF Operations Orchestrator Client properties to the VCF Automation schema properties. VCF Automation day 2 actions support three types of bindings.

Binding type	Description
in request	The default value binding type. When selected, the input property is displayed in the request form and its value must be provided by the user at the request time.
with binding action	This option is available only for reference type inputs such as: • `VC:VirtualMachine` • `VC:Folder` The user selects an action that performs the binding. The selected action must return the same type as the input parameter. The correct property definition is `${properties.someProperty}`.
direct	This option is available for input properties that use primitive data types. When selected, the property, with the suitable type, is mapped directly from the schema of the input property. The user selects the property from the schema tree. Properties with different types are inactive.

In this use case, the binding is an VCF Operations Orchestrator Client action that makes the connection between VCF Operations Orchestrator Client `VC:VirtualMachine` input type used in the workflow and the VCF Automation`Cloud.vSphere.Machine` resource type. By setting up the binding, you make the day 2 action seamless for the user requesting the vMotion action on a vSphere VM machine. The system provides the name in the workflow so that the user does not have to do it.

a) After selecting the **Migrate virtual machine with vMotion** workflow, navigate to the **Property Binding** pane.
b) Select the binding of the `vm` input property.
c) Under **Binding**, select **with binding action**.

   The **findVcVmByVcAndVmUuid** action is automatically selected. This action comes preconfigured with your VCF Operations Orchestrator Client integration in VCF Automation.

d) Click **Save**.

3. To save the changes to your day 2 action, click **Create**.
4. To account for the other input parameters in the workflow, you can customize the request form that users see when they request the action.
   a) From **Resource Actions**, select your recently created day 2 action.
   b) Click **Edit Request Parameters**.
   You can customize how the request page is presented to users.

Default Field Name	Appearance	Values	Constraints
Destination resource pool for the virtual machine. Default is the current resource pool.	• Label = Target resource pool • Display type = Value Picker		
Destination host to which to migrate the virtual machine	• Label = Target host • Display type = Value Picker		Required = Yes
Priority of the migration task	Label = Priority of the task	Value options • Value source = Constant In the text box, enter a comma-separated list. `lowPriority|Low,defaultPriority|Default,highPriority|High`	Required = Yes
(Optional) Only migrate the virtual machine if its power on state matches the specified state	Delete this text box. vMotion can move machines in any power state.		

   c) Click **Save**.

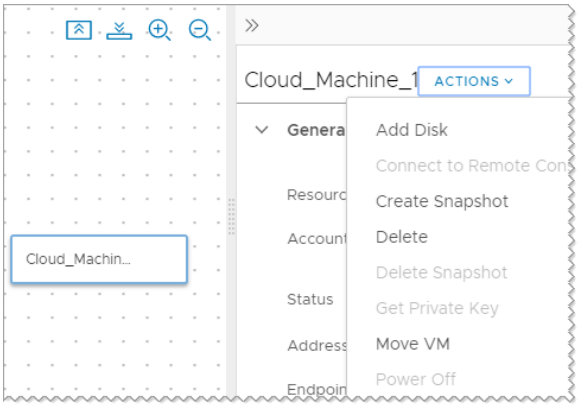
5. To limit when the action is available, you can configure the conditions.
   For example, you only want the vMotion action to be available when the machine has four or fewer CPUs.
   a) Toggle on **Requires condition**.
   b) Enter the condition.

Key	Operator	Value
${properties.cpuCount}	lessThan	4

If you need complex conditions, see How to build advanced conditions for VCF Automation custom actions.

c) Click **Update**.

6. Verify that the Move VM action is available for deployed machines that match the criteria.
   a) Select **Deployments**.
   b) Locate a deployment that includes a deployed machine that matches the defined criteria.
   c) Open the deployment and select the machine.
   d) Click actions in the right pane and verity that the `Move VM` action exists.

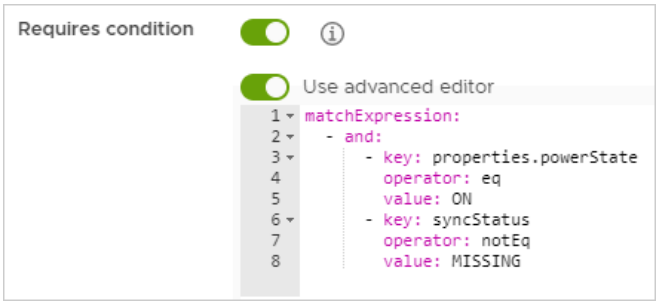


   e) Run the action.

How to build advanced conditions for VCF Automation custom actions

As an alternative to the simple conditions list in VCF Automation, the advanced editor lets you assemble more complex criteria expressions to control when the action is available.

When creating a new resource action, select **Requires condition** and **Use advanced editor**. Then, enter the criteria expression that you want.



The expression is a clause or list of clauses, each of which is in key-operator-value format. The preceding figure shows criteria where the target must be powered on and present.

Clauses

Clause	Description	Example

and	All subclauses need to be true for the expression result to be true.	Evaluate as true only when both properties.powerState is ON and syncStatus is not MISSING.
		`matchExpression:`
		`  - and:`
		`  - key: properties.powerState`
		`  operator: eq`
		`  value: ON`
		`  - key: syncStatus`
		`  operator: notEq`
		`  value: MISSING`

or	One or more of the subclauses need to be true for the expression result to be true.	Evaluate as true whether properties.powerState is ON or OFF.
		`matchExpression:`
		`  - or:`
		`  - key: properties.powerState`
		`  operator: eq`
		`  value: ON`
		`  - key: properties.powerState`
		`  operator: eq`
		`  value: OFF`

Operators

Operator	Description	Example

eq	Equal. Look for an exact match.	Evaluate as true when properties.powerState is ON.
		`matchExpression:`
		`  - and:`
		`  - key: properties.powerState`
		`  operator: eq`
		`  value: ON`

notEq	Not equal. Avoid an exact match.	Evaluate as true when properties.powerState is not OFF.
		`matchExpression:`
		`  - and:`
		`  - key: properties.powerState`
		`  operator: notEq`
		`  value: OFF`

Operator	Description	Example

hasAny	Look for a match in a collection of objects.	Evaluate as true when the storage.disks array includes a 100 IOPS EBS object.
		`matchExpression:`
		`- key: storage.disks`
		`operator: hasAny`
		`value:`
		`matchExpression:`
		`- and:`
		`- key: iops`
		`operator: eq`
		`value: 100`
		`- key: service`
		`operator: eq`
		`value: ebs`

in	Look for a match in a set of values.	Evaluate as true when properties.powerState is either OFF or SUSPEND.
		`matchExpression:`
		`- and:`
		`- key: properties.powerState`
		`operator: in`
		`value: OFF, SUSPEND`

notIn	Avoid matching a set of values.	Evaluate as true when properties.powerState is neither OFF nor SUSPEND.
		`matchExpression:`
		`- and:`
		`- key: properties.powerState`
		`operator: notIn`
		`value: OFF, SUSPEND`

greaterThan	Look for a match over a given threshold. Only applies to numeric values.	Evaluate as true when the first object in the storage.disks array has IOPS over 50.
		`matchExpression:`
		`- and:`
		`- key: storage.disks[0].iops`
		`operator: greaterThan`
		`value: 50`

lessThan	Look for a match under a given threshold. Only applies to numeric values.	Evaluate as true when the first object in the storage.disks array has IOPS under 200.
		`matchExpression:`
		`- and:`
		`- key: storage.disks[0].iops`
		`operator: lessThan`
		`value: 200`

greaterThanEquals	Look for a match at or above a given threshold. Only applies to numeric values.	Evaluate as true when the first object in the storage.disks array has IOPS of 100 or higher.
		`matchExpression:`
		`- and:`
		`- key: storage.disks[0].iops`
		`operator: greaterThanEquals`
		`value: 100`

Operator	Description	Example

lessThanEquals	Look for a match at or below a given threshold. Only applies to numeric values.	Evaluate as true when the first object in the storage.disks array has IOPS of 100 or lower.
		`matchExpression:`
		`  - and:`
		`  - key: storage.disks[0].iops`
		`  operator: lessThanEquals`
		`  value: 100`

matchesRegex	Use a regular expression to look for a match.	Evaluate as true when the properties.zone is us-east-1a or us-east-1c.
		`matchExpression:`
		`  - and:`
		`  - key: properties.zone`
		`  operator: matchesRegex`
		`  value: (us-east-1)+(a|c){1,2}`

Examples

The following criteria expression evaluates as true when properties.tags includes a tag of key `key1` and value `value1`.

The outer expression uses `hasAny` because properties.tags is an array, and you want to evaluate as true whenever key1=value1 appears in any of the key-value pairs in the array.

In the inner expression, there are two clauses, one for the key field and one for the value field. The properties.tags array holds key-value tagging pairs, and you need to match both the key and value fields.

```
matchExpression:
  - key: properties.tags
    operator: hasAny
    value:
      matchExpression:
        - and:
            - key: key
              operator: eq
              value: key1
            - key: value
              operator: eq
              value: value1
```

The following criteria expression is similar to the previous example, but now evaluates as true whenever properties.tags includes either a tag of key1=value1 or key2=value2.

```
matchExpression:
  - or:
      - key: properties.tags
        operator: hasAny
        value:
          matchExpression:
            - and:
                - key: key
                  operator: eq
                  value: key1
                - key: value
                  operator: eq
                  value: value1
      - key: properties.tags
        operator: hasAny
        value:
```

```
matchExpression:
  - and:
      - key: key
        operator: eq
        value: key2
      - key: value
        operator: eq
        value: value2
```

Virtual Machine reconciliation after vMotion migration

After migrating a virtual machine through vSphere vMotion, an event is triggered which perform a reconciliation in VCF Automation.

You can use vSphere vMotion to migrate your virtual machines without powering them off and as such avoiding having any downtime in your infrastructure.

VCF Automation supports the following migration scenarios:

- Migrating virtual machines in the vCenter instance.
- Migrating virtual machines across vCenter instances.

> **Note:**
>
> Migrating virtual machines outside of VCF Automation using a migration tool like HCX is not supported. See Broadcom knowledge base article 330182.

After initiating the migration from the vSphere Client, the data collection in VCF Automation registers the migration as started and triggers an event to reconcile the migration changes. You can use the **Compute post migration reconcile status** event topic to subscribe to this event. For more information on the event topic, go to Event topics provided with VCF Automation for VM Apps in.

After the migration finishes successfully, all the changes to the virtual machine are reconciled and can be viewed in the cloud template in which the virtual machine is added.

Requirements

Before performing a vMotion migration, be aware of the following requirements:

- When performing a migration across vCenter instances, verify that both instances are added as cloud zones to the same project used in the cloud template.
- When performing a migration across vCenter instances, be aware that tags associated with the virtual machine in the source vCenter are not reconciled after migrating to the target vCenter.
- When performing a migration across vCenter instances, day 2 network reconfiguration and scaling out are not supported.
- Migration reconciliation for existing NSX networks is supported. However, this covers only the network itself. Additional NSX constructs such as on-demand networks, security groups, and load balancers are not reconciled.
- Verify that the virtual machine you want to migrate is not part of a cluster or is sharing resources with other virtual machines such as networks or load balancers.
- When migrating a virtual machine that uses a First Class Disk (FCD) there are certain limitations on the day 2 actions that can be performed on the virtual machine.

Note:
When migrating a virtual machine across vCenter instances, an equivalent disk is created in the target vCenter and disk can be either a normal disk or FCD. Regardless of the outcome, VCF Automation continues to identify the newly created disk as a FCD after reconciliation.

Migration type	Resize	Resize Boot Disk	Resize Disk	Add Disk
Migrated in the same vCenter	Supported	Supported	Supported.	Supported
Migrated across vCenter instances	Supported	Supported	Supported. Resizes the FCD which is now associated with the new disk created in the target vCenter. The disk file is updated to point to the new location.	Not supported.

Migration process and reconciliation

1. Log in to the vSphere Client.
2. Select the virtual machine you want migrate.
3. Expand the **Actions** drop-down and select **Migrate**.
4. Select the relevant migration type for your use case.

   Note:
   To migrate a virtual machine across vCenter instances, select **Cross vCenter Server export**.
5. Configure the remaining migration options by selecting a compute resource, storage, network, and vMotion priority.

   Note:
   For migrations across vCenter instances, you must also provide the required details of the target vCenter.
6. To begin the migration, click **Finish**.
7. Log in to VCF Automation.
8. Navigate to **Resources** > **Deployments** and select the cloud template where your virtual machine is used.
9. Select the virtual machine and expand the **Custom properties section** drop-down.
10. The **migrationStatus** custom property displays the current status of the migration. The possible values of the property can be `STARTED`, `SUCCESS`, or `FAILED`, depending on the current progress and outcome of the migration.
11. The **migrationMessage** custom property can display messages relevant to the migration. For successful migrations, this property is empty. If the migration encounters an error, this property displays a message associated with the specific error.
12. After the migration finishes successfully, the changes to the virtual machine are now visible in the cloud template.

    Note:
    For example, if you migrated the virtual machine to a new host, this change is now reflected in the **Compute host** property under the **General** drop-down.

More VCF Automation template examples

Cloud template code in VCF Automation can be almost limitless in combination and application.

Often, an example of successful code is your best starting point for further development. When following an example, make substitutions in the template samples to apply your own settings in terms of resource types, values, and so on.

Network, security group, and load balancer resource examples in VCF Automation

You can use networking, security, and load balancer resources and settings in VCF Automation cloud templates.

- More about network resources in VCF Automation cloud templates

- More about security group and tag resources in VCF Automation cloud templates
- More about load balancer resources in VCF Automation cloud templates

These examples illustrate network, security, and load balancer resources within basic cloud template designs.

Networks

Resource scenario	Example cloud template design code

vSphere machine with multiple NICs connected to vSphere and NSX networks with DHCP IP assignment	`resources:`
	`demo-machine:`
	`type: Cloud.vSphere.Machine`
	`properties:`
	`  image: ubuntu`
	`  flavor: small`
	`  networks:`
	`  - network: ${resource["demo-vSphere-Network"].id}`
	`  deviceIndex: 0`
	`  - network: ${resource["demo-NSX-Network"].id}`
	`  deviceIndex: 1`
	`demo-vSphere-Network:`
	`type: Cloud.vSphere.Network`
	`properties:`
	`  networkType: existing`
	`demo-NSX-Network:`
	`type: Cloud.NSX.Network`
	`properties:`
	`  networkType: outbound`

NSX private network using the `vlanIds` property to specify an array of 3 VLANs - 123, 456, and 7	`formatVersion: 1`
	`inputs: {}`
	`resources:`
	`Cloud_Machine_1:`
	`type: Cloud.Machine`
	`properties:`
	`  image: test`
	`  flavor: test`
	`  networks:`
	`  - network: '${resource.Cloud_NSX_Network_1.id}'`
	`Cloud_NSX_Network_1:`
	`type: Cloud.NSX.Network`
	`properties:`
	`  networkType: private`
	`  vlanIds:`
	`  - 123`
	`  - 456`
	`  - 7`

Resource scenario	Example cloud template design code

Add a private network with a static IP address for an Azure VM deployment	`formatVersion: 1`
	`inputs: {}`
	`resources:`
	`Cloud_Azure_Machine_1:`
	`type: Cloud.Machine`
	`properties:`
	`  image: photon`
	`  flavor: Standard_B1ls`
	`  networks:`
	`  - network: '${resource.Cloud_Network_1.id}'`
	`  assignment: static`
	`  address: 10.0.0.45`
	`  assignPublicIpAddress: false`
	`Cloud_Network_1:`
	`type: Cloud.Network`
	`properties:`
	`  networkType: existing`

You can use a static IP assignment with VCF Automation IPAM (internal as supplied with VCF Automation or external based on the VCF Automation IPAM SDK such as for one of the Infloblox plug-ins available in the VMware Marketplace). Other uses of `assignment: static` are not supported, as described in the *Caveats* section of More about network resources in VCF Automation cloud templates.	`resources:`
	`demo_vm:`
	`type: Cloud.vSphere.Machine`
	`properties:`
	`  image: 'photon'`
	`  cpuCount: 1`
	`  totalMemoryMB: 1024`
	`  networks:`
	`  - network: ${resource.demo_nw.id}`
	`  assignment: static`
	`demo_nw:`
	`type: Cloud.vSphere.Network`
	`properties:`
	`  networkType: existing`

Resource scenario	Example cloud template design code
Add or edit NAT and DNAT port forwarding rules in a `Cloud.NSX.NAT` resource for an existing deployment.	<pre>resources: gw: type: Cloud.NSX.Gateway properties: networks: - ${resource.akout.id} nat: type: Cloud.NSX.Nat properties: networks: - ${resource.akout.id} natRules: - translatedInstance: ${resource.centos.networks[0].id} index: 0 protocol: TCP kind: NAT44 type: DNAT sourceIPs: any sourcePorts: 80 translatedPorts: 8080 destinationPorts: 8080 description: edit - translatedInstance: ${resource.centos.networks[0].id} index: 1 protocol: TCP kind: NAT44 type: DNAT sourceIPs: any sourcePorts: 90 translatedPorts: 9090 destinationPorts: 9090 description: add gateway: ${resource.gw.id} centos: type: Cloud.vSphere.Machine properties: image: WebTinyCentOS65x86 flavor: small customizationSpec: Linux networks: - network: ${resource.akout.id} assignment: static akout: type: Cloud.NSX.Network properties: networkType: outbound constraints: - tag: nsxt-nat-1-M2</pre>

Resource scenario	Example cloud template design code

Public cloud machine to use an internal IP instead of a public IP. This example uses a specific network ID.
Note: The `network:` option is used in the `networks:` setting to specify a target network ID. The `name:` option in the `networks:` setting has been deprecated and should not be used.
	`resources:`
	`wf_proxy:`
	`type: Cloud.Machine`
	`properties:`
	`  image: ubuntu 16.04`
	`  flavor: small`
	`  constraints:`
	`  - tag: 'platform:vsphere' networks:- network: '${resource.wf_net.id}'assignPublicIpAddress: false`

Routed network using the NSX network resource type.	`Cloud_NSX_Network_1:`
	`type: Cloud.NSX.Network`
	`properties:`
	`  networkType: routed`

Add a tag to a machine NIC resource in the cloud template.	`formatVersion: 1`
	`inputs: {}`
	`resources:`
	`Cloud_Machine_1:`
	`type: Cloud.vSphere.Machine`
	`properties:`
	`flavor: small`
	`image: ubuntu`
	`networks:`
	` - name: '${resource.Cloud_Network_1.name}'`
	` deviceIndex: 0`
	` tags:`
	`  - key: 'nic0'`
	`  value: null`
	`  - key: internal`
	`  value: true`
	` - name: '${resource.Cloud_Network_2.name}'`
	` deviceIndex: 1`
	` tags:`
	`  - key: 'nic1'`
	`  value: null`
	`  - key: internal`
	`  value: false`

Tag NSX logical switches for an outbound network.
Tagging is supported for NSX and VMware Cloud on AWS.
	`Cloud_NSX_Network_1:`
	`type: Cloud.NSX.Network`
	`properties:`
	`  networkType: outbound`
	`  tags:`
	`  - key: app`
	`  value: opencart`

Security groups

Resource scenario	Example cloud template design code

Existing security group with a constraint tag applied to a machine NIC.
To use an existing security group, enter *existing* for the `securityGroupType` property.
You can assign tags to a `Cloud.SecurityGroup` resource to allocate existing security groups by using tag constraints. Security groups that do not contain tags cannot be used in the cloud template design.
Constraint tags must be set for `securityGroupType: existing` security group resources. Those constraints must match the tags set on the existing security groups.
Constraint tags cannot be set for `securityGroupType: new` security group resources.
	`formatVersion: 1`
	`inputs: {}`
	`resources:`
	`allowSsh_sg: type: Cloud.SecurityGroup`
	`properties: securityGroupType: existingconstraints:- tag: allowSsh`
	`compute:`
	`type: Cloud.Machine`
	`properties:`
	`  image: centos`
	`  flavor: small`
	`  networks:`
	`    - network: '${resource.prod-net.id}'`
	`      securityGroups:`
	`        - '${resource.allowSsh_sg.id}'`
	`prod-net:`
	`type: Cloud.Network`
	`properties:`
	`  networkType: existing`

On-demand security group with two firewall rules illustrating the `Allow` and `Deny` access options.	`resources:`
	`Cloud_SecurityGroup_1: type: Cloud.SecurityGroup`
	`properties: securityGroupType: new`
	`  rules:`
	`    - ports: 5000`
	`      source:`
	`'fc00:10:000:000:000:56ff:fe89:48b4' access: Allow`
	`      direction: inbound name: allow_5000`
	`      protocol: TCP`
	`    - ports: 7000`
	`      source:`
	`'fc00:10:000:000:000:56ff:fe89:48b4' access: Deny`
	`      direction: inbound name: deny_7000`
	`      protocol: TCP`
	`Cloud_vSphere_Machine_1:`
	`type: Cloud.vSphere.Machine`
	`properties:`
	`  image: photon`
	`  cpuCount: 1`
	`  totalMemoryMB: 256`
	`  networks:`
	`    - network: '${resource.Cloud_Network_1.id}'`
	`      assignIPv6Address: true`
	`      assignment: static`
	`      securityGroups:`
	`        - '${resource.Cloud_SecurityGroup_1.id}'`
	`Cloud_Network_1:`
	`type: Cloud.Network`
	`properties:`
	`  networkType: existing`

Resource scenario	Example cloud template design code
Complex cloud template with 2 security groups, including: • 1 existing security group • 1 on-demand security group with multiple firewall rule examples • 1 vSphere machine • 1 existing network This sample illustrates different combinations of protocols and ports, services, IP CIDR as source and destination, IP range as source or destination, and the options for any, IPv6, and (::/0). For machine NICs, you can specify the connected network, and security group(s). You can also specify the NIC index or an IP address.	formatVersion: 1 inputs: {} resources: DEMO_ESG :*existing security group - security group 1)***type: Cloud.SecurityGroup** properties: constraints: - tag: BlockAll **securityGroupType:** **existing**(*designation of existing for security group 1*) DEMO_ODSG: (*on-demand security group - security group 2)*) **type: Cloud.SecurityGroupproperties:rules:**(*multiple firewall rules in this section*) - **name: IN-ANY**(*rule 1*) source: any service: any direction: inbound **access: Deny- name: IN-SSH**(*rule 2*) source: any service: SSH direction: inbound **access: Allow- name: IN-SSH-IP**(*rule 3*) source: 33.33.33.1-33.33.33.250 protocol: TCP ports: 223 direction: inbound **access: Allow- name: IPv-6-ANY-SOURCE**(*rule 4*) source: '::/0' protocol: TCP ports: 223 direction: inbound **access: Allow- name: IN-SSH-IP**(*rule 5*) source: 44.44.44.1/24 protocol: UDP ports: 22-25 direction: inbound **access: Allow- name: IN-EXISTING-SG**(*rule 6*) **source: '${resource["DEMO_ESG"].id}'** protocol: ICMPv6 direction: inbound **access: Allow- name: OUT-ANY**(*rule 7*) destination: any service: any direction: outbound **access: Deny- name: OUT-TCP-IPv6**(*rule 8*) destination: '2001:0db8:85a3::8a2e:0370:7334/64' protocol: TCP ports: 22 direction: outbound **access: Allow- name: IPv6-ANY-DESTINATION**(*rule 9*) destination: '::/0' protocol: UDP ports: 23 direction: outbound **access: Allow- name: OUT-UDP-SERVICE**(*rule 10*) destination: any service: NTP direction: outbound **access: AllowsecurityGroupType:** **new**(*designation of on-demand for security group 2*) DEMO_VC_MACHINE: (*machine resource*) **type:** **Cloud.vSphere.Machine** properties:

Load balancers

Resource scenario	Example cloud template design code

Specify a load balancer logging level, algorithm, and size.	Sample NSX load balancer showing use of logging level, algorithm, and size:
	`resources:`
	`Cloud_LoadBalancer_1: type: Cloud.NSX.LoadBalancer`
	`properties:`
	`name: myapp-lb`
	`network: '${appnet-public.name}'`
	`instances: '${wordpress.id}'`
	`routes:`
	`- protocol: HTTP port: '80' loggingLevel: CRITICALalgorithm: LEAST_CONNECTIONtype: MEDIUM`

Associate a load balancer with a named machine or a named machine NIC. You can specify either `machine ID` or `machine network ID` to add the machine to the load balancer pool. The instances property supports both machines (`machine by ID`) and NICs (`machine by network ID`).
In the first example, the deployment uses the `machine by ID` setting to load balance the machine when it is deployed on any network.
In the second example, the deployment uses the `machine by network ID` setting to load balance the machine only when the machine is deployed on the named machine NIC.
The third example shows both settings used in the same `instances` option.
	You can use the `instances` property to define a machine ID or a machine network ID:
	• Machine ID
	`Cloud_LoadBalancer_1:`
	`type: Cloud.LoadBalancer`
	`properties:`
	`network: '${resource.Cloud_Network_1.id}'`
	`instances: '${resource.Cloud_Machine_1.id}'`
	• Machine network ID
	`Cloud_LoadBalancer_1:`
	`type: Cloud.LoadBalancer`
	`properties:`
	`network: '${resource.Cloud_Network_1.id}'`
	`instances:`
	`'${resource.Cloud_Machine_1.networks[0].id}'`
	• One machine specified for load balancer inclusion and another machine NIC specified for load balancer inclusion:
	`instances:`
	`- resource.Cloud_Machine_1.id`
	`- resource.Cloud_Machine_2.networks[2].id`

Add health check settings to an NSX load balancer. Additional options include `httpMethod`, `requestBody`, and `responseBody`.	`myapp-lb: type: Cloud.NSX.LoadBalancer`
	`properties:`
	`name: myapp-lb`
	`network: '${appnet-public.name}'`
	`instances: '${wordpress.id}'`
	`routes:`
	`- protocol: HTTP`
	`port: '80'`
	`algorithm: ROUND_ROBIN`
	`instanceProtocol: HTTP`
	`instancePort: '80' healthCheckConfiguration :protocol: HTTPport: '80'urlPath: /mywordpresssite/wp-admin/install.phpintervalSeconds: 60timeoutSeconds: 10unhealthyThreshold: 10healthyThreshold: 2`
	`connectionLimit: '50'`
	`connectionRateLimit: '50'`
	`maxConnections: '500'`
	`minConnections: ''`
	`internetFacing: true{code}`

Resource scenario	Example cloud template design code
On-demand network with a 1-arm load balancer.	inputs: {} resources: mp-existing: type: Cloud.Network properties: name: mp-existing networkType: existing mp-wordpress: type: Cloud.vSphere.Machine properties: name: wordpress count: 2 flavor: small image: tiny customizationSpec: Linux networks: - network: '${resource["mp-private"].id}' mp-private: type: Cloud.NSX.Network properties: name: mp-private networkType: private constraints: - tag: nsxt mp-wordpress-lb: type: Cloud.LoadBalancer properties: name: wordpress-lb internetFacing: false network: '${resource.mp-existing.id}' instances: '${resource["mp-wordpress"].id}' routes: - protocol: HTTP port: '80' instanceProtocol: HTTP instancePort: '80' healthCheckConfiguration: protocol: HTTP port: '80' urlPath: /index.pl intervalSeconds: 60 timeoutSeconds: 30 unhealthyThreshold: 5 healthyThreshold: 2

Resource scenario	Example cloud template design code
Existing network with a load balancer.	formatVersion: 1 inputs: count: type: integer default: 1 resources: ubuntu-vm: type: Cloud.Machine properties: name: ubuntu flavor: small image: tiny count: '${input.count}' networks: - network: '${resource.Cloud_NSX_Network_1.id}' Provider_LoadBalancer_1: type: Cloud.LoadBalancer properties: name: OC-LB routes: - protocol: HTTP port: '80' instanceProtocol: HTTP instancePort: '80' healthCheckConfiguration: protocol: HTTP port: '80' urlPath: /index.html intervalSeconds: 60 timeoutSeconds: 5 unhealthyThreshold: 5 healthyThreshold: 2 network: '${resource.Cloud_NSX_Network_1.id}' internetFacing: false instances: '${resource["ubuntu-vm"].id}' Cloud_NSX_Network_1: type: Cloud.NSX.Network properties: networkType: existing constraints: - tag: nsxt24prod

More about network resources in VCF Automation cloud templates

As you create or edit your VCF Automation cloud templates, use the most appropriate network resources for your objectives. Learn about the NSX and cloud-agnostic network options that are available in the cloud template.

Select one of the available network resource types based on machine and related conditions in your VCF Automation cloud template.

Cloud agnostic network resource

You add a cloud agnostic network by using the **Cloud Agnostic** > **Network** resource on the cloud template **Design** page. The resource displays in the cloud template code as a `Cloud.Network` resource type. The default resource displays as:

```
Cloud_Network_1:
  type: Cloud.Network
```

```
  properties:
    networkType: existing
```

Use a cloud agnostic network when you want to specify networking characteristics for a target machine type that is not, or might not, be connected to an NSX network.

The cloud agnostic network resource is available for these resource types:

- Cloud agnostic machine
- vSphere
- Google Cloud Platform (GCP)
- Amazon Web Services (AWS)
- Microsoft Azure
- VMware Cloud on AWS (VMC)

The cloud agnostic network resource is available for these network type (`networkType`) settings:

- public
- private
- outbound
- existing

vSphere network resource

You add a vSphere network by using the **vSphere** > **Network** resource on the cloud template **Design** page. The resource displays in the cloud template code as a `Cloud.vSphere.Network` resource type. The default resource displays as:

```
Cloud_vSphere_Network_1:
  type: Cloud.vSphere.Network
  properties:
    networkType: existing
```

Use a vSphere network when you want to specify networking characteristics for a vSphere machine type (`Cloud.vSphere.Machine`).

The vSphere network resource is only available for a `Cloud.vSphere.Machine` machine type.

The vSphere resource is available for these network type (`networkType`) settings:

- public
- private
- existing

For examples, see Using network settings in network profiles and cloud templates in VCF Automation.

NSX network resource

You add an NSX network by using the **NSX** > **Network** resource on the cloud template **Template** page. The resource displays in the cloud template code as a `Cloud.NSX.Network` resource type. The default resource displays as:

```
Cloud_NSX_Network_1:
  type: Cloud.NSX.Network
  properties:
    networkType: existing
```

Use an NSX network when you want to attach a network resource to one or more machines that have been associated to an NSX cloud account. The NSX network resource allows you to specify NSX networking characteristics for a vSphere machine resource that is associated to an NSX cloud account.

The `Cloud.NSX.Network` resource is available for these network type (`networkType`) settings:

- public
- private
- outbound

- existing
- routed - Routed networks are only available for NSX-V and NSX.

If you want multiple outbound or routed networks to share the same NSX Tier-1 router or NSX-V Edge Service Gateway (ESG), connect a single NSX gateway resource (`Cloud.NSX.Gateway`) to the connected networks in the template prior to initial deployment. If you add the gateway after deployment as a day 2 or iterative development operation, each network creates its own router.

You can use the NSX NAT resource in the template to support NAT and DNAT port forwarding rules.

Machine tags are defined in the cloud template and apply to the machine resource if it is deployed on vCenter. Machine tags are also applied to the NSX network if the machine resource is connected to an NSX network, including an NSX global network. NSX global networks are networks that are defined by the NSX global manager and apply to one or more NSX local managers. Note that machine tagging is different than machine NIC (network interface) tagging.

Cloud agnostic network resource with Azure, AWS, or GCP deployment intent

Public cloud provider VMs can require specific cloud template property combinations that are not necessarily required in NSX or vSphere-based machine deployments. For examples of cloud template code that support some of these scenarios, see Network, security group, and load balancer resource examples in VCF Automation.

NSX gateway resource

You can reuse or share a single NSX Tier-1 router or NSX-V Edge Service Gateway (ESG) within a single deployment by using a gateway resource (`Cloud.NSX.Gateway`) in the cloud template. The gateway resource represents the Tier-1 or ESG and can be connected to multiple networks in the deployment. The gateway resource can be used with outbound or routed networks only.

The `Cloud.NSX.Gateway` resource allows you to share the NSX Tier-1 router or NSX-V Edge Service Gateway (ESG) among connected outbound or routed networks in a deployment.

The gateway is often attached to a single outbound or routed network. However, if the gateway is attached to multiple networks, the networks must be of the same type, for example all outbound or all routed. The gateway can be connected to multiple machines or load balancers that are connected to the same outbound or routed networks. The gateway must be connected to a load balancer on the shared on-demand network so that it can reuse the NSX Tier-1 router or NSX-V Edge Service Gateway (ESG) created by the gateway.

To allow multiple outbound or routed networks to share the same T1 router or Edge, connect a single `Cloud.NSX.Gateway` gateway resource to all the networks initially. All the intended networks and the single gateway must be connected together before you deploy the cloud template, otherwise each network creates its own router.

For an NSX network that contains an associated compute gateway resource, the gateway settings are applied to all associated networks in the deployment. A single NSX Tier-1 logical router is created for each deployment and shared by all on-demand networks and load balancers in the deployment. A single NSX-V Edge is created for each deployment and shared by all the on-demand networks and load balancers in the deployment.

You can attach the gateway resource to a network as an iterative deployment update. However, doing so does not create a Tier-1 or Edge router - the initial network deployment creates the router.

For NSX networks that do not use an associated gateway resource, multiple on-demand networks in the cloud template continue to create multiple Tier-1 logical routers in the deployment.

If the gateway contains NAT rules, you can reconfigure or delete the NAT or DNAT rules for the Tier 1 router or Edge router. If the gateway is initially deployed with no NAT rules, it has no available day 2 actions.

NSX NAT resource

The `Cloud.NSX.NAT` resource allows DNAT rules and port forwarding to be attached to all the connected outbound networks by way of the gateway resource. You can attach a NAT resource to a gateway resource for which the DNAT rules need to be configured.

Note:

The `Cloud.NSX.Gateway` resource was originally available for DNAT rules. However, use of the `Cloud.NSX.Gateway` as a means of defining DNAT rules and port forwarding has been deprecated. It does remain available for backward compatibility. Use the `Cloud.NSX.NAT` cloud template resource for DNAT rules and port forwarding. A warning appears in the cloud template if you attempt to use the `Cloud.NSX.Gateway` resource type with NAT rule specifications.

The `Cloud.NSX.NAT` resource supports DNAT rules and port forwarding when connected to an outbound NSX-V or NSX network.

The NAT rules setting in the resource is `natRules:`. You can attach the NAT resource to the gateway resource to configure the `natRules:` entries on the gateway. DNAT rules that are specified in the resource use the associated machines or load balancers as their target.

You can reconfigure a machine NIC or compute gateway in an existing deployment to modify its `natRules:` settings by adding, reordering, editing or deleting DNAT port forwarding rules. You cannot use DNAT rules with clustered machines. You can specify DNAT rules for individual machines within the cluster as part of a day 2 operation.

VLAN segments and private NSX networks

You can specify VLAN segments for private NSX on-demand network when the network segments are used with a Policy API-type of NSX cloud account.

External IPAM integration properties for Infoblox

For information about cloud template properties that are available for use with your Infoblox external IPAM integrations, see *Using Infoblox-specific properties and extensible attributes for IPAM integrations in cloud templates*.

Caveats for using a static IP assignment in a cloud template

You can use a static IP assignment in a VCF Automation cloud template only when by using VCF Automation IPAM, meaning IPAM that is either the VCF Automation-supplied *internal* IPAM or IPAM derived from an *external* provider plug-in that has been created by using the VCF Automation IPAM SDK - for example one of the Infobox plug-ins that are available for download from the VMware Marketplace. Using a static IP assignment (`assignment:static`) is not supported within a cloud template when using a **Network Configure** event topic (which is used by either an VCF Automation extensibility (ABX) action or a VCF Operations orchestrator workflow). Unsupported static IP assignments cause deployment failure.

For an example of a static IP assignment in a cloud template, see Network, security group, and load balancer resource examples in VCF Automation.

Address value in General section of deployed cloud template

When examining a deployed cloud template, the **Address** value in the **General** section of the template is the primary IP address of the machine. The primary address is often the public or otherwise accessible machine address. For vSphere deployments, the primary IP address is calculated by VCF Automation. All IP addresses for all NICS, including their public, private, IPv6, static, and dynamic properties, are considered and ranked to determine the primary IP address. For non- vSphere deployments, the primary IP address of the machine is calculated by each cloud vendor’s ranking system.

Available day 2 operations

For a list of common day 2 operations that are available for cloud template and deployment resources, see https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-0/building-your-cloud-applications/getting-started-with-the-tools-for-building-applications/getting-started-with-the-vcf-automation-catalog/request-a-catalog-item-in-vcf-automation-for-vm-apps/map-head-resources-tab-service-broker/service-broker-deployment-actions.html.

For an example of how to move from one network to another, see How to move a deployed machine to another network in VCF Automation for VM Apps.

Learn more

For related information and examples that illustrate sample network resources and settings, see Network, security group, and load balancer resource examples in VCF Automation.

For information about defining network resources, see Network resources in VCF Automation.

For information about defining network profiles, see Learn more about network profiles in VCF Automation.

More about security group and tag resources in VCF Automation cloud templates

As you create or edit your VCF Automation cloud templates, use the most appropriate security resource options to meet your objectives.

Cloud agnostic security group resource

You add a security group resource by using the **Cloud Agnostic** > **Security Group** resource on the **Template** page. The resource displays in the cloud template code as a `Cloud.SecurityGroup` resource type. The default resource displays as:

```
Cloud_SecurityGroup_1:
  type: Cloud.SecurityGroup
  properties:
    constraints: []
    securityGroupType: existing
```

You specify a security group resource in a cloud template design as either existing (`securityGroupType: existing`) or on-demand (`securityGroupType: new`).

You can add an existing security group to your cloud template or you can use an existing security group that has been added to a network profile.

For NSX, as well as NSX with the policy manager switch enabled in combination with VMware Cloud on AWS, you can add an existing security group or define a new security group as you design or modify your cloud template. On-demand security groups are supported for NSX, and VMware Cloud on AWS when used with NSX policy manager.

For all cloud account types except Microsoft Azure, you can associate one or more security groups to a machine NIC. A Microsoft Azure virtual machine NIC (*machineName*) can only be associated to one security group.

By default, the security group property `securityGroupType` is set to `existing`. To create an on-demand security group, enter `new` for the `securityGroupType` property. To specify firewall rules for an on-demand security group, use the `rules` property in the `Cloud.SecurityGroup` section of the security group resource.

Existing security groups

Existing security groups are created in a source cloud account resource such as NSX or Amazon Web Services. They are data collected by VCF Automation from the source. You can select an existing security group from a list of available resources as part of a VCF Automation network profile. In a cloud template design, you can specify an existing security group either inherently by its membership in a specified network profile or specifically by name using the `securityGroupType: existing` setting in a security group resource. If you add a security group to a network profile, add at least one capability tag to the network profile. On-demand security group resources require a constraint tag when used in a cloud template design.

You can associate a security group resource in your cloud template design to one or more machine resources.

> **Note:**
> If you intend to use a machine resource in your cloud template design to provision to a Microsoft Azure virtual machine NIC (*machineName*), you should only associate the machine resource to a single security group.

On-demand security groups

You can define on-demand security groups as you define or modify a cloud template design by using the `securityGroupType: new` setting in the security group resource code.

You can use an on-demand security group for NSX, as well as Amazon Web Services when used with NSX Policy type, to apply a specific set of firewall rules to a networked machine resource or set of grouped resources. Each security group can contain multiple named firewall rules. You can use an on-demand security group to specify services or protocols and ports. Note that you can specify either a service or a protocol but not both. You can specify a port in addition to a protocol. You cannot specify a port if you specify a service. If the rule contains neither a service or a protocol, the default service value is Any.

You can also specify IP addresses and IP ranges in firewall rules. Some firewall rule examples are shown in Network, security group, and load balancer resource examples in VCF Automation.

When you create firewall rules in an NSX on-demand security group, the default allows the specified network traffic. The default also allows other network traffic. To control network traffic, you must specify an access type for each rule. The rule access types are:

- Allow (default) - allows the network traffic that is specified in this firewall rule.
- Deny - blocks the network traffic that is specified in this firewall rule. Actively tells the client that the connection is rejected.
- Drop - rejects the network traffic that is specified in this firewall rule. Silently drops the packet as if the listener is not online.

For an example design that uses an `access: Allow` and an `access: Deny` firewall rule, see Network, security group, and load balancer resource examples in VCF Automation.

> **Note:**
> A cloud administrator can create a cloud template design that contains only an NSX on-demand security group and can deploy that design to create a reusable existing security group resource that members of the organization can add to network profiles and cloud template designs as an existing security group.

Firewall rules support either IPv4 or IPv6 format CIDR values for source and destination IP addresses. For an example design that uses IPv6 CIDR values in a firewall rule, see Network, security group, and load balancer resource examples in VCF Automation.

Using constraints for security group placement

You can control where security groups are provisioned by using constraint tags, as shown in the following example:

```
Cloud_SecurityGroup_4:
  type: Cloud.SecurityGroup
  properties:
    securityGroupType: new
    constraints:
        - tag: na
```

In this example, an on-demand security group is provisioned in endpoints (local NSX Manager cloud accounts) that have the `na` capability tag.

- If multiple endpoints satisfy the constraint, the endpoint whose cloud zone has the highest provisioning priority is selected.
- If no endpoint satisfies the constraint, the deployment fails.
- If the security group is attached to another resource, the respective endpoint of that resource must satisfy the security group constraints in addition to any placement constraints on the resource itself.

On-demand and existing security groups for VMware Cloud on AWS

You can define an on-demand security group for a VMware Cloud on AWS machine in a cloud template by using the `securityGroupType: new` setting in the security group resource code.

A sample code snippet for an on-demand security group is shown below:

```
resources:
  Cloud_SecurityGroup_1:
    type: Cloud.SecurityGroup
    properties:
```

```
      name: vmc-odsg
      securityGroupType: new
      rules:
        - name: datapath
          direction: inbound
          protocol: TCP
          ports: 5011
          access: Allow
          source: any
```

You can also define an existing security group for a networked VMware Cloud on AWS machine and optionally include constraint tagging, as shown in the following examples:

```
  Cloud_SecurityGroup_2:
    type: Cloud.SecurityGroup
    properties:
      constraints: [xyz]
      securityGroupType: existing

Cloud_SecurityGroup_3:
  type: Cloud.SecurityGroup
  properties:
    securityGroupType: existing
     constraints:
        - tag: xyz
```

Iterative cloud template development is supported.

- If a security group is associated with one or more machines in the deployment, a delete action displays a message stating that the security group cannot be deleted.
- If a security group is not associated with any machine in the deployment, a delete action displays a message stating that the security group will be deleted from this deployment and the action cannot be undone. An existing security group is deleted from the cloud template, while an on-demand security group is destroyed.

Using app isolation policies in on-demand security group firewall rules

You can use an app isolation policy to only allow internal traffic between the resources that are provisioned by the cloud template. With app isolation, the machines provisioned by the cloud template can communicate with each other but cannot connect outside the firewall. You can create an app isolation policy in the network profile. You can also specify app isolation in a cloud template design by using an on-demand security group with a deny firewall rule or a private or outbound network.

An app isolation policy is created with a lower precedence. If you apply multiple policies, the policies with the higher weight will take precedence.

When you create an application isolation policy, an auto-generated policy name is generated. The policy is also made available for reuse in other cloud template designs and iterations that are specific to the associated resource endpoint and project. The app isolation policy name is not visible in the cloud template but it is visible as a custom property on the project page (**Infrastructure** > **Administration** > **Projects**) after the cloud template design is deployed.

For the same associated endpoint in a project, any deployment that requires an on-demand security group for app isolation can use the same app isolation policy. Once the policy is created, it is not deleted. When you specify an app isolation policy, VCF Automation searches for the policy within the project and relative to the associated endpoint - If it finds the policy it reuses it, if it does not find the policy, it creates it. The app isolation policy name is only visible after its initial deployment in the project's custom properties listing.

Using security groups in iterative cloud template development

When changing security group constraints during iterative development, where the security group is not associated to a machine in the cloud template, the security group updates in the iteration as specified. However, when the security group is already associated to a machine, redeployment fails. You must detach existing security groups and/

or `securityGroupType` resource properties from associated machines during iterative cloud template development and then re-associate between each redeployment. The needed workflow is as follows, assuming that you have initially deployed the cloud template.

1. In the VCF Automation template designer, detach the security group from all its associated machines in the cloud template.
2. Redeploy the template by clicking **Update an existing deployment**.
3. Remove the existing security group constraint tags and/or `securityGroupType` properties in the template.
4. Add new security group constraint tags and/or `securityGroupType` properties in the template.
5. Associate the new security group constraint tags and/or `securityGroupType` property instances to the machines in the template.
6. Redeploy the template by clicking **Update an existing deployment**.

Available day 2 operations

For a list of common day 2 operations that are available for cloud template and deployment resources, see What actions can I run on deployments or supported resources in VCF Automation for VM Apps.

Learn more

For information about using a security group for network isolation, see Security resources in VCF Automation.

For information about using security groups in network profiles, see Learn more about network profiles in VCF Automation and Using security group settings in network profiles and cloud template designs in VCF Automation.

For examples of using security groups in cloud templates, see Network, security group, and load balancer resource examples in VCF Automation.

More about load balancer resources in VCF Automation cloud templates

As you create or edit your VCF Automation cloud templates, use the most appropriate load balancer resources for your objectives.

You can use NSX and cloud-agnostic load balancer resources in a cloud template to control load balancing in a deployment.

The cloud-agnostic load balancer can be deployed across multiple clouds. A cloud-specific load balancer can specify advanced settings and features that are available only to a specific cloud/topology. Cloud-specific properties are available in the NSX load balancer (Cloud.NSX.LoadBalancer) resource type. If you add these properties on a cloud-agnostic load balancer (Cloud.LoadBalancer), they are ignored if, for example, an Amazon Web Services or Microsoft Azure load balancer is provisioned, but are respected if an NSX load balancer is provisioned. Choose one of the available load balancer resource types based on conditions in your VCF Automation cloud template.

You cannot connect a load balancer resource directly to a security group resource in the design canvas.

Cloud agnostic load balancer resource

Use a cloud agnostic load balancer when you want to specify networking characteristics for any type of target machine.

You add a cloud agnostic load balancer by using the **Cloud Agnostic** > **Load Balancer** resource on the cloud template design page. The resource displays in the cloud template code as a `Cloud.LoadBalancer` resource type. The default resource displays as:

```
Cloud_LoadBalancer_1: type: Cloud.LoadBalancer
properties:
routes: []
network: ''
instances: []
internetFacing: false
```

NSX load balancer resource

Use an NSX load balancer when your cloud template contains characteristics that are specific to NSX (either Policy API or Manager API methods). You can attach one or more load balancers to an NSX network or to machines that are associated to an NSX network.

You add an NSX load balancer by using the **NSX** > **Load Balancer** resource. The resource displays in the cloud template code as a `Cloud.NSX.LoadBalancer` resource type. The default resource displays as:

```
Cloud_NSX_LoadBalancer_1: type: Cloud.NSX.LoadBalancer
properties:
routes: []
network: ''
instances: []
```

Load balancer options in cloud template code

Adding one or more load balancer resources to your cloud template allows you to specify the following settings. Some examples are available at Network, security group, and load balancer resource examples in VCF Automation.

The HTTP protocol is supported for all on-demand load balancers.

The HTTPS protocol is supported only for on-demand load balancers that are associated to an NSX cloud account whose NSX mode is set to **Policy**. NSX cloud accounts whose NSX mode is set to **Manager** cannot use the HTTPS protocol.

- Machine specification
  You can specify named machine resources to participate in a load balancing pool. Alternatively you can specify that a specific machine NIC participate in the load balancer pool.
  This option is available for the **NSX** load balancer resource (`Cloud.NSX.LoadBalancer`) only.
  - resource.*`Cloud_Machine_1`*.id
    Specifies that the load balancer include the machine identified in the cloud template code as *`Cloud_Machine_1`*.
  - resource.*`Cloud_Machine_2.networks[2]`*.id
    Specifies that the load balancer only include the machine identified in the cloud template code as *`Cloud_Machine_2`* when it is deployed to machine NIC *`Cloud_Machine_2.networks[2]`*.
- Logging level
  The logging level value specifies a severity level for the error log. The options are NONE, EMERGENCY, ALERT, CRITICAL, ERROR, WARNING, INFO, DEBUG, and NOTICE. The logging level value applies to all load balancers in the cloud template. This option is specific to NSX. For load balancers that have a parent, the parent logging level setting overrides any logging level setting in its children.
  For related information, see topics such as *Add Load Balancers* in the NSX product documentation.
- Type
  Use a load balancer type to specify a scaling size. The default is small. This option is specific to NSX. For load balancers that have a parent, the parent type setting overrides any type setting in its children.
  - Small
    Correlates to compact in NSX-V and small in NSX.
  - Medium
    Correlates to large in NSX-V and medium in NSX.
  - Large
    Correlates to quad-large in NSX-V and large in NSX.
  - Extra Large
    Correlates to xlarge in NSX-V and large in NSX.
    For related information, see topics such as *Scaling Load Balancer Resources* in the NSX product documentation.

  This option is available for the **NSX** load balancer resource (`Cloud.NSX.LoadBalancer`).
- Algorithm (server pool)
  Use an algorithm balancing method to control how incoming connections are distributed among the server pool members. The algorithm can be used on a server pool or directly on a server. All load balancing algorithms skip servers that meet any of the following conditions:

- The Admin state is set to DISABLED.
- The Admin state is set to GRACEFUL_DISABLED and there is no matching persistence entry.
- The active or passive health check state is DOWN.
- The connection limit for the maximum server pool concurrent connections is reached.

This option is specific to NSX.

- IP_HASH
  Selects a server based on a hash of the source IP address and the total weight of all the running servers.
  Correlates to IP-HASH in NSX.
- LEAST_CONNECTION
  Distributes client requests to multiple servers based on the number of connections already on the server. New connections are sent to the server with the fewest connections. Ignores the server pool member weights even if they are configured.
  Correlates to LEASTCONN in NSX-V and LEAST_CONNECTION in NSX.
- ROUND_ROBIN
  Incoming client requests are cycled through a list of available servers capable of handling the request. Ignores the server pool member weights even if they are configured. Default.
  Correlates to ROUND_ROBIN in NSX.
- WEIGHTED_LEAST_CONNECTION
  Each server is assigned a weight value that signifies how that server performs relative to other servers in the pool. The value determines how many client requests are sent to a server compared to other servers in the pool. This load balancing algorithm focuses on using the weight value to distribute the load among the available server resources fairly. By default, the weight value is 1 if the value is not configured and slow start is enabled.
  Correlates to WEIGHTED_LEAST_CONNECTION in NSX.There is no correlation in NSX-V.
- WEIGHTED_ROUND_ROBIN
  Each server is assigned a weight value that signifies how that server performs relative to other servers in the pool. The value determines how many client requests are sent to a server compared to other servers in the pool. This load balancing algorithm focuses on fairly distributing the load among the available server resources.
  Correlates to WEIGHTED_ROUND_ROBIN in NSX. There is no correlation in NSX-V.
- URI
  The left part of the URI is hashed and divided by the total weight of the running servers. The result designates which server receives the request. This ensures that a URI is always directed to the same server if no server goes up or down. The URI algorithm parameter has two options `uriLength=<len>` and `uriDepth=<dep>`. The length parameter range should be `1<=len<256`. The depth parameter range should be `1<=dep<10`. Length and depth parameters are followed by a positive integer number. These options can balance servers based on the beginning of the URI only. The length parameter indicates that the algorithm should only consider the defined characters at the beginning of the URI to compute the hash. The depth parameter indicates the maximum directory depth to be used to compute the hash. One level is counted for each slash in the request. If both parameters are specified, the evaluation stops when either is reached.
  Correlates to URI in NSX-V. There is no correlation in NSX.
- HTTPHEADER
  HTTP header name is looked up in each HTTP request. The header name in parentheses is not case-sensitive. If the header is absent or does not contain any value, the round robin algorithm is applied. The HTTPHEADER algorithm parameter has one option `headerName=<name>`.
  Correlates to HTTPHEADER in NSX-V. There is no correlation in NSX.
- URL
  URL parameter specified in the argument is looked up in the query string of each HTTP GET request. If the parameter is followed by an equal sign = and a value, then the value is hashed and divided by the total weight of the running servers. The result designates which server receives the request. This process is used to track user identifiers in requests and ensure that a same user ID is always sent to the same server as long as no server goes up or down. If no value or parameter is found, then a round robin algorithm is applied. The URL algorithm parameter has one option `urlParam=<url>`.
  Correlates to URL in NSX-V. There is no correlation in NSX.

For related information, see topics such as *Add a Server Pool for Load Balancing* in the NSX product documentation.

- Health monitor

  Use the health monitor options to test whether a server is available. Active health monitoring for HTTP, ICMP, TCP, and UDP protocols is supported. Passive health monitoring is available for NSX only.

  This option is specific to NSX.

  – httpMethod

    HTTP method to use to detect server status for the health check request. Methods are GET, HOST, OPTIONS, HEAD, or PUT.

  – requestBody

    Health check request body content. Used, and required, by HTTP, TCP, and UDP protocols.

  – responseBody

    Health check expected response body content. If the received string matches this response body, the server is considered healthy. Used, and required, by HTTP, TCP, and UDP protocols.

    Note:
    If you use the UDP monitor protocol, the `UDP Data Sent` and `UDP Data Expected` parameters are required. The `requestBody` and `responseBody` properties map to these parameters.

  This option is available for the NSX load balancer resource (`Cloud.NSX.LoadBalancer`).

  For related information, see topics such as *Configure an Active Health Monitor* in the NSX product documentation.

- Health check

  Use health check options to specify how the load balancer performs its heath checks.

  This option is only available for the NSX load balancer resource (`Cloud.NSX.LoadBalancer`).

  For a sample of available heath check settings, see Network, security group, and load balancer resource examples in VCF Automation.

NSX network types and load balancer options

Load balancer options depend on the network that the load balancer resource is associated to in the cloud template. You can configure a load balancer relative to the network type and network conditions.

- On-demand network

  If the load balancer computes are attached to an on-demand network, a new Tier-1 router is created and attached to the Tier-0 router specified in the network profile. The load balancer is then attached to the Tier-1 router. The Tier-1 router VIP advertisement is enabled if the VIP is on an existing network. If an on-demand network is configured for DHCP, the on-demand network and load balancer share the Tier-1 router.

- Existing network

  If the load balancer is attached to an existing network, the load balancer is created with the Tier-1 router of the existing network. A new load balancer is created if there is no load balancer attached to the Tier-1 router. If the load balancer already exists, new virtual servers are attached to it. If the existing network is not attached to a Tier-1 router, a new Tier-1 router is created and attached to a Tier-0 router defined in the network profile, the Tier-1 router VIP advertisement is not enabled.

  VCF Automation does not support an NSX two-arm load balancer (inline load balancer) on two different existing networks. Note that in a two-arm load balancer scenario, the VIP uplink is on an existing network while the pool member machines are connected to an on-demand network. To specify load balancing when using an existing network, you must configure a one-arm load balancer where the same existing network is used for the load balancer VIP and the pool member machines. However, if you are using a load balancer that you've selected in the network profile, you can load balance between machines on two different existing networks if there is connectivity between the two existing networks.

- Network isolation defined in the network profile

  For network types of `outbound` or `private`, you can specify network isolation settings in a network profile to emulate a new security group. Because machines are attached to an existing network and isolation settings are defined in the profile, this option is similar to a load balancer created on an existing network. The difference is that to enable the data path, the Tier-1 uplink port IP is added to the isolation security group.

You can specify load balancer settings for NSX-associated networks by using an NSX load balancer resource in the cloud template design.

NSX load-balancer profile persistence

VCF Automation cloud templates and deployments support NSX load balancer persistence profile options for SOURCE_IP and COOKIE settings. For information about how to configure these settings in NSX so that they can be consumed by VCF Automation cloud templates and deployments, see *Add a Persistence Profile* in NSX documentation.

Reconfiguring logging level or type settings when multiple load balancers share an NSX Tier 1 or NSX-V Edge

When using a cloud template that contains multiple load balancers which share a Tier-1 router in the NSX endpoint or an Edge router in the NSX-V endpoint, reconfiguring the logging level or type settings in one of the load balancer resources does not update the settings for the other load balancers. Mismatched settings cause inconsistencies in NSX. To avoid inconsistencies when reconfiguring these logging level and/or type settings, use the same reconfiguration values for all the load balancer resources in the cloud template which share a Tier 1 or Edge in their associated NSX endpoint.

Available day 2 operations

When you scale in or scale out a deployment that contains a load balancer, the load balancer is configured to include newly added machines or to stop load balancing machines that are targeted for tear down.

Learn more

For information about defining load balancer settings in a network profile, see Learn more about network profiles in VCF Automation.

For examples of template designs that include load balancers, see Network, security group, and load balancer resource examples in VCF Automation.

vSphere resource examples in VCF Automation for VM Apps

These code examples illustrate vSphere machine resources within VCF Automation cloud templates.

Resource	Example Cloud Template

Custom properties for a vSphere virtual machine and a content library with custom properties	`formatVersion: 1`
	`inputs: {}`
	`resources:`
	`  Cloud_vSphere_Machine_1:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  image: nimbus-mapping`
	`  flavor: nimbus-flavor`
	`  ovfProperties:`
	`  - key: hostname`
	`  value: ubuntuguest`
	`  - key: instance-id`
	`  value: test-id-ovf`
	`  - key: password`
	`  value: test-pwd`
	`  - key: public-keys`
	`  value: test-pk`
	`  - key: seedfrom`
	`  value: test-sf`
	`  - key: user-data`
	`  value: test-ud`

vSphere virtual machine with CPU, memory, and operating system	`resources:`
	`  demo-machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  name: demo-machine`
	`  cpuCount: 1`
	`  totalMemoryMB: 1024`
	`  image: ubuntu`

vSphere machine with a datastore resource	`resources:`
	`  demo-vsphere-disk-001:`
	`  type: Cloud.vSphere.Disk`
	`  properties:`
	`  name: DISK_001`
	`  type: 'HDD'`
	`  capacityGb: 10`
	`  dataStore: 'datastore-01'`
	`  provisioningType: thick`

Resource	Example Cloud Template

vSphere machine with an attached disk	`resources:`
	`  demo-vsphere-disk-001:`
	`  type: Cloud.vSphere.Disk`
	`  properties:`
	`  name: DISK_001`
	`  type: HDD`
	`  capacityGb: 10`
	`  dataStore: 'datastore-01'`
	`  provisioningType: thin`
	`  demo-machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  name: demo-machine`
	`  cpuCount: 2`
	`  totalMemoryMB: 2048`
	`  imageRef: >-`
	`  https://packages.vmware.com/photon/4.0/Rev1/ova/photon-ova-4.0-ca7c9e9330.ova`
	`  attachedDisks:`
	`  - source: '${demo-vsphere-disk-001.id}'`

vSphere machine with a dynamic number of disks	`inputs:`
	`  disks:`
	`  type: array`
	`  title: disks`
	`  items:`
	`  title: disks`
	`  type: integer`
	`  maxItems: 15`
	`resources:`
	`  Cloud_Machine_1:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  image: Centos`
	`  flavor: small`
	`  attachedDisks:`
	`  '${map_to_object(resource.Cloud_Volume_1[*].id, "source")}'`
	`  Cloud_Volume_1:`
	`  type: Cloud.Volume`
	`  allocatePerInstance: true`
	`  properties:`
	`  capacityGb: '${input.disks[count.index]}'`
	`  count: '${length(input.disks)}'`

vSphere machine from a snapshot image. Append a forward slash and the snapshot name. The snapshot image can be a linked clone.	`resources:`
	`  demo-machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  imageRef: 'demo-machine/snapshot-01'`
	`  cpuCount: 1`
	`  totalMemoryMB: 1024`

Resource	Example Cloud Template

vSphere machine in a specific folder in vCenter	`resources:`
	`  demo-machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  name: demo-machine`
	`  cpuCount: 2`
	`  totalMemoryMB: 1024`
	`  imageRef: ubuntu`
	`  resourceGroupName: 'myFolder'`

vSphere machine with multiple NICs	`resources:`
	`  demo-machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  image: ubuntu`
	`  flavor: small`
	`  networks:`
	`  - network: '${network-01.name}'`
	`  deviceIndex: 0`
	`  - network: '${network-02.name}'`
	`  deviceIndex: 1`
	`  network-01:`
	`  type: Cloud.vSphere.Network`
	`  properties:`
	`  name: network-01`
	`  network-02:`
	`  type: Cloud.vSphere.Network`
	`  properties:`
	`  name: network-02`

vSphere machine with an attached tag in vCenter	`resources:`
	`  demo-machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  flavor: small`
	`  image: ubuntu`
	`  tags:`
	`  - key: env`
	`  value: demo`

vSphere machine with a customization spec	`resources:`
	`  demo-machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  name: demo-machine`
	`  image: ubuntu`
	`  flavor: small`
	`  customizationSpec: Linux`

Resource	Example Cloud Template
vSphere machine with remote access	inputs: username: type: string title: Username description: Username default: testUser password: type: string title: Password default: VMware@123 encrypted: true description: Password for the given username resources: demo-machine: type: Cloud.vSphere.Machine properties: flavor: small imageRef: >- https://cloud-images.ubuntu.com/releases/16.04/release-20170307/ubuntu-16.04-server-cloudimg-amd64.ova cloudConfig: | ssh_pwauth: yes chpasswd: list: | ${input.username}:${input.password} expire: false users: - default - name: ${input.username} lock_passwd: false sudo: ['ALL=(ALL) NOPASSWD:ALL'] groups: [wheel, sudo, admin] shell: '/bin/bash' runcmd: - echo "Defaults:${input.username} !requiretty" >> /etc/sudoers.d/${input.username}

Documented VCF Automation template example

By including a thorough set of comments, this example lets you review the structure and purpose of the sections in a VCF Automation template.

```
# ****************************************************************************
#
# This WordPress cloud template is enhanced with comments to explain its
# parameters.
#
# Try cloning it and experimenting with its YAML code. If you're new to
# YAML, visit yaml.org for general information.
#
# The cloud template deploys a minimum of 3 virtual machines and runs scripts
# to install packages.
#
# ****************************************************************************
#
# ----------------------------------------------------------------------------
```

```
# Templates need a descriptive name and version if
# source controlled in git.
# ------------------------------------------------------------------------
name: WordPress Template with Comments
formatVersion: 1
version: 1
#
# ------------------------------------------------------------------------
# Inputs create user selections that appear at deployment time. Inputs
# can set placement decisions and configurations, and are referenced
# later, by the resources section.
# ------------------------------------------------------------------------
inputs:
#
# ------------------------------------------------------------------------
# Choose a cloud endpoint. 'Title' is the visible
# option text (oneOf allows for the friendly title). 'Const' is the
# tag that identifies the endpoint, which was set up earlier, under the
# Infrastructure tab.
# ------------------------------------------------------------------------
  platform:
    type: string
    title: Deploy to
    oneOf:
      - title: AWS
        const: aws
      - title: Azure
        const: azure
      - title: vSphere
        const: vsphere
    default: vsphere
#
# ------------------------------------------------------------------------
# Choose the operating system. Note that the
# Infrastructure must also have an AWS, Azure, and vSphere Ubuntu image
# mapped. In this case, enum sets the option that you see, meaning there's
# no friendly title feature this time. Also, only Ubuntu is available
# here, but having this input stubbed in lets you add more operating
# systems later.
# ------------------------------------------------------------------------
  osimage:
    type: string
    title: Operating System
    description: Which OS to use
    enum:
      - Ubuntu
#
# ------------------------------------------------------------------------
# Set the number of machines in the database cluster. Small and large
# correspond to 1 or 2  machines, respectively, which you see later,
# down in the resources section.
# ------------------------------------------------------------------------
  dbenvsize:
    type: string
    title: Database cluster size
    enum:
      - Small
```

```
      - Large
#
# ---------------------------------------------------------------------
# Dynamically tag the machines that will be created. The
# 'array' of objects means you can create as many key-value pairs as
# needed. To see how array input looks when it's collected,
# open the cloud template and click TEST.
# ---------------------------------------------------------------------
  Mtags:
    type: array
    title: Tags
    description: Tags to apply to machines
    items:
      type: object
      properties:
        key:
          type: string
          title: Key
        value:
          type: string
          title: Value
#
# ---------------------------------------------------------------------
# Create machine credentials. These credentials are needed in
# remote access configuration later, in the resources section.
# ---------------------------------------------------------------------
  username:
    type: string
    minLength: 4
    maxLength: 20
    pattern: '[a-z]+'
    title: Database Username
    description: Database Username
  userpassword:
    type: string
    pattern: '[a-z0-9A-Z@#$]+'
    encrypted: true
    title: Database Password
    description: Database Password
#
# ---------------------------------------------------------------------
# Set the database storage disk size.
# ---------------------------------------------------------------------
  databaseDiskSize:
    type: number
    default: 4
    maximum: 10
    title: MySQL Data Disk Size
    description: Size of database disk
#
# ---------------------------------------------------------------------
# Set the number of machines in the web cluster. Small, medium, and large
# correspond to 2, 3, and 4 machines, respectively, which you see later,
# in the WebTier part of the resources section.
# ---------------------------------------------------------------------
  clusterSize:
    type: string
```

```
    enum:
      - small
      - medium
      - large
    title: Wordpress Cluster Size
    description: Wordpress Cluster Size
#
# ---------------------------------------------------------------------
# Set the archive storage disk size.
# ---------------------------------------------------------------------
  archiveDiskSize:
    type: number
    default: 4
    maximum: 10
    title: Wordpress Archive Disk Size
    description: Size of Wordpress archive disk
#
# ---------------------------------------------------------------------
# The resources section configures the deployment of machines, disks,
# networks, and other objects. In several places, the code pulls from
# the preceding interactive user inputs.
# ---------------------------------------------------------------------
resources:
#
# ---------------------------------------------------------------------
# Create the database server. Choose a cloud agnostic machine 'type' so
# that it can deploy to AWS, Azure, or vSphere. Then enter its property
# settings.
# ---------------------------------------------------------------------
  DBTier:
    type: Cloud.Machine
    properties:
#
# ---------------------------------------------------------------------
# Descriptive name for the virtual machine. Does not become the hostname
# upon deployment.
# ---------------------------------------------------------------------
      name: mysql
#
# ---------------------------------------------------------------------
# Hard-coded operating system image to use. To pull from user input above,
# enter the following instead.
# image: '${input.osimage}'
# ---------------------------------------------------------------------
      image: Ubuntu
#
# ---------------------------------------------------------------------
# Hard-coded capacity to use. Note that the
# Infrastructure must also have AWS, Azure, and vSphere flavors
# such as small, medium, and large mapped.
# ---------------------------------------------------------------------
      flavor: small
#
# ---------------------------------------------------------------------
# Tag the database machine to deploy to the cloud vendor chosen from the
# user input. Tags are case-sensitive, so 'to_lower' forces the tag to
# lowercase to ensure a match with a site's tagging convention. It's
```

```
# important if platform input were to contain any upper case characters.
# ---------------------------------------------------------------------------
      constraints:
        - tag: '${"env:" + to_lower(input.platform)}'
#
# ---------------------------------------------------------------------------
# Also tag the database machine with any free-form tags that were created
# during user input.
# ---------------------------------------------------------------------------
      tags: '${input.Mtags}'
#
# ---------------------------------------------------------------------------
# Set the database cluster size by referencing the dbenvsize user
# input. Small is one machine, and large defaults to two.
# ---------------------------------------------------------------------------
      count: '${input.dbenvsize == "Small" ? 1 : 2}'
#
# ---------------------------------------------------------------------------
# Add a variable to connect the machine to a network resource based on
# a property binding to another resource. In this case, it's the
# 'WP_Network' network that gets defined further below.
# ---------------------------------------------------------------------------
      networks:
        - network: '${resource.WP_Network.id}'
#
# ---------------------------------------------------------------------------
# Enable remote access to the database server. Reference the credentials
# from the user input.
# ---------------------------------------------------------------------------
      remoteAccess:
        authentication: usernamePassword
        username: '${input.username}'
        password: '${input.userpassword}'
#
# ---------------------------------------------------------------------------
# You are free to add custom properties, which might be used to initiate
# an extensiblity subscription, for example.
# ---------------------------------------------------------------------------
      ABC-Company-ID: 9393
#
# ---------------------------------------------------------------------------
# Run OS commands or scripts to further configure the database machine,
# via operations such as setting a hostname, generating SSH private keys,
# or installing packages.
# ---------------------------------------------------------------------------
      cloudConfig: |
        #cloud-config
        repo_update: true
        repo_upgrade: all
        packages:
        - mysql-server
        runcmd:
        - sed -e '/bind-address/ s/^#*/#/' -i /etc/mysql/mysql.conf.d/mysqld.cnf
        - service mysql restart
        - mysql -e "CREATE USER 'root'@'%' IDENTIFIED BY 'mysqlpassword';"
        - mysql -e "GRANT ALL PRIVILEGES ON *.* TO 'root'@'%';"
        - mysql -e "FLUSH PRIVILEGES;"
```

```
      attachedDisks: []
#
# --------------------------------------------------------------------------
# Create the web server. Choose a cloud agnostic machine 'type' so that it
# can deploy to AWS, Azure, or vSphere. Then enter its property settings.
# --------------------------------------------------------------------------
  WebTier:
    type: Cloud.Machine
    properties:
#
# --------------------------------------------------------------------------
# Descriptive name for the virtual machine. Does not become the hostname
# upon deployment.
# --------------------------------------------------------------------------
      name: wordpress
#
# --------------------------------------------------------------------------
# Hard-coded operating system image to use. To pull from user input above,
# enter the following instead:
# image: '${input.osimage}'
# --------------------------------------------------------------------------
      image: Ubuntu
#
# --------------------------------------------------------------------------
# Hard-coded capacity to use. Note that the
# Infrastructure must also have AWS, Azure, and vSphere flavors
# such as small, medium, and large mapped.
# --------------------------------------------------------------------------
      flavor: small
#
# --------------------------------------------------------------------------
# Set the web server cluster size by referencing the clusterSize user
# input. Small is 2 machines, medium is 3, and large defaults to 4.
# --------------------------------------------------------------------------
      count: '${input.clusterSize== "small" ? 2 : (input.clusterSize == "medium" ? 3 : 4)}'
#
# --------------------------------------------------------------------------
# Set an environment variable to display object information under the
# Properties tab, post-deployment. Another example might be
# {env.blueprintID}
# --------------------------------------------------------------------------
      tags:
        - key: cas.requestedBy
          value: '${env.requestedBy}'
#
# --------------------------------------------------------------------------
# You are free to add custom properties, which might be used to initiate
# an extensiblity subscription, for example.
# --------------------------------------------------------------------------
      ABC-Company-ID: 9393
#
# --------------------------------------------------------------------------
# Tag the web server to deploy to the cloud vendor chosen from the
# user input. Tags are case-sensitive, so 'to_lower' forces the tag to
# lowercase to ensure a match with your site's tagging convention. It's
# important if platform input were to contain any upper case characters.
# --------------------------------------------------------------------------
```

```
      constraints:
        - tag: '${"env:" + to_lower(input.platform)}'
#
# ---------------------------------------------------------------------
# Add a variable to connect the machine to a network resource based on
# a property binding to another resource. In this case, it's the
# 'WP_Network' network that gets defined further below.
# ---------------------------------------------------------------------
      networks:
        - network: '${resource.WP_Network.id}'
#
# ---------------------------------------------------------------------
# Run OS commands or scripts to further configure the web server,
# with operations such as setting a hostname, generating SSH private keys,
# or installing packages.
# ---------------------------------------------------------------------
      cloudConfig: |
        #cloud-config
        repo_update: true
        repo_upgrade: all
        packages:
        - apache2
        - php
        - php-mysql
        - libapache2-mod-php
        - mysql-client
        - gcc
        - make
        - autoconf
        - libc-dev
        - pkg-config
        - libmcrypt-dev
        - php-pear
        - php-dev
        runcmd:
        - mkdir -p /var/www/html/mywordpresssite && cd /var/www/html && wget https://wordpress.org/
latest.tar.gz && tar -xzf /var/www/html/latest.tar.gz -C /var/www/html/mywordpresssite --strip-components 1
        - i=0; while [ $i -le 10 ]; do mysql --connect-timeout=3 -h ${DBTier.networks[0].address} -u root -
pmysqlpassword -e "SHOW STATUS;" && break || sleep 15; i=$((i+1)); done
        - mysql -u root -pmysqlpassword -h ${DBTier.networks[0].address} -e "create database wordpress_blog;"
        - mv /var/www/html/mywordpresssite/wp-config-sample.php /var/www/html/mywordpresssite/wp-config.php
        - pecl channel-update pecl.php.net
        - pecl update-channels
        - pecl install mcrypt
        - sed -i -e s/"define( 'DB_NAME', 'database_name_here' );"/"define( 'DB_NAME',
 'wordpress_blog' );"/ /var/www/html/mywordpresssite/wp-config.php && sed -i -e s/"define( 'DB_USER',
 'username_here' );"/"define( 'DB_USER', 'root' );"/ /var/www/html/mywordpresssite/wp-config.php && sed -
i -e s/"define( 'DB_PASSWORD', 'password_here' );"/"define( 'DB_PASSWORD', 'mysqlpassword' );"/ /var/www/
html/mywordpresssite/wp-config.php && sed -i -e s/"define( 'DB_HOST', 'localhost' );"/"define( 'DB_HOST',
 '${DBTier.networks[0].address}' );"/ /var/www/html/mywordpresssite/wp-config.php
        - sed -i '950i extension=mcrypt.so' /etc/php/7.4/apache2/php.ini
        - service apache2 reload
#
# ---------------------------------------------------------------------
# Create the network that the database and web servers connect to.
# Choose a cloud agnostic network 'type' so that it can deploy to AWS,
# Azure, or vSphere. Then enter its property settings.
```

```
# ---------------------------------------------------------------------
  WP_Network:
    type: Cloud.Network
    properties:
#
# ---------------------------------------------------------------------
# Descriptive name for the network. Does not become the network name
# upon deployment.
# ---------------------------------------------------------------------
      name: WP_Network
#
# ---------------------------------------------------------------------
# Set the networkType to an existing network. You could also use a
# constraint tag to target a specific, like-tagged network.
# The other network types are private or public.
# ---------------------------------------------------------------------
      networkType: existing
#
# *********************************************************************
#
# VMware hopes that you found this commented template useful. Note that
# you can also access an API to create templates, or query for input
# schema that you intend to request. See the following Swagger
# documentation.
#
# www.mgmt.cloud.vmware.com/blueprint/api/swagger/swagger-ui.html
#
# *********************************************************************
```

Attaching an existing disk in VCF Automation for VM Apps

Your VCF Automation template code can attach an existing, deployed disk to a newly created machine.

Use the following cloud template code example as a guideline.

```
formatVersion: 1
inputs:
  disk:
    type: string
    $data: /data/availableDisks?resourceType=Cloud.Volume
resources:
  Cloud_Machine_1:
    type: Cloud.Machine
    properties:
      image: ubuntu
      flavor: small
      attachedDisks:
        - source: ${input.disk}
```

The highlighted line is an API call that lists the disks that are available for attachment, disks unattached to any machine.

The resourceType parameter lets VCF Automation list available disks by cloud account type.

Cloud.Volume	Disks on any cloud that you have added
Cloud.vSphere.Disk	Only vSphere disks
Cloud.AWS.Volume	Only Amazon Web Services (AWS) disks
Cloud.Azure.Disk	Only Microsoft Azure disks
Cloud.GCP.Disk	Only Google Cloud Platform (GCP) disks

When using this feature, there's no design canvas component for the existing disk that you attach. Even though there's no component on the design canvas, the attached disk appears under the storage section of the deployed machine, and under the Resources tab.

Cores per socket and CPU count in VCF Automation for VM Apps

VCF Automation template code lets you specify a number of cores per socket for a vSphere machine resource.

You can specify the number of cores per virtual socket or the total number of sockets. For example, your licensing terms might restrict software that is licensed per socket or available operating systems might only recognize a certain number of sockets so that additional CPUs must be provisioned as additional cores.

Add the `coreCount` property to a cloud template in the vSphere machine resource.

The `coreCount` value must be less than or equal to the CPU count (`cpuCount`) value specified in the flavor mapping or in the vSphere machine resource code in the cloud template.

The `coreCount` property is optional and available only for vSphere machine resources.

An example vSphere machine resource snippet is shown below.

```
Cloud_vSphere_Machine_1:
  type: Cloud.vSphere.Machine
  properties:
    cpuCount: 8
    coreCount: 4
```

Puppet-enabled cloud template with username and password access

In this example, you add Puppet configuration management to a cloud template deployed on a vCenter compute resource with username and password access.

- Set up a Puppet Enterprise instance on a valid network.
- Add your Puppet Enterprise instance to VCF Automation using the Integrations feature. See Configure Puppet Enterprise integration in VCF Automation for VM Apps
- Set up a vSphere account and a vCenter compute resource.

This procedure shows an example of how you might create a Puppet enabled deployable resource that requires username and password authentication. Username and password access means that the user must manually log in from the compute resource to the Puppet primary machine in order to invoke Puppet configuration management.

Optionally, you can configure remote access authentication which sets up configuration management in a cloud template so that the compute resource handles authentication with the Puppet primary machine. With remote access enabled, the compute resource automatically generates a key to satisfy password authentication. A valid username is still required.

See AWS Puppet configuration management cloud template examples and vCenter Puppet configuration cloud template examples for more examples of how you can configure different Puppet scenarios in VCF Automation blueprints.

1. Add a Puppet configuration management component to a vSphere compute resource on the canvas for the desired cloud template.
   a) Select **Infrastructure** > **Manage** > **Integrations**.
   b) Click **Add Integration** and select Puppet.
   c) Enter the appropriate information on the Puppet configuration page.

Confguration	Description	Example Value
Hostname	Host name or IP address of the Puppet primary machine	Puppet-Ubuntu
SSH Port	SSH port for communication between VCF Automation and Puppet primary machine. (Optional)	NA

Confguration	Description	Example Value
Autosign secret	The shared secret configured on the Puppet primary machine that nodes should provide to support autosign certificate requests.	User specific
Location	Indicate whether the Puppet primary machine is on a private or public cloud. **Note:** Cross cloud deployment is supported only if there is connectivity between the deployment compute resource and the Puppet primary machine.	
Cloud proxy	Not required for public cloud accounts, such as Microsoft Azure or Amazon Web Services. If you are using a vCenter based cloud account, select the appropriate cloud proxy for your account.	NA
Username	SSH and RBAC user name for Puppet primary machine.	User specific. YAML value is '${input.username}'
Password	SSH and RBAC password for Puppet primary machine.	User specific YAML value is '${input.password}'
Use sudo commands for this user	Select to use sudo commands for the procidd.	true
Name	Puppet primary machine name.	PEMasterOnPrem
Description		

2. Add the username and password properties to the Puppet YAML as shown in the following example.
3. Ensure that the value for the remoteAccess property to the Puppet cloud template YAML is set to `authentication: username and password` as shown in the example below.

vCenter username and password YAML code

The following example shows the representative YAML code for adding username and password authentication on a vCenter compute resource.

```
inputs:
  username:
    type: string
    title: Username
    description: Username to use to install Puppet agent
    default: puppet
  password:
    type: string
    title: Password
    default: VMware@123
    encrypted: true
    description: Password for the given username to install Puppet agent
resources:
  Puppet-Ubuntu:
    type: Cloud.vSphere.Machine
    properties:
      flavor: small
      imageRef: >-
        https://cloud-images.ubuntu.com/releases/16.04/release-20170307/ubuntu-16.04-server-
cloudimg-amd64.ova
      remoteAccess:
        authentication: usernamePassword
```

```
        username: '${input.username}'
        password: '${input.password}'
    Puppet_Agent:
      type: Cloud.Puppet
      properties:
        provider: PEMasterOnPrem
        environment: production
        role: 'role::linux_webserver'
        username: '${input.username}'
        password: '${input.password}'
        host: '${Puppet-Ubuntu.*}'
        useSudo: true
        agentConfiguration:
          certName: '${Puppet-Ubuntu.address}'
```

AWS Puppet configuration management cloud template examples

There are several options for configuring cloud templates to support Puppet based configuration management on AWS compute resources.

Puppet management on AWS with username and password

Example of...	Sample Blueprint YAML
authentication of cloud configuration on any supported Amazon Machine Image.	<pre>inputs: username: type: string title: Username default: puppet password: type: string title: Password encrypted: true default: VMware@123 resources: Webserver: type: Cloud.AWS.EC2.Instance properties: flavor: small image: centos cloudConfig: | #cloud-config ssh_pwauth: yes chpasswd: list: | ${input.username}:${input.password} expire: false users: - default - name: ${input.username} lock_passwd: false sudo: ['ALL=(ALL) NOPASSWD:ALL'] groups: [wheel, sudo, admin] shell: '/bin/bash' ssh-authorized-keys: - ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDytVL+Q6/+vGbmkXoRpX dmettem@dmettem-m01.vmware.com runcmd: - echo "Defaults:${input.username} !requiretty" >> /etc/ sudoers.d/${input.username} Puppet_Agent: type: Cloud.Puppet properties: provider: PEOnAWS environment: production role: 'role::linux_webserver' host: '${Webserver.*}' osType: linux username: '${input.username}' password: '${input.password}' useSudo: true</pre>
Authentication of cloud configuration on a custom Amazon Machine Image with an existing user.	<pre>inputs: username: type: string title: Username default: puppet password: type: string title: Password encrypted: true default: VMware@123 resources: Webserver:</pre>

Puppet management on AWS with generated PublicPrivateKey

Example of...	Sample Blueprint YAML

remoteAccess.authentication authentication on AWS with generatedPublicPrivateKey acces.	`inputs: {}`
	`resources:`
	`  Machine:`
	`  type: Cloud.AWS.EC2.Instance`
	`  properties:`
	`  flavor: small`
	`  imageRef: ami-a4dc46db`
	`  remoteAccess:`
	`  authentication: generatedPublicPrivateKey`
	`  Puppet_Agent:`
	`  type: Cloud.Puppet`
	`  properties:`
	`  provider: puppet-BlueprintProvisioningITSuite`
	`  environment: production`
	`  role: 'role::linux_webserver'`
	`  host: '${Machine.*}'`
	`  osType: linux`
	`  username: ubuntu`
	`  useSudo: true`
	`  agentConfiguration:`
	`  runInterval: 15m`
	`  certName: '${Machine.address}'`
	`  useSudo: true`

vCenter Puppet configuration cloud template examples

There are several options for configuring cloud templates to support Puppet based configuration management on vCenter compute resources.

Puppet on vSphere with username and password authentication

The following example shows example YAML code for Puppet on a vSphere OVA with username and password authentication.

Table 1835:

Example of...	Sample Blueprint YAML
YAML code for Puppet on a vSphere OVA with username and password authentication.	inputs: username: type: string title: Username default: puppet password: type: string title: Password encrypted: true default: VMware@123 resources: Puppet_Agent: type: Cloud.Puppet properties: provider: PEonAWS environment: dev role: 'role::linux_webserver' username: '${input.username}' password: '${input.password}' useSudo: true host: '${Webserver.*}’ osType: linux agentConfiguration: runInterval: 15m certName: ‘${Machine.address}' Webserver: type: Cloud.vSphere.Machine properties: cpuCount: 1 totalMemoryMB: 1024 imageRef: >- https://cloud-images.ubuntu.com/releases/16.04/release-20170307/ ubuntu-16.04-server-cloudimg-amd64.ova cloudConfig: | #cloud-config ssh_pwauth: yes chpasswd: list: | ${input.username}:${input.password} expire: false users: - default - name: ${input.username} lock_passwd: false sudo: ['ALL=(ALL) NOPASSWD:ALL'] groups: [wheel, sudo, admin] shell: '/bin/bash' ssh-authorized-keys: - ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDytVL +Q6+vGbmkXoRpX dmettem@dmettem-m01.vmware.com runcmd: - echo "Defaults:${input.username}
YAML code for Puppet on a vSphere OVA with username and password authentication on the compute resource.	inputs: username: type: string title: Username default: puppet password:

Example of...	Sample Blueprint YAML
YAML code for Puppet on a vCenter with remote access enabled password authentication on the compute resource.	<pre>inputs: username: type: string title: Username description: Username to use to install Puppet agent default: puppet password: type: string title: Password default: VMware@123 encrypted: true description: Password for the given username to install Puppet agent resources: Puppet-Ubuntu: type: Cloud.vSphere.Machine properties: flavor: small imageRef: >- https://cloud-images.ubuntu.com/releases/16.04/ release-20170307/ubuntu-16.04-server-cloudimg-amd64.ova remoteAccess: authentication: usernamePassword username: '${input.username}' password: '${input.password}' Puppet_Agent: type: Cloud.Puppet properties: provider: PEMasterOnPrem environment: production role: 'role::linux_webserver' username: '${input.username}' password: '${input.password}' host: '${Puppet-Ubuntu.*}' useSudo: true agentConfiguration: certName: '${Puppet-Ubuntu.address}'</pre>

Puppet on vSphere with generated PublicPrivateKey authentication

Table 1836:

Example of...	Sample Blueprint YAML

YAML code for Puppet on a vSphere OVA with generated PublicPrivateKey authentication on the compute resource.	`inputs: {}`
	`resources:`
	`  Machine:`
	`  type: Cloud.vSphere.Machine`
	`  properties:`
	`  flavor: small`
	`  imageRef: >-`
	`https://cloud-images.ubuntu.com/releases/16.04/release-20170307/`
	`ubuntu-16.04-server-cloudimg-amd64.ova`
	`  remoteAccess:`
	`  authentication: generatedPublicPrivateKey`
	`  Puppet_Agent:`
	`  type: Cloud.Puppet`
	`  properties:`
	`  provider: puppet-BlueprintProvisioningITSuite`
	`  environment: production`
	`  role: 'role::linux_webserver'`
	`  host: '${Machine.*}'`
	`  osType: linux`
	`  username: ubuntu`
	`  useSudo: true`
	`  agentConfiguration:`
	`  runInterval: 15m`
	`  certName: '${Machine.address}'`
	`  - echo "Defaults:${input.username}`

Special VCF Automation properties

VCF Automation supports a small number of properties that might be useful outside of production environments or in other special situations.

> **CAUTION:**
> The following properties should only be applied in cases where guest OS customization isn't being tested or expected.

awaitIp	By default, VCF Automation provisioning status isn't reported as Finished until the guest OS is fully powered on and configuration has completed. Use of `awaitIp: false` allows provisioning to finish even though full configuration did not occur. CAUTION: Use of this setting completes the provisioning process sooner but might result in an unconfigured machine with no IP address.
awaitHostName	Similar to awaitIp, use of `awaitHostName: false` allows provisioning to finish even though the machine might not have been configured with a host name.

Other ways to create templates in VCF Automation for VM Apps

In addition to building an VCF Automation template from a blank canvas, you can take advantage of existing code.

Cloud template cloning

To clone a template, go to **Design**, select a source, and click **Clone**. You clone a cloud template to create a copy based on the source, then assign the clone to a new project or use it as starter code for a new application.

Uploading and downloading

You can upload, download, and share cloud template YAML code in any way that makes sense for your site. You can also modify template code by using external editors and development environments.

> **Note:**
> A good way to validate shared template code is to inspect it in the VCF Automation code editor on the design page.

Integrating VCF Automation with a repository

An integrated git source control repository can make cloud templates available to qualified users as the basis for a new deployment. See How do I use Git integration in VCF Automation for VM Apps.

Extending and automating application life cycles with extensibility

You can extend your application life cycles by using either extensibility actions or VCF Operations Orchestrator Client workflows with extensibility subscriptions.

With VCF Automation Extensibility, you can assign an extensibility action or VCF Operations Orchestrator Client workflow to an event by using subscriptions. When the specified event occurs, the subscription initiates the action or workflow to run, and all subscribers are notified.

Extensibility Actions

Extensibility actions are small, lightweight scripts of code used to specify an action and how that action is to perform. You can import extensibility actions from pre-defined VCF Automation action templates or from a ZIP file. You can also use the action editor to create custom scripts for your extensibility actions. When multiple action scripts are linked together in one script, you create an action flow. By using action flows, you can create a sequence of actions. For information on using action flows, see What is an action flow.

VCF Operations Orchestrator Client Workflows

By integrating VCF Automation with your existing VCF Operations Orchestrator Client environment, you can use workflows in your extensibility subscriptions.

Extensibility action subscriptions

You can assign an extensibility action to an VCF Automation subscription to extend your application life cycle.

> **Note:**
> The following subscriptions are use case examples and do not cover all extensibility action functionality.

How do I integrate VCF Automation with ServiceNow using extensibility actions

Using extensibility actions you can integrate VCF Automation with an Enterprise ITSM, like ServiceNow.

- Before configuring this integration, filter all event subscriptions with the conditional cloud template property: `event.data["customProperties"]["enable_servicenow"] === "true"`

  > **Note:**
  > This property exists on cloud templates that require a ServiceNow integration.
- Download and install Python.

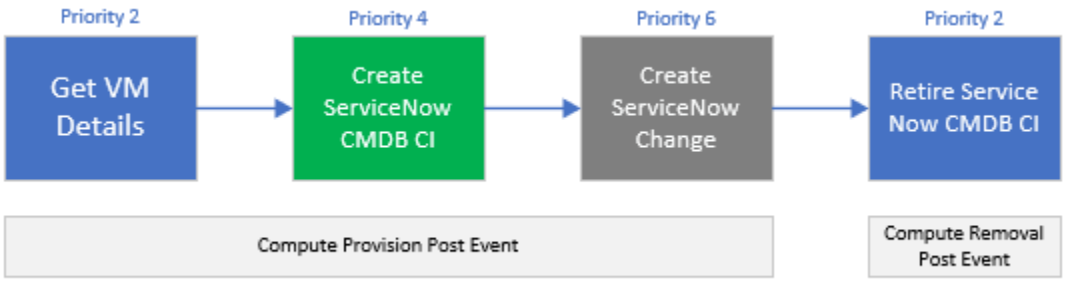
For more information on filtering subscriptions, see Create an extensibility subscription.

Enterprise users commonly integrate their Cloud Management Platform with an IT Service Management (ITSM) and Configuration Management Database (CMDB) platform for compliance. Following this example, you can integrate VCF Automation with ServiceNow for CMDB and ITSM by using extensibility action scripts.

> **Note:**
> You can also integrate ServiceNow with VCF Automation by using VCF Operations Orchestrator Client workflows. For information about integrating ServiceNow by using workflows, see How do I integrate VCF Automation for ITSM with ServiceNow using VCF Operations Orchestrator Client workflows.

To create this integration, you use four extensibility action scripts. The first three scripts are initiated in sequence during provisioning, at the compute provision post event. The fourth script triggers at the compute removal post event.

For more information on event topics, refer to Event topics provided with VCF Automation for VM Apps in.



Get VM Details

The Get VM details script acquires additional payload details required for CI creation and an identity token that is stored in Amazon Web Services Systems Manager Parameter Store (SSM). Also, this script updates `customProperties` with additional properties for later use.

Create ServiceNow CMDB CI

The Create ServiceNow CMDB CI script passes the ServiceNow instance URL as an input and stores the instance in SSM to meet security requirements. This script also reads the ServiceNow CMDB unique record identifier response (sys_id). It passes it as an output and writes the custom property `serviceNowSysId` during creation. This value is used to mark the CI as retired when the instance is destroyed.

> **Note:**
> Additional permissions might need to be allocated to your VCF AutomationAmazon Web Services role to allow Lambda to access the SSM Parameter Store.

Create ServiceNow Change

This script finishes the ITSM integration by passing the ServiceNow instance URL as an input and storing the ServiceNow credentials as SSM to meet security requirements.

Create ServiceNow Change

The retire ServiceNow CMDB CI script prompts the ServiceNow to stop and marks the CI as retired based on the custom property `serviceNowSysId` that was created in the creation script.

1. Open a command-line prompt from your Virtual Machine.
2. Run the Get VM details script.

```
from botocore.vendored import requests
import json
import boto3
client = boto3.client('ssm','ap-southeast-2')

def handler(context, inputs):
    baseUri = inputs['url']
```

```
    casToken = client.get_parameter(Name="casToken",WithDecryption=True)

    url = baseUri + "/iaas/login"
    headers = {"Accept":"application/json","Content-Type":"application/json"}
    payload = {"refreshToken":casToken['Parameter']['Value']}

    results = requests.post(url,json=payload,headers=headers)

    bearer = "Bearer "
    bearer = bearer + results.json()["token"]

    deploymentId = inputs['deploymentId']
    resourceId = inputs['resourceIds'][0]

    print("deploymentId: "+ deploymentId)
    print("resourceId:" + resourceId)

    machineUri = baseUri + "/iaas/machines/" + resourceId
    headers = {"Accept":"application/json","Content-Type":"application/json", "Authorization":bearer }
    resultMachine = requests.get(machineUri,headers=headers)
    print("machine: " + resultMachine.text)

    print( "serviceNowCPUCount: "+ json.loads(resultMachine.text)["customProperties"]["cpuCount"] )
    print( "serviceNowMemoryInMB: "+ json.loads(resultMachine.text)["customProperties"]["memoryInMB"] )

    #update customProperties
    outputs = {}
    outputs['customProperties'] = inputs['customProperties']
    outputs['customProperties']['serviceNowCPUCount'] = int(json.loads(resultMachine.text)
["customProperties"]["cpuCount"])
    outputs['customProperties']['serviceNowMemoryInMB'] = json.loads(resultMachine.text)
["customProperties"]["memoryInMB"]
    return outputs
```

3. Run the CMDB configuration item creation action.

```
from botocore.vendored import requests
import json
import boto3
client = boto3.client('ssm','ap-southeast-2')

def handler(context, inputs):

    snowUser = client.get_parameter(Name="serviceNowUserName",WithDecryption=False)
    snowPass = client.get_parameter(Name="serviceNowPassword",WithDecryption=True)
    table_name = "cmdb_ci_vmware_instance"
    url = "https://" + inputs['instanceUrl'] + "/api/now/table/{0}".format(table_name)
    headers = {'Content-type': 'application/json', 'Accept': 'application/json'}
    payload = {
        'name': inputs['customProperties']['serviceNowHostname'],
        'cpus': int(inputs['customProperties']['serviceNowCPUCount']),
        'memory': inputs['customProperties']['serviceNowMemoryInMB'],
        'correlation_id': inputs['deploymentId'],
        'disks_size': int(inputs['customProperties']['provisionGB']),
        'location': "Sydney",
        'vcenter_uuid': inputs['customProperties']['vcUuid'],
        'state': 'On',
        'sys_created_by': inputs['__metadata']['userName'],
```

```
            'owned_by': inputs['__metadata']['userName']
            }
        results = requests.post(
            url,
            json=payload,
            headers=headers,
            auth=(snowUser['Parameter']['Value'], snowPass['Parameter']['Value'])
        )
        print(results.text)

        #parse response for the sys_id of CMDB CI reference
        if json.loads(results.text)['result']:
            serviceNowResponse = json.loads(results.text)['result']
            serviceNowSysId = serviceNowResponse['sys_id']
            print(serviceNowSysId)

            #update the serviceNowSysId customProperty
            outputs = {}
            outputs['customProperties'] = inputs['customProperties']
            outputs['customProperties']['serviceNowSysId'] = serviceNowSysId;
            return outputs
```

4. Run the Creation action script.

```
from botocore.vendored import requests
import json
import boto3
client = boto3.client('ssm','ap-southeast-2')

def handler(context, inputs):
    snowUser = client.get_parameter(Name="serviceNowUserName",WithDecryption=False)
    snowPass = client.get_parameter(Name="serviceNowPassword",WithDecryption=True)
    table_name = "change_request"
    url = "https://" + inputs['instanceUrl'] + "/api/now/table/{0}".format(table_name)
    headers = {'Content-type': 'application/json', 'Accept': 'application/json'}
    payload = {
        'short_description': 'Provision CAS VM Instance'
              }
    results = requests.post(
        url,
        json=payload,
        headers=headers,
        auth=(snowUser['Parameter']['Value'], snowPass['Parameter']['Value'])
    )
    print(results.text)
```

VCF Automation is successfully integrated with ITSM ServiceNow.

When desired, you can retire your CI by using the CMDB configuration item retire action:

```
from botocore.vendored import requests
import json
import boto3
client = boto3.client('ssm','ap-southeast-2')

def handler(context, inputs):
    snowUser = client.get_parameter(Name="serviceNowUserName",WithDecryption=False)
    snowPass = client.get_parameter(Name="serviceNowPassword",WithDecryption=True)
    tableName = "cmdb_ci_vmware_instance"
    sys_id =inputs['customProperties']['serviceNowSysId']
```

```
url = "https://" + inputs['instanceUrl'] + "/api/now/"+tableName+"/{0}".format(sys_id)
headers = {'Content-type': 'application/json', 'Accept': 'application/json'}
payload = {
    'state': 'Retired'
    }

results = requests.put(
    url,
    json=payload,
    headers=headers,
    auth=(inputs['username'], inputs['password'])
)
print(results.text)
```

How do I tag virtual machines during provisioning by using extensibility actions

You can use extensibility actions along with subscriptions to automate and simplify tagging VMs.

- Access to cloud administrator credentials.
- Amazon Web Services role for Lambda functions.

As a cloud administrator, you can create deployments that are automatically tagged with specified inputs and outputs by using extensibility actions and extensibility subscriptions. When a new deployment is created against the project containing the tag VM subscription, the deployment event triggers the Tag VM script to run and the tags are automatically applied. This saves time and promotes efficiency while allowing for easier deployment management.

1. Navigate to **Extensibility** > **Library** > **Actions** > **New Action** and create an action with the following parameters.

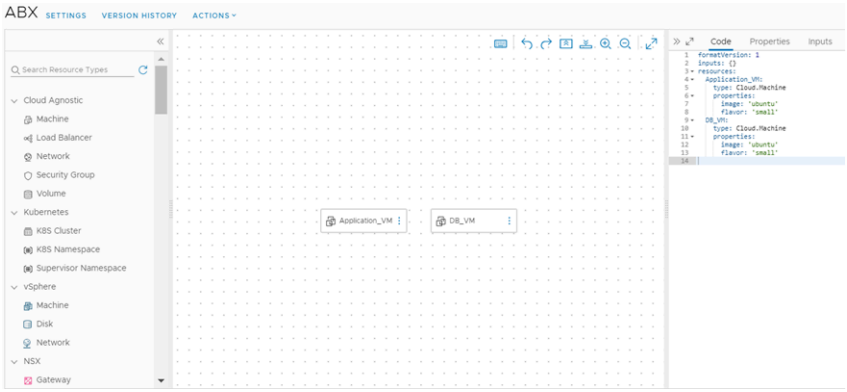
Parameter	Description
Action Name	Extensibility action name, preferably with `TagVM` as a prefix or suffix.
Project	Project to test the extensibility action against.
Action Template	`Tag VM`
Runtime	Python
Script Source	Write Script

2. Enter `Handler` as the **Main function**.
3. Add tagging inputs for testing the extensibility action.
   For example, `resourceNames = ["DB_VM"]` and `target = world`.
4. To save your action, click **Save**.
5. To test your action, click **Test**.
6. To exit the action editor, click **Close**.
7. Navigate to **Extensibility** > **Subscriptions**.
8. Click **New Subscription**.
9. Enter the following subscription details.

Detail	Setting
Event Topic	Select an event topic related to the tagging phase of the VM. For example, Compute Allocation. **Note:** Tags must be part of the event parameters of the selected event topic.
Blocking	Set the timeout for the subscription to 1 minute.

Detail	Setting
Action/Workflow	Select an extensibility action runnable type, and select your custom extensibility action.

10. To save your custom extensibility action subscription, click **Save**.
11. Navigate to **Design** > **Cloud Templates**, and create a cloud template from a blank canvas.
12. Add two virtual machines to the cloud template: `Application_VM` and `DB_VM`.



13. To deploy the VMs, click **Deploy**.
14. During deployment, verify that the event is initiated and the extensibility action is run.
15. To verify that the tags are applied correctly, navigate to **Resources** > **Resources** > **Virtual Machines**.

How can I configure a network interface controller name by using extensibility actions

You can configure the interface name of a network interface controller (NIC) by using IaaS API calls applied through extensibility actions.

- You can only configure the NIC interface name prior to provisioning a compute resource. Therefore, only the **Compute Provision** event topic can be selected for relevant extensibility subscriptions.
- You can only configure NIC interface names for NICs that use Microsoft Azure as a provider.

To configure the interface name of a NIC, you must make `GET` and `PATCH` calls to the VCF Automation IaaS API. By making a `GET` call to https://*your_vRA_fqdn*/iaas/api/machines/*{id}*, you can retrieve the NIC link for the compute resource you want to modify. Then you can make a `PATCH` call to https://*your_vRA_fqdn*/iaas/api/machines/*{id}*/network-interfaces/*{nicId}*, which includes the NIC interface name as a payload, to add the new name for your NIC.

The following scenario uses a sample Python script that can be used for NIC interface name configuration. For your own use cases, you can use a different script and script language, such as Node.js.

1. Create the extensibility action.
   a) Navigate to **Extensibility** > **Actions**.
   b) Click **New Action**.
   c) Enter a name and project for the extensibility action and **Next**.
   d) Add the NIC configuration script.

   The following is a sample Python script:

```
import json

def handler(context, inputs):

    # Get the machine info, which contains machine nic link
    response = context.request('/iaas/api/machines/'+inputs["resourceIds"][0], "GET", {})

    # Build PATCH machine nic payload here
    name = "customized-nic-02";
    data = {'name':name};
```

```
    # Convert machine data string to json object
    response_json = json.loads(response["content"])

    # Patch machine nic
    response_patch = context.request(response_json["_links"]["network-interfaces"]["hrefs"][0] + "?
apiVersion=2021-07-15", 'PATCH', data)

    # return value is empty since we are not changing any compute provisioning parameters
    outputs = {}
    return outputs
```

The preceding sample script performs two primary operations through the IaaS API. First, the script uses a `GET` call to retrieve the NIC link and then uses a `PATCH` call to apply the interface name. In this sample, the NIC interface name is hard-coded into the script as `"customized-nic-02"`.

e) To finish editing the extensibility action, click **Save**.

2. Create an extensibility subscription.
   a) Navigate to **Extensibility** > **Subscriptions**.
   b) Click **New Subscription**.
   c) Enter a name for the extensibility subscription.
   d) Under **Event Topic**, select **Compute Provision** as the event topic for the extensibility subscription.
   e) Under **Action/workflow**, select the extensibility action you created for NIC configuration.
   f) Enable event blocking.

      By enabling blocking, you make sure that the provisioning process is blocked until the extensibility action finishes its run.
   g) To finish editing the extensibility subscription, click **Save**.

The new extensibility subscription runs when a compute provision event is triggered and configures the NIC interface name for the compute resources to be provisioned.

Learn more about extensibility actions in VCF Automation for VM Apps

Action-based extensibility uses streamlined scripts of code within VCF Automation to automate extensibility actions.

Action-based extensibility provides a lightweight and flexible run-time engine interface where you can define small scriptable actions and configure them to initiate when events specified in extensibility subscriptions occur.

You can create these extensibility action scripts of code within VCF Automation, or on your local environment, and assign them to subscriptions. Extensibility action scripts are used for more lightweight and simple automation of tasks and steps. For more information on integrating VCF Automation with a VCF Operations Orchestrator Client server, see Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps.

Action-based extensibility provides:

- An alternative to VCF Operations Orchestrator Client workflows, using small and reusable scriptable actions, for lightweight integrations and customizations.
- A way to reuse action templates, which contain reusable parameterized actions.

You can create extensibility actions by either writing a user-defined action script code or importing a predefined script code as a .ZIP package. Action-based extensibility supports Node.js, Python, and PowerShell run-time environments. The Node.js and Python run-times rely on Amazon Web Services Lambda. Therefore, you must have an active subscription with Amazon Web Services Identity and Access Management (IAM), and configure Amazon Web Services as an endpoint in VCF Automation.

> **Note:**
> Extensibility actions are project-specific.

How do I create extensibility actions

With VCF Automation, you can create extensibility actions for use in extensibility subscriptions.

- Membership in an active and valid project.

- Configured Amazon Web Services role for Lambda functions. For example, `AWSLambdaBasicExecutionRole`.
- Cloud administrator role or `iam:PassRole` permissions enabled.

Extensibility actions are highly customizable, lightweight, and flexible ways to extend application life cycles by using user-defined script code and action templates. Action templates contain predefined parameters that help set up the foundation of your extensibility action.

There are two methods of creating an extensibility action:

- Writing user-defined code for an extensibility action script.

  Note:
  Writing user-defined code in the extensibility action editor might require an active Internet connection.
- Importing a deployment package as a ZIP package for an extensibility action. For information on creating a ZIP package for extensibility actions, see Create a ZIP package for Python runtime extensibility actions, Create a ZIP package for Node.js runtime extensibility actions, or Create a ZIP package for PowerShell runtime extensibility actions.

The following steps describe the procedure for creating an extensibility action that uses Amazon Web Services as a FaaS provider.

1. Select **Extensibility** > **Library** > **Actions**.
2. Click **New Action**.
3. Enter a name for your action and select a project.
4. Optional: Add a description for your action.
5. Click **Next**.
6. Search and select an action template.

   Note:
   To create a custom action without using an action template, select **Custom script**.

   New configurable parameters appear.
7. Select **Write script** or **Import package**.
8. Select the action runtime.
9. Enter an **Main function** name for the action's entry point.

   Note:
   For actions imported from a ZIP package, the main function must also include the name of the script file that contains the entry point. For example, if your main script file is titled `main.py` and your entry point is `handler (context, inputs)`, the name of the main function must be *`main.handler`*.
10. Define the input and output parameters of the action.
11. Optional: Add secrets or extensibility action constants to your default inputs.

    Note:
    For more information on secrets and extensibiltity action constants, see How can I create secrets for use in extensibility actions and How can I create extensibility action constants.
12. Optional: Add application dependencies to the action.

    Note:
    For PowerShell scripts, you can define your application dependencies so they are resolved against the PowerShell Gallery repository. To define your application dependencies so, they are resolvable from the public repository use the following format:

```
@{
                Name = 'Version'
}

e.g.
```

```
@{
    Pester = '4.3.1'
}
```

Note:
For actions imported from a ZIP package, application dependencies are added automatically.

13. To define timeout and memory limits, enable the **Set custom timeout and limits** option.

    The maximum timeout value for scripting type actions is 15 minutes. The maximum timeout value for flow type actions is five hours. For flow type actions, you can also leave the timeout text box blank in which case there is no timeout limit. If you do not set a timeout limit, the action run ends when all flow elements finish their tasks or an error occurs.

14. To test your action, click **Save** and then **Test**.

After your extensibility action is created and verified, you can assign it to a subscription.

Note:
Extensibility subscriptions use the latest released version of an extensibility action. After creating a new version of an action, click **Versions** on the top-right of the editor window. To release the version of the action you want to use in your subscription, click **Release**.

Create a ZIP package for Python runtime extensibility actions

You can create a ZIP package that contains the Python script and dependencies used by your VCF Automation extensibility actions.

If you are using Python 3.3 or earlier, download and configure the PIP package installer. See Python Package Index.

There are two methods of building the script for your extensibility actions:

- Writing your script directly in the extensibility action editor in VCF Automation.
- Creating your script on your local environment and adding it, with any relevant dependencies, to a ZIP package.

By using a ZIP package, you can create a custom preconfigured template of action scripts and dependencies that you can import to VCF Automation for use in extensibility actions.

Furthermore, you can use a ZIP package in scenarios where modules associated with dependencies in your action script cannot be resolved by the VCF Automation service, such as when your environment lacks Internet access.

You can also use a ZIP package to create extensibility actions that contain multiple Python script files. Using multiple script files can be useful for organizing the structure of your extensibility action code.

1. On your local machine, create a folder for your action script and dependencies.
   For example, `/home/user1/zip-action`.
2. Add your main Python action script or scripts to the folder.
   For example, `/home/user1/zip-action/main.py`.
3. Optional: Add any dependencies for your Python script to the folder.
   a) Create a `requirements.txt` file that contains your dependencies. See Requirements Files.
   b) Open a Linux shell.

      Note:
      The runtime of action-based extensibility in VCF Automation is Linux-based. Therefore, any Python dependencies compiled in a Windows environment might make the generated ZIP package unusable for the creation of extensibility actions. Therefore, you must use a Linux shell.

   c) Install your `requirements.txt` file in the script folder by running the following command:

```
pip install -r requirements.txt --target=/home/user1/zip-action
```

4. In the assigned folder, select your script elements and, if applicable, your `requirements.txt` file and compress them to a ZIP package.

   Note:
   Both your script and dependency elements must be stored at the root level of the ZIP package. When creating the ZIP package in a Linux environment, you might encounter a problem where the package content is not stored at the root level. If you encounter this problem, create the package by running the `zip -r` command in your command-line shell.

   ```
   cd your_script_and_dependencies_folder
   zip -r ../your_action_ZIP.zip *
   ```

Use the ZIP package to create an extensibility action script. See How do I create extensibility actions.

Create a ZIP package for Node.js runtime extensibility actions

You can create a ZIP package that contains the Node.js script and dependencies used by your VCF Automation extensibility actions.

There are two methods of building the script for your extensibility actions:

- Writing your script directly in the extensibility action editor in VCF Automation.
- Creating your script in your local environment and adding it, with any relevant dependencies, to a ZIP package.

By using a ZIP package, you can create a custom preconfigured template of action scripts and dependencies that you can import to VCF Automation for use in extensibility actions.

Furthermore, you can use a ZIP package in scenarios where modules associated with dependencies in your action script cannot be resolved by the VCF Automation service, such as when your environment lacks Internet access.

Also, you can use packages to create extensibility actions that contain multiple Node.js script files. Using multiple script files can be useful for organizing the structure of your extensibility action code.

1. On your local machine, create a folder for your action script and dependencies.
   For example, `/home/user1/zip-action`.
2. Add your main Node.js action script or scripts to the folder.
   For example, `/home/user1/zip-action/main.js`.
3. Optional: Add any dependencies for your Node.js script to the folder.
   a) Create a `package.json` file with dependencies in your script folder. See Creating a package.json file and Specifying dependencies and devDependencies in a package.json file.
   b) Open a command-line shell.
   c) Navigate to the folder that you created for the action script and dependencies.

      ```
      cd /home/user1/zip-action
      ```

   d) Install your `package.json` file in the script folder by running the following command:

      ```
      npm install --production
      ```

      Note:
      This command creates a `node_modules` directory in your folder.
4. In the assigned folder, select your script elements and, if applicable, your `node_modules` directory and compress them to a ZIP package.

   Note:
   Both your script and dependency elements must be stored at the root level of the ZIP package. When creating the ZIP package in a Linux environment, you might encounter a problem where the package content is not stored at the root level. If you encounter this problem, create the package by running the `zip -r` command in your command-line shell.

   ```
   cd your_script_and_dependencies_folder
   zip -r ../your_action_ZIP.zip *
   ```

Use the ZIP package to create an extensibility action script. See How do I create extensibility actions.

Create a ZIP package for PowerShell runtime extensibility actions

You can create a ZIP package that contains your PowerShell script and dependency modules for use in extensibility actions.

Verify that you are familiar with PowerShell and PowerCLI. You can find a Docker image with PowerShell Core, PowerCLI 10, PowerNSX, and several community modules and script examples at Docker Hub.

There are two methods of building the script for your extensibility actions:

- Writing your script directly in the extensibility action editor in VCF Automation.
- Creating your script on your local environment and adding it, with any relevant dependencies, to a ZIP package.

By using a ZIP package, you can create a custom preconfigured template of action scripts and dependencies that you can import to VCF Automation for use in extensibility actions.

> **Note:**
> You do not need to define PowerCLI cmdlets as dependencies or bundle them into a ZIP package. PowerCLI cmdlets come preconfigured with the PowerShell runtime of your VCF Automation service.

Furthermore, you can use a ZIP package in scenarios where modules associated with dependencies in your action script cannot be resolved by the VCF Automation service, such as when your environment lacks Internet access.

You can also use a ZIP package to create extensibility actions that contain multiple PowerShell script files. Using multiple script files can be useful for organizing the structure of your extensibility action code.

1. On your local machine, create a folder for your action script and dependencies.
   For example, `/home/user1/zip-action`.
2. Add your main PowerShell script with a `.psm1` extension to the folder.
   The following script presents a simple PowerShell function called `main.psm1`:

```
function handler($context, $payload) {

  Write-Host "Hello " $payload.target



  return $payload
```

   > **Note:**
   > The output of a PowerShell extensibility action is based on the last variable displayed in the body of the function. All other variables in the included function are discarded.

3. Optional: Add a proxy configuration to your main PowerShell script by using `context` parameters. See Using context parameters to add a proxy configuration in your PowerShell script.
4. Optional: Add any dependencies for your PowerShell script.

   > **Note:**
   > Your PowerShell dependency script must use the `.psm1` extension. Use the same name for the script and the subfolder where the script is saved.

   a) Log in to a Linux PowerShell shell.

   > **Note:**
   > The runtime of action-based extensibility in VCF Automation is Linux-based. Any PowerShell dependencies compiled in a Windows environment might make the generated ZIP package unusable. Any installed third-party dependencies must be compatible with the VMware Photon OS as PowerShell scripts run on Photon OS.

   b) Navigate to the `/home/user1/zip-action` folder.
   c) Download and save the PowerShell module containing your dependencies, by running the `Save-Module` cmdlet.

```
Save-Module -Name <module name> -Path ./
```

d) Repeat the previous substep for any additional dependency modules.

Important:
Verify that each dependency module is located in a separate subfolder. For more information on writing and managing PowerShell modules, see How to Write a PowerShell Script Module.

5. In the assigned folder, select your script elements and, if applicable, your dependency module subfolders and compress them to a ZIP package.

Note:
Both your script and dependency module subfolders must be stored at the root level of the ZIP package. When creating the ZIP package in a Linux environment, you might encounter a problem where the package content is not stored at the root level. If you encounter this problem, create the package by running the `zip -r` command in your command-line shell.

```
cd your_script_and_dependencies_folder
zip -r ../your_action_ZIP.zip *
```

Use the ZIP package to create an extensibility action script. See How do I create extensibility actions.

Using context parameters to add a proxy configuration in your PowerShell script

You can enable network proxy communication in your PowerShell script by using `context` parameters.

Certain PowerShell cmdlets might require that you set a network proxy as an environment variable in your PowerShell function. Proxy configurations are provided to the PowerShell function with the `$context.proxy.host` and `$context.proxy.host` parameters.

You can add these `context` parameters in the beginning of your PowerShell script.

```
$proxyString = "http://" + $context.proxy.host + ":" + $context.proxy.port
$Env:HTTP_PROXY = $proxyString
$Env:HTTPS_PROXY = $proxyString
```

If the cmdlets support the `-Proxy` parameter, you can also pass the proxy value directly to the specific PowerShell cmdlets.

Configure cloud-specific extensibility actions

You can configure extensibility actions to work with your cloud accounts.

A valid cloud account is required.

When creating an extensibility action, you can configure and link it to various cloud-based accounts:

- Microsoft Azure
- Amazon Web Services

1. Select **Extensiblity** > **Library** > **Action**.
2. Click **New Action**.
3. Enter the action parameters as necessary.
4. In the **FaaS provider** drop-down menu, select your cloud account provider or select **Auto Select**.

Note:
If you select **Auto**, the action automatically defines the FaaS provider.

5. Click **Save**.

Your extensibility action is linked for use with the configured cloud account.

Configure on-premises extensibility actions

You can configure your extensibility actions to use an on-premises FaaS provider instead of an Amazon Web Services or Microsoft Azure cloud account.

By using an on-premises FaaS provider for your extensibility actions, you can use on-premises services like LDAP, CMDB, or vCenter data centers in your VCF Automation extensibility subscriptions.

1. Select **Extensibility** > **Library** > **Actions**.
2. Click **New Action**.
3. Enter a name and project for the extensibility action.
4. Optional: Enter a description for the extensibility action.
5. Click **Next**.
6. Create or import your extensibility action script.
7. Click the **FaaS provider** drop-down menu and select **On Prem**.
8. To save the new extensibility action, click **Save**.

Use the created extensibility action in your VCF Automation extensibility subscriptions.

How can I create secrets for use in extensibility actions

You can add encrypted inputs to your extensibility action by using project level secrets.

- Create a secret to use with the extensibility action. See Secret properties in VCF Automation for VM Apps.
- Verify that the extensibility action and the secret you want to use together are associated with the same project.

With secrets, you can add encrypted input values to your extensibility actions. Encryption is useful for use cases where your inputs are used to manage sensitive data, such as passwords and certificates. Secrets are available for all FaaS providers and runtimes.

> **Note:**
> You can also add encrypted input values by using action constants. See How can I create extensibility action constants.

Access to secrets depends on the project that they are created in. Secrets created in Project A, for example, are accessible only to users included in Project A.

Secrets use the `context.getSecret()` function to decrypt the secret value when it is added to your script. This function uses the name of the secret as a parameter. For example, you might use an secret named `abxsecret` as an encrypted input parameter in your action. To add this input parameter to your action script, you must use `context.getSecret(inputs["abxsecret"])`.

1. On the **Extensibility** > **Actions** page, select an existing extensibility action or create a new extensibility action.
2. Under **Default Inputs**, select the type **Secret**.
3. Search for your secret and add it to the extensibility action inputs.
4. Add the secret to the script of the extensibility action by using the `context.getSecret()` function.
5. To test your secret, click **Test**.

How can I create extensibility action constants

You can create and store constants for use in extensibility actions.

With extensibility action constants, you can add encrypted input values to your extensibility actions. Encryption is useful for use cases where your inputs are used to manage sensitive data, such as passwords and certificates. Constants are available for all FaaS providers and runtimes.

> **Note:**
> Unlike secrets, extensibility action constants can only be used for extensibility secrets. For more information on secrets, see How can I create secrets for use in extensibility actions.

Extensibility action constants are accessible to all users included in your organization.

Constants use the `context.getSecret()` function to run as part of your script. This function uses the name of constant as a parameter. For example, you might use an extensibility action constant named `abxconstant` as an encrypted input parameter in your action. To add this input parameter to your action script, you must use `context.getSecret(inputs["abxconstant"])`.

1. Create a extensibility action constant.
   a) Navigate to **Extensibility** > **Library** > **Actions**.
   b) Select **Action Constants**.
   c) To create a constant, click **New Action Constant**.
   d) Enter a name and value for the constant, and click **Save**.
2. Add your constant to a extensibility action.
   a) Select an existing extensibility action or create a new extensibility action.
   b) Under **Default Inputs**, tick the **Secret** check box.
   c) Search for your constant and add it to the extensibility action inputs.
   d) Add the constant to the script of the extensibility action by using the `context.getSecret()` function.
   e) To test your extensibility action constant, click **Test**.

Create shared extensibility actions

As an VCF Automation administrator, you create extensibility actions that can be shared across projects without exporting and importing the action.

Create two or more projects in your VCF Automation organization.

For information on exporting and importing extensibility actions, see Export and import extensibility actions.

1. Select **Extensibility** > **Library** > **Actions**.
2. Click **New Action**.
3. Enter a name for your extensibility action.
4. Optional: Enter a description for your extensibility action.
5. Select a project in which your extensibility action is created.
6. Tick the **Share with all projects in this organization** checkbox.
7. Click **Next**.
8. Create or import your action script, and save your extensibility action.

   Note:
   You can enable or disable sharing from **Settings**. If the extensibility action is used in subscriptions, you cannot disable sharing. To disable sharing, you must remove the extensibility action from your subscriptions.
9. Create an extensibility subscription, add the shared extensibility action, and set the subscription scope to **Any Project**.

   Note:
   For more information on creating extensibility subscriptions, see Create an extensibility subscription.

   The extensibility subscription is triggered by matching events in any of your projects.

You can also import shared extensibility actions as a content source in the VCF Automation catalog. When you select the source project, enter the project that the extensibility action was created in.

Configuring the Log Analytics Workspace for Azure based extensibility actions

You can use the Log Analytics Workspace to gather useful logging data about your Azure based extensibility actions. Before using the Log Analytics Workspace, you must register the relevant resource provider.

Verify that your Microsoft Azure account has the **Contributor** or **Owner** role. These roles include the permission to use the `/register/action` operation required for registering a resource provider. You can view your active roles by navigating to your subscription and selecting **My permissions**.

Log Analytics enables you to edit and run log queries on data collected by Azure Monitor Logs, and then interactively analyze the results. You can use Log Analytics queries to retrieve records that match specific criteria to help

identify trends and patterns and provide multiple data insights. Before using Log Analytics, you must register the **Microsoft.OperationalInsights** resource provider to your Azure subscription.

1. Log in to the Microsoft Azure portal.
2. In the search box, enter `Subscriptions`.
3. Select the **Subscriptions** service.
4. Select the subscription associated with your extensibility actions.
5. From the left-side navigation pane, select **Resource providers**.
6. In the **Filter by name** search box, enter **Microsoft.OperationalInsights**.
7. To finish registering the ,resource provider, click **Register**.

   The status of the **Microsoft.OperationalInsights** resource provider changes from **NotRegistered** to **Registered**.

Azure logging for Python-based extensibility actions

You can now use Microsoft Azure 3.x logging functions in your extensibility action script.

Extensibility actions in VCF Automation now use the Microsoft Azure 3.x Scripting API which replaces the previous 1.x version. Microsoft Azure 3.x Scripting API is Linux-based and runs in a container environment.

Because of this version change, logging functions inserted into the script of extensibility actions that use Microsoft Azure as a FaaS (Function as a Service) provider work differently. The next two script samples demonstrate the different logging functions used in the two API versions.

Microsoft Azure 1.x script sample.

```
def handler(context, inputs):
    greeting = "Hello, {0}!".format(inputs["target"])
    print(greeting)

    outputs = {
      "greeting": greeting
    }

    return outputs
```

Microsoft Azure 3.x script sample.

```
import logging

def handler(context, inputs):
    greeting = "Hello, {0}!".format(inputs["target"])
    logging.info(greeting)

    outputs = {
      "greeting": greeting
    }

    return outputs
```

The preceding sample demonstrates that the 3.x version adds the `import logging` function at the beginning of the script while replacing the `print()` function with the `logging.info()` function. To continue using logging with extensibility actions created in the Microsoft Azure 1.x API, you must change the logging functions in your script so it matches the Microsoft Azure 3.x sample.

Export and import extensibility actions

With VCF Automation, you can export and import extensibility actions for use in different projects.

An existing extensibility action.

1. Export an extensibility action.
   a) Navigate to **Extensibility** > **Library** > **Actions**.
   b) Select an extensibility action and click **Export**.
      The action script and its dependencies are saved on your local environment as a ZIP file.
2. Import an extensibility action.
   a) Navigate to **Extensibility** > **Library** > **Actions**.
   b) Click **Import**.
   c) Select the exported extensibility action and assign it to a project.
   d) Click **Import**.

   Note:
   If the imported extensibility action is already assigned to the specified project, you are prompted to select a conflict resolution policy.

What is an action flow

Action flows are a set of extensibility action scripts that are used to extend life cycles and automation further.

All action flows begin with `flow_start` and end with `flow_end`. You can link several extensibility action scripts together, by using the following action flow elements:

- Sequential - Multiple extensibility action scripts running sequentially.
- Fork - Multiple extensibility action scripts or flows that split pathways to contribute to the same output.
- Join - Multiple extensibility action scripts or flows that join together and contribute to the same output.
- Conditional - Multiple extensibility action scripts or flows that run after a condition is satisfied.

Sequential action flows

Multiple extensibility action scripts running sequentially.

```
version: "1"
flow:
 flow_start:
  next: action1
action1:
  action: <action_name>
  next: action2
action2:
  action: <action_name>
  next: flow_end
```

Note:
You can loop back to a previous action by assigning it as the `next:` action. For instance, in this example, instead of `next: flow_end`, you can enter `next: action1` to rerun action1 and restart the sequence of actions.

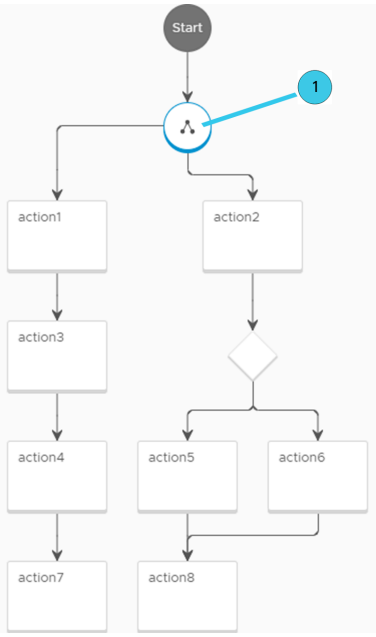


Fork action flows

Multiple extensibility action scripts or flows that split pathways to contribute to the same output.

```
version: "1"
flow:
 flow_start:
  next: forkAction
forkAction:
  fork:
   next: [action1, action2]
action1:
 action: <action_name>
 next: action3
action3:
  action: <action_name>
  next: action4
action4:
  action: <action_name>
   next: action7
action7:
  action: <action_name>
action2:
  action: <action_name>
```

Note:
You can loop back to a previous action by assigning it as the `next:` action. For example, instead of `next: flow_end` to end your action flow, you can enter `next: action1` to rerun action1 and restart the sequence of actions.



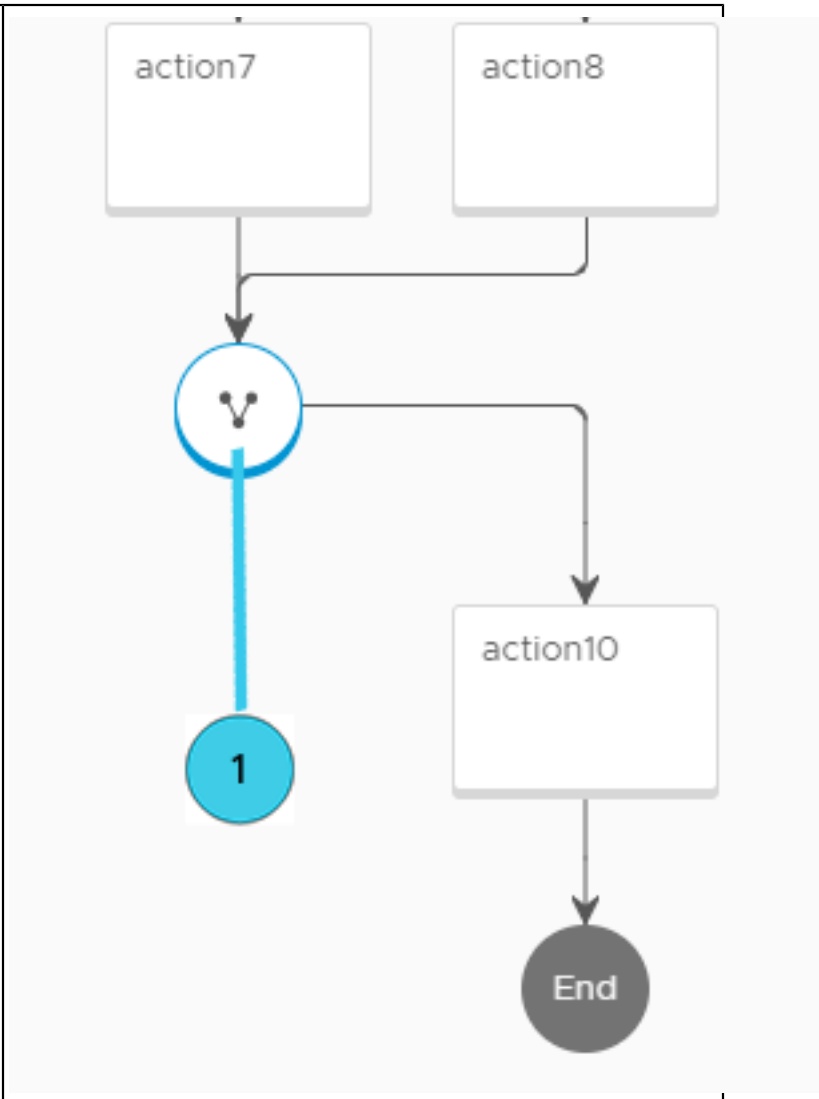
1 Fork Element

Join action flows

Multiple extensibility action scripts or flows that join pathways together and contribute to the same output.

```
version: "1"
action7:
 action: <action_name>
 next: joinElement
action8:
 action: <action_name>
 next: joinElement
joinElement:
 join:
   type: all
   next: action10
action10:
 action: <action_name>
 next: flow_end
```

Note:
You can loop back to a previous action by assigning it as the `next:` action. For instance, in this example, instead of `next: flow_end`, you can enter `next: action1` to rerun action1 and restart the sequence of actions.



1 Join Element

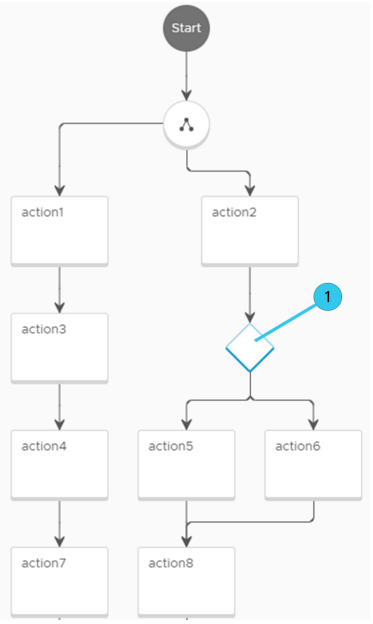
Conditional action flows

Multiple extensibility action scripts or flows that run when a condition is satisfied using a switch element.

In some cases, the condition must be equal to `true` in order for the action to run. Other cases, as seen in this example, require parameter values to be met before an action can run. If none of the conditions are met the action flow fails.

```
version: 1
id: 1234
name: Test
inputs: ...
outputs: ...
flow:
 flow_start:
  next: forkAction
forkAction:
  fork:
   next: [action1, action2]
action1:
   action: <action_name>
   next: action3
action3:
   action: <action_name>
   next: action4
action4:
   action: <action_name>
   next: action7
action7:
   action: <action_name>
   next: joinElement
action2:
  action: <action_name>
  next: switchAction
switchAction:
  switch:
  "${1 == 1}": action5
  "${1 != 1}": action6
action5:
  action: <action_name>
  next: action8
action6:
  action: <action_name>
  next: action8
action8:
 action: <action_name>
```

Note:
You can loop back to a previous action by assigning it as the `next:` action. For example, instead of `next: flow_end` to end your action flow, you can enter `next: action1` to rerun action1 and restart the sequence of actions.



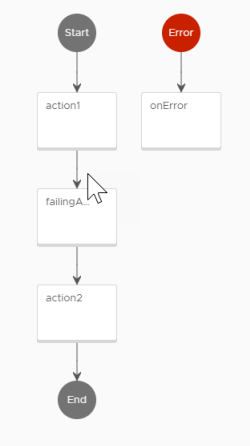
1 Switch element

How do I use an error handler with action flows

You can configure your action flow to issue an error at specified stages of the flow by using an error handler element.

An error handler element requires two inputs:

- Specified error message of the failed action.
- Action flow inputs.



If an action in your flow fails and the action flow contains an error handler element, an error message is issued alerting you of the action failure. The error handler is an action on its own. The following script is an example of an error handler that can be used in an action flow.

```
def handler(context, inputs):

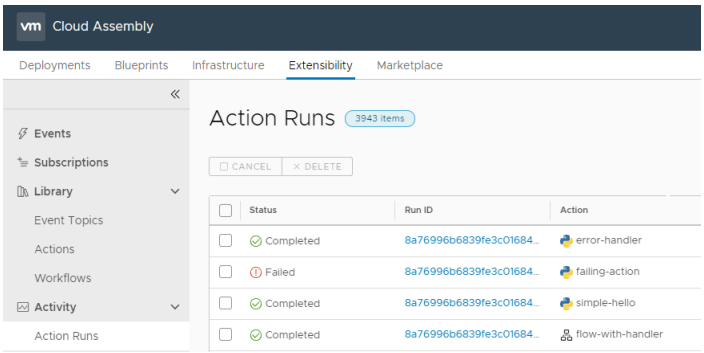
    errorMsg = inputs["errorMsg"]
    flowInputs = inputs["flowInputs"]

    print("Flow execution failed with error {0}".format(errorMsg))
    print("Flow inputs were: {0}".format(flowInputs))

   outputs = {
      "errorMsg": errorMsg,
      "flowInputs": flowInputs
  }

   return outputs
```

You can view the successful and failed runs on the Action Runs window.



In this example, the flow-with-handler action flow, which contains an error handler element, was run successfully. However, one of the actions in the flow failed, which then initiated the error handler to issue an error.

How do I track action runs

The action runs tab shows you a log of subscription triggered extensibility actions and their status.

You can view the log of action runs using **Extensibility** > **Activity** > **Action Runs**. You can also filter the list of action runs by one or more properties at once.

Troubleshooting failed extensibility action runs

If your extensibility action run fails, you can perform troubleshooting steps to correct it.

When an action run fails you might receive an error message, a failed status, and a failed log. If your action run fails, it is either due to a deployment or code failure.

Problem	Solution
Deployment Failure	These failures are a result of problems related to the cloud account configuration, action deployment, or other dependencies that can prevent the action from deploying. Ensure that the project you used is defined within the configured cloud account and granted permissions to run functions. Before initiating the action again, you can test the action against a specific project within the action's details page.
Code Failure	These failures are a result of invalid scripts or code. Use the Action run logs to troubleshoot and correct the invalid scripts.

Extensibility workflow subscriptions
You can use your VCF Operations Orchestrator Client hosted workflows with VCF Automation to extend application lifecycle.

How do I modify virtual machine properties using a VCF Operations Orchestrator Client workflow subscription
You can use an existing VCF Operations Orchestrator Client workflow to modify virtual machine properties and add virtual machines to the active directory.

- Cloud administrator user role
- Existing VCF Operations Orchestrator Client on-premises workflows.
- Successful integration and connection to the VCF Operations Orchestrator Client client server.

The event topic parameters define the format of the payload for Event Broker Service (EBS) messages. To receive and use EBS message payload inside a workflow, you must define the `inputProperties` workflow input parameters.

1. Select **Extensiblity** > **Subscriptions**.
2. Click **New Subscription**.
3. Create a subscription with the following parameters:

Parameter	Value
Name	`RenameVM`
Event topic	Select an event topic suitable for the desired VCF Operations Orchestrator Client integration. For example, compute allocation.
Blocking/Non-blocking	Non-blocking
Action/workflow	Select a VCF Operations Orchestrator Client runnable type. Select the desired workflow. For example, Set VM name.

4. To save your subscription, click **Save**.
5. Assign and activate your subscription by creating a cloud template or deploying an existing cloud template.

Verify that the workflow initiated successfully by one of the following methods:

- Verify the workflow runs log, **Extensibility** > **Activity** > **Workflow Runs**.
- Open the VCF Operations Orchestrator Client client and check workflow status by navigating to the workflow and verifying the status or by opening the specific logs tab.

How do I integrate VCF Automation for ITSM with ServiceNow using VCF Operations Orchestrator Client workflows
Using VCF Operations Orchestrator Client hosted workflows, you can integrate VCF Automation with ServiceNow for ITSM compliance.

- A standalone or clustered VCF Operations Orchestrator Client environment.

- A VCF Operations Orchestrator Client integration in VCF Automation. For information on integrating a standalone VCF Operations Orchestrator Client with VCF Automation, see Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps.

Enterprise users commonly integrate their Cloud Management Platform with an IT Service Management (ITSM) and Configuration Management Database (CMDB) platform for compliance. Following this example, you can integrate VCF Automation with ServiceNow for CMDB and ITSM using VCF Operations Orchestrator Client hosted workflows. When using VCF Operations Orchestrator Client integrations and workflows, capability tags are especially useful if you have multiple instances for different environments.

> **Note:**
> You can also integrate ServiceNow with VCF Automation using extensibility action scripts. For information about integrating ServiceNow using extensibility action scripts, see How do I integrate VCF Automation with ServiceNow using extensibility actions.

In this example, the ServiceNow integration is composed of three top-level workflows. Each workflow has their own subscriptions so that you can update and iterate each component individually.

- Event subscription entry point - Basic logging, identifies the requesting user and vCenter VM, if applicable.
- Integration workflow - Separates objects and feeds inputs into the technical workflow, handles logging, property, and output updates.
- Technical workflow - Downstream system integration for ServiceNow API to create the CMDB CI, CR, and VCF Automation IaaS API with additional virtual machine properties outside of the payload.

1. Create and save a configuration file in VCF Operations Orchestrator Client that contains common configuration used in multiple workflows.
2. Save your VCF Automation API token in the same location, as the configuration file from Step 1.

   > **Note:**
   > The VCF Automation API token has an expiration.

3. Create a workflow in VCF Operations Orchestrator Client with the provided script element. This script references and locates a REST Host. It also standardizes REST actions that use an optional parameter of a token, which is added as an extra authorization header.

```
var configPath = "CS"
var configName = "environmentConfig"
var attributeName = "CASRestHost"

//get REST Host from configuration element
var restHost =
 System.getModule("au.com.cs.example").getRestHostFromConfig(configPath,configName,attributeName)

var ConfigurationElement =
 System.getModule("au.com.cs.example").getConfigurationElementByName(configName,configPath);
System.debug("ConfigurationElement:" + ConfigurationElement);
var casToken = ConfigurationElement.getAttributeWithKey("CASToken")["value"]
if(!casToken){
 throw "no CAS Token";
}
//REST Template
var opName = "casLogin";
var opTemplate = "/iaas/login";
var opMethod = "POST";

// create the REST operation:
var opLogin = System.getModule("au.com.cs.example").createOp(restHost,opName,opMethod,opTemplate);

//cas API Token
var contentObject = {"refreshToken":casToken}
postContent = JSON.stringify(contentObject);
```

```
var loginResponse = System.getModule("au.com.cs.example").executeOp(opLogin,null,postContent,null) ;

try{
 var tokenResponse = JSON.parse(loginResponse)['token']
 System.debug("token: " + tokenResponse);
} catch (ex) {
 throw ex + " No valid token";
}

//REST Template  Machine Details
var opName = "machineDetails";
var opTemplate = "/iaas/machines/" + resourceId;
var opMethod = "GET";

var bearer = "Bearer " + tokenResponse;

var opMachine = System.getModule("au.com.cs.example").createOp(restHost,opName,opMethod,opTemplate);

// (Rest Operation, Params, Content, Auth Token)
var vmResponse = System.getModule("au.com.cs.example").executeOp(opMachine,null,"",bearer) ;

try{
 var vm = JSON.parse(vmResponse);
} catch (ex) {
 throw ex + " failed to parse vm details"
}

System.log("cpuCount: " + vm["customProperties"]["cpuCount"]);
System.log("memoryInMB: " + vm["customProperties"]["memoryInMB"]);

cpuCount =  vm["customProperties"]["cpuCount"];
memoryMB = vm["customProperties"]["memoryInMB"];
```

This script sends the output `cpuCount` and `memoryMB` to the parent workflow and updates the existing `customProperties` properties. These values can be used in subsequent workflows when creating the CMDB.

4. Add the ServiceNow CMDB Create CI script element to your workflow. This element locates the ServiceNow REST Host using the configuration item, creates a REST operation for the `cmdb_ci_vmware_instance` table, creates a string of content object based on workflow inputs for post data, and outputs the returned `sys_id`.

```
var configPath = "CS"
var configName = "environmentConfig"
var attributeName = "serviceNowRestHost"
var tableName = "cmdb_ci_vmware_instance"

//get REST Host from configuration element
var restHost =
 System.getModule("au.com.cs.example").getRestHostFromConfig(configPath,configName,attributeName)

//REST Template
var opName = "serviceNowCreatCI";
var opTemplate = "/api/now/table/" + tableName;
var opMethod = "POST";

// create the REST operation:
var opCI = System.getModule("au.com.cs.example").createOp(restHost,opName,opMethod,opTemplate);
```

```
//cmdb_ci_vm_vmware table content to post;
var contentObject = {};
contentObject["name"] = hostname;
contentObject["cpus"] = cpuTotalCount;
contentObject["memory"] = MemoryInMB;
contentObject["correlation_id"]=  deploymentId
contentObject["disks_size"]=  diskProvisionGB
contentObject["location"] = "Sydney";
contentObject["vcenter_uuid"] = vcUuid;
contentObject["state"] = "On";
contentObject["owned_by"] = owner;

postContent = JSON.stringify(contentObject);
System.log("JSON: " + postContent);

// (Rest Operation, Params, Content, Auth Token)
var ciResponse = System.getModule("au.com.cs.example").executeOp(opCI,null,postContent,null) ;

try{
 var cmdbCI = JSON.parse(ciResponse);
} catch (ex) {
 throw ex + " failed to parse ServiceNow CMDB response";
}

serviceNowSysId = cmdbCI['result']['sys_id'];
```

5. Using the output from the child workflow, create a properties object using the existing `customProperties` and overwrite the `serviceNowSysId` property with the value from ServiceNow. This unique id is used in the CMDB to mark an instance as retired on destroy.

VCF Automation is successfully integrated with ITSM ServiceNow.

Learn more about workflow subscriptions

By using an VCF Operations Orchestrator Clientintegration with VCF Automation, you can extend the life cycles of applications with workflows.

VCF Automation includes an embedded VCF Operations Orchestrator Client deployment. You can use the workflow library of the embedded VCF Operations Orchestrator Client deployment in your subscriptions. You can create, modify, and delete workflows by using the VCF Operations Orchestrator Client client.

You can also integrate an external VCF Operations Orchestrator Client deployment in VCF Automation. See Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps.

Best practices for creating VCF Operations Orchestrator Client workflows

A workflow subscription is based on a specific event topic and the event parameters of that topic. To ensure that the subscriptions initiate the VCF Operations Orchestrator Client workflows, you must configure them with the correct input parameters so that they work with the event data.

Workflow Input Parameters

Your custom workflow can include all the parameters or a single parameter that consumes all the data in the payload.

To use a single parameter, configure one parameter with a type of `Properties` and name `inputProperties`.

Workflow Output Parameters

Your custom workflow can include output parameters that are relevant to subsequent events necessary for a reply event topic type.

If an event topic expects a reply, the workflow output parameters must match the parameters of the reply schema.

How do I track workflow runs

The **Workflow Runs** window displays the logs of the subscription triggered workflows and their status.

You can view the logs of your workflow runs by navigating to **Extensibility** > **Activity** > **Workflow Runs**.

Troubleshooting failed workflow subscriptions

If your workflow subscription fails, you can perform troubleshooting steps to correct it.

Failed workflow runs can cause your workflow subscription not to start or complete successfully. Workflow run failure can result from several common problems.

Problem	Cause	Solution
Your VCF Operations Orchestrator Client workflow subscription did not start or complete successfully.	You configured a workflow subscription to run a custom workflow when the event message is received, but the workflow does not run or complete successfully.	1. Verify that the workflow subscription is saved correctly. 2. Verify that the workflow subscription conditions are configured correctly. 3. Verify that VCF Operations Orchestrator Client contains the specified workflow. 4. Verify that the workflow is configured correctly within VCF Operations Orchestrator Client.
Your approval request VCF Operations Orchestrator Client workflow subscription did not run.	You configured a pre-approval or post-approval workflow subscription to run a VCF Operations Orchestrator Client workflow. The workflow does not run when a machine that matches the defined criteria is requested in the service catalog.	To successfully run an approval workflow subscription, you must verify that all the components are configured correctly. 1. Verify that the approval policy is active and correctly applied. 2. Verify that your workflow subscription is correctly configured and saved. 3. Review the event logs for messages related to approvals.
Your approval request VCF Operations Orchestrator Client workflow subscription was rejected.	You configured a pre-approval or post-approval workflow subscription that runs a specified VCF Operations Orchestrator Client workflow, but the request is rejected on the external approval level. One possible cause is an internal workflow run error in VCF Operations Orchestrator Client. For example, the workflow is missing or the VCF Operations Orchestrator Client server is not running.	1. Review the logs for messages related to approvals. 2. Verify that the VCF Operations Orchestrator Client server is running. 3. Verify that VCF Operations Orchestrator Client contains the specified workflow.

Learn more about extensibility subscriptions

You can extend your application life cycles by using either extensibility actions or VCF Operations Orchestrator Client workflows with extensibility subscriptions.

When a triggering event occurs in your environment, the subscription is initiated and the specified workflow or extensibility action is run. You can view system events on the event log, workflow runs in the workflow runs window, and action runs in the action run window. Subscriptions are project-specific, meaning they are linked to cloud templates and deployments through the specified project.

Extensibility terminology

As you work with extensibility subscriptions within VCF Automation, you might encounter some terminology that is specific to the subscriptions and event broker service.

Table 1837: Extensibility Terminology

Term	Description
Event Topic	Describes a set of events that have the same logical intent and the same structure. Every event is an instance of an event topic. You can assign blocking parameters to certain event topics.
Event	Indicates a change in the state in the producer or any of the entities managed by it. The event is the entity that records information about the event occurrence.
Event Broker Service	The service that dispatches messages published by a producer to the subscribed consumers.
Payload	The event data that contains all the relevant properties related to that Event Topic.
Subscription	Indicates that a subscriber is interested in being notified about an event by subscribing to an event topic and defining the criteria that triggers the notification. Subscriptions link either extensibility actions or workflows to triggering events used to automate parts of the applications life cycle.
Subscriber	The users notified by the events published to the event broker service based on the subscription definition. The subscriber can also be called a consumer.
System Administrator	A user with privileges to create, read, update, and delete tenant workflow subscriptions and system workflow subscriptions using VCF Automation.
Workflow Subscription	Specifies the event topic and conditions that trigger a VCF Operations Orchestrator Client workflow.
Action Subscription	Specifies the event topic and conditions that trigger an extensibility action to run.
Workflow	A VCF Operations Orchestrator Client workflow that is integrated within VCF Automation. You can link these workflows to events within subscriptions.
Extensibility Action	A streamlined script of code that can run after an event is triggered in a subscription. Extensibility actions are similar to workflows, but are more lightweight. Extensibility actions can customized from within VCF Automation.
Action Runs	Accessible through the **Action Runs** tab. An action run is a detailed log of extensibility actions that have run in response to triggering events.

Blocking event topics

Some event topics support blocking events. The behavior of an extensibility subscription depends on whether the topic supports these event types and how you configure the subscription.

VCF Automation extensibility subscriptions can use two broad types of event topics: non-blocking and blocking event topics. The event topic type defines the behavior of the extensibility subscription.

Non-Blocking Event Topics

Non-blocking event topics only allow you to create non-blocking subscriptions. Non-blocking subscriptions are triggered asynchronously and you cannot rely on the order that the subscriptions are triggered in.

Blocking Event Topics

Some event topics support blocking. If a subscription is marked as blocking, all messages that meet the set conditions are not received by any other subscriptions with matching conditions until the runnable item of the blocking subscription is run.

Blocking subscriptions run in priority order. The highest priority value is 0 (zero). If you have more than one blocking subscription for the same event topic with the same priority level, the subscriptions run in a reverse alphabetical order based on the name of the subscription. After all blocking subscriptions are processed, the message is sent to all the non-blocking subscriptions at the same time. Because the blocking subscriptions run synchronously, the changed event payload includes the updated event when the subsequent subscriptions are notified.

You can use blocking event topics to manage multiple subscriptions that are dependent on each other.

For example, you can have two provisioning workflow subscriptions where the second subscription depends on the results of the first subscription. The first subscription changes a property during provisioning, and the second subscription records the new property, such as a machine name, in a file system. The ChangeProperty subscription is prioritized as 0 and the RecordProperty is prioritized as 1 because the second subscription uses the results of the first subscription. When a machine is provisioned, the ChangeProperty subscription begins running. Because the RecordProperty subscription conditions are based on a post-provisioning condition, an event triggers the RecordProperty subscription. However, because the ChangeProperty workflow is a blocking workflow, the event is not received until it is finished. When the machine name is changed and the first workflow subscription is finished, the second workflow subscription runs and records the machine name in the file system.

Recovery Runnable Item

For blocking event topics, you can add a recovery runnable item to the subscription. The recovery runnable item in a subscription runs if the primary runnable item fails. For example, you can create a workflow subscription where the primary runnable item is a workflow that creates records in a CMDB system such as ServiceNow. Even if the workflow subscription fails, some records might be created in the CMDB system. In this scenario, a recovery runnable item can be used to clean up the records left in the CMDB system by the failed runnable item.

For use cases that include multiple subscriptions that are dependent on each other, you can add an `ebs.recover.continuation` property to the recovery runnable item. With this property, you can direct if the Extensibility service must continue with the next subscription in your chain, if the current subscription fails.

Event topics provided with VCF Automation for VM Apps in VCF Automation includes predefined event topics.

Event Topics

Event topics are the categories that group similar events together. When assigned to a subscription, event topics define which event triggers the subscription. The following event topics are provided by default with VCF Automation. All topics can be used to add or update custom properties or tags of the resource. If a VCF Operations Orchestrator Client workflow or extensibility action fails, the corresponding task fails as well.

Table 1838: VCF Automation Event Topics

Event Topic	Blockable	Description
`Approval Events`	Yes	Issued when an action that requires an approval is triggered.
`Template configuration`	No	Issued when a cloud template configuration event, such as the creation or deletion of a cloud template, occurs. This event topic can be useful for notifying external systems of such events.

Event Topic	Blockable	Description
`Template version configuration`	No	Issued when a new cloud template versioning event occurs, such as the creation, release, de-release, or restoration of a version. This event topic can be useful with integrations of third-party version control systems.
`Compute allocation`	Yes	Issued before the allocation of `resourcenames` and `hostselections`. Both of these properties can be modified at this stage. Issued once for a cluster of machines.
`Compute gateway post provisioning`	Yes	Issued after a compute gateway resource is provisioned.
`Compute gateway post removal`	Yes	Issued after a compute gateway is removed.
`Compute gateway provisioning`	Yes	Issued before a compute gateway is provisioned.
`Compute gateway removal`	Yes	Issued before a compute gateway is removed.
`Compute initial power on`	Yes	Issued after a resource is provisioned at the hypervisor layer, but before the resource is powered on for the first time. Currently, this event topic is only supported for vSphere. Events are sent for each machine in a cluster.
`Compute nat post provisioning`	Yes	Issued after a compute NAT resource is provisioned.
`Compute nat post removal`	Yes	Issued after a compute NAT resource is removed.
`Compute nat provisioning`	Yes	Issued before a compute NAT is provisioned.
`Compute nat removal`	Yes	Issued before a compute NAT is removed.
`Compute post provision`	Yes	Issued after a resource is provisioned. Events are sent for each machine in a cluster.
`Compute post removal`	Yes	issued after a compute resource is removed. Events are sent for each machine in a cluster.
`Compute provision`	Yes	Issued before the resource is provisioned at the hypervisor layer. Events are sent for each machine in a cluster. **Note:** You can change the allocated IP address.
`Compute removal`	Yes	Issued before the resource is removed. Events are sent for each machine in a cluster.

Event Topic	Blockable	Description
`Compute reservation`	Yes	Issued at the time of reservation. Issued once for a cluster of machines. **Note:** You can change the placement order.
`Compute post migration reconcile status`	No	When a managed virtual machine is moved by using vMotion in vCenter or across vCenter instances, the VM is then reconciled in VCF Automation. This event topic is issued to publish details of this virtual machine. The event topic contains the status of the VM reconciliation in VCF Automation and the destination details of the moved VM.
`Custom resource post provision`	Yes	Issued for post provisioning events triggered by custom resource operations. **Note:** This event topic is usable only for VCF Operations Orchestrator Client-based custom resources.
`Custom resource pre provision`	Yes	Issued for pre provisioning events triggered by custom resource operations. **Note:** This event topic is usable only for VCF Operations Orchestrator Client-based custom resources.
`Deployment action completed`	Yes	Issued after a deployment action is finished.
`Deployment action requested`	Yes	Issued before a deployment action is finished.
`Deployment completed`	Yes	Issued after the deployment of a cloud template or catalog request.
`Deployment onboarded`	No	Issued when a new deployment is onboarded.
`Deployment requested`	Yes	Issued before the deployment of a cloud template or catalog request.
`Deployment resource action completed`	Yes	Issued after the deployment of a resource action.
`Deployment resource action requested`	Yes	Issued before the deployment of a resource action. **Note:** In case of an Expire action, you can continue to subscribe to `Deployment resource action requested` or `Deployment resource action completed` and events will be triggered.
`Deployment resource completed`	Yes	Issued after the provisioning of a deployment resource.

Event Topic	Blockable	Description
`Deployment resource requested`	Yes	Issued before the provisioning of a deployment resource.
`Disk allocation`	Yes	Issued for the preallocation of disk resources.
`Disk attach`	Yes	Issued before a disk is attached to a machine. `Disk attach` is a read and write event. Disk properties supported for write-back are: • `diskFullPaths` • `diskDatastoreNames` • `diskParentDirs` All three vSphere specific disk properties are required for updates. All other properties are read-only. **Note:** Write-back is optional for vSphere First Class Disks.
`Disk detach`	Yes	Issued after a disk is detached from a machine. `Disk detach` is a read-only event.
`Disk post removal`	Yes	Issued after a disk resource is deleted.
`Disk post resize`	Yes	Issued after a disk resource is resized.
`Idem Service post event`	No	Issued when an idem task is completed or failed
`Load balancer post provision`	Yes	Issued after the provisioning of a load balancer.
`Load balancer post removal`	Yes	Issued after the removal of a load balancer.
`Load balancer provision`	Yes	Issued before provisioning a load balancer.
`Load balancer removal`	Yes	Issued before removing a load balancer.
`Network Configure`	Yes	Issued when the network is configured during compute allocation. **Note:** The Network Configure event topic supports multiple IP addresses/ NICs. Using a static IP assignment (`assignment:static`) is not supported within a cloud template when using a Network Configure event topic.
`Network post provisioning`	Yes	Issued after a network resource is provisioned.
`Network post removal`	Yes	Issued after a network resource is removed.
`Network provisioning`	Yes	Issued before a network resource is provisioned.
`Network removal`	Yes	Issued before a network resource is removed.
`Post unregistered provisioned approval`	Yes	Issued after a provisioned machine is unregistered.

Event Topic	Blockable	Description
`Pre unregistered provisioned approval`	Yes	Issued before a provisioned machine is unregistered.
`Project Lifecycle Event Topic`	No	Issued when a project is created, updated, or deleted.
`Provider resource action cud`	No	Issued when a resource action is created, updated, or deleted.
`Provisioning request`	Yes	Issued when a request is sent to the provisioning service. This can be an allocation request, provisioning request, or a day 2 operation, such as removal. Allocation requests are issued once for each cluster of machines. Provisioning and day 2 requests are issued for each machine in a cluster.
`Security group post provision`	Yes	Issued after a security group is provisioned.
`Security group post removal`	Yes	Issued after a security group is removed.
`Security group provisioning`	Yes	Issued before a security group is provisioned.
`Security group removal`	Yes	Issued before a security group is removed.

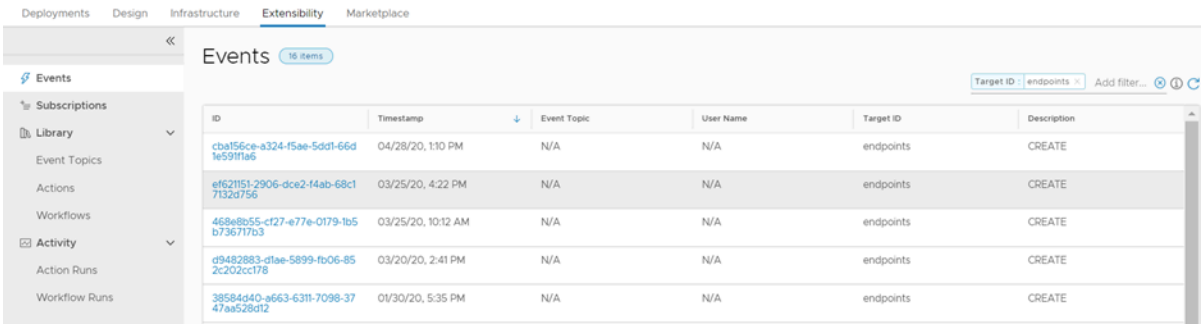
Event Parameters

After you add an event topic, you can view the parameters of that event topic. These event parameters define the structure of the event's payload, or `inputProperties`. Certain event parameters cannot be modified and are marked as read-only. You can identify these read-only parameters by clicking the info icon to the right of the parameter.

Extensibility event log

The extensibility events page displays a list of all events that have occurred within your environment.

You can view the extensibility event logs by navigating to **Extensibility** > **Events**. You can also filter the list of events by one or more properties. To view additional details of an individual event, select the event's ID.



Create an extensibility subscription

By using a VCF Operations Orchestrator Client integration, or extensibility actions with VCF Automation, you can create subscriptions to extend your applications.

- Verify that you have the cloud administrator user role.
- If you are using VCF Operations Orchestrator Client workflows:
  - The library of the embedded VCF Operations Orchestrator Client Client or the library of any integrated external VCF Operations Orchestrator Client instance.
- If you are using extensibility actions:

– Existing extensibility action scripts. For more information, see How do I create extensibility actions.

Extensibility subscriptions allow you to extend your applications by triggering workflows or actions at specific life-cycle events. You can also apply filters to your subscriptions to set Boolean conditions for the specified event. For example, the event and workflow or action only triggers if the Boolean expression is `'true'`. This is helpful for scenarios where you want to control when events, actions, or workflows are triggered.

1. Select **Extensibility** > **Subscriptions**.
2. Click **New Subscription**.
3. Enter the details of your subscription.
4. Set the **Organization scope** for the subscription.
5. Select an **Event Topic**.
6. Optional: Set conditions for the event topic.

   Note:
   Conditions can be created by using a JavaScript syntax expression. This expression can include Boolean operators, such as `"&&" (AND)`, `"||" (OR)`, `"^" (XOR)`, and `"!" (NOT)`. You can also use arithmetic operators, such as `"==" (equal to)`, `"!=" (not equal to)`, `">=" (greater than or equal)`, `"<=" (less than or equal)`, `">" (greater than)`, and `"<" (lesser than)`. More complex Boolean expressions can be built out of simpler expressions. To access the event payload according to the specified topic parameters, use `'event.data'` or any of the event header properties: `sourceType`, `sourceIdentity`, `timeStamp`, `eventType`, `eventTopicId`, `correlationType`, `correlationId`, `description`, `targetType`, `targetId`, `userName`, and `orgId`.
7. Under **Action/workflow**, select a runnable item for your extensibility subscription.
8. Optional: If applicable, configure the blocking behavior for the event topic.
9. Optional: To define the project scope of the extensibility subscription, deselect **Any Project** and click **Add Projects**.

   Note:
   If the organization scope of the subscription is set to **Any tenant organization**, the project scope is always set to **Any Project** and the project scope cannot be changed. You can only change the project scope if the organization scope is set to the provider organization.
10. To save your subscription, click **Save**.

Your subscription is created. When an event, categorized by the selected event topic occurs, the linked VCF Operations Orchestrator Client workflow or extensibility action is initiated and all subscribers are notified.

After creating your subscription, you can create or deploy a cloud template to link and use the subscription. You can also verify the status of the workflow or extensibility action run in the **Extensibility** tab in VCF Automation. For subscriptions containing VCF Operations Orchestrator Client workflows, you can also monitor runs and workflow status from the VCF Operations Orchestrator Client Client.

Using extensibility subscriptions to manage deployment expiry

You can manage expired deployments and their resources by using the `Expire` action alongside existing event topics.

After a deployment lease in your environment expires, you can use extensibility event topics to perform tasks, such as stopping the back up or monitoring of any deployment resources. To perform these day 2 operations, the VCF Automation API uses a system-level `Expire` action. This action is triggered automatically by the system whenever a deployment lease in your organization expires. The `Expire` action trigger precedes the power off event for any resources associated with that deployment.

Note:
In previous product releases, the power off event was triggered at the deployment level after lease expiry. Now the power off event is triggered at the resource level for each deployment resource that is in the powered on state.

The `Expire` action is included in the payload of existing event topics, such as **Deployment action requested** and **Deployment action completed**, and uses the `deploymentid` parameter to perform pre-expiry and post-expiry tasks associated with the deployment resources.

Note:
The `Expire` action is triggered approximately 10 to 15 minutes after your deployment lease expires. The system does not trigger lease expiry events prior to the actual lease expiry. The `Expire` action is a system-level action and users cannot trigger the events associated with it manually.

For the current use case, you are using the **Deployment action requested** event topic along with the `Expire` action to back up a virtual machine in your deployment as a template. For this case, the back up is performed by using a VCF Operations Orchestrator Client workflow but the same task can also be performed by using an extensibility action as the runnable item of the subscription.

1. Navigate to **Extensibility** > **Subscriptions** and click **New Subscription**.
2. Enter a name for the subscription.
3. Under **Status**, verify that the subscription is enabled.
4. Under **Event Topic**, select the **Deployment action requested** event topic.
5. Toggle on the **Condition** option and add a filter for the expiry action:

   ```
   event.data.actionName == 'Expire'
   ```

   Note:
   The **Deployment action requested** event topic can be triggered by different deployment day 2 operations, such as changing the deployment lease duration. Adding the lease expiry action filter guarantees that the subscription is triggered only for expiry events.
6. Under **Action/workflow**, add the VCF Operations Orchestrator Client workflow.

   The schema of this sample workflow includes a scriptable task and a workflow element which includes the **Clone virtual machine, no customization** workflow which comes preconfigured with VCF Operations Orchestrator Client. The scriptable task element includes the following sample script:

   ```
   System.log("Lease expiry action triggered to clone a VM...")

   System.log("Deployment Id is: " + inputProperties.deploymentId);
   inputHeaders = new Properties();
   deploymentId = inputProperties.deploymentId;
   pathUriVariable = "/deployment/api/deployments/" +deploymentId +"/resources";
   var restClient = vRAHost.createRestClient();
   var request = restClient.createRequest("GET", pathUriVariable, null);
   var keys = inputHeaders.keys;
   for(var key in keys){
       request.setHeader(keys[key], inputHeaders.get(keys[key]));
   }
   var response = restClient.execute(request);
   System.log("Content as string: " + response.contentAsString);
   var content = response.contentAsString;
   var obj = JSON.parse(content);

   var object = new Properties(obj);
   var contentJson = object.content;
   for (var i = 0; i < contentJson.length; i++) {
       var resources = contentJson[i];

       var resourceProperties = resources.properties;
       System.log("Resource name is: " + resourceProperties.resourceName)
       resourceName = resourceProperties.resourceName;
   }

   var query = "xpath:name='" + resourceName + "'";
         var vms=Server.findAllForType("VC:VirtualMachine", query);
         vcVM=vms[0];
   ```

```
System.log("VM input is: " + vcVM);
dataStoreOutput = datastore
template= true;
name="test-vm-name"
```

7. Decide whether to set the subscription as blocking or non-blocking.

   Note:
   Making the subscription blocking means that the power off event for the deployment resources is triggered only after the runnable item, in this case the lease expiry workflow, finishes its run successfully. Making the subscription non-blocking means that power off event is triggered for the deployment resources regardless of the status of the workflow run.

8. To finish editing the subscription, click **Save**.

After the extensibility subscription is triggered by the lease expiry event and the workflow run is successful, navigate to the vSphere Client and validate that your virtual machine is converted to a template.

Troubleshooting an extensibility subscription

Troubleshoot extensibility subscription failures.

When your subscription fails, it is commonly a result of errors with your workflow or extensibility action script.

View topic parameters and payload

You can use a dump subscription topic parameters script to view the specific parameters and payload of your virtual machine at any given event stage.

Primarily, this script is useful for debugging and verifying available inputs for your VCF Operations Orchestrator Client workflow. To view all parameters of your virtual machine, use the following script with your workflow:

```
function dumpProperties(props,lvl){
         var keys = props.keys;
         var prefix = ""
     for (var i=0; i<lvl; i++){
                  prefix = prefix + "";
          }
          for (k in keys){
                 var key = keys[k];
                 var value = props.get(keys[k])
                 if ("Properties" == System.getObjectType(value)){
                  System.log(prefix + key + "[")
                  dumpProperties(value,(lvl+2));
                  System.log(prefix+ "]")
                 } else{
                   System.log( prefix + key + ":" + value)
                 }
                 }
   }

   dumpProperties(inputProperties, 0)

   customProps = inputProperties.get("customProperties")
```

Subscription version history

If your subscription fails, you can view the version history.

Viewing Subscription Version History

The **Version History** tab of the subscription editor can show you the change history of your subscription, including the user and date of the change. You can also compare different subscription versions by clicking **Compare to**. If your subscription fails or is running incorrectly, the version history can help identify the cause.

Adding Content to the Catalog in VCF Automation for VM Apps

The requirements and process for setting up your VCF Automation catalog depends on the content that you are providing to your users.

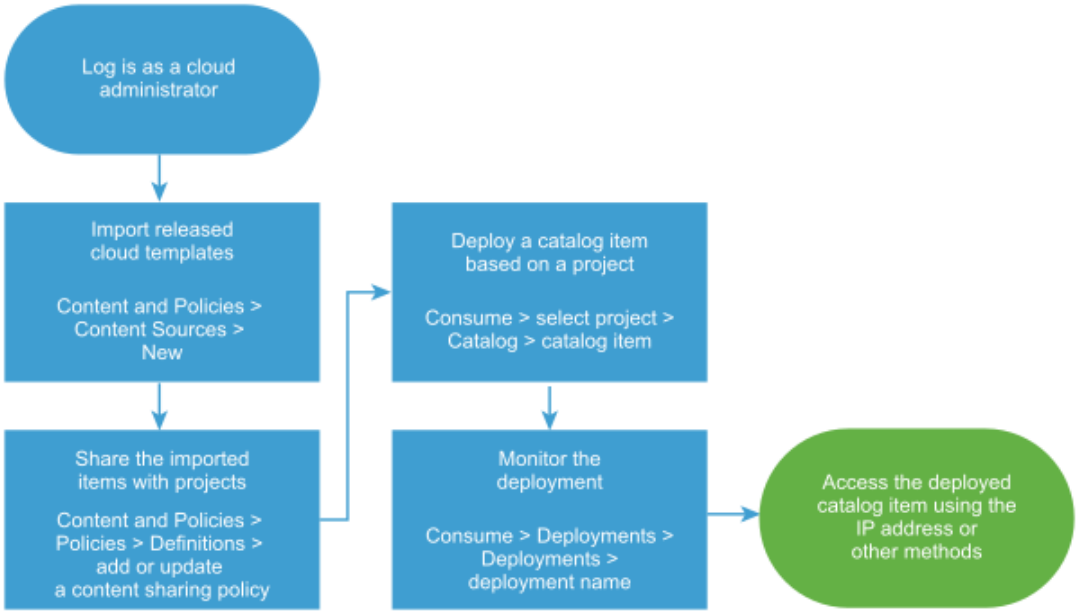
Each process is provided as an end-to-end procedure. Identify the content that you are providing and add each relevant type. Ensure that the imported content is working properly outside of VCF Automation before you add it to the catalog.

After you add the content sources, the templates are refreshed every six hours. Any changes to the templates in your external sources are reflected in the catalog after a refresh.

Add VCF Automation templates to the VCF Automation catalog

As a cloud administrator, you can make VCF Automation templates available in the VCF Automation catalog by adding an VCF Automation content source and sharing the templates. The templates are the specifications for services or applications that you can deploy to your cloud providers.

Verify that the cloud templates that you are importing are deployable and released in VCF Automation before you import them.



After you import the VCF Automation templates, you share them with project members so that they can deploy the templates. At the request time, the template is deployed to cloud zone account region or datastore that supports the template requirements.

1. Import VCF Automation templates.
   a) Select **Content and Policies** > **Content Sources**.
   b) Click **New**, and then click **Template**.
   c) Enter the **Name** for this content source.
   d) Select the **Source project** and then click **Validate**.

      The validation process tests the connection and provides the number of released templates that are associated with the project in VCF Automation.

e) Click **Create and Import**.

The Content Sources page lists your new source and the number of discovered and imported items.

2. Share the imported items with a project.

   If you want to share the template with more than one project, you must create a separate content sharing policy for each project.

   a) Select **Content and Policies** > **Policies** > **Definitions**, and create a new content sharing policy.
   b) Enter a name for the content sharing policy.
   c) In the **Scope** list, select the target project.
   d) In the **Content sharing** section, click **Add Items**, and then select one or more cloud templates to share with the project.

      The list of possible templates includes the templates associated with the current project in VCF Automation and any templates for other projects where sharing is enabled.

      You can select all the items imported from a content sources or you can select individual items. To select only particular templates, select **All Content** in the Content Sources drop-down menu.
   e) In the **Users** section, select the users and user groups that you want to have access to the content.

      You can share the content with all users and groups in the project, or you can select individual users and groups.
   f) Click **Create**.

   The VCF Automation templates are added to the catalog where the project members can request them.

3. Verify that the template is available in the catalog to the members of the selected projects with whom you shared the content.

   a) Click the **Consume** tab.
   b) In the **Projects** drop-down menu, select the project with which you shared the cloud template.

      You can select multiple projects.
   c) On the **Catalog** page, locate the imported template, and review the projects to ensure that the project you configured is included.
   d) Click **Request** and provide any required information.

      If the template has more than one released version, select the version that you want to deploy.
   e) Click **Submit**.

      The provisioning process begins and the Deployments page opens with your current request at the top.

4. Monitor the provisioning process to ensure successful deployment.

   a) Select **Deployments** > **Deployments** and locate your deployed catalog item.
   b) Monitor the card status until it is successful.

The released VCF Automation templates are imported into VCF Automation, shared in the catalog, and deployable.

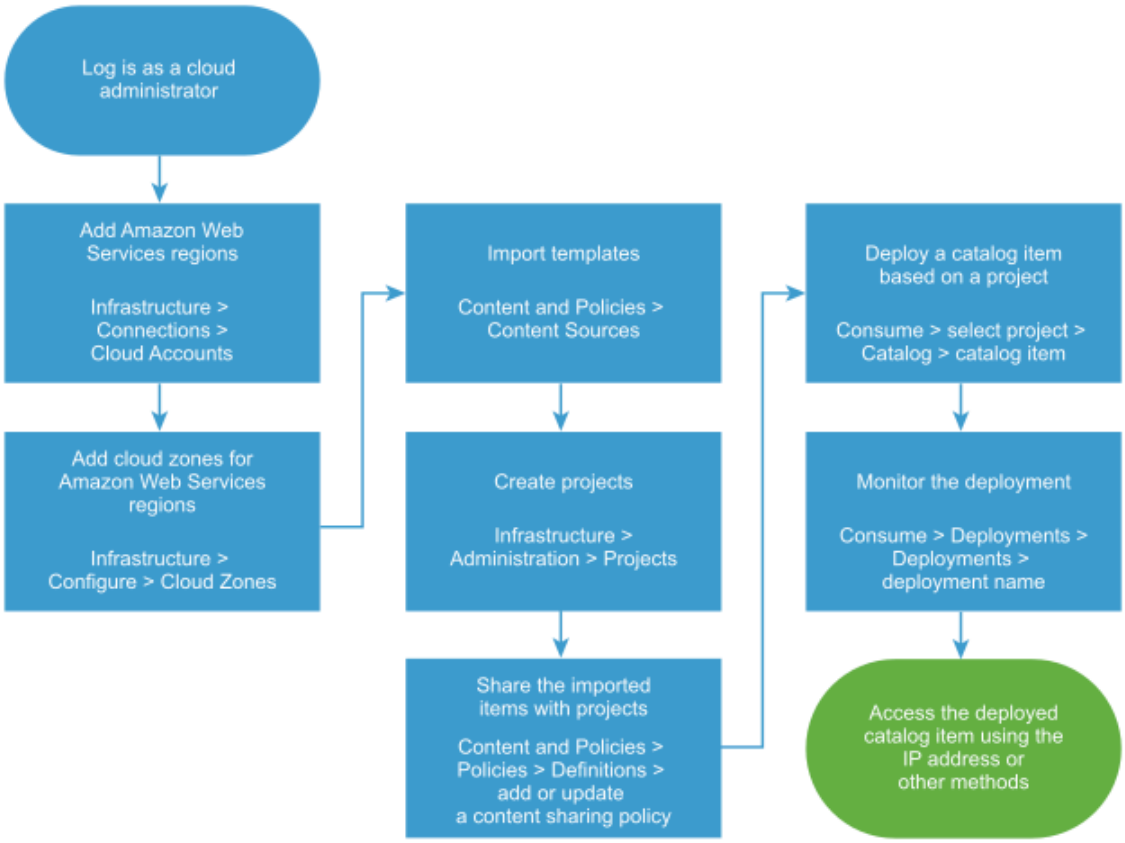
- If the deployment fails, click the deployment name and begin troubleshooting. If you are a VCF Automation cloud administrator, you can also do more extensive troubleshooting in VCF Automation.
- If you want to control how long a deployment can exist, create a lease. See Setting up policies in VCF Automation for VM Apps.
- To provide more or fewer user inputs at request time, you can create a custom form. See Customize an icon and request form in VCF Automation for VM Apps.

Add Amazon CloudFormation templates to the VCF Automation catalog

As a cloud administrator, you can populate the VCF Automation catalog with Amazon CloudFormation templates by adding one or more Amazon S3 buckets as content sources and sharing them with project members. The templates are the specifications for the services or applications that you can deploy to Amazon Web Services.

- Ensure that you know to name of the S3 bucket that contains your CloudFormation templates.

- If you are adding a private bucket, you must know the access key and the secret key.
- Ensure that the CloudFormation template you want to import meets the size limit. The maximum allowed size is 150KB.



You can only add one bucket as a content source. To add multiple buckets, you create a content source for each bucket.

After you add the templates, you entitle project members to deploy the cloud templates. At the request time, the cloud template is deployed to the cloud account region that you define when you add the content source.

1. To deploy your CloudFormation templates, you must have at least one Amazon Web Services cloud account and select the regions.
   a) Select **Infrastructure** > **Connections** > **Cloud Accounts**.
   b) Click **Add Cloud Account** and then click **Amazon Web Services**.
   c) Enter the 20-digit **Access Key ID** and corresponding **Secret Access Key**.
   d) To verify the credentials, click **Validate**.
   e) Enter an account name.

      Provide a name that you can identify when you share templates with projects.

f) Select one or more regions in this account that you want to deploy templates to.
g) Click **Create**.

2. Define cloud zones for the Amazon Web Services cloud account regions.
   a) Select **Infrastructure** > **Configure** > **Cloud Zones**, and then click **New Cloud Zone**.
   b) Select the **Account/region**, the **Name**, and the **Placement policy**.
   c) Click the **Compute** tab and verify or modify the resources that are included in the cloud zone.
   d) Click **Create**.
3. Import the templates.
   a) Select **Content and Policies** > **Content Sources**.
   b) Click **New**, and then click **AWS CloudFormation Template**.
   c) Enter the **Name** for this content source.
   d) Add the S3 bucket information.
   e) Click **Validate**.

      If the bucket is public, the validation process verifies the name and the number of templates. If the bucket is private, the validation process verifies the name, the keys, and the number of templates.
   f) Select the **Deployment Target** Amazon Web Services cloud account and a region.
   g) Click **Create and Import**.
4. Add a project so that you can share the templates with project members.
   a) In VCF Automation, select **Infrastructure** > **Administration** > **Projects**, and then click **New Project**.
   b) Enter the project information on the **Summary** tab.
   c) Click the **Users** tab and then click **Add Users**.

      To add project users, the individuals or the groups must already be active service organization users.
   d) If this project supports only CloudFormation templates, ignore the Provisioning tab.

      CloudFormation templates are deployed to the target account and region that you defined when you imported the templates. If the project members can deploy other templates or content, you must add the target cloud zones for the content to the project.
   e) Click **Create**.

   The new project is added to your projects. It is also added to your associated VCF Automation instance. If the project is for VMware Cloud Templates, you can add cloud zones in VCF Automation. If the project is for templates, you do not need to add cloud zones.
5. Share the imported templates with a project.

   If you want to share the templates with more than one project, you must create a separate content sharing policy for each project.
   a) Select **Content and Policies** > **Policies** > **Definitions**, and create a new content sharing policy.
   b) Enter a name for the content sharing policy.
   c) In the **Scope** list, select the project that includes the users who should be able to deploy the templates.
   d) In the **Content sharing** section, click **Add Items**, and then select one or more Amazon Web Services content sources to share with the project.
   e) In the **Users** section, select the users and user groups that you want to have access to the content.
   f) Click **Create**.

   The templates are added to the catalog where the project members can request them.
6. Verify that the template is available in the catalog to the members of the selected projects.
   a) Click the **Consume** tab.
   b) In the **Projects** drop-down menu, select the project with which you shared the CloudFormation template.

      You can select multiple projects.

c) On the **Catalog** page, locate the imported CloudFormation template, and review the projects to ensure that the project you configured is included.
d) Click **Request** and provide any required information.
e) Click **Submit**.

The provisioning process begins and the Deployments page opens with your current request at the top.

7. Monitor the provisioning process to ensure successful deployment.
   a) Select **Deployments** > **Deployments** and locate your deployed catalog item.
   b) Monitor the card status until it is successful.

The templates are imported into VCF Automation and shared in the catalog.

- If the deployment fails, click the deployment name and begin troubleshooting. If you are a VCF Automation cloud administrator, you can also do more extensive troubleshooting in VCF Automation.
- If you want to control how long a deployment can exist, create a lease. See Setting up policies in VCF Automation for VM Apps.
- To provide more or fewer user inputs at request time, you can create a custom form. See Customize an icon and request form in VCF Automation for VM Apps.

Add VCF Operations Orchestrator Client workflows to the VCF Automation catalog

As a cloud administrator, you can add VCF Operations orchestrator workflows to the catalog. The workflows are created in VCF Operations Orchestrator Client to accomplish a simple or complex task.

Verify that you have VCF Operations Orchestrator Client workflows that can perform required tasks.

In addition to the regular input parameters, the workflows can include composite types as input parameters.

1. If you do not have a VCF Operations orchestrator integration configured in VCF Automation, you can add the integration in VCF Automation.
   a) Select **Infrastructure** > **Connections** > **Integrations**.
   b) Click **Add Integration** and then click **Orchestrator**.
   c) Enter a name that identifies this instance when you create the content source.
   d) Enter the URL for your VCF Operations Orchestrator Client instance.
   e) Click **Validate**.
   f) Click **Add**.
2. Import the workflow.
   a) Select **Content and Policies** > **Content Sources**.
   b) Click **New**, and then click **Orchestrator**.
   c) Enter the **Name** for this content source so that you can identify it when you share the content.
   d) Click **Add** and select the workflows that you want to make available in VCF Automation.
   e) Click **Create and Import**.
3. Share the imported workflow with a project.
   a) Select **Content and Policies** > **Policies** > **Definitions**, and create a new content sharing policy.
   b) Enter a name for the content sharing policy.
   c) In the **Scope** list, select the project that includes the users who should be able to deploy the workflows.
   d) In the **Content sharing** section, click **Add Items** and then select one or more workflows to share with the project members.

      You can select all the items imported from a content source or you can select individual items. To select only particular workflows, select **All Content** in the Content Sources drop-down menu.

e) In the **Users** section, select the users and user groups that you want to have access to the content.
f) Click **Create**.

4. Verify that the workflow is available in the catalog to members of the selected project.
   a) Click the **Consume** tab.
   b) In the **Projects** drop-down menu, select the project with which you shared the workflow.

      You can select multiple projects.
   c) On the **Catalog** page, locate the imported workflow, and review the projects to ensure that the project you configured is included.
   d) Click **Request** and provide any required information.
   e) Click **Submit**.

   The provisioning process begins and the Deployments page opens with your current request at the top.
5. Monitor the provisioning process to ensure that the workflow runs successfully.
   a) Select **Deployments** > **Deployments** and locate your deployed request.
   b) Monitor the card status until it is successful.

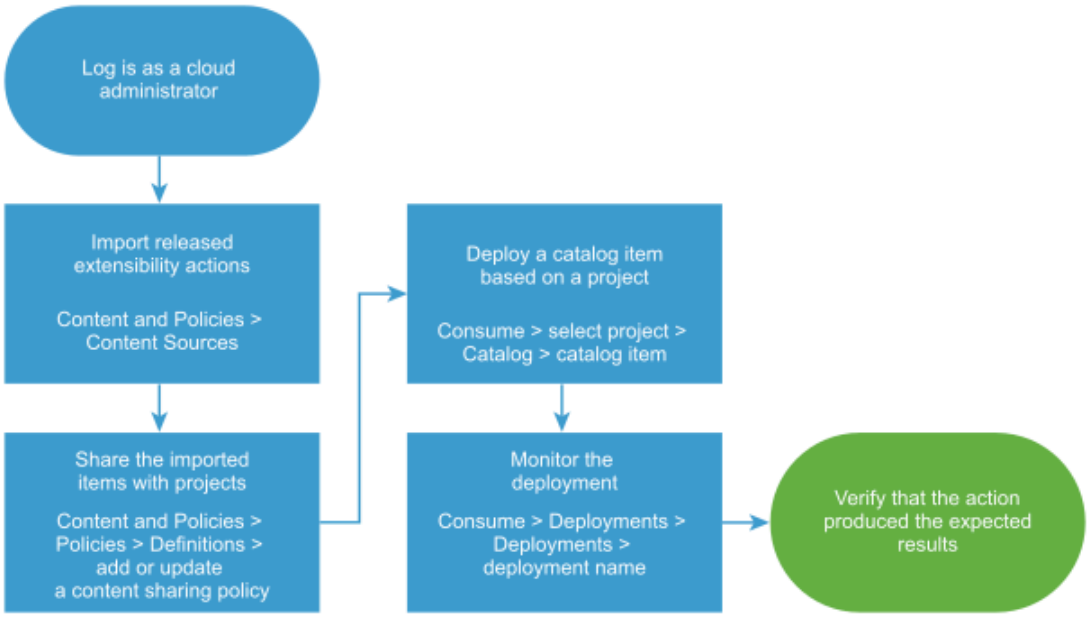
The VCF Operations Orchestrator Client workflows are imported into VCF Automation and shared in the catalog.

- If the deployment fails, click the deployment name and begin troubleshooting. If you are a VCF Automation cloud administrator, you can also do more extensive troubleshooting in VCF Automation.
- If you want to control how long a deployment can exist, create a lease. See Setting up policies in VCF Automation for VM Apps.
- To provide more or fewer user inputs at request time, you can create a custom form. See Customize an icon and request form in VCF Automation for VM Apps. If a workflow includes data grids, do not change the column IDs in the custom form. Use the IDs provided in the workflow.

Add extensibility actions to the VCF Automation catalog

As a cloud administrator, you can add VCF Automation extensibility actions to VCF Automation as a content source. The extensibility actions are created and managed in VCF Automation.

Verify that the actions you are adding are associated with a project, and that they are released.



The actions are small scripts that perform lightweight tasks or steps. For example, rename a virtual machine or assign an IP address.

1. Import the released extensibility actions.
   a) Select **Content and Policies** > **Content Sources**, and click **New**.
   b) Click **New**, and then click **Extensibility actions**.
   c) Enter the **Name** for this content source.
   d) Select the **Source project** and then click **Validate**.

      The validation process verifies the number of released extensibility actions that are associated with the project in VCF Automation.
   e) Click **Create and Import**.
2. Share the imported actions with a project.
   a) Select **Content and Policies** > **Policies** > **Definitions**, and create a new content sharing policy.
   b) Enter a name for the content sharing policy.
   c) In the **Scope** list, select the project that includes the users who should be able to deploy the extensibility actions.
   d) In the **Content sharing** section, click **Add Items** and then select one or more actions to share with the project.

      You can select all the items imported from a content source or you can select individual items. To select only particular extensibility actions, select **All Content** in the Content Sources drop-down menu.
   e) In the **Users** section, select the users and user groups that you want to have access to the content.

      You can share the content with all users and groups in the project, or you can select individual users and groups.
   f) Click **Create**.

   The actions are added to the catalog where the project members can request them.
3. Verify that the action is available in the catalog to the members of the selected projects.
   a) Click the **Consume** tab.
   b) In the **Projects** drop-down menu, select the project with which you shared the extensibility action.

      You can select multiple projects.
   c) On the **Catalog** page, locate the imported extensibility action, and review the projects to ensure that the project you configured is included.
   d) Click **Request** and provide any required information.
   e) Click **Submit**.

   The provisioning process begins and the Deployments page opens with your current request at the top.
4. Monitor the provisioning process to ensure that the action runs successfully.
   a) Select **Deployments** > **Deployments** and locate your deployed request.
   b) Monitor the card status until it is successful.

The extensibility actions are imported into VCF Automation and shared in the catalog.

- If the deployment fails, click the deployment name and begin troubleshooting. If you are a VCF Automation cloud administrator, you can also do more extensive troubleshooting in VCF Automation.
- If you want to control how long a deployment can exist, create a lease. See Setting up policies in VCF Automation for VM Apps.
- To provide more or fewer user inputs at request time, you can create a custom form. See Customize an icon and request form in VCF Automation for VM Apps.

Customize an icon and request form in VCF Automation for VM Apps

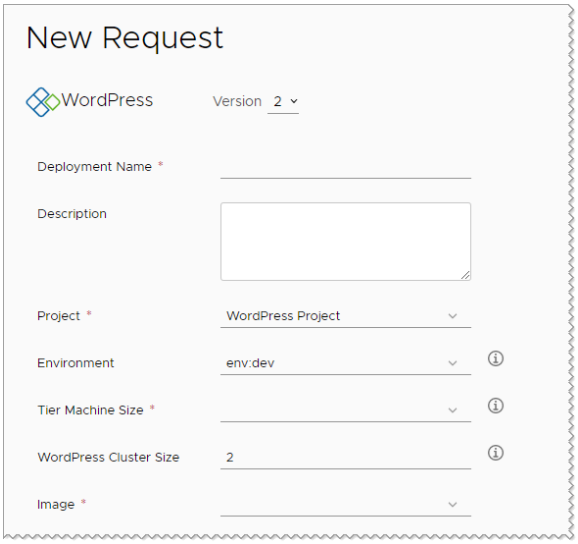
In VCF Automation, you can customize the icon that represents the content in the catalog, limit the number of deployed instances for a catalog item, and customize the request form for imported templates. When customizing the request form, you can also design the input parameters that allow the user requesting a catalog item to provide the values. You can customize how the custom options are presented in the form.

- To add an icon, verify that you have an image that does not exceed 100 KB. The optimal size is no larger than 100x100 pixels.

- This use case assumes that you imported the WordPress use case cloud template from VCF Automation, or that you have a cloud template or template that includes input parameters.

The icon that you provide helps you and your catalog consumers use visual queues to identify specific items. You are not required to customize a form if all you want is a custom icon. Nor are you required to customize the icon when you create a custom form.

When creating the custom form, the WordPress cloud template is used as the example in this use case. If you don't customize the request form, it is a simple list of parameters. See the following example.



In this use case, you customize the following options:

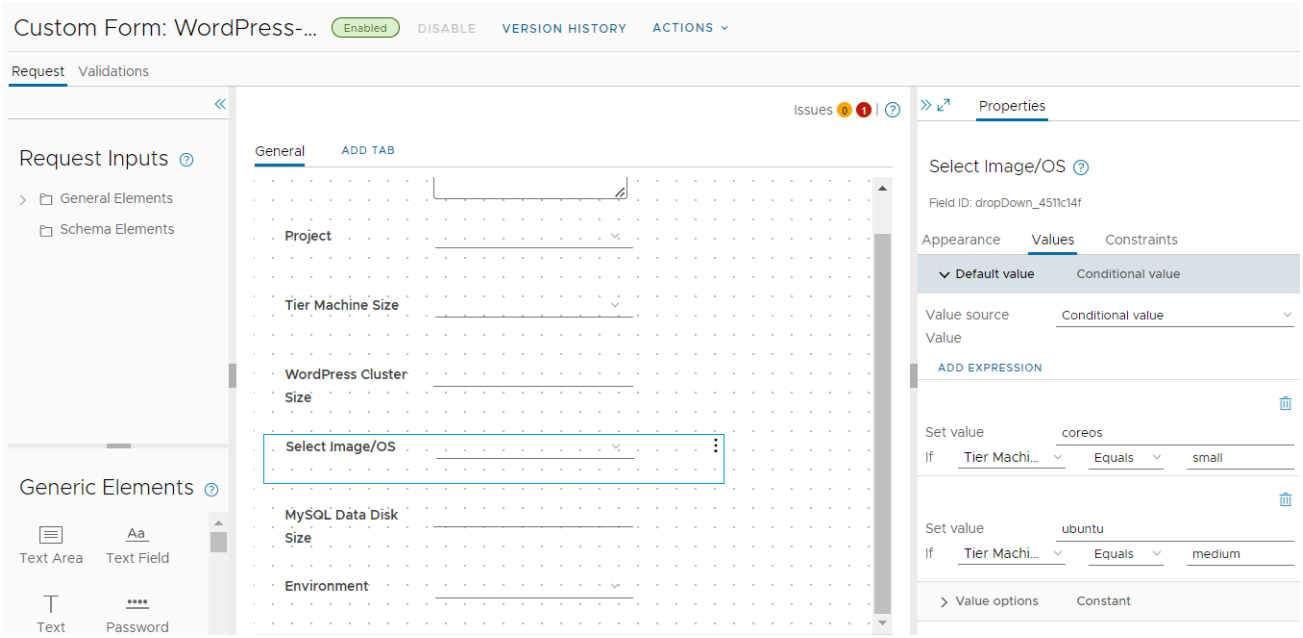
- Reduce the maximum number of WordPress Cluster Size from 5 to 3.
- Specify operating system based on Node Size. For example, if size is small, then the operating system is coreos. If it medium, then the operating system is ubuntu.
- Set the MySQL Data Disk Size value to 5 and hide the option from the requesting users.

1. Select **Content and Policies** > **Content**.
2. Locate the WordPress cloud template, click the menu to the left of the name, and select **Configure item**.
   a) Set the maximum number of deployment instances for this catalog item.

      If you select a value greater than one, the **Deployment count** field is added to the request form. This option allows the requesting user to do bulk deployments.
   b) Add a custom icon.

   If all you want is a custom icon, you can stop here.
3. Locate the WordPress template, click the arrow to the left of the name, locate the version you want to edit, and select **Customize form**.

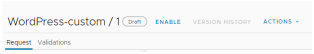
   If the template has input properties, they are listed in the Request Inputs pane on the left, and are added to the canvas.



4. Edit the form using the values provided in the following table.

For this field in the screenshot	Appearance	Values	Constraints
WordPress Cluster Size			Maximum value • Value source = Constant • Max value = 3
Select Image/OS		Default value • Value source = Conditional value • Expression = Set value = coreos If Tier Machine Size Equals small • Expression = Set value = ubuntu If Tier Machine Sieze Equals medium	
MySQL Data Disk Size	Visibility • Value source = Constant • Visible = No	Default value • Value source = Constant • Default value = 5	

5. Click and drag the fields to rearrange them on the form.
6. Optional: To create a new version of the form, click **Version**.
7. To turn on the custom form, click **Enable**.



8. Click **Save**.

The request form is now similar to the following example.



Notice that the Wordpress Cluster Size field indicates an error. The limit is 3, but the user entered a value of 4.

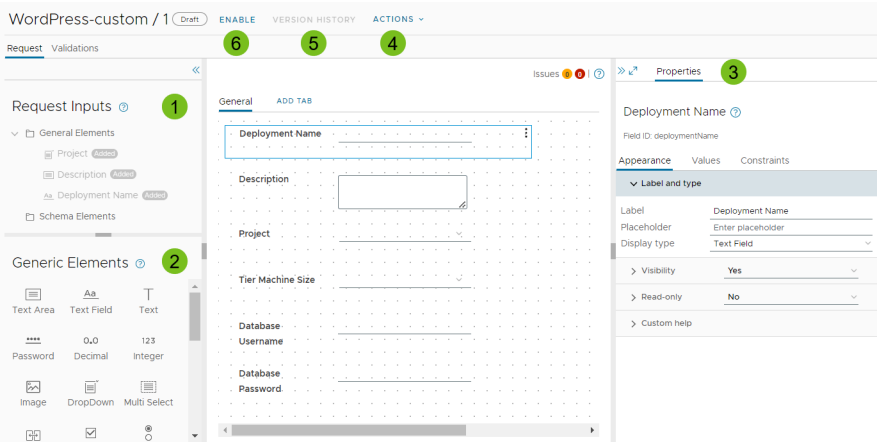
Request the item in the catalog and verify that the presentation and behavior is what you expected.

Learn more about custom forms in VCF Automation for VM Apps

To create useful forms based on input parameters, you can use VCF Automation to design how the information appears at request time, how the parameters values are populated, and add any specialized constraints.

Custom request form designer

You use the form designer to create your custom form.



To create a custom form:

1. Notice that request inputs that are already on the canvas.
2. Drag any custom elements onto the design canvas.
3. Configure each element using the properties pane.
   For more about the fields properties, see Custom form designer field properties in VCF Automation for VM Apps.
4. Use the Actions menu options to import or export the form, or import or export a CSS file. The following sections provide more information.
5. View the snapshot history for the form. For more information about versioning custom forms, see How do I version custom forms in VCF Automation for VM Apps.
6. Enable the form.

The custom form designer supports data validation by adding constraints to a field. For constraints options that are applied as you create a form, see Custom form designer field properties in VCF Automation for VM Apps. For a constraint example, see Customize an icon and request form in VCF Automation for VM Apps.

Catalog items can have a single custom form at a time. If you edit a catalog item, for example a cloud template, that already has a custom form defined, the changes are not reflected in the custom form. To be able to see the changes that you made to the cloud template, you must delete the old custom form, and create a new one. Similarly, if you edit a

template with an active custom form, the change is not reflected in the form, so you must delete the form element that you updated and add it back in.

Importing and exporting custom forms between templates

You might find, after you develop a custom form, that you want to use part or all of it with another template. You can export a form from one template and import it into another template, and then continue customizing the form for the new template.

To share the custom forms, you can click **Actions** on the custom form designer and select one of the following options.

Table 1839: Action menu options for importing and exporting custom forms

Action Menu Item	Description
Import form	Imports a JSON or YAML file.
Export form	Exports your current custom form as a JSON file.
Export form as YAML	Exports your current custom form as YAML. Export the file as YAML when you want to move a custom from one VCF Automation instance to another. For example, from your test environment to your production environment. If you prefer to edit the form as YAML, you can export the form, edit it, and then import it back into the template.

Adding your own style sheet to a custom form

You can use a custom cascading style sheet to refine how the text appears on the screen. You must create the CSS file outside of VCF Automation. But you can export and import a CSS file from one template to another.

Table 1840: Action menu options for importing and exporting CSS files

Action Menu Item	Description

Import CSS	Imports a CSS file that enhances the catalog request form.
	The file might be similar to the following example.
	
	`#input_<field_ID> {`
	`font-size: 20px;`
	`font-weight: bold;`
	`color: red;`
	`width: 600px;`
	`}`
	`#input_<field_ID> {`
	`font-size: 20px;`
	`font-weight: bold;`
	`font-style: italic;`
	`width: 600px;`
	`}`
	
	In this example, replace `<field_ID>` with the actual field IDs from the custom form. You can locate the values by selecting the field in the form, and then you can see the value in the properties pane, beneath the field name. For example, Field ID: `deploymentName` or Field ID: `textField_fe7cf66a`.

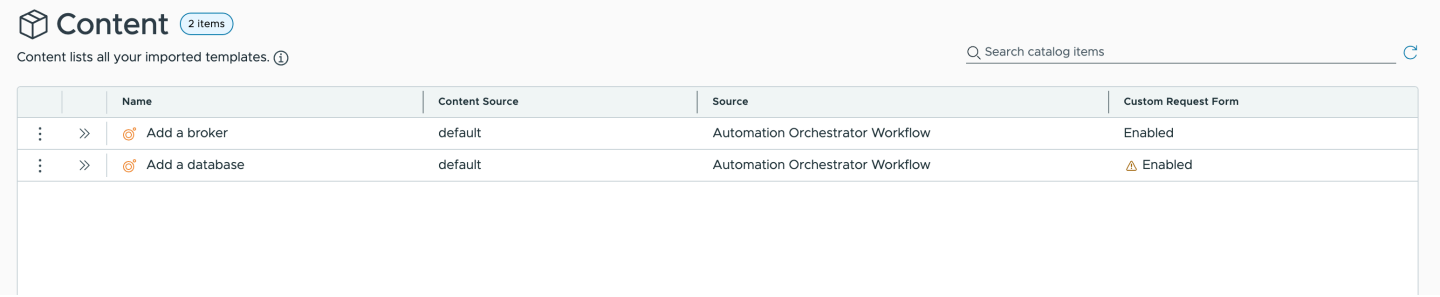
Export CSS	Exports your customized CSS.
Remove CSS	Discards your custom CSS. The discarded CSS is not recoverable.

Validatable and non-validatable action inputs

VCF Operations Orchestrator Client actions used in custom forms might contain inputs that can't be validated when the form is called using the API. When you save a form which contains unvalidatable actions, a warning appears. To save the form, you must confirm that the actions were developed according to best practices.

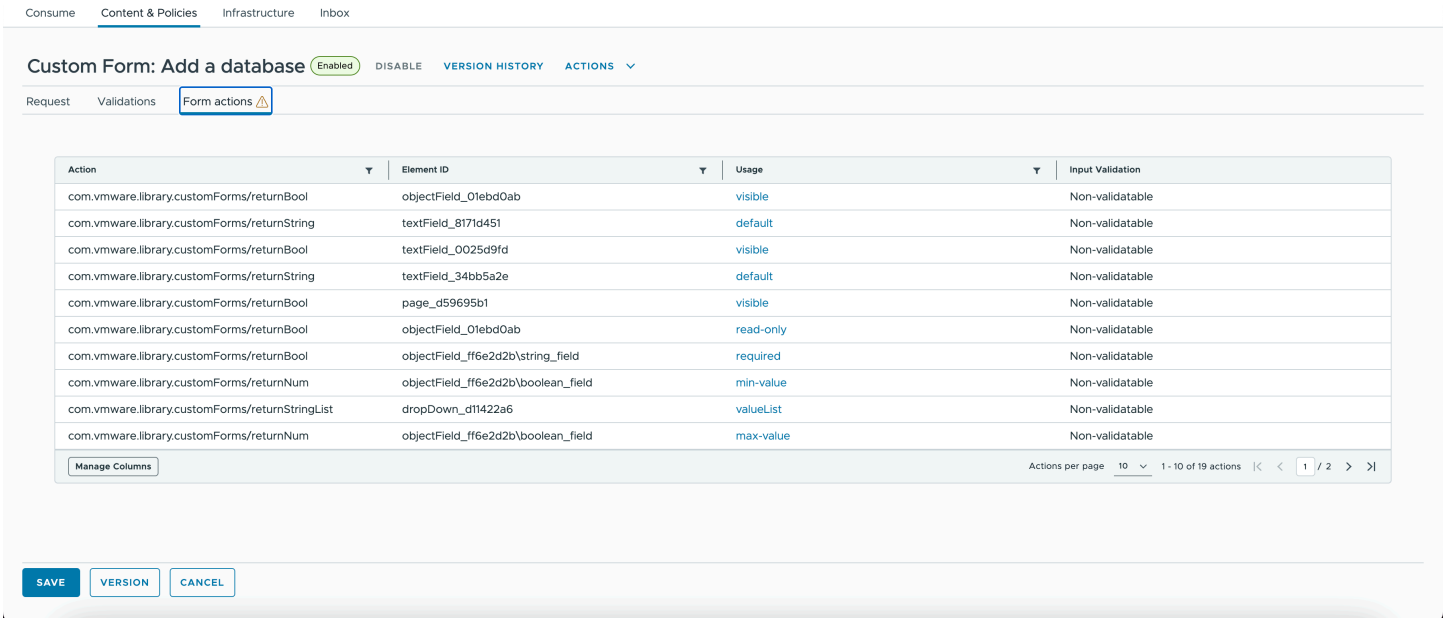
In certain cases, the VCF Operations Orchestrator Client action might not have a custom form. For example, if the template that contains `$dynamicEnum/$dynamicDefault`, or the VCF Operations Orchestrator Client workflow has a form that was customized in VCF Operations Orchestrator Client before it was imported into the catalog.

To determine which custom forms contain actions that cannot be validated, go to **Content and Policies** > **Content** in VCF Automation.



- If the custom request form status is Enabled or Disabled, the content item already has a customized form. A warning icon next to the form status indicates that the form contains actions that cannot be validated.
- If the custom request form status is empty, the form has not been customized VCF Automation. In that case, even if the catalog item has actions that are non-validatable, no warning appears. To see if the content item contains any non-validatable actions, customize the form.

The **Form Actions** tab in the custom form designer shows a list of all actions that are used in the form. You can review the non-validatable actions and verify that their code contains input validation, if necessary.



Adding external validation to a custom form

You can also validate catalog request forms by adding an external validation. External validations are run once the user submits a form request from the UI or the API. External validations use VCF Operations Orchestrator Client action scripts that return a string value when the input parameter value validation fails. The returned string value is displayed as an error message in the catalog request form. If the input parameter value is valid, the external validation must return nothing.

You can add an external validation from the **Validations** tab in the form editor. You drag an Orchestrator validation element to the canvas and add an VCF Operations Orchestrator Client action that you want to use for validation. External validations can be related to more than one field at a time.

Custom form designer field properties in VCF Automation for VM Apps

The field properties in VCF Automation determine how the fields looks and what default values are presented to the user. You can also use the properties to define rules that ensure that the users provide a valid entry when they request the item in the catalog.

You configure each field individually. Select the field and edit the field properties.

Value source

For many of the properties, you can select from various value source options. Not all source options are available for all field types or properties.

- **Constant.** The value does not change. Depending on the property, the value might be a string, an integer, a regular expression, or selected from a limited list, for example, Yes or No. For example, you can provide 1 as a default value integer, select No for the Read-only property, or provide the regular expression to validate a field entry.
- **Conditional value.** The value is based on one or more conditions. The conditions are processed in the order listed. If more than one condition is true, the last condition that is true determines the behavior of the field for that property. For example, you can create a condition that determines if a field is visible based on the value in another field.
- **External source.** The value is based on the results of a VCF Operations Orchestrator Client action. For example, calculate cost based on a scripted VCF Operations Orchestrator Client action. For an example, see Using VCF Operations orchestrator actions in the custom form designer in VCF Automation for VM Apps.
- **Bind field.** The value is the same as the field to which it is bound. The available fields are limited to the same field type. For example, you bind default value for an authentication needed check box field to another check box field. When one target field check box is selected in the request form, the check box on the current field is selected.
- **Computed value.** The value is determined based on how the operator processes the selected fields and values. Text fields use the concatenate operator. Integer fields use the selected add, subtract, multiply or divide operations. For example, you can configure an integer field to convert megabytes to gigabytes using the multiply operation.

Field appearance

You use the appearance properties to determine whether the field appears on the form and what label and custom help you want to provide to your catalog users.

Table 1841: Appearance Tab Options

Option	Description
Label and type	Provide a label and select a display type. The available display types depend on the element. Some some elements support multiple text types and others only support integers. Possible values: • Array Input • Checkbox • Combobox • Data Grid • Date Time • Decimal • Drop Down • Dual List • File Upload The maximum allowed size of all files in the custom form is 768KB. When provisioning a custom resource, the maximum allowed size of the form input request is 768KB. • Image • Integer • Link • Multi Select • Multi Value Picker • Object Field • Password To ensure that passwords are encrypted in the deployment request details page, the input property in the cloud template must include `encrypted:true`. • Radio Group • Text • Text Area • Text Field • Tree Picker The reference type must be an inventory object. • Value Picker Drop-down and data grid fields include a **Placeholder** setting. The entered value appears as an internal label or instructions in the drop-down menu, or as a general label or instructions in the data grid.
Visibility	Show or hide a field on the request form. • **Constant.** Select Yes to display the field on the form. Select No to hide the field. • **Conditional value.** Visibility is determined by the first expression that is true. For example, a field is visible if a check box is selected on a form. • **External source.** Visibility is determined by the results of the selected VCF Operations Orchestrator Client action. • **Bind field.** Visibility of the field is bound to another check box field. If the visibility for a field is set to No, the field constraints, like regular expressions and required fields, are disregarded.

Option	Description
Read-only	Prevent users from changing the field values. • **Constant.** Select Yes to display the value but prevent changes. Select No to allow changes. • **Conditional value.** Status is determined by the first expression that is true. For example, a field is read-only if the value in a storage field is greater than 2GB. • **External source.** Status is determined by the results of the selected VCF Operations Orchestrator Client action.
Rows per page	For data grid elements only. Enter the number of rows.
Custom help	Provide information about the field to your users. This information appears in signpost help for the field. You can use simple text or HTML, including href links. For example, `<a href="https://docs.vmware.com/en/VMware-Aria-Automation/index.html">VMware Aria Automation documentation</a>`.

Field values

You use the values properties to provide any default values.

Table 1842: Values Tab Options

Option	Description
Columns	For the data grid element only. Provide the label, ID, and value type for each column in your table. The default value for the data grid must include the header data that matches the defined columns. For example, if you have user_name ID for one column and user_role ID for another, then the first row is user_name,user_role.
Default value	Populates the field with a default value based on the value source. Possible value sources depend on the field. • **Constant.** The entered string. • **Conditional value.** The default value is determined by the first expression that is true. For example, the default value of a storage field is 1GB if the memory field is less than 512MB. • **External source.** Value is based on the results of the selected VCF Operations Orchestrator Client action. • **Bind field.** Value is the same as the selected field. • **Computed value.** Value is based on the results of the provided field values and the selected operator. For example, the default value of memory in MB is based on the memory in GB multiplied by 1024.
Value option	Populates a drop-down, multi-select, radio group, or value picker fields. • **Constant.** The format for the list is Value|Label,Value|Label,Value|Label. For example, `2|Small,4|Medium,8|Large`. • **External source.** Value is based on the results of the selected VCF Operations Orchestrator Client action.
Step	For integer or decimal fields, define the incremental or decremental values. For example, if the default value is 1 and you set the step value to 3, then the allowed values are 4, 7, 10, and so on.

Field constraints

You use the constraint properties to ensure that the requesting user provides valid values in the request form.

Table 1843: Constraints Tab Options

Option	Description
Required	The requesting user must provide a value for this field. • **Constant.** Select Yes to require that the requesting user provides a value. Select no if the field is optional. • **Conditional value.** Whether the field is required is determined by the first expression that is true. For example, this field is required if the operating system family starts with Darwin in another field. • **External source.** Status is based on the results of the selected VCF Operations Orchestrator Clientaction.
Regular expression	Provide a regular expression that validates the value and a message that appears when the validation fails. The regular expressions must follow JavaScript syntax. • **Constant.** Provide a regular expression. For example, for an email address, the regular expression might be `^[A-Za-z 0-9._%+-]+@[A-Zaz0-9.-]+\\.[A-Za-z]{2,}$` and the validation error message is `The email address format is not valid. Please try again.` • **Conditional value.** The regular expression that is used is determined by the first expression that is true.
Minimum value	Specify a minimum numeric value. For example, a password must have at least 8 characters. Provide an error message. For example, `The password must be at least 8 characters.` • **Constant.** Enter the integer. • **Conditional value.** The minimum value is determined by the first expression that is true. For example, a minimum CPU value is 4 if the operating system does not equal Linux. • **External source.** Value is based on the results of the selected VCF Operations Orchestrator Client action.
Maximum value	Maximum numeric value. For example, a field is limited to 50 characters. Provide an error message. For example, `This description cannot exceed 50 characters.` • **Constant.** Enter the integer. • **Conditional value.** The maximum value is determined by the first expression that is true. For example, a maximum storage value is 2GB if the deployment location equals AMEA. • **External source.** Value is based on the results of the selected VCF Operations Orchestrator Client action.
Match field	This field value must match the selected field value. For example, a password confirmation field must match the password field.

Using the data grid element in the custom form designer in VCF Automation for VM Apps

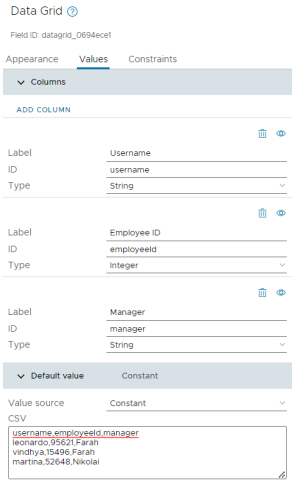
If you use a data grid element in a custom form, the data that is presented in the table might be manually provided.

Provided CSV data example

In this use case, you have a table of values that you provide in the custom request form. You provide the information in the table as a constant value source. The source is based on a CSV data structure where the

first row defines the grid headers. The headers are the column IDs separated by a comma. Each additional row is the data that appears in each row in the table.

1. Add the Data Grid generic element to the design canvas.
2. Select the data grid and define the values in the properties pane.



Label	ID	Type
Username	username	String
Employee ID	employeeId	Integer
Manger	manager	String

Define the CSV values.

```
username,employeeId,manager
leonardo,95621,Farah
vindhya,15496,Farah
martina,52648,Nikolai
```

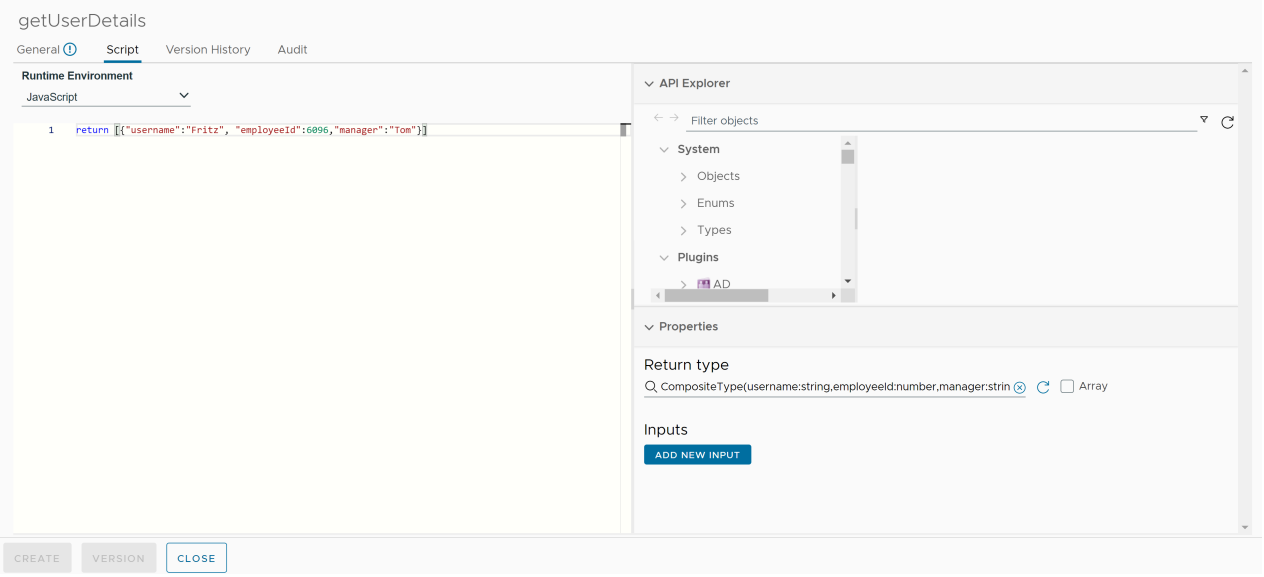
3. Verify that the data grid displays the expected data in the request form.



External Source Example

This example uses the previous example but the values are based on a VCF Operations orchestrator action. Although this is a simple action example, you can use a more complex action where you retrieve this information from another database or system.

1. In VCF Operations orchestrator, configure an action, `getUserDetails`, with an array similar to the following example.



a. On the General tab, enter the name `getUserDetails` and provide a Module name.
b. On the Script tab, use the following script example.

```
return [{"username":"Fritz", "employeeId":6096,"manager":"Tom"}]
```

c. In the Return type area, clear the initial selection and click **New Composite Type**.
d. Define a new composite type named `UserDetails` and add the following fields, then click **Create**.

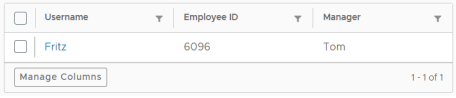
Field	Type
username	string
employeeId	number
manager	string

e. In the Return type area, click **Array**.
f. Version and save the action.

2. In VCF Automation, add the data grid and use the Values tab to configure the data grid columns with the following values.

Label	ID	Type
Username	username	String
Employee ID	employeeId	Integer
Manager	manager	String

3. In the Default value, Value source list, select **External source**.
4. In Select action, enter `getUserDetails` and select the action you created in VCF Operations orchestrator.
5. Save the form.
6. In the catalog, verify the table in the request form.



Inspecting complex values in data grids

You can examine data grid rows in a separate modal, where complex values are displayed as expandable tree nodes.

The values in the first column of the data grid are links. Click the link to view detailed information about the value in the row you want to examine.

The following table provides information about how different value types are displayed in the table.

Value type	Displayed as
Boolean	`true/false` string
Non-complex value	`label:value` pair
Complex value	1. If the value is an array of objects, `{objects.count} objects` is displayed. If the array has only 1 element, `object` is displayed in the cell. 2. If the object has 2 or fewer properties, none of which is complex, the `label:value` pairs are separated by a comma and displayed in the cell as `Object, object`. 3. If the object has more than 2 properties, or has a complex property, `object` is displayed in the cell.
Array of complex values	`{length of array} objects`
Empty value	hyphen (-)

If the data grid contains only one column, the hide/show control in the footer is not displayed.

Using VCF Operations orchestrator actions in the custom form designer in VCF Automation for VM Apps
When you customize a VCF Automation request form, you can base the behavior of some fields on the results of a VCF Operations orchestrator action.

There are several ways that you can use VCF Operations orchestrator actions. You might have an action that pulls the data from a third source, or you can use a script that defines the size and cost.

The first example is based on manually added fields so that you understand the underlying process. The second example uses the same premise, but instead relies on a template field.

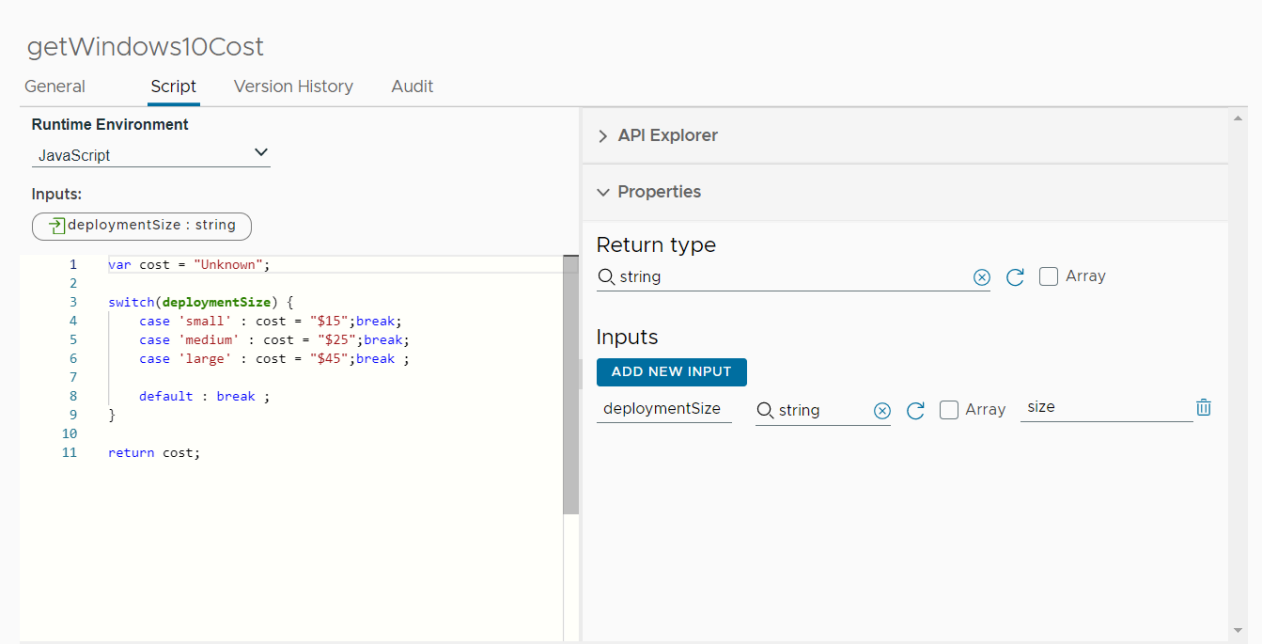
The third example is based on a custom option that is added to the catalog request form, where the user selects a folder based on the results from a custom VCF Operations orchestrator action.

The fourth example uses VCF Operations orchestrator workflow attributes to customize a field in the request form.

Size and cost as manually added fields example

In this use case, you want the catalog user to select a virtual machine size, and then display the cost of that machine per day. To do this example, you have a VCF Operations orchestrator script that correlates the size and cost. You then add a size field and a cost field to the template custom form. The size field determines the value that appears in the cost field.

1. In VCF Operations orchestrator, configure an action named `getWindows10Cost`.



2. Add a script.
   You can use the following example script.

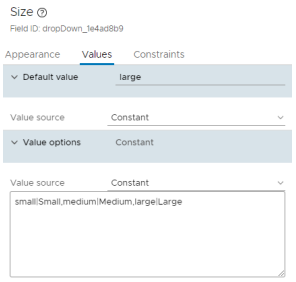
```
var cost = "Unknown";

switch(deploymentSize) {
 case 'small' : cost = "$15";break;
 case 'medium' : cost = "$25";break;
 case 'large' : cost = "$45";break ;

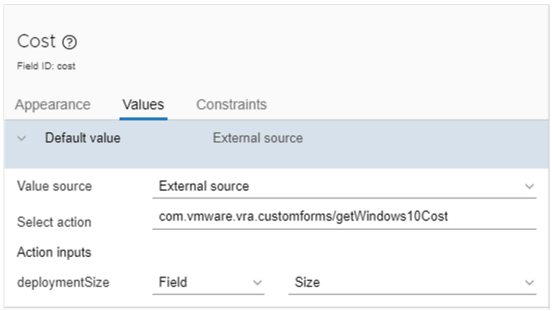
 default : break ;
}

return cost;
```

3. Add `deploymentSize` as an input string.
4. In VCF Automation, add and configure a Size field to a template custom form.
   Configure the size field as a drop-down element with Small, Medium, and Large values.



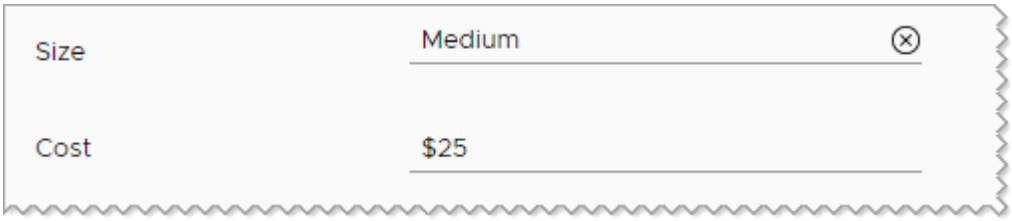
   On the **Values** tab, configure the following property values.
   - Default value = `Large`
   - Value options
     - Value source = `Constant`
     - Value definition = `small|Small,medium|Medium,large|Large`
5. Add the cost field as a text field to display the cost as defined in the VCF Operations orchestrator action based on the value selected in the size field.



On the **Values** tab, configure the following property values.

- Default value = External source
- Select action = <your VCF Operations orchestrator actions folder>/getWindows10Cost
- Action inputs
  - deploymentSize. This value was configured in the action as the input.
  - Field
  - Size. This is the field that you previously created.

6. Enable the custom form and save it.
7. To verify that it is working, request the item in the catalog. You should see the Cost field populated based on the selected Size value.



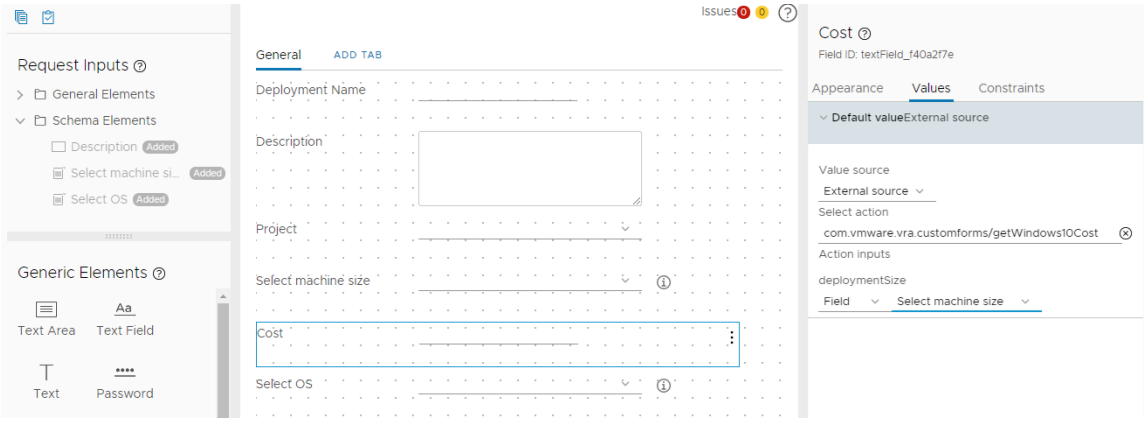
Cost based on schema element example

In this use case, you want the catalog user to see the cost of that machine per day based on the flavor property in the template. To do this example, you use the VCF Operations orchestrator script from the previous example. But in this use case the cost is based on the flavor size that your user selected in the custom form when they request the VCF Automation catalog item.

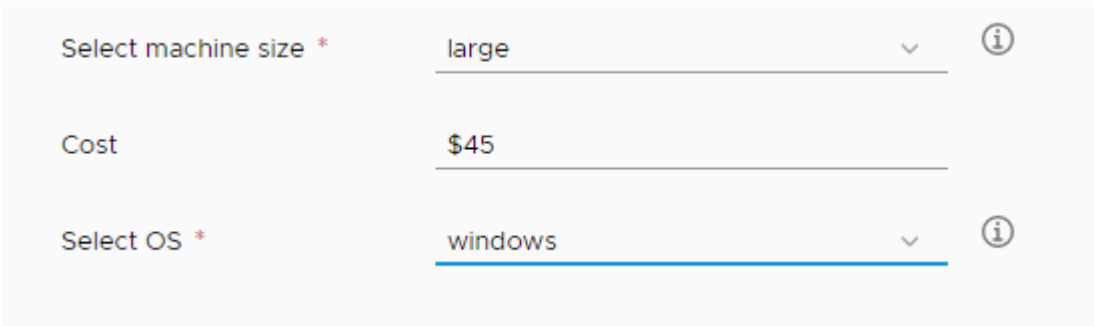
The simple example template includes a size input field where the user selects the flavor property.

```
formatVersion: 1
inputs:
  size:
    type: string
    enum:
      - small
      - medium
      - large
    description: Size of Nodes
    title: Select machine size
  image:
    type: string
    enum:
      - ubuntu
      - centos
      - windows
    description: OS image
    title: Select OS
resources:
  Cloud_vSphere_Machine_1:
    type: Cloud.vSphere.Machine
    properties:
      image: '${input.image}'
      flavor: '${input.size}'
```

The custom form uses the field, named `Select machine size` in this example.



The cost deploymentSize input is based on the Select machine size field.



Destination folder based on a custom action example

In this use case, you want the catalog user to select from the folders that are available to them when they move a machine from one folder to another. To do this example, you create a custom action in VCF Operations orchestrator that returns folders available to the user requesting the action from the catalog. You then customize the **Destination folder** field in the catalog request form.

In the catalog, the **Move virtual machines to folder** workflow that is used in this example is exposed as a custom resource action created in VCF Automation.

1. In VCF Operations orchestrator, create an action named `getFolderForUser`.
   a. On the **Script** tab, add action inputs, select the return type of the output, and add a script. You can use the following example script.

```
var parentTargetDirName = "users";
var sdkConnection = VcPlugin.findSdkConnectionForUUID(vcUuid);
var rootFolder = sdkConnection.getAllVmFolders(null, "xpath:matches(name,'" +
 parentTargetDirName + "')")[0];
var result = new Array();
for each(var folder in rootFolder.childEntity) {
    if (folder instanceof VcFolder && folder.permission.length > 0) {
        var entityPrivilege =
 sdkConnection.authorizationManager.hasUserPrivilegeOnEntities([folder], username,
 ["System.Read"])[0];
        if (entityPrivilege.privAvailability[0].isGranted) {
            result.push(folder);
        }
    }
}
return result;
```

   b. When you finish editing the action, click **Save**.
2. In VCF Automation, create a custom resource action named `ChangeFolder`.
   a. Select the **Cloud.vSphere.Machine** resource type.
   b. Select the **Move virtual machines to folder** workflow.
3. Customize the request form that users see when they request the action.
   a. Open the `ChangeFolder` action.
   b. Click **Edit Request Parameters**.
   c. Customize how the **Destination folder** field is presented to users.

Value options	Sample value
Value source	External source
Select action	getFolderForUser
Action inputs `username Field`	Click **Select string** > **Request info fields** > **Requested by**. Click **Select**.
Action inputs `vcUuid`	Click **Select string** > **Request info fields** > **Resource properties** > **vCenter UUID**. Click **Select**.

   d. Click **Save**.

When a user requests the action in the catalog, they can select a destination folder for the virtual machine from the folders that are available to them.

Hostname based on workflow attributes example

In this use case, you want to customize the URL field in the request form using VCF Operations orchestrator workflow attributes. To do this example, you need a VCF Operations orchestrator workflow with an attribute or variable that has either a constant value or is bound to a configuration element. The attribute or variable must have an initial value set because the binding is done before you run the workflow.

1. In VCF Operations orchestrator, create or clone an existing workflow. For example, clone the **Add a REST Host** workflow.

a. On the **Variables** tab, add a variable and bind it to a configuration element.

Value options	Sample value
Name	*hostname*
Type	*string*
Configuration	Enable the **Bind to configuration** option. In the **Configuration** text box, select the configuration that you want to bind the variable to. For example, select the custom **vRO Configuration: hostname** element.

b. Save the workflow.

2. Import the workflow in VCF Automation.
3. Customize the request form.
   a. Click the **URL** element in the form designer.
   b. On the **Values** tab, set the value source to **Bind field**.
   c. Click **Value Field** > **Request info fields**.
   d. Select the *hostname* variable or any of the available workflow attributes.
4. Save and enable the form.

When a user requests the workflow from the catalog, the URL field is populated based on the *hostname* variable.

If you switch to a different project, the workflow attributes are recalculated because the workflow might come from a different VCF Operations orchestrator integration.

Using value picker and multi value picker elements in the custom form designer in VCF Automation for VM Apps
When you create a custom form, you can add elements where the user selects a value from a search results list. Using the value picker, the user selects a single value. Using the multi value picker, the user selects one or more values.

The value picker and multi value picker work with the Reference Type that is defined on the custom form Appearance tab. The Reference type is a VCF Operations orchestrator resource. For example, AD:UserGroup or VC:Datastore. By defining the reference type, when the user enters a search string, the results are limited to the resources that have the matching parameter.

For the pickers, you can then further limit the possible values by configuring an external source.

Working with the Value Picker

The value picker appears in the form as a search option when users request the item in the catalog. The user enters a string and the picker provides list based on how you configured it.

You can use the picker based on the following use cases. The most valuable use of the value picker is pairing it with an external source value.

- Value picker with a constant value source.
  Use this method when you want the requesting user to select from a predefined static list of values. Similar to the combobox, drop down, multiselect, and radio group elements, this method provides search results in a list based on the defined constant values and labels.
- Value picker with no defined value source.
  Use this method when you want the requesting user to search the VCF Operations orchestrator inventory for a specific object with the configured reference type. For example, the reference type is VC:Datastore and you want the users to select the datastore from the retrieved list.
- Value picker with an external value source.
  Use this method when you want the requesting user to select from results that are based on a VCF Operations orchestrator action. The following script provides an example of a basic VCF Operations orchestrator action that works with the value picker. In this example, the reference type is Properties.

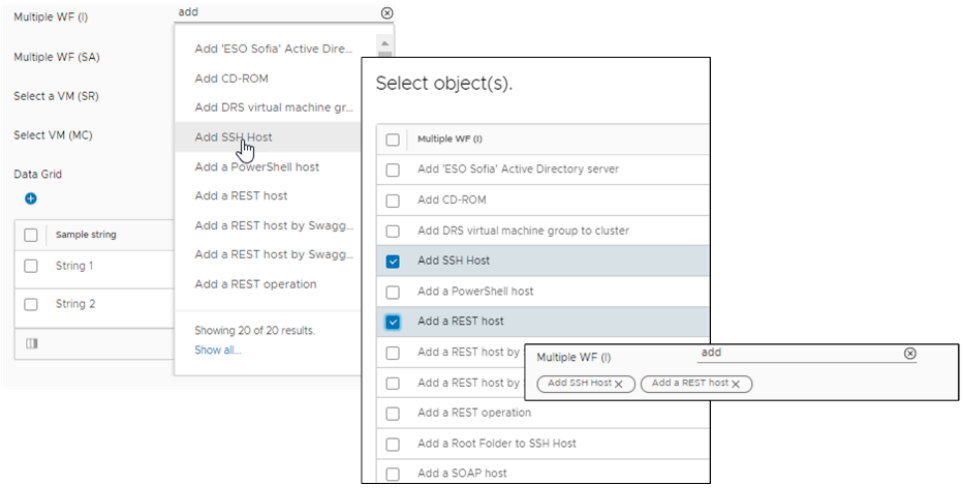
```
var res = [];
```

```
res.push(new Properties({label: 'label1',value: 'value1'}));
res.push(new Properties({label: 'label2',value: 'value2'}));
res.push(new Properties({label: 'label3',value: 'value3'}));
return res;
```

Note:
The Properties input cannot be input to a workflow, only an intermediate value in the custom form.

Working with the Multi Value Picker

The multi value picker appears in the request form as a search option, similar to the value picker, but where you can select one or more values. The user enters a string and the picker provides list based on how you configured the element properties.



You can use the multi value picker based on the following use cases in addition to the use cases described for the value picker. The most valuable use of the multi value picker is using it with a reference data type and a VCF Operations orchestrator reference.

- Multi value picker with a complex data type and constant value source.
  Use this method when you want the requesting user to select one or more values from a predefined static list of values. Similar to the data grid, this method provides search results in a list based on the defined constant values and labels.
- Multi value picker with a complex data type and an external source.
  Use this method when you want the requesting user to select one or more values from a list of values based on a VCF Operations orchestrator action. You can use this method with VCF Operations orchestrator composite types.
- Multi value picker with a reference data type and a VCF Operations orchestrator reference type. Use this method when you want the requesting user to search the VCF Operations orchestrator inventory for a specific object with the configured reference type. For example, the reference type is VC:Datastore and you want the users to select the datastore from the retrieved list. Or, if you have a workflow filter configured, you can use Workflow as the reference. To be retrieved, the filter must return values in a property array, not a string array. An example of a workflow filter is provided in the next section. In this example, the filtering is done in the UI when the user enters a search term.
- Multi value picker with a reference data type, a VCF Operations orchestrator reference type, and an external source.
  Use this method when you want the requesting user to select from results that are first filtered by the reference type and then based on a VCF Operations orchestrator action. This combination more thoroughly refines the results and populates the request form more quickly. Just as the reference type results must return a property array, so must the external source action. In this example, the filtering is done in VCF Operations orchestrator and might improve the speed with which the list is populated, particularly if you have a large number of VCF Operations orchestrator actions.

Limit the VCF Operations orchestrator results for a multi value picker element results list

To limit the number of actions returned when the user searches for an action, you can create a filter action and bind the filter results to the search term.

1. In VCF Operations orchestrator, create an action named filterWorkflow.
   a. Select **Library** > **Actions**, and click **New Action**.
   b. On the **General** tab, enter or select the following values.

Option	Value
Name	`filterWorkflow`
Module	`com.vmware.library.workflow`

   c. Click the **Script** tab and add the following script.

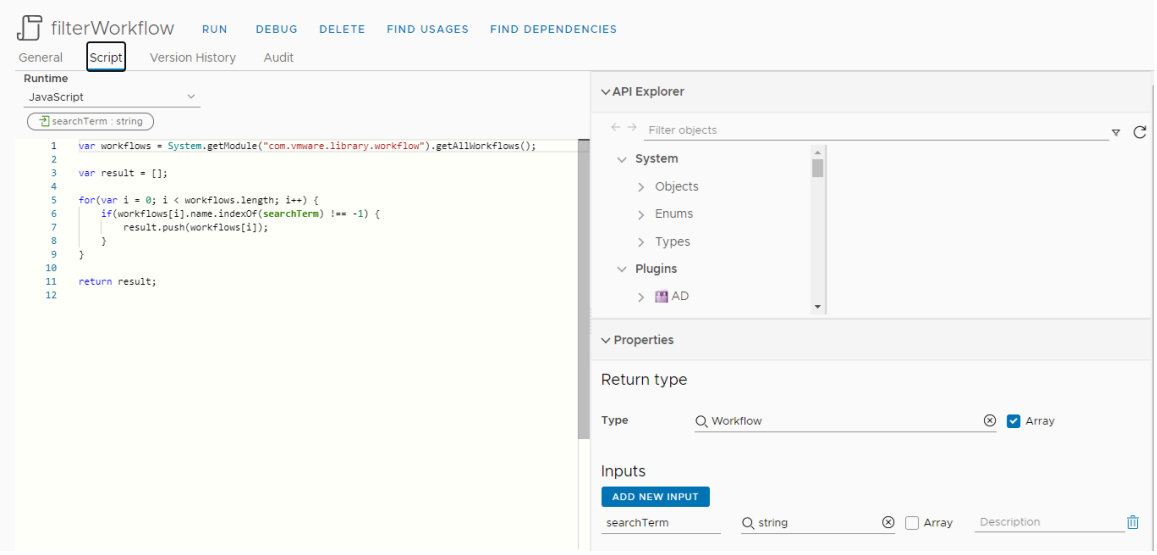
```
var workflows = System.getModule("com.vmware.library.workflow").getAllWorkflows();

var result = [];

for(var i = 0; i < workflows.length; i++) {
    if(workflows[i].name.indexOf(searchTerm) !== -1) {
        result.push(workflows[i]);
    }
}

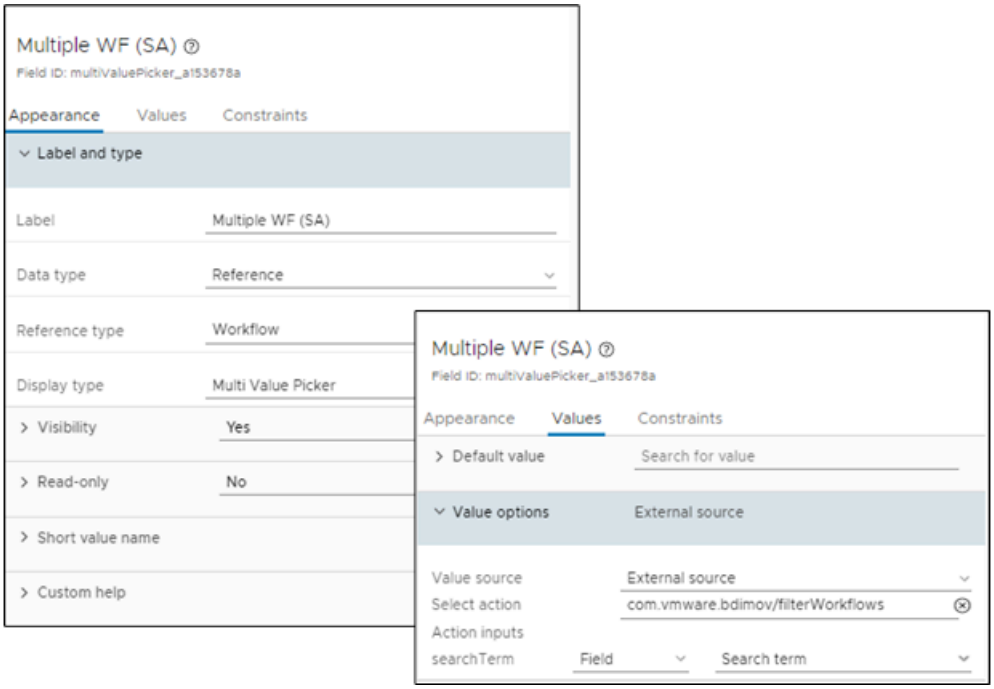
return result;
```

   d. Configure the following properties.



Properties Option	Value
Return type	Enter `Workflow` and select **Array**. You can use any of the returned types when you run the search. The selected reference type in the custom form must match it. If this procedure, continue to use Workflow.
Inputs	Enter `searchTerm`. Notice that the input `searchTerm` matches the string used in the script.

   e. Click **Create**.
2. Configure the multi value picker properties in the custom form designer in VCF Automation.



a. In VCF Automation, select **Content and Policies** > **Content** and click the vertical dots to the left of the template that you are modifying and click **Customize form**.
b. Add or select the multi value picker element in the design canvas.
c. In the Properties pane, click **Appearance** and configure the following values.

Property	Value
Data type	Reference
Reference type	Enter `Workflow`. Remember, this value is the return type selected for the filterWorkflow action in VCF Operations orchestrator, and it must be an array.
Display type	Multi Value Picker

d. Click the **Values** tab and configure the following values.

Property	Value
Value options > Value source	External source
Select action	Select the filter action. In this example, select `filterWorkflows`.
Action inputs searchTerm	Select `Field` and `Search term`.

3. Test the filter by requesting the catalog item.
   You must ensure that the filter returns the expected values in the multi value picker list, and that the catalog item deploys correctly.

How do I version custom forms in VCF Automation for VM Apps

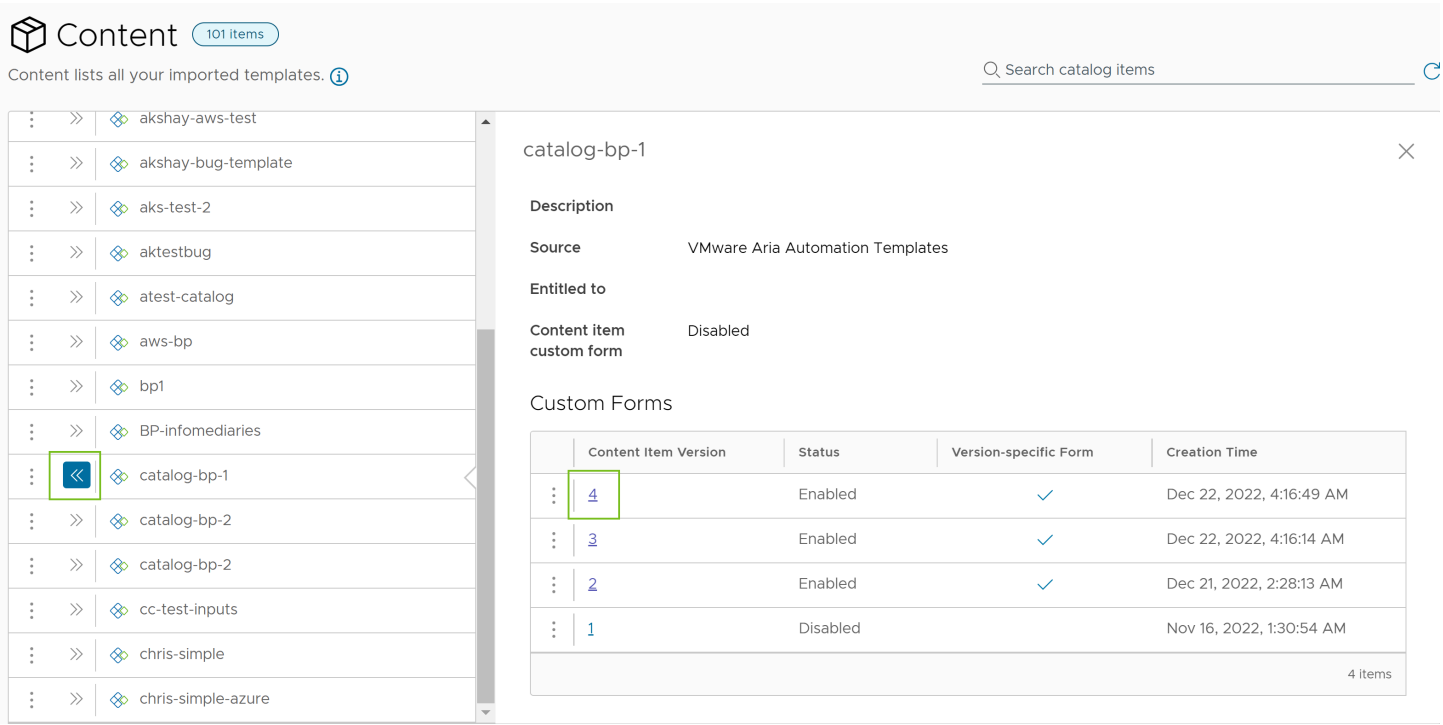
The VCF Automation custom forms maintain a version history record for custom forms that you create for VCF Automation templates. You can create snapshots of your custom forms, compare different form versions, and revert to a previous version.

In addition to the version-specific form configuration, the catalog item keeps a base, or root, version of the form, which provides a reference to the basic form design. You can update just the base form, just the version-specific form, or both, depending on your use case.

Once you have versions of your base custom form, you must edit the version-specific form for the changes to appear for your catalog users. Updating the base only doesn't affect the actual request form in the catalog. You can choose to keep the base up-to-date with the latest version-specific form or not.

Versioning the template itself removes the version-specific custom forms. You can recreate old form versions from the base form or from saved copies of the YAML.

To view what versions are available for your custom forms, locate your imported template at **Content and Policies** > **Content**. To update the base form, click the vertical ellipsis next to the custom form and click Customize form. To update a form version, click the arrow button to the left of the template name to expand the details view, and then click the version that you want to edit.



Versioning a custom form

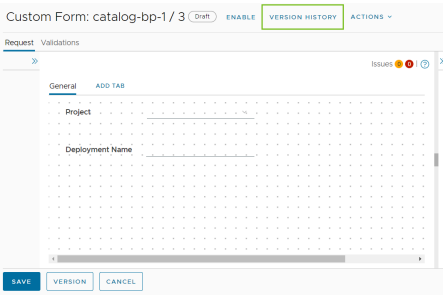
When editing a custom form for the first time, you notice that the **Version** button is dimmed at first. To version the custom form you are editing, you must create the form or save your changes first. Then, you click **Version** and provide a name. When you are done editing the form, save your changes again.

When you version custom forms, the system validates your changes and prevents you from creating a new version that is identical to the previous version.

Comparing form versions

When you have multiple versions for a custom form, you can inspect the differences between them.

From the custom form editor, click **Version History**.



On the versioning timeline on the left, select a version and click **Diff**. From the **Compare to** dropdown select another version. You can view the comparison between the two versions side-by-side or inline.



Use the **View** tab on the **Version History** page to preview the custom form versions.

Restoring an earlier version

When you develop a custom form, you might have to restore an earlier working version of your form.

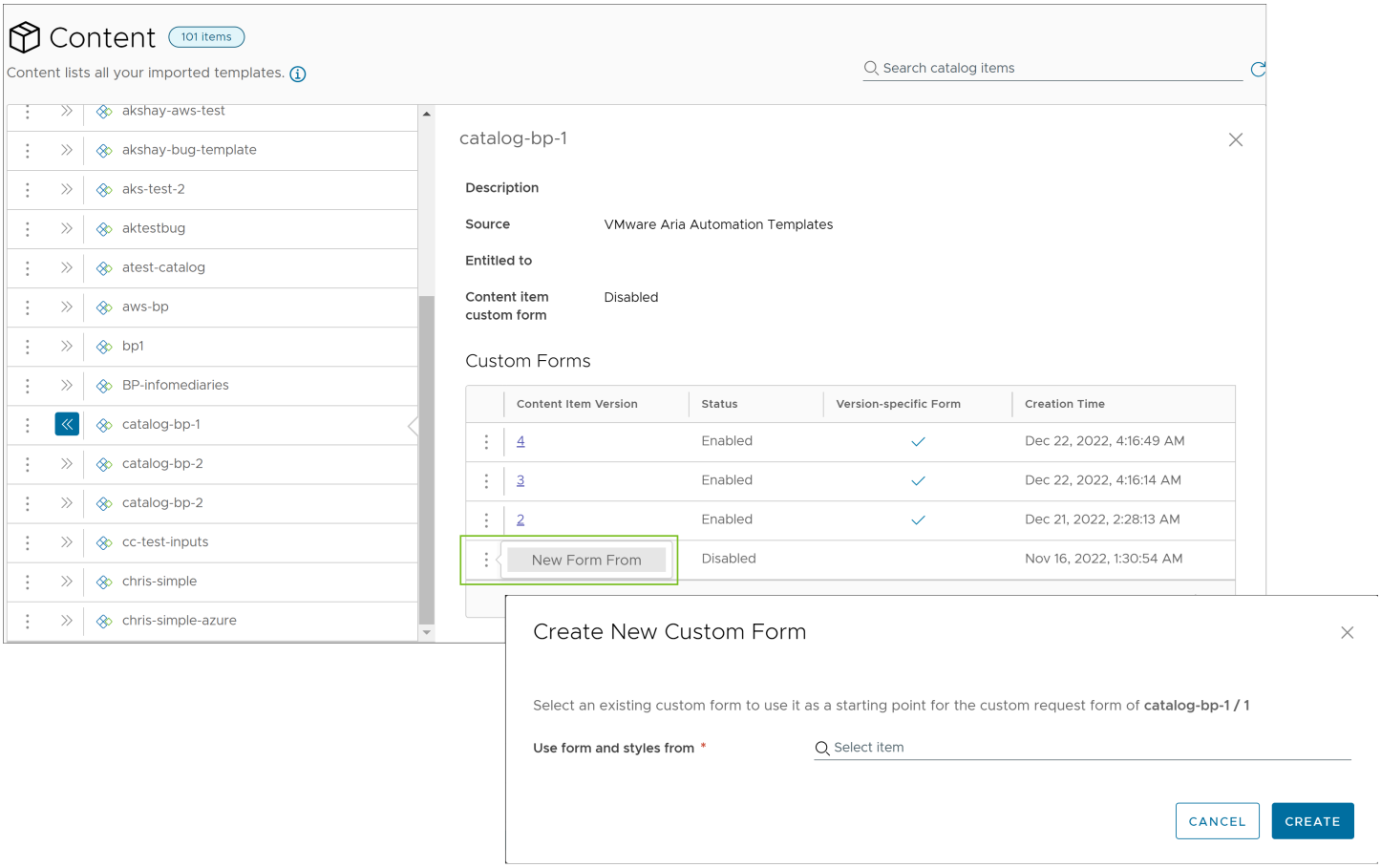
From the versioning timeline on the **Version History** page, select your desired version and click **Restore**.

The snapshot you selected is applied to the current custom form.

Cloning a custom form

You can create a custom form that is based on another existing custom form or custom form version.

1. On the **Content** page, locate the template for which you want to create a new form version.
2. Click **New Form From** in the action menu to the left of the version name.
3. Select the catalog item whose form you want to copy.
   If the item has multiple custom form versions, you can select which version to copy.
4. Click **Create**.



Troubleshooting VCF Automation custom forms
If you encounter problems when using custom forms in VCF Automation, you can refer to this troubleshooting topic to understand the problem or solve it, if there is a workaround.

External actions are not executed correctly

Problem: External actions in your custom forms are not executed correctly, for example, a field might not autopopulate with the default value or a dropdown menu might appear without options.

Cause: External actions used in a custom form field are only executed when all dependency fields with constraints are properly populated with valid values. When a field breaks some of its constraints, such as required, min/max value, or length, and pattern constraints, any directly dependent input field with an external action is evaluated only after the field is within the valid requirements. Transitively dependent fields with an external action are not evaluated unless the field is valid. If a valid field is updated and becomes invalid, then all dependent actions are not run.

Solution: Ensure that all dependency fields with constraints are populated with valid values. For example, ensure that all required dependencies are filled and all constraints on the dependencies are satisfied.

Form requests fail

Problem: You get form errors, for example, `Some data cannot be retrieved. If the problem persists, contact your system administrator. Failed request: <action name>`.

Cause: Your form might include one or more fields that contain a regex constraint that doesn't comply with Java and JavaScript.

Solution: Ensure that the regex in your fields is both Java and JavaScript compliant.

Unable to import VCF Operations Orchestrator Client workflows

Problem: When you import an VCF Operations Orchestrator Client workflow into the VCF Automation catalog content sources, you get an error message, for example, `Error downloading catalog item '/workflow/<workflowId>' (Error: Content provider error).`

Solution: Verify that the workflow does not contain any inputs or fields with the id `project`, which is a system property.

Administering VM Apps Organizations in VCF Automation

As an organization administrator, you monitor and manage critical infrastructure and user management aspects of a VCF Automation for VM Apps deployment.

The tasks described in this section are vital to keeping a VCF Automation deployment operating appropriately. These tasks include user and group management, and monitoring system logs.

In addition, it describes how to configure and manage multi-organization deployments.

While some VCF Automation administration tasks are completed from within VCF Automation, others require the use of related products such as VCF Operations and fleet lifecycle. Users should familiarize themselves with these products and their functionality before completing applicable tasks.

Integrating VCF Automation for VM Apps with other applications

Integrations enable you to add external systems to VCF Automation.

Integrations include VCF Operations orchestrator, configuration management and other external systems such as GitHub, Ansible, Puppet, and external IPAM providers such as Infoblox.

Note:

How do I use Git integration in VCF Automation for VM Apps

VCF Automation supports integration with various flavors of Git repositories so that you can manage VMware cloud templates and action scripts under source control. This functionality facilitates auditing and accountability of processes around deployment.

VCF Automation supports different flavors of Git integration as described in the following list. Each of these options is a separate integration.

- GitHub cloud, GitHub Enterprise on-premises
- GitLab cloud GitLab Enterprise on-premises
- BitBucket on-premises

You must have an appropriate local Git repository configured with access for all designated users in order to set up Git integration with VCF Automation. Also, you must save your cloud templates in a specific structure in order for them to be detected by Git. To create an integration with GitLab or GitHub, select **Infrastructure** > **Connections** > **Integrations** in VCF Automation and then make the appropriate selection. You will need the url and token for the target repository.

When Git integration is configured with an existing repository, all cloud templates associated with selected projects become available to qualified users. You can use these templates with an existing deployment or as the basis of a new deployment. When you add a project, you must select some properties regarding where and how it is stored in Git.

You can save actions to a Git repository directly from VCF Automation. You can version action scripts either directly to Git, or you can create versions in VCF Automation. If you create a version of an action in VCF Automation, then it is automatically saved to Git as a version. Cloud templates are a bit more complicated, because you cannot directly add them to a Git integration from VCF Automation. You must save them directly to a Git instance, and then you can retrieve them from Git when working with the cloud template management page in VCF Automation.

Before you Begin

You must create and save your cloud templates in a specific structure in order for them to be detected by GitLab or GitHub.

- Configure and store cloud templates to be integrated with GitLab correctly. Only valid templates are imported into GitLab.
  - Create one or more designated folders for the cloud templates.
  - All cloud templates must be stored within `blueprint.yaml` files.
  - Ensure that the top of your templates include the `name:` and `version:` properties.
- Extract an API key for the applicable repository. In your Git account, select your login in the upper right corner, and click **Developer Settings**. Navigate to **Personal Access Tokens**, and then name your token and set an expiration date. Then, select API and create the token. Copy the resulting value and save it.

The following guidelines must be observed for all cloud templates used with Git integration.

- Each cloud template must reside in a separate folder.
- All cloud templates must be named `blueprint.yaml`.
- All cloud template YAML files must use `name` and `version` fields.
- Only valid cloud templates are imported.
- If you update a draft cloud template imported from Git, and its content differs from that in the top version, the draft will not be updated in subsequent syncs and a new version is created. If you want to update a template and also allow further sync's from Git, then you must create a new version after final changes.

Configure GitLab cloud template integration in VCF Automation

This procedure demonstrates configuring GitLab integration in VCF Automation so that you can work with cloud templates in the repository and automatically download saved templates that are associated with designated projects. To use cloud templates with GitLab, you must create a connection to an appropriate GitLab instance, and then save the desired templates to that instance.

- Extract an API key for the applicable repository. In your GitLab account, select your login in the upper right corner, and navigate to the Settings menu. Select Access Tokens, then name your token, set an expiration date. Then, select API and create the token. Copy the resulting value and save it.

You must have an appropriate local Git repository configured with access for all designated users in order to set up Git integration with VCF Automation. Also, you must create and save your cloud templates in a specific structure in order for them to be detected by GitLab.

- Configure and store cloud templates to be integrated with GitLab correctly. Only valid templates are imported into GitLab. See How do I use Git integration in VCF Automation for VM Apps.

When GitLab integration is configured with an existing repository, all cloud templates associated with selected projects become available to qualified users. You can use these templates with an existing deployment or as the basis of a new deployment. When you add a project, you must select some properties regarding where and how it is stored in GitLab.

> **Note:**
> You cannot push new or updated cloud templates to the Git repository from VCF Automation. Also, you cannot push new templates to the repository from VCF Automation. To add cloud templates to a repository, developers must use the Git interface.

If you update a draft cloud template imported from Git, and its content differs from that in the top version, the draft will not be updated in subsequent syncs and a new version is created. If you want to update a cloud template and also allow further sync's from Git, then you must create a new version after final changes.

After you set up your cloud templates for use with GitLab and collect required information, you must set up integration with your GitLab instance. Then, you can import the designated cloud templates into GitLab.

1. Set up integration with your GitLab environment in VCF Automation.
   a) Select **Infrastructure** > **Integrations** > **Add New** and choose GitLab.
   b) Enter the **URL** for your GitLab instance. For a software as a service GitLab instance, in most cases, it will be gitlab.com.
   c) Enter the **Token**, also known as an API key, for the specified GitLab instance. See the prerequisites above for information about extracting the token from your GitLab instance.
   d) Add an appropriate Name and Description.
   e) Click **Validate** to verify the connection.
   f) Add capability tags if desired. See Using capability tags in VCF Automation for more information.
   g) Click **Add**.
2. Configure the GitLab connection to accept cloud templates in an appropriate repository.
   a) Select **Infrastructure** > **Integrations** and choose the appropriate GitLab integration.
   b) Select **Projects**.
   c) Select **New Project** and create a name for the project.
   d) Enter the **Repository** path within GitLab. Typically, this is the user name of the main account appended to the repository name.
   e) Enter the appropriate GitLab **Branch** that you want to use.
   f) If applicable, enter a **Folder** name. If left blank, all folders are available.
   g) Enter an appropriate **Type**. If applicable, enter a folder name. If left blank, all folders are available.
   h) Click **Next** to finish adding the repository.

      When you click **Next**, an automated synchronization task is initiated that imports cloud templates into the platform.

      When the synchronization tasks are complete, a message indicates that the cloud templates have been imported.

You can now retrieve cloud templates from GitLab.

Configure GitHub integration in VCF Automation for VM Apps
You can integrate the GitHub cloud-based repository hosting service in VCF Automation

- You must have access to GitHub.
- Configure and store cloud templates to be integrated with GitHub correctly. Only valid cloud templates are imported into GitHub. See How do I use Git integration in VCF Automation for VM Apps.

You need a valid GitHub token to configure GitHub integration in VCF Automation See the GitHub documentation for information about creating and locating your token.

1. Select **Infrastructure** > **Connections** > **Integrations** and click **Add Integration**.
2. Select GitHub.
3. Enter the required information on the GitHub configuration page.
4. Click **Validate** to check the integration.
5. If you need to add tags to support a tagging strategy, enter capability tags.
6. Click **Add**.
7. Configure the GitHub connection to accept cloud templates in an appropriate repository.
   a) Select **Infrastructure** > **Integrations** and choose the appropriate GitHub integration.
   b) Select **Projects**.
   c) Select **New Project** and create a name for the project.
   d) Enter the **Repository** path within GitHub. Typically, this is the user name of the main account appended to the repository name.
   e) Enter the appropriate GitHub **Branch** that you want to use.
   f) If applicable, enter a **Folder** name. If left blank, all folders are available.
   g) Enter an appropriate **Type**.
   h) Click **Next** to finish adding the repository.

      An automated synchronization task is initiated that imports cloud templates into the platform.

When the synchronization tasks are complete, a message indicates that the cloud templates have been imported.

GitHub is available for use in VCF Automation blueprints.

You can now retrieve cloud templates from GitHub.

Configure Bitbucket integration in VCF Automation for VM Apps
VCF Automation supports integration with Bitbucket for use as a Git-based repository for ABX action scripts and VMware cloud templates.

- Set up an on premises Bitbucket Server deployment with one or more ABX or cloud template-based projects that you want to use with your deployments. Bitbucket Cloud is currently not supported.
- Create or designate VCF Automation project to associate your Bitbucket integration.
- Cloud template files to be synched to a Bitbucket integration must be named `blueprint.yaml`.

In VCF Automation, you can work with two types of repository items using Bitbucket integration: VMware cloud templates or ABX action scripts. You must synch projects that you want to work with before using a Bitbucket integration. ABX actions support write back to the Bitbucket repository, but you cannot write back cloud templates from the integration. If you want to create new versions of cloud template files, you must do so manually.

1. Select **Infrastructure** > **Connections** > **Integrations** and click **Add Integration**.
2. Select Bitbucket.
3. Enter the Summary information and Bitbucket credentials on the Bitbucket new integration Summary page.
4. To check the integration, click **Validate**.
5. If you use add tags to support a tagging strategy, enter capability tags.
6. Click **Add**.
7. Select the Projects tab on the main page for the Bitbucket integration to associate a project with this Bitbucket integration.
8. Select the Project to associate with this Bitbucket integration.
9. Click **Next** to add a Repository to Bitbucket project and indicate the type of repository you are adding and then specify the **Repository** name and **Branch**, as well as the **Folder**.
10. Click **Add**.

    If you want to add one or more repositories to a project, click **Add Repository**.

Bitbucket integration is configured with the specified repository configuration, and you can view and work with ABX actions and cloud templates contained in configured repositories. When you add a project to a Bitbucket integration, a synch operation runs to pull the latest versions of ABX action scripts and cloud template files from the designated repository. The History tab on the Bitbucket integration page shows records of all synch operations for the integration. By default, files are automatically synched every 15 minutes,but you can manually synch a file by selecting it and clicking **SYNCH** at any time.

You can work with ABX actions on the VCF Automation Extensibility page, and you can work with cloud templates on the Design page. If you save a changed version of an ABX action on the Extensibility area of VCF Automation, the new version of the script is created and written back to the repository.

Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps

You can configure a VCF Operations orchestrator integration, so that you can use workflows as part of extensibility and cloud templates.

- Verify that you have cloud administrator credentials. See Credentials required for working with cloud accounts in VCF Automation for VM Apps.
- Verify you are running the latest VCF Operations Orchestrator Client release.

1. Select **Infrastructure** > **Connections** > **Integrations**.
2. Click **Add integration**.
3. Select **Orchestrator**.
4. Enter a name for the VCF Operations Orchestrator Client integration.
5. Optional: Enter a description for the VCF Operations Orchestrator Client integration.
6. Under **VCF Operations Orchestrator instance**, select the type of Orchestrator instance.
   a) **Embedded VCF Operations Orchestrator** is the default VCF Operations Orchestrator instance that is part of your VCF Automation deployment.
   b) **User-deployed VCF Operations Orchestrator** is an external VCF Operations Orchestrator instance.

      If you select this option, you must provide the fully qualified domain name (FQDN) of your external VCF Operations Orchestrator Client instance.

      For example, `https://my_Orchestrator_FQDN.com:443`.

      Then, to validate the integration, click **Validate**.
7. Optional: If prompted to do so, review the certificate information, and click **Accept**.
8. Optional: Add capability tags.

   Note:

   Capability tags can be used to manage multiple VCF Operations Orchestrator Client integrations.
9. Click **Add**.
   The VCF Operations Orchestrator Client integration is saved.
10. To verify that the integration is configured and that the workflows are added, select **Orchestrator** > **Library** > **Workflows**.

Activate or deactivate VCF Operations Orchestrator Client integrations in VCF Automation for VM Apps
You can manually activate or deactivate your VCF Operations Orchestrator Client integration so you can perform maintenance while the integration is still running.

Configure one or more VCF Operations Orchestrator Client integrations in VCF Automation. See Configure an VCF Operations Orchestrator Client integration in VCF Automation for VM Apps.

You can deactivate your VCF Operations Orchestrator Client integration to perform maintenance. While deactivated, your VCF Operations Orchestrator Client integration is still in a **RUNNING** state so you can continue to perform tasks such as resource monitoring and data collection.

Note:
In addition to manual disabling, the VCF Operations Orchestrator Client Gateway service performs periodic health status checks to verify if your VCF Operations Orchestrator Client integrations are active or not. Any inactive VCF Operations Orchestrator Client integrations are deactivated automatically and are set to the **DISCONNECTED** state. You will be unable to perform tasks such as data collection or resource monitoring on disconnected integrations.

After disabling a VCF Operations Orchestrator Client integration, or having the integration be disconnected by the health status checker, workflows will only run on remaining integrations that are activated. If your environment includes multiple activated VCF Operations Orchestrator Client integrations which are not managed through project constraints or capability tags, a random VCF Operations Orchestrator Client integration will be selected to run your workflow.

Note:
Since the VCF Operations Orchestrator Client integration is selected randomly, you must ensure that information required to run a given operation is available on all integrations. For content entities such as workflows, this means that they should be synchronized across all integrations. For inventory objects there is no guarantee that they will have the same object identifier on all integrations, so trying to run a workflow that includes such an inventory object as a input parameter might fail.

1. Deactivate your VCF Operations Orchestrator Client integration.
   a) Navigate to **Infrastructure** > **Connections** > **Integrations**.
   b) Select the VCF Operations Orchestrator Client integration you want to deactivate.
   c) Under **Orchestrator Server Credentials**, toggle off the **Enable endpoint** option.
   d) Click **Validate**.
   e) After successful validation, click **Save**.
2. Perform the necessary maintenance tasks on the deactivated VCF Operations Orchestrator Client integration.
3. Activate your VCF Operations Orchestrator Client integration.
   a) Navigate to **Infrastructure** > **Connections** > **Integrations**.
   b) Select the previously deactivated VCF Operations Orchestrator Client integration.
   c) Under **Server Credentials**, toggle on the **Enable endpoint** option.
   d) Click **Validate**.
   e) After successful validation, click **Save**.

Data collection for VCF Operations Orchestrator Client integrations
VCF Automation performs periodic data collection for your VCF Operations Orchestrator Client integrations.

Data collection events for VCF Operations Orchestrator Client integrations are triggered every 10 minutes. The data collection gathers data about the workflows included in the library of each VCF Operations Orchestrator Client integration.

> **Important:**
> Verify that you version up a workflow when you are finished editing it. Changes to non-versioned up workflows are not picked up by the data collector.

You can find information about the last data collection performed on a VCF Operations Orchestrator Client integration by navigating **Infrastructure** > **Connections** > **Integrations** and selecting the specific integration. You can also trigger a manual data collection event by clicking **Start Data Collection**.

For more information on VCF Automation data collection, see How does data collection work in VCF Automation.

What is configuration management in VCF Automation for VM Apps

VCF Automation supports integration with Puppet Enterprise, Ansible Open Source, and Ansible Tower so that you can manage deployments for configuration and drift.

Puppet Integration

To integrate Puppet-based configuration management, you must have a valid instance of Puppet Enterprise installed on a public or private cloud with a vSphere workload. You must establish a connection between this external system and your VCF Automation instance. Then you can make Puppet configuration management available to VCF Automation by adding it to appropriate blueprints.

The VCF Automation blueprint service Puppet provider installs, configures, and runs the Puppet agent on a deployed compute resource. The Puppet provider supports both SSH and WinRM connections with the following prerequisites:

- SSH connections:
  - The user name must be either a super user or a user with sudo permissions to run commands with `NOPASSWD`.
  - Deactivate `requiretty` for the given user.
  - cURL must be available on the deployment compute resource.
- WinRM connections:
  - PowerShell 2.0 must be available on the deployment compute resource.
  - Configure the Windows template as described in the VCF Operations orchestrator documentation.

The DevOps administrator is responsible for managing the connections to a Puppet master and for applying Puppets roles, or configuration rules, to specific deployments. Following deployment, virtual machines configured to support configuration management are registered with the designated Puppet Master.

When virtual machines are deployed, users can add or delete a Puppet Master as an external system or update projects assigned to the Puppet Master. Finally, appropriate users can de-register deployed virtual machines from the Puppet Master when the machines are decommissioned.

Ansible Open Source Integration

When setting up an Ansible integration, install Ansible Open Source in accordance with the Ansible installation instructions. See the Ansible documentation for more information about installation.

Ansible enables host key checking by default. If a host is reinstalled with a different key in the `known_hosts` file, an error message appear. If a host is not listed in the `known_hosts` file, you must supply the key on start-up. You can deactivate host key checking with the following setting in the `/etc/ansible/ansible.cfg` or `~/.ansible.cfg` file:

```
[defaults]
host_key_checking = False
localhost_warning = False

[paramiko_connection]
record_host_keys = False

[ssh_connection]
#ssh_args = -C -o ControlMaster=auto -o ControlPersist=60s
ssh_args = -o UserKnownHostsFile=/dev/null
```

To avoid the host key checking errors, set `host_key_checking` and `record_host_keys` to False including adding an extra option `UserKnownHostsFile=/dev/null` set in `ssh_args`.In addition, if the inventory is empty initially, Ansible warns that the host list is empty. This causes the playbook syntax check to fail.

Ansible vault enables you to store sensitive information, such as passwords or keys, in encrypted files rather than as plain text. Vault is encrypted with a password. In VCF Automation, Ansible uses Vault to encrypt data such as ssh passwords for host machines. It assumes that the path to the Vault password has been set.

You can modify the `ansible.cfg` file to specify the location of the password file using the following format.

```
vault_password_file = /path to/file.txt
```

You can also set the `ANSIBLE_VAULT_PASSWORD_FILE` environment variable so that Ansible automatically searches for the password. For example, `ANSIBLE_VAULT_PASSWORD_FILE=~/.vault_pass.txt`

VCF Automation manages the Ansible inventory file, so you must ensure that the VCF Automation user has rwx access on the inventory file.

```
cat ~/var/tmp/vmware/provider/user_defined_script/$(ls -t ~/var/tmp/vmware/provider/user_defined_script/ |
 head -1)/log.txt
```

If you want to use a non-root user with VCF Automation open-source integration, the users require a set of permissions to run the commands used by the VCF Automation open-source provider. The following commands must be set in the user's sudoers file.

```
Defaults:myuser !requiretty
```

If the user is not part of an admin group that has no askpass application specified, set the following command in the user's sudoers file.

```
myuser ALL=(ALL) NOPASSWD: ALL
```

If you encounter errors or other problems when setting up Ansible integration, refer to the `log.txt` file at `'cat~/var/tmp/vmware/provider/user_defined_script/$(ls -t ~/var/tmp/vmware/provider/user_defined_script/ | head -1)/'` on the Ansible Control Machine.

Ansible Tower Integration

Supported Operating System Types

- Red Hat Enterprise Linux 8.0 or later 64-bit (x86), supports only Ansible Tower 3.5 and greater.

- Red Hat Enterprise Linux 7.4 or later 64-bit (x86).
- CentOS 7.4 or later 64-bit (x86).

The following is a sample inventory file, which is generated during an Ansible Tower installation. You may need to modify it for VCF Automation integration uses.

```
[root@cava-env8-dev-001359 ansible-tower-setup-bundle-3.5.2-1.el8]# pwd

/root/ansible-tower-install/ansible-tower-setup-bundle-3.5.2-1.el8

[root@cava-env8-dev-001359 ansible-tower-setup-bundle-3.5.2-1.el8]# cat inventory

[tower]

localhost ansible_connection=local




[database]




[all:vars]

admin_password='VMware1!'




pg_host=''

pg_port=''




pg_database='awx'

pg_username='awx'

pg_password='VMware1!'




rabbitmq_port=5672

rabbitmq_vhost=tower

rabbitmq_username=tower

rabbitmq_password='VMware1!'

rabbitmq_cookie=cookiemonster
```

```
# Needs to be true for fqdns and ip addresses

rabbitmq_use_long_name=false



# Isolated Tower nodes automatically generate an RSA key for authentication;

# To deactivate this behavior, set this value to false

# isolated_key_generation=true
```

Configure Puppet Enterprise integration in VCF Automation for VM Apps

VCF Automation supports integration with Puppet Enterprise configuration management.

When you add Puppet Enterprise to VCF Automation as an external system, by default it is available on all projects. You can restrict it to specific projects.

To add a Puppet Enterprise integration, you must have the Puppet master name and the hostname or IP address of the master.

You can find Puppet logs at the following location in case you need to check them for errors or information purposes.

Description	Log Location
Log for create and install related events	Logs are on the deployed machine at `~/var/tmp/vmware/provider/user_defined_script/$(ls -t~/var/tmp/vmware/provider/user_defined_script/ | head -1)/`. Refer to the **log.txt** file for full logs. For detailed Puppet agent logs, refer to https://puppet.com/docs/puppet/4.8/services_agent_unix.html#logging
Log for Puppet delete and run related tasks	Logs are on the PE at `~/var/tmp/vmware/provider/user_defined_script/$(ls -t ~/var/tmp/vmware/provider/user_defined_script/ | head -1)/`. Refer to the **log.txt** file for full logs.

1. Select **Infrastructure** > **Connections** > **Integrations** and click **Add Integration**.
2. Select Puppet.
3. Enter the required information on the Puppet configuration page.

   For Puppet integration to work properly, the provided credentials must be valid for both the SSH and the API account. Also, the specified OS and application user accounts must have the same username and password.
4. Click **Validate** to check the integration.
5. Click **Add**.

Puppet is available for use with cloud templates.
Add Puppet components to the desired cloud templates.

1. On the VCF Automation Design tab select Puppet under the Content Management heading on the cloud template menu and drag the Puppet component to the canvas.

2. Enter Puppet Properties on the pane to the right.

Property	Description
Master	Enter the name of the Puppet primary machine used with this cloud template.
Environment	Select the environment for the Puppet primary machine.
Role	Select the Puppet role to be used with this cloud template.
Agent Run Interval	The frequency at which you want the Puppet agent to poll the Puppet primary machine for configuration details to be applied to deployed virtual machines related to this cloud template.

3. Click the Code tab on the right pane to view the YAML code for the Puppet configuration properties.

When you add a Puppet component to a cloud template, you can add the `installMaster` property to the YAML file to point to a Puppet install master, also known as a compile master. The value of this property can be the IP address or the hostname of the Puppet compile master. Using this property provides access to enhanced capabilities for deployed Puppet virtual machines and also supports additional day two actions.

```
Puppet_Agent:
  type: Cloud.Puppet
  properties:
    account: PEIntegrationAccount
    environment: production
    role: 'role::linux_webserver'
    host: '${CentOS-Puppet.*}'
    username: root
    password: password123!
    installMaster: my-pe-compile-master.example.com
    agentConfiguration:
      certName: '${CentOS-Puppet.address}'
    osType: linux
    count: 1
```

Note:
Though the user defined here is root, the cloud template can be configured with any user that is included in the sudoers list.

In some cases, VCF Automation passes some machine related information to Puppet virtual machines as facts by default. Custom facts are not supported for Windows machines. On Linux machines some information is passed by default, and users can pass additional information using custom properties.

There are some limitations on what is passed to Puppet machines under Linux. Custom properties on host resources and on the Puppet agent are passed to Puppet virtual machines. Custom properties on network resources are not passed to the virtual machine. Items passed include simple properties, boolean properties as well as custom named and complex types such as nested maps with arrays.

The following example shows how various custom resources can be called on host resources:

```
resources:
Puppet-Host:
  type: Cloud.AWS.EC2.Instance
  properties:
    customer_specified_property_on_ec2_resource: "property"
    customer_specified_property_on_network_resource_that_should_also_be_a_fact_and_is_boolean: true
    CustomerNameStuff: "zone A"
    try_map:
    key: value
    keytwo: value
    nested_array:
```

```
      - one
      - two
      - true
    try_array:
      - one
      - two
      -three:
         inner_key: value
```

If a Puppet purge command results in errors, in most cases, VCF Automation will ignore purge errors for nodes and proceed with deletion of the node. Even if a certificate is not found for a specific node, VCF Automation will proceed with deletion. If VCF Automation cannot proceed with the node deletion for some reason, you can click Delete on the Deployments page Actions menu to open a dialog that will enable you to proceed with the node deletion. A similar workflow is executed when you remove a Puppet integration from a cloud template and then apply the template to the deployment. This workflow triggers a node purge operation that is handled as described above.

Integration with Puppet Enterprise requires a public IP address. If there is no public IP address configured for the Puppet Enterprise machine, the IP address of the first NIC is used.

If the NIC of a Puppet provisioned machine running on a vSphere machine has multiple IP addresses, you can use the `primaryAddress` YAML property in cloud templates to specify which IP address to use for connections. When the `primaryAddress` property is assigned to a NIC, then the IP address of this NIC is used by Puppet. Only one NIC can be designated as primary. See the following YAML snippet for an example of how the `primaryAddress` property is used.

```
BaseVM:
type: Cloud.vSphere.Machine
properties:
 image: photon
 count: 2
 customizationSpec: Linux
 cpuCount: 1
 totalMemoryMB: 1024
 networks:
  - network: '${resource.dev.id}'
    deviceIndex: 0
    primaryAddress: true
    assignment: static
  - network: '${resource.prod.id}'
    deviceIndex: 1
    assignment: static
```

If the `primaryAddress` property is not set for any virtual machine NIC, the cloud template logic will default to the current behavior for IP address selection.

If you want to set up Puppet integration for a non-root user with Sudo privileges, that user should be enabled to run the following commands.

- The user should be enabled to create a new Puppet facts directory and file:

```
sudo mkdir -p /etc/puppetlabs/facter/facts.d
     sudo tee /etc/puppetlabs/facter/facts.d/puppet_cloudassembly_facts.json
```

- The user should be enabled to run Puppet:

```
sudo /opt/puppetlabs/bin/puppet resource service puppet ensure=stopped
     sudo /opt/puppetlabs/bin/puppet agent --test --color=false --detailed-exitcode
```

- The user should be enabled to delete Certificate Signing Request(CSR) files, including the csr_attributes.yaml and CSR pem:

```
sudo rm /etc/puppetlabs/puppet/csr_attributes.yaml
     sudo rm -f /etc/puppetlabs/puppet/ssl/certificate_requests/*
```

Configure Ansible Open Source integration in VCF Automation for VM Apps

VCF Automation supports integration with Ansible Open Source configuration management. After configuring integration, you can add Ansible components to new or existing deployments.

- The Ansible control machine must use an Ansible version.
- Ansible log verbosity must be set to default of zero.
- The user must have read/write access to the directory where the Ansible inventory file is located. In addition, the user must have read/write access to the inventory file, if it exists already.
- If you are using a non-root user with the sudo option, ensure that the following is set in the sudoers file:
  `Defaults:`*`user_name`* `!requiretty`
  and
  `username` `ALL=(ALL) NOPASSWD: ALL`
- Ensure that host key checking is deactivated by setting `host_key_checking = False` at `/etc/ansible/ansible.cfg` or `~/.ansible.cfg`.
- Ensure that the vault password is set by adding the following line to the `/etc/ansible/ansible.cfg` or `~/.ansible.cfg` file:

  ```
  vault_password_file = /path/to/password_file
  ```

  The vault password file contains the password in plain text and is used only when cloud templates or deployments provide the username and password combination to use between ACM and the node as show in the following example.

  ```
  echo 'myStr0ng9@88w0rd' > ~/.ansible_vault_password.txt
  echo 'ANSIBLE_VAULT_PASSWORD_FILE=~/.ansible_vault_password.txt' >> ~/.profile        # Instead of this
   way, you can also set it setting 'vault_password_file=~/.ansible_vault_password.txt' in either /etc/
  ansible/ansible.cfg or ~/.ansible.cfg
  ```

- To avoid host key failures while trying to run playbooks, it is recommended that you include the following settings in `/etc/ansible/ansible config`.

  ```
  [paramiko_connection]
  record_host_keys = False

  [ssh_connection]
  ssh_args = -o UserKnownHostsFile=/dev/null

  # If you already have options set for ssh_args, add the additional
  # options at the end as in the following example.
  # ssh_args = -C -o ControlMaster=auto -o ControlPersist=60s
  ```

When you integrate Ansible Open Source with VCF Automation, you can configure it to run one or more Ansible playbooks in a given order when a new machine is provisioned to automate configuration management. You specify the desired playbooks in the cloud template for a deployment.

When setting up an Ansible integration, you must specify the Ansible Open Source host machine as well as the inventory file path that defines information for managing resources. In addition, you must provide a name and password to access the Ansible Open Source instance. Later, when you add an Ansible component to a deployment, you can update the connection to use key-based authentication.

By default, Ansible uses ssh to connect to the physical machines. If you are using Windows machines as specified in the cloud template with the osType Windows property, the `connection_type` variable is automatically set to `winrm`.

Initially, Ansible integration uses the user/password or user/key credentials provided in the integration to connect to the Ansible Control Machine. Once the connection is successful, the provided playbooks in the cloud template are validated for syntax.

If the validation is successful, then an execution folder is created on the Ansible Control Machine at `~/var/tmp/vmware/provider/user_defined_script/`. This is the location from which scripts run to add the host to the inventory, create the host vars files including setting up the authentication mode to connect to the host, and finally run the playbooks. At this point, the credentials provided in the cloud template are used to connect to the host from the Ansible Control Machine.

Ansible integration supports physical machines that do not use an IP address. For machines provisioned on public clouds such as AWS, Azure, and GCP, the address property in the created resource is populated with the machine's public IP address only when the machine is connected to a public network. For machines not connected to a public network, the Ansible integration looks for the IP address from the network attached to the machine. If there are multiple networks attached, Ansible integration looks for the network with the least deviceIndex; that is, the index of the Network Interface Card (NIC) attached to the machine. If the deviceIndex property is not specified in the blueprint, the integration uses the first network attached.

See What is configuration management in VCF Automation for VM Apps for more details on configuring Ansible Open Source for integration in VCF Automation.

1. Select **Infrastructure** > **Connections** > **Integrations** and click **Add Integration**.
2. Click Ansible.
   The Ansible configuration page appears.
3. Enter the Hostname, Inventory File Path and other required information for the Ansible Open Source instance.
4. Click **Validate** to check the integration.
5. Click **Add**.

Ansible is available for use with cloud templates.

Add Ansible components to the desired cloud templates.

1. On the cloud template canvas page, select Ansible under the Configuration Management heading on the cloud template options menu and drag the Ansible component to the canvas.
2. Use the panel on the right to configure the appropriate Ansible properties such as specifying the playbooks to run.

In Ansible, users can assign a variable to a single host, and then use it later in playbooks. Ansible Open Source integration enables you to specify these host variable in cloud templates. The `hostVariables` property must be in proper YAML format, as expected by the Ansible control machine, and this content will be placed at the following location:

parent_directory_of_inventory_file`/host_vars/`*host_ip_address*`/vra_user_host_vars.yml`

The default location of the Ansible inventory file is defined in the Ansible account as added on the Integrations page in VCF Automation. The Ansible integration will not validate the `hostVariable` YAML syntax in the cloud template, but the Ansible Control Machine will throw an error when you run a playbook in the case of incorrect format or syntax.

The following cloud template YAML snippet shows an example useage of the `hostVariables` property.

```
Cloud_Ansible_1:
type: Cloud.Ansible
properties:
host: '${resource.AnsibleLinuxVM.*}'
osType: linux
account: ansible-CAVA
username: ${input.username}
password: ${input.password}
maxConnectionRetries: 20
groups:
- linux_vms
playbooks:
provision:
- /root/ansible-playbooks/install_web_server.yml
hostVariables: |
message: Hello ${env.requestedBy}
project: ${env.projectName}
```

Ansible integrations expect authentication credentials to be present in a cloud template in one of the following ways:

- User name and password in the Ansible resource.

- User name and privateKeyFile in the Ansible resource.
- Username in Ansible resource and privatekey in the compute resource by specifying remoteAccess to `generatedPublicPrivateKey`.

When you create an Ansible Open Source integration, you must provide login information for the integration user to connect with the Ansible control machine using SSH. To run playbooks with an integration, you can specify a different user in the integration YAML code. The `username` property is mandatory and required to connect to the virtual machine where Ansible will make changes. The `playbookRunUsername` property is optional and can be provided to execute the playbook on the Ansible node. The default value of `playbookRunUsername` is the Ansible endpoint integration username.

If you specify a different user, that user should have write access to the Ansible hosts file and should have permission to create private key files.

When you add an Ansible Open Source tile to a cloud template, VCF Automation creates the host entry for the attached virtual machine. By default, VCF Automation will use the virtual machine's resource name to create the host entry, but you can specify any name using the `hostName` property in the blueprint YAML. In order to communicate with the machine, VCF Automation will create the host variable `ansible_host: IP Address` for the host entry. You can override the default behaviour to configure communication using FQDN, by specifying the keyword `ansible_host` under `hostVariables` and providing FQDN as its value. The following YAML code snippet shows an example of how hostname and FQDN communication can be configured:

```
Cloud_Ansible:
type: Cloud Ansible
properties:
osType: linux
username: ubuntu
groups:
- sample
hostName: resource name
host: name of host
account: name of account
hostVariables:
ansible_host:Host FQDN
```

In this example you override the default `ansible_host` value by providing the FQDN. This may be useful for users who want Ansible Open Souce to connect to the host machine using the FQDN.

The default value of `hostVariables` in the YAML will be `ansible_host:IP_address` and the IP address is used to communicate with the server.

If the YAML count property is more than 1 for Ansible Open Source, the hostname could be mapped to any of the respective virtual machine's properties.The following example shows mapping for a virtual machine resource named Ubuntu-VM if we want its address property to be mapped to the hostname.

```
hostname: '${resource.Ubuntu-VM.address[count.index]}'
```

In cloud templates, ensure that the path to the Ansible playbook is accessible to the user specified in the integration account. You can use an absolute path to specify the playbook location, but it is not necessary. An absolute path to the user's home folder is recommended so that the path remains valid even if the Ansible integration credentials change over time.

Users with cloud administrator privileges can change the project of a deployment containing Ansible Open Source and Ansible Tower resources. The functionality is available as a day 2 action on deployment level.

To change the project for an Ansible deployment, select the Change Project option from the Actions menu of the deployment as shown on the VCF Automation Deployments page, and then choose the target project and click **Submit** on the displayed dialog.

Configure Ansible Automation Platform/Ansible Tower Integration in VCF Automation for VM Apps

You can integrate Ansible Automation Platform, fomerly Ansible Tower, with VCF Automation to support configuration management of deployed resources. After configuring integration, you can add Ansible Automation Platform virtual components to new or existing deployments from the cloud template editor.

- Grant non-administrator users the appropriate permissions to access Ansible Automation Platform. There are two options that work for most configurations. Choose the one that is most appropriate for your configuration.
  - Grant users Inventory Administrator and Job Template Administrator roles at the organization level.
  - Grant users Administrator permission for a particular inventory and the Execute role for all job templates used for provisioning.
- You must configure the appropriate credentials and templates in Ansible Automation Platform for use with your deployments. Templates can be job templates or workflow templates. Job templates define the inventory and playbook for use with a deployment. There is a 1:1 mapping between a job template and a playbook. Playbooks use a YAML-like syntax to define tasks that are associated with the template. For most typical deployments, use machine credentials for authentication.

  Workflow templates enable users to create sequences consisting of any combination of job templates, project syncs, and inventory syncs that are linked together so that you can execute them as a single unit. The Ansible Automation Platform Workflow Visualizer helps users to design workflow templates. For most typical deployments, you can use machine credentials for authentication.

  If you are working with Ansible Automation Platform, you must define an execution environment on the Ansible Controller to satisfy ansible-runner dependencies. See the Ansible documentation for more information about execution environments and container images.

  a. Log in to Ansible Automation Platform and navigate to the Templates section.
  b. Select Adding a new job template.
     - Select the credential that you already created. These are the credentials of the machine to be managed by Ansible Automation Platform. For each job template, there can be one credential object.
     - For the Limit selection, select Prompt on Launch. This ensures that the job template runs against the node being provisioned or de-provisioned from VCF Automation. If this option is not selected, a Limit is not set error will appear when the blueprint that contains the job template is deployed.
  c. Select Adding a new workflow template.
     - Select the credentials that you already created and then define the inventory. Using Workflow Visualizer, design the workflow template.

     For the Limit box of workflow or job templates, generally you can select Prompt on Launch. This selection ensures that the job or workflow template runs against the node being provisioned or de-provisioned from VCF Automation.
- You can view the execution of the Job templates or workflow templates invoked from VCF Automation on the Ansible Tower Jobs tab.

1. Select **Infrastructure** > **Connections** > **Integrations** and click **Add Integration**.
2. Click Ansible Tower.

   The Ansible configuration page appears.
3. Enter an appropriate **Name** and **Description** for the integration.
4. Enter the **Hostname**, which can be an IP address, and other required information for the Ansible Automation Platform instance.
5. Enter the UI-based authentication **Username** and **Password** for the applicable Ansible Automation Platform instance.
6. Click **Validate** to verify the integration.

   In VCF Automation 9.1.1, you are prompted to accept the certificate. The trusted certificate is then reused for all subsequent API calls to that Ansible host.
7. Click **Add**.

Ansible Tower is available for use in cloud templates.

Add Ansible Automation Platform components to the desired cloud templates. You must specify the applicable job template with execute permission for the user specified in the integration account.

1. On the cloud template canvas page, select Ansible under the Configuration Management heading on the blueprint options menu and drag the Ansible Automation Platform component to the canvas.
2. Use the panel on the right to configure the appropriate Ansible Automation Platform properties such as job templates.

When you add an Ansible Automation Platform tile to a cloud template, VCF Automation creates the host entry for the attached virtual machine in the Ansible Automation Platform. By default, VCF Automation will use the virtual machine's resource name to create the host entry, but you can specify any name using the `hostName` property in the blueprint YAML. To communicate with the machine, VCF Automation will create the host variable `ansible_host: IP Address` for the host entry. You can override the default behavior to configure communication using FQDN, by specifying the keyword `ansible_host` under `hostVariables` and providing FQDN as its value. The following YAML code snippet shows an example of how hostname and FQDN communication can be configured:

```
Cloud_Ansible_Tower_1:
  type: Cloud Ansible Tower
  properties:
      host: name of host
      account: name of account
      hostName: resource name
      hostVariables:
          ansible_host:Host FQDN
```

In this example, you override the default `ansible_host` value by providing the FQDN. This may be useful for users who want Ansible Tower to connect to the host machine using the FQDN.

The default value of `hostVariables` in the YAML will be `ansible_host:IP_address` and the IP address is used to communicate with the server.

If the YAML count property is more than 1 for Ansible Automation Platform, the hostname could be mapped to any of the respective virtual machine's properties.The following example shows mapping for a virtual machine resource named Ubuntu-VM if we want its address property to be mapped to the hostname.

```
hostname: '${resource.Ubuntu-VM.address[count.index]}'
```

When you add an Ansible Automation Platform component to a cloud template, and you can specify the job template to call in the cloud template YAML. You can also specify workflow templates or a combination of job templates and workflow templates. If you don't specify the template type, by default VCF Automation assumes that you are calling a job template.

The following YAML snippet shows an example of how a combination of job and workflow templates can be called in an Ansible Tower cloud template.

```
Cloud_Ansible_1:
type: Cloud.Ansible.Tower
properties:
  host: '${resource.CentOS_Machine.*}'
  account:
  maxConnectionRetries: 2
  maxJobRetries: 2
  templates:
    provision:
      - name: My workflow
        type: workflow
      - name: My job template
```

We added the `maxConnectionsRetries` and `maxJobRetries` to handle Ansible related failures. The cloud templates accepts the custom value and, in case no value is provided, it uses the default value. For `maxConnectionRetries`, the default value is 10, and for `maxJobRetries` the default value is 3.

Note:
Earlier versions of VCF Automation supported the execution of job templates only using the jobTemplate schema in the cloud template. jobTemplate is now deprecated and might be removed in future releases. For now, using the jobTemplate property will continue to work as expected. To run workflow templates and use additional features, it is recommended to use the templates schema.

VCF Automation templates for Ansible Automation Platform integrations include the `useDefaultLimit` property with a true or false value to define where Ansible templates are executed. Ansible templates can be job templates or workflow templates. If this value is set to true, the specified templates are run against the machine specified in the Limit box on the Ansible Templates page. If the value is set to false, the templates are run against the provisioned machine, but users should check the Prompt on Launch checkbox on the Ansible Automation Platform Templates page. By default, the value of this property is false. The following YAML example shows how the `useDefaultLimit` property appears in cloud templates.

```
templates:
provision:
  - name: ping aws_credentials
    type: job
    useDefaultLimit: false
    extraVars: '{"rubiconSurveyJob" : "checkSurvey"}'
```

In addition, as the preceding example shows, you can use the `extraVars` property to specify extra variables or survey variables. This capability can be useful for running templates that require input. If a user has maintained the survey variable, then you must pass the variable in the `extraVars` section of the cloud template to avoid errors.

Users with cloud administrator privileges can change the project of a deployment containing Ansible Open Source and Ansible Automation Platform resources. The functionality is available as a day 2 action on deployment level.

To change the project for an Ansible deployment, select the Change Project option from the Actions menu of the deployment as shown on the VCF Automation Deployments page, and then choose the target project and click **Submit** on the displayed dialog.

Users must make two changes to make it work:

- Use `ansible_host` (e.g., FQDN) and `hostName` in the cloud template.
- In AWX, turn on the update on launch flag; that is, sync to the vCenter for new hosts before running the playbook. The sync will merge FQDN host entries added by VCF Automation and imported by VMware inventory plug-in and assign hosts to groups. Inventory groups are created from VM tag values using the sync source variables above.

Note:
Though VMware does not formally support the use of AWX, it will function in the case described herein.

See the following cloud template for an example implementation.

```
# Created by Quickstart wizard.
name: RHEL 8
version: 0.0.1
formatVersion: 1
inputs:
image:
  type: string
  description: Select an OS Version
  default: RHEL 8 Base
  enum:
    - RHEL 8 Base
    - RHEL 7 Base
AWX:
  type: string
  description: Choose AWX Environment
  enum:
```

```
    - LabAWX
    - FA/CC-AWX
envrionmnetTag:
  type: string
  description: Choose VM Environment
  enum:
    - cel
    - mag
    - wdr
purposeTag:
  type: string
  description: Choose Server Purpose
  default: ''
  enum:
    - ''
    - mariadb
    - oracle
authGroupTag:
  type: string
  description: Choose Authentication Group
  default: ''
  enum:
    - ''
    - dbo_linux
    - oracle
    - postgres
hostname:
  type: string
  description: Desired hostname
  default: changeme
cpuCount:
  type: integer
  description: Number of virtual processors
  default: 1
totalMemoryMB:
  type: integer
  description: Machine virtual memory size in Megabytes
  default: 1024
disk1Size:
  type: integer
  description: A SIZE of 0 will disable the disk and it will not be provisioned.
  default: 0
disk2Size:
  type: integer
  description: A SIZE of 0 will disable the disk and it will not be provisioned.
  default: 0
neededip:
  type: string
  description: Enter an available IP Address
  title: Needed-IP-Address
vlan:
  type: string
  description: Enter in needed vlan
  title: Enter VLAN ID example "vl500"
resources:
Cloud_Ansible_Tower_1:
  type: Cloud.Ansible.Tower
```

```
  metadata:
    layoutPosition:
      - 0
      - 0
  properties:
    host: ${resource.Cloud_vSphere_Machine_1.*}
    account: ${input.AWX}
    hostName: ${input.hostname}
    hostVariables:
      ansible_host: ${input.hostname}.dcl.wdpr.mycompany.com
    templates:
      provision:
        - name: Linux-Role
Cloud_vSphere_Machine_1:
  type: Cloud.vSphere.Machine
  metadata:
    layoutPosition:
      - 0
      - 1
  properties:
    image: ${input.image}
    Infoblox.IPAM.Network.dnsSuffix: dcl.wdpr.mycompany.com
    Infoblox.IPAM.Network.dnsView: Internal
    customizationSpec: Rhel7Base
    name: ${input.hostname}
    cpuCount: ${input.cpuCount}
    totalMemoryMB: ${input.totalMemoryMB}
    attachedDisks: ${map_to_object(resource.Cloud_Volume_1[*].id + resource.Cloud_Volume_2[*].id, "source")}
    networks:
      - network: ${resource.Cloud_vSphere_Network_1.id}
        assignment: static
        address: ${input.neededip}
    tags:
      - key: Server-Team
        value: ${input.envrionmnetTag}
      - key: Server-Team
        value: ${input.purposeTag}
      - key: Server-Team
        value: ${input.authGroupTag}
Cloud_Volume_1:
  type: Cloud.Volume
  metadata:
    layoutPosition:
      - 0
      - 2
  properties:
    count: '${input.disk1Size == 0 ? 0 : 1 }'
    capacityGb: ${input.disk1Size}
Cloud_Volume_2:
  type: Cloud.Volume
  metadata:
    layoutPosition:
      - 0
      - 3
  properties:
    count: '${input.disk2Size == 0 ? 0 : 1}'
    capacityGb: ${input.disk2Size}
```

```
Cloud_vSphere_Network_1:
  type: Cloud.vSphere.Network
  metadata:
    layoutPosition:
      - 1
      - 0
  properties:
    networkType: existing
    constraints:
      - tag: ${input.vlan}
```

How do I create an Active Directory integration in VCF Automation for VM Apps

VCF Automation supports integration with Active Directory servers to provide out of the box creation of computer accounts in a specified Organizational Unit (OU) within an Active Directory server prior to provisioning a virtual machine. Active Directory supports an LDAP connection to the Active Directory server.

- Active Directory integration requires an LDAP connection to the Active Directory server.
- If you are configuring an integration with Active Directory in the cloud, you must have a Microsoft Azure or Amazon Web Services account.
- You must have a project configured with appropriate cloud zones, and image and flavor mappings to use with the Active Directory integration.
- The desired OU on your Active Directory must be pre-created before you associated your Active Directory integration with a project.
- The user configured for the Active Directory integration must have permissions to create/delete/search for computer objects in the configured OU.

An Active Directory policy that is associated with a project is applied to all virtual machines provisioned within the scope of that project. Users can specify one or more tags to selectively apply the policy to virtual machines that are provisioned to the cloud zones with matching capability tags.

When an Active Directory integration is created, some properties are set during the creation of the computer object in Active Directory and cannot be changed. In particular, the following default properties cannot be changed:

```
WORKSTATION_TRUST_ACCOUNT    0x1000
PASSWD_NOTREQD (No password is required)    0x0020
```

For on-premises deployments, Active Directory integration enables you to set up a health check feature that shows the status of the integration and the underlying ABX integration on which it relies, including the required extensibility cloud proxy. Prior to applying an Active Directory policy, VCF Automation checks the status of the underlying integrations. If the integration is healthy, VCF Automation creates the deployed computer objects in the specified Active Directory. If the integration is unhealthy, the deploy operation skips the Active Directory phase during provisioning.

1. Select **Infrastructure** > **Connections** > **Integrations** and then **New Integration**.
2. Click **Active Directory**.
3. On the **Summary** tab, enter the appropriate LDAP host and environment names.

   The specified LDAP host is used to validate the Active Directory integration, and it is also used for subsequent deployments if no alternative hosts are specified and invoked due to errors or unavailability.
4. Enter the name and password for the LDAP server.
5. Enter the appropriate Base DN that specifies the root for the desired Active Directory resources.

   Note:
   You can specify only one DN per Active Directory integration.

6. Click **Validate** to ensure that the integration is functional.
7. Enter a Name and Description of this integration.
8. Click **Save**.
9. Click the **Project** tab to add a project to the Active Directory integration.

   On the **Add Projects** dialog, you must select a project name and a relative DN, which is a DN that exists within the Base DN specified on the Summary tab.
10. Under the Extended Options selection, provide a comma separated list of **Alternate Hosts** that will be used if the initially selected server is unavailable during deployment. The primary server is always used for initial validation of the integration.

    Note:
    If the format of the primary host is LDAP, LDAPS is not supported for alternative hosts.
11. Enter the time in seconds to wait for the initial server to respond before trying an alternate server in the **Connection Timeout** box.
12. Click **Save**.

You can now associate the project with Active Directory integration to a cloud template. When a machine is provisioned using this cloud template, it is pre-staged in the specified Active Directory and Organizational Unit.

Initially, Active Directory integrations are deployed to a default OU with little user restrictions. The OU is set by default when you map an Active Directory integration to a project. You can add a property called `FinalRelativeDN` to blueprints to change the OU for Active Directory deployments. This property enables you to specify the OU to use with an Active Directory deployment.

```
formatVersion: 1
inputs: {}
resources:
Cloud_vSphere_Machine_1:
 type: Cloud.vSphere.Machine
 properties:
   image: CenOS8
   flavor: tiny
   activeDirectory:
     finalRelativeDN: ou=test
     securityGroup: TestSecurityGroup
```

As shown in the preceding YAML example, users can add a property to an Active Directory integration deployment that adds a computer account to the security group so that appropriate permissions are assigned to access the shared resource over a network. The Active Directory virtual machine is initially deployed to a fixed OU but when the machine is ready to release, it is moved to a different OU with the appropriate policy as applicable for users.

If a computer account is moved to a different OU after deployment, VCF Automation attempts to delete the accounts on the initial OU. Deletion of computer accounts succeeds only in the case of virtual machines moved to a different OU within the same domain.

You can also implement a tag-based health check for on-premises Active Directory integrations as follows.

1. Create an Active Directory integration as described in the preceding steps.
2. Click the **Project** tab to add a project to the Active Directory integration.
3. Select a project name and a relative DN on the Add Projects dialog. The relative DN must exist within the specified base DN.

   There are two switches on this dialog that enable you to control Active Directory configuration from cloud templates. Both switches are off by default.
   - Override - This switch enables you to override Active Directory properties, specifically the relative DN in cloud templates. When switched on, you can change the OU specified in the `relativeDN` property in the cloud template. When provisioned the machine will be added to the OU specified in the `relativeDN` property in the cloud template. The following example shows the cloud template hierarchy in which this property appears.

```
activeDirectory:
  relativeDN: OU=ad_integration_machine_override
```

   - Ignore - This switch enables you to ignore the Active Directory configuration for the project. When switched on, it adds a property to the cloud template called `ignoreActiveDirectory` for the associated virtual machine. When this property is set to true means that the machine is not added to the Active Directory when deployed.
4. Add appropriate tags. These tags are applicable to the cloud zone to which the Active Directory policy may apply.
5. Click Save.

The Status of the Active Directory integration is displayed for each integration on the **Infrastructure** > **Connections** > **Integrations** page in VCF Automation.

You can associate the project with Active Directory integration with a cloud template. When a machine is provisioned using this template, it is pre-staged in the specified Active Directory and OU.

Integrating with VCF Operations

VCF Automation can work with VCF Operations to perform advanced workload placement, provide deployment health and virtual machine metrics, and display pricing.

Number and type of integrations

You can integrate one VCF Automation instance with one VCF Operations instance and eachVCF Operations instance can only be connected to one VCF Automation instance.

Basic requirements for integration

To integrate with VCF Operations, go to **Infrastructure** > **Connections** > **Integrations**. To add the integration, you need the VCF Operations URL and credentials for the login account described in the next section. In addition, VCF Automation and VCF Operations need to manage the same vSphere endpoint.

VCF Operations integration credentials with VCF Automation

This section describes the credentials that are required to integrate with VCF Operations. These credentials are established and configured in VCF Operations, not in VCF Automation.

Provide a local login account to VCF Operations with the following read privileges.

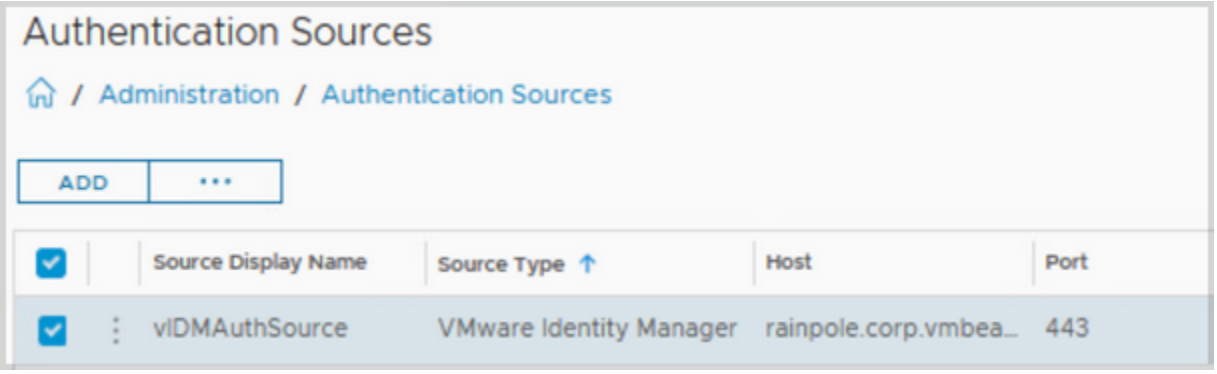
- Adapter Instance vCenter Adapter > VC Adapter Instance for *`vCenter-FQDN`*

Login account for integration

In VCF Operations, you need a local or non-local VCF Operations login account for the integration to use. The account requires read-only privileges to the vCenter adapter instance for the vSphere endpoint. Note that a non-local account might need to be imported in VCF Operations and have its read-only role assigned.

For the integration, the username format for non-local account login is *`username`*@*`domain`*@*`authenticated-source`*, such as `jdoe@rainpole.local@vIDMAuthSource`.

If you are using Workspace ONE Access for source authentication, your account login is *username* without any domain information. Authenticated sources are defined during VCF Operations server initial setup.



See the following sections for details. For pricing information, see Using Pricing Cards in VCF Automation.

Advanced workload placement using VCF Operations
VCF Automation and VCF Operations can work together to optimally place deployment workloads.

You enable workload placement at the vSphere based cloud zone level. Only Distributed Resource Scheduler (DRS) enabled clusters of a cloud zone are eligible for advanced placement using VCF Operations.

- VCF Automation placement—The VCF Automation placement engine is application intent based. It considers tag-based constraints, project membership and the associated cloud zones, and affinity filters related to network, storage, and compute. Resource placement depends on all of these factors plus the presence of other, related target resources in the same deployment.
- VCF Operations placement— VCF Operations considers operational intent for optimal placement. Operational intent can take past workloads and future, what-if predictions into account.
  When using advanced workload placement, you must apply VCF Automation tagging in order to implement business intent decisions, instead of using the VCF Operations business intent options.

When integrating with VCF Operations, VCF Automation continues to follow its application intent model and its related constraints to filter for target placement. Then, from within those results, it uses the VCF Operations recommendation to further refine placement.

In the absence of a recommendation

If you enable advanced workload placement, and VCF Operations analysis returns no recommendations, you may configure VCF Automation to fall back to its default, application intent placement.

Limitations on workload placement
Certain limitations apply when using VCF Operations to place workloads.

- VCF Operations does not support workload placement on resource pools in vCenter Server.
- If VCF Operations is down, the timeout used for workload placement to call VCF Operations might expire.
- Placement doesn't cross multiple cloud zones. VCF Automation sends one cloud zone to VCF Operations for placement recommendations within that single cloud zone.

How to enable workload placement
To enable workload placement, there are steps to take for vSphere, VCF Operations, and VCF Automation.

1. In VCF Automation, connect to your vCenter cloud account.
   The options are under **Infrastructure** > **Connections** > **Cloud Accounts**.
2. In vCenter, verify that DRS enabled clusters exist and are set to fully automated.
3. In VCF Operations, verify that the same vCenter is being managed.
4. In VCF Automation, add the VCF Operations integration.
   The options are under **Infrastructure** > **Connections** > **Integrations**.

To add the integration, you need the VCF Operations primary node URL below, plus the login username and password.
https://*operations-manager-IP-address-or-FQDN*/suite-api
After entering the values, click VALIDATE.

5. Synchronize the integration to the vCenter by clicking SYNC.
   Also synchronize any time that VCF Automation and VCF Operations begin managing a new vCenter.
6. In VCF Automation, create a cloud zone for the vCenter account.
   The options are under **Infrastructure** > **Configure** > **Cloud Zones**.
7. Under the cloud zone Summary tab, set the Placement Policy to ADVANCED.
8. Under the Placement Policy, select whether to have VCF Automation fall back to its default placement if VCF Operations returns no recommendations.

Troubleshooting workload placement

If VCF Operations isn't recommending workload placements the way that you expect, review the deployment request details in VCF Automation or VCF Automation.

1. Go to **Infrastructure** > **Activity** > **Requests**, and click the request.
2. In Request Details, look at the allocation phases.
   Look for targets that were successfully or unsuccessfully identified.
3. In Request Details, at the upper right, enable Dev Mode.
4. Follow the request path to locate filter blocks.
5. Click a filter block, and review the following section.

```
filterName: ComputePlacementPolicyAffinityHostFilter
  ? computeLinksBefore
  ? computeLinksAfter
  ? filteredOutHostsReasons
```

Entry	Description
computeLinksBefore	List of potential placement hosts based on VCF Automation algorithms.
computeLinksAfter	Selected placement host.
filteredOutHostsReasons	Messages describing why a host was selected or rejected. When VCF Operations selects the host, the following message appears. `advance policy filter: Filtered hosts based on recommendation from vROPS.`

Learn more about workload placement

To find the best infrastructure on which to place a deployment, VCF Automation makes several filtering decisions. VCF Automation integration with VCF Operations may further refine the placement decision.

VCF Operations can help to optimally place workloads provided that you have enabled the Advanced placement policy option in your vSphere based cloud zones.

In addition, the vSphere cloud accounts for the cloud zones must be monitored by VCF Operations.

Phase 1: Reservation

The VCF Automation reservation phase is the same whether or not you enable Advanced placement with VCF Operations.

1. Reservation starts with a cloud template linked to a project. That project is in turn linked to cloud zones.
2. The cloud zones consist of compute resource hosts, pools, and clusters, and attached storage.
   Initially, any cloud zone in the project may be a potential placement target.
3. VCF Automation filters out cloud zones that don't have enough healthy resources for the deployment.
   For example, if too many resources are powered off or in maintenance, that cloud zone is filtered out.
4. VCF Automation filters out cloud zones that can't meet business requirements.

For example, the deployment might exceed a pricing or budget limit for the zone.

5. VCF Automation filters out cloud zones that can't meet cluster requirements.
   For example, the cloud zone resources might have CPU or memory usage limits that are too low for the deployment.
6. VCF Automation filters out cloud zones that have no affinity with application constraints.
   Affinity requires that cloud template or project-level constraint tags match capability tags found somewhere in the cloud zone resources.
   For example, if the cloud template or project includes a storage constraint to use storage tagged `pci`, a cloud zone where none of the storage resources have that capability tag would be filtered out.
7. VCF Automation selects cloud zones with the best provisioning priority.
8. If the project-level placement policy is something other than Default, VCF Automation selects a cloud zone that supports the non-default placement policy.
   In this release, Spread is the only non-default. Spread distributes the load by selecting the cloud zone with the lower ratio of virtual machines to hosts. Default simply deploys to the first available zone.
   The project placement policy is only a factor during the cloud zone reservation phase. It has no effect on, nor relation to, the cloud zone placement policy in the allocation phase.

When finished, the reservation phase selects one cloud zone and its resources. VCF Automation reserves the first available zone that remains qualified after passing the preceding filters.

Phase 2: Allocation

VCF Automation inspects the reserved cloud zone compute resources and linked storage.

1. Within the cloud zone, VCF Automation filters out resources that are in a maintenance or powered-off state.
   Note that there are still enough healthy resources for the deployment. Otherwise, the entire cloud zone would have been filtered out during the reservation phase.
2. VCF Automation filters out resources that don't match cluster-level constraints found in the cloud template or project.
   For example, a resource in the cloud zone might be tagged `test` under **Infrastructure** > **Resources** > **Compute**.
   If the cloud template or project includes a constraint tag to use a `dev` resource, the `test` resource is filtered out.
   In addition, storage or network profiles in the cloud zone might be tagged in ways that don't match cluster-level storage or network constraints in the cloud template or project.
3. VCF Automation filters out resources based on affinity settings that are defined in vCenter.
   For example, there might be a rule in vCenter where the presence of a virtual machine in one cluster might block another cluster from being used.
4. VCF Automation filters out resources that don't match any remaining custom constraints found in the cloud template or project.
   For example, if the cloud template includes a constraint to use a `ubuntu` tagged image, a cloud zone where none of the image mappings are tagged `ubuntu` would be filtered out.
5. VCF Automation looks for the best possible compute and storage according to the cloud zone placement policy.
   VCF Automation engages VCF Operations only when the following two conditions are true:
   – The cloud zone placement policy is set to Advanced.
   – After filtering through step 4, at least one DRS enabled cluster and the storage linked to it remain qualified.
   Otherwise, VCF Automation proceeds with its own placement algorithm without input from VCF Operations.

VCF Operations placement recommendation

If qualified for input from VCF Operations, VCF Automation contacts VCF Operations for a recommendation of the best possible compute and storage for the deployment. VCF Automation sends the following data to VCF Operations:

- The qualified target DRS enabled clusters and their attached datastores or datastore cluster
- The resource count or cluster size of the deployment
- CPU and memory requirements for the virtual machines in the deployment
- Disk requirements for the virtual machines in the deployment

From the qualified targets, if VCF Operations can return an optimal placement for each of the virtual machines, VCF Automation allocates compute and storage according to the VCF Operations recommendation.

If VCF Operations couldn't find a recommendation, or VCF Automation couldn't find any DRS enabled cluster and storage, VCF Automation checks the fallback setting of the cloud zone:

- With Fallback
  VCF Automation allocates compute and storage that remains qualified even without a VCF Operations recommendation.
- Without Fallback
  VCF Automation cancels the request and does not proceed with provisioning.

Phase 3: Provisioning

VCF Automation deploys the requested virtual machines, storage, and network through the adapter for the placement target selected at the end of the allocation phase.

The placement target consists of compute hosts, clusters, or resource pools, and attached storage datastore or datastore cluster.

Continuous optimization using VCF Operations

When you add the VCF Automation adapter in VCF Operations, VCF Operations automatically creates a new custom datacenter (CDC) for VCF Automation based workloads.

With continuous optimization, you take advantage of workload rebalancing and relocation, and use VCF Automation with VCF Operations beyond initial workload placement. As virtualization resources move or come under heavier or lighter load, VCF Automation provisioned workloads can move as needed.

- Continuous optimization automatically creates a new CDC in VCF Operations.
  There is one new CDC for each VCF AutomationvSphere cloud zone.
- The newly created CDC contains every VCF Automation managed cluster associated with the cloud zone.

  Note:
  Do not manually create a mixed CDC of VCF Automation and non- VCF Automation clusters.
- You use VCF Operations to run continuous optimization for the newly created VCF Automation based CDC.
- Workloads can only be rebalanced or relocated within the same cloud zone or CDC.
- Optimization never creates a new VCF Automation or VCF Operations placement violation.
  - If you have existing placement violations, optimization can fix VCF Operations operational intent issues.
  - If you have existing placement violations, optimization cannot fix VCF Operations business intent issues.
    For example, if you used VCF Operations to manually move a virtual machine to a cluster that doesn't support your constraints, VCF Operations doesn't detect a violation nor try to resolve it.
- This release obeys operational intent at the CDC level. All member VCF Automation clusters are optimized to the same settings.
  To set a different operational intent for clusters, you must configure them in separate VCF Automation CDCs, associated with separate vSphere cloud zones. Having different test and production clusters might be one example situation.
- VCF Automation application intent and the constraints defined in VCF Automation are honored during any optimization rebalance or relocation operations.
- VCF Operations placement tags cannot be applied to VCF Automation provisioned workloads.

In addition, scheduled optimization involving multiple machines is supported. Regularly scheduled optimizations are not all-or-nothing processes. If conditions interrupt machine movement, successfully relocated machines stay relocated, and the next VCF Operations cycle attempts to relocate the remainder as is usual for VCF Operations. Such a partially completed optimization causes no negative effect in VCF Automation.

How to enable continuous optimization

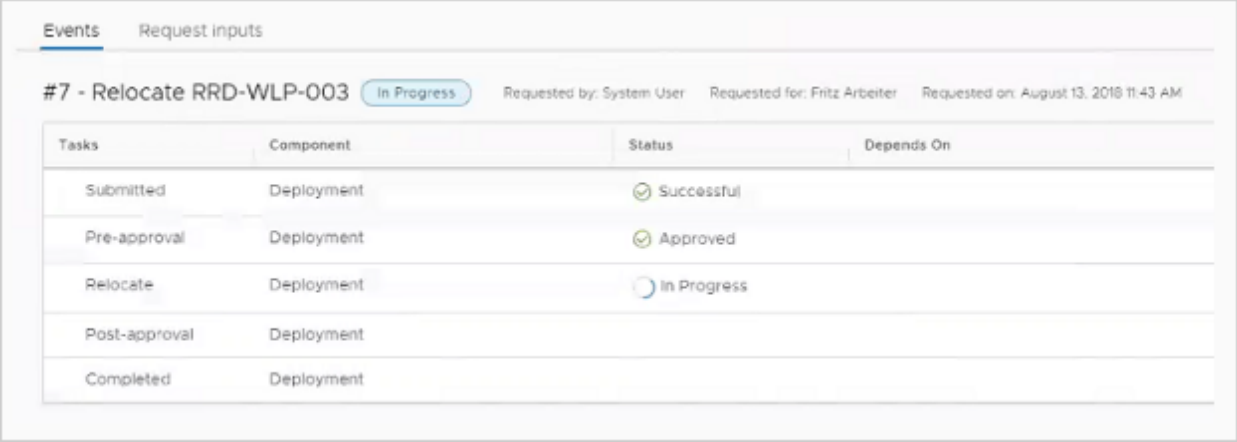
When you add the VCF Automation adapter in VCF Operations, VCF Operations automatically creates a new, dedicated datacenter for VCF Automation based workloads.

Other than adding the integration within VCF Automation, there are no separate installation steps for continuous optimization. You may begin configuring and using VCF Operations for workload relocation in the new datacenter. See the Continuous optimization example .

Continuous optimization example

The following example shows a rebalancing workflow for VCF Automation continuous optimization with VCF Operations.

1. From the VCF Operations home page, click **Workload Optimization**.
2. Select the automatically created VCF Automation datacenter.
3. Under **Operational Intent**, click **Edit**, and select **Balance**.
   You cannot select or edit Business Intent, which is disabled when the datacenter is for VCF Automation optimization.
4. Under **Optimization Recommendation**, click **Optimize Now**.
   VCF Operations displays a before-and-after diagram of the proposed operation.
5. Click **Next**.
6. Click **Begin Action**.
7. In VCF Automation, monitor the operation in progress by clicking **Resources** > **Deployments** and looking at event status.



When rebalancing finishes, VCF Automation refreshes. The Compute Resources page shows that machines have moved.

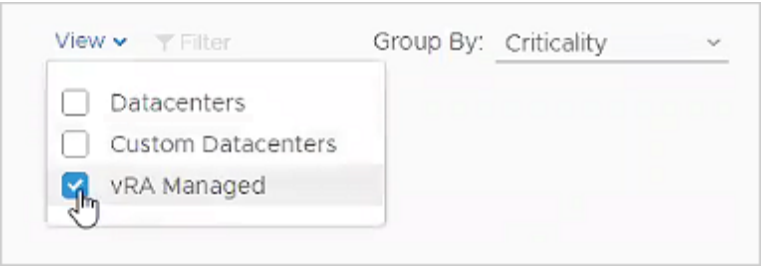
In VCF Operations, the next data collection refreshes the display to show that optimization is complete.

In VCF Operations, you can review the operation by clicking **Administration** > **History** > **Recent Tasks**.

Locate VCF Automation managed datacenters

You can use VCF Operations to display only the VCF Automation managed datacenters.

1. From the VCF Operations home page, click **Workload Optimization**.
2. Near the top right, click the **View** drop-down.
3. Select only the VCF Automation managed datacenters.



Deployment monitoring based on VCF Operations

VCF Automation can show VCF Operations data about your deployments.

Reviewing the filtered set of metrics directly in VCF Automation saves you the task of accessing or searching VCF Operations. Although you cannot launch in context to VCF Operations, you are of course free to log in and use VCF Operations for additional data as needed.

Enable VCF Operations data

For VCF Automation to show VCF Operations data, specific integrations must be present. The integrations require you to supply the address and login credentials for VCF Automation, VCF Operations, and vCenter.

1. In VCF Operations, go to **Data Sources** > **Integrations**, and verify or add your vCenter account integration.
2. In VCF Automation, go to **Infrastructure** > **Connections** > **Cloud Accounts**, and verify or add your vCenter account.
   VCF Operations and VCF Automation must be connected to the same vCenter.
3. In VCF Automation, identify or create a local user with proper permissions, for example, a local user with the Organization Owner role.
4. In VCF Operations, go to **Data Sources** > **Integrations**, and add the VCF Automation for VM Apps organization.
   a) Leave the **VCF Automation Org ID** text box empty.
   b) Add the credentials of the local user account that you created in step 3.
5. In VCF Automation, go to **Infrastructure** > **Connections** > **Integrations**, and add the VCF Operations integration.

   Enter the VCF Operations address in the following form:

   https://*operations-manager-IP-address-or-FQDN*/suite-api

In VCF Automation, click **Resources** > **Deployments**, select a deployment on your vCenter, and verify that the Monitor tab appears.

Health and alerts provided by VCF Operations

When monitoring is enabled, VCF Automation retrieves VCF Operations Health and associated alerts about your deployments.

To access monitoring, click a deployment and select the **Monitor** tab. If the tab is missing, see Enable VCF Operations data.

To see alerts, highlight the deployment name at the top of the component tree in the left panel.

- You can review the severity and text of the alerts.
- To focus on areas of concern, filter and sort on data in the columns.
- Only Health badges and Health alerts appear. Other alert types such as Efficiency or Risk are not supported.

Metrics provided by VCF Operations

When monitoring is enabled, VCF Automation retrieves VCF Operations metrics about your deployments.

To access monitoring, click a deployment and select the **Monitor** tab. If the tab is missing, see Enable VCF Operations data.

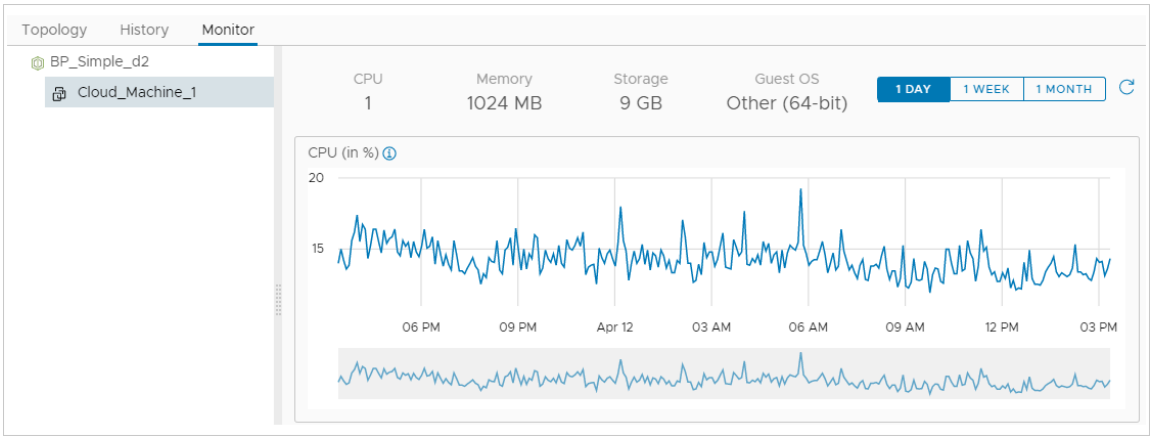
To see metrics, expand the component tree on the left, and highlight a virtual machine.

- Metrics are not cached. They come directly from VCF Operations and might take a few moments to load.
- Only virtual machine metrics appear. Metrics from other components such as vCloud Director, Software, or XaaS are not supported.
- Only vSphere virtual machine metrics appear. Other cloud providers such as AWS or Azure are not supported.

Metrics appear as timeline graphs that show highs and lows for the following measures.

- CPU
- Memory
- Storage IOPS
- Network MBPS

To reveal the specific metric name, click the blue information icon at the upper left corner of the timeline.

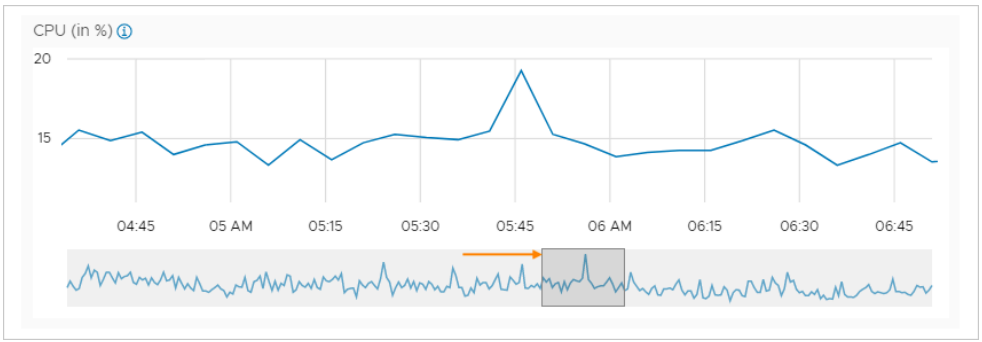


Acting on data provided by VCF Operations

When metrics provided by VCF Operations expose a problem, you can identify trouble areas directly in VCF Automation.

To see metrics provided by VCF Operations, click a deployment and select the **Monitor** tab. If the tab is missing, see Enable VCF Operations data.

Metrics for the past day, week, or month are available. To zoom in on an area of concern, select a small area in the lower, shaded part under any metric timeline:



Resource management and deployment optimization using VCF Operations metrics in VCF Automation

In an integrated VCF Automation and VCF Operations environment, you can access insights and alerts for VCF Automation objects that are monitored by VCF Operations.

The **Insights** dashboard and **Alerts** tab pages provide the real-time capacity and related awareness information that you need to make management decisions in VCF Automation without needing to open VCF Operations. The information is supplied by the associated VCF Operations application.

Working with the insights dashboard and with resource alerts

The **Insights** dashboard conveys information about capacity consumption across all computes within the cloud zone and grouped by projects. It can also show project deployments that are in need of optimization.

The **Alerts** pages displays potential capacity and performance concerns for objects such as cloud zones, projects, deployments, and virtual machines. They also contain information for project owners as to which of their deployments can be optimized. Each deployment link opens the **Optimize** tab in the deployment, where specific guidance is provided.

The following diagram illustrates the relationship between VCF Automation resources and deployments, and the data that the associated VCF Operations application provides in VCF Automation.

Working with the insights dashboard

The **Insights** dashboard, which is available on each cloud zone page, provides the following VCF Operations metrics:

- CPU, memory, and storage utilization usage as a percentage of capacity

- Capability consumption summary
- CPU and memory demand and usage history
- Consumption across projects
- Reclaimable resource capacity, with cost savings, for deployments and projects in a cloud zone

It also provides an option to alert project owners of deployments that can be optimized.

The **Insights** dashboard is available for vSphere and VMware Cloud on AWS cloud zones, provided that the cloud accounts are configured in both VCF Automation and VCF Operations and are being monitored in VCF Operations.

Working with alerts

The **Alerts** pages provide the following filtering categories. Filtering categories are supplied by the associated VCF Operations application.

- Severity
- Status
- Impact
- Type
- Subtype
- Resource

Each filter can be further refined using quick filters. For example, the resource filter can be further refined by its quick filter types of cloud zone, virtual machine, deployment, and project resource.

You use combinations of filters and quick filters to control which alerts are available for display.

Some **Alerts** provide information about, and a link to, deployments that can be optimized. An individual alert can provide the option to contact the project owner, examine an Insights dashboard, or take possible actions.

Alerts are available for vSphere and VMware Cloud on AWS resource objects.

Configuring a Provider-Specific External IPAM Integration for VCF Automation for VM Apps

You can use an external IPAM provider to manage IP address assignments for your cloud template deployments. This tutorial describes how to configure external IPAM integration in VCF Automation using Infoblox as the external IPAM provider.

In this procedure, you use an existing IPAM provider package, in this case an Infoblox package, and an existing running environment to build a provider-specific IPAM integration point. You configure an existing network and create a network profile to support IP address allocation from the external IPAM provider. Finally, you create a cloud template that is matched to the network and network profile and deploy networked machines using IP values obtained from the external IPAM provider.

Information about how to obtain and configure the IPAM provider package, and how to configure a running environment that accesses a cloud extensibility proxy to support the IPAM provider integration, is included as reference.

The values you see in this sample workflow are example values. You won't be able to use them verbatim in your environment. Think about where you would make your own substitutions to fit your organization's needs.

Add required extensible attributes in the Infoblox application for integration with VCF Automation for VM Apps
Before you can download and deploy the Infoblox provider package (infoblox.zip) for integration with VCF Automation for VM Apps from either the Infoblox website or from the VMware Marketplace, you must add required extensibility attributes in Infoblox.

- Verify that you have an account with Infoblox and that you have the correct access credentials to your organization's Infoblox account.

- Confirm that the Infoblox WAPI version is supported. IPAM integration with Infoblox depends on Infoblox WAPI version v2.7. Infoblox appliances that support WAPI v2.7 are supported.
- Review Using Infoblox-specific properties and extensible attributes for IPAM integrations in VMware Aria Automation cloud templates.

This procedure is applicable if you are creating an external IPAM integration point for Infoblox integration with VCF Automation.

Before you can use the `infoblox.zip` download, you must log in to your Infoblox account, using your organization account administrator credentials, and pre-create the following Infoblox extensible attributes:

- `VMware NIC index`
- `VMware resource ID`

1. Log in to your Infoblox account using administrator credentials.x

   These are the same administrator user name and password credentials that you specify when you create an external IPAM integration point in VCF Automation using the **Infrastructure** > **Connections** > **Integrations** menu sequence.

2. Use the procedure described in the Infoblox documentation to create the following required extensible attributes in your Infoblox application.

   - `VMware NIC index`- type Integer
   - **VMware resource ID**- type String

   The procedure is described in the *Adding Extensible Attributes* section of the Infoblox documentation topic About Extensible Attributes. Also see Managing Extensible Attributes.

After you add the required attributes, you can resume the process of downloading and deploying the Infloblox package as described in Download and deploy an external IPAM provider package for use in VCF Automation.

Download and deploy an external IPAM provider package for use in VCF Automation for VM Apps
Before you can define an external IPAM integration point in VCF Automation for VM Apps, you need a configured IPAM provider package.

- Verify that you have cloud administrator credentials. See Credentials required for working with cloud accounts in VCF Automation for VM Apps.
- Verify that you have the cloud administrator user role. See What are the VCF Automation for VM Apps user roles.
- Verify that you have an account with the external IPAM provider, for example Infoblox or Bluecat, and that you have the correct access credentials to your organization's account with the IPAM provider.
- If you are using Infobox as your external IPAM provider, verify that you have added the required extensible attributes to your Infoblox account before continuing. See Add required extensible attributes in the Infoblox application for integration with VCF Automation for VM Apps.

  Note:

  A certificate chain issue exists relative to how the Python element in the Infoblox plug-in handles SSL handshakes. For information about the issue and required actions to resolve the issue, see Knowledge Base Article vRA Cloud Infoblox Plugin throws a certificate chain error during authentication process (88057).

You can download a provider-specific integration package from your IPAM provider's website or the VMware Marketplace.

Note:

This example uses the VMware-supplied Infoblox packageInfoblox.zip, which is available for download from VMware Marketplace as follows:

- Infoblox plugin version 1.5- (**Aria Automation Infoblox Plugin 1.5**) Compatible with VCF Automation and vRealize Automation 8.9.1 and later and containing all the functionality of previous versions. In this version, a local Infoblox property overrides a global Infoblox property. This allows theInfoblox.IPAM.Network.dnsSuffixproperty to be configurable for multiple NICs. For related information, see Using Infoblox-specific properties and extensible attributes for IPAM integrations in VMware Aria Automation cloud templates.

The Infoblox version 1.5 plug-in is supported for use withVMware Aria Automationand vRealize Automation 8.9.1 and later. It is not supported for use with vRealize Automation 8.9 or earlier.

- Infoblox plug-in version 1.4- (**vRA Cloud Infoblox Plugin 1.4**) Compatible withVMware Aria Automationand vRealize Automation 8.3.x forward, and providing all the functionality of previous versions. With this version, you can use the same host name with a different DNS suffix for two NICs. See plug-in release notes for additional details.
- Infoblox plug-in version 1.3- Compatible withVMware Aria Automationand vRealize Automation 8.3.x and providing additional network data collection filters. See Control network data collection by using Infoblox filters in VMware Aria Automation.

  The Infoblox v1.3 plug-in may be used with vRealize Automation 8.1 or 8.2, but only in select situations and with caution as described in KB article Infoblox 1.3 Compatibility with vRealize Automation 8.x (82142).
- vRA Cloud Infoblox plugin version 1.2- Compatible with vRealize Automation 8.1.x and 8.2.x
- vRA Cloud Infoblox plugin version 1.1- Compatible with vRealize Automation 8.1.x
- vRA Cloud Infoblox plugin version 1.0- Compatible with vRealize Automation 8.0.1.x with or without an internet connection to the global network.
- vRA Cloud Infoblox plugin version 0.4- Compatible with vRealize Automation 8.0.0.x and 8.0.1.x when there is an internet connection with the global network.

This example uses the VMware-supplied Infoblox packageInfoblox.zip, which is available for download from the VMware Marketplaceas follows. While all packages are compatible with earlier versions, VMware recommends that you use the latest Infoblox package for the currentVMware Aria Automationversions. You can also search on the phrase **Aria Automation Infoblox Plugin** in VMware Marketplace, for example **Aria Automation Infoblox Plugin 1.5**.

- Infoblox plugin version 1.5- (**Aria Automation Infoblox Plugin 1.5**) Compatible with the currentVMware Aria Automationiteration and containing all the functionality of previous versions. In this version, a local Infoblox property overrides a global Infoblox property. This allows the Infoblox.IPAM.Network.dnsSuffix property to be configurable for multiple NICs. For related information, see Using Infoblox-specific properties and extensible attributes for IPAM integrations in VMware Aria Automation cloud templates.
- Infoblox plug-in version 1.4- (**vRA Cloud Infoblox Plugin 1.4**) Compatible with the current product version and providing all the functionality of previous versions. With this version, you can use the same host name with a different DNS suffix for two NICs. See plug-in release notes for details.
- Infoblox plug-in version 1.3- Compatible with the current product version and providing additional network data collection filters. See Control network data collection by using Infoblox filters in VMware Aria Automation.
- Infoblox plugin version 1.2
- Infoblox plugin version 1.1
- Infoblox plugin version 1.0
- Infoblox plugin version 0.4

IPAM integration with Infoblox depends on Infoblox WAPI version v2.7. All Infoblox appliances that support WAPI v2.7 are supported.

For information about how to create an IPAM integration package for other IPAM providers, if one does not already exist in the VMware Marketplace, see How do I use the IPAM SDK to create a provider-specific external IPAM integration package for VMware Aria Automation.The IPAM provider package contains scripts that are packaged with metadata and other configurations. The scripts contain the source code used for the operations thatVMware Aria Automationperforms in coordination with the external IPAM provider. Example operations includeAllocate an IP address for a virtual machine,Fetch a list of IP ranges from the provider, andUpdate the MAC address of a host record in the provider.

1. Navigate to the correct download page for the Infoblox plug. See above for links to a specific Infoblox plug-in version.

   See above for the Infoblox plugin options that are available in the VMware Marketplace.
2. Log in and download the plug-in package.If you have not already done so, add the required extensible attributes in Infoblox. See Add required extensible attributes in the Infoblox application for integration with VMware Aria Automation.

The package is now available for you to deploy by using the **IntegrationsAdd IntegrationIPAMManage ProvidersImport package**menu sequence as described in Add an external IPAM integration for Infoblox in VMware Aria Automation.

Send email notifications to users in VCF Automation for VM Apps

As a cloud administrator, you can configure VCF Automation to send users notifications when specific events in VCF Automation and VCF Automation occur.

- Verify that you configured an outbound email server. See Add an email server in VCF Automation to send notifications.

You can send notifications for several types of events, called scenarios, such as the successful completion of a catalog request or a required approval.

Email messages are sent to users in the following scenarios.

Scenario	Description
Deployment Lease Deletion	A deployment lease expired and the deployment is scheduled for deletion. The message is sent to the deployment owner 15–30 minutes before the deployment is destroyed.
Deployment Lease Expired	A deployment lease expired and the deployment is about to be deleted.
Deployment Lease Expiring	A deployment lease expires soon. The message is sent to the deployment owner three days before the lease expires.
Deployment Request Approved	A request is approved. The message is sent to the user who requested the deployment.
Deployment Request Rejected	A request is rejected. The message is sent to the user who requested the deployment.
Deployment Request Waiting for Approval (Notification sent to requester)	A request awaits approval. The message is sent to the user who requested the deployment.
Pending Approval Request (Notification sent to approver)	A request requires approval. The message is sent to the user who must approve the request.
Pending Work Item (Notification sent to assignees)	A request requires user input to proceed. The message is sent to the user who must provide input.
Pending Work Item (Notification sent to requester)	A request requires user input to proceed. The message is sent to the user who requested the deployment.
Work Item Answered	A request that required user input was answered and the deployment proceeded.
Work Item Rejected	A request that required user input was rejected and the deployment did not proceed.

1. Log in to VCF Automation as an administrator.
2. Select **Content and Policies** > **Notifications** > **Scenarios**.
3. Select one or more events to trigger user notifications.

Users are subscribed to the notifications that you enabled.
When configuring notifications in VCF Automation, consider:

- If a user email is changed in Active Directory, it might take at least 15 minutes before the email address is updated in VCF Automation. In the meantime, notifications might be sent to the old email address.
- Notification recipients are limited to 100 users per notification. For Active Directory groups, the limit is 10 groups per notification, where each group consists of a maximum of 500 users. If a group consists of more than 500 users, the notification is only sent to 500 users, and not to the rest of the users.

Add an email server in VCF Automation to send notifications

As a cloud administrator, you configure an email server if you want to send messages to users about events in VCF Automation and VCF Automation. The messages are a courtesy that improves the experience of your consumers.

- Verify that you know the credentials required to configure the email server. You must provide the server name and an email account that you want to be the message sender. If your email server requires authentication, you must also provide the user name and password.

This email server is for outbound messages only.

Email messages are sent to users in the following scenarios.

- A deployment lease expires soon. The message is sent to the deployment owner three days before the lease expires.
- A deployment lease expired and the deployment is about to be deleted. The message is sent to the deployment owner 15–30 minutes before it is destroyed.

1. Select **Content and Policies** > **Notifications** > **Email Server**.
2. Enter the information for each setting.

   If you need assistance on a particular setting, consult the signpost help.
3. To verify the configured settings, click **Test Connection**.
4. To save, click **Create**.

As the administrator, monitor the leases to ensure that the messages are sent to the deployment owners at the correct time.

Customize notifications in VCF Automation

As a cloud administrator, you can customize the email notifications that the users in your organization receive about events in VCF Automation and VCF Automation.

You can change the text and the branding elements of the email, such as the logo, so that it matches the requirements of your organization.

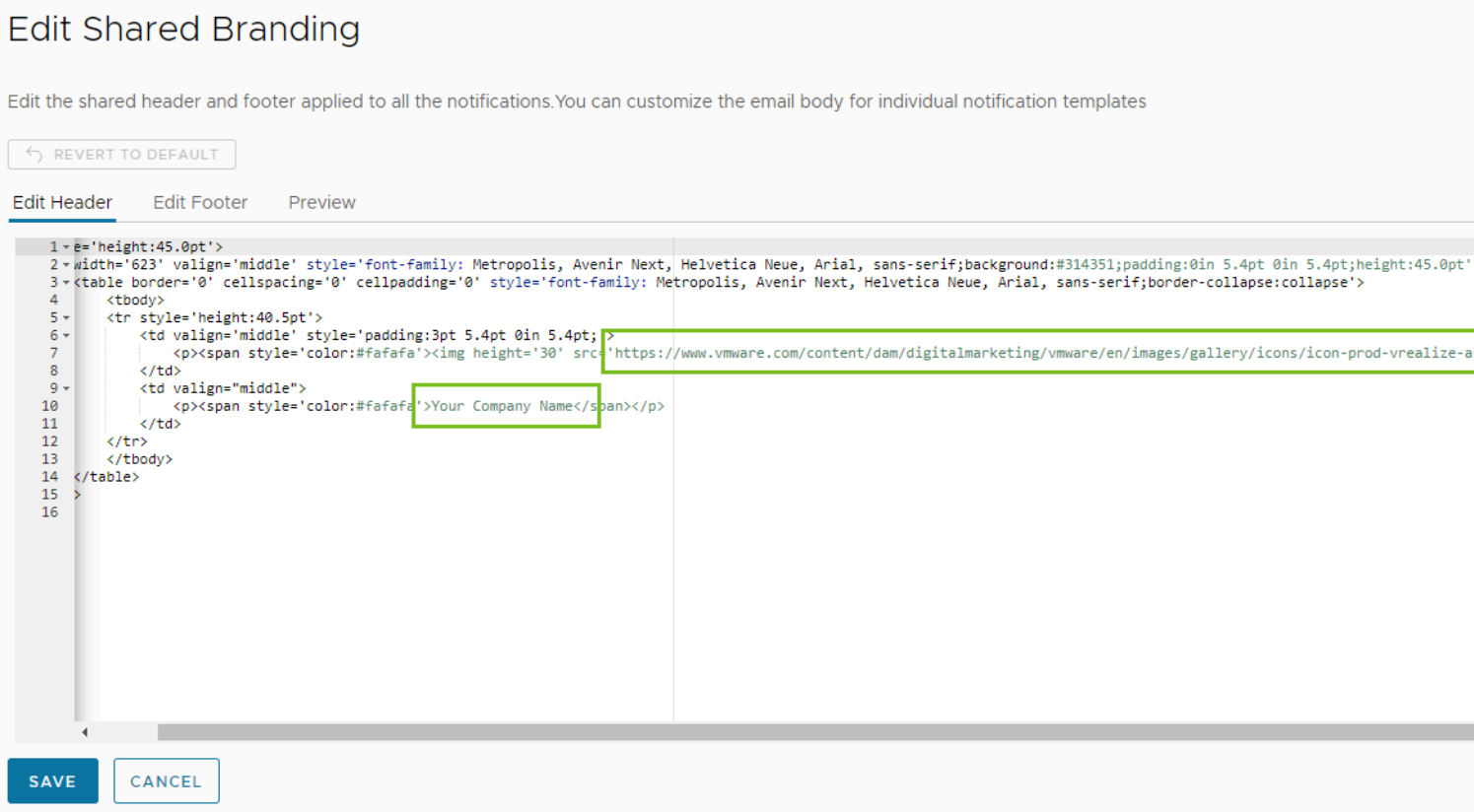
Customize the branding of email notifications

As a cloud administrator, you can customize the way that email notifications look.

You can change the background color, edit the text, and add a new logo in the header and the footer of email notifications. Branding is applied across all notification templates.

In this scenario, you want to replace the default logo and company name in the notifications your users receive with the logo of your company.

1. Select **Content and Policies** > **Notifications** > **Scenarios**.
2. Click **Edit shared branding**.
3. On the **Edit header** tab, add the URL of your logo and the name of your organization.



4. To review you changes, click the **Preview** tab.
5. Click **Save**.

Customize the subject and body of email notifications
You can edit the subject line and the message body of each notification template and use the deployment attributes that are available for the scenario.

As a cloud administrator, you want to craft an email notification that informs your developers when a cloud service is provisioned and provides details about the deployment.

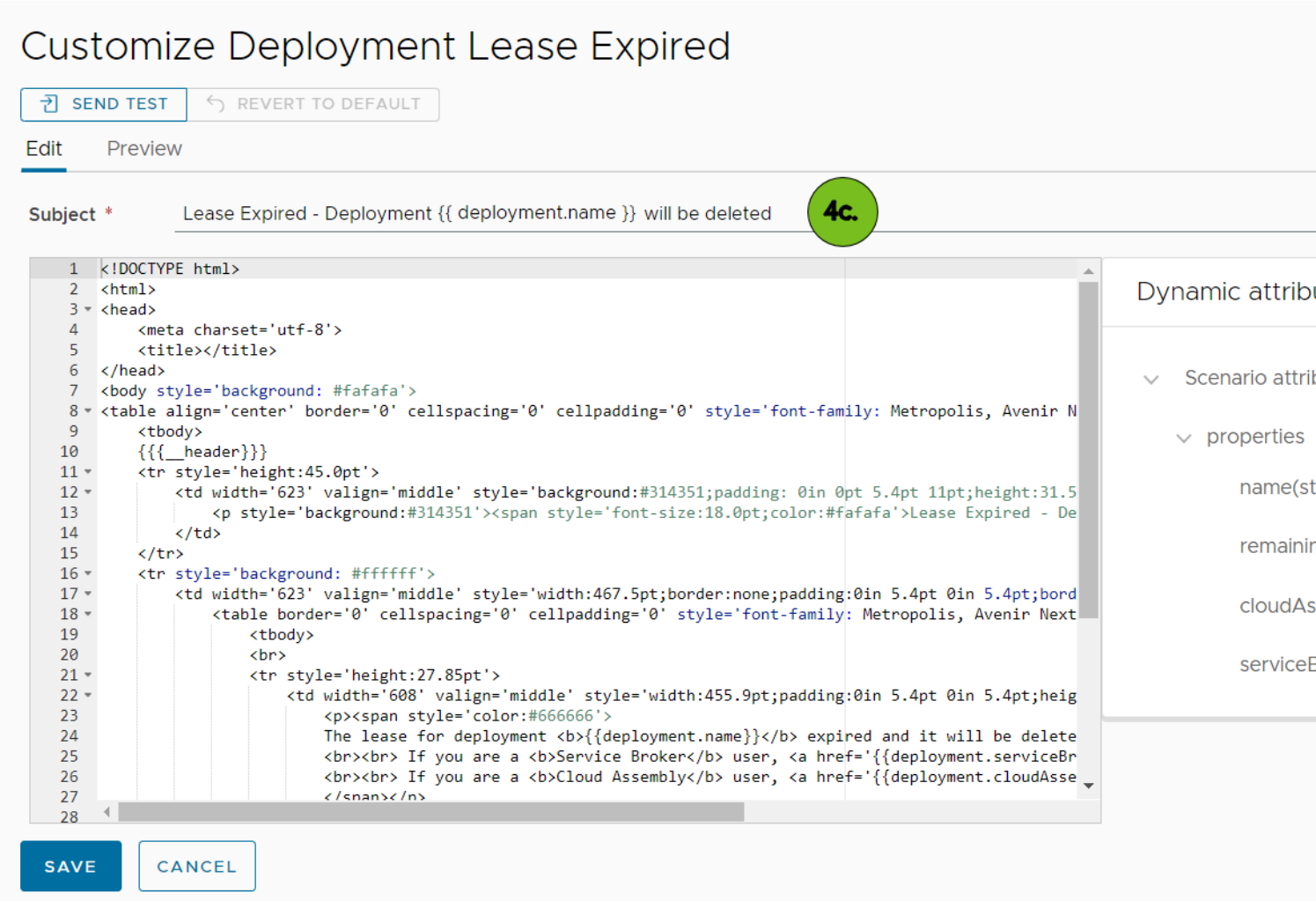
In this scenario, you add the deployment name to the subject of an email notification.

1. Select **Content and Policies** > **Notifications** > **Scenarios**.
2. Click the arrow on the left of the notification scenario you want to customize.
   A preview of the notification email appears on the right.
3. Click **Customize**.
4. On the **Edit** tab, update the subject line and the message body of the notification as necessary.
   a) Click **Scenario attributes** > **properties**.

      The available attributes are specific to each notification scenario.

b) Click the copy icon next to the `name(string)` property.
c) Paste the property in the **Subject** field.

You can also add the dynamic attributes in the message body. Using custom properties is not supported.



5. To review you changes, click the **Preview** tab.
   You see the placeholder text for the deployment name in the subject.
6. Click **Save**.
7. Optional: Test the updated notification.
   a) Click **Send test**.
   b) Enter an email to send the test email to.
   c) Click **Send test**.

Using the ServiceNow ITSM Plug-in for VCF Automation

The ServiceNow ITSM Plug-in for VCF Automation® is an integration between VCF Automation (VMware Cloud Foundation Automation) and the ServiceNow platform.

The ServiceNow ITSM Plug-in for VCF Automation (henceforth referred to as "**Plug-in**") imports the VCF Automation catalog items in the ServiceNow platform. ServiceNow users can use these catalog items to perform various operations in ServiceNow, including the deployment of virtual machine to various other day-2 operations.

> **Important:**
>
> The ServiceNow ITSM Plug-in for VCF Automation is supported only for VM Apps organizations in VCF Automation. You cannot use the plug-in in All Apps organizations.

Installation Prerequisites

Before you install ServiceNow ITSM Plug-in for VCF Automation, make sure that your environment meets the supported compatibility requirements and the MID server is installed.

Compatibility Matrix

The following table lists the version compatibility between the ServiceNow ITSM Plug-in for VCF Automation, VCF Automation, and ServiceNow releases.

Table 1844: ServiceNow ITSM Plug-in for VCF Automation Compatibility Matrix

ServiceNow ITSM Plug-in for VCF Automation	Supported VCF Automation Version	Supported VMware Aria Automation Version	Supported ServiceNow Releases
9.0.1	9.0 VM App Orgs only 9.0.1 VM App Orgs Only	8.18.1	Yokohama and Zurich

Note:

- **Please follow the steps if you are using ServiceNow ITSM Plug-in for VCF Automation on Zurich or newer version.**
- By default, the CMDB schedule job will run only once a day. The schedule period is configurable and you can configure it based on workload size.
- ServiceNow ITSM Plug-in for VCF Automation does not support the ServiceNow on-premises versions.
- ServiceNow ITSM Plug-in for VCF Automation only supports VM Apps organizations.

Installation of MID Server

Before installation of ServiceNow ITSM Plug-in for VCF Automation, you must configure a MID server. The configured MID Server acts as a communication bridge between the ServiceNow platform and the VCF Automation.

Refer to the ServiceNow documentation to install a MID Server.

- For the Zurich release, see ServiceNow Zurich Release Documentation.
- For the Yokohama release, see ServiceNow Yokohama Release Documentation.

Note:

- For better performance, use a dedicated MID Server for the ServiceNow ITSM Plug-in for VCF Automation.
- Please provide `x_vmw_cloudservice.vra_catalog_admin` role to the MID Server user created while installing a MID Server.

MID Server policies setup

ServiceNow has enhanced security with the TLS MID Server certificate check policies. If VCF Automation has an untrusted certificated connection, the connection will be refused. This procedure modifies the TLS MID Server certificate check policies and disables the check for untrusted certificates to connect them insecurely.

For security considerations and detailed information, refer to the MID Server TLS or SSL certificate check policy information.

Configuration of MID Server policies for ServiceNow

This procedure modifies the TLS MID Server certificate check policies and disables the check for untrusted certificates to connect them insecurely. You do not need to perform the steps provided in this section if the endpoint has CA signed certificate.

You must be the system administrator in ServiceNow.

1. Log in to ServiceNow as a system administrator.
2. In the left pane, search for **MID security policy**.
3. In the Intranet record, set the **Certificate Chain Check** to **false**.
4. Register VCF Automation from the Manage Endpoint. See Register ServiceNow ITSM Plug-in for VCF Automation with VCF Automation / Aria Automation.

Assign Script Writer role to Admin role (mandatory step for Zurich and later versions)

ServiceNow has introduced a new Scripting Governance Tool in Zurich release. This feature enabled a better control to restrict write access across all script-like fields. To allow the plugin to write catalog client scripts dynamically, Zurich onwards, ServiceNow ITSM Plug-in for VCF Automation would require Admin role to have "snc_required_script_writer_permission" role assigned.

1. Login with **system administrator**
2. Search "**sys_user_role.LIST**" in filter navigation
3. Search for the record with name "**admin**"
4. Go to the related list "**Contains roles**"
5. Edit to add "**snc_required_script_writer_permission**" role
6. Save

Install the user criteria scoped API plug-in

The User Criteria Scoped API plug-in enables you to create, modify, or delete user criteria records using scripts. The User Criteria Scoped API plug-in is a paid ServiceNow plug-in and requires a subscription. Contact ServiceNow Support for more information.

You must be the system administrator in ServiceNow.

1. Log in to ServiceNow as the system administrator.
2. Navigate to **System Definition > Plug-ins**.
3. In the **Search** box, enter **User criteria scoped API** to search the plug-in.

   The User Criteria Scoped API plug-in dialog box appears.
4. Click **Install**, and then click **Activate**.

   After the plug-in is installed, the success message appears.

Create a Data Source Choice in ServiceNow

The ServiceNow ITSM Plug-in for VCF Automation uses Identification and Reconcile Engine which requires to have discovery source element for the application.

- You must be the system administrator in ServiceNow.
- Set the application scope to `Global`.

1. Log in to ServiceNow as a system administrator.
2. In the left pane, go to **System Definition > Choice Lists**.
3. Click **New**.
4. Select the **Table** as **Configuration Item [cmdb_ci]**.
5. In the **Element** text box, enter the value as **discovery_source**.
6. In the **Label** and **Value** text boxes, enter **ITSM 8.x**.
7. Click **Submit** to create a data source choice.

Enable Application Access

The ServiceNow ITSM Plug-in for VCF Automation requires access to certain ServiceNow tables to perform operations. The plug-in creates, reads, updates, and deletes the records in these tables using a script.

You must be the system administrator in ServiceNow.

1. Log in to ServiceNow as a system administrator.
2. In the left pane, go to **System Definition** > **Tables**.
3. To provide `question_choice` table access:
   a) In the **Search** box, search for **question_choice** table name.
   b) Click the label name link to open table details.
   c) On the **Application Access** tab, select the **can read**, **can create**, **can update**, and **can delete** check boxes.
   d) Click **Update** to provide permission.
4. To provide `sc_cat_item_category` table access:
   a) In the **Search** box, search for **sc_cat_item_category** table name.
   b) Click the label name link to open table details.
   c) On the **Application Access** tab, select the **can read**, **can create**, **can update**, and **can delete** check boxes.
   d) Click **Update** to provide permission.
5. If by default `item_option_new` table access is not available, then provide table access:
   a) In the **Search** box, search for **item_option_new** table name.
   b) Click the label name link to open table details.
   c) On the **Application Access** tab, select the **can read**, **can create**, **can update**, and **can delete** check boxes.
   d) Click **Update** to provide permission.
6. If by default the `sc_cat_item_user_criteria_mtom` table access is not available, then provide table access:
   a) In the **Search** box, search for **sc_cat_item_user_criteria_mtom** table name.
   b) Click the label name link to open table details.
   c) On the **Application Access** tab, select the **can read**, **can create**, and **can update** check boxes.
   d) Click **Update** to provide permission.

Install and Update the ServiceNow ITSM Plug-in for VCF Automation

The ServiceNow System Administrator can install or update the ServiceNow ITSM Plug-in for VCF Automation from the ServiceNow store.

> **Note:**
>
> The ServiceNow ITSM Plug-in for VCF Automation supports only incremental upgrades of the plug-in. Upgrading the plug-in directly from any of the earlier versions to the latest plug-in version is not recommended. To upgrade the plug-in from the earlier versions to the latest version, you can refer to the desired user guide of the earlier versions.

Installation of ServiceNow ITSM Plug-in for VCF Automation

You can install the ServiceNow ITSM Plug-in for VCF Automation from ServiceNow Store. The installation is free, you must have a valid VCF Automation product license. If you have previously installed this plugin then you can skip this topic.

- You must have access to the ServiceNow Store.
- You must be the system administrator in ServiceNow.
- The **user criteria scoped API plug-in** must be installed. For details, see Install the user criteria scoped API plug-in.
- A data source choice list must be created. See Create a Data Source Choice in ServiceNow.

1. Log in to the ServiceNow Store https://store.servicenow.com/.
2. Search for VCF Automation Plugin.
3. Click the application to view the details. In the top-right corner of the form, click **Manage Entitlements**.
4. Select the ServiceNow instances to install the application. Click **OK**.

   The application is now available on the selected ServiceNow instances for installation.
5. Log in to the ServiceNow instance as a system administrator.
6. In the left pane, click **System Applications > All Available Applications > All**.
7. Search for the VCF Automation Plugin.
8. Click **Install**.
9. Select **Install with demo data**, and complete the installation.

   The application is successfully installed message displays.

   Note:

   Demo data is responsible for importing groups, table records, schedule jobs, and system properties related to an application. Selecting the demo data check box is mandatory. If the check box is not selected, then a new demo installation cannot be completed and affects the application flow.

Update from VMware Aria Automation Plug-in 8.18.1 to VCF Automation Plug-in 9.0

If you have previously installed the VMware Aria Automation Plug-in 8.18.1 version from the ServiceNow store, you can update it to VCF Automation Plug-in 9.0 version in your instance.

- You must be the system administrator in ServiceNow.
- You must deactivate the following scheduled jobs:
  - Import Project
  - Import Catalog
  - Import Resources and CMDB
  - Delete Queue
  - Reconcile Actions
- The **user criteria scoped API plug-in** must be installed. For details, see Install the user criteria scoped API plug-in.
- A data source choice list must be created. See Create a Data Source Choice in ServiceNow.
- You must set the **Replace on upgrade** flag as **True** for the table records having customizations by following the below steps:
  - In the filter Navigator, type the **sys_update_xml.LIST** table name and press Enter.
  - On the **Customer Updates** page, click the gear icon. The **Personalize List Columns** window will display.
  - Under the **Available** column list, add the following to the **Selected** column and click **OK**.
    - Replace on upgrade
    - Application
  - In the **Application** column, search for **VCF Automation Plug-in**. The number of table records will be listed.
  - Place the pointer on the **Type**, and click **vertical ellipsis** > **Group By Type**. A list of records will be listed based on Type.
  - Expand the **System Property** and set the **Replace on upgrade** to **True** for all the system properties.

Note:

- You must set the “**Replace on upgrade**” as **True** for any table records that contain customizations.
- For more information on overwriting your customizations, see Overwrite customizations during an upgrade.
- If you have skipped updating the “**Replace on upgrade**” flag for table records having customizations before upgrading, then the records will fall into Skipped and Skipped Manual Merge. For more information, see Troubleshoot.

1. Log in to ServiceNow as a system administrator.
2. In the left pane, click **System Applications** > **All Available Applications** >**All**.
3. Search for VCF Automation Plugin
4. Select the latest release version and click **Update**.
5. Select the **Load Demo Data** checkbox and click **Update**. The current version will be updated to latest release version.
6. Make sure that all the scheduled jobs are in the Inactive state.
7. Provide the required access for the question choice table. See Enable Application Access.
8. Navigate to **Manage Endpoints** and open each registered endpoint.
9. **Important:**

   Please make sure to change the **Aria / VCFA** dropdown value from **VRA 8.18.X** to **VCFA 9.0** if you are upgrading from VMware Aria Automation to VCF Automation.

   Click the **Validate and Save** button.
10. Run the scheduled jobs:
    a) In the left pane, click **VCF Automation** >**Job Schedules**.
    b) Activate all the scheduled jobs that were deactivated initially.
    c) Run the **Import Project** first then run the **Import Catalog** schedule job.
    d) Run the **Import Resources and CMDB** schedule job.

Configuration of the ServiceNow ITSM Plug-in for VCF Automation

After the ServiceNow ITSM Plug-in for VCF Automation is installed, the system administrator is required to carry out the configuration steps described in this chapter.

Managing the ServiceNow Users

In the ServiceNow ITSM Plug-in for VCF Automation, the system administrator must set up users and groups, create the required user roles, and complete other required configuration tasks.

Note:

It is important to set up the users in both Plug-in and VCF Automation with the same **user_Id/user_name** in ServiceNow as that of VCF Automation.

Adding the ServiceNow User Roles

For the ServiceNow ITSM Plug-in for VCF Automation, the system administrator is required to create two types of users.

The categorization of users is achieved by assigning the required roles to each user.

Users	Description	Required Roles
admin-user	Responsible for all administrative tasks including application configuration, end-point registration, user entitlement management, and so on.	• `catalog_admin` • `agent_admin` • `x_vmw_cloudservice.vra_catalog_admin` • `x_vmw_cloudservice.vra_end_user` • `cmdb_read`
end user	Uses the VCF Automation Service Portal in ServiceNow.	• `x_vmw_cloudservice.vra_end_user` • `cmdb_read`
approver user	Approve or rejects the catalog item requests.	• `x_vmw_cloudservice.vra_end_user` • `cmdb_read` • `approver_user`

1. Log in to ServiceNow as a system administrator.
2. In the left pane, click **System Security > Users and Groups > Users**.
3. Select the user and open the user record.
4. In the **Roles** tab, click **edit**.
5. Add the roles for the users from the **Collection List** and move the role to the **Roles List**.
6. Click **Save** to assign the role.
7. Click **Update** to save the user record.
8. Repeat steps from 2 to 6 for each user to assign roles.

Setting Up ServiceNow Groups

The ServiceNow system administrator can add users to the approval and support group. Users in the approval group are responsible for approving or rejecting the deployment requests made by the users. The incidents created by the failure of the catalog request submissions are assigned to users of the support group.

- You can add the number of users to the approval group by adding users to the **Automation-ApprovalGroup** to approve or reject the requests.
- You can add the number of users to the support group by adding users to the **Automation-SupportGroup** to approve or reject the requests.

> **Note:**
>
> If you do not add any users to the approval group, the requests get auto-approved.

Configuration of VCF Automation Endpoint

You must set up an administrative user in VCF Automation. The ServiceNow ITSM Plug-in for VCF Automation connects to VCF Automation using the credentials of this user.

This user must have these roles in VCF Automation:

- Organization Member
- Project Member
- Service Broker User
- Assembler Administrator

Note:

- All catalog item requests from ServiceNow are serviced by this user in the VCF Automation.
  In the VCF Automation, the requests and corresponding deployments display this user as the owner. However, in the ServiceNow, requests, and corresponding deployments display the ServiceNow user who initiated the request. This is achieved by correlating the email address from the user records across two systems.
- You must add the integration user to each project as a project admin to use the project in ServiceNow ITSM Plug-in for VCF Automation

Create Day 2 Policies

Create a day-2 action policy for the integration user. The integration user must be added to the deployment criteria using the created by condition to limit the policy to integration user. This policy can be used to limit the day-2 actions for the integration user.

To create policies, see the VCF Automation documentation.

Note:

Use the Deployment Criteria option to limit the day-2 actions for the integration user.

Verify Email Address Mapping

The VCF Automation and ServiceNow can have different Authentication Providers. It is important to set up the users in both systems with the same email address.

The correlation is required to assign ownership of the deployments and resources.

Register ServiceNow ITSM Plug-in for VCF Automation with VCF Automation / Aria Automation

To establish the connection between the VCF Automation/ VMware Aria Automation and the ServiceNow ITSM Plug-in for VCF Automation , the ServiceNow system administrator or Plug-in administrator must register the Plugin to a VCF Automation/ VMware Aria Automation instance. To register the ServiceNow ITSM Plug-in for VCF Automation with VCF Automation/VMware Aria Automation, a MID server is required and its status must be Up and Validated.

- You must be the system administrator or Plug-in admin in ServiceNow.
- MID Server must be configured and its status must be Up and Validated.

1. Log in to ServiceNow as a system administrator or Plugin admin user.
2. From the left pane, go to **Manage Endpoints > New** and provide the endpoint details.
3. **Important:**

   Please note that you will be presented with the set of fields depending upon the value provided in Aria / VCFA field. This field lets plugin understand if the endpoint is with VMware Aria Automation or VCF Automation.

   Enter the details for the following fields :

Field	Description
Name	Name of the endpoint.
Aria / VCFA	Option that defines the endpoint as **VRA 8.18.X** or **VCFA 9.0**
Host Url	URL address for the **VCF Automation** or **VMware Aria Automation** appliance.
MID Server	Name of the MID Server that you created for use.
Active	Import jobs run only if the Active box is checked.
Username	User name as configured in the VMware Aria Automation.
Password	User password as configured in the VMware Aria Automation.
Organization Name	Refers to the Tenant Organization name for VCF Automation
Organization Id	Refers to the Tenant Organization ID for VCF Automation

Field	Description
Refresh Key	Refers to the API token generated for the integration user at VCF Automation

4. Click **Save and Import** to register and import the appliance data into the VCF Automation Plugin.

   Note:

   If an endpoint is deactivated and reactivated again, then run the scheduled jobs again to re-import. Else, the user cannot perform any day 2 actions. See Configure and Run Import Jobs.

   While registering the endpoint, the telemetry pop-up will be displayed for consent. For more information, see Telemetry.

   All catalog item requests from ServiceNow are serviced by this user in VCF Automation.

   In VCF Automation, the requests and corresponding deployments display this user as the owner. However, in ServiceNow, requests and corresponding deployments display the ServiceNow user who initiated the request. This is achieved by correlating the email address from the user records across two systems.

Generating API Token for an Integration User on VCF Automation

ServiceNow Plug-in integrates with VCF Automation using API token. This section helps users to generate the API token for an integration user on VCF Automation. The generated API token then can be used to create an endpoint from ServiceNow to VCF Automation.

- Integration user must have the roles mentioned in Configuration of VCF Automation for on-premise Version.
- In case if you are upgrading from 8.18.1 to 9.0, make sure to follow the below steps using the same integration user with which the endpoint was created.

1. Log in to the desired VCF Automation organization using an **integration user.**
2. Navigate to the **User Profile** and select **My Account**.
3. Under **My Account**, click **API Tokens**.
4. Click **New** to generate a new token.
5. Enter a unique **Client Name** and click **Create**.
6. Copy the generated API token and use it to create or update an endpoint in ServiceNow using Register ServiceNow ITSM Plug-in for VCF Automation with VCF Automation / Aria Automation.
   - Make sure to copy the API token immediately after creation. It will no longer be visible once you click the **OK** button.
   - The API Token has a default expiry of 90 days. ServiceNow Plug-in will notify the ServiceNow User about the details of the expiry of an API Token. Please repeat above steps when token expires.
7. Please take a note of the Org ID and Org Name under the **User Profile** section that will be needed while creating an endpoint in ServiceNow.

Configure Event-based data sync in VCF Automation (Optional)

You can optimize the plugin's data import process by configuring event-based data synchronization using the **VCF Operations Orchestrator** interface and creating subscriptions in **VCF Automation**. The event data is migrated from VCF Automation to ServiceNow through **VCF Operations Orchestrator** by configuring it.

- You must have access to the VCF Automation and VCF Operations Orchestrator.
- Required ServiceNow Plug-in 8.13 version or above.
- ServiceNow user being used in **step 4** below needs to have **x_vmw_cloudservice.vra_catalog_admin** role.

1. Login to the **VCF Operations Orchestrator**.
2. Navigate to the **Add a REST Host** workflow.
3. Click **Run** to execute the workflow.
4. Provide the values to the following fields in the workflow page:

Tabs	Fields
Host Properties	**Name**: Choose a descriptive name for the host (example: demo-host).
	URL: Enter the URL of the ServiceNow instance you want to connect to.
Host Authentication	**Host Authentication Type**: Select Basic
User Credential	**Session Mode**: Choose Shared
	User Name: Enter the user name of the ServiceNow user.
	Password: Enter the password of the ServiceNow user.

5. Run the workflow. Wait till the workflow execution is completed successfully.
6. Navigate to the **Add a REST Operation** workflow.
7. Click **Run** to execute the workflow.
8. Provide the values to the following fields in the workflow page:

Fields	Inputs
Parent Host	Select the previously added host (example: "demo-host")
Name	Assign a name to the operation, such as "demo-operation
Template URL	Provide `api/now/import/x_vmw_cloudservice_aria_automation_web_s`
HTTP Method	Choose **POST**
Content	Select **application/json**

9. Run the workflow. Wait till the workflow execution is completed successfully.
10. You must download and import the VCF Operations Orchestrator package, and configure the **Event Based Data Sync for ServiceNow** workflow by following the below steps:
    a) To download the **VCF Operations Orchestrator** package, click here.
    b) To import the package into **VCF Operations Orchestrator**, see Import a Package in the VCF Operations Orchestrator Client.
    c) Open the **Event Based Data Sync for ServiceNow** workflow, which is imported with the VCF Operations Orchestrator package.
    d) Select the **Variables** tab and open the **restOperation** variable.
    e) Provide the entity of the **Add a Rest Operation** workflow created earlier. Refer step **6** to **9**.
    f) Click **Save** and validate the workflow if required.

Configure Event Subscriptions in VCF Automation

Configuring subscriptions in VCF Automation establishes a communication channel from VCF Operations Orchestrator to ServiceNow so that event data received from VCF Automation through VCF Operations Orchestrator will be sent to ServiceNow.

You must have access to **VCF Automation** and **VCF Operations Orchestrator**.

1. Log in to the **VCF Automation/ VMware Aria Automation**.
2. Go to **Assembler** (Not required for VCF Automation).
3. Select the **Extensibility** tab.
4. On the left pane, click **Subscriptions**.
5. Click **NEW SUBSCRIPTION** to create a new subscription.
6. Provide the values to the following fields on the **NEW SUBSCRIPTION** page:

Fields	Inputs
Name	Provide **Deployment Complete**.
Status	Enable statistics if desired (Optional).
Event Topic	Choose **Deployment Complete**.
Workflow	Add the **Event Based Data Sync for ServiceNow** workflow that was configured earlier. See Configure Event-based data sync in VCF Automation.
Project Scope	Enable the **Any Project** toggle button.

7. Click **Save**.
8. Repeat the above steps to create subscriptions for the following events:
   a) Deployment Action Complete.
   b) Deployment Resource Complete
   c) Deployment Resource Action Complete
   d) Provider Resource Action CUD

   Note:

   - After successfully configuring events using the VCF Operations Orchestrator interface and creating subscriptions in VCF Automation, the Event Based Data Sync for ServiceNow workflow will handle the events based on the specified subscriptions, allowing the ServiceNow users to receive the updates from VCF Automation.
   - The plug-in is compatible with event-based data synchronization out of the box hence no configuration steps are required.

Configuring and Using the Service Portal

The Service Portal is available as a default portal for deployment. You can use the Service Portal to view Service Catalog, deployment, and resources. You can also perform various day-2 actions on associated deployments and resources.

Setting Up the Service Portal Prefix

The ServiceNow system administrator must ensure that the Service Portal prefix value in their instance is the same as the prefix value in the `x_vmw_cloudservice.servicePortalPrefix` system property.

You must update the `x_vmw_cloudservice.servicePortalPrefix` system property. For information, refer system properties table in Appendix.

Configuring the Deployment Widget

To view the VCF Automation deployments on the Service Portal, the ServiceNow system administrator must add the VCF Automation deployment widget on a Service Portal page.

- You must be the system administrator in ServiceNow.
- The application scope must be set to Global.
- The ServiceNow Plug-in Service Portal Prefix value must be the same as the Service Portal prefix value on your instance. See Setting Up the Service Portal Prefix.

1. Log in to ServiceNow as a system administrator.
2. In the left pane, click **Service Portal > Service Portal Configuration**.
3. Click **Designer**.

   The Service Portal Designer page opens with a list of all the available pages.
4. Search for **index**, and then open the **Home Page**.
5. Select the required column to insert the widget.
6. On the top right, click **Edit**.
7. Ensure that the minimum width size for the column is as follows:

Screen	Size
xs	12
sm	12
md	6
lg	6

8. Click **Save**.
9. On the left pane, in the **Filter Widgets** text box, search for **CAS_SP_deployments**.
10. Drag and drop the widget to the column.

    If the ServiceNow Plugin is registered and VCF Automation required roles are assigned, the VCF Automation deployments will display. See Adding the ServiceNow User Roles.

    The VCF Automation Deployments widget is displayed on the Service Portal home page with the five most recent deployments with their status.

Add the View and Modify Resource Widget on the Request Summary Page

The **View and Modify Resource** widget adds a button to the **Request Summary** page to allow users to go back to the deployment details page.

- You must be the system administrator in ServiceNow.
- The application scope must be set to Global.

1. Log in to ServiceNow as a system administrator.
2. In the left pane, click **Service Portal > Service Portal Configuration**.
3. Click **Designer**.

   The Service Portal Designer page opens with a list of all the available pages.
4. In **Search**, search for **order_status**.
5. Select and open the **Request Summary** page.
6. Select the bottom column to insert the widget.

   Note:

   If an empty column is not available then, you can insert a new column.
7. On the left pane, in the **Filter Widgets** text box, search for **CAS_SP_returnToDeployment**.
8. Drag and drop the widget to the column.

   The View and modify resources widget is added to the **Request Summary** page.

Configure the Deployment Counts in the Deployment List

The ServiceNow system administrator or the plug-in admin can configure the number of deployments that must display on a single page in the **Deployment List** widget. By default, users can view 5 deployments on one page.

You must update the `x_vmw_cloudservice.deploymentWindowSize` system property to configure the deployment counts. For more information, refer system properties table in Appendix.

Configure the Deployment Counts in the Deployments Widget

By default, users can view 5 deployments in the Deployments widget on Service Portal. The ServiceNow system administrator or the plug-in admin can configure the number of deployment entries that must display in the Deployments widget.

You must update the `x_vmw_cloudservice.DeploymentLimit` system property to configure the deployment counts. For more information, refer system properties table in Appendix.

Using the Service Portal

This chapter describes how to use the Service Portal for the ServiceNow plug-in.

Viewing the Deployments

The Deployment List page displays the list of deployments entitled to the user. Users can view the status of the deployments they own. From the Deployment List page, the user can view the deployment status, expiry date, number of resources associated with deployment, a project associated with the deployment, owner, and so on.

- You must have the required roles assigned to view the Service Portal. See Adding the ServiceNow User Roles.
- The Deployment widget must be added. See Configuring the Deployment Widget.

1. Log in to the ServiceNow portal.
2. From the left pane, go to **Service Portal > Service Portal Home**.
3. On the **Deployments** widget, click **View All**.

   The **Deployment List** page opens with a list of all the available deployments.
4. Click the desired deployment to view the deployment details.

   On the **Deployment List** page, you can search deployment using the search bar only for the following fields:

   - Name
   - Status
   - Description
   - Project
   - Endpoint

   Note:

   For better consistency between requested item comments and the actual deployment status field, you can have one of the following values:

   - CREATE_SUCCESSFUL
   - CREATE_FAILED
   - CREATE_INPROGRESS

   Note:

   The out-of-box business rule “**Assign from Stock**” on the RITM table will set `install_status` of deployments to `pending_install` on the deployments table based on the absence of the ServiceNow plug-in for Hardware Asset Management. For more information, see KB article.

Deployment Actions

You can perform actions that are associated with deployments and are entitled to a user. You can perform one action at a time. If any deployment action is in progress, then a progress bar appears, and all the other day-2 actions are hidden until the in-progress action is complete. The progress bar shows the status of in-progress action.

You must have the required roles to view the VCF Automation catalog items on Service Portal. See Adding the ServiceNow User Roles.

1. Log in to the ServiceNow portal.
2. From the left pane, go to **Service Portal > Service Portal Home**.
3. On the **Deployments** widget, click **View All**.

   The **Deployment List** page opens with a list of all the available deployments.
4. You can perform deployment action using one of the following methods:
   - From the **Deployments List** page: click the vertical ellipsis before the desired deployment name and select the action.
   - From the **Deployment details** page:
     a. Select the desired deployment.
     b. On the deployment header, from the **Actions** drop-down, select the required deployment action.
5. If the approval group is configured, the request will be sent for approval, and the status will be updated to Request Approval.

   Note:

   If the approval group is not configured or the users are not added to the approval group, then the request will get auto-approved. The ServiceNow ITSM Plug-in for VCF Automation submits the request on the VCF Automation instance.

Resource Actions

From the Deployments details page, you can view the details of resources. You can view the resource properties including, custom properties, network information, general properties, owner information, routes, and storage information. You can also perform all the entitled resource actions.

You must have the required roles to view the VCF Automation catalog items on Service Portal. See Adding the ServiceNow User Roles.

1. Log in to the ServiceNow portal.
2. From the left pane, go to **Service Portal > Service Portal Home**.
3. On the **Deployments** widget, click **View All**.

   The **Deployment List** page opens with a list of all the available deployments.
4. Click the desired deployment to view the deployment details.
5. On the **Resources** pane, search and select the desired resource to view the details.
6. Click the **Actions** drop-down button next to the resource name and select the required resource action.

   If the approval group is configured, the request will be sent for approval, and the status will be updated to Request Approval.

   Note:

   If the approval group is not configured or the users are not added to the approval group, then the request will get auto-approved. The ServiceNow ITSM Plug-in for VCF Automation submits the request on the VCF Automation instance.

Configuring and Using the Service Catalog on Native UI

Using the ServiceNow ITSM Plug-in for VCF Automation, you can configure the Service Catalog on the Native UI.

Configuring the Service Catalog on Native UI

The system administrator can create Service Catalog for catalog items to use the Native UI for the ServiceNow ITSM Plug-in for VCF Automation.

You must be the ServiceNow system administrator.

1. Log in to ServiceNow as a system administrator.
2. Go to **Self Service** > **Service Catalog**.
3. Click the plus (+) icon.
4. Search for **VCF Automation Items** category for VCF Automation Catalog items and click **Add here**.
5. Search for **VCF Automation Actions** category for deployment and resource day-2 actions and click **Add here**.

Using the Service Catalog
This chapter describes how to use the Service Catalog for the ServiceNow ITSM Plug-in for VCF Automation.

Viewing and Requesting the Catalog Items

The VCF Automation Items page displays the list of catalog items entitled to the user. User can create a request for catalog items.

You must have the required roles assigned to view the Service Portal. See Adding the ServiceNow User Roles.

1. Log in to ServiceNow.
2. Go to **Self Service > Service Catalog**.
3. Click **VCF Automation Items**.

   The **VCF Automation Items** page opens the catalog items related categories.
4. Click the desired Project to view the catalog items.
5. Select the required catalog item to request.
6. Fill in all mandatory fields and click **Order Now**.

Performing the Actions

From the VCF Automation Actions page, you can perform deployment and machine actions entitled to you.

You must have the required roles assigned to view the Service Portal. See Adding the ServiceNow User Roles.

1. Log in to ServiceNow.
2. Go to **Self Service > Service Catalog**.
3. Click **VCF Automation Actions**.

   The **VCF Automation Actions** page opens the deployments and resource actions related categories.
4. Click **Deployment Action**:
   a) Select the required deployment action.

      The deployment action form appears.
   b) Fill in all mandatory fields and click **Order Now**.
5. Click **Resource Action**:
   a) Select the required resource action.

      The resource action form appears.
   b) Fill in all mandatory fields and click **Order Now**.

      Note:

      For best practice, it is recommended to perform only one action at a time.

Create the User Criteria

The user criteria control the accessibility of categories and catalog items in the Service Catalog. The ServiceNow users and end-users require user criteria to access projects/catalog items. The ServiceNow system administrator can create user criteria in the VCF Automation Plugin

- You must create entitlement using the user criteria. To create the user criteria records in the Service Catalog, see Create User Criteria Record.
- To assign the user criteria to catalog items and projects in the Service Catalog, see Apply User Criteria to Items and Categories.
- The restrictions for user criteria to view individual projects/catalog items are directly applied by adding items in **Available For** or **Not Available For** related lists in the User Criteria form.
- The system administrator must provide access to projects/ catalog items for a specific user by following the below steps:

1. Create **User Criteria** in the Service Catalog.
2. Add **Groups** and **Users** in User Criteria.
3. Add the user criteria in **Available For** if a specific user requires to access a specific project/catalog item.
4. Add the user criteria in **Not Available For** if the specific user does not want to access a specific project/catalog item.

> **Note:**
>
> The user cannot access the catalog item when the user is on both **Available For** and **Not Available For** lists because the **Not Available For** settings will override **Available For** settings.

Advanced Configuration

The ServiceNow system administrators can configure the additional features to enhance the ServiceNow ITSM Plug-in for VCF Automation.

Configure and Run Import Jobs

The Configure and Run Import Jobs section describes the jobs that are automatically scheduled at the time of installation. You can update the default schedule as required. You can also run these jobs manually (on-demand).

However, exercise caution while changing the default configuration or manually running the jobs as data corruption or loss can occur.

> **Note:**
>
> The Scheduled Jobs and System logs functionality will only be visible to the ServiceNow system administrator.

> **CAUTION:**
>
> Import jobs must be configured and run in the order they appear in the following table. Manual (on-demand) execution of these jobs must also follow this order. Failure to follow this order can cause the data corruption or loss.

This table describes the import jobs.

Scheduled Import Jobs	Description
`Generate API Token`	Generates an authentication token when VCF Automation is registered. This job runs automatically and updates the API token.
`Import Project`	Imports project from VCF Automation into ServiceNow as projects.
`Import Catalog`	Imports catalog items from VCF Automation into ServiceNow as catalog items.

Scheduled Import Jobs	Description
`Import Resources and CMDB`	Imports deployments and resources from VCF Automation into ServiceNow so that users can view the resources they own on the portal. By default, the job execution time is set to 23.59.00 PDT at midnight. The system administrator can change the execution time as per their requirement. VMware recommends that you must not run this job frequently and always run it during off-peak hours.
`Delete Queue`	Deletes the scheduled import queues that are older than the interval specified.
`Reconcile Actions`	While importing the day-2 actions from the VCF Automation services, if any day-2 actions are removed from VCF Automation services, then they will be removed from the ServiceNow.
`Thread Limit`	Runs every second to process a limited number of topics from the Job Queue table. Note that the user need not activate, deactivate, or run this job manually. This job is automated to activate or deactivate based on certain conditions.
`Thread Limit Maximum`	By default, the scheduled job runs every Friday at midnight and can be configured based on requirements. This scheduled job will maximize the VCF Automation thread limit to the default value of 16 (value set from the system property `x_vmw_cloudservice.ThreadUsed_Maximum`) considering less load on the system during weekends. This allows the customer to avoid configuring the VCF Automation thread system property manually.
`Thread Limit Minimum`	By default, the scheduled job runs every Sunday at midnight and can be configured based on requirements. This scheduled job will minimize the VCF Automation thread limit to the default value of 8 (value set from the system property `x_vmw_cloudservice.ThreadUsed_Minimum`) when the system is under normal load. This allows the customer to avoid configuring the VCF Automation thread system property manually.
`Telemetry Data Push`	This scheduled job will run only when the user has provided their consent as **Yes** in the telemetry pop-up and the "`x_vmw_cloudservice.customer_consent_value`" system property value is `true`. The scheduled job will import the telemetry data from ServiceNow ITSM Plug-in for VCF Automation. By default, this scheduled job runs every midnight.

Email and Ticketing Properties

The ServiceNow system administrator or plug-in admin can configure email notification properties and support tickets in the ServiceNow ITSM Plug-in for VCF Automation.

The system administrator or plug-in admin can configure the email notification for these functionalities:

- Manage Endpoint: An email is sent to the users responsible to create, update, and delete the endpoint.
- Approval request: If the email settings are configured, then:
  - An email is sent to both requester and approval group once the request is created.
  - An email is sent to requester if the submitted request is approved or rejected.
  - An email is sent to requester if more information is required by the approval group.
- Deployment Requested Items (RITM) to VCF Automation: If the email settings are configured, then:
  - An email is sent to requester once the RITM is submitted.
  - If the approval group is configured, an email is sent to the approval group once the RITM is submitted.
  - An email is sent to requester if RITM is successful.
  - An email is sent to requester if RITM is failed and a support ticket is created.
- Day-2 actions: If the email settings are configured, then:
  - An email is sent to requester if a Day-2 deployment or resource action is successful.
  - An email is sent to requester if a Day-2 deployment or resource action is failed and a support ticket is created.

The system administrator or plug-in admin can configure the properties to create support tickets:

- A support ticket is created if a RITM request is failed.
- A support ticket is created if a Day-2 deployment or resource action is failed.

Configure the Email and Ticketing Properties

The system administrator or plug-in admin can configure the system properties for email notifications and support tickets if there is a failure of the Day 2 action and RITM request.

You must be the system administrator or plug-in admin in ServiceNow.

1. Log in to ServiceNow as a system administrator or Plugin admin.
2. From the left pane, go to **VCF Automation**> **Email Properties**.
3. On the **VCF Automation Email and Ticketing properties** page, select or deselect the checkboxes to configure these properties:

Field	Default Values	Description
Trigger the email notifications on Manage Endpoints.	Yes	To receive emails updates from Manage Endpoints including create, update, and delete of endpoints.
Trigger the email notifications to group on the approval requests.	Yes	To send the email notification to the approval group for the approval of the request.
Trigger the email notifications on request approval.	No	To receive an email notification if the approval group, approves the request.
Trigger the email notifications on the reject of the approval request.	Yes	To receive an email notification if the approval group, rejects the request.
Trigger the email notification when more details need to be provided on approval requests.	Yes	To receive an email notification when the approval group requests more information.
Trigger the email notifications on the successful completion of catalog item requests.	Yes	To receive email notification on the successful completion of the catalog item request.
Trigger the email notifications on successful/failure of Day 2 actions.	No	To receive email notification on the successful completion or failure of Day 2 actions.
Create the incident on the failure of catalog item submission in ServiceNow ITSM Plug-in for VCF Automation	Yes	To create an incident (support ticket) if a catalog request submission is failed in ServiceNow ITSM Plug-in for VCF Automation
Create the incident on the failure of job queue topics after the retry is failed.	No	To create an incident (support ticket) if a topic is failed in the job queue after the retry is failed.

4. Click **Save** to configure.

Note:

- To receive the Out-of-Box (OOB) emails and not the plug-in emails then disable the email properties for the ServiceNow ITSM Plug-in for VCF Automation
- To receive plug-in emails and not the OOB emails then apply the appropriate conditions at the places where OOB emails are initiated.

System Properties

The system properties chapter contains the most commonly used system properties in the ServiceNow ITSM Plug-in for VCF Automation.

Configure the Support Group for Request Failure Incidents

If a deployment request initiated from the ServiceNow fails in VCF Automation or if a day-2 action fails, the ServiceNow ITSM Plug-in for VCF Automation creates a support ticket and assigns it to the support group. The system administrator can configure the system properties for support groups.

You must update the `x_vmw_cloudservice.support_group_for_ITSM_8.x` system property to configure the support group.

Retriggering Time for Catalog Item Request if MID Server is Down

The system administrator and plug-in admin can configure a retriggering time for catalog requests. The system uses the configured time to resend the catalog request when the MID server is down. The system resends the request until the request is successfully submitted to VCF Automation

You must update the `x_vmw_cloudservice.TimetoRe-TriggerCatalogRequest` system property. For more information, refer system properties table in Appendix.

Cancel Time for Catalog Item Request if MID Server is Down

The ServiceNow system administrator and plug-in admin can define a cancel time for catalog requests when the mid-server is down.

- You must update the `x_vmw_cloudservice.cancelCatalogRequest` system property to configure the cancel time if the mid-server is down. For more information, refer system properties table in Appendix.
- The catalog requests will be declined if the request is not submitted to VCF Automation before the defined cancelation time.

Re-triggering Time for Catalog Item Request Status Check

The ServiceNow system administrator and plug-in admin can configure a time to automatically check the catalog request status. Once the catalog request is submitted to VCF Automation, the system uses this configured time to automatically check the request status and then update the status appropriately.

You must update the `x_vmw_cloudservice.CSWorkflowTimerToCheckRequestStatus` system property to configure the time for Catalog request status.

Cancel Time for Catalog Item Request Status Check

The ServiceNow system administrator and plug-in admin can define an end time to stop verifying the status of the request.

You must update the `x_vmw_cloudservice.CSTimeToCancelRequestWorkflow` system property to configure the time to end the status of the catalog request.

Configure the Multi-Level Approval for Day-0 and Day-2 Catalog Item Requests

The ServiceNow administrator can configure the multi-level approval for the day-0 and day-2 catalog item requests.

- You must be the system administrator in ServiceNow.
- The scope must be selected as the VCF Automation Plugin.

1. Log in to ServiceNow as a system administrator.
2. From the left pane, click **Workflow** > **Workflow Editor**.

   The Workflow Editor opens in a new browser tab.

3. Search for **Automation Day0 Items Approval SubWorkflow** for day-0 catalog item and **Automation Day2 Action Approval SubWorkFlow** for day-2 actions, and then open the workflow.
4. Check out the workflow to add more activities to the sub-workflow.
5. From the **Core** tab, add the **Approval - Group** to the workflow.
6. In the **Approvers** section, select the groups.
7. Update the connecting arrows for **Accept** and **Reject**.
8. Click **Update**.
9. Click the **menu** icon and select **Publish** to save.
10. Repeat the steps to add more approval activities in the sub-workflow.

    Note:

    - By default, three approval activities are added to the sub-workflow. The administrator can add desired approval groups to these activities.
    - The system administrator must add all the approval group names in the `x_vmw_cloudservice.Automation-ApprovalGroup` system property.

Create a Custom Workflow for Specific Catalog Items
By default, a VCF Automation Request Workflow is assigned to each catalog item in the ServiceNow ITSM Plug-in for VCF Automation. The ServiceNow system administrator can create their custom workflow and assign it to catalog items.

Create a Sub-Workflow

The system administrator is required to create a sub-workflow. This workflow contains the approval group activity and approval group information.

You must be the system administrator in ServiceNow.

1. Log in to ServiceNow as a system administrator.
2. From the left pane, go to **Administration > Workflow Editor**.

   The Workflow Editor opens in the new browser tab.
3. Select and open the **Automation Day0 Items Approval SubWorkflow**.
4. Click the **menu** icon and select **Copy**.
5. On the **Workflow Name** dialogue box, enter the new workflow name, and then click **OK**.

   The New workflow is created.
6. Configure the required workflow activity.

   Note:

   The result of sub-workflow must be either Approved or Rejected.
7. Click the **menu** icon and select **Publish** to save.

Create the Main Workflow

The workflow created in this section is assigned to the catalog items.

You must be the system administrator in ServiceNow.

1. Log in to ServiceNow as a system administrator.
2. From the left pane, click **Administration** > **Workflow Editor**.

   The Workflow Editor opens in the new browser tab.

3. Select and open the VCF Automation Request Workflow.
4. Click the **menu** icon and select **Copy**.
5. On the **Workflow Name** dialogue box, enter the new workflow name, and then click **OK**.

   The new workflow is created. After you have created the main workflow and sub-workflow, link the main workflow with the sub-workflow.

6. In the **Workflow Activity Properties** section, update the workflow created in Create a Sub-Workflow.
7. Click **Update** to link.
8. Click the **menu** icon and select **Publish** to save.

Update the Approval Group System Property

The ServiceNow system administrator is required to update the approval group system property to add more approval groups used in the subworkflow.

You must update the `x_vmw_cloudservice.Automation-ApprovalGroup` system property. For more information, refer system properties table in Appendix.

Configure the Custom Sub-Workflow

The ServiceNow system administrator can update the existing VCF Automation Approval Sub-Workflow with their custom workflow to send approvals to specific approval groups.

You must be the ServiceNow system administrator.

The ServiceNow system administrator can set conditions or criteria using JavaScript in ServiceNow to achieve the following type of context:

A custom workflow can be created to ensure that the requester belongs to either entitlement A or entitlement B or entitlement C. If the user is in entitlement A, the request must go to approval group A. If the user is in entitlement B, the request must go to approval group B. If the user is in entitlement C, the request must go to approval group C.

1. Log in to ServiceNow as a system administrator.
2. From the left pane, go to **Administration** > **Workflow Editor**.

   The Workflow Editor opens in a new browser tab.
3. Search for **Automation Day0 Items Approval SubWorkflow** and then open the Workflow.
4. Check out the workflow to add more approval group activities to subworkflow.
5. In the **Approval group activity** section, select the **Advance** check box.
6. On the **Additional groups script** box, add the appropriate conditions or criteria in javascript.

   Note:

   For this JavaScript code use the Manage Entitlement table system name that is `x_vmw_cloudservice_cas_catalog_entitlement`.
7. Save and publish the workflow.

Update the System Property of the Workflow Activity

The ServiceNow system administrator is required to update the workflow turnstile activity sys_ids for the system properties if the system administrator checks out or publishes the workflow.

- You must update `x_vmw_cloudservice.Request re-trigger turnstile activity` and `x_vmw_cloudservice.Request Iteration turnstile activity sys_id` system property. For more information, refer system properties table in Appendix.
- To access the sys_ids, see Access the Sys IDs.

Access the Sys IDs

Each turnstile activity has a unique sys_id and a new turnstile activity is created when the system administrator updates the workflow. This section provides information to view the required sys_ids in the ServiceNow ITSM Plug-in for VCF Automation

You must be the system administrator in ServiceNow.

1. Log in to ServiceNow as a system administrator.
2. In the left pane search for **wf_activity.LIST**.

   The Workflow activity window opens in a new tab.
3. In the **Name** column, search for **Check for iterations**.

   The list of available activities is displayed.
4. Select the activity for which **Published** is set to **True**.
5. Right-click on the activity and select **Copy sys_id**.

   Note: Update this sys_id for `x_vmw_cloudservice.Request Iteration turnstile activity sys_id` system property. See Update the System Property of the Workflow Activity.
6. In the **Name** column, search for **Check for Request re-trigger**.

   The list of available activities is displayed.
7. Select the activity for which **Published** is set to **True**.
8. Right-click on the activity and select **Copy sys_id**.

   Note: Update this sys_id for `x_vmw_cloudservice.Request re-trigger turnstile activity sys_id` system property.

   See Update the System Property of the Workflow Activity.

Supported and Unsupported Functionality

This section provides information for supported and unsupported functionalities for the ServiceNow ITSM Plug-in for VCF Automation.

Catalog Item Support and Limitation

- A single ServiceNow instance supports multiple VCF Automation instances. The Catalog Items from the multiple instances are consolidated.
- Approval groups to approve or reject the deployment requests are supported.
- Value Picker only supports Properties Array as the return type when the value is fetched from the VCF Automation Orchestrator Actions.
- Under **Custom Resources**, for **Add Action**, the **Requires condition** must be disabled.
- For **Custom** forms, field properties such as the **Match** field are not supported in the Constraints.
- For the Service Portal and Native UI, the placeholder functionality is not supported.
- The ServiceNow ITSM Plug-in for VCF Automation supports only the following field types in Cloud Template (YAML):
  - String
  - Integer
  - Number
  - Boolean
  - Object
  - Data grid
- The ServiceNow ITSM Plug-in for VCF Automation does not support the **Multi-value Picker** field type. When the Import catalog job runs, it skips the catalog items with the **Multi-value Picker** field type.

- In the ServiceNow ITSM Plug-in for VCF Automation, the dependent fields having minimum and maximum values with the VCF Automation action are not supported.
- In the ServiceNow ITSM Plug-in for VCF Automation, the stepwise increment option (stepper) is not supported for decimal and integer data type constraints.
- The ServiceNow ITSM Plug-in for VCF Automation does not support the Text Element if fetched from External Source.
- The ServiceNow ITSM Plug-in for VCF Automation does not support the Deployment Count feature.
- All the functionalities of the ServiceNow ITSM Plug-in for VCF Automation available on the Service Portal are supported on the Employee Center.
- In ServiceNow, it is recommended to have numeric values for the version field of the catalog item.
- On vRA, the **Date Time** field value for **Objectfield** and **Datagrid** must be in the UTC time zone.
- For catalog items, the price calculation is not supported for complex datatypes such as **Objectfield**, **Datagrid**, **Value Picker**, etc.
- If you have created the following field in the ServiceNow ITSM Plug-in for VCF Automation then while performing import in ServiceNow, this field will not be created:
  - Field ID: **project**
- **Data grid Limitations using Multi-Row Variable Set**
  - The table describes unsupported functionalities of the data grid in VCF Automation and ServiceNow:

Functionalities	VCF Automation	ServiceNow
Rows Limit	You can add multiple rows in the data grid as per your requirement.	Supports limited rows in the data grid. You can add only 50 rows.
Rows per Page	You can configure Pagination in the custom form for the data grid.	Pagination is not supported for the data grid.
Placement of data grid	You can drag and place the data grid on any tab as per your requirement.	The placement of the data grid will be visible only at the end of the tab.
Filter Icon	The filter icon is visible for the data grid in the catalog form.	The filter icon is not visible for the data grid in the catalog form.
Custom help	Custom help is supported for the data grid.	Custom help is not supported for the data grid.
Manage Columns	Manage Columns option is available for the data grid.	Manage Columns option is not available for the data grid.
Hide Columns	You can hide the columns in the custom form.	Hiding columns is not supported for the data grid.
Data grid name	Supports multiple underscores(_) in the data grid name.	Multiple underscores(_) are not supported in the data grid name.

- **Supported Data type Matrix for Day-0**
  This section provides information on supported data types for **Appearance**, **Default values**, and **Constraints** in ServiceNow ITSM Plug-in for **VCF Automation**.
  - **Appearance**

    Note:

    - The support for data types such as Drop down, Multi-select, and Dual list as shown in the below table is also supported for String, Integer, and Decimal.
    - The Conditional visibility and Read-only is supported for the following datatypes using the operators:
      - **Drop down** and **Text field** using **Equals**, **NotEquals**, and **Within** operators.
      - **Checkbox** using **Equals** and **NotEquals** operators.
      - **Integer** and **Decimal** using **Equals**, **NotEquals**, **Greater than**, **Lesser than**, **Greater than equal to**, and **Lesser than equal to.**
    - The External Source visibility and Read-only can be set by the direct **VCF Operations Orchestrator** action and Input in Action inputs as Constant and Field using the following data types: **Text field**, **Text Area**, **Decimal**, **Integer**, **Drop down**, and **Check box.**

- '**N/A**' in the below table represents that data types are not supported in ServiceNow since they are not available in VCF Automation.

By default, the following VCF Automation field types are supported:

Appearance									
Label and Type		Visibility				Read-Only			Custom help
Data Types	Placeholder	Constant	Conditional	External Source	Bind field	Constant	Conditional	External Source	
Array Input	×	×	×	×	×	×	×	×	×
Dropdown	×	✓	✓	✓	×	✓	✓	✓	✓
Image	N/A	×	×	×	×	N/A	N/A	N/A	×
Link	N/A	×	×	×	×	N/A	N/A	N/A	×
Dual List	N/A	✓	✓	✓	×	✓	×	×	✓
Decimal	N/A	✓	✓	✓	×	✓	✓	✓	✓
File upload	N/A	✓	✓	✓	×	✓	✓	✓	✓
Integer	N/A	✓	✓	✓	×	✓	✓	✓	✓
Multi select	N/A	✓	✓	✓	×	✓	×	×	✓
Multi value picker	N/A	×	×	×	×	×	×	×	×
Object field	N/A	✓	✓	✓	×	✓	✓	✓	✓
Password	N/A	✓	✓	✓	×	✓	✓	✓	✓
Radio group	N/A	✓	✓	✓	×	✓	✓	✓	✓
Text area	✓	✓	✓	✓	×	✓	✓	✓	✓
Text field	✓	✓	✓	✓	×	✓	✓	✓	✓
Text	N/A	✓	✓	✓	×	N/A	N/A	N/A	✓
Value picker	×	✓	✓	✓	×	✓	×	×	✓
Check box	N/A	✓	✓	✓	×	✓	✓	✓	✓
Data grid	×	✓	✓	✓	×	✓	✓	✓	×
Combo box	×	×	×	×	×	×	×	×	×
Date Time	N/A	✓	✓	✓	×	✓	✓	✓	✓
Tree Picker	×	×	×	×	×	×	×	×	×

- You need to enter or click outside the area of the text field after entering a word that matches the read-only and visibility of fields and tabs.
- The External Source option from visibility and read-only is not supported for an empty field value.
- The ServiceNow ITSM Plug-in for VCF Automation does not support request info fields for read-only and visibility of fields and tabs.
- **Default Values Support**

  Note:

  - The support for data types such as Drop down, Multi-select, and Dual list as shown in the below table is also supported for String, Integer, and Decimal.
  - The Conditional visibility and Read-only is supported for the following datatypes using the operators:
    - **Drop down** and **Text field** using **Equals**, **NotEquals**, and **Within** operators.

  - **Checkbox** using **Equals** and **NotEquals** operators.
  - **Integer** and **Decimal** using **Equals**, **NotEquals**, **Greater than**, **Lesser than**, **Greater than equal to**, and **Lesser than equal to.**
- The External Source default value and Value Options can be set by the direct **VCF Operations Orchestrator** action and Input in Action inputs as Constant and Field using the following data types: **Text field**, **Text area**, and **Drop down**.
- The Computed default value is supported for the following field type selections:**Integer**, **Decimal**, **Text field**, **Drop down**, and **Text area**.
- **'N/A'** in the below table represents the selected combination of default value and value options that are not supported in ServiceNow since they are not available in VCF Automation.

The supported VCF Automation field types for default value and value options are as follows:

Default Value and Value Options															
Data Types	**Default Value - Constant**			**Default Value - External Source**			**Default Value - Computed**			**Default Value - Conditional**			**Default Value - Bind fields**		
	Value Options			**Value Options**			**Value Options**			**Value Options**			**Value Options**		
	No Optio	**Consta**	**Externa Source**	**No Optio**	**Consta**	**Externa Source**	**No Options**	**Consta**	**Externa Source**	**No Options**	**Consta**	**Externa Source**	**No Options**	**Consta**	**External Source**
Array input	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Check box	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A
Combo box	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Data grid	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Date time	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	✓	N/A	N/A	×	N/A	N/A
Decimal	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A
Drop down	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dual list	✓	✓	✓	✓	✓	✓	N/A	×	×	×	×	×	N/A	×	×
File upload	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Image	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Integer	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A
Link	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Multi select	✓	✓	✓	✓	✓	✓	N/A	×	×	×	×	×	N/A	×	×
Multi value picker	×	×	×	×	×	×	N/A	×	×	N/A	×	×	N/A	×	×
Object field	×	N/A	N/A	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Password	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	✓	N/A	N/A	×	N/A	N/A
Radio group	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Text	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Text area	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A
Text field	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	N/A
Tree picker	×	N/A	N/A	×	N/A	N/A	N/A	N/A	N/A	×	N/A	N/A	×	N/A	N/A

Default Value and Value Options															
Data Types	Default Value - Constant			Default Value - External Source			Default Value - Computed			Default Value - Conditional			Default Value - Bind fields		
	Value Options			Value Options			Value Options			Value Options			Value Options		
	No Option	Consta	Externa Source	No Option	Consta	Externa Source	No Options	Consta	Externa Source	No Options	Consta	Externa Source	No Options	Consta	External Source
Value picker	✓	✓	✓	✓	✓	✓	N/A	N/A	N/A	×	×	×	×	×	×

- In **request info fields**, the following value fields are supported for the **Bind field** and **External source** value source:
  - Catalog item name
  - Content source type
  - Current organization
- In Bind fields and Computed default value, once the child **text field** value is modified and when you change the parent **text field** value, the modified child text field value will also reflect the same as that of the parent field.
- The operator containing an empty value is not supported for the computed default value.
- **Support for Constraints**

Note:

- The support for data types such as Drop down, Multi-select, and Dual list as shown in the below table is also supported for String, Integer, and Decimal.
- '**N/A**' in the below table represents that data types are not supported in ServiceNow since they are not available in **VCF Automation**.

The following VCF Automation field types are supported for Constraints:

Constraints											
	Required			Regular Expression		Minimum Value			Maximum Value		
Data Types	Constant	Conditional	External Source	Constant	Conditional	Constant	Conditional	External Source	Constant	Conditional	External Source
Array input	×	×	×	×	×	×	×	×	×	×	×
Check box	✓	×	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Combo box	×	×	×	×	×	×	×	×	×	×	×
Data grid	✓	×	×	✓	×	✓	×	×	✓	×	×
Date time	×	×	×	N/A	N/A	✓	×	×	✓	×	×
Decimal	✓	×	×	×	×	✓	×	×	✓	×	×
Drop down	✓	×	×	×	×	×	×	×	×	×	×
Dual list	✓	×	×	×	×	✓	×	×	✓	×	×
File upload	✓	×	×	N/A	N/A	✓	N/A	×	✓	N/A	×
Image	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Integer	✓	×	×	×	×	✓	×	×	✓	×	×
Link	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Multi select	✓	×	×	×	×	✓	×	×	✓	×	×

Constraints											
	Required			Regular Expression		Minimum Value			Maximum Value		
Data Types	Constant	Conditional	External Source	Constant	Conditional	Constant	Conditional	External Source	Constant	Conditional	External Source
Multi value picker	×	×	×	N/A	N/A	×	×	×	×	×	×
Object field	✓	×	×	✓	×	✓	×	×	✓	×	×
Password	✓	×	×	✓	×	✓	×	×	✓	×	×
Radio group	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Text	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Text area	✓	×	×	✓	×	✓	×	×	✓	×	×
Text field	✓	×	×	✓	×	✓	×	×	✓	×	×
Tree picker	×	×	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Value picker	✓	×	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

- **Custom Categories support**:
  - The ServiceNow ITSM Plug-in for VCF Automation supports creating custom categories and linking catalog items to custom categories.
  - To create a new category, go to the filter Navigator, and type **Categories**> **Service Catalog**> **Maintain Categories**.
- To link a newly created category to catalog items, follow the steps:
  - In filter Navigator>**Catalog items**.
  - On **Catalog items**, under the **Categories** tab, click **Edit**.
  - In **Edit Members**, search for the newly created categories.
  - Add the newly created categories to the **Categories List** and click **Save**. The new categories are linked to the catalog items.
- **Support of custom fields for Day-0 catalog item**:
  - The container for custom fields facilitates you to create a custom field with orders ranging from conditions **10002-10099** under variables.
  - You can also change the name of the custom container if required.
  - You can create customized client scripts for custom fields.
  - You can create a new custom field by clicking the **New** button under variables, on catalog items in the ServiceNow ITSM Plug-in for VCF Automation.
  - While creating new **custom fields**, if the user changes VCF Automation scope to global scope on catalog items, then the **New** button under variables will not be visible to create new custom fields.
  - Custom fields can be created only on the Day-0 catalog item.
  - You can view the custom fields on RITM. The reference of the RITM number is displayed under requested items by navigating to deployments and vice versa.
  - While creating a client script, in **UI type**, you must select **All** so that the script works on both the Native UI and Service Portal.
  - Custom fields cannot be sent to **VCF Automation/VMware Aria Automation**.
  - The **Application Scope** must be **Global** for creating a Custom field and Variable set else the catalog items request will fail in ServiceNow.
  - **Note:**

    The **Application Scope** must be **Global** for creating a Custom field and Custom variable set.
- **Support for custom variable set for day-0 catalog items**:
  - The **Application Scope** must be **Global** for creating a custom variable set.

- **Deployment Name field support**:
  - You can view the **Deployment Name** field in the plug-in for both custom form and non-custom form catalog items. The **Deployment Name** field displays as a Read-only field and can be enabled or disabled from VCF Automation.
  - If the catalog item does not have a Deployment name field in VCF Automation, then the Deployment name will be displayed after the Version field.
  - The ServiceNow ITSM Plug-in for VCF Automation supports the Deployment name field for external source and default values.

Custom property Support and Limitation

- The following input types are supported for YAML:
  - String
  - Integer
  - Boolean
- Read-only and Default values are not supported for YAML.

Property group Support and Limitation

- The ServiceNow ITSM Plug-in for VCF Automation supports the following input types to create a new property for property groups:
  - String
  - Integer
  - Boolean
- The table represents the supported options for the following input-type fields in Property groups:

Field Options	Field Values	Input Type: String	Input Type: Integer	Input Type: Boolean
Default Value Source: Constant	Default Value	✓	✓	✓
	Encrypted	✓	×	×
	Read Only	✓	✓	✓
More Options	Format	✓	×	×
	Minimum Length	✓	✓	×
	Maximum Length	✓	✓	×
	Pattern	✓	×	×
Enumerated list type	Values	✓	✓	×
	Keys and Values	✓	✓	×
	External Source	×	×	×
Enum	Value	✓	✓	×

Day 2 Action Support and Limitation

- Configuration of email notification for endpoint, entitlement, approval request, RITM request on VCF Automation, day-2 actions, and resource sharing are available.
- A support ticket is created within ServiceNow on the failure of catalog request submission and day-2 action.
- After you have removed the disk from a machine, it takes time to reflect the same in the ServiceNow ITSM Plug-in for VCF Automation.
- The following day-2 actions are available from the ServiceNow ITSM Plug-in for VCF Automation:

Resource Actions	Deployment Actions
Add Disk	Change Lease
Compute Relocate	Delete
Create Snapshot	Power Off

Resource Actions	Deployment Actions
Delete	Power On
Delete Snapshot	Change Owner
Power Off	Edit Deployment
Power On	Change Project
Reboot	
Remove Disk	
Reset	
Resize	
Resize Boot Disk	
Resize Disk	
Restart	
Revert to Snapshot	
Shutdown	
Suspend	

- You can change the owners of deployments for day-2 actions with the available users displayed in the **Owner Type** dropdown only if the same users are available on VCF Automation and ServiceNow :
  - For On-premise, the **user_name** of users will be displayed if the same user_name is available on vRA and ServiceNow.
  - Adding **Groups** is not supported in the **Owner Type**.
- Edit deployment day-2 action does not support changing the icon for a deployment.
- The Run Puppet Task day-2 action is not supported for Puppet deployments.
- Custom day-2 Actions and Custom Resources are supported.
- Custom day-2 Action is supported for:
  - Deployments
  - Built-in Resources
  - Custom Resources
  - Form-based day-2 actions
  - Formless day-2 actions

The ServiceNow ITSM Plug-in for VCF Automation supports the following field types for custom resources and custom day-2 actions through property binding:

Field Types	Custom Resource	Custom Day-2 Actions		
		Direct Binding	In Request	Binding Action
Text area	✓	✓	✓	×
Text field	✓	✓	✓	×
Text	✓	✓	✓	×
Password	✓	×	✓	×
Decimal	✓	✓	✓	×
Integer	✓	✓	✓	×
Drop down	✓	✓	✓	×
Check box	✓	✓	✓	×
Value picker	✓	×	✓	✓
Date time	✓	×	×	×

- The request form submission fails in ServiceNow for the data types that are not supported in Custom day-2 actions.

- Data type as Complex and display as Object field is not supported in the plug-in for day-2 catalog items.
- The ServiceNow ITSM Plug-in for VCF Automation does not support the Update day-2 action for:
  - Custom Resources
  - Built-in resources in the Native UI
- In ServiceNow, to import day-2 actions for resources and deployments, you can now set the following properties as either `True` or `False`:
  - `x_vmw_cloudservice.fetch_deployments_day2_actions`: The default value is `True`, when you set the property as `False`, the day-2 actions for deployments will not be imported.
  - `x_vmw_cloudservice.fetch_resources_day2_actions`: The default value is `True`, when you set the property as `False`, the day-2 actions for resources will not be imported.
- **Supported data type matrix for Day-2**
  This section provides information on supported data types for **Appearance**, **Default values**, and **Constraints** in ServiceNow ITSM Plug-in for **VCF Automation**.
  - **Appearance Support**

    Note:

    - The data type support of Drop down, Multi-select, and Dual list as shown in the below table is also supported for **String**, **Integer**, and **Decimal**.
    - The Conditional visibility and Read-only is supported for the following datatypes using operators:
      - **Drop-down** and **Text field** using **Equals**, **NotEquals**, and **Within** operators.
      - **Checkbox** using **Equals** and **NotEquals** operators.
      - **Integer** and **Decimal** using **Equals**, **NotEquals**, **Greater than**, **Lesser than**, **Greater than equal to**, and **Lesser than equal to**.
    - The Signpost position for custom help is not supported.
    - '**N/A**' in the below table represents that data types are not supported in ServiceNow since they are not available in VCF Automation.

- By default, the following VCF Automation field types are supported:

Appearance									
Label and Type		Visibility				Read-Only			Custom help
Data Types	Placeholder	Constant	Conditional	External Source	Bind field	Constant	Conditional	External Source	
Array Input	×	×	×	×	×	×	×	×	×
Dropdown	×	✓	✓	×	×	✓	×	×	✓
Image	N/A	×	×	×	×	N/A	N/A	N/A	×
Link	N/A	×	×	×	×	N/A	N/A	N/A	×
Dual List	N/A	×	×	×	×	×	×	×	×
Decimal	N/A	✓	✓	×	×	✓	×	×	✓
File upload	N/A	✓	✓	×	×	✓	✓	✓	✓
Integer	N/A	✓	✓	×	×	✓	×	×	✓
Multi select	N/A	×	×	×	×	×	×	×	×
Multi value picker	N/A	×	×	×	×	×	×	×	×
Object field	N/A	×	×	×	×	×	×	×	×
Password	N/A	✓	×	×	×	✓	×	×	✓
Radio group	N/A	×	×	×	×	×	×	×	×
Text area	×	✓	✓	×	×	✓	×	×	✓

Appearance									
Label and Type		Visibility				Read-Only			Custom help
Data Types	Placeholder	Constant	Conditional	External Source	Bind field	Constant	Conditional	External Source	
Text field	×	✓	✓	×	×	✓	×	×	✓
Text	N/A	✓	×	×	×	N/A	N/A	N/A	✓
Value picker	×	✓	✓	×	×	✓	×	×	✓
Check box	N/A	✓	✓	×	×	✓	×	×	✓
Data grid	×	×	×	×	×	×	×	×	×
Combo box	×	×	×	×	×	×	×	×	×
Date Time	N/A	✓	×	×	×	✓	×	×	✓

- You need to enter or click outside the area of the text field after entering a word that matches the visibility of the fields and tabs.
- The visibility of fields and tabs for constraints is not supported for the following:
  - Minimum
  - Maximum
  - Regex
- **Default Values Support**

  Note:

  - The data type support of Drop down, Multi-select, and Dual list as shown in the below table is also supported for String, Integer, and Decimal.
  - The External Source default value and value options can be set by the direct **VCF Operations Orchestrator** action and Input in Action inputs as Constant and Field using the following data types: **Text field**, **Text area**, and **Drop down**.
  - **'N/A'** in the below table represents the selected combination of default value and value options that are not supported in ServiceNow since they are not available in VCF Automation.

The supported VCF Automation field types for default value and value options are as follows:

Default Value and Value Options															
	Default Value - Constant			Default Value - External Source			Default Value - Computed			Default Value - Conditional			Default Value - Bind fields		
Data Types	Value Options			Value Options			Value Options			Value Options			Value Options		
	No Options	Constant	External Source	No Options	Constant	External Source	No Options	Constant	External Source	No Options	Constant	External Source	No Options	Constant	External Source
Check box	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	×	N/A	N/A	✓	N/A	N/A
Date time	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Decimal	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Drop down	✓	✓	✓	✓	✓	✓	×	×	×	×	×	×	✓	✓	✓
Integer	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Value picker	✓	✓	✓	✓	✓	✓	N/A	N/A	N/A	×	×	×	×	×	×
Password	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Text area	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Text field	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A	×	N/A	N/A	✓	N/A	N/A
Text	✓	N/A	N/A	✓	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A

Default Value and Value Options															
	Default Value - Constant			Default Value - External Source			Default Value - Computed			Default Value - Conditional			Default Value - Bind fields		
Image	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
Multi select	×	×	×	×	×	×	N/A	×	×	×	×	×	N/A	×	×
Dual list	×	×	×	×	×	×	N/A	×	×	×	×	×	N/A	×	×
Radio Group	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Data Grid	×	N/A	N/A	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Object Field	×	N/A	N/A	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Multi Value Picker	×	×	×	×	×	×	N/A	×	×	N/A	×	×	N/A	×	×
Combo box	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Array Input	×	N/A	N/A	×	N/A	N/A	N/A	N/A	N/A	×	N/A	N/A	N/A	N/A	N/A
Link	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A	×	N/A	N/A
File Upload	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

- In request info fields, the following value fields are supported for the Bind field and External source value source:
  - Catalog item name
  - Content source type
  - Current organization
  - All fields under resource properties
- **Support for Constraints**

  Note:

  - The data type support of Drop down, Multi-select, and Dual list as shown in the below table is also supported for String, Integer, and Decimal.
  - '**N/A**' in the below table represents that data types are not supported in ServiceNow since they are not available in VCF Automation.
- The supported VCF Automation field types are supported for Constraints:

Constraints											
	Required			Regular Expression		Minimum Value			Maximum Value		
Data Types	**Constant**	**Conditional**	**External Source**	**Constant**	**Conditional**	**Constant**	**Conditional**	**External Source**	**Constant**	**Conditional**	**External Source**
Array input	×	×	×	×	×	×	×	×	×	×	×
Check box	✓	×	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Combo box	×	×	×	×	×	×	×	×	×	×	×
Data grid	×	×	×	×	×	×	×	×	×	×	×
Date time	✓	×	×	N/A	N/A	×	×	×	×	×	×
Decimal	✓	×	×	×	×	✓	×	×	✓	×	×

Constraints											
	Required			Regular Expression		Minimum Value			Maximum Value		
Data Types	Constant	Conditional	External Source	Constant	Conditional	Constant	Conditional	External Source	Constant	Conditional	External Source
Drop down	✓	×	×	×	×	×	×	×	×	×	×
Dual list	×	×	×	×	×	×	×	×	×	×	×
File upload	✓	×	×	N/A	N/A	×	×	×	×	×	×
Image	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Integer	✓	×	×	×	×	✓	×	×	✓	×	×
Link	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Multi select	×	×	×	×	×	×	×	×	×	×	×
Multi value picker	×	×	×	N/A	N/A	×	×	×	×	×	×
Object field	×	×	×	×	×	×	×	×	×	×	×
Password	✓	×	×	✓	×	✓	×	×	✓	×	×
Radio group	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Text	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Text area	✓	×	×	✓	×	✓	×	×	✓	×	×
Text field	✓	×	×	✓	×	✓	×	×	✓	×	×
Tree picker	×	×	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Value picker	✓	×	×	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Supported Cloud Providers

- VMware vSphere®, Amazon Web Services™ (AWS), Google Cloud Platform™, and Microsoft® Azure® Cloud are supported.

Other Supported Functionality and Limitations

- Defining the permissions for Admin, Catalog Admin, and User is supported.
- Reconciliation of CMDB to keep the CI item updated is supported.
- The ServiceNow RITM number and Requester email ID or user name will be sent to VCF Automation for Day-0 and Day-2 catalog requests.
- If the ServiceNow users are not able to view deployments, ensure that the user is using the email address as a user name in the VCF Automation and it must be the same as the email address of the user in ServiceNow.
- **Project Group Entitlements**:
  - In ServiceNow, on the Project record, under the **Users** section, you can add multiple groups in the **Project Group**, so that all the members of the group can view the deployments and resources related to a particular Project.
  - Groups must have `cmdb_read` and `x_vmw_cloudservice.vra_end_user` roles to access deployments and resources, and also to perform the day-2 actions.

- The ServiceNow ITSM Plug-in for VCF Automation is tested on the following web browsers and versions:

Web Browser	Minimum Supported Version
Google Chrome	141.0.7390.123 (Official Build) (64-bit)
Microsoft Edge	141.0.3537.85 (Official build) (64-bit)
Mozilla Firefox	144.0 (aarch64)

Telemetry

The telemetry feature facilitates users to provide consent whether to send the data from the ServiceNow ITSM Plug-in for **VCF Automation** to **VMware**by Broadcom.

The following data will be sent to VMware by Broadcom after the user provides the consent:

- Organization Id
- Projects
- Deployments
- Catalog items

To provide consent to send data or not, the user must navigate to the **Manage Endpoint** table and click the existing endpoint, or while registering a new endpoint, the consent pop-up will be displayed.

- To provide consent, click **Yes**. The scheduled job will start sending telemetry data to VMware by Broadcom and the consent pop-up will not be prompted again for the user.
- To not provide consent, click **No**. The telemetry data will not be sent to VMware by Broadcom and a consent pop-up will be prompted after every 30 days for further consent.

1. If the user responds to the consent pop-up as **Yes** but do not wish to send the telemetry data to VMware by Broadcom, then the user needs to perform the following steps:
   a) Navigate to `x_vmw_cloudservice.telemetry_last_trigger` system property.
   b) Open the system property and in the **value** option, remove the date and time, and click **Update**.
   c) Search for the `x_wm_cloudservice.customer_consent_value` system property.
   d) Open the system property and set the value as **false**.
   e) When the next consent pop-up is prompted, click **No**, so that telemetry data will not be sent to VMware by Broadcom.
2. If the user responds to the consent pop-up as **No** but intends to send the telemetry data to VMware, then the user needs to perform the following steps:
   a) Navigate to `x_vmw_cloudservice.telemetry_last_trigger` system property.
   b) Open the system property and in **value**, remove the date and time, and click **Update**.
   c) Navigate to the **Manage Endpoint** table, and open the existing endpoint.
   d) When the consent pop-up is prompted, click **Yes**, so that telemetry data will be sent to VMware by Broadcom.

Known Issues

The following issues are known in the ServiceNow ITSM Plug-in for VCF Automation:

- The ServiceNow ITSM Plug-in for VCF Automation displays Boolean type fields as checkboxes in catalog items.
- While deleting an endpoint make sure that data is in the VMware application scope. If not, an error message displays for the cross scope data deletion post endpoint is deleted and data is removed successfully.
- In the Native UI, if a day-2 action catalog item contains the check box or label fields then ServiceNow displays the cross-scope info messages while loading day-2 catalog items and also on RITM after submitting the day-2 catalog items. This is a ServiceNow-known issue.
- On the Service Portal, duplicate data type fields are displayed in the RITM if the catalog item has multiple versions.
- On the Service Portal, for the visibility feature, the Dropdown field list on the catalog item will have an extra empty option.

- When day-2 actions for multiple deployments are processed simultaneously, the duplicate variables may be generated in respective day-2 catalog items. This is a ServiceNow-known issue and occurs rarely.
- In ServiceNow, if you hide all fields under a tab, then the tab will also not be visible even if the visibility condition is matched.
- The catalog form request with empty fields within property groups will fail after submission because the property groups are mandatory in VCF Automation.
- In the Data grid, if the input field with type number/integer is empty on the order form, then the catalog request fails after submitting in ServiceNow.
- In ServiceNow, when you modify the value of any dependent field containing conditional default value and conditional read-only, then the dependent field will reflect the default value instead of the modified value.
- In ServiceNow, when you perform the price estimate calculations without filling in any mandatory fields in the form, the price estimate can give an error message that can be viewed in the error logs.
- The “**Update**” button for price estimate calculations is always active even if any of the field’s inputs are not modified.
- In ServiceNow, if the mandatory field has a default value and conditional read-only, then the read-only condition will not be applied to the mandatory field.
- For getting an empty value from **VCF Operations Orchestrator**, make sure your VCF Operations Orchestrator action script code considers both **“Empty”** and **“Null”** conditions while checking the input parameters.
- In the plug-in, the below-mentioned configurations for the **Dual List** and **Multi-Select** data type are not recommended:
  - Default value is not present in the **Value Options**.
  - **Value Options** are empty and the default value source is **External**.

Troubleshoot

The troubleshooting topics provide solutions to potential problems that you might encounter when using ServiceNow ITSM Plug-in for VCF Automation.

Unable to perform Day-2 Action
ServiceNow users are not able to perform day-2 actions.

From the deployment and resource actions, ServiceNow users are not able to perform day-2 operations.

The registered endpoint is activated after the deactivation in ServiceNow.

Run the import jobs for the deployment and resources. See Configure and Run Import Jobs.

Complete Hover Message or Field Name is not Displayed
Large hover messages or field names are getting truncated.

Not able to see the full value of translated text and Question text.

By default field length of the `translated_text` and `question_text` is 250 and 255 respectively. If the length is greater than the desired length, the field length will be truncated.

The ServiceNow system administrator must increase the field length for translated_text and question_choice. You must import the catalog items again after updating the length.

Deployments and Resources are not displayed
The deployments and resources are not visible in ServiceNow.

Even after running the scheduled job, deployments and resources are not displayed in ServiceNow.

This problem occurs as some fields in the `CMDB_ci_appl` table have mandatory conditions.

1. Right-click on the header bar > **configure** and open the `CMDB_ci_appl`table.
2. Check if the following mandatory field is set to `true`:
   a. Locate the mandatory fields that are set to `true`, make them non-mandatory, and save them.
   b. Check if you are allowed to make any changes in the `CMDB_ci_appl` table. If not, then you need to make changes in the `x_vmw_cloudservice_deployments` table.
3. To make changes, perform the following:
   a. In the `x_vmw_cloudservice_deployments` table, locate the mandatory field that is set to `true`.

b. Navigate to **Dictionary Override** > **Create new record** in the selected `x_vmw_cloudservice_deployments` table.
c. Select the **Override mandatory** check box and click **Submit**.

Note:

If there is a mandatory condition in the `x_vmw_cloudservice_deployments` column that does not have any condition set, then update it to mandatory `false` or create a dictionary override.

ServiceNow Users are not able to View Deployments
Deployments and resource owners are not able to view their deployments and resources.

ServiceNow end users are not able to view their deployments and resources.

The user's user name is not same in VCF Automation.

Ensure that the user is using the same **user_id**/**user_name** of users in ServiceNow as that of **VCF Automation**

End user of Automation-Approval Group is not able to approve functionality on other's plug-in request
The ServiceNow ITSM Plug-in for VCF Automation end users are not able to approve/reject/comment on other users plug-in requests as a part of the approval functionality. These end users are already added to the Approval group to use approval functionality.

End user of Automation-ApprovalGroup is not able to approve functionality on other's plug-in request.

This problem occurs due to a change in the ServiceNow certification process. Now user requires to manually create the ACL records.

1. Create a read ACL in Global scope on the `sc_req_item` OOTB table for `x_vmw_cloudservice.vra_end_user` (application end user) role.
2. Create a read ACL in Global scope on the `sysapproval_approver` OOTB table for `x_vmw_cloudservice.vra_end_user` (application end user) role.

ServiceNow Users facing string limit issues when exceeded to the max
ServiceNow users may face string limit issues when the maximum limit is exceeded. For such cases, change the system property as follows:

ServiceNow users can face issues when the string limit exceeds 32 MB.

This problem occurs when the string limit exceeds the max, the user will get an error log "`String object would exceed the maximum permitted size of 33554432`".

The user must change the system Property "`x_vmw_cloudservice.apiPageSize`" value from `200` to `100`.

After/Before the Upgrade, if the plug-in behaves abnormally OR the user forgets to set the "Replace on upgrade" flag then check the Skipped and Skipped Manual Merge records in the Upgrade History
The records will fall into Skipped and Skipped Manual Merge if the user forgets to update the **Replace on upgrade** flag.

The plug-in will behave abnormally when records will fall into skipped and skipped manual merge if the user skips setting the "**Replace on upgrade**" flag for customized records.

The records will fall into Skipped and Skipped Manual Merge in table `sys_upgrade_history_log`.

You need to perform the following steps:

1. Navigate to plug-in and click **View upgrade history**. The **Upgrade Details** page will open.
2. Under the **Skipped Changes to Review** tab, find all the Skipped Manual Merge, and Skipped records that are listed.
3. Open each one of the records and click the **Resolve Conflicts** button to verify if there are any conflicts in the **Customized** record compared to the **Base System** record.
4. Go back to the **Upgrade Details** page and click the **Revert to Base System** button.

Note:

If you have any customized record for the plug-in, you must copy your customizations and mark that record to **Revert to base System** and update again as per your customization.

Expected variables are not visible on the RITM table for end user
The expected variables on the RITM table are not visible for end user.

The RITM view may differ from the order form for the visible variables.

The end user does not have write access to the variables field on the RITM table.

End user must create a new field-level ACL for write access to the variables field on the RITM table with a required role as `x_vmw_cloudservice.vra_end_user`.

Appendix

This chapter contains a list of the available custom tables, CMDB tables, and system properties used in ServiceNow ITSM Plug-in for VCF Automation.

ServiceNow Custom Table used in ServiceNow ITSM Plug-in for VCF Automation

These custom tables are used in ServiceNow.

Label	Name	Extended Table
Project	x_vmw_cloudservice_cas_project	Category
Catalog Items	x_vmw_cloudservice_cas_catalog_items	Catalog Item
Deployments	x_vmw_cloudservice_deployments	Application
Resources	x_vmw_cloudservice_machines	Virtual Machine Instance
Deployments Import Set	x_vmw_cloudservice_vra_deployments_import_set	Import Set Row
Resources Import Set	x_vmw_cloudservice_vra_machines_import_set	Import Set Row
Job Queue	x_vmw_cloudservice_cas_queue	
Manage Endpoints	x_vmw_cloudservice_register_cas_organization	
Day-2 Actions	x_vmw_cloudservice_vra_cloud_actions	

ServiceNow CMDB Tables in ServiceNow ITSM Plug-in for VCF Automation

These custom tables are used in ServiceNow.

Table	Custom Field	Label
x_vmw_cloudservice_deployments	• blueprint_version • bluprint_id • code • deployment_id • last_updated_time • project_id • catalog_id • organization_id • catalog_version • operational_status • servicenow_owner • Shared_groups • vra_owner • vra_requester • entitled_users • unit • created_at • expired • expires_on • last_updated_at • last_updated_by • u_catalog_item • u_project	• Blueprint Version • Bluprint Id • Code • Deployment Id • Last Update Time • Project Id • Catalog Id • Organization Id • Catalog Version • Operational status • ServiceNow Owner • Entitled Groups • Owner • Requester • Entitled Users • Unit • Created At • Expired • Expires On • Last Updated At • Last Updated by • Catalog Item • Project
x_vmw_cloudservice_machines	• createdat • deploymentid • entitled_users • machine_id • machine_name • name • network_schema • operational_status • project • storage • syncstatus • status • vra_owner • vra_requester • resource_name • encrypted • attachment_status • address • endpointtype • account • volumetype • node_uuid • environment • workflow_name • started_on • id • account_region • network_type • capacity_gb • zone • region • externallink • role	• CreatedAt • DeploymentId • Entitled Users • Resource Id • Resource Name • Name • Network Schema • Operational status • Project • Storage Schema • SyncStatus • Status • Owner • Requester • Resource Name • Encrypted • Attachment status • Address • endpointType • Account • volumeType • Node UUID • Environment • Workflow name • Started on • Id • Account/Region • Network Type • Capacity(GB) • Availability Zone • Region • externalLink • Role

System Properties in ServiceNow ITSM Plug-in for VCF Automation

The catalog admin or system administrator of ServiceNow ITSM Plug-in for VCF Automation can update the system properties.

System Properties	Description	Updated by
x_vmw_cloudservice.cancelCatalogRequest	Update the value in hours to cancel the catalog request if mid server is down.	Plug-in admin or System administrator
x_vmw_cloudservice.CAS.log.header	Contains header for the log file.	Plug-in admin or System administrator
x_vmw_cloudservice.CAS.logging.verbosity	Contains log verbosity for the ServiceNow ITSM Plug-in for VCF Automation.	Plug-in admin or System administrator
x_vmw_cloudservice.cloud_API_endpoint_URL	The API URL for different geographical locations can be different.	Plug-in admin or System administrator
x_vmw_cloudservice.Request Iteration turnstile activity sys_id	This system property allows you to add Request Iteration turnstile activity sys_id	Plug-in admin or System administrator
x_vmw_cloudservice.Request re-trigger turnstile activity sys_id	This system property allows you to add Request re-trigger turnstile activity sys_id	Plug-in admin or System administrator
x_vmw_cloudservice.Request re-trigger turnstile activity sys_id	Create the support group for the ServiceNow ITSM Plug-in for VCF Automation when an incident is created from RITM assign it to them.	Plug-in admin or System administrator
x_vmw_cloudservice.TimetoRe-TriggerCatalogRequest	This system property allows you to update time in minutes to retrigger the catalog request if the MID server is down. The system resends the request untill the requests is successfully submitted to the VCF Automation.	Plug-in admin or System administrator
x_vmw_cloudservice.itsm_application_version	Version of the ServiceNow ITSM Plug-in for VCF Automation	Plug-in admin or System administrator
x_vmw_cloudservice.Automation-ApprovalGroup	This system property has a default value as Automation-ApprovalGroup. It allows you to add multiple approval group names.	Plug-in admin or System administrator
x_vmw_cloudservice.SetApiVersionDate	Allows you to lock the API at a given point in time and control when you upgrade to a new API version. Update the API version date in the yyyy-mm-dd format. For example: 2019-01-15.	Plug-in admin or System administrator
x_vmw_cloudservice.deploymentWindowSize	This system property allows you to update the number of Deployments in the Deployment List Widget of the Service Portal. The default value is 20.	Plug-in admin or System administrator
x_vmw_cloudservice.DeploymentLimit	This system property allows you to update the number of Deployments in the Deployment Widget of the Service Portal. The default value is 5.	Plug-in admin or System administrator
x_vmw_cloudservice.servicePortalPrefix	This system property allows you to update the customized Service Portal prefix. The default value is sp.	Plug-in admin or System administrator
x_vmw_cloudservice.Servicenow UI	This system property allows you to select the choices to see UI specific Module in ServiceNow.	Plug-in admin or System administrator
x_vmw_cloudservice.deploymentCategory	This system property allows you to set parent category of Deployment day 2 Action Catalog Items.	Plug-in admin or System administrator
x_vmw_cloudservice.machineCategory	This system property allows you to set the parent category of Resource day-2 Action Catalog Items.	Plug-in admin or System administrator
x_vmw_cloudservice.parent_category	It contains the sys_id of the category for Service Portal. The user needs to update the sys_id of category in use or sys_id category created.	Plug-in admin or System administrator
x_vmw_cloudservice.apiPageSize	This system property is used to set the page size of the API call to retrieve records from VCF Automation. The default value for the page size is 20.	Plug-in admin or System administrator

System Properties	Description	Updated by
x_vmw_cloudservice.resourceActionResource	This system property contains the deployment field sysId of the Resource Day2 Action.	Plug-in admin or System administrator
x_vmw_cloudservice.deploymentActionDeployment	This system property contains the deployment field sys Id of the Deployment Day-2 Action.	Plug-in admin or System administrator
x_vmw_cloudservice.cas_valuepicker_ui_macro_sysid	This property contains the default sysId of the UI Macro CAS_Valuepicker_UI_Macro.	Plug-in admin or System administrator
x_vmw_cloudservice.cas_valuepicker_widget_sysid	This property contains the default sysId of the widget CAS_SP_valuePicker.	Plug-in admin or System administrator
x_vmw_cloudservice.apiPageSize	The default value is 200. You can reduce the default value. VCF Automation API supports pagination of a maximum of 200 pages only.	System administrator
x_vmw_cloudservice.ThreadUsed	The default value is 8 and can be configured based on the requirements and resources available on ServiceNow. Specified threads will be used to process the job queue (data parsing). You need to set an appropriate value to the system property ensuring that not all ServiceNow worker threads are utilized for processing the job queue. **Note**: if ServiceNow instance has 8 threads, then make sure you set the system property value as 4 to restrict the threads usage to 50%.	System administrator
x_vmw_cloudservice.ThreadUsed_Maximum	The default value is 16 and can be configured based on the requirements and resources available on the ServiceNow instance. This property allows you to dedicate maximum ServiceNow threads towards importing the data into ServiceNow. The scheduled job "`Thread Limit Maximum`" will set the value of "`x_vmw_cloudservice.ThreadUsed_Maximum`" to system property "`x_vmw_cloudservice.Thread`" when the system is under less load. **Note:** Ensure that the maximum value set to the system property must be equal to or less than the number of worker threads available on the ServiceNow instance. Also setting this system property with the number of worker threads available on the ServiceNow instance will impact other ServiceNow operations too.	Plug-in admin or System administrator
x_vmw_cloudservice.ThreadUsed_Minimum	The default value is 8 and can be configured based on the requirements and resources available on the ServiceNow instance. This property allows you to dedicate minimum ServiceNow threads towards importing the data into ServiceNow. The scheduled job "`Thread Limit Minimum`" will set the value of "`x_vmw_cloudservice.ThreadUsed_Minimum`" to system property "`x_vmw_cloudservice.Thread`" when the system is under normal load. **Note**: The value set to the system property allows the number of threads for ServiceNow ITSM Plug-in for VCF Automation data import. Make sure you set the system property with the proper value so that ServiceNow instance can have sufficient worker threads to use for other operations that are not under the ServiceNow ITSM Plug-in for VCF Automation.	Plug-in admin or System administrator

System Properties	Description	Updated by
x_vmw_cloudservice.API Timeout	The default value is 60000 milliseconds. Sets API call timeout. All API calls wait for the provided time (in milliseconds) to receive the response.	System administrator
x_vmw_cloudservice.catalogLoader	System property checks for the value. If the value is set to **true**, the loader displays. For **false**, the loader is not displayed. The loader is displayed only on Catalog items during form load.	System administrator
x_vmw_cloudservice.telemetry_last_trigger	This system property stores the Date and Time when the user responds to the consent pop-up when triggered for telemetry data collection from ServiceNow ITSM Plug-in for VCF Automation to VMware.	System administrator
x_vmw_cloudservice.customer_consent_value	This system property value will be populated based on the user response to telemetry data collection. If the user responds to the telemetry pop-up as “**Yes**”, the system property value will be **true**. If the user responds to the telemetry pop-up as “**No**”, the system property value will be **false**.	System administrator
x_vmw_cloudservice.telemetry_consent_provided_user	This system property stores the sys-id of the user who provides consent for the telemetry data collection.	System administrator
x_vmw_cloudservice.telemetry_Collector_Url	This system property describes the URL for telemetry data collection.	System administrator
x_vmw_cloudservice.telemetry_collector_id	This system property represents the Collector ID for telemetry data collection.	System administrator
x_vmw_cloudservice.consent_request_trigger_gap	This system property defines after how many days the consent pop-up must be displayed again on the **Manage Endpoint** if the user has provided consent as **No** for the last telemetry pop-up.	System administrator